

Arithmetic Values, Endpoint Control and Exact Receiving Maps

Complete proof continuation and original providers

Split-Zero research programme

19 September 2026

This reader gives the complete accepted calculations received and extended on 19 September 2026, with their original provider proofs and human-source citations. The accompanying three cumulative LaTeX successors retain every byte of their preceding editions by reversible insertion. This edition has no newly assigned deposited DOI.

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Part I

September 19: arithmetic values and their exact receiving maps

The current arithmetic-baseline scalar is evaluated by the complete proofs below. Earlier statements that this scalar was uncomputed record their historical state and are superseded by the new value theorem and MRI1–3. The original absolute directional allocations and projected signs retain the exact attained metrics specified in MRI4–11. Every earlier source paragraph remains byte-recoverable; no former edition is overwritten.

Signed arithmetic value: full received proof and finite repairs

1 Signed arithmetic baseline: evaluated return at order q

18 September 2026. New written research derivation on the original fixed-packet domain. The proof and its imported inputs are identified below; this is not an independent formal verification of the entire programme.

1.1 M1. The original assigned quantity

Retain the programme's stipulated quartet and its complete multiplicity:

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad h(s) = \prod_{z \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s-z)^{m_0}.$$

The original source and reference are

$$v_h = 2\xi/h, \quad w_h(y) = \frac{|v_h(1/2 + iy)|^2}{2\pi}, \quad m_k = w_h^{*k}, \quad \sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}.$$

Their complete masses are retained; in particular $\int \sigma = \sqrt{2\pi}$. The admitted tensor orders and packet dimension are exactly

$$k = 4\ell + 1 \geq 5, \quad e = 1 + k(m_0 - 1), \quad q = e(k + 1)^2.$$

The full relation polynomial is

$$\chi_k(S) = \prod_{a, b_0=0}^k \left[S - \frac{k}{2} - (2a - k)\delta - i(2b_0 - k)\gamma \right]^e.$$

No numerical off-critical zero is supplied or asserted by this calculation. The fixed original packet is the same one used in the incoming arithmetic construction.

For its remainder map $J_N = \text{rem}_{\chi_k}$, write

$$G_N[\nu] = (J_N H_N[\nu]^{-1} J_N^*)^{-1},$$

$$\mathcal{B}_k[\nu] = \log \det G_{q-1}[\nu] + \log \det G_q[\nu] - \log \det G_{2q-1}[\nu] - \log \det G_{2q}[\nu].$$

The assigned scalar is

$$\boxed{\delta_k = \delta_{k,1}^{\text{ar}} = \mathcal{B}_k[m_k] - \mathcal{B}_k[\sigma].} \quad (\text{M1})$$

The original full arithmetic unit remains in w_h , in the programme's inclusion, and in the fixed coefficient identifications. The source-coordinate map is explicitly $P(S) \mapsto P(k/2 + iy)$, with inverse $y = (S - k/2)/i$; it carries the complete relation to $i^q Q(y)$, where $Q(y) = i^{-q} \chi_k(k/2 + iy)$. Monic phases have modulus one. A fixed change of quotient coordinates contributes the same squared determinant at all four endpoints, so its four-sign contribution is exactly zero rather than an assumed isometry.

1.2 M2. The evaluated coefficient

Define the fixed one-factor quantities

$$\alpha = \pi/2, \quad M_h(\theta) = \int_{\mathbb{R}} e^{\theta y} w_h(y) dy, \quad \mu_h(\theta) = \frac{M'_h(\theta)}{M_h(\theta)}, \quad 0 \leq \theta \leq \alpha,$$

$$\boxed{I_h = \int_0^{\pi/2} \mu_h(\theta)^2 d\theta.} \quad (\text{M2})$$

The RTM envelope gives finite first and second boundary-tilted moments, and the preceding central calculation proves $0 < I_h < \infty$. In particular this coefficient is a fixed integral of the original one-factor source, not a limit of growing minimum matrices. Equivalently, retaining the exact evenness factors,

$$I_h = \int_0^{\pi/2} \left[\frac{\int_0^\infty y \sinh(\theta y) w_h(y) dy}{\int_0^\infty \cosh(\theta y) w_h(y) dy} \right]^2 d\theta. \quad (\text{M3})$$

Retain the programme's already evaluated universal constant

$$\boxed{\mathcal{C}_\Gamma = 2\mathcal{L}(2, \pi) - 8 \log 2 + 4 > 0.} \quad (\text{M4})$$

Its complete fixed endpoint integral is specified in M4 below. Put $d = 4m_0 - 2$.

The evaluated signed return is

$$\boxed{\delta_k = -(4m_0 - 2)\mathcal{C}_\Gamma q + \frac{\log 2}{\pi} I_h k^2 + o_h(q).} \quad (\text{M5})$$

More precisely, there is a fixed-source constant C_h , on the cofinal range established in the proof, such that

$$\boxed{\left| \delta_k + (4m_0 - 2)\mathcal{C}_\Gamma q - \frac{\log 2}{\pi} I_h k^2 \right| \leq C_h q \mathcal{R}(q),} \quad (\text{M6})$$

$$\mathcal{R}(q) = \frac{\log \log(q + 16)}{\log(q + 2)} + q^{-1/200} \log^2(q + 2) \longrightarrow 0.$$

The exponent in this explicit remainder is deliberately conservative. It comes from the averaged original zeta tail, the full Gamma projection bound, the Schur localization, and the quantified convex-secant argument; it is not a fitted numerical error.

Consequently the requested coefficients are

$$\boxed{d_{\log}(h) = 0,} \quad (\text{M7})$$

$$\boxed{d_0(h) = \begin{cases} -2\mathcal{C}_\Gamma + \frac{\log 2}{\pi} I_h, & m_0 = 1, \\ -(4m_0 - 2)\mathcal{C}_\Gamma, & m_0 \geq 2. \end{cases}} \quad (\text{M8})$$

This follows from the actual dimension, not a redefinition of scale:

$$\frac{k^2}{q} = \frac{k^2}{[1 + k(m_0 - 1)](k + 1)^2} \longrightarrow \begin{cases} 1, & m_0 = 1, \\ 0, & m_0 \geq 2. \end{cases}$$

Thus $\delta_k = q d_0(h) + o_h(q)$. There is no additional term between orders q and $q \log k$, including no $q \log \log k$ term. For higher multiplicity, the separately evaluated central term in (M5) is of order k^2 , but the displayed full-return remainder is only claimed to be $o(q)$; (M5) is not an assertion that every term of the total return through order k^2 has been evaluated there.

1.3 M3. How the two previously unfinished values are completed

Let $\mathfrak{M} = M_h(\alpha)$, $b = \mu_h(\alpha)$, and let $\theta(t)$ invert μ_h on $[0, b]$. Define

$$D(t) = \log \mathfrak{M} - \log M_h(\theta(t)) - (\alpha - \theta(t))t,$$

$$W_k(y) = kD(|y|/k)1_{|y| \leq kb}.$$

The previous two source calculations have exactly the same rate function, with opposite notational sign: $F_h^{\text{cen}}(t) = -D(|t|)$. They already calculate $\int W_k = k^2 I_h$. The present calculation evaluates

the determinant response to this rate, rather than relabeling its Lebesgue integral as a determinant coefficient.

The literal endpoint decomposition is

$$\begin{aligned}\omega_k(y) &= \begin{cases} \sigma(y), & |y| \leq kb, \\ \mathfrak{M}^{-k} m_k(y), & |y| > kb, \end{cases} \\ \delta_k^{\text{out}} &= \mathcal{B}_k[\omega_k] - \mathcal{B}_k[\sigma], \quad \delta_k^{\text{cen, end}} = \mathcal{B}_k[m_k] - \mathcal{B}_k[\omega_k], \\ \boxed{\delta_k &= \delta_k^{\text{out}} + \delta_k^{\text{cen, end}}.} \end{aligned} \tag{M9}$$

The mass factor is legitimate through the exact relation $G_N[A\nu] = AG_N[\nu]$: its logarithm contributes $(q + q - q - q) \log A = 0$. The outer density is fixed throughout the central endpoint change.

Outer value. The boundary-tilted convolution is averaged against its actual zeta factor, using a single-large-summand decomposition, the proved uniform local Fourier estimate, and Ivić's short-interval second-moment bound for $E(t+U) - E(t)$. The resulting Lebesgue averaged approximation is carried into the full source and relation determinants by an explicit Meixner–Pollaczek projection bound and the two-sided determinant inequality (O20). This yields

$$\delta_k^{\text{out}} = \mathcal{B}_k[(1 + y^2)^{-d} \sigma] - \mathcal{B}_k[\sigma] + o(q).$$

The actual rational model is then evaluated: its source contribution is

$$4dq \log q + (8d \log 2 - 4d)q + o(q),$$

while its full relation contribution, with its original signs, is

$$-4dq \log q - 2dq \mathcal{L}(2, \pi) + o(q).$$

Therefore

$$\boxed{\delta_k^{\text{out}} = -d\mathcal{C}_\Gamma q + o_h(q).} \tag{M10}$$

The error is quantified in (O35). The relation calculation uses an exact convex function of the original determinant; the quantitative appendix proves the shrinking-secant estimate without differentiating a free-energy error.

Central value. The established localization gives the original central endpoint change as two bands of Gamma monic-norm losses, with $O_h(k \log^2 q)$ error. An explicit Gaussian regularization of the convex potentials $\alpha|y| + tW_k(y)$ is returned through the same Taylor-jet Schur quotients with $o(k^2)$ error. For these regularized potentials the full mass- n equilibrium is constructed explicitly and the monic squared norm is proved uniformly to be

$$H_n(t) = \pi A_{n,t} e^{-\lambda_{n,t}} [1 + O_h(n^{-1})], \quad q \leq n \leq 2q.$$

The exact Robin variation is

$$\partial_t \lambda_{n,t} = \frac{1}{\pi} \int_{-A_{n,t}}^{A_{n,t}} \frac{\widetilde{W}_k(y)}{\sqrt{A_{n,t}^2 - y^2}} dy = \frac{k^2 I_h}{2\pi n} + O_h(k^4/n^3).$$

Summing the two original bands retains the weights $1, 2, \dots, 2, 1$, so that $\sum_{r=0}^q c_r/(q+r) = 2 \log 2 + O(1/q)$. Consequently

$$\boxed{\delta_k^{\text{cen, end}} = \frac{\log 2}{\pi} I_h k^2 + o_h(k^2).} \tag{M11}$$

The stronger error is in (C32). Appendix A gives the exact matrix jumps, Airy normalization, and the shrinking-strip Cauchy estimate needed for uniformity. No fixed-potential theorem is presumed uniform on this changing family.

Equations (M9)–(M11) prove (M5). Their explicit remainders prove (M6), and the exact dimension gives (M7)–(M8).

1.4 M4. The universal constant has no remaining growing determinant

The EIQ convention is

$$V_{\alpha_E, \beta}(x) = \beta\sqrt{x} - \alpha_E \log x,$$

with $\mathcal{F}$ equal to the supremum of logarithmic energy minus this potential. At $(\alpha_E, \beta) = (2, \pi)$, take the unique $0 < \kappa < 1$, $0 < u < v$ satisfying

$$uK(\kappa) = 2, \quad vE(\kappa) = 4, \quad \kappa^2 = 1 - u^2/v^2. \quad (\text{M12})$$

Here K, E are the complete elliptic integrals in the original real convention. Put

$$A_E = u^2, \quad B_E = v^2, \quad m_E = (A_E + B_E)/2, \quad R_s = \sqrt{(A_E + s)(B_E + s)}, \quad R_0 = uv,$$

$$C_E = (u + v)^2/4, \quad \eta_E = (v - u)/(v + u), \quad \zeta_s = \frac{B_E - A_E}{2(m_E + s + R_s)},$$

$$L_s = (m_E + s - R_s) \log C_E - (m_E - R_0) - 2R_s \log(1 - \eta_E \zeta_s).$$

Then the same original logarithmic moment is

$$\boxed{\mathcal{L}(2, \pi) = \frac{1}{4} \int_0^\infty \frac{L_0 - L_s}{\sqrt{s} R_s} ds.} \quad (\text{M13})$$

The integral converges at both ends. This is precisely EIQ24–33, not a new fitted constant. The source proves $\mathcal{C}_\Gamma > 0$ independently of numerical quadrature. A high-precision non-interval diagnostic gives

$$\mathcal{C}_\Gamma \approx 0.654099894824426, \quad \log 2/\pi \approx 0.220635600152652.$$

The exact expressions (M2), (M4), and (M12)–(M13) govern the result; these decimals are not used to prove it. Appendix I1–I4 gives an explicit finite truncation and quadrature error for the fixed source functional I_h .

1.5 M5. Return to the complete original action

Retain the source's exact identity

$$J_k^{\text{action}} = \mathcal{B}_k[\sigma] + \delta_k + W_k^{\text{mono}} - P_{k,-} - P_{k,+}.$$

The superscript on W_k^{mono} distinguishes the original monic-window scalar from the central insertion $W_k(y)$. Substitution of the evaluated correction gives

$$\boxed{\begin{aligned} J_k^{\text{action}} &= \mathcal{B}_k[\sigma] - (4m_0 - 2)\mathcal{C}_\Gamma q + \frac{\log 2}{\pi} I_h k^2 \\ &\quad + W_k^{\text{mono}} - P_{k,-} - P_{k,+} + \mathcal{E}_{h,k}, \quad |\mathcal{E}_{h,k}| \leq C_h q \mathcal{R}(q). \end{aligned}} \quad (\text{M14})$$

The Gamma baseline remains the exact baseline, not just its positive leading q^2 term. The correction is $O_h(q)$, and therefore does not cancel that q^2 term. The independent proper-source metric, directional allocations, and retained-class projected action are not assigned values by this scalar calculation.

The assignment recorded as section 15.1 is thus evaluated through its requested q -scale precision. No purity or Frobenius value has been substituted for the arithmetic minima. The derivation does not assert programme closure or the existence of an off-critical zero.

2 The outer arithmetic return: its value, not only a bound

This is a new derivation on the fixed original packet. The source inputs are the RTM envelopes, the consolidated polynomial-prefactor comparison, the original Gamma normalization, and the original source/relation determinant identity. The external analytic input is the established short-interval mean-square bound for the zeta mean-square error in Ivić [I, (2.1)]. All subsequent comparisons below are made on the specified polynomial subspaces and attained quotients.

2.1 O1. Objects and exact scalar bookkeeping

Retain

$$\alpha = \pi/2, \quad m = m_0 \geq 1, \quad d = 4m - 2, \quad p = 2d - \frac{1}{2}, \quad a = 2d + \frac{1}{2} = p + 1, \\ k = 4\ell + 1, \quad e = 1 + k(m - 1), \quad q = e(k + 1)^2, \quad Q(y) = i^{-q} \chi_k(k/2 + iy).$$

The degree of the real even monic polynomial Q is exactly q , with all original repetitions. Its roots have modulus at most $R = k\sqrt{\delta^2 + \gamma^2}$.

Write

$$M(\theta) = \int e^{\theta y} w_h(y) dy, \quad \mathfrak{M} = M(\alpha), \quad b = (\log M)'(\alpha), \quad f(y) = \mathfrak{M}^{-1} e^{\alpha y} w_h(y).$$

Thus f is an auxiliary probability density of mean b and variance $V_b > 0$, and the original convolution is recovered exactly by

$$\boxed{\mathfrak{M}^{-k} m_k(y) = e^{-\alpha y} f^{*k}(y).} \quad (\text{O1})$$

The original measure is not replaced by this auxiliary probability. Equation (O1) records the complete inverse factor.

Let

$$\omega_k(y) = \begin{cases} \sigma(y), & |y| \leq kb, \\ \mathfrak{M}^{-k} m_k(y), & |y| > kb, \end{cases} \quad \tau_d(y) = (1 + y^2)^{-d} \sigma(y).$$

The outer endpoint return is

$$\delta_k^{\text{out}} = \mathcal{B}_k[\omega_k] - \mathcal{B}_k[\sigma].$$

We prove

$$\boxed{\delta_k^{\text{out}} = -d\mathcal{C}_\Gamma q + o_h(q), \quad \mathcal{C}_\Gamma = 2\mathcal{L}(2, \pi) - 8 \log 2 + 4.} \quad (\text{O2})$$

Here $\mathcal{L}$ is the original EIQ logarithmic moment, with its original potential and normalization. Its fixed integral is reproduced in the main proof.

For a scalar $A > 0$, the identity on every affine remainder fibre gives $G_N[A\nu] = AG_N[\nu]$. Therefore

$$\mathcal{B}_k[A\nu] - \mathcal{B}_k[\nu] = (q + q - q - q) \log A = 0. \quad (\text{O3})$$

This is the scalar cancellation used below. No scalar is removed from an individual determinant.

2.2 O2. The actual one-factor tail

The retained arithmetic density is exactly

$$w_h(y) = \frac{(y^2 + 1/4)^2}{\sqrt{\pi}} \frac{|\zeta(1/2 + iy)|^2}{[(y - \gamma)^2 + \delta^2][(y + \gamma)^2 + \delta^2]^{2m}} \sigma(y).$$

Consequently, as $y \rightarrow +\infty$,

$$\boxed{f(y) = c_f y^{-a} |\zeta(1/2 + iy)|^2 (1 + O_h(y^{-1})), \quad c_f = \frac{\sqrt{2/\pi}}{\mathfrak{M}}.} \quad (\text{O4})$$

Indeed the rational factor is $y^{-2d}(1 + O_h(y^{-2}))$, and $e^{\alpha y}\sigma(y) = \sqrt{2}y^{-1/2}(1 + O(y^{-1}))$. The Gamma mass and the factor $\sqrt{\pi}$ are thus retained.

The original envelope yields

$$0 \leq f(y) \leq C_h(1 + |y|)^{-p}, \quad p \geq 7/2. \quad (\text{O5})$$

Its negative tail has the additional factor $e^{-2\alpha|y|}$. In particular the variance and a 9/4-moment are finite.

Put $Z(y) = |\zeta(1/2 + iy)|^2$ and

$$\mathcal{L}_\zeta(y) = \log y + c_\zeta, \quad c_\zeta = 2\gamma_E - \log(2\pi).$$

The classical mean-square formula gives, uniformly for $|z| \leq T^{19/20}$,

$$\int_T^{2T} f(y - z) dy \leq C_h T^{1-a} \log T. \quad (\text{O6})$$

Only this upper bound is used in the convolution comparison; no pointwise lower bound for Z is used. For example, the usual error bound $E(T) = O(T^{1/2} \log^2 T)$, or any established $o(T \log T)$ bound, implies $\int_{T/2}^{3T} Z \ll T \log T$. Equations (O4) and (O6) then follow on the indicated shifted interval. The same estimate at a finite number of comparable dyadic intervals will always be understood literally.

2.3 O3. Uniform total variation at the boundary tilt

The incoming Fourier proof gives, for every real x ,

$$\left| f^{*n}(nb + x) - \frac{e^{-x^2/(2nV_b)}}{\sqrt{2\pi nV_b}} \right| \leq C_h n^{-(1+1/4)/2}. \quad (\text{O7})$$

Translation in Fourier inversion has modulus one, so the absolute error at the mean has this same bound at every shifted point.

Let g_n be the Gaussian density in (O7), centered at nb . Integrate (O7) on $|x| \leq \sqrt{n} n^{1/24}$. This costs $O_h(n^{-1/12})$. On its complement, Chebyshev's inequality for each of the two probability measures costs $O_h(n^{-1/12})$. Therefore

$$\boxed{\|f^{*n} - g_n\|_{L^1(\mathbb{R})} \leq C_h n^{-1/12}}. \quad (\text{O8})$$

This is a total-variation estimate for the actual tilted convolution; all masses in (O1) remain unchanged.

2.4 O4. A single large summand, in the averaged norm actually needed

Throughout this section take

$$q^{9/10} \leq T \leq 128q.$$

For each fixed original multiplicity and packet, this gives

$$c_h T^{1/3} \leq k \leq C_h T^{5/9}. \quad (\text{O9})$$

Set $R_T = T^{19/20}$, $f_s = f1_{|y| \leq R_T}$, and $f_l = f - f_s$. These are subprobability densities, not normalized substitutes.

Expand $(f_s + f_l)^{*k}$. Its terms with at least two large summands have total mass at most

$$C_h k^2 R_T^{-2(p-1)}. \quad (\text{O10})$$

Relative to $kT^{1-a} \log T = kT^{-p} \log T$, this is

$$O_h(kT^p R_T^{-2(p-1)} / \log T) = O_h(T^{-2/3})$$

on the cofinal range; the worst exponent occurs at $p = 7/2$, and is $5/9 + 7/2 - (19/10)(5/2) = -25/36$.

The all-small convolution has exponentially small mass in $[T, 2T]$. To check this directly, use $\lambda = (\log T)/R_T$. Taylor's formula and the finite second moment give

$$\int e^{\pm \lambda y} f_s(y) dy \leq \exp(C_h \lambda + C_h \lambda^2 T).$$

Thus either tail beyond $T/2$ is at most

$$\exp[-c_h T^{1/20} \log T], \quad (\text{O11})$$

for all sufficiently large original k . The inequalities $k/T \rightarrow 0$ and $k \log T / R_T \rightarrow 0$, from (O9), justify the last assertion.

The remaining term is exactly $k f_l * f_s^{*(k-1)}$. Split its second variable z into $|z| \leq R_T$, $R_T < |z| \leq T/2$, and $|z| > T/2$. On the middle region, (O5) and the variance of the original sum give the integrated bound

$$C_h k^2 T^{1-p} / R_T^2. \quad (\text{O12})$$

For clarity, $f_s^{*(k-1)} \leq f^{*(k-1)}$, and on $|z| > R_T$ one has $|z - (k-1)b| \geq R_T/2$. Its mass is therefore at most $C_h k / R_T^2$. The first density is at most $C_h T^{-p}$ on the relevant shifted window. Relative to the leading mass, (O12) is $O_h(kT/R_T^2)$, hence a negative fixed power of T . The final region is bounded by (O11), using the full mass of f_l .

On $|z| \leq R_T$, the condition $y-z > R_T$ holds for every $T \leq y \leq 2T$. Replacing the truncated rest convolution by the full one has total-variation cost at most $(k-1) \int_{|u| > R_T} f(u) du$. The important bound for its convolution into this window is (O6), not a pointwise bound charged over the whole interval. The resulting relative error is at most $C_h k R_T^{1-p}$.

Replace this full rest convolution by its Gaussian using (O8) and the same shifted-window estimate (O6). Its relative error is $O_h(k^{-1/12})$. The Gaussian mass outside $|z| \leq R_T$ is exponentially small. Finally (O4), uniformly on the retained shifted window, allows replacement of $(y-z)^{-a}$ by y^{-a} at relative integrated cost $O_h(R_T/T)$.

Put $n = k-1$, $G = \sqrt{2nV_b}$, and

$$Z_G(t) = \frac{1}{\sqrt{\pi}G} \int_{\mathbb{R}} e^{-(u/G)^2} Z(t+u) du.$$

The preceding decomposition proves

$$\left| \int_T^{2T} \left| f^{*k}(y) - k c_f y^{-a} Z_G(y - nb) \right| dy \right| \leq C_h k T^{1-a} \log T [k^{-1/12} + T^{-1/20}]. \quad (\text{O13})$$

Every discarded event in deriving this inequality has the displayed quantitative bound. No pointwise regular-variation assertion about the zeta factor has been made.

2.5 O5. The zeta factor is averaged with an adequate error

The external input [I, (2.1)] is the established bound

$$\int_T^{2T} |E(t+U) - E(t)|^2 dt \ll_{\varepsilon} TU \log^3 T + T^{1/2+\varepsilon} U^2 + T^{1+\varepsilon} U^{1/2}, \quad 1 \ll U \leq \frac{1}{2} \sqrt{T}, \quad (\text{O14})$$

where

$$E(T) = \int_0^T Z(t) dt - T[\log(T/2\pi) + 2\gamma_E - 1].$$

This bound concerns the actual E , not a conjectural pointwise short-interval estimate.

Integration by parts against the Gaussian, subtracting the constant $E(t)$ under the odd kernel, gives

$$\mathcal{Z}_G(t) - \mathcal{L}_\zeta(t) = \frac{2}{\sqrt{\pi}G^3} \int u e^{-(u/G)^2} [E(t+u) - E(t)] du + O(G/T) + O(T^{-100}). \quad (\text{O15})$$

One can restrict $|u| \leq G \log T$ before this operation. The omitted Gaussian tails are bounded by any chosen inverse power, using polynomial bounds already implied by (O4)–(O5). On that range the arguments are positive and comparable to T .

For $|u| \geq 1$, apply (O14), using a finite dyadic cover after a negative translation. Minkowski's inequality yields

$$T^{-1/2} \|\mathcal{Z}_G - \mathcal{L}_\zeta\|_{L^2[T, 2T]} \ll_\varepsilon G^{-1/2} \log^{3/2} T + T^{-1/4+\varepsilon} + T^\varepsilon G^{-3/4} + G/T. \quad (\text{O16})$$

The part $|u| < 1$ needs no fourth-moment input: by positivity of Z , its error increment is bounded by a length-four increment of E , plus $O(\log T)$. Equation (O14) at this fixed length gives a contribution $O(T^\varepsilon G^{-3})$, already smaller than the right side of (O16).

In the present range, $c_h T^{1/6} \leq G \leq C_h T^{5/18}$. Thus the use of (O14), including $G \log T < \sqrt{T}/2$, is valid on a cofinal range. Fix $\varepsilon = 1/1000$. Apply Cauchy–Schwarz to (O16), account for the shift $nb = O_h(k)$, and combine with (O13). With, for example, $\eta_0 = 1/200$, one obtains

$$\int_T^{2T} \left| \frac{\mathfrak{M}^{-k} m_k(y)}{\frac{k}{\sqrt{\pi} \mathfrak{M}} \mathcal{L}_\zeta(y) (1+y^2)^{-d} \sigma(y)} - 1 \right| dy \leq C_h T^{1-\eta_0}. \quad (\text{O17})$$

The original evenness supplies the corresponding negative interval. This is the averaged estimate subsequently used in full determinants. It is not a pointwise replacement of the arithmetic density.

2.6 O6. A uniform bound for the original Gamma projection kernel

Let $K_N^\sigma(y, z) = \sum_{j=0}^N \varphi_j(y) \overline{\varphi_j(z)}$. Then

$$K_N^\sigma(y, y) \sigma(y) \leq 20 \log(N+2), \quad y \in \mathbb{R}, \quad N \geq 0. \quad (\text{O18})$$

Here is a derivation preserving the normalization. In the orthonormal Meixner–Pollaczek convention of Koelink–Van der Jeugt [K, Proposition 2.1], take $\lambda = 1/4$, angle $\pi/2$, and $x = y/2$. Their weight is $w_\lambda(x) = 2^{2\lambda} |\Gamma(\lambda + ix)|^2 / (2\pi)$, and the exact map is $\varphi_j(y) = 2^{\lambda-1/2} p_j^{(\lambda)}(y/2)$. Hence the original diagonal density is one half of their diagonal density.

For $0 < t < 1$, put $R_t = 4t/(1-t)^2$. Their Poisson kernel on the diagonal is

$$\frac{(1+t)^{2ix} (1-t)^{-2\lambda-2ix}}{\Gamma(2\lambda)} {}_2F_1(\lambda + ix, \lambda + ix; 2\lambda; -R_t).$$

Euler's integral (DLMF 15.6.1) is valid because $\Re(2\lambda) > \Re(\lambda + ix) > 0$. Its Gamma denominator is exactly $|\Gamma(\lambda + ix)|^2$, which cancels the displayed weight. Every remaining imaginary power has modulus one. Thus the weighted kernel is at most

$$\frac{2^{2\lambda}}{2\pi} (1-t)^{-2\lambda} \int_0^1 u^{\lambda-1} (1-u)^{\lambda-1} (1+R_t u)^{-\lambda} du.$$

For $t \geq 1/2$, split at $1/R_t$ and $1/2$. At $\lambda = 1/4$, the integral is at most $R_t^{-\lambda} (16 + 2 \log R_t)$. The original half-weight factor and $(1-t)^{-2\lambda} R_t^{-\lambda} = (4t)^{-\lambda}$ therefore bound the original Poisson density by $(16 + 2 \log R_t) t^{-1/4} / (4\pi)$.

All diagonal summands are nonnegative. Take $t = (N+1)/(N+2)$, multiply this bound by $t^{-N} \leq e$, and use $R_t \leq 4(N+2)^2$. This proves the enlarged explicit bound (O18).

For every fixed subspace $S \subset \mathcal{P}_{2q}$, its Gamma-orthogonal projector satisfies

$$K_S(y, y) \leq K_{2q}^\sigma(y, y). \quad (\text{O19})$$

This includes each complete original relation space, rather than only monomials.

2.7 O7. A determinant sandwich on the entire subspace

Let S be any finite polynomial subspace, in a Gamma-orthonormal frame, and let $r > 0$ have the required integrable moments and logarithms. Expansion of the Gram determinant gives

$$\det \text{Gram}_S[r\sigma] = \frac{1}{s!} \int |\det[\phi_i(y_j)]|^2 \prod_j r(y_j) d\sigma(y_j).$$

At $r = 1$ the right-hand measure has total mass one, and its one-point density is $K_S(y, y)d\sigma(y)$. Jensen's inequality, and separately $\log \det A \leq \text{Tr}(A - I)$, give

$$\boxed{\int \log r K_S d\sigma \leq \log \det \text{Gram}_S[r\sigma] \leq \int (r - 1) K_S d\sigma.} \quad (\text{O20})$$

All determinant cross terms remain in the squared determinant. This is why a Lebesgue averaged tail estimate can be used without assuming an arithmetic-kernel limit.

2.8 O8. Patch a sublinear interval through a proved Schur comparison

Set $K = q^{9/10}$,

$$A_k = \frac{k\mathcal{L}_\zeta(q)}{\sqrt{\pi\mathfrak{M}}} > 0, \quad \widehat{\omega}_k(y) = \begin{cases} A_k \tau_d(y), & |y| \leq K, \\ \omega_k(y), & |y| > K. \end{cases} \quad (\text{O21})$$

This operation has an $o(q)$ signed determinant cost; it is not a deletion of the inner interval.

Indeed the consolidated whole-line polynomial bounds give, on degree at most $2q$,

$$q^{-C_h} \|P\|_\sigma^2 \leq \|P\|_{\omega_k}^2, \quad \|P\|_{\widehat{\omega}_k}^2, \quad A_k \|P\|_{\tau_d}^2 \leq q^{C_h} \|P\|_\sigma^2. \quad (\text{O22})$$

Polynomial lower weights are transferred by Jensen and the original Gamma multiplication bound. The scalar A_k is also between fixed powers of q .

Use the first r Taylor coefficients at zero, restricted to each actual source or relation subspace. Its image has dimension at most r ; its full kernel consists of the elements vanishing to order r . The generating function of the original Gamma polynomials and Cauchy's estimate give

$$\int_{-K}^K |P|^2 d\sigma \leq 4^{1-r} e^2 (2q+1)^2 \sqrt{2q+2} [2(2q+2)]^{2K} \|P\|_\sigma^2 \quad (\text{O23})$$

on that kernel. Choose $r = \lceil C_h(K+1) \log(q+2) \rceil$, with the constant enlarged so that the right-hand fraction, after (O22), is at most q^{-20} . This has $r = o(q)$.

Factor the determinant into its restriction to this kernel and its attained quotient on the actual Taylor image. The kernel's relative log-determinant change is $O(q^{-18})$. The quotient, of dimension at most r , costs $O_h(r \log q)$ by (O22). This proves at every original endpoint and on every original relation space

$$|\log \det H_S[\widehat{\omega}_k] - \log \det H_S[\omega_k]| = O_h(q^{9/10} \log^2 q) = o(q). \quad (\text{O24})$$

The fixed coordinate determinant in this Schur factorization cancels between the two metrics. The Taylor observation is only an auxiliary comparison; it does not replace the programme's observation or its relation ideal.

2.9 O9. Evaluate the patched density in the full determinants

Put

$$r_k(y) = \widehat{\omega}_k(y) / (A_k \tau_d(y)).$$

It is exactly one on $[-K, K]$. On each dyadic outer interval below $128q$, equation (O17) says that it differs in L^1 from

$$a_q(y) = \mathcal{L}_\zeta(|y|) / \mathcal{L}_\zeta(q)$$

by at most $C_h T^{1-\eta_0}$. Here a_q stays between two fixed positive constants. The polynomial lower and upper bounds on r_k imply on this region

$$|\log r_k - \log a_q| \leq C_h \log q |r_k - a_q|. \quad (\text{O25})$$

This follows by separating $|r_k - a_q| \leq a_q/2$ from its complement; no pointwise zeta lower bound is used.

Equations (O18)–(O19) show that the errors in both sides of the determinant sandwich (O20) are $O_h(q^{1-\eta_0} \log^2 q) = o(q)$. For the deterministic factor a_q , on $q/\log^4 q \leq |y| \leq 128q$ it differs from one by $O(\log \log q / \log q)$, so use the exact kernel mass $\dim S \leq 2q + 1$. On the remaining interval $K \leq |y| \leq q/\log^4 q$, use its length and (O18). Both costs are $o(q)$.

The whole remaining tail is retained. From (O22) and the pointwise envelopes, $r_k(y) \leq q^{C_h}(1+y^2)^d$ and $|\log r_k(y)| \leq C_h[\log q + \log(1+y^2)]$. Repeated Gamma multiplication, with an extra power y^q , bounds the integral of these functions against $K_{2q}^\sigma d\sigma$ on $|y| > 128q$ by $e^{-c_h q}$ times a fixed power of q . For example

$$\int_{|y| > 128q} y^{2d} |P|^2 d\sigma \leq (128q)^{-2q} [2(3q + d + 1)]^{2(q+d)} \|P\|_\sigma^2$$

for $\deg P \leq 2q$, and then sum an orthonormal frame. Thus

$$\boxed{\int |r_k - 1| K_S d\sigma = o_h(q), \quad \int |\log r_k| K_S d\sigma = o_h(q)} \quad (\text{O26})$$

uniformly for all the subspaces needed here.

The reference for our desired comparison is τ_d , not σ . Remove this fixed denominator by the exact multiplication map

$$P \mapsto (y + i)^d P.$$

It is an isometry from $\mathcal{P}_{N-d}$ in σ onto $(y + i)^d \mathcal{P}_{N-d}$ in τ_d , because $|y + i|^{2d} = (1 + y^2)^d$. With the factor r_k , the same map transports $r_k \sigma$ to $r_k \tau_d$. On a relation space use instead $QP \mapsto Q(y + i)^d P$. The omitted fixed quotient has dimension at most d , and its two attained metrics have logarithmic relative cost $O_h(d \log q)$ by (O22). All these are fixed subspaces and exact minimum quotients.

Apply (O20) and (O26) after this multiplication. We obtain the complete source and relation comparisons, each with error $o_h(q)$. Summing their original signs, using (O3) for A_k , and returning the patch by (O24), proves

$$\boxed{\delta_k^{\text{out}} = \mathcal{B}_k[\tau_d] - \mathcal{B}_k[\sigma] + o_h(q)}. \quad (\text{O27})$$

The zeta smoothing has therefore been carried through the original minima and their full cross blocks. It is not only a density estimate.

2.10 O10. Evaluate the fixed rational model: the source part

In $\mathcal{P}_N$, use the monic frame

$$1, y, \dots, y^{d-1}, (y + i)^d, y(y + i)^d, \dots, y^{N-d}(y + i)^d.$$

Its determinant is one. The last block has exactly the Gamma Gram on $\mathcal{P}_{N-d}$. The first d divided derivatives at $-i$ identify its attained quotient; their matrix on the first d monomials has determinant one. Its Gram eigenvalues are bounded above by fixed trial lifts and below by a negative fixed power of $N + 1$. The latter follows from the polynomial lower comparison $\tau_d \succeq C_d(N + 1)^{-2d} \sigma$ and the Gamma jet covariance estimate at the fixed point $-i$. Therefore

$$\log \det H_N[\tau_d] = \sum_{j=0}^{N-d} \log \gamma_j + O_d(\log(N + 2)). \quad (\text{O28})$$

This includes the fixed jet quotient rather than setting its determinant to one.

It follows, at the four original source signs, that

$$\boxed{\sum_N \epsilon_N \log \frac{\det H_N[\tau_d]}{\det H_N[\sigma]} = 4dq \log q + (8d \log 2 - 4d)q + o(q).} \quad (\text{O29})$$

To check the coefficients directly, the relative determinant at N is $-\sum_{j=N-d+1}^N \log \gamma_j + O(\log N)$, and $\log \gamma_j = 2j \log j - 2j + O(\log(j+2))$. A fixed shift in j costs only $O_d(\log q)$.

2.11 O11. Evaluate the fixed rational model: the full relation part

For a relation rank $n = q$ or $q+1$, define

$$\begin{aligned} D_{k,n}(t) &= \det \text{Gram}\{Q, yQ, \dots, y^{n-1}Q\}_{(1+y^2)^{qt}\sigma}, \\ F_{k,n}(t) &= q^{-2} [\log D_{k,n}(t) - \log D_{k,n}(0) - 2qnt \log q]. \end{aligned} \quad (\text{O30})$$

Andreief's identity on these exact columns proves convexity of $F_{k,n}$: its second derivative is a variance of the sum of the logarithms of $1 + y_j^2$, multiplied by a positive scalar. This is the whole relation Gram, not a product of independent trial norms.

We verify its leading limit at every fixed $|t| \leq 1/8$:

$$\boxed{F_{k,n}(t) \longrightarrow \frac{1}{2} \{\mathcal{F}(2+2t, \pi) - \mathcal{F}(2, \pi)\}.} \quad (\text{O31})$$

Here $\mathcal{F}$ is exactly the original GEL energy, the negative of the EIQ minimum energy.

The complete root comparison needed for (O31) is as follows. Set $Y = 2^{-16}q$. Outside $[-Y, Y]$, opposite-root pairing gives

$$\left| \log \frac{|Q(y)|^2 (1+y^2)^{qt}}{|y|^{2q(1+t)}} \right| \leq \frac{qR^2}{Y^2 - R^2} + q|t| \log(1+Y^{-2}) = O_h(1). \quad (\text{O32})$$

Inside, use $(y^2 + R^2)^q (1+y^2)^{qt}$. For $t \geq 0$ its maximum is at Y . For $t < 0$, its only possible interior stationary point is a minimum, as differentiation gives numerator $(1+t)y^2 + 1 + tR^2$; hence its maximum is at zero or at Y . The zero-endpoint bound R^{2q} is negligible relative to $q^{2q(1+t)}$, uniformly for $|t| \leq 1/8$, because $R^2/q^{7/4} \rightarrow 0$. The Y -endpoint bound is at most $[(5/4)2^{-28}]^q q^{2q(1+t)}$. The same complete shifted-Legendre estimate as in the original root comparison, now on $[q, 2q]$, shows that both inner integrals are exponentially small relative to the full reference-power norm, for every polynomial of degree at most q . Thus the full quadratic forms are comparable with log width $O_h(1)$; their rank- n log determinants differ by $O_h(q)$. Their leading q^2 -scaled limits agree.

For the reference power norm, use the literal parity map $x = (y/q)^2$. The two determinant blocks have sizes $\lceil n/2 \rceil$, $\lfloor n/2 \rfloor$, and starting powers $q(1+t)$, $q(1+t)+1$, respectively. Each block has exactly the GEL3 Jacobian and source factor. GEL10 gives the energy at $(2+2t, \pi)$, since the block size divided by q tends to $1/2$. The two q^2 -coefficients add to $1/2$. The change of the exact scaling power is precisely $2qnt \log q$, the term subtracted in (O30). This proves (O31). The quantitative appendix Q1–Q3 strengthens it to a uniform error of $O_h(\log(q+2)/q)$ on the same fixed parameter interval. Its proof retains the original Gamma ensemble and uses an explicitly accounted circle regularization only to bound each finite configuration.

The EIQ variational calculation proves $\partial_\alpha \mathcal{F}(\alpha, \beta) = \mathcal{L}(\alpha, \beta)$, continuously. Thus the right side of (O31) is differentiable at zero, with derivative $\mathcal{L}(2, \pi)$. Convexity now determines the shrinking secant from $-d/q$ to zero. To be explicit, enclose that secant between secants on two fixed small intervals to the left and right of zero, apply (O31) there, and then shrink the fixed intervals. This gives

$$\boxed{\log \frac{D_{k,n}(-d/q)}{D_{k,n}(0)} = -2dn \log q - dq \mathcal{L}(2, \pi) + O_h(\sqrt{q \log(q+2)}), \quad n = q, q+1.} \quad (\text{O33})$$

For the stated rate use the uniform free-energy remainder and a convex secant of width $\sqrt{\log(q+2)/q}$, as proved in Q3. No error term from a q^2 -asymptotic has been differentiated.

The low rank-one relation is retained. Its relative norm is at most one and, by Jensen and Gamma multiplication applied to Q , at least $[1 + 4(q+1)^2]^{-d}$. Its log contribution is therefore $O_d(\log q)$. The other low relation is empty. Applying the actual signs in the source/relation identity gives a total relation contribution to the baseline of

$$-4dq \log q - 2dq\mathcal{L}(2, \pi) + o_h(q). \quad (\text{O34})$$

Combining this with (O29) proves

$$\mathcal{B}_k[\tau_d] - \mathcal{B}_k[\sigma] = -d[2\mathcal{L}(2, \pi) - 8 \log 2 + 4]q + o_h(q).$$

Together with (O27), this proves the value (O2).

The errors in the density and determinant transfer were quantified before taking their limits. The sublinear patch costs $O_h(q^{9/10} \log^2 q)$, the averaged zeta error costs $O_h(q^{199/200} \log^2 q)$, and the residual factor $\mathcal{L}_\zeta(y)/\mathcal{L}_\zeta(q)$ costs $O_h(q \log \log(q+16)/\log(q+2))$. The remaining model error is $O_h(\sqrt{q \log(q+2)})$ by Q3. Enlarging the fixed-source constant absorbs the smaller terms and gives

$$\left| \delta_k^{\text{out}} + d\mathcal{C}_\Gamma q \right| \leq C_h \left[q \frac{\log \log(q+16)}{\log(q+2)} + q^{199/200} \log^2(q+2) \right] \quad (\text{O35})$$

on the cofinal range specified in the proof. Both terms divided by q tend to zero. In particular no $q \log k$, $q \log \log k$, or other contribution exceeding order q remains hidden in this outer return.

3 Quantitative free-energy remainder used in the outer value

This appendix supplies a uniform remainder for the convex-secant step. It uses the original EIQ equilibrium and proves the finite-size upper bound without differentiating an asymptotic error.

3.1 Q1. A finite-configuration bound from the actual equilibrium

Let $V(x) = \beta\sqrt{x} - \alpha_E \log x$ on $(0, \infty)$, with (α_E, β) in a fixed compact neighborhood of $[7/4, 9/4] \times \{\pi\}$. Let ρ be its original EIQ probability density, $U(x) = \int \log|x - y|\rho(y)dy$, and λ its Euler constant. Thus

$$2U(x) - V(x) \leq -\lambda \quad (x > 0), \quad \mathcal{F}(\alpha_E, \beta) = -\lambda - \iint \log|x - y|\rho(x)\rho(y)dxdy. \quad (\text{Q1})$$

The EIQ endpoint and density formulas show that the supports lie in one fixed compact subinterval of $(0, \infty)$, their lengths stay positive, and $\|\rho\|_\infty$ is uniformly bounded. Their logarithmic energies and entropies relative to the reference probability

$$d\omega(x) = \frac{1}{2}x^{-1/2}e^{-\sqrt{x}}dx$$

are uniformly finite. These assertions follow directly from the positive density formula with its square-root endpoints and its smooth parameter dependence.

For a real configuration $x_1, \dots, x_n > 0$, replace each point by uniform mass on the complex circle $x_j + \varepsilon e^{i\theta}$, and let ν_ε be their average. This is an auxiliary embedding of the finite configuration into $\mathbb{C} = \mathbb{R}^2$, not a change in the original ensemble integration. The circle mean identity gives

$$\frac{1}{2\pi} \int_0^{2\pi} \log|z - \varepsilon e^{i\theta}|d\theta = \log \max\{|z|, \varepsilon\}. \quad (\text{Q2})$$

It follows by the harmonic mean-value property outside the circle, its value at zero inside, and continuity at the circle. Applying (Q2) twice gives an off-diagonal mean at least $\log|x_i - x_j|$; the diagonal self-energy is exactly $\log \varepsilon$.

The logarithmic energy of a compactly supported real signed measure of total mass zero on $\mathbb{R}^2$ is nonpositive in the $+\log$ convention:

$$-\iint \log|z - w|d\eta(z)d\eta(w) = \frac{1}{2} \int_0^\infty \frac{1}{t} \iint e^{-t|z-w|^2} d\eta(z)d\eta(w)dt \geq 0. \quad (\text{Q3})$$

The equality follows from the same scalar integral used in EIQ20; total mass zero cancels its constant. The Gaussian Fourier transform on $\mathbb{R}^2$ makes every inner integral nonnegative. All logarithmic integrals for the circle measures and the bounded EIQ density are finite, so the identity follows by truncation and dominated convergence. This proves the particular two-dimensional comparison used here rather than assuming a new equilibrium problem in the plane.

Apply (Q3) to $\nu_\varepsilon - \rho(x)dx$. By (Q2),

$$0 \leq \int U(z)d\nu_\varepsilon(z) - \frac{1}{n} \sum_j U(x_j) \leq 2\|\rho\|_\infty \varepsilon.$$

Consequently

$$\frac{2}{n^2} \sum_{i < j} \log|x_i - x_j| + \frac{\log \varepsilon}{n} \leq \frac{2}{n} \sum_j U(x_j) - \iint \log|x - y|\rho(x)\rho(y)dxdy + 4\|\rho\|_\infty \varepsilon.$$

Subtract $n^{-1} \sum_j V(x_j)$, use (Q1), and take $\varepsilon = 1/n$. We obtain the uniform pointwise inequality

$$\boxed{2 \sum_{i < j} \log|x_i - x_j| - n \sum_j V(x_j) \leq n^2 \mathcal{F}(\alpha_E, \beta) + n \log n + Cn.} \quad (\text{Q4})$$

Repeated configuration points cause no difficulty: their original left side is $-\infty$. No original integration region is removed in this bound.

3.2 Q2. Apply it to the original Gamma ensemble

The exact GEL coordinate map $y^2 = q^2x$ has reference factor

$$B_q(x) = 2qe^{\sqrt{x}}\sigma(q\sqrt{x}),$$

and ensemble

$$J_{n,q}(n\alpha_E) = \frac{1}{n!} \int \prod_{i < j} (x_i - x_j)^2 \prod_j x_j^{n\alpha_E} B_q(x_j) d\omega(x_j). \quad (\text{Q5})$$

All Gamma masses and coordinate powers remain those in GEL3.

The original GEL5 bound is

$$B_q(x) \leq Cq^{3/2} \exp[-(\pi q/2 - 3/2)\sqrt{x}].$$

Thus (Q4), integrated against the probability ω^n , proves the upper estimate with potential coefficient $\beta_n = \pi q/(2n) - 3/(2n)$. Replacing this coefficient by $\pi q/(2n)$ costs $O(n)$ in the exponent, because the EIQ derivative $\partial_\beta \mathcal{F} = -\int \sqrt{x}\rho$ is uniformly bounded here. The factor $(Cq^{3/2})^n/n!$ costs at most $O(n \log(q+n))$.

For the lower estimate, restrict to the actual equilibrium support for $(\alpha_E, \pi q/(2n))$, use GEL6, and change the integration measure to ρ^n . Jensen's inequality gives the equilibrium energy minus the entropy and the missing diagonal term. The uniform bounds in Q1 show that these costs are $O(n)$, apart from the explicit Gamma prefactor and $n!$, whose costs are $O(n \log(q+n))$.

Therefore, uniformly when q/n remains in a fixed neighborhood of two and α_E remains in the stated compact interval,

$$\left| \log J_{n,q}(n\alpha_E) - n^2 \mathcal{F}\left(\alpha_E, \frac{\pi q}{2n}\right) \right| \leq Cn \log(q+n). \quad (\text{Q6})$$

This gives an explicit order for the remainder in the already established free-energy limit, on exactly the parameters required by the rational-model comparison.

3.3 Q3. Quantify the shrinking secant on the full original relation Gram

Use the convex function $F_{k,n}(t)$ in (O30), for relation rank $n = q$ or $q+1$. The full paired-root comparison (O32) has logarithmic quadratic-form width $O_h(1)$. Its rank- n determinant cost is $O_h(q)$. Apply (Q6) to the two literal parity blocks. Their sizes are $q/2 + O(1)$, and their potential parameters differ from $(2+2t, \pi)$ by $O(1/q)$. EIQ parameter smoothness and the exact scaling powers in (O30) therefore give, uniformly for $|t| \leq 1/8$,

$$|F_{k,n}(t) - F_*(t)| \leq C_h \frac{\log(q+2)}{q}, \quad F_*(t) = \frac{1}{2}[\mathcal{F}(2+2t, \pi) - \mathcal{F}(2, \pi)]. \quad (\text{Q7})$$

The derivative $F'_*(0) = \mathcal{L}(2, \pi)$, and F''_* is bounded on this fixed interval by the original smooth endpoint formulas.

Set $\varepsilon_q = \sqrt{\log(q+2)/q}$. On its cofinal range $d/q < \varepsilon_q/2$, convexity places the secant on $[-d/q, 0]$ between secants on fixed-width adjacent intervals contained in $[-2\varepsilon_q, 2\varepsilon_q]$. The error from (Q7), divided by their length, is $O_h(\log(q+2)/(q\varepsilon_q))$. The difference of the corresponding F_* secants from $F'_*(0)$ is $O_h(\varepsilon_q)$. Thus

$$\frac{F_{k,n}(0) - F_{k,n}(-d/q)}{d/q} = \mathcal{L}(2, \pi) + O_h\left(\sqrt{\frac{\log(q+2)}{q}}\right).$$

Return the exact powers from (O30):

$$\log \frac{D_{k,n}(-d/q)}{D_{k,n}(0)} = -2dn \log q - dq \mathcal{L}(2, \pi) + O_h(\sqrt{q \log(q+2)}). \quad (\text{Q8})$$

This estimates a secant of the complete attained relation determinant. It does not differentiate an error term and does not assign independently favorable values at its endpoints.

4 The central endpoint contribution: evaluation of its coefficient

This part evaluates the central Gamma quantity left by the consolidated localization. It does not replace the arithmetic source-band kernel by an assumed constant. Instead it proves a uniform monic-norm estimate for an explicitly smoothed convex family and evaluates its Robin variation. The smoothing is returned through fixed Taylor-jet quotients with an error smaller than k^2 .

4.1 C1. The quantity inherited from the exact localization

Retain

$$M(\theta) = \int e^{\theta y} w_h(y) dy, \quad L = \log M, \quad \mu = L', \quad V = L'', \quad \alpha = \pi/2, \quad b = \mu(\alpha) > 0.$$

The inverse of μ is denoted by $\theta(t)$. Put

$$D(t) = L(\alpha) - L(\theta(t)) - (\alpha - \theta(t))t, \quad 0 \leq t \leq b, \\ W_k(y) = kD(|y|/k)1_{|y| \leq kb}, \quad I_h = \int_0^\alpha \mu(\theta)^2 d\theta = \frac{1}{k^2} \int_{\mathbb{R}} W_k(y) dy > 0. \quad (\text{C1})$$

The equality of the two expressions for I_h was derived in the incoming central calculation. The whole function W_k is nonnegative, Lipschitz with constant α , and bounded above by $\Lambda_k = k[L(\alpha) - L(0)]$.

For a positive weight w and a nonnegative insertion W , write

$$D_N[w, W] = \log \det H_N[w] - \log \det H_N[e^{-W}w], \\ \Phi_{q,k}[w, W] = D_{2q-1}[w, W] + D_{2q}[w, W] - D_{q-1}[w, W] - D_q[w, W]. \quad (\text{C2})$$

The established exact-source localization gives

$$\delta_k^{\text{cen, end}} = \Phi_{q,k}[\sigma, W_k] + O_h(k \log^2 q). \quad (\text{C3})$$

Its original relation determinants and all four source signs are part of that result. We now prove

$$\boxed{\Phi_{q,k}[\sigma, W_k] = \frac{\log 2}{\pi} I_h k^2 + o_h(k^2)}. \quad (\text{C4})$$

This supplies a value for the central contribution on the simple stratum, not only an $O(q)$ enclosure.

4.2 C2. An explicit analytic regularization with a controlled determinant cost

Let

$$a = a_k = \frac{k}{[\log(q+2)]^4}, \quad G_a(u) = \frac{e^{-u^2/a^2}}{\sqrt{\pi}a}.$$

We work on the cofinal range $a \geq 1$. Define the actual comparison potentials

$$Q_{k,t}(y) = \alpha|y| + tW_k(y), \quad \tilde{Q}_{k,t} = G_a * Q_{k,t}, \quad 0 \leq t \leq 1, \\ \widetilde{W}_k = G_a * W_k, \quad \tilde{Q}_{k,t} = \tilde{Q}_{k,0} + t\widetilde{W}_k. \quad (\text{C5})$$

The convolution here is an auxiliary comparison of real potentials, not a replacement of the original arithmetic class. Its return into (C2) is proved next.

The exact scalar properties are

$$0 \leq \tilde{Q}_{k,0}(y) - \alpha|y| \leq \alpha a / \sqrt{\pi}, \quad \|\widetilde{W}_k - W_k\|_\infty \leq \alpha a / \sqrt{\pi}, \quad (\text{C6})$$

$$\int \widetilde{W}_k = \int W_k = k^2 I_h, \quad \int y^2 \widetilde{W}_k(y) dy = O_h(k^4). \quad (C7)$$

The first inequalities follow respectively from convexity of absolute value and from the exact Lipschitz constant of W_k , since $\int |u| G_a(u) du = a/\sqrt{\pi}$. Fubini proves the area identity. The second moment is the original compactly supported moment plus $(a^2/2)k^2 I_h$. These are moments of the auxiliary rate function, not nonexistent fourth moments of the boundary-tilted arithmetic density.

We claim the following determinant comparison:

$$\boxed{\Phi_{q,k}[\sigma, W_k] = \Phi_{q,k}[e^{-\widetilde{Q}_{k,0}}, \widetilde{W}_k] + O_h(k \log q [a + \log q])}. \quad (C8)$$

Its error is $o(k^2)$, by the specified choice of a .

Here are the fixed-fibre details. Put $g = e^{-\widetilde{Q}_{k,0}}/\sigma$. The original Gamma bounds and (C6) give

$$c_h e^{-C_h a} \leq g(y) \leq C_h (1 + y^2)^{1/4}.$$

On every polynomial of degree at most $2q$, both before and after multiplication of the weight by e^{-tW_k} , these imply

$$c_h e^{-C_h a} \|P\|_{e^{-tW_k}\sigma}^2 \leq \|P\|_{ge^{-tW_k}\sigma}^2 \leq C_h (q+1)^{1/2} \|P\|_{e^{-tW_k}\sigma}^2. \quad (C9)$$

For the upper bound use Jensen and the consolidated full-tail second-moment bound

$$\int y^2 |P|^2 e^{-tW_k} d\sigma \leq 272(2q+1)^2 \|P\|_{e^{-tW_k}\sigma}^2.$$

Thus the logarithmic width in (C9) is $O_h(a + \log q)$, at both endpoints of the nonlinear insertion.

Let $K' = kb + 10a\sqrt{\log q}$. Outside this interval the Gaussian tail gives $\widetilde{W}_k \leq \Lambda_k q^{-100}$. Restrict the first r Taylor coefficients at zero to each source space, with

$$r = \lceil C_h \{ (K' + 1) \log(q+2) + \Lambda_k + a + \log(q+2) \} \rceil = O_h(k \log q). \quad (C10)$$

The constant can be chosen so that the Gamma central restriction fraction on its full kernel is at most $e^{-3\Lambda_k - 4C_h a} q^{-50}$, by the explicit generating-function estimate (O23). The actual Taylor-image quotient has dimension at most r .

First compare the loss $D_N[g\sigma, W_k]$ with $D_N[\sigma, W_k]$. On the Taylor kernel, both insertion losses are $O(q^{-20})$: the unsuppressed central norm fraction is bounded by the preceding estimate times $e^{C_h a}$ and a fixed power of q . On the two attained Taylor-image quotients, (C9) bounds the difference of the two endpoint losses by $r O_h(a + \log q)$. This is a comparison of two complete Schur losses, not a bound on two unrelated trial metrics.

Second compare the insertions W_k and $\widetilde{W}_k$ on the same background $g\sigma$. Their pointwise relative factor is between $e^{-C_h a}$ and $e^{C_h a}$. On the Taylor kernel its deviation from one costs at most $e^{\Lambda_k + 2C_h a}$ times the chosen central restriction fraction, plus $O(\Lambda_k q^{-100})$ outside. Hence its kernel log-determinant cost is again negligible. Its attained quotient costs at most $r C_h a$. The fixed frame determinant cancels in each comparison. Sum the four original endpoint signs. This proves (C8), with every preceding source direction included in the Schur minimum.

4.3 C3. The regularized potentials have an exact positive curvature measure

Define the even finite positive measure

$$d\kappa_{k,t}(u) = 2\alpha(1-t)\delta_0(du) + \frac{t}{kV(\theta(|u|/k))} 1_{|u| < kb} du. \quad (C11)$$

Its mass is exactly 2α : on the positive half-line the substitution $s = \mu(\theta)$ gives $ds = V(\theta)d\theta$. Its first moment is zero. Moreover

$$\boxed{Q_{k,t}(y) = \frac{1}{2} \int |y - u| d\kappa_{k,t}(u)}. \quad (C12)$$

To verify the constant as well as the derivative, both sides equal $\alpha|y|$ for $|y| \geq kb$. On $0 < y < kb$, both derivatives are $(1-t)\alpha + t\theta(y/k)$. Evenness gives the other half-line.

Let $\kappa_{k,t}^a = G_a * \kappa_{k,t}$. Then

$$\begin{aligned} \tilde{Q}_{k,t}''(y) &= \kappa_{k,t}^a(y) > 0, & \int \kappa_{k,t}^a &= 2\alpha, \\ \int u^2 \kappa_{k,t}^a(u) du &= 2tk^2 I_h + \alpha a^2 = O_h(k^2). \end{aligned} \quad (C13)$$

The original unsmoothed curvature has compact support $[-kb, kb]$. These moment assertions therefore do not require any additional tilted arithmetic moments.

The analytic continuation is explicit:

$$\tilde{Q}_{k,t}(z) = \frac{1}{2} \int \left[(z-u) \operatorname{erf}\left(\frac{z-u}{a}\right) + \frac{a}{\sqrt{\pi}} e^{-(z-u)^2/a^2} \right] d\kappa_{k,t}(u). \quad (C14)$$

It is entire, real on the real axis, even, and strictly convex there. Its derivative tends to $\pm\alpha$ at the two real infinities.

4.4 C4. Construct the equilibrium at the actual monic degree

Fix an integer n with $q \leq n \leq 2q$, and a parameter $t \in [0, 1]$. There is exactly one $A = A_{n,t} > 0$ satisfying

$$\boxed{n = \frac{1}{2\pi} \int_{-A}^A \sqrt{A^2 - u^2} \kappa_{k,t}^a(u) du.} \quad (C15)$$

The right side is continuous, strictly increasing from zero to infinity; its derivative is the positive integral with numerator A . This proves existence, uniqueness, and smooth dependence on t .

The same equation is

$$\int_{-A}^A \frac{\tilde{Q}_{k,t}'(s)}{\sqrt{A^2 - s^2}} ds = 0, \quad \int_{-A}^A \frac{s \tilde{Q}_{k,t}'(s)}{\sqrt{A^2 - s^2}} ds = 2\pi n. \quad (C16)$$

The first identity is parity. Integration by parts gives the second directly from (C15).

Write $Q_t = \tilde{Q}_{k,t}$ temporarily and define

$$\begin{aligned} h_{n,t}(z) &= \frac{1}{\pi} \int_{-A}^A \frac{Q_t'(s) - Q_t'(z)}{s - z} \frac{ds}{\sqrt{A^2 - s^2}}, \\ \rho_{n,t}(y) &= \frac{\sqrt{A^2 - y^2}}{2\pi} h_{n,t}(y), \quad |y| < A. \end{aligned} \quad (C17)$$

The divided difference is removable at $s = z$. Thus $h_{n,t}$ is analytic, and strict convexity makes it positive on the real axis. The density vanishes as a square root at each endpoint.

For a direct verification of its mass and Euler equation, let $R_A(z) = \sqrt{z^2 - A^2}$, with $R_A(z)/z \rightarrow 1$. The function

$$G(z) = \frac{R_A(z)}{2\pi} \int_{-A}^A \frac{Q_t'(s)}{(z-s)\sqrt{A^2 - s^2}} ds$$

has $G(z) = n/z + O(z^{-2})$ by (C16), and jump $G_+ - G_- = -2\pi i \rho_{n,t}$. Hence it is the Cauchy transform of this density. Its real boundary value is $Q_t'/2$, proving

$$\boxed{Q_t(y) - 2 \int \log |y - s| \rho_{n,t}(s) ds = \lambda_{n,t}, \quad |y| \leq A.} \quad (C18)$$

Outside, the derivative of the left side is $R_A(y)h_{n,t}(y)$, negative to the left and positive to the right. The full exterior inequality follows. Positivity of the logarithmic energy of a zero-mass

difference, as proved in EIQ20–23, then proves that this is the unique mass- n equilibrium, not just a candidate supported on one interval.

Its radius has the uniform estimate

$$A_{n,t} = \frac{\pi n}{\alpha} + O_h(k^2/n) = 2n + O_h(k^2/n). \quad (\text{C19})$$

Indeed, if $m_2 = \int u^2 \kappa_{k,t}^a(u) du$, then (C15) and $A - \sqrt{A^2 - u^2} \leq u^2/A$ on $|u| \leq A$, together with $A \leq u^2/A$ outside, give

$$0 \leq \frac{\alpha A}{\pi} - n \leq \frac{m_2}{2\pi A}.$$

Use (C13). In particular $kb/A \rightarrow 0$ and $a/A \rightarrow 0$, uniformly throughout the stated degree and parameter ranges.

4.5 C5. Uniformity of the monic-norm estimate for this moving family

Let $H_n(t)$ be the squared norm of the monic degree- n polynomial in the weight $e^{-Q_t(y)} dy$. We prove

$$H_n(t) = \pi A_{n,t} e^{-\lambda_{n,t}} [1 + O_h(n^{-1})], \quad q \leq n \leq 2q, \quad 0 \leq t \leq 1. \quad (\text{C20})$$

The uniformity in this statement is essential. We verify it below for the explicit family (C14); a theorem for one fixed analytic potential is not simply applied to a changing potential.

4.5.1 C5.1. Positivity and analytic control in the narrow central strip

For $|y|, |u| < A$, put

$$\mathcal{G}_A(y, u) = \log \left| \frac{A^2 - yu + \sqrt{A^2 - y^2} \sqrt{A^2 - u^2}}{A(y - u)} \right|.$$

Integration of the elementary arcsine resolvent in (C17) gives

$$\rho_{n,t}(y) = \frac{1}{2\pi^2} \int_{-A}^A \mathcal{G}_A(y, u) \kappa_{k,t}^a(u) du. \quad (\text{C21})$$

For example, set $y = A \cos \theta$, $u = A \cos \phi$; the kernel is $\log |\sin((\theta + \phi)/2) / \sin((\theta - \phi)/2)|$. Its sine expansion is $2 \sum_{j \geq 1} \sin(j\theta) \sin(j\phi) / j$, and integration against $A \sin \theta$ gives $\pi \sqrt{A^2 - u^2}$, confirming the mass in (C15).

There are constants $c, C > 0$, independent of n, k, t on the cofinal fixed-packet range, such that

$$\rho_{n,t}(y) \geq c \quad (|y| \leq A/3), \quad |\rho'_{n,t}(z)| \leq C/a + C/A \quad (|\Re z| \leq A/3, |\Im z| \leq a/10). \quad (\text{C22})$$

For the real lower bound, pair in (C17) the intervals $[A/2, 3A/4]$ and their negatives. There Q'_t is exponentially close to α and $-\alpha$, respectively, uniformly by (C14), while $|Q'_t(y)| \leq \alpha$. Their positive divided differences supply $h_{n,t}(y) \geq c/A$, hence the first assertion.

For the analytic derivative bound, decompose the kernel in (C21) into $-\log|y - u|$ and the logarithm of $(A^2 - yu + \sqrt{A^2 - y^2} \sqrt{A^2 - u^2})/A$. The latter has derivative bounded by C/A in the indicated central region, uniformly for $|u| \leq A$. The missing tails of κ^a have exponentially small mass and are analytic there. The logarithmic convolution with the Gaussian has an entire real-analytic continuation J_a , whose derivative is

$$J'_a(z) = \int_0^\infty e^{-a^2 v^2/4} \sin(zv) dv.$$

This follows either by differentiating the Gaussian logarithmic convolution or by solving $J''_a + 2zJ'_a/a^2 = 2/a^2$, $J'_a(0) = 0$. On $|\Im z| \leq a/10$, its absolute value is at most C/a . Integrating against the unsmoothed curvature measure of exact mass 2α proves the second assertion in (C22).

Choose a fixed sufficiently small $\varepsilon_0 > 0$. Equation (C22) then proves

$$\Re \rho_{n,t}(x + iv) \geq c/2 \quad (|x| \leq A/3, |v| \leq \varepsilon_0 a). \quad (\text{C23})$$

Thus the central strip has an explicitly growing physical height proportional to a , not a fixed scaled width that the varying family might fail to possess.

4.5.2 C5.2. Uniform soft edges and the rest of the lens

After scaling $y = Az$, the equilibrium probability density is $(A/n)\rho_{n,t}(Az)$. Near either scaled endpoint ± 1 , its analytic square-root coefficient converges, with three derivatives on fixed slightly larger disks, to the coefficient for $\pi|z|$. The error is

$$O_h((k^2 + a^2)/n^2) + O_h(e^{-c_h n^2/a^2}). \quad (\text{C24})$$

Here is the estimate behind this assertion. On $\Re y \geq A/2$, $|\Im y| \leq A/20$, (C14) gives $Q'_t(y) = \alpha + O(e^{-c_h A^2/a^2})$, with the corresponding differentiated estimates; the negative side is identical. In the integral (C17), split the real variable at $|s| = A/2$. On the outer pieces the divided differences of this error are exponentially small, including the removable point. On the middle piece the denominator stays a fixed distance from the endpoint disks. The difference from $\alpha \operatorname{sgn} s$ is odd, so its zeroth moment cancels. Its first absolute moment is $O_h(k^2 + a^2)$, by (C12)–(C14). Taylor expansion of the remaining denominator therefore gives an error $O_h((k^2 + a^2)/A^3)$ in the physical h , and the same estimate for fixed scaled derivatives. Multiplication by A^2/n , and (C19), prove (C24).

The limiting scaled density is $\pi^{-1} \operatorname{arcosh}(1/|z|)$ on $(-1, 1)$. Its analytic edge coefficient in $(2\pi)^{-1} \sqrt{1 - z^2} \tilde{h}(z)$ has value $\tilde{h}(\pm 1) = 2$. Thus fixed endpoint disks have uniformly nonzero edge coefficients and uniformly bounded analytic derivatives. The same integral argument on fixed scaled compact sets away from zero proves uniform analytic convergence there. In particular the positivity needed for the rest of the lens persists in a fixed sufficiently small scaled neighborhood.

Choose the upper and lower lens boundaries at physical height $\varepsilon_0 a$ over the middle third. In the overlap with the region away from zero, let them rise along fixed-slope segments and enter the fixed scaled endpoint disks transversally. This uses a bounded number of line segments and smooth arcs, with uniformly controlled angles. By (C23), and the uniform analytic convergence away from zero, the off-diagonal lens jumps after the equilibrium transformation are $O(e^{-c_h a})$ outside the endpoint disks. The full exterior Euler inequality gives exponentially small real jumps outside those disks. At large real distance this is integrable uniformly: $Q'_t(y) \geq \alpha/2$ and the equilibrium resolvent is at most $n/(y - A)$.

4.5.3 C5.3. The Riemann–Hilbert calculation and its error coefficient

Use the exact orthogonal-polynomial matrix problem of Fokas–Its–Kitaev: its jump is $\begin{pmatrix} 1 & e^{-Q_t} \\ 0 & 1 \end{pmatrix}$, its first entry is the monic polynomial, and the coefficient of z^{-n-1} in its Cauchy-transform entry is $-H_n(t)/(2\pi i)$. These statements follow directly from the Cauchy expansion and the defining orthogonality. They are also the normalization used in [D, Theorem 31].

Scale by the actual A , and let g be the logarithmic transform of the equilibrium probability. Its Euler constant in the scaled convention is

$$\ell_{n,t} = -\lambda_{n,t}/n - 2 \log A. \quad (\text{C25})$$

The equilibrium transformation and lens factorization produce the constant interval jump $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Its exact solution is

$$N(z) = \frac{1}{2} \begin{pmatrix} \beta + \beta^{-1} & (\beta - \beta^{-1})/i \\ -(\beta - \beta^{-1})/i & \beta + \beta^{-1} \end{pmatrix}, \quad \beta(z) = \left(\frac{z-1}{z+1} \right)^{1/4},$$

with $\beta \rightarrow 1$ at infinity. In particular $N_{12}(z) = i/(2z) + O(z^{-2})$.

In each fixed endpoint disk the analytic phase has a simple soft edge: its $3/2$ -power coefficient is bounded above and away from zero by (C24). The Airy fundamental matrix, with its exact constant Stokes jumps, gives the local solution. Multiplying its leading analytic factor by N matches the exterior solution. The Airy asymptotic, uniform on the fixed disk boundary, gives the matching error $I + O_h(1/n)$. The coefficient bound is uniform because the phase and its inverse conformal coordinate have the bounds just proved. This is precisely the standard Airy construction in [D, Section 3]; the new moving-family bounds needed to use it are (C22)–(C24).

For completeness the small-norm step remains valid despite the narrow middle of the lens. On the scaled contour its Cauchy boundary operator has norm at most $C_h A/a$. A direct coarse proof splits it into the finitely many segments, arcs, and rays. Self terms have the ordinary line or circle Cauchy bound. Intersecting pieces have fixed transverse angles and the usual Carleman-kernel bound (obtainable by the weight $s^{-1/2}$). The other cross kernels have separation at least a fixed multiple of a/A , giving the displayed bound. No uniform separation independent of k is assumed.

Write the final jump as $I + w$. Its L^1, L^2, L^∞ norms are bounded by

$$C_h/n + C_h e^{-c_h a}.$$

Therefore

$$\|C_-(\cdot w)\| \leq \frac{C_h A}{a} (n^{-1} + e^{-c_h a}) = o(1). \quad (\text{C26})$$

The singular-integral equation is solved by its convergent Neumann series. Its solution ν obeys $\|\nu - I\|_2 = O_h(a^{-1}) + O_h((n/a)e^{-c_h a})$. The coefficient of z^{-1} in the final error matrix is $-\frac{1}{2\pi i} \int \nu w$, hence is $O_h(1/n)$: use the L^1 bound for w and Cauchy–Schwarz for $(\nu - I)w$. The term $e^{-c_h a}$ is smaller than every inverse power of q , because $a = k/\log^4(q+2)$.

Undoing the exact transformations now gives

$$H_n(t) = \pi A^{2n+1} e^{n\ell_{n,t}} [1 + O_h(1/n)] = \pi A e^{-\lambda_{n,t}} [1 + O_h(1/n)],$$

by (C25). This proves (C20), including its normalization and the required uniformity in n, k, t . Appendix A1–A3 writes the exact matrix transformations, all four Airy jumps, the analytic matching factor, and the finite-piece Cauchy-operator bound explicitly. No fixed-potential remainder is differentiated or presumed uniform without these estimates.

4.6 C6. The Robin variation has an exact arithmetic-source integral

Differentiate the Euler equality (C18) at a fixed interior point, then integrate against the arcsine probability on the current interval $[-A, A]$. The derivative of the equilibrium density has total mass zero; its endpoint contributions vanish because the density has square-root decay. The arcsine logarithmic potential is exactly $\log(A/2)$ on this whole interval. Therefore its pairing with that zero-mass derivative vanishes, giving the exact identity

$$\partial_t \lambda_{n,t} = \frac{1}{\pi} \int_{-A_{n,t}}^{A_{n,t}} \frac{\widetilde{W}_k(y)}{\sqrt{A_{n,t}^2 - y^2}} dy. \quad (\text{C27})$$

The parameter derivatives exist from (C15)–(C17). This differentiates an exact finite equilibrium equation, not an asymptotic error.

By (C7), (C19), and the Gaussian tail outside $[-kb, kb]$,

$$\partial_t \lambda_{n,t} = \frac{k^2 I_h}{2\pi n} + O_h(k^4/n^3) + O_h(q^{-20}), \quad (\text{C28})$$

uniformly in the same range. To verify the error, on $|y| \leq A/2$ expand $(A^2 - y^2)^{-1/2} = A^{-1} + O(y^2/A^3)$ and use the second moment in (C7). Beyond this interval the Gaussian convolution

of the compact rate is exponentially small, including the integrable arcsine endpoint singularity. Finally $A^{-1} = (2n)^{-1} + O_h(k^2/n^3)$ by (C19).

Also $\log(A_{n,1}/A_{n,0}) = O_h(k^2/n^2)$. Integrate (C28) over t , and use (C20) only at its two endpoints. It follows that the actual monic norm loss is

$$\log \frac{H_n(0)}{H_n(1)} = \frac{k^2 I_h}{2\pi n} + O_h\left(\frac{k^4}{n^3} + \frac{k^2}{n^2} + \frac{1}{n}\right). \quad (\text{C29})$$

This is the missing value of the central high-degree norm response, with the coupling to all preceding polynomials already included in $H_n(t)$.

4.7 C7. Sum the original bands, with the endpoint weights intact

A source determinant is the product of its successive monic norms. Thus the four signs in (C2) give exactly

$$\Phi_{q,k}[e^{-\tilde{Q}_{k,0}}, \tilde{W}_k] = \sum_{r=0}^q c_r \log \frac{H_{q+r}(0)}{H_{q+r}(1)}, \quad c_0 = c_q = 1, \quad c_r = 2 \quad (0 < r < q). \quad (\text{C30})$$

No determinant has been replaced by an arithmetic mean. Substitute (C29):

$$\Phi_{q,k}[e^{-\tilde{Q}_{k,0}}, \tilde{W}_k] = \frac{k^2 I_h}{2\pi} \sum_{r=0}^q \frac{c_r}{q+r} + O_h(1 + k^2/q + k^4/q^2).$$

The displayed sum is the exact trapezoidal sum for $2 \int_1^2 x^{-1} dx$, with error $O(q^{-1})$. Hence

$$\Phi_{q,k}[e^{-\tilde{Q}_{k,0}}, \tilde{W}_k] = \frac{\log 2}{\pi} I_h k^2 + O_h(1 + k^2/q + k^4/q^2). \quad (\text{C31})$$

Return the smoothing by (C8), and then the original arithmetic central endpoint by (C3). The completed central result is

$$\delta_k^{\text{cen,end}} = \frac{\log 2}{\pi} I_h k^2 + O_h\left(\frac{k^2}{\log^3(q+2)} + k \log^2(q+2) + 1 + \frac{k^4}{q^2}\right). \quad (\text{C32})$$

The error is $o_h(k^2)$, and in particular $o_h(q)$, on every fixed multiplicity stratum. The positive coefficient in (C32), including its factor $\log 2/\pi$, is now evaluated. It is not the unweighted negative area coefficient from the preceding source-only calculation.

5 Explicit matrix construction for the uniform central monic norm

This appendix expands the matrix step in C5. The object is the already constructed even analytic potential $Q_t = \tilde{Q}_{k,t}$, at an actual degree $q \leq n \leq 2q$. Its support radius is $A = A_{n,t}$, and its physical Robin constant is $\lambda = \lambda_{n,t}$.

5.1 A1. Exact scaling and equilibrium transformation

Set $V_n(z) = Q_t(Az)/n$. The scaled equilibrium probability is

$$\varrho(x) = \frac{A}{n} \rho_{n,t}(Ax), \quad -1 < x < 1,$$

and

$$g(z) = \int_{-1}^1 \log(z-s) \varrho(s) ds, \quad \ell = -\lambda/n - 2 \log A.$$

Its boundary equation is $g_+ + g_- - V_n = \ell$. Write $\hat{h}(z) = A^2 h_{n,t}(Az)/n$, so that $\varrho(x) = (2\pi)^{-1} \sqrt{1-x^2} \hat{h}(x)$.

The exact orthogonal-polynomial matrix for $e^{-nV_n(x)} dx$ has jump $Y_+ = Y_- \begin{pmatrix} 1 & e^{-nV_n} \\ 0 & 1 \end{pmatrix}$, and

$$Y(z) z^{-n\sigma_3} = I + Y_1/z + O(z^{-2}), \quad (Y_1)_{12} = -\frac{A^{-2n-1} H_n(t)}{2\pi i}. \quad (\text{A1})$$

The second equality follows from expanding the Cauchy transform: all lower moments vanish by the defining monic orthogonality, and its first nonzero moment is the squared monic norm.

Define

$$T(z) = e^{-n\ell\sigma_3/2} Y(z) e^{-n(g(z)-\ell/2)\sigma_3}. \quad (\text{A2})$$

Then $T \rightarrow I$. Near the right edge put

$$\phi(z) = \frac{1}{2} \int_1^z \hat{h}(s) \sqrt{s^2 - 1} ds. \quad (\text{A3})$$

On $z > 1$, this is one half of $V_n - 2g + \ell$, and is positive. On the upper side of $(-1, 1)$,

$$\phi_+(x) = -i\pi \int_x^1 \varrho(s) ds, \quad g_+ - g_- = -2\phi_+.$$

Thus the interval jump of T is

$$\begin{pmatrix} e^{2n\phi_+} & 1 \\ 0 & e^{-2n\phi_+} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ e^{-2n\phi_+} & 1 \end{pmatrix} J_0 \begin{pmatrix} 1 & 0 \\ e^{2n\phi_+} & 1 \end{pmatrix}, \quad J_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (\text{A4})$$

Inside the upper lens multiply T on the right by $\begin{pmatrix} 1 & 0 \\ -e^{2n\phi} & 1 \end{pmatrix}$; inside the lower lens multiply by the same matrix with the minus sign replaced by plus. The resulting matrix S has the constant jump J_0 on the interval, lower-triangular jumps with entry $e^{2n\phi}$ on the two lens boundaries, and upper-triangular entry $e^{-2n\phi}$ on the exterior real rays. The phases on the left are their even-potential continuations; integer n removes the full-mass $2\pi i$ branch increment.

The explicit global solution of the remaining constant interval jump is

$$N(z) = \frac{1}{2} \begin{pmatrix} \beta + \beta^{-1} & (\beta - \beta^{-1})/i \\ -(\beta - \beta^{-1})/i & \beta + \beta^{-1} \end{pmatrix}, \quad \beta(z) = \left(\frac{z-1}{z+1} \right)^{1/4}, \quad \beta(\infty) = 1. \quad (\text{A5})$$

In particular $(N_1)_{12} = i/2$. All of these matrices act on the original two-by-two polynomial/Cauchy-transform problem; they do not alter the programme's arithmetic quotient.

5.2 A2. Airy matrix and all four constant jumps

Let $\omega = e^{2\pi i/3}$, and define entire column vectors

$$F(\zeta) = \begin{pmatrix} \text{Ai}(\zeta) \\ \text{Ai}'(\zeta) \end{pmatrix}, \quad G(\zeta) = -\omega^2 \begin{pmatrix} \text{Ai}(\omega^2 \zeta) \\ \omega^2 \text{Ai}'(\omega^2 \zeta) \end{pmatrix}.$$

With $c_A = \sqrt{2\pi}e^{-\pi i/4}$, define the four sectors

$$\Psi(\zeta) = c_A \begin{cases} [F, G], & 0 < \arg \zeta < 2\pi/3, \\ [F - G, G], & 2\pi/3 < \arg \zeta < \pi, \\ [G, G - F], & -\pi < \arg \zeta < -2\pi/3, \\ [F, G - F], & -2\pi/3 < \arg \zeta < 0. \end{cases} \quad (\text{A6})$$

Orient the positive ray outwards and the other three rays inwards. Direct column multiplication gives the jumps

$$\begin{array}{l|l} \arg \zeta = 0 & \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \\ \arg \zeta = \pm 2\pi/3 & \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \\ \arg \zeta = \pi & J_0. \end{array} \quad (\text{A7})$$

The exact Airy connection identity $\text{Ai}(\zeta) + \omega \text{Ai}(\omega \zeta) + \omega^2 \text{Ai}(\omega^2 \zeta) = 0$ and its derivative verify that all sector columns are the appropriate rotated Airy solutions. Their Wronskian gives $\det \Psi = 1$. These are DLMF 9.2.8–9.2.12 in the displayed convention.

Their sectorial asymptotic is

$$\Psi(\zeta) = \zeta^{-\sigma_3/4} C_A [I + O(\zeta^{-3/2})] e^{-(2/3)\zeta^{3/2}\sigma_3}, \quad C_A = \frac{e^{-\pi i/4}}{\sqrt{2}} \begin{pmatrix} 1 & i \\ -1 & i \end{pmatrix}. \quad (\text{A8})$$

For example, in the upper right sector the two columns have leading vectors $(1, -1)^\top$ and $(i, i)^\top$, after the common factors are extracted. This verifies the constant in (A8), rather than leaving it as an unspecified normalization. The other sectors follow from the just-written connection identities. DLMF 9.7 supplies the uniform Airy remainder in the closed subsectors used for the disk boundaries.

Near $z = 1$, set

$$f(z) = \left(\frac{3}{2} \phi(z) \right)^{2/3}.$$

It is analytic with a simple zero at one. Since $\hat{h}(1) \rightarrow 2$,

$$f'(1) = \left(\frac{\hat{h}(1)}{\sqrt{2}} \right)^{2/3}$$

is bounded above and away from zero. The three-derivative bounds in (C24), on a slightly larger fixed disk, give a uniformly invertible conformal coordinate after the disk radius is fixed sufficiently small. The lens arcs within this disk can be chosen as the inverse images of the two Airy rays; their intersection angles with the disk boundary are uniformly nonzero.

The exact local matrix is

$$P(z) = E(z) \Psi(n^{2/3} f(z)) e^{n\phi(z)\sigma_3}, \quad E(z) = N(z) C_A^{-1} f(z)^{\sigma_3/4} n^{\sigma_3/6}. \quad (\text{A9})$$

It has precisely the jumps in (A4): conjugation of the two triangular jumps in (A7) produces $e^{-2n\phi}$ and $e^{2n\phi}$, while $\phi_- = -\phi_+$ leaves the interval jump equal to J_0 . The factor E is analytic across the interval because

$$C_A J_0 C_A^{-1} = \text{diag}(-i, i).$$

Its apparent endpoint singularity is removable: the fourth-root singularity of N is canceled by that of $f^{\sigma_3/4}$. Equation (A8) gives, uniformly on the fixed boundary circle,

$$P(z)N(z)^{-1} = I + O_h(n^{-1}). \quad (\text{A10})$$

The left-edge construction is obtained by the exact reflection $z \mapsto -z$ and conjugation by σ_3 . Reflection reverses orientations; σ_3 inverts all three constant jumps in (A7), and $N(-z) = \sigma_3 N(z) \sigma_3$. Thus the same matching estimate holds at the left edge, with no new metric or omitted endpoint.

5.3 A3. The narrow contour has an adequate finite Cauchy bound

Outside the two endpoint disks put $R = SN^{-1}$; inside put $R = SP^{-1}$. The central portions of the lens have scaled distance $\varepsilon_0 a/A$ from the real line. The remaining portions are a fixed finite number of segments and smooth arcs chosen in the uniformly analytic regions, as described in C5. All intersection angles are bounded away from zero.

The boundary Cauchy operator on this contour satisfies

$$\|C_-\|_{2 \rightarrow 2} \leq C_h A/a. \quad (\text{A11})$$

One explicit coarse justification is to split the contour into its finitely many pieces. On each line or circle piece the usual Cauchy bound follows from the Fourier Hilbert transform or its circular version. Adjacent pieces meeting at a fixed angle have cross kernel bounded by $C/(s+t)$; the weighted Schur test with weight $s^{-1/2}$ gives the finite Carleman bound $C\pi$. Nonadjacent bounded pieces, separated by at least ca/A , have Hilbert–Schmidt norm at most their length product’s square root divided by $2\pi ca/A$. Cross terms with an unbounded exterior ray have a square-integrable $1/(1+s)$ tail; two opposite rays again use the Carleman bound. Summing this finite block matrix gives (A11). Thus a uniform lower separation independent of k has not been assumed.

The two circle jumps are $I + O(1/n)$ by (A10). Every remaining jump is $I + O(e^{-c_h a})$, with integrable tails, by the real positivity and the analytic-strip estimate (C23). Therefore, for $w = J_R - I$,

$$\|w\|_1 + \|w\|_2 + \|w\|_\infty \leq C_h/n + C_h e^{-c_h a}.$$

Since $A \asymp n$ and $a \rightarrow \infty$, the operator $C_-(\cdot w)$ has norm less than one half on a cofinal range. Its Neumann solution ν has

$$\|\nu - I\|_2 \leq 2\|C_-\| \|w\|_2 = O_h(a^{-1}) + O_h((n/a)e^{-c_h a}).$$

The exact Cauchy solution then has coefficient

$$R_1 = -\frac{1}{2\pi i} \int \nu(s)w(s)ds = O_h(n^{-1}), \quad (\text{A12})$$

by the L^1 bound for w and Cauchy–Schwarz for the remaining product. This establishes the error coefficient actually needed for the monic norm.

Finally, $(N_1)_{12} = i/2$, and undoing (A2) shows $(Y_1)_{12} = e^{n\ell}[(N_1)_{12} + (R_1)_{12}]$. Equation (A1) yields

$$H_n(t) = \pi A^{2n+1} e^{n\ell} [1 + O_h(n^{-1})] = \pi A e^{-\lambda} [1 + O_h(n^{-1})]. \quad (\text{A13})$$

This proves the exact constant and the growing-family uniformity used in (C20).

6 Finite evaluation of the remaining fixed source functional

The coefficient I_h in the result is a fixed one-factor integral, not a growing determinant or its unevaluated asymptotic limit. This appendix gives a finite truncation and quadrature error for that integral using exactly the original RTM envelope. It does not require a numerical off-critical packet to be invented.

Retain

$$e^{\theta y} w_h(y) \leq C(1 + |y|)^{-p}, \quad |\theta| \leq \alpha = \pi/2, \quad p = 8m_0 - 9/2 > 3.$$

Let

$$m_* := \int_{-1}^1 w_h(y) dy > 0, \quad C_1 = \frac{2C}{p-2}, \quad C_2 = \frac{2C}{p-3}, \quad B_\mu = C_1/m_*. \quad (\text{II})$$

These constants retain the original source mass. For $R \geq 1$, define finite integrals

$$M_R(\theta) = \int_{-R}^R e^{\theta y} w_h(y) dy, \quad \mu_R(\theta) = \frac{\int_{-R}^R y e^{\theta y} w_h(y) dy}{M_R(\theta)}.$$

Evenness gives $M_R(\theta) \geq m_*$ on $[0, \alpha]$. The original envelope gives

$$|M_h(\theta) - M_R(\theta)| \leq T_0(R) := \frac{2C}{p-1}(1+R)^{1-p},$$

$$|M'_h(\theta) - M'_R(\theta)| \leq T_1(R) := \frac{2C}{p-2}(1+R)^{2-p}.$$

The full and truncated first moments are at most C_1 in absolute value. Comparing the two exact quotients therefore gives

$$\boxed{|\mu_h(\theta) - \mu_R(\theta)| \leq \varepsilon_R := \frac{T_1(R)}{m_*} + \frac{C_1 T_0(R)}{m_*^2}.} \quad (\text{I2})$$

Both means have absolute value at most B_μ . Their squared difference is consequently at most $2B_\mu \varepsilon_R$.

The derivative of the truncated mean is its actual positive tilted variance. The second-moment envelope gives

$$0 \leq \mu'_R(\theta) \leq C_2/m_*, \quad \left| \frac{d}{d\theta} \mu_R(\theta)^2 \right| \leq 2B_\mu C_2/m_*.$$

For $N \geq 1$, take the literal midpoint values

$$\theta_j = \frac{\alpha(j+1/2)}{N}, \quad I_{h;R,N} = \frac{\alpha}{N} \sum_{j=0}^{N-1} \mu_R(\theta_j)^2. \quad (\text{I3})$$

The midpoint error for a Lipschitz function of constant L on $[0, \alpha]$ is at most $L\alpha^2/(4N)$, as follows by integrating the distance to each midpoint. Hence

$$\boxed{|I_h - I_{h;R,N}| \leq 2\alpha B_\mu \varepsilon_R + \frac{\alpha^2 B_\mu C_2}{2Nm_*}.} \quad (\text{I4})$$

This error tends to zero at the explicit rate $O_h(R^{2-p} + N^{-1})$. The finite integrals in (I3) use the original, explicitly given xi/zeta density on a bounded interval; any numerical errors in those finite integrals must be added when performing a certified numerical evaluation. The formula does not replace them by artificial packet data.

Equations (M3), (M13), and (I4) specify the coefficient by fixed real integrals with convergent error control. There is no remaining limit of growing arithmetic minimum matrices in $d_0(h)$. No numerical sign is asserted for the simple-stratum coefficient without evaluating its actual fixed-source functional.

7 Sources, calculations, and verification record

7.1 Original programme inputs

The original fixed-packet definitions, source and relation minima, and the complete receiver are retained from the supplied programme. `SOURCES.json` records exact physical line ranges and SHA-256 hashes for every source extract. The two preceding central calculations and their consolidation are preserved unchanged in `inputs/`, including the exact preceding ZIP.

[P1] *06_Foundations_and_Original_Six_Readers(1).tex*, physical lines 111601–111947: SXR original arithmetic source, all monic minima, four endpoint signs, and complete relation/quotient exchange.

[P2] The same collection, physical lines 112323–112945: RTM original tilted source and full convolution mass, and GTM original whole-line comparison and Schur transfers.

[P3] *01_Current_Cumulative_and_Proofs(1).tex*, physical lines 4350–5254: EIQ explicit equilibrium, its variational derivatives and endpoint integral; GEL exact Gamma ensemble, source scaling, and the constant $\mathcal{C}_\Gamma$.

[P4] *Untitled 1775.md*, physical lines 9383–9400 and 6437–6456: the assigned signed-coefficient precision and the complete original action receiver. Its previous successful finite approximations were not treated as values for their unevaluated centers.

[P5] *CONSOLIDATED_LOCALIZATION.md*: the original central/outer endpoint split, full nonlinear central transfer, Taylor-jet restriction bounds, relation suppression, and the $O_h(k^2)$ central bound. Its two remaining value questions are evaluated in the present outer and central proofs.

7.2 External primary references

[I] Aleksandar Ivić. *On the mean square of the Riemann zeta-function in short intervals*. arXiv:0803.0132v3, 15 January 2009. Primary text. The actual input is the established second-moment estimate for $E(t + U) - E(t)$ stated after (2.1), with its original range. The paper’s conjectural fourth-moment and pointwise estimates are not used.

[K] H. T. Koelink and J. Van der Jeugt. *Bilinear generating functions for orthogonal polynomials*. arXiv:q-alg/9704016. Primary text. The input is the orthonormal Meixner–Pollaczek measure and Proposition 2.1, equation (2.9). The change to the programme’s y coordinate and Gamma mass is written in O6.

[D] P. Deift, T. Kriecherbauer, K. T.-R. McLaughlin, S. Venakides, and X. Zhou. *Uniform asymptotics for orthogonal polynomials*. Proceedings of the International Congress of Mathematicians 1998, volume III, pp. 491–501. Publisher text. The exact polynomial/Cauchy-transform matrix formulation and the equilibrium, lens, and Airy method are used. The moving-family strip, edge, and contour estimates are derived separately in C5 and Appendix A; uniformity is not inferred from a fixed-potential statement.

[N] NIST Digital Library of Mathematical Functions. §5.11 for Gamma asymptotics; §15.6.1 for the hypergeometric integral; §9.2 and §9.7 for Airy connection formulas, Wronskians, and sectorial asymptotics. All substitutions and matrix constants used here are displayed in the proof.

7.3 What was executed

The self-contained `checks/verify_exact.py` uses explicit exceptions rather than Python assertions. Its 42 exact checks and five deliberately false-formula controls produce identical output in ordinary Python and under `python -O`. They cover finite coefficient arithmetic, original endpoint weights, full Schur determinants and nonzero cross blocks, scalar mass cancellation, curvature/area algebra, the Airy sector jumps, the exact norm scaling, and the cancellation of the outer logarithmic terms.

The controls reject dropping the relation return, discarding the Schur cross block, assigning the unweighted source area as the determinant coefficient, losing the coordinate Jacobian, and

deleting an Airy connection factor. These are exact auxiliary diagnostics, not formal verification of every analytic estimate.

The central numerical diagnostic uses an explicitly artificial quadratic rate obtained from a Gaussian tilted-mass function, not an off-critical xi packet. It evaluates the four original Gamma determinant endpoints. With $v = 0.1$, its ratios $\Phi/(k^2 I)$ are:

k	$q = (k+1)^2$	Computed ratio
5	36	0.219020284196899
9	100	0.220053135775213
17	324	0.220462759517792
25	676	0.220549730730594
41	1764	0.220597259596785
65	4356	0.220614166598438

The derived limiting value is $\log 2/\pi \approx 0.220635600152652$. These non-interval numerical results are a diagnostic of the scale and factor, not the proof of that limit.

Three separate high-precision quadratures of the original EIQ endpoint integral, with increasing precision and different tail splits, give

$$C_\Gamma \approx 0.6540998948244260328147439735.$$

The calculation uses a cancellation-free form of the original integrand and an explicit reciprocal tail coordinate. The stored decimals are not an interval certificate. The analytic expression and the original proved positivity, rather than these decimals, govern the theorem.

No numerical off-critical zero, unspecified original period, or numerical value for the unspecified original functional I_h was invented. No independent peer review, Lean certification, or full-corpus re-verification is claimed. The current result is the written derivation and its stated supporting checks, with the original broader programme questions left separate.

8 Receiving calculation: exact central moments and finite Robin error

This calculation retains the source order $s = 1$, the actual complete packet and all the definitions of M1–M14. In particular

$$q = [1 + k(m_0 - 1)](k + 1)^2, \quad \chi_k(S) = \prod_{u,v=0}^k [S - k/2 - (2u - k)\delta - i(2v - k)\gamma]^{1+k(m_0-1)}.$$

The coordinate is still $S = k/2 + iy$. The polynomial $Q(y) = i^{-q}\chi_k(k/2 + iy)$, every repeated root, and both integration half-lines occur in the source/relation determinants. None of the calculations below assigns an allocation to a proper source, an observation kernel, or a projected arithmetic class.

8.1 The phase correction and its exact effect

In Appendix A1 the displayed boundary convention is $g_+ + g_- - V_n = \ell$. With that convention the real exterior identity is

$$\phi(x) = \frac{1}{2}\{V_n(x) - 2g(x) + \ell\}, \quad x > 1. \quad (\text{SAI1})$$

The minus sign before ℓ in the received prose was incorrect. Here is the verification including its constant. By C17 and A3,

$$V'_n(z) - 2g'(z) = \widehat{h}(z)\sqrt{z^2 - 1} = 2\phi'(z).$$

Both 2ϕ and $V_n - 2g + \ell$ vanish at $x = 1$ by the Euler equation. Integration on $x > 1$ proves SAI1. In the exact transformation

$$T = e^{-n\ell\sigma_3/2} Y e^{-n(g-\ell/2)\sigma_3},$$

the upper-right real exterior jump is consequently $e^{-n(V_n-2g+\ell)} = e^{-2n\phi}$. Thus A4–A13 use the correct jump and need no sign change. In particular

$$(Y_1)_{12} = e^{n\ell}\{i/2 + (R_1)_{12}\}, \quad H_n = \pi A^{2n+1} e^{n\ell}\{1 + 2(R_1)_{12}/i\}.$$

Since $n\ell = -\lambda - 2n \log A$, the original norm factor is $\pi A e^{-\lambda}$. This calculation follows the polynomial/Cauchy-transform formulation of Fokas, Its and Kitaev as explicitly stated in Deift, Kriecherbauer, McLaughlin, Venakides and Zhou, *Uniform asymptotics for orthogonal polynomials*, ICM 1998, Theorem 31 and the subsequent transformation (<https://ems.press/content/book-chapter-files/27262?nt=1>). The Airy identities are those cited in A2 from <https://dlmf.nist.gov/9.2> and <https://dlmf.nist.gov/9.7>. Their moving-family estimates remain C22–C26 and A11–A12, rather than a fixed-potential uniformity assumption.

8.2 All even moments of the original central rate

Write $\alpha = \pi/2$, $L = \log M_h$, $\mu = L'$, $b = \mu(\alpha)$, and let θ be the inverse of μ on $[0, b]$. The variance bounds ACW4–ACW5 imply $0 < v_* \leq \mu' \leq V_*$, so this inverse exists through its endpoints. Define, without changing the source measure,

$$D(t) = L(\alpha) - L(\theta(t)) - (\alpha - \theta(t))t, \quad W_k(y) = kD(|y|/k) \mathbf{1}_{\{|y| \leq kb\}}.$$

Differentiation gives $D'(t) = \theta(t) - \alpha$ and $D(b) = 0$; hence $D(t) = \int_t^b (\alpha - \theta(u)) du$. For every integer $j \geq 0$ there is the exact identity

$$\int_{\mathbb{R}} y^{2j} W_k(y) dy = \frac{k^{2j+2}}{(j+1)(2j+1)} \int_0^\alpha \mu(\theta)^{2j+2} d\theta. \quad (\text{SAI2})$$

Indeed, evenness and the literal substitution $y = kt$ give

$$2k^{2j+2} \int_0^b t^{2j} \int_t^b (\alpha - \theta(u)) du dt = \frac{2k^{2j+2}}{2j+1} \int_0^b u^{2j+1} (\alpha - \theta(u)) du.$$

The substitution $u = \mu(\theta)$ has Jacobian $\mu'(\theta)$. Integration by parts using $(\mu^{2j+2})' = (2j+2)\mu^{2j+1}\mu'$ has zero boundary terms: $\mu(0) = 0$ and $\alpha - \theta = 0$ at $\theta = \alpha$. This proves SAI2. Every integral is finite because $0 \leq \mu \leq b$; it makes no assertion about higher moments of the boundary-tilted density.

In particular set

$$Z = k^2 I_h, \quad J_{4,h} = \int_0^\alpha \mu(\theta)^4 d\theta, \quad M_2 = \frac{k^4}{6} J_{4,h} + \frac{a^2}{2} Z, \quad a = \frac{k}{\log^4(q+2)}. \quad (\text{SAI3})$$

For $\widetilde{W}_k = G_a * W_k$, Gaussian mass, mean and variance give exactly

$$\int \widetilde{W}_k = Z, \quad \int y^2 \widetilde{W}_k(y) dy = M_2.$$

More generally,

$$\int y^{2j} \widetilde{W}_k(y) dy = \sum_{r=0}^j \binom{2j}{2r} \frac{(2r)!}{4^r r!} a^{2r} \int y^{2j-2r} W_k(y) dy.$$

This last formula follows by expanding $(y+u)^{2j}$, using the vanishing odd Gaussian moments and $\int u^{2r} G_a(u) du = (2r)! a^{2r} / (4^r r!)$. Thus every rate moment used in a controlled expansion has an explicit fixed-source integral and its exact return under smoothing.

8.3 A finite radius and Robin-variation estimate

For $q \leq n \leq 2q$ and $0 \leq t \leq 1$, let $A = A_{n,t}$ be the unique solution C15 for the actual Gaussian-convolved curvature $\kappa_{k,t}^a$. Its total mass is 2α and its second moment is exactly $2tZ + \alpha a^2$. Put

$$A_* = \frac{\pi n}{\alpha} = 2n, \quad B_2 = 2Z + \alpha a^2.$$

The radius bounds, with no omitted tail of the curvature, are

$$A_* \leq A_{n,t} \leq A_* + \frac{B_2}{2\alpha A_*}. \quad (\text{SAI4})$$

To prove them, write C15 as

$$\frac{\alpha A}{\pi} - n = \frac{1}{2\pi} \left[\int_{|u| \leq A} (A - \sqrt{A^2 - u^2}) \kappa_{k,t}^a(u) du + A \int_{|u| > A} \kappa_{k,t}^a(u) du \right].$$

Every term is nonnegative. Inside the interval, $A - \sqrt{A^2 - u^2} \leq u^2/A$; outside, $A \leq u^2/A$. The entire right side is at most $B_2/(2\pi A)$. This proves the lower bound, and then the upper bound after $A \geq A_*$ is inserted.

Keep the explicit finite guards $a \geq 1$ and $K = kb \leq A_*/4$. They hold on a cofinal range for every fixed packet and fixed complete multiplicity. Define

$$\begin{aligned} E_\lambda(n) &= \frac{M_2 + ZB_2/(2\alpha)}{\pi A_*^3} \\ &\quad + Z \left(\frac{1}{\sqrt{\pi}a} + \frac{1}{\pi A_*} \right) \exp\left(-\frac{A_*^2}{16a^2}\right). \end{aligned} \quad (\text{SAI5})$$

Then the exact Robin variation C27 obeys

$$\left| \partial_t \lambda_{n,t} - \frac{Z}{\pi A_*} \right| \leq E_\lambda(n). \quad (\text{SAI6})$$

Here is the complete estimate on both regions. On $|y| \leq A/2$,

$$0 \leq \frac{1}{\sqrt{A^2 - y^2}} - \frac{1}{A} \leq \frac{y^2}{A^3}.$$

For $0 \leq z \leq 1/4$, the scalar inequality behind this line is $(1 - z)^{-1/2} \leq 1 + z$: after squaring it becomes $z(1 - z - z^2) \geq 0$. Integrating yields at most $M_2/(\pi A^3)$.

For $|y| \geq A/2$, since W_k is nonnegative, supported in $[-K, K]$, and has mass Z ,

$$\widetilde{W}_k(y) \leq \frac{Z}{\sqrt{\pi}a} \exp\left(-\frac{(|y| - K)^2}{a^2}\right) \leq \frac{Z}{\sqrt{\pi}a} e^{-A^2/(16a^2)}.$$

The first remaining contribution integrates this bound against the arcsine measure on $A/2 < |y| < A$, whose total mass is at most one. The second, subtracting the unweighted tail mass in $Z/(\pi A)$, is at most $Ze^{-A^2/(16a^2)}/(\pi A)$. For this last statement use Tonelli and the Gaussian tail $\text{erfc}(x) \leq e^{-x^2}$, $x \geq 0$. A direct proof is to differentiate their difference: its derivative is $2e^{-x^2}(1/\sqrt{\pi} - x)$; the difference starts at zero, first increases, then decreases to zero, and stays nonnegative. These estimates account for the entire line, including $|y| > A$. Finally SAI4 gives

$$\left| \frac{Z}{\pi A} - \frac{Z}{\pi A_*} \right| \leq \frac{ZB_2}{2\pi\alpha A_*^3}.$$

Together they prove SAI5–SAI6.

8.4 A finite error for the actual monic norm

The contour of A1–A3 is the one belonging to the potential C14 at this exact n, k, t . Let $w = J_R - I$ be its error jump. For the same oriented contour and arclength measure, set

$$B = \|C_-\|_{L^2 \rightarrow L^2}, \quad u_j = \|w\|_{L^j} \quad (j = 1, 2), \quad u_\infty = \|w\|_\infty, \quad r_{n,t} = \frac{1}{\pi} \left(u_1 + \frac{Bu_2^2}{1 - Bu_\infty} \right).$$

Impose the explicit guards $Bu_\infty < 1$ and $r_{n,t} < 1$. They hold on the cofinal range constructed in C22–C26: $B \leq C_h A/a$ and $u_1 + u_2 + u_\infty \leq C_h(n^{-1} + e^{-c_h a})$. The singular-integral equation $\nu = I + C_-(\nu w)$ gives

$$\|\nu - I\|_2 \leq \frac{Bu_2}{1 - Bu_\infty}.$$

In its Cauchy solution $R_1 = -(2\pi i)^{-1} \int \nu w$, the constant and remaining parts are bounded respectively by u_1 and $Bu_2^2/(1 - Bu_\infty)$. SAI1 therefore proves

$$\left| \log \frac{H_n(t)}{\pi A_{n,t} e^{-\lambda_{n,t}}} \right| \leq -\log(1 - r_{n,t}). \quad (\text{SAI7})$$

The quantity inside the logarithm is positive because it is the norm of its actual nonzero monic minimizer. Thus no complex logarithm branch is selected to obtain this inequality. The derived bounds give $r_{n,t} = O_h(n^{-1})$, uniformly; SAI7 also records the finite Neumann denominator, rather than discarding it when the limiting order is taken.

Write $f(x) = -\log(1 - x)$, and put

$$E_H(n) = E_\lambda(n) + \frac{B_2}{2\alpha A_*^2} + f(r_{n,0}) + f(r_{n,1}). \quad (\text{SAI8})$$

The two endpoint radii lie in the interval SAI4, so their logarithmic ratio has absolute value at most $B_2/(2\alpha A_*^2)$. Integrating SAI6 in t , and subtracting the two estimates SAI7, proves

$$\left| \log \frac{H_n(0)}{H_n(1)} - \frac{k^2 I_h}{2\pi n} \right| \leq E_H(n). \quad (\text{SAI9})$$

This is the same actual monic minimum as in C29, with an explicit decomposition of its equilibrium, radius and contour errors. The lower polynomial space and its Schur inverse have already been included in that minimum.

8.5 Exact weighted-band summation and complete-action receiver

Keep $c_0 = c_q = 1$ and $c_r = 2$ for $0 < r < q$, exactly as in SXR10. The two overlapping original source bands give

$$S_q = \sum_{r=0}^q \frac{c_r}{q+r} = 2(H_{2q}^{\text{harm}} - H_q^{\text{harm}}) + \frac{1}{2q}.$$

There is the sharpened finite inequality

$$0 < S_q - 2 \log 2 < \frac{1}{8q^2}. \quad (\text{SAI10})$$

Indeed integration of exponentials gives

$$S_q - 2 \log 2 = \int_0^\infty (e^{-qt} - e^{-2qt}) \left(\coth(t/2) - \frac{2}{t} \right) dt.$$

For $u > 0$, $0 < \coth u - 1/u < u/3$. The first inequality follows from $(u \cosh u - \sinh u)' = u \sinh u > 0$. For the second, expand $(u^2 + 3) \sinh u - 3u \cosh u$: the coefficient of u^{2j+1} is $4j(j-1)/(2j+1)!$,

positive for $j \geq 2$, and the earlier coefficients vanish. Integrating the resulting upper bound $(e^{-qt} - e^{-2qt})t/6$ proves SAI10.

Consequently, for the regularized four-endpoint expression $\tilde{\Phi}$ in C30, SAI9 proves the finite return

$$\left| \tilde{\Phi} - \frac{\log 2}{\pi} k^2 I_h \right| \leq \sum_{r=0}^q c_r E_H(q+r) + \frac{k^2 I_h}{16\pi q^2}. \quad (\text{SAI11})$$

The smoothing and arithmetic localization are the attained Taylor-image comparisons C8 and L22–L35; their full kernel restriction and quotient factors remain present. Adding their proved errors to SAI11 gives exactly C32, with a sharper harmonic-sum contribution. Thus the central endpoint value, rather than only its source-density area, is

$$\delta_k^{\text{cen,end}} = \frac{\log 2}{\pi} I_h k^2 + O_h \left(\frac{k^2}{\log^3(q+2)} + k \log^2(q+2) + 1 + \frac{k^4}{q^2} \right).$$

The outer proof O1–O35 uses the actual convolution on $|y| > kb$, both half-lines, the full relation polynomial, and the original four signs. Its bound is added before any scale comparison. Therefore the accepted receiver is

$$\begin{aligned} J_k^{\text{action}} &= \mathcal{B}_k[\sigma] - (4m_0 - 2)\mathcal{C}_\Gamma q + \frac{\log 2}{\pi} I_h k^2 \\ &\quad + W_k^{\text{mono}} - P_{k,-} - P_{k,+} + \mathcal{E}_{h,k}, \end{aligned} \quad (\text{SAI12})$$

with the explicit sum of the central bound above and O35 as a bound for $|\mathcal{E}_{h,k}|$. Here $\mathcal{B}_k[\sigma]$ is its full existing value. It is not replaced by its leading positive term; neither penalty nor the monic-window scalar has been dropped. This new scalar evaluation changes the previously uncomputed $s = 1$ arithmetic-baseline entry. It does not compute the independent absolute allocations or same-class projected signs.

Outer arithmetic value: independent complete reconstruction

This receiving proof concerns the outer part of the arithmetic density. The original source, packet, relation ideal, four endpoint signs and Gamma mass are retained. The original EIQ equilibrium and RTM one-factor bounds are the proved programme providers specified in the accompanying acceptance record; no arithmetic allocation or projected sign is inferred from a full-quotient determinant.

9 Objects and exact receiver

Fix the original stipulated quartet $\rho = 1/2 + \delta + i\gamma$, $0 < \delta < 1/2$, $\gamma > 2$, with original multiplicity $m \geq 1$. Define

$$h(s) = \prod_{z \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s - z)^m, \quad w_h(y) = \frac{|(2\xi/h)(1/2 + iy)|^2}{2\pi}, \quad m_k = w_h^{*k}.$$

Keep $k = 4\ell + 1$, $e = 1 + k(m - 1)$, $q = e(k + 1)^2$, $c = k/2$, and

$$\chi_k(S) = \prod_{a, b_0=0}^k [S - c - (2a - k)\delta - i(2b_0 - k)\gamma]^e, \quad Q(y) = i^{-q} \chi_k(c + iy).$$

Thus Q is real, even and monic of degree q ; all its roots, including every repetition, have modulus at most $R = k(\delta^2 + \gamma^2)^{1/2}$. Write

$$\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}, \quad \int_{\mathbb{R}} \sigma = \sqrt{2\pi}, \quad \alpha = \pi/2, \\ M(\theta) = \int e^{\theta y} w_h(y) dy, \quad \mathfrak{M} = M(\alpha), \quad b = (\log M)'(\alpha) > 0.$$

For a positive density ν , $H_N[\nu]$ is the Gram of $1, y, \dots, y^N$, J_N is remainder modulo the full Q , and

$$G_N[\nu] = (J_N H_N[\nu]^{-1} J_N^*)^{-1}, \quad \mathcal{B}[\nu] = \log \det G_{q-1}[\nu] + \log \det G_q[\nu] - \log \det G_{2q-1}[\nu] - \log \det G_{2q}[\nu]. \quad (\text{OR1})$$

The monic frame consisting of the first q monomials followed by $Q, yQ, \dots, y^{N-q}Q$ has determinant one. Its Schur complement proves, with its cross blocks retained,

$$\det G_N[\nu] = \frac{\det H_N[\nu]}{\det \text{Gram}_\nu(Q, yQ, \dots, y^{N-q}Q)}. \quad (\text{OR2})$$

The polynomial substitution $P(S) \mapsto P(c + iy)$ is invertible, has coefficient determinant of modulus one, and carries $\chi_k P$ to $i^q Q P(c + iy)$; this proves that (OR1)–(OR2) compute the original quotient. A positive scalar A multiplies every G_N by A . Each quotient has dimension q , hence $\mathcal{B}[A\nu] = \mathcal{B}[\nu]$.

Set

$$d = 4m - 2, \quad p = 2d - \frac{1}{2}, \quad a = p + 1, \quad \tau_d = (1 + y^2)^{-d} \sigma, \quad \omega_k(y) = \begin{cases} \sigma(y), & |y| \leq kb, \\ \mathfrak{M}^{-k} m_k(y), & |y| > kb. \end{cases} \quad (\text{OR3})$$

The notation a in (OR3) is a tail exponent, distinct from the integer indexing a root in χ_k .

Theorem 9.1. *On the original fixed-packet cofinal range, with all original finite guards retained,*

$$\mathcal{B}[\omega_k] - \mathcal{B}[\sigma] = -d\mathcal{C}_\Gamma q + \mathcal{E}_k, \quad \mathcal{C}_\Gamma = 2\mathcal{L}(2, \pi) - 8 \log 2 + 4, \quad (\text{OR4})$$

where $\mathcal{L}(\alpha_E, \beta) = \int \log x \rho_{\alpha_E, \beta}(x) dx$ is the original EIQ logarithmic moment and

$$|\mathcal{E}_k| \leq C_h \left[\frac{q \log \log(q + 16)}{\log(q + 2)} + q^{199/200} \log^2(q + 2) \right] = o_h(q). \quad (\text{OR5})$$

In particular the outer term has no nonzero contribution of magnitude $q \log k$ or $q \log \log k$. This theorem is about the outer density in (OR3); it does not delete the central density in m_k .

10 The growing convolution, including both tails

The exact identity following from the definition of the convolution is

$$f(y) = \mathfrak{M}^{-1} e^{\alpha y} w_h(y), \quad \mathfrak{M}^{-k} m_k(y) = e^{-\alpha y} f^{*k}(y). \quad (\text{OR6})$$

The auxiliary f has mass one, mean b and variance $V > 0$. The original density is recovered with the entire factor in (OR6). From the completed-zeta definition one obtains directly

$$w_h(y) = \frac{(y^2 + 1/4)^2}{\sqrt{\pi}} \frac{|\zeta(1/2 + iy)|^2}{[(y - \gamma)^2 + \delta^2]^{2m} [(y + \gamma)^2 + \delta^2]^{2m}} \sigma(y).$$

Stirling's formula [4], on this fixed vertical line, gives

$$f(y) = c_f y^{-a} Z(y) (1 + O_h(y^{-1})), \quad c_f = \frac{\sqrt{2/\pi}}{\mathfrak{M}}, \quad Z(y) = |\zeta(1/2 + iy)|^2, \quad y \rightarrow +\infty. \quad (\text{OR7})$$

The RTM envelope gives $f(y) \leq C_h (1 + |y|)^{-p}$, $p \geq 7/2$; on the negative half-line it has an additional $e^{-2\alpha|y|}$ factor. Thus its 9/4-moment is finite. The mean-square zeta formula gives, whenever $|z| \leq T^{19/20}$,

$$\int_T^{2T} f(y - z) dy \leq C_h T^{1-a} \log T. \quad (\text{OR8})$$

Indeed $y - z$ lies in a fixed enlargement of $[T, 2T]$, the power factor in (OR7) is $O(T^{-a})$, and the zeta integral over that enlargement is $O(T \log T)$.

The proved Fourier local-limit estimate in the supplied localization provider is uniform in the evaluation point:

$$\sup_x \left| f^{*n}(nb + x) - (2\pi nV)^{-1/2} e^{-x^2/(2nV)} \right| \leq C_h n^{-5/8}. \quad (\text{OR9})$$

Its Fourier proof bounds an L^1 Fourier difference; the factor introduced by translation has modulus one. Integrating (OR9) on $|x| \leq n^{1/2+1/24}$ costs $O(n^{-1/12})$. Chebyshev's inequality for both probability densities bounds the two complementary masses by $O(n^{-1/12})$. If g_n is the Gaussian centered at nb , this proves

$$\|f^{*n} - g_n\|_1 \leq C_h n^{-1/12}. \quad (\text{OR10})$$

For $q^{9/10} \leq T \leq 128q$, the exact definition of q implies

$$c_h T^{1/3} \leq k \leq C_h T^{5/9}. \quad (\text{OR11})$$

Put $A = T^{19/20}$, $f_s = f 1_{\{|y| \leq A\}}$ and $f_l = f - f_s$. Neither subprobability density is normalized. In the binomial convolution expansion, terms with at least two large summands have total mass at most $C_h k^2 A^{2-2p}$. Dividing by $L_T = kT^{-p} \log T$ gives

$$C_h k T^p A^{2-2p} / \log T \leq C_h T^{-25/36} / \log T. \quad (\text{OR12})$$

The exponent is largest at $p = 7/2$; larger original multiplicity only improves it.

For $\lambda = (\log T)/A$, Taylor's formula on the truncated support gives

$$\int e^{\pm \lambda y} f_s(y) dy \leq \exp(C_h \lambda + C_h \lambda^2 T).$$

Consequently the mass of f_s^{*j} , $j \leq k$, outside $[-T/2, T/2]$ is at most $2 \exp[-c_h T^{1/20} \log T]$, eventually. Here $k/T \rightarrow 0$ and $k\lambda \rightarrow 0$ by (OR11), so the two positive exponent terms are strictly smaller than the Chernoff exponent. This bounds the all-small convolution.

The remaining convolution is $k f_l * f_s^{*(k-1)}$. On $A < |z| \leq T/2$, $f_l(y - z) \leq C_h T^{-p}$, whereas the rest density has mass at most $C_h k/A^2$ by its variance and $A \geq 2(k-1)b$. Its window integral is at most $C_h k^2 T^{1-p}/A^2$; divided by L_T , it is $O_h(kT/A^2)$, a fixed negative power of T . On $|z| > T/2$ use

the preceding Chernoff bound and the full mass of f_l . On $|z| \leq A$, one has $y - z > A$. Replacing the small rest convolution by the full one has total variation at most $C_h k A^{1-p}$. The window estimate (OR8) bounds its contribution by $L_T O_h(k A^{1-p})$. Replacing that full rest convolution by g_{k-1} , on the same region, costs $L_T O_h(k^{-1/12})$ by (OR8) and (OR10). The complementary Gaussian region is exponentially small, including the part near $z = y$, since its center is $O_h(k)$ and its variance is $O_h(k)$. Finally (OR7) and $|z| \leq A$ permit replacing $(y - z)^{-a}$ by y^{-a} , with relative integrated error $O_h(A/T)$. Thus, with $n = k - 1$ and $G = \sqrt{2nV}$,

$$\int_T^{2T} \left| f^{*k}(y) - k c_f y^{-a} \mathcal{Z}_G(y - nb) \right| dy \leq C_h L_T (k^{-1/12} + T^{-1/20}), \quad (\text{OR13})$$

$$\mathcal{Z}_G(t) = \frac{1}{\sqrt{\pi}G} \int e^{-(u/G)^2} Z(t + u) du.$$

This estimate uses an averaged zeta norm; it asserts no pointwise regular variation across zeta zeros.

11 The arithmetic smoothing error

Let $c_\zeta = 2\gamma_E - \log(2\pi)$ and $\mathcal{L}_\zeta(t) = \log t + c_\zeta$. The established theorem of Ivić [1, Section 2, (2.1) and its stated E -analogue] gives, for $1 \ll U \leq \sqrt{T}/2$,

$$\int_T^{2T} |E(t + U) - E(t)|^2 dt \ll_\epsilon T U \log^3 T + T^{1/2+\epsilon} U^2 + T^{1+\epsilon} U^{1/2}, \quad (\text{OR14})$$

where $E(t) = \int_0^t Z(v) dv - t(\log(t/(2\pi)) + 2\gamma_E - 1)$. The cited result is a second-moment theorem, without a pointwise or fourth-moment conjecture. After truncation at $|u| \leq G \log T$, integration by parts and subtraction of $E(t)$ under the odd kernel give

$$\mathcal{Z}_G(t) - \mathcal{L}_\zeta(t) = \frac{2}{\sqrt{\pi}G^3} \int u e^{-(u/G)^2} (E(t + u) - E(t)) du + O(G/T) + O(T^{-100}). \quad (\text{OR15})$$

Polynomial bounds from (OR7) justify the Gaussian tails. Minkowski's inequality applied to the square roots of the three terms in (OR14) yields respectively

$$G^{-1/2} \log^{3/2} T, \quad T^{-1/4+\epsilon}, \quad T^\epsilon G^{-3/4}$$

after division by $\sqrt{T}$. The Gaussian integrals giving these powers are $G^{-3} \int_0^\infty u^{3/2} e^{-(u/G)^2} du$, $G^{-3} \int_0^\infty u^2 e^{-(u/G)^2} du$, and $G^{-3} \int_0^\infty u^{5/4} e^{-(u/G)^2} du$. Negative u is reduced by translation and a bounded dyadic cover. For $|u| \leq 1$, positivity of Z bounds the error increment by the length-four increment plus $O(\log T)$; (OR14) at length four gives the smaller term $O(T^\epsilon G^{-3})$. Since $c_h T^{1/6} \leq G \leq C_h T^{5/18}$, all uses of (OR14) lie in its stated range eventually. Taking $\epsilon = 1/1000$, Cauchy–Schwarz, (OR11)–(OR15), and the exact (OR6) imply

$$\int_T^{2T} \left| \frac{\mathfrak{M}^{-k} m_k(y)}{[k/(\sqrt{\pi}\mathfrak{M})] \mathcal{L}_\zeta(y) (1 + y^2)^{-d} \sigma(y)} - 1 \right| dy \leq C_h T^{199/200}. \quad (\text{OR16})$$

The shift $nb = O_h(k)$ changes the logarithmic main term by $O_h(k/T)$. Division in (OR16) is legitimate because its power-log denominator on this window is comparable to $kT^{-a} \log T$; the exponential factors cancel exactly. The original evenness of m_k, σ, τ_d supplies the negative half-line.

12 Full polynomial metrics and their attained quotients

For the Gamma-orthonormal polynomials φ_j , define $K_N^\sigma(y, y) = \sum_{j=0}^N |\varphi_j(y)|^2$. The normalization of Koelink–Van der Jeugt [2, Proposition 2.1, (2.9)] is transported by

$$x = y/2, \quad \lambda = 1/4, \quad \phi = \pi/2, \quad \varphi_j(y) = 2^{\lambda-1/2} p_j^{(\lambda)}(y/2; \pi/2).$$

The weighted diagonal density in y is one half its x version. Euler's integral [3, (15.6.1)], with its $\Gamma(2\lambda)$ conversion from the regularized hypergeometric function, cancels the weight's $|\Gamma(\lambda + ix)|^2$. Its absolute integrand is

$$u^{\lambda-1}(1-u)^{\lambda-1}(1+R_t u)^{-\lambda}, \quad R_t = 4t/(1-t)^2.$$

For $t \geq 1/2$, splitting at $1/R_t$ and $1/2$ bounds this integral by $R_t^{-1/4}(16 + 2 \log R_t)$. Taking $t = (N+1)/(N+2)$, using positivity of each diagonal summand and $t^{-N} \leq e$, proves

$$K_N^\sigma(y, y)\sigma(y) \leq 20 \log(N+2). \quad (\text{OR17})$$

For any subspace $S \subset \mathcal{P}_{2q}$, orthogonal projection gives $K_S \leq K_{2q}^\sigma$ and $\int K_S d\sigma = \dim S$.

In a Gamma-orthonormal frame of S , expansion of the squared determinant proves

$$\det H_S[r\sigma] = \frac{1}{s!} \int |\det[\varphi_i(y_j)]|^2 \prod_{j=1}^s r(y_j) d\sigma(y_j).$$

The unweighted right-hand measure has mass one and one-point density $K_S d\sigma$. Jensen's inequality and $\log \lambda \leq \lambda - 1$, applied to the Gram eigenvalues, give

$$\int \log r K_S d\sigma \leq \log \frac{\det H_S[r\sigma]}{\det H_S[\sigma]} \leq \int (r-1) K_S d\sigma. \quad (\text{OR18})$$

This is a statement about the full Gram determinant, including all cross blocks.

Let $K = q^{9/10}$ and

$$A_k = \frac{k\mathcal{L}_\zeta(q)}{\sqrt{\pi}\mathfrak{M}}, \quad \hat{\omega}_k = \begin{cases} A_k \tau_d, & |y| \leq K, \\ \omega_k, & |y| > K, \end{cases} \quad r_k = \hat{\omega}_k / (A_k \tau_d). \quad (\text{OR19})$$

The proved localization bounds, obtained by placing the $(k-3)$ -fold tilted convolution in its central Gaussian interval and retaining the three-factor RTM lower tail, give

$$c_h k^{-1/2} (1+y^2)^{-M_*} \leq \omega_k(y)/\sigma(y) \leq C_h k^{p+1}. \quad (\text{OR20})$$

They start at $|y| = kb$; inside, the quotient is exactly one. Here $M_* = 21 + 4m$. In detail, the interval $[(k-3)b - \sqrt{(k-3)V}, (k-3)b]$ has tilted probability bounded below by a positive fixed constant, by (OR9). For $y \geq kb$ the remaining positive argument is at most $2y$, eventually. Inserting the original three-factor estimate $w_h^{*3}(u) \geq c_h e^{-\alpha|u|}(1+|u|)^{-42-8m}$ in that part of the exact convolution proves (OR20); evenness handles the negative side. This derivation retains $\mathfrak{M}^{k-3}$, which becomes the fixed factor $\mathfrak{M}^{-3}$ after (OR6).

On $\mathcal{P}_{2q}$, Gamma multiplication has norm at most $2(2q+1)$. Jensen's inequality applied to the convex function $(1+x)^{-M_*}$, in the probability $|P|^2 d\sigma / \|P\|_\sigma^2$, and (OR20), therefore prove

$$q^{-C_h} \|P\|_\sigma^2 \leq \|P\|_{\omega_k}^2, \quad \|P\|_{\hat{\omega}_k}^2, \quad A_k \|P\|_{\tau_d}^2 \leq q^{C_h} \|P\|_\sigma^2. \quad (\text{OR21})$$

The constants here can be enlarged to absorb fixed finite source constants; the assertion applies on the explicitly cofinal original sequence. The same inequality holds for every restriction, and for a quotient by taking the minimum in each unchanged affine fibre.

For completeness the small central replacement in (OR19) has its own finite metric comparison. The exact generating function

$$\sum_{j=0}^{\infty} \frac{b_j(z)}{j!} t^j = (1+t^2)^{-1/4} \exp(z \arctan t), \quad \gamma_j = \|b_j\|_\sigma^2 = \sqrt{2\pi} (2j)! / 4^j$$

follows by summing the original three-term recurrence. Cauchy's coefficient estimate at $|t| = (2q+1)/(2q+2)$ and then at $|z| = 2K$ shows, for any P vanishing to order r ,

$$\int_{-K}^K |P|^2 d\sigma \leq A_{q,K,r} \|P\|_{\sigma}^2, \quad A_{q,K,r} = 4^{1-r} e^2 (2q+1)^2 \sqrt{2q+2} [2(2q+2)]^{2K}. \quad (\text{OR22})$$

Choose the actual integer r so that $A_{q,K,r} q^{2C_h} \leq q^{-20}$; one may take $r = \lceil C'_h(K+1) \log(q+2) \rceil$. On any actual source or relation space use its own restricted Taylor map to $\mathbb{C}^r$, its full kernel and its attained image. The kernel metric changes by relative $O(q^{-20})$; its log determinant changes by $O(q^{-19})$. The attained image has dimension at most r , so (OR21) costs at most $C_h r \log q$. Factoring the original Gram by this exact short sequence proves

$$\left| \log \frac{\det H_S[\widehat{\omega}_k]}{\det H_S[\omega_k]} \right| \leq C_h q^{9/10} \log^2(q+2) \quad (\text{OR23})$$

for each of the original source and relation spaces. No quotient frame is discarded; its fixed determinant cancels in the ratio.

On each dyadic interval $K \leq |y| \leq 128q$, (OR16) bounds $r_k - a_q$ in L^1 , where $a_q(y) = \mathcal{L}_{\zeta}(|y|)/\mathcal{L}_{\zeta}(q)$. This a_q is bounded above and below by fixed positive constants. The pointwise polynomial bounds in (OR20) imply

$$|\log r_k - \log a_q| \leq C_h \log(q+2) |r_k - a_q|. \quad (\text{OR24})$$

To verify this inequality, when $|r_k - a_q| \leq a_q/2$ use the bounded derivative of $\log$ on that interval. On the complement the right-hand difference is bounded below by a fixed positive constant and the logarithms are $O_h(\log q)$. Thus (OR17) bounds both error integrals in (OR18) by $C_h q^{199/200} \log^2(q+2)$. For $q/\log^4 q \leq |y| \leq 128q$, the deterministic $a_q - 1$ and its logarithm are $O(\log \log q / \log q)$; use the exact kernel mass at most $2q+1$. On $K \leq |y| \leq q/\log^4 q$, use (OR17) and the interval length. The full remaining tails are bounded by repeated Gamma multiplication: for $\deg P \leq 2q$,

$$\int_{|y| > 128q} y^{2d} |P|^2 d\sigma \leq (128q)^{-2q} [2(3q+d+1)]^{2(q+d)} \|P\|_{\sigma}^2. \quad (\text{OR25})$$

Together with (OR20), this gives an exponentially small tail for both $r_k - 1$ and $\log r_k$, after summing a Gamma-orthonormal frame. Both integration regions are retained. Writing

$$E_q = C_h \left[q^{199/200} \log^2(q+2) + q \log \log(q+16) / \log(q+2) \right],$$

we have therefore proved, uniformly in the required subspaces,

$$\int (|r_k - 1| + |\log r_k|) K_S d\sigma \leq E_q. \quad (\text{OR26})$$

The desired reference is τ_d , and its exact transport is $P \mapsto (y+i)^d P$. It is an isometry from $\mathcal{P}_{N-d}$ in σ onto $(y+i)^d \mathcal{P}_{N-d}$ in τ_d , since $|y+i|^{2d} = (1+y^2)^d$. For the factor r_k the same identity holds. On a relation space the map is $QP \mapsto Q(y+i)^d P$, preserving the complete factor Q . Its omitted quotient is the actual image of the first d divided derivatives of P at $-i$, hence has dimension at most d . Its two metrics have logarithmic relative cost $O_h(d \log q)$ by (OR21). Apply (OR18), (OR26) on the exact image, then include this quotient. Sum the original signs using (OR2), undo (OR23), and cancel A_k only after each quotient's dimension has been retained. This proves

$$\mathcal{B}[\omega_k] - \mathcal{B}[\sigma] = \mathcal{B}[\tau_d] - \mathcal{B}[\sigma] + O_h(E_q). \quad (\text{OR27})$$

13 The rational model and its finite free-energy error

Let $\rho_{\alpha_E, \beta}$ be the original EIQ density for $V(x) = \beta\sqrt{x} - \alpha_E \log x$. Its Euler inequality is $2U - V \leq -\lambda$, and $\mathcal{F} = -\lambda - \iint \log|x-y|\rho(x)\rho(y) dx dy$. The EIQ endpoint formulas give smooth dependence on both positive parameters and support in a fixed positive compact interval on the parameter set used below. The density is bounded there, has square-root edges, finite logarithmic energy and finite entropy relative to $d\omega(x) = \frac{1}{2}x^{-1/2}e^{-\sqrt{x}}dx$.

Here is an independent finite-configuration estimate underlying the shrinking secant. Replace each configuration point x_j by uniform mass on the complex circle $x_j + \epsilon e^{i\theta}$; denote the average of these measures by ν_ϵ . The circle mean identity is

$$\frac{1}{2\pi} \int_0^{2\pi} \log|z - \epsilon e^{i\theta}| d\theta = \log \max(|z|, \epsilon). \quad (\text{OR28})$$

It follows from harmonic mean value outside and inside the circle, with continuity at its boundary. For compactly supported zero-mass signed measures with finite logarithmic energy,

$$-\iint \log|z-w| d\eta(z) d\eta(w) = \frac{1}{2} \int_0^\infty \iint e^{-t|z-w|^2} d\eta(z) d\eta(w) \frac{dt}{t} \geq 0. \quad (\text{OR29})$$

The scalar logarithm identity proves the equality after truncation; the zero mass cancels the constant. The Fourier transform of the planar Gaussian proves the nonnegative sign. Absolute logarithmic integrability of the present circle measures and bounded density justifies passage to the limit. Use (OR29) with $\eta = \nu_\epsilon - \rho dx$. The diagonal circle energy is $\log \epsilon$, off-diagonal circle means are at least $\log|x_i - x_j|$, and (OR28) bounds the increase of the cross energy by $2\|\rho\|_\infty \epsilon$. The Euler inequality now yields

$$2 \sum_{i < j} \log|x_i - x_j| - n \sum_j V(x_j) \leq n^2 \mathcal{F} + n \log n + 4n\|\rho\|_\infty \quad (\epsilon = 1/n). \quad (\text{OR30})$$

The literal parity coordinate is $y^2 = q^2 x$. With $B_q(x) = 2qe^{\sqrt{x}}\sigma(q\sqrt{x})$, its exact ensemble is

$$J_{n,q}(n\alpha_E) = \frac{1}{n!} \int \prod_{i < j} (x_i - x_j)^2 \prod_j x_j^{n\alpha_E} B_q(x_j) d\omega(x_j). \quad (\text{OR31})$$

The original Gamma bound gives $B_q(x) \leq Cq^{3/2}e^{-(\pi q/2 - 3/2)\sqrt{x}}$. Apply (OR30) with $\beta_n = \pi q/(2n) - 3/(2n)$, and integrate against the probability ω^n . The explicit prefactor and $n!$ cost $O(n \log(q+n))$. Changing β_n to $\pi q/(2n)$ costs $O(n)$, since $\partial_\beta \mathcal{F} = -\int \sqrt{x} \rho dx$ is uniformly bounded here. For the lower bound restrict to the actual equilibrium support, use the original Gamma lower bound there, and change measure to ρ^n . Jensen's inequality gives the same $n^2 \mathcal{F}$, with $O(n)$ missing-diagonal energy and entropy, plus $O(n \log(q+n))$ prefactors. Hence

$$|\log J_{n,q}(n\alpha_E) - n^2 \mathcal{F}(\alpha_E, \pi q/(2n))| \leq Cn \log(q+n), \quad (\text{OR32})$$

uniformly when q/n is near 2, and α_E is near $[7/4, 9/4]$.

For the original relation rank $r = q, q+1$, define

$$\begin{aligned} D_{k,r}(t) &= \det \text{Gram}_{(1+y^2)^{qt}\sigma}(Q, yQ, \dots, y^{r-1}Q), \\ F_{k,r}(t) &= q^{-2} \{\log[D_{k,r}(t)/D_{k,r}(0)] - 2qrt \log q\}. \end{aligned} \quad (\text{OR33})$$

The squared-Vandermonde integral proves convexity by differentiating twice: the second derivative of its logarithm is a nonnegative variance. The paired-root comparison on $|y| \geq 2^{-16}q$, and the complete inner-interval estimate in the accompanying root-comparison proof, give logarithmic quadratic-form width $O_h(1)$ uniformly for $|t| \leq 1/8$. Its determinant cost is $O_h(q)$, because the rank is r . For the reference power $|y|^{2q(1+t)}\sigma$, its even and odd blocks have sizes

$$n_0 = \lceil r/2 \rceil, \quad n_1 = \lfloor r/2 \rfloor, \quad a_0 = q(1+t), \quad a_1 = q(1+t) + 1.$$

Their parameters satisfy

$$(a_i/n_i, \pi q/(2n_i)) = (2 + 2t, \pi) + O(1/q) \quad (i = 0, 1). \quad (\text{OR34})$$

The exact scaling factor of each block is $q^{2a_i n_i + 2n_i(n_i - 1)}$. Its change is exactly $2qrt \log q$ after the two blocks are added. Applying (OR32) and EIQ smoothness therefore gives

$$\sup_{|t| \leq 1/8} |F_{k,r}(t) - F_*(t)| \leq C_h \frac{\log(q+2)}{q}, \quad F_*(t) = \frac{1}{2}[\mathcal{F}(2+2t, \pi) - \mathcal{F}(2, \pi)]. \quad (\text{OR35})$$

Here $F'_*(0) = \mathcal{L}(2, \pi)$ and F''_* is bounded. Take $\epsilon = \sqrt{\log(q+2)/q}$ with $d/q < \epsilon/2$ and $2\epsilon < 1/8$. The convex secant over $[-d/q, 0]$ is bounded below and above by adjacent secants of length ϵ . Equation (OR35) changes these latter slopes by at most $C_h \log(q+2)/(q\epsilon)$; bounded F''_* changes their centers from $F'_*(0)$ by $O_h(\epsilon)$. Returning the exact powers in (OR33) proves

$$\log \frac{D_{k,r}(-d/q)}{D_{k,r}(0)} = -2dr \log q - dq \mathcal{L}(2, \pi) + O_h(\sqrt{q \log(q+2)}), \quad r = q, q+1. \quad (\text{OR36})$$

No derivative of an asymptotic remainder was taken.

For the source determinant, use the monic frame

$$1, \dots, y^{d-1}, (y+i)^d, \dots, y^{N-d}(y+i)^d.$$

Its determinant is one and the last block is exactly the Gamma Gram on $\mathcal{P}_{N-d}$. The first d divided derivatives at $-i$ give the quotient and have determinant one on the first d monomials. Their minimum Gram is bounded above by the fixed monomial trial lifts. It is bounded below by a negative fixed power of $N+2$: use $\tau_d \succeq [1 + 4(N+1)^2]^{-d} \sigma$ on $\mathcal{P}_N$, then the above generating function on fixed circles about $-i$ bounds every Gamma jet covariance by a fixed power of $N+2$. There are only d rows, so the reciprocal covariance lower bound is again polynomial. Consequently, retaining this quotient determinant,

$$\log \frac{\det H_N[\tau_d]}{\det H_N[\sigma]} = - \sum_{j=N-d+1}^N \log \gamma_j + O_d(\log(N+2)). \quad (\text{OR37})$$

The exact $\gamma_j = \sqrt{2\pi}(2j)!/4^j$ gives $\log \gamma_j = 2j \log j - 2j + O(\log(j+2))$. Insert the four source endpoints and their original signs:

$$\sum_{N=q-1, q, 2q-1, 2q} \epsilon_N \log \frac{\det H_N[\tau_d]}{\det H_N[\sigma]} = 4dq \log q + (8d \log 2 - 4d)q + O_d(\log q). \quad (\text{OR38})$$

For the low relation, rank zero contributes 0; rank one has norm ratio between $[1 + 4(q+1)^2]^{-d}$ and 1, by the same Jensen estimate applied to the full Q . It contributes $O_d(\log q)$. Using (OR36) at the two high relation ranks $q, q+1$, and the minus sign of the relation determinant in (OR2), the net relation contribution to the baseline is

$$-4dq \log q - 2dq \mathcal{L}(2, \pi) + O_h(\sqrt{q \log(q+2)}). \quad (\text{OR39})$$

Adding (OR38) and (OR39) proves

$$\mathcal{B}[\tau_d] - \mathcal{B}[\sigma] = -d[2\mathcal{L}(2, \pi) - 8 \log 2 + 4]q + O_h(\sqrt{q \log(q+2)}). \quad (\text{OR40})$$

Together with (OR27), this proves (OR4)–(OR5).

14 Integration scope

All constants denoted C_h, c_h are fixed-source constants with existence proved by the indicated estimates. They have not been evaluated numerically for an unspecified off-critical quartet. The cofinal assertion retains the original packet cutoffs and uses, in addition, the following literal finite inequalities wherever the corresponding step is applied:

$$\begin{aligned}
 q &\geq 2^{16}, \quad Y = 2^{-16}q, \quad R \leq Y/4, \quad R^2/q^{7/4} \leq (5/4)2^{-28}, \\
 K = q^{9/10} &\geq kb, \quad T^{19/20} \geq 2(k-1)b, \quad 2T^{19/20} < T, \quad G \log T < \sqrt{T}/2, \\
 \epsilon_k &\leq 1/2, \quad \epsilon_{k-3} \leq e^{-1/2}/2, \quad k \geq V/b^2, \quad r \leq 2q+1, \\
 \epsilon &= \sqrt{\log(q+2)/q}, \quad d/q < \epsilon/2, \quad 2\epsilon < 1/8, \quad q^{9/10} \leq T \leq 128q.
 \end{aligned} \tag{OR41}$$

Here ϵ_n is the original relative local-limit error from the localization provider and r is the selected Taylor rank in (OR22). The Chernoff step also takes its explicit eventual range $C_h k/T + C_h k \log T/T^{19/20} \leq 1/4$, enlarging the fixed-source constants if needed. The root-comparison appendix retains its displayed exponentially small $\epsilon_q \leq 1/2$. Every guard holds eventually on each original fixed-multiplicity sequence: $q \asymp_h k^2$ for $m = 1$, $q \asymp_h k^3$ for $m > 1$, and $R = O_h(k)$. Thus these are finite-domain specifications, not new assumptions replacing any missing estimate.

The fixed multiplicity, every root and repetition, both real half-lines, the source-to-remainder map and its complete relation Gram were used throughout. The scalar A_k cancelled only at the full quotient receiver (OR1). The estimated complete quotient determinant does not determine the seven absolute allocations or projected arithmetic signs. Those quantities retain their original restricted metrics and same-class pairings. The exact metric maps established here are (OR2), the Taylor short sequence used in (OR23), and multiplication by $(y+i)^d$ used in (OR27).

Finite comparison with the complete original root polynomial

Scope and sources. This calculation supplies the full polynomial quadratic-form comparison used between O32 and the reference-power reduction in O31 of `Signed_Arithmetic_Value_20260918/proof/OUTER_VALUE.md`. It uses the original Gamma mass and lower bound RTM1–2 in that packet's `sources/RTM_GTM_original_providers.tex`. The Legendre normalization and recurrence are the classical ones recorded in NIST DLMF, Table 18.3.1 and Section 18.9; their use and the resulting finite extrapolation inequality are detailed below. This note establishes the root-comparison step. It makes no assertion that the separate reference-ensemble free-energy calculation has thereby been proved.

Original objects and finite range. Retain the original fixed multiplicity $m \geq 1$, actual quartet parameters $0 < \delta < 1/2$, $\gamma > 2$, and original orders $k \equiv 1 \pmod{4}$. Write

$$r_h = \sqrt{\delta^2 + \gamma^2}, \quad e = 1 + k(m-1), \quad q = e(k+1)^2, \quad R = kr_h, \quad c = 2^{-16}, \quad Y = cq.$$

The original real even monic polynomial is $Q(y) = i^{-q}\chi_k(k/2 + iy)$, of exact degree q , with every repetition retained and every root of modulus at most R . In particular there are exactly $q/2$ opposite-root pairs. Set

$$\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}, \quad c_\Gamma = \sqrt{2}e^{-7/6}, \quad M_\Gamma = \sqrt{2\pi}, \quad B = \frac{5}{4}2^{-28}.$$

The exact mass and original lower bound are

$$\int_{\mathbb{R}} \sigma(y) dy = M_\Gamma, \quad \sigma(y) \geq c_\Gamma e^{-\pi|y|/2} (1 + |y|)^{-1/2}. \quad (\text{RC1})$$

For the explicit estimates below use the finite guard

$$q \geq 2^{16}, \quad R \leq Y/4, \quad \frac{R^2}{q^{7/4}} \leq B. \quad (\text{RC2})$$

This guard is attained on every original fixed packet; it is not an extra asymptotic assumption. Indeed $q \geq (k+1)^2 \geq k^2$, so

$$\frac{R}{Y} \leq \frac{r_h}{ck}, \quad \frac{R^2}{q^{7/4}} \leq r_h^2 k^{-3/2}.$$

Thus (RC2) follows from the completely explicit sufficient condition

$$k \geq \max \left\{ 256, 2^{18}r_h, (r_h^2/B)^{2/3} \right\}, \quad k \equiv 1 \pmod{4}. \quad (\text{RC3})$$

Any earlier finite guards imposed on the original packet remain in force. The enlargement here only makes the cofinal range explicit.

Both full forms. For every real $t \in [-1/8, 1/8]$ and every complex polynomial $P \in \mathcal{P}_q$, define

$$A_t(P) = \int_{\mathbb{R}} |P(y)|^2 |Q(y)|^2 (1 + y^2)^{qt} \sigma(y) dy, \quad B_t(P) = \int_{\mathbb{R}} |P(y)|^2 |y|^{2q(1+t)} \sigma(y) dy. \quad (\text{RC4})$$

All integrals converge, and both forms are positive definite. The compared linear map is the identity $\mathcal{P}_q \rightarrow \mathcal{P}_q$ in the original y coordinate; in the relation space it is the exact map $P \mapsto QP$, with every coefficient and root of Q retained. The nonintegral reference power occurs only in the comparison metric. No relation polynomial is replaced in the original form.

Lemma 14.1 (Exterior comparison). *Define*

$$H_{q,h} = \frac{qR^2}{Y^2 - R^2} + \frac{q}{8} \log(1 + Y^{-2}). \quad (\text{RC5})$$

For $|y| \geq Y$ and $|t| \leq 1/8$ the quotient of the original weight in A_t by the reference weight in B_t belongs to $[e^{-H_{q,h}}, e^{H_{q,h}}]$. Moreover

$$H_{q,h} \leq c^{-2} \left(\frac{16}{15} r_h^2 + \frac{1}{8q} \right). \quad (\text{RC6})$$

Proof. Write $Q(y) = \prod_{j=1}^{q/2} (y^2 - z_j^2)$, where $|z_j| \leq R$. For $|w| < 1$ the absolutely convergent power series for $\log(1 - w)$ gives $|\log|1 - w|| \leq |w|/(1 - |w|)$. Consequently

$$\left| \log \frac{|Q(y)|^2}{|y|^{2q}} \right| \leq 2 \sum_{j=1}^{q/2} \frac{|z_j|^2}{y^2 - |z_j|^2} \leq \frac{qR^2}{y^2 - R^2}.$$

The logarithm contributed by the other factor is exactly $qt \log(1 + y^{-2})$. This proves (RC5) on each exterior half-line. Since $R/Y \leq 1/4$, $R^2/q \leq r_h^2$, and $\log(1 + Y^{-2}) \leq Y^{-2}$, (RC6) follows. $\square$

Lemma 14.2 (Uniform weight bound on the retained inner interval). *For $|y| \leq Y$ and $|t| \leq 1/8$,*

$$|Q(y)|^2 (1 + y^2)^{qt} \leq B^q q^{2q(1+t)}, \quad |y|^{2q(1+t)} \leq B^q q^{2q(1+t)}. \quad (\text{RC7})$$

Proof. Opposite-root pairing gives $|Q(y)|^2 \leq (y^2 + R^2)^q$. For $u = y^2$, the derivative of $\log[(u + R^2)(1 + u)^t]$ has the sign of $(1 + t)u + 1 + tR^2$. Its slope is strictly positive. Thus its maximum on $[0, Y^2]$ occurs at an endpoint, for every allowed t ; any interior stationary point is a minimum. At $u = 0$, division by $q^{2(1+t)}$ gives at most $R^2/q^{7/4} \leq B$. At $u = Y^2$, the quotient equals

$$\frac{Y^2 + R^2}{q^2} \left(\frac{1 + Y^2}{q^2} \right)^t.$$

Here $(1 + Y^2)/q^2 = q^{-2} + c^2 < 1$ by (RC2), so this expression is largest at $t = -1/8$. Since $1 + Y^2 \geq Y^2$ and $R^2/Y^2 \leq 1/16$, its value is at most $(17/16)c^{7/4} \leq (5/4)2^{-28} = B$. Raising the two endpoint bounds to the q th power proves the first claim. The reference bound follows from $c^{2(1+t)} \leq c^{7/4} < B$. $\square$

Lemma 14.3 (Complete shifted-Legendre extrapolation). *Every $P \in \mathcal{P}_q$ and every $|y| \leq Y$ satisfy*

$$|P(y)|^2 \leq \frac{(q+1)^2}{q} 100^q \int_q^{2q} |P(u)|^2 du. \quad (\text{RC8})$$

Proof. Use the classical Legendre polynomials $L_j(x) = (2^j j!)^{-1} (d/dx)^j (x^2 - 1)^j$. Integration by parts j times proves orthogonality to every polynomial of degree below j , because $(x^2 - 1)^j$ has zeros of order j at both endpoints. Their leading coefficient is $(2j)!/[2^j (j!)^2]$. Applying the same integration by parts with L_j itself gives

$$\int_{-1}^1 L_j(x)^2 dx = \frac{(2j)!}{2^{2j} (j!)^2} \int_{-1}^1 (1 - x^2)^j dx = \frac{2}{2j+1};$$

the middle integral is $2^{2j+1} (j!)^2 / (2j+1)!$, obtained by a beta integral or successive integration by parts. These are the normalizations in DLMF Table 18.3.1 [5]. Their recurrence $(j+1)L_{j+1} = (2j+1)xL_j - jL_{j-1}$ (DLMF Section 18.9 [6]) also follows directly by matching the leading coefficient and using the just-proved orthogonality: xL_j has only L_{j+1} and L_{j-1} components by parity and orthogonality, and their coefficients are $(j+1)/(2j+1)$ and $j/(2j+1)$. For $|x| \leq 4$ it proves

$|L_j(x)| \leq 10^j$ by induction: $L_0 = 1$, $|L_1| \leq 4$, and the next value is at most $8 \cdot 10^j + 10^{j-1} \leq 10^{j+1}$. The functions

$$\ell_j(u) = \sqrt{\frac{2j+1}{q}} L_j(2u/q - 3), \quad 0 \leq j \leq q,$$

are therefore a complete orthonormal basis for $\mathcal{P}_q$ on $[q, 2q]$. If $|y| \leq Y$, then $|2y/q - 3| < 4$. Expanding P in that entire basis and applying Cauchy–Schwarz gives

$$|P(y)|^2 \leq \left(\int_q^{2q} |P(u)|^2 du \right) \frac{1}{q} \sum_{j=0}^q (2j+1) 100^j.$$

Finally the sum is at most $100^q(q+1)^2$. All polynomial directions, including the endpoint degree q , were included in this calculation. $\square$

The two inner integrals, with their original mass. For nonzero P the original Gamma lower bound on $[q, 2q]$ gives

$$B_t(P) \geq \frac{c_\Gamma e^{-\pi q}}{\sqrt{1+2q}} q^{2q(1+t)} \int_q^{2q} |P(u)|^2 du. \quad (\text{RC9})$$

Here $2q(1+t) > 0$, so its power has its minimum at $u = q$. Combining (RC7)–(RC9) with the entire mass M_Γ proves

$$\begin{aligned} \int_{-Y}^Y |P(y)|^2 |Q(y)|^2 (1+y^2)^{qt} \sigma(y) dy &\leq \varepsilon_q B_t(P), \\ \int_{-Y}^Y |P(y)|^2 |y|^{2q(1+t)} \sigma(y) dy &\leq \varepsilon_q B_t(P), \end{aligned} \quad (\text{RC10})$$

where the same completely explicit number works for both:

$$\varepsilon_q = \frac{\sqrt{2\pi}}{c_\Gamma} \frac{(q+1)^2 \sqrt{1+2q}}{q} (100e^\pi B)^q. \quad (\text{RC11})$$

In particular $100e^\pi B < 1/16$ and $\sqrt{2\pi}/c_\Gamma < 6$, so, for $q \geq 1$, $\varepsilon_q \leq 42q^2 16^{-q}$. This is less than $1/2$ throughout (RC2). Thus both inner intervals have been quantitatively retained.

Theorem 14.4 (Finite forms and full determinant return). *Throughout (RC2), uniformly in $|t| \leq 1/8$ and $P \in \mathcal{P}_q$,*

$$e^{-H_{q,h}}(1 - \varepsilon_q) B_t(P) \leq A_t(P) \leq (e^{H_{q,h}} + \varepsilon_q) B_t(P). \quad (\text{RC12})$$

For every fixed linearly independent frame $P_1, \dots, P_n$ in $\mathcal{P}_q$, the two full Gram matrices satisfy

$$\left| \log \frac{\det \text{Gram}_{A_t}(P_1, \dots, P_n)}{\det \text{Gram}_{B_t}(P_1, \dots, P_n)} \right| \leq n \{H_{q,h} - \log(1 - \varepsilon_q)\}. \quad (\text{RC13})$$

In particular this is $O_h(q)$ for the original frames $1, y, \dots, y^{n-1}$ with $n = q$ and $n = q+1$.

Proof. On the union of the two exterior half-lines use the pointwise bounds (RC5). For the lower bound, discard only the nonnegative original inner integral and retain the estimate $B_t(\{|y| \geq Y\}) \geq (1 - \varepsilon_q) B_t(P)$. For the upper bound, add the first inequality of (RC10) to the exterior upper bound. This proves (RC12), without changing either full norm.

Since (RC12) holds for every linear combination of the frame, it is a Loewner comparison of the complete Gram matrices, including all cross blocks. Conjugate by the positive inverse square root of the reference Gram. Every eigenvalue of the resulting positive matrix lies in the interval from (RC12); their product is the displayed determinant ratio. The lower endpoint has logarithm $-H_{q,h} + \log(1 - \varepsilon_q)$. The upper logarithm is $H_{q,h} + \log(1 + \varepsilon_q e^{-H_{q,h}})$, which is at most $H_{q,h} - \log(1 - \varepsilon_q)$. This proves (RC13). The bound (RC6) and exponentially decreasing (RC11) give the claimed uniform $O_h(q)$, with its finite constant and domain displayed. $\square$

Exact consequence for O30–O31. Let $D_{k,n}^{\text{ref}}(t)$ denote the full Gram determinant on $1, y, \dots, y^{n-1}$ for the second weight in (RC4), and put

$$F_{k,n}^{\text{ref}}(t) = q^{-2} \{ \log D_{k,n}^{\text{ref}}(t) - \log D_{k,n}^{\text{ref}}(0) - 2qnt \log q \}.$$

The determinant denoted $D_{k,n}(t)$ in O30 is exactly the A_t determinant, because its columns are QP_j in the original weighted Gamma norm. Applying (RC13) twice, at t and at 0, proves

$$|F_{k,n}(t) - F_{k,n}^{\text{ref}}(t)| \leq \frac{2n}{q^2} \{ H_{q,h} - \log(1 - \varepsilon_q) \} = O_h(q^{-1}). \quad (\text{RC14})$$

This uniform finite bound proves precisely the root-comparison reduction required in O31, including its q^2 scaling. The zero-endpoint sentence in O32 is valid on a cofinal range, but the finite guard $R^2/q^{7/4} \leq B$ must be stated to use the displayed uniform endpoint constant. The Y -endpoint constant $[(5/4)2^{-28}]^q$ is valid, as proved above.

Primitive-endpoint control and exact CPQ integration

Abstract

The original PRT comparison between primitive endpoints has three quotient directions. Its preceding finite estimate had logarithmic width $O(q \log a)$. The original full-remainder AKS estimate and the supplied LRC conductor bounds imply a native Loewner growth bound of width $O_{\text{actual}}(q)$ before any fixed flag restriction or attained quotient. The three original PRT returns are therefore $O(q)$, as is the difference of the original common losses. In particular the primitive-endpoint difference has zero $q \log a$ coefficient, with an $O(q)$ remainder, on every retained rank stratum. This resolves that specific logarithmic ambiguity. It does not evaluate either absolute directional allocation, the independent proper-source standard return, or the retained-class projected signs; the requested directional completion target remains open.

The programme origin and its existing attribution are retained [13, 14]. This additive integration retains the complete supplied calculation below and appends the exact provider correspondence and a sharper finite endpoint receiver. The original supplied bytes remain in the accompanying input directory.

15 Original domain and scope

Fix the stipulated simple quartet, its actual full unit, the original five-orbit period and its logarithm branch. Retain

$$\begin{aligned} a &= k + 4 \equiv 1 \pmod{4}, \quad k \geq 9, \quad q = (a + 1)^2, \quad Q = (a - 7)^2, \\ \Delta &= 16a - 48, \quad m = \Delta - v, \quad s \in \{1, a\}, \quad c = a/2, \quad c' = a/2 - 4, \\ E_A(z) &= \sum_{u,w=0}^8 a_{uw} e^{\beta_{8,u,w} z}, \quad \mu_t = E_A^{(t)}(0), \quad v = \text{ord}_0 E_A \leq 80, \\ \mu_v &\neq 0, \quad \alpha_A = v!/\mu_v, \quad \beta_{a,u,w} = c + (2u - a)\delta + i(2w - a)\gamma, \\ &0 < \delta < 1/2, \quad \gamma > 2. \end{aligned} \tag{PEQ1}$$

Here m is a coefficient dimension, not a zero multiplicity. The lower grid has degree $a - 8$ and source order s , not order $a - 8$. The independent proper source keeps degree k , its own centre and orders $1, k$. All original five-orbit and PRT coefficient conditions remain. Neither a new large-root separation guard nor a numerical period choice is introduced by the bound below. The constants depend on the fixed actual conductor; joint degree/period uniformity is not asserted.

Use exactly

$$\begin{aligned} dm_s(y) &= \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\ h_{n,s} &= M_s n! (s/2)_n, \quad \phi_{n,s,c}(S) = p_{n,s}((S - c)/i) / \sqrt{h_{n,s}}, \\ p_{n+1,s} &= y p_{n,s} - n(n - 1 + s/2) p_{n-1,s}, \quad p_0 = 1, \quad p_1 = y. \end{aligned} \tag{PEQ2}$$

These classical orthogonal-polynomial formulae retain the exact Meixner–Pollaczek dictionary $p_{n,s}(y) = n! P_n^{(s/4)}(y/2; \pi/2)$ [12, eqs. (18.19.8)–(18.19.9), (18.23.7)]. In particular the change $y = 2x$ keeps the full source mass. Write $D_N = D_{a,N-v}^{(s)}$ for the original divided-native metric at $N \geq q - 1$, and $\mathcal{D}_N = \text{diag}(D_N, 4)$. This abbreviation retains the cutoff shift v .

The new evaluated logarithmic conclusion is

$$\boxed{\mathcal{L}_0 - \mathcal{L}_1 = O_{\text{actual}}(q), \quad \frac{\mathcal{L}_0 - \mathcal{L}_1}{q \log a} \longrightarrow 0.} \tag{PEQ3}$$

No existence of an asymptotic expansion for either absolute loss is assumed. The same conclusion applies to the PRT-related complementary differences specified below. Every result is restricted to the simple-quartet metric lane.

16 A full-numerator estimate in the original coordinates

Set $R = a\sqrt{\delta^2 + \gamma^2}$ and $b = s/2$. The exact AKS3 envelope at degree a is

$$\begin{aligned} t_q &= \pi/2 - 1/q, \quad A_q = 2 \log 3/t_q, \quad S_q = \sum_{j=0}^{q-1} A_q^j, \\ B_{s,q} &= [(9/25) \sin(1/q)]^{-b} \exp(2R(\log 3 + t_q/2)), \\ Z_{a,s} &= 1 + B_{s,q} S_q^2 \sum_{n=q}^{2q} \frac{n!}{(b)_n} (25/16)^n. \end{aligned} \tag{PEQ4}$$

These are scalar envelopes, not boundary subspaces or coefficient frames. The original $q \geq 36$ requirement is satisfied in (PEQ1).

Let $E_N[n, \lambda] = \phi_{n,s,c}(\beta_\lambda)$ be the literal evaluation matrix on the full original grid, and $K_N = E_N^* E_N$. Let v_j be the remainder-coefficient column of $\phi_{q+j,s,c}$ in the low original orthonormal frame, and $E_0 = E_{q-1}$. The added evaluation row is $v_j^\top E_0$, so

$$K_N = E_0^* \left(I + \sum_{j=0}^{N-q} v_j v_j^* \right) E_0. \tag{PEQ5}$$

The ordinary transpose and the subsequent conjugation are required. The full original AKS18–24 calculation gives

$$I \preceq I + \sum_{j=0}^{N-q} v_j v_j^* \preceq Z_{a,s} I.$$

Consequently

$$\boxed{K_{q-1} \preceq K_N \preceq Z_{a,s} K_{q-1}, \quad q-1 \leq N \leq 2q.} \tag{PEQ6}$$

This imports the full remainder operator bound, not an observation-rank estimate.

For its normalization, AKS applies the literal remainder map to the original generating identity

$$\sum_{n \geq 0} p_{n,s}(y) z^n / n! = (1 + z^2)^{-s/4} e^{y \arctan z}.$$

The divided-difference integral at the complete root list proves $|[\omega_1, \dots, \omega_{j+1}] e^v| \leq |v|^j e^{R|v|} / j!$. The whole Newton polynomial, expanded with every coefficient, satisfies

$$\left\| \prod_{i=1}^j (y - \omega_i) \right\|_s \leq \sqrt{M_s} (\cos t)^{-s/4} (2/t)^j j! e^{Rt/2}.$$

Cauchy extraction on $|z| = 4/5$ and $t = t_q$, followed by the exact normalization $h_{n,s}$, gives

$$\|v_{n-q}\|^2 \leq B_{s,q} S_q^2 \frac{n!}{(s/2)_n} (25/16)^n.$$

Adding the complete positive rank-one matrices through $n = 2q$ yields (PEQ4). The original mass cancels only between this squared source norm and its matching $h_{n,s}$.

The explicit envelope has size

$$\log Z_{a,s} = O_h(q), \quad \limsup_{a \rightarrow \infty} \frac{\log Z_{a,s}}{q} \leq 2 \log \frac{25 \log 3}{4\pi}. \tag{PEQ7}$$

Indeed $A_q \rightarrow 4 \log 3 / \pi > 1$, hence $\log S_q = q \log A_q + O(1)$. For $s \geq 1$,

$$\frac{n!}{(s/2)_n} \leq \frac{n!}{(1/2)_n} = \frac{4^n}{\binom{2n}{n}} \leq 2n + 1.$$

The sum in (PEQ4) is at most $(q+1)(4q+1)(25/16)^{2q}$. Finally $\log B_{s,q} = O_h(a \log q) = o(q)$ in both original orders. This proves (PEQ7), a bound on the scalar envelope, not an individual directional eigenvalue.

Use the original numerator matrices C_a, Z_a of NCT/NPK/IGI. Their attained form is

$$\widehat{D}_N = Z_a^* [K_N - K_N C_a^\top (\overline{C}_a K_N C_a^\top)^{-1} \overline{C}_a K_N] Z_a. \quad (\text{PEQ8})$$

Equivalently its quadratic form is the minimum of $(Z_a x + C_a^\top \eta)^* K_N (Z_a x + C_a^\top \eta)$ over the entire lower coefficient vector η . Applying (PEQ6) to every such vector and then minimizing proves

$$\widehat{D}_{q-1} \preceq \widehat{D}_N \preceq Z_{a,s} \widehat{D}_{q-1}. \quad (\text{PEQ9})$$

In particular neither of the two cross factors or the old lower inverse in (PEQ8) has been removed.

17 Transport the entire bound through the original conductor

At $D = N - v$ the original conductor is the square map in the original native frames,

$$A_{D,s} : \mathcal{P}_N / \mathcal{P}_{v-1} \longrightarrow \mathcal{P}_D, \quad [P] \longmapsto \mathcal{T}_A P.$$

Its exact WCF/LRC matrix is

$$\begin{aligned} A_{D,s} &= \text{diag}(\rho_i)^{-1} T_D(g) \text{diag}(\rho_{i+v}), & \rho_n^2 &= n! / (s/2)_n, \\ g(t) &= t^{-v} e^{-4w(t)} E_A(w(t)), & w(t) &= -i \arctan t, & g(0) &= (-i)^v \mu_v / v!. \end{aligned} \quad (\text{PEQ10})$$

The nilpotent Toeplitz matrix uses its literal cutoff D . No interior zero, pole direction, full unit value or branch factor is replaced. Both source centres are the original ones in (PEQ1).

Here are the complete finite LRC6–9 constants used for the transport. Define

$$\begin{aligned} A_{\text{abs}} &= \sum_{u,w} |a_{uw}|, & F_0 &= 4 + 4\sqrt{\delta^2 + \gamma^2}, & B_0 &= 2^{v+8} A_{\text{abs}} e^{2\pi\gamma}, \\ B_2 &= B_0 \{F_0(F_0 + 1) + 4vF_0 + 4v(v+1)\}, & H &= B_0(1 + \sqrt{3}) + 2\sqrt{10}B_2, \\ H_s &= \begin{cases} \max(1, H\sqrt{2v+1}), & s=1, \\ \max(1, H), & s=a, \end{cases} & \kappa_s &= \begin{cases} 1 + \sqrt{8/3}, & s=1, \\ 2, & s=a. \end{cases} \end{aligned} \quad (\text{PEQ11})$$

Write $d = D + 1$ and $\mathbf{H}_d = \sum_{n=1}^d 1/n$. For $1 \leq r \leq bd$ and $b = 1$ or 4 , put

$$\begin{aligned} F_{D,s,r,b} &= r \log H_s + 4(bd - r) \log \kappa_s, & F_{D,s,0,b} &= 0, \\ J_{D,s} &= d \log^+ \frac{1}{|g(0)|} + \frac{1}{2}(s/2 - 1)_+ v \mathbf{H}_d, \\ I_{D,s,r,b} &= b J_{D,s} + F_{D,s,bd-r,b}, & I_{D,s,0,b} &= 0. \end{aligned} \quad (\text{PEQ12})$$

Here b is a copy count only in (PEQ12); it is not the $s/2$ abbreviation in (PEQ4). LRC proves the exact original exterior bounds

$$\log \| \wedge^r (A_{D,s}^\top)^{\oplus b} \| \leq F_{D,s,r,b}, \quad \log \| \wedge^r (A_{D,s}^{-\top})^{\oplus b} \| \leq I_{D,s,r,b}.$$

Only rank one and one copy are needed below. In the admitted cutoff range $d \geq 2$, the explicit rank-one inverse bound is nondecreasing in D .

For the exact map on a complex-linear functional, multiplying its germ by E_A sends its native value column f to $A_{D,s}^\top f$, with zero first v coordinates in the upper polynomial source. The original lower exponential columns are carried by this same map to $C_a^\top$ -numerator columns. Thus the map bijects the full affine lower fibres for D_N and $\widehat{D}_N$. On every representative,

$$e^{-2I_{D,s,1,1}} \|f\|^2 \leq \|A_{D,s}^\top f\|^2 \leq e^{2F_{D,s,1,1}} \|f\|^2.$$

Taking the two attained minima gives

$$\boxed{e^{-2I_{N-v,s,1,1}} D_N \preceq \widehat{D}_N \preceq e^{2F_{N-v,s,1,1}} D_N.} \quad (\text{PEQ13})$$

The transpose is the original ordinary dual transpose. Its singular values equal those of $A_{D,s}$; no bilinear residue pairing is substituted for a Hermitian norm. The actual coefficient numerator is $Z_a x$, so the equality of value coordinates is the NCT/NPK square, not an independently assigned scale. In particular α_A remains in (PEQ10) through $g(0)$.

Theorem 17.1 (Native growth before any flag compression). *Set*

$$D_- = q - v - 1, \quad D_+ = 2q - v, \quad \boxed{\mathfrak{E}_{a,s} = \log Z_{a,s} + 2I_{D_+,s,1,1} + 2F_{D_-,s,1,1}.} \quad (\text{PEQ14})$$

Then for every original $q - 1 \leq N \leq 2q$,

$$\boxed{D_{q-1} \preceq D_N \preceq e^{\mathfrak{E}_{a,s}} D_{q-1}, \quad \mathfrak{E}_{a,s} = O_{\text{actual}}(q).} \quad (\text{PEQ15})$$

The same scalar bounds hold on the original four-copy metric, and on every fixed restriction or attained quotient of either metric, in its original value frame.

Proof. Native row extension adds a positive rank-one form after the entire old lower minimization; hence $D_N \succeq D_{q-1}$. The upper estimate follows from the complete chain

$$D_N \preceq e^{2I_{D_+,s,1,1}} \widehat{D}_N \preceq e^{2I_{D_+,s,1,1}} Z_{a,s} \widehat{D}_{q-1} \preceq e^{\mathfrak{E}_{a,s}} D_{q-1}.$$

No two unrelated minimum sections have been identified in this chain. For the size, (PEQ7) controls the first term. All constant conductor data in (PEQ11) are fixed. The terms linear in D in (PEQ12) are $O_{\text{actual}}(q)$, and its harmonic term is $O_{\text{actual}}(av \log q) = o(q)$ since v is fixed. This proves the asserted order for both original source orders.

Taking four copies preserves the scalar Loewner inequality without multiplying its logarithm by four. Restriction follows by substitution in a fixed frame. For a fixed quotient Y/X , apply the inequalities to every representative of the same affine fibre and minimize. The resulting quotient Gram is exactly

$$X_Q^* \mathcal{D}_N X_Q - X_Q^* \mathcal{D}_N X (X^* \mathcal{D}_N X)^{-1} X^* \mathcal{D}_N X_Q,$$

with both cross factors and the entire denominator retained. The scalar inequality and monotonicity therefore pass to that form, irrespective of the condition number of the chosen coefficient frame. $\square$

On the PZU domain $v = 0$, $D_- = q - 1$, $D_+ = 2q$, and $g(0) = E_A(0) = 1/\alpha_A$. These are exact simplifications; α_A is not set to one. Its possibly large fixed-period magnitude remains inside $J_{D,s}$. At an admitted period with $v > 0$, no low coordinate or derivative condition has been removed. The bound is not uniform in varying periods.

18 Evaluate the primitive-endpoint difference above the quadratic scale

Retain the PRT/CPQ coefficient subspaces

$$\begin{aligned} \mathcal{B} &= \text{im } B_{\text{cof}}, \quad \mathbb{J} = \ker \mathbb{B}, \quad W = \mathcal{B} + \mathbb{J}, \\ S_e &= \text{im}[B_{\text{cof}}, F_e], \quad K_e = S_e \cap W, \quad T_e = S_e + W \quad (e = 0, 1). \end{aligned} \quad (\text{PEQ16})$$

The defining primitive endpoint e is distinct from the later variable norm cutoff N . The original PRT6–10 matrix L has rank exactly three and gives the literal identity $S_1 = \mathcal{B} + F_0 \ker L \subset S_0$. With the original three lifts $F_0 Y$ it proves the exact sequence

$$0 \longrightarrow K_0/K_1 \longrightarrow S_0/S_1 \longrightarrow T_0/T_1 \longrightarrow 0. \quad (\text{PEQ17})$$

Write $r_\partial = \dim(T_0/T_1) \in \{0, 1, 2, 3\}$. Then the three ranks are respectively $3 - r_\partial, 3, r_\partial$. This retains the exact rank of the original three-column image, which may depend on the packet, period and degree; no generic rank is assigned.

Give these spaces their full PRT11–12 attained Grams

$$G_A(N) = \text{Gram}_{\mathcal{D}_N}(S_0/S_1), \quad G_B(N) = \text{Gram}_{\mathcal{D}_N}(K_0/K_1), \quad G_C(N) = \text{Gram}_{\mathcal{D}_N}(T_0/T_1).$$

The original frame factors in the Gram factorizations are fixed in N . They have not been normalized to one.

Lemma 18.1 (Fixed-rank quotient return). *For any of the fixed restrictions or quotients in the preceding theorem, let $G_X(N)$ be its attained form of rank r_X . With the unchanged return*

$$\mathcal{R}f = -f(q-1) - f(q) + f(2q-1) + f(2q),$$

one has the finite inequality

$$\boxed{0 \leq \mathcal{R} \log \det G_X \leq 2r_X \mathfrak{E}_{a,s}.} \quad (\text{PEQ18})$$

Proof. Put $f_X(N) = \log \det G_X(N) - \log \det G_X(q-1)$. The full scalar Loewner bounds, positive congruence and multiplication of the r_X eigenvalues give $0 \leq f_X(N) \leq r_X \mathfrak{E}_{a,s}$. Minimum monotonicity makes f_X increasing. Thus

$$\mathcal{R} \log \det G_X = [f_X(2q-1) - f_X(q-1)] + [f_X(2q) - f_X(q)]$$

is nonnegative and at most $2r_X \mathfrak{E}_{a,s}$. For $r_X = 0$, the return is exactly zero by the original empty determinant convention. $\square$

Set $U_X = \mathcal{R} \log \det G_X$ for $X = A, B, C$. Substitution of their actual ranks yields

$$\boxed{0 \leq U_A \leq 6\mathfrak{E}_{a,s}, \quad 0 \leq U_B \leq 2(3 - r_\partial)\mathfrak{E}_{a,s}, \quad 0 \leq U_C \leq 2r_\partial\mathfrak{E}_{a,s}.} \quad (\text{PEQ19})$$

These are new $O(q)$ bounds for the existing three attained matrices. They do not replace those matrices by three arbitrary rank-one directions.

PRT14–17 already proves the exact identities

$$\begin{aligned} V_{\text{conn},0} - V_{\text{conn},1} &= U_A, & V_{h,0} - V_{h,1} &= U_B, \\ V_{s,0} - V_{s,1} &= U_A - U_B, & V_{t,0} - V_{t,1} &= U_C, \\ V_{\text{rem},1} - V_{\text{rem},0} &= U_C, & \mathcal{L}_0 - \mathcal{L}_1 &= U_C - U_A + U_B. \end{aligned} \quad (\text{PEQ20})$$

Their fixed frame constants cancel only after the four signs are taken. Applying (PEQ19) to these exact identities gives

Theorem 18.2 (No primitive-choice logarithmic coefficient). *On the original domain (PEQ1),*

$$\boxed{\begin{aligned} |\mathcal{L}_0 - \mathcal{L}_1| &\leq 6\mathfrak{E}_{a,s} = O_{\text{actual}}(q), \\ 0 &\leq V_{\text{rem},1} - V_{\text{rem},0} \leq 2r_\partial\mathfrak{E}_{a,s}, \\ -2(3 - r_\partial)\mathfrak{E}_{a,s} &\leq V_{s,0} - V_{s,1} \leq 6\mathfrak{E}_{a,s}. \end{aligned}} \quad (\text{PEQ21})$$

Every difference in (PEQ20) divided by $q \log a$ tends to zero. The aq and every other scale strictly larger than q are unchanged by the primitive endpoint. This assertion does not assume absence of intermediate scales, a particular arithmetic subsequence, or convergence of either absolute directional return.

Proof. The two positive terms $U_B + U_C$ in the loss difference have total rank three by (PEQ17), and hence sum at most $6\mathfrak{E}_{a,s}$. The negative term $-U_A$ is at least $-6\mathfrak{E}_{a,s}$. This proves the first bound, rather than charging three unrelated rank-three errors. The other inequalities follow from their indicated differences. Since $\mathfrak{E}_{a,s} = O(q)$ and $\log a \rightarrow \infty$, all the normalized conclusions follow. They are uniform over the four possible ranks at each fixed period. At $r_\partial = 0$, $U_C = 0$ exactly; at $r_\partial = 3$, $U_B = 0$ exactly. $\square$

The old PRT18–20 finite envelope $\log A_{a,s}^{\text{end}}$ remains valid. In every previous directed interval replace it by

$$\mathfrak{E}_{a,s}^{\text{best}} = \min\{\log A_{a,s}^{\text{end}}, \mathfrak{E}_{a,s}\}. \quad (\text{PEQ22})$$

Thus the new interval is an intersection, never a weakening of an earlier finite one. The order $O(q)$ in the new result follows from the explicit new term.

19 Correlated receiving identities and the uncompleted terminal target

Let $Z_e = V_b + V_{\text{rem},e} + V_{s,e} + V_A$. Here Z_e is precisely the original complementary return sum, not the scalar $Z_{a,s}$ in (PEQ4). PRT gives, with the same original maps,

$$\begin{aligned} Z_1 - Z_0 &= \mathcal{L}_0 - \mathcal{L}_1 = U_C - U_A + U_B, \\ \mathcal{P}_0^{\text{rest}} - \mathcal{P}_1^{\text{rest}} &= U_C - U_A + U_B, \\ 4S_K &= \mathcal{L}_0 + Z_0 = \mathcal{L}_1 + Z_1, \quad S_K + S_R = \mathcal{T}. \end{aligned} \quad (\text{PEQ23})$$

The V_b, V_A, V_I terms do not depend on the primitive endpoint, since their coefficient subspaces and frames are unchanged. The two labelled occurrences in a signed flag expression remain labelled even when the underlying subspaces coincide. The independent degree- k proper-source form has not been replaced by a target-order form, and its standard-window value has not been evaluated by (PEQ23). The known prefix correction remains $-V_A^{\text{prefix}}$.

The new estimate resolves the former possibility that switching between the two primitive constructions could change a $q \log a$ coefficient. It permits either construction to be used for coefficients above the q scale, with the exact PRT map and the $O(q)$ error recorded. It does not determine those coefficients. In particular no rank-proportional allocation is implied by (PEQ18).

The prescribed signed profile is still the clipped sum over the actual angular data, with the original eight labels and zeros. CRV's $O(q)$ radial error and UIT's $O(q)$ attained/trial error are not replaced or claimed as new. The present native Loewner ratio is $e^{O(q)}$, not a small relative error. It therefore cannot be used to infer a retained-class pairing at the arithmetic $k\delta F_{33}$ scale.

For the coordinator's exact retained-class data, keep

$$U = K_{13}/F_{13}, \quad V = K_{23}/F_{23}, \quad P_K = \bar{U}V, \quad P_B = (1 - \bar{U})(1 - V), \quad P_\times = \bar{U} + V - 2\bar{U}V.$$

Its already evaluated full current implies

$$\frac{\Phi_X}{2F_{33}} = k[\delta\Re P_X - (-1)^r \gamma \Im P_X] + (-1)^r \varepsilon_{N,\lambda} \Im P_X.$$

The same positive $\varepsilon_{N,\lambda}$ occurs in all three terms, and each absolute replacement error still includes $|\Im P_X|$. No upper bound for that amplification is supplied here. Both leakage blocks remain in $(\Lambda AI)^* Q_N + H_N A_{KB}$; neither is identified with their sum.

Finally $J = \mathcal{B} + \mathcal{W} - \mathcal{P}$ retains its complete original cancellation and its positive $C_B q^2$ baseline. Since (PEQ23) changes no allocation sum, the $O(q)$ primitive comparison creates no new negative term in that action. The closing criterion is neither proved nor disproved by this partial result. The outstanding terminal targets are the absolute original angular clipped sums, the independent proper-source standard return, and the individual retained-class projected signs at their stipulated scale.

20 Exact provider correspondence and finite endpoint refinement

The incoming calculation is received into the frozen LRC transport proof, LT1–28 and XIM1–23, and the original PRT1–20/CPQ construction. Their source locations and exact byte pins accompany this manuscript in `SOURCE_PINS.json`. The domain is the fixed actual simple quartet, the original five-orbit coefficient construction and logarithm branch, $k = 4\ell + 1 \geq 9$, $a = k + 4$, and $s = 1, a$. Every original coefficient condition needed to define B_{cof} , $\mathbb{B}$ and F_e is retained. The estimates here add no root separation condition. If these results are used alongside a theorem requiring a larger degree or a quantitative root guard, that theorem’s guard remains in force.

There are two distinct uses of the letter a and two distinct uses of Q in the old PRT notation. The exact substitution into this manuscript is

PRT notation	notation here	
$j = k + 4$	a	
$Q = (j + 1)^2$	$q = (a + 1)^2$	
$q = (k + 1)^2$	$(a - 3)^2$	
$a = \dim(T_0/T_1)$	r_{∂}	
$D = N - v$	$D = N - v$	
$c'_j = j/2 - 4$	$c' = a/2 - 4$	(PEI1)

In particular PRT’s proper-source size $(k + 1)^2$ is not this manuscript’s lower-grid size $Q = (a - 7)^2$. The coefficient spaces and linear maps are preserved by the identity map on their original coefficients; PEI1 merely specifies the substitutions in their formulae. The proper-source orders remain $1, k$, and the target orders remain $1, a$.

20.1 The full affine-fibre square at every original cutoff

Here is the specific typed map used in the native inequality. Let $\mathcal{N}_v$ be the subspace of numerator coefficient columns $\nu \in \mathbb{C}^q$ for which $\sum_{\beta} \nu_{\beta} \beta^r = 0$ for $0 \leq r < v$. The injective map $C_a^{\text{T}} : \mathbb{C}^Q \rightarrow \mathcal{N}_v$ is defined by the coefficient identity

$$\sum_{\beta} (C_a^{\text{T}} \eta)_{\beta} e^{\beta z} = E_A(z) \sum_{\beta'} \eta_{\beta'} e^{\beta' z}. \quad (\text{PEI2})$$

The roots on the right are the full degree- $a - 8$ grid. Their addition to every degree-eight conductor root gives the original degree- a root, so PEI2 retains every convolution coefficient. If its left side vanishes, the lower exponential sum vanishes wherever $E_A \neq 0$ and then identically. Its Vandermonde is nonsingular, so $\eta = 0$. The first v moments vanish because E_A vanishes to order v .

Let $Z_a : \mathbb{C}^m \rightarrow \mathcal{N}_v$ be the original complement, so $[C_a^{\text{T}}, Z_a]$ is the fixed numerator frame. For its column b define $F_b(z) = \sum_{\beta} (Z_a)_{\beta b} e^{\beta z} / E_A(z)$, an analytic germ at zero. At $D = N - v$ put

$$\begin{aligned} (H_D)_{n, \beta'} &= \phi_{n, s, c'}(\beta'), \\ (J_D)_{n, b} &= \Lambda_{F_b}(\phi_{n, s, c'}), \quad \Lambda_F(S'^r) = F^{(r)}(0), \quad 0 \leq n \leq D, \\ B_D &= A_{D, s}^{\text{T}} : \mathbb{C}^{D+1} \longrightarrow \mathbb{C}^{D+1}. \end{aligned} \quad (\text{PEI3})$$

The source and target of B_D are respectively the complex-linear functional value coordinates in the original lower polynomial frame and the upper frame with its first v zero coordinates removed. It is an isomorphism since the diagonal of $A_{D, s}$ is $g(0)\rho_{i+v}/\rho_i \neq 0$. Leibniz’s rule gives $\Lambda_{F_b}(\mathcal{T}_A P) = \Lambda_{E_A F_b}(P)$. The same equation holds for each lower exponential functional. Applying it to every upper basis polynomial proves the literal square

$$\mathbb{E}_N[Z_a, C_a^{\text{T}}] = \begin{bmatrix} 0_{v \times (m+Q)} \\ B_D[J_D, H_D] \end{bmatrix}. \quad (\text{PEI4})$$

This is XIM12, including its ordinary transpose, both centres and the cutoff shift. Every lower coefficient $\eta \in \mathbb{C}^Q$ on one side is carried to that identical coefficient on the other side. Thus

$$\begin{aligned} x^* D_N x &= \min_{\eta \in \mathbb{C}^Q} \|J_D x + H_D \eta\|_2^2, \\ x^* \widehat{D}_N x &= \min_{\eta \in \mathbb{C}^Q} \|B_D(J_D x + H_D \eta)\|_2^2 \\ &= \min_{\eta \in \mathbb{C}^Q} (Z_a x + C_a^\top \eta)^* K_N (Z_a x + C_a^\top \eta). \end{aligned} \quad (\text{PEI5})$$

There is no requirement that the minimizing η agree between these lines. The second equality follows before minimization from PEI4 and the zero first rows.

For clarity, if $K > 0$, H has full column rank and J is the retained lift matrix, expansion about $\eta_* = -(H^* K H)^{-1} H^* K J x$ gives

$$\begin{aligned} (Jx + H\eta)^* K (Jx + H\eta) &= x^* \{J^* K J - J^* K H (H^* K H)^{-1} H^* K J\} x \\ &\quad + (\eta - \eta_*)^* H^* K H (\eta - \eta_*). \end{aligned} \quad (\text{PEI6})$$

This proves the minimum and retains both cross blocks and the complete inverse. It is the complex Hermitian Schur calculation; the corresponding classical block identity is recorded by Boyd and Vandenberghe, Appendices A.5.5 and C.4.1 [11]. Here PEI6 proves the complex form directly. Since $[H_D, J_D]$ has full column rank, these attained forms are positive definite.

For any η , increasing N appends nonnegative squared coordinates to $\|J_D x + H_D \eta\|^2$. The old rows of J_D, H_D are unchanged because their germs, centres and source order are fixed. Taking the minimum over the same full coefficient space therefore proves $D_{N+1} \succeq D_N$. No assertion about the equality of successive minimum sections is needed.

20.2 A sharper finite bound using each high endpoint

For integers $q \leq N \leq 2q$, define the finite original AKS envelope

$$Z_{a,s}(N) = 1 + B_{s,q} S_q^2 \sum_{n=q}^N \frac{n!}{(s/2)_n} (25/16)^n, \quad Z_{a,s}(q-1) = 1. \quad (\text{PEI7})$$

All constants on the right are those of (PEQ4). The full positive remainder sum in (PEQ5), truncated at its actual last row, proves $K_{q-1} \preceq K_N \preceq Z_{a,s}(N) K_{q-1}$. Equation PEI5 then proves the same scalar comparison of the attained numerator forms. For $q \leq N \leq 2q$ put

$$E_{a,s}(N) = \log Z_{a,s}(N) + 2I_{N-v,s,1,1} + 2F_{q-v-1,s,1,1}, \quad E_{a,s}(q-1) = 0. \quad (\text{PEI8})$$

The value zero at $q-1$ is the exact identity of the endpoint with itself, not the positive expression applied outside its stated range. Applying the two rank-one bounds for B_D to every representative in PEI5 gives the explicit chain

$$\begin{aligned} D_N &\preceq e^{2I_{N-v,s,1,1}} \widehat{D}_N \\ &\preceq e^{2I_{N-v,s,1,1}} Z_{a,s}(N) \widehat{D}_{q-1} \\ &\preceq e^{E_{a,s}(N)} D_{q-1}. \end{aligned} \quad (\text{PEI9})$$

It holds alongside $D_{q-1} \preceq D_N$. At positive rank one the inverse allowance, with $d = D + 1 \geq 2$, is

$$I_{D,s,1,1} = d \log^+(1/|g(0)|) + \frac{1}{2}(s/2 - 1)_+ v H_d + (d - 1) \log H_s + 4 \log \kappa_s. \quad (\text{PEI10})$$

Every term in this expression is nondecreasing in D . Consequently $E_{a,s}(N) \leq \mathfrak{E}_{a,s}$ for all the displayed cutoffs. This also checks explicitly the monotonicity of the inverse allowance used in the incoming manuscript.

Let $A_{a,s}^{\text{end}}$ be the exact old PRT18 scalar, with the substitution PEI1. It retains the original reconstruction norm, mass, Vandermonde factors and conductor. Set

$$e_{a,s}(N) = \min\{\log A_{a,s}^{\text{end}}, E_{a,s}(N)\}, \quad B_{a,s}^{\text{end}} = e_{a,s}(2q-1) + e_{a,s}(2q). \quad (\text{PEI11})$$

Both original scalar inequalities hold on the same native form, so their intersection gives

$$D_{q-1} \preceq D_N \preceq e^{e_{a,s}(N)} D_{q-1}, \quad 0 \leq B_{a,s}^{\text{end}} \leq 2\mathfrak{E}_{a,s}^{\text{best}}. \quad (\text{PEI12})$$

Taking four copies is a direct sum of these inequalities and keeps the same scalar exponent.

Let $X \subset Y$ be any fixed pair of the original coefficient spaces and let $G_{Y/X}(N)$ be its original attained metric in a fixed value frame, of rank $t = \dim(Y/X)$. Applying PEI12 to every representative of each fixed affine fibre and taking its attained minimum preserves both inequalities. Congruence by the inverse positive square root of $G_{Y/X}(q-1)$ shows that each of its t generalized eigenvalues lies in $[1, e^{e_{a,s}(N)}]$. Multiplication, followed by logarithm, proves

$$0 \leq f_{Y/X}(N) := \log \frac{\det G_{Y/X}(N)}{\det G_{Y/X}(q-1)} \leq t e_{a,s}(N). \quad (\text{PEI13})$$

The minima are increasing in N , by the same pointwise argument. The original signed endpoint functional therefore satisfies

$$0 \leq \mathcal{R} \log \det G_{Y/X} = [f_{Y/X}(2q-1) - f_{Y/X}(q-1)] + [f_{Y/X}(2q) - f_{Y/X}(q)] \leq t B_{a,s}^{\text{end}}. \quad (\text{PEI14})$$

The rank-zero determinant remains one and its return zero. This is a finite refinement of (PEQ18); no new asymptotic assumption enters it.

20.3 All primitive CPQ receivers with the correlated ranks

The maps in (PEQ17) send a class to the same coefficient class. Their exactness follows explicitly as follows. For the first map, $K_0 \cap S_1 = K_1$ proves injectivity. If $x \in S_0$ lies in $T_1 = S_1 + W$, write $x = s + w$. Then $w = x - s \in K_0$ and its class maps to $[x]$. Every element of $T_0 = S_0 + W$ has the class of its S_0 summand modulo T_1 , proving surjectivity. PRT6–8 uses the rank-three original relation map, so $\dim(S_0/S_1) = 3$. This proves that the ranks in PEI14 are exactly 3, $3 - r_\partial$ and r_∂ .

For a fixed inclusion $X \subset Y$, let P_X, P_Y be the original frames and let V be the quotient lifts. The actual invertible coefficient matrix R is defined by $[P_X, V] = P_Y R$. Taking the determinant of PEI6 in this full frame gives

$$\det G_{Y/X}(N) = |\det R|^2 \frac{\det(P_Y^* \mathcal{D}_N P_Y)}{\det(P_X^* \mathcal{D}_N P_X)}. \quad (\text{PEI15})$$

The factor $|\det R|^2$ is retained. It disappears only under $\mathcal{R}$, since it is independent of N and $\mathcal{R}1 = 0$. Applying PEI15 to $S_e/\mathcal{B}$, $K_e/\mathcal{B}$, S_e/K_e , T_e/W and $\mathcal{U}/T_e$ proves exactly the six original signed identities (PEQ20), including their orders of subtraction. Thus the complete sharper receiving intervals are

$$\begin{aligned} & 0 \leq U_A \leq 3B_{a,s}^{\text{end}}, \\ & 0 \leq U_B \leq (3 - r_\partial)B_{a,s}^{\text{end}}, \quad 0 \leq U_C \leq r_\partial B_{a,s}^{\text{end}}, \\ & 0 \leq V_{\text{conn},0} - V_{\text{conn},1} \leq 3B_{a,s}^{\text{end}}, \\ & 0 \leq V_{h,0} - V_{h,1} \leq (3 - r_\partial)B_{a,s}^{\text{end}}, \\ & -(3 - r_\partial)B_{a,s}^{\text{end}} \leq V_{s,0} - V_{s,1} \leq 3B_{a,s}^{\text{end}}, \\ & 0 \leq V_{t,0} - V_{t,1} = V_{\text{rem},1} - V_{\text{rem},0} \leq r_\partial B_{a,s}^{\text{end}}, \\ & |\mathcal{L}_0 - \mathcal{L}_1| = |Z_1 - Z_0| = |\mathcal{P}_0^{\text{rest}} - \mathcal{P}_1^{\text{rest}}| \leq 3B_{a,s}^{\text{end}}. \end{aligned} \quad (\text{PEI16})$$

For the last line, $U_B + U_C$ lies in $[0, 3B_{a,s}^{\text{end}}]$, and U_A lies in that same interval, so their difference has absolute value at most $3B_{a,s}^{\text{end}}$. This uses the exact sequence rather than three unrelated ranks. At $r_\partial = 0$, $U_C = 0$ exactly; at $r_\partial = 3$, $U_B = 0$ exactly.

Since $B_{a,s}^{\text{end}} \leq 2\mathfrak{E}_{a,s}^{\text{best}} = O(q)$, the incoming $6\mathfrak{E}_{a,s}^{\text{best}}$ bound and all its above- q conclusions follow. The complete identity $4S_K = \mathcal{L}_0 + Z_0 = \mathcal{L}_1 + Z_1$ is unchanged. The prefix term remains $-V_A^{\text{prefix}}$. No independent proper-source return has been replaced by a target-order return.

For an absolute rank- t restriction or attained quotient, PEI13 contains the factor t . At $t = O(a)$ its allowed logarithmic variation remains $O(aq)$. Accordingly PEI16 supplies no value for an absolute common loss, S_K , S_R , an individual complementary return, the independent proper-source standard-window return, or an individual retained-class projected sign. Those quantities remain the original quantities with the original coefficient and metric maps. The current result advances the precision of their primitive-choice comparison and its correlated receiving identities.

20.4 Explicit asymptotic coefficient of the proved envelope

The finite formula also gives a useful size record without assigning an allocation coefficient. Put $G = \log^+(1/|g(0)|)$ and $z_* = 2 \log(25 \log 3 / (4\pi))$. For each of the two order families, H_s, κ_s are the fixed constants of (PEQ11). From (PEQ7), PEI10, and $d_+ = 2q - v + 1$, $d_- = q - v$, one obtains

$$\limsup_{a \rightarrow \infty} \frac{\mathfrak{E}_{a,s}}{q} \leq z_* + 4G + 4 \log H_s + 8 \log \kappa_s. \quad (\text{PEI17})$$

Indeed $d_+/q \rightarrow 2$, $(d_+ - 1)/q \rightarrow 2$ and $(d_- - 1)/q \rightarrow 1$; the harmonic term divided by q is bounded by $O(av \log q/q) \rightarrow 0$. The remaining rank-one constants divided by q vanish. These limits give respectively $4G$, $4 \log H_s$, and $8 \log \kappa_s$ in the displayed bound. At $s = a$, $H_s = \max\{1, H\}$ and $\kappa_s = 2$; at $s = 1$, their literal values in (PEQ11) remain. The right side is a bound for the proved metric envelope and is not an evaluated coefficient of any absolute determinant.

Source and verification record

The finite source amplification is read from the supplied complete AKS3,15–24 and transported to literal IGI6–10 coordinates. The conductor formulas are read from LRC6–10 with its exact WCF transpose map, and NCT/NPK supplies its original affine-fibre meaning. The three-dimensional coefficient maps, original rank strata and all receiving identities are read directly from PRT1–20. The kernel-current input is the supplied EC37–41. These are written programme proofs, not external claims of numerical period certification. The integration checks the complete AKS envelope, the LRC/XIM affine-fibre map and the PRT coefficient sequence against their pinned provider sources. No Lean execution, actual-period computation or public edition is claimed by this local source. The native affine-fibre and nested-flag calculations are proved explicitly in PEI2–16. No numerical diagnostic is used as proof of an unresolved directional coefficient.

Original endpoint retractions and reflected-pair period calculation

21 Objects, source conventions, and the receiving maps

Keep a complete stipulated quartet

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad \mathcal{R} = \{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2,$$

each of order m . Set

$$h(s) = \prod_{\alpha \in \mathcal{R}} (s - \alpha)^m, \quad v_h = 2\xi/h, \quad U_h = j_h v_h, \quad E_h = \mathbb{C}[s]/(h).$$

For $k \equiv 1 \pmod{4}$, $k \geq 5$, retain

$$e = 1 + k(m-1), \quad c = k/2, \quad q = e(k+1)^2, \quad \chi(S) = \prod_{a,b=0}^k (S - c - (2a-k)\delta - i(2b-k)\gamma)^e.$$

The spaces and operators are $E_k = \mathbb{C}[S]/(\chi)$, $A = M_S$, $A_\Sigma = \sum_i M_{s_i}$, and

$$\eta_k : E_k \longrightarrow E_h^{\otimes k}, \quad [P] \longmapsto U_h^{\otimes k} P(A_\Sigma) 1. \quad (\text{ET1})$$

These are the original maps OPG1–5, not altered unit values [16, OPG1–5]. The exact source is

$$w_h(y) = |v_h(1/2 + iy)|^2/(2\pi), \quad m_k = w_h^{*k}, \quad \sigma(y) = |\Gamma(1/4 + iy/2)|^2/(2\pi). \quad (\text{ET2})$$

All norms below keep their stated measures. In particular, a coordinate congruence used to calculate eigenvalues does not change either metric.

22 The complete finite endpoint retraction

Write $\zeta(\alpha + z) = z^m z_\alpha(z)$, $z_\alpha(0) \neq 0$. The finite singular germ attached to a local class u is exactly

$$\mathfrak{b}_\alpha u = x^{-\alpha} \text{Res}_{z=0} \frac{e^{zL} u(z)}{2z^m z_\alpha(z)} dz = x^{-\alpha} \sum_{j=0}^{m-1} [z^{m-1-j}] \frac{u(z)}{2z_\alpha(z)} \frac{L^j}{j!}, \quad L = \log(1/x). \quad (\text{ET3})$$

The equality follows by multiplying the convergent Taylor germs and selecting the coefficient of z^{-1} . Thus no unknown lower local coefficient has been suppressed. This verifies the finite-germ formula E1 of the incoming note [15, E1]. Identifying it with a particular infinite-dimensional analytic connecting map additionally uses that map's specified source extension; the construction here works directly with the displayed residue and does not change its domain.

The linear reversal $J_m(z^r) = L^{m-1-r}/(m-1-r)!$ satisfies $J_m M_z = \partial_L J_m$, including the top nilpotent power. Multiplication by the entire unit commutes with M_z . Hence ET3 intertwines local multiplication by $\alpha + z$ with $\alpha + \partial_L$. For an ordered tuple α , put $\beta = \sum_i \alpha_i$, $n = k(m-1)$. Diagonal multiplication is

$$\mathfrak{d}_\alpha \left(\bigotimes_i x_i^{-\alpha_i} L_i^{j_i} / j_i! \right) = x^{-\beta} \frac{(\sum_i j_i)!}{\prod_i j_i!} \frac{L^{\sum_i j_i}}{(\sum_i j_i)!}. \quad (\text{ET4})$$

The product rule proves exact intertwining with $\sum_i (\alpha_i + \partial_{L_i})$. In particular the factor $n!/((m-1)!)^k$ on the longest polynomial does not disappear.

Let $A_*(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)/2$ and $h_\alpha(s) = h(s)/(s-\alpha)^m$. The identity $\xi = A_*\zeta$ gives, in the original local algebra,

$$\frac{U_h(\alpha + z)}{2z_\alpha(z)} = \frac{A_*(\alpha + z)}{h_\alpha(\alpha + z)} \pmod{z^m}. \quad (\text{ET5})$$

All these units are invertible: the stipulated roots are nontrivial, the Gamma function has no zeros and no pole at these points [19, §5.2(i)], and the full orders have already been divided out. Define the actual polynomial

$$C_{\alpha}(L) = \prod_{i=1}^k \left(\sum_{j=0}^{m-1} [z^{m-1-j}] \frac{A_*(\alpha_i + z)}{h_{\alpha_i}(\alpha_i + z)} \frac{L^j}{j!} \right) = \sum_{j=0}^n c_j L^j / j!. \quad (\text{ET6})$$

Its top coefficient is

$$c_n = \frac{n!}{((m-1)!)^k} \prod_i \frac{U_h(\alpha_i)}{2z_{\alpha_i}(0)} \neq 0. \quad (\text{ET7})$$

With $D_{\alpha} = \mathfrak{d}_{\alpha} \otimes_i \mathfrak{b}_{\alpha_i}$,

$$D_{\alpha} \eta_k P = x^{-\beta} P(\beta + \partial_L) C_{\alpha}(L). \quad (\text{ET8})$$

For an arbitrary output $x^{-\beta} \sum b_j L^j / j!$, set $B(T) = \sum_{r=0}^n b_{n-r} T^r$ and $H(T) = \sum_{r=0}^n c_{n-r} T^r$. Reversal intertwines ∂_L with multiplication by T , so ET8 is precisely

$$B(T) = P(\beta + T) H(T) \pmod{T^{n+1}}.$$

The complete inverse is finite:

$$P(\beta + T) = H(T)^{-1} B(T), \quad H(T)^{-1} = c_n^{-1} \sum_{j=0}^n \left(-\frac{H(T) - c_n}{c_n} \right)^j \pmod{T^{n+1}}. \quad (\text{ET9})$$

Choose and record one ordered tuple at every original centre $\beta_{ab} = c + (2a - k)\delta + i(2b - k)\gamma$. Every such tuple exists because the two signs in a quartet root are independent. Restrict to that tuple, apply ET9, and use the ordinary Chinese remainder inverse for the pairwise coprime polynomials $(S - \beta_{ab})^e$. This defines the exact map

$$R_{\partial} : E_h^{\otimes k} \longrightarrow E_k, \quad R_{\partial} \eta_k = I, \quad R_{\partial} A_{\Sigma} = A R_{\partial}. \quad (\text{ET10})$$

These two identities follow component by component from ET9 and from the derivative intertwining. They also prove injectivity of ET1 without a dimension argument. All unselected tuples remain in the domain and belong to the specified kernel. For any earlier equivariant retraction R of this same inclusion, the exact comparison is

$$R_{\partial} - R = (D\eta_k)^{-1} D(I - \eta_k R), \quad (\text{ET11})$$

where D is the direct sum of the selected maps, with the Chinese remainder target understood. Indeed its right side is $R_{\partial} - R_{\partial} \eta_k R = R_{\partial} - R$. This is an explicit map through the old complementary module; neither complement is asserted orthogonal.

The complete Jordan list of that kernel can be recovered directly. Put $d = m - 1$ and give the auxiliary polynomial basis L^j , $0 \leq j \leq d$, squared lengths $1/\binom{d}{j}$. This auxiliary inner product is used only to prove a finite representation identity; it is not substituted for ET2. The operators

$$N = \partial_L, \quad H = 2L\partial_L - d, \quad R = L(d - L\partial_L)$$

satisfy $N^* = R$, $H^* = H$, $[H, N] = -2N$, $[H, R] = 2R$, $[R, N] = H$. Their sums act on the ordered tensor. If $Rv = 0$, $Hv = \ell v$, then

$$RN^j v = j(\ell - j + 1)N^{j-1}v, \quad \|N^j v\|^2 = j(\ell - j + 1)\|N^{j-1}v\|^2.$$

These identities follow by induction from the commutators and adjointness. Taking a highest nonzero weight in any invariant orthogonal complement therefore constructs a string of length $\ell + 1$; its orthogonal complement is again invariant because both adjoints belong to the operator family. Induction on the finite dimension decomposes the complete tensor into these strings. If

$p_r = [t^r](1 + \dots + t^d)^k$, the weight multiplicities and their symmetry give exactly $p_r - p_{r-1}$ strings of length $kd - 2r + 1$, for $0 \leq r \leq \lfloor kd/2 \rfloor$, with $p_{-1} = 0$. In particular the longest string is unique. Diagonal multiplication ET4 intertwines all three displayed operators with those of degree kd , by the product rule. It is onto by its nonzero top value, and kills each shorter string: a highest weight vector of smaller weight maps to a vector killed by R in that same smaller weight, where the degree- kd target has no such vector.

At β_{ab} , there are exactly $M_{ab} = \binom{k}{a}\binom{k}{b}$ ordered tuples, by choosing the two sign sets independently. ET10 removes one longest string from their direct sum. Its kernel consequently has $M_{ab} - 1$ strings of length e and

$$M_{ab}(p_r - p_{r-1}) \quad \text{strings of length } e - 2r \quad (1 \leq r \leq \lfloor (e-1)/2 \rfloor). \quad (\text{ET11a})$$

Multiplication by the full local units commutes with the original nilpotent action, so it changes the explicit splitting but preserves this complete list.

23 Actual period coordinates and single-class survival

This section uses the original simple lane $m = 1$. Put

$$x = \delta + i\gamma, \quad h_c(w) = w^4 + \beta w^2 + \eta, \quad \beta = 2(\gamma^2 - \delta^2), \quad \eta = (\delta^2 + \gamma^2)^2, \\ e_x(w) = \frac{h_c(w)}{(w-x)h'_c(x)} = \frac{w^3 + xw^2 + (x^2 + \beta)w + x(x^2 + \beta)}{h'_c(x)}. \quad (\text{ET12})$$

Use its column $\mathbf{v}_x$ in the unchanged frame $1, w, w^2, w^3$. The recovered provider gives

$$P(u) = e^{a/u} G_u \mathcal{R}(u^{-1}), \quad G_u = \text{diag}(u^{a/5} g_a)_{a=1}^4, \quad g_a = 5^{a/5-1} e^{\pi i a/5} \Gamma(a/5), \quad (\text{ET13})$$

$$\mathfrak{a} = 1/160 + \beta/24 + \eta/2, \quad \det \mathcal{R}(z) = 1, \quad \mathcal{R}(0) = \begin{pmatrix} 1 & 0 & -\beta/3 & 0 \\ 0 & 1 & 0 & -2\beta/3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} =: J_\infty. \quad (\text{ET14})$$

The exact coefficients and their convergent bound are PCL2–6 [16, PCL1–6, RPC8]:

$$R_{ar,n} = \sum_{\substack{p,h \geq 0 \\ 2p+4h=5n+r+1-a}} \frac{\beta^p \eta^h}{3^p p! h!} (-1)^{p+h-n} 5^{p+h-n} (a/5)_{p+h-n}, \quad C_1 = \sum_{n \geq 1} \|R_n\|_1 < \infty, \quad (\text{ET15})$$

$$\|\mathcal{R}(z) - J_\infty\|_1 \leq C_1 |z| \quad (|z| \leq 1).$$

The original logarithm branch remains in G_u . Let $\zeta_5 = e^{2\pi i/5}$, $D = \text{diag}(\zeta_5^a)$ and

$$\mathcal{P}_k = \frac{1}{5} \sum_{j=0}^4 (D^{-j})^{\otimes k}. \quad (\text{ET16})$$

OPG5 and its Fourier conjugacy show that this is the actual observation projector in the stated Fourier coordinates, not a new observation [16, OPG5, PQO5]. It is an orthogonal projector for the Euclidean metric in those coordinates. Writing $b = P(u)\mathbf{v}_x$, the original extremal class has output

$$U_h(\rho)^k \mathcal{P}_k b^{\otimes k}. \quad (\text{ET17})$$

Since $P(u)$ is invertible, $b \neq 0$. For $5 \mid k$, choose any nonzero entry b_a . The all- a ordered tensor entry is nonzero and survives ET16. Thus ET17 is nonzero on the original cofinal sequence $k = 20n + 5$. If two entries b_a, b_b are nonzero, choose $j \in \{0, \dots, 4\}$ with $ka + j(b-a) \equiv 0 \pmod{5}$. For $k \geq 4$, the entry with j labels b and $k-j$ labels a exists and proves survival.

With $p_a = |b_a|^2/\|b\|^2$, exact character orthogonality gives

$$f_k(b) := \frac{\|\mathcal{P}_k b^{\otimes k}\|^2}{\|b\|^{2k}} = \frac{1}{5} \sum_{j=0}^4 \left(\sum_a p_a \zeta_5^{ja} \right)^k. \quad (\text{ET18})$$

If at least two entries are nonzero, $r_b = \max_{1 \leq j \leq 4} |\sum_a p_a \zeta_5^{ja}| < 1$, since distinct fifth roots cannot all have the same argument. Consequently $|f_k(b) - 1/5| \leq 4r_b^k/5$. If exactly one entry is nonzero, $f_k(b)$ is one for $5 \mid k$ and zero otherwise. These statements verify E15–18 in their actual period metric, with the common unit cancelling only in the ratio.

24 A new reflected-pair observation calculation

Retain $b_+ = P(u)\mathbf{v}_x$, $b_- = P(u)\mathbf{v}_{-x}$ and

$$a = \|b_+\|^2, \quad b_0 = \|b_-\|^2, \quad z_j = b_+^* D^j b_-, \quad a_j = b_+^* D^j b_+, \quad b_j = b_-^* D^j b_-.$$

The exact output Gram on the original reflected pair is

$$\mathcal{O}_k = |U_h(\rho)|^{2k} \frac{1}{5} \begin{pmatrix} \sum_{j=0}^4 a_j^k & \sum_{j=0}^4 z_j^k \\ \sum_{j=0}^4 \bar{z}_j^k & \sum_{j=0}^4 b_j^k \end{pmatrix}. \quad (\text{ET19})$$

The lower left uses the summed conjugates; replacing j by $-j$ proves Hermitian symmetry directly. To prove ET19, insert ET16 between each pair of tensor vectors and use $\langle v^{\otimes k}, w^{\otimes k} \rangle = \langle v, w \rangle^k$. The original reflection has the same unit at ρ and $1 - \rho$ because both h and ξ satisfy the same reflection; this is why the displayed unit is common. No cross term is discarded.

Here is an explicit nonalignment guard derived from the actual PCL coefficients. Put

$$A_0 = x/h'_c(x), \quad B_0 = 1/h'_c(x), \quad c_5 = \min_j |1 + \zeta_5^j| = (\sqrt{5} - 1)/2,$$

$$V = \|\mathbf{v}_x\|_1 = \|\mathbf{v}_{-x}\|_1, \quad U_* = \max \left\{ 1, \frac{16C_1 V}{c_5 \min\{|A_0|, |B_0|\}} \right\}. \quad (\text{ET20})$$

The last two entries of $J_\infty \mathbf{v}_x$ are (A_0, B_0) , and those of $J_\infty \mathbf{v}_{-x}$ are $(A_0, -B_0)$, because $h'_c(-x) = -h'_c(x)$. For $|u| \geq U_*$, every perturbation of those four entries is at most $d = c_5 \min(|A_0|, |B_0|)/16$. For each j , the two-row minor between these two columns after multiplying the second by D^j has unperturbed modulus $|A_0 B_0| |1 + \zeta_5^j|$. Its perturbation is at most

$$2d(|A_0| + |B_0|) + 2d^2 \leq |A_0 B_0| (c_5/4 + c_5^2/128) < \frac{1}{2} c_5 |A_0 B_0|. \quad (\text{ET21})$$

Thus all five minors are nonzero. Multiplication by the original nonzero diagonal $e^{a/u} G_u$ does not change their nonvanishing. In particular b_+ and $D^j b_-$ are independent for every j . This also guarantees two nonzero entries in both columns.

At a fixed admitted period satisfying ET20 define the actual numbers

$$r_+ = \max_{1 \leq j \leq 4} |a_j|/a < 1, \quad r_- = \max_{1 \leq j \leq 4} |b_j|/b_0 < 1, \quad r_\times = \max_{0 \leq j \leq 4} |z_j|/\sqrt{ab_0} < 1. \quad (\text{ET22})$$

Strictness in the third inequality follows from ET21 and the equality case of Cauchy–Schwarz. These are finite sums of the four original coordinates; no unknown asymptotic Gram replaces them. On the finite guards $4r_+^k \leq 1$ and $4r_-^k \leq 1$, ET19 gives the lower determinant bound

$$\frac{\det \mathcal{O}_k}{|U_h(\rho)|^{4k} (ab_0)^k} \geq \frac{(1 - 4r_+^k)(1 - 4r_-^k)}{25} - r_\times^{2k}. \quad (\text{ET23})$$

For example the right side is positive whenever

$$4r_+^k \leq \frac{1}{4}, \quad 4r_-^k \leq \frac{1}{4}, \quad 25r_\times^{2k} < \frac{9}{16}. \quad (\text{ET24})$$

Taking logarithms supplies a finite threshold directly from ET22; if a radius is zero its condition is already satisfied. This proves eventual injectivity of the original observation on the entire reflected pair at every fixed admitted period in ET20. It strengthens single-class nonvanishing without asserting a characteristic-changing comparison.

There is a quantitative full-pair version of the $1/5$ calculation. The unprojected period Gram is

$$\mathcal{H}_k = |U_h(\rho)|^{2k} \begin{pmatrix} a^k & z_0^k \\ \bar{z}_0^k & b_0^k \end{pmatrix}.$$

Define

$$\epsilon_k = \frac{\frac{4}{5} \max(r_+^k, r_-^k) + \frac{4}{5} r_\times^k}{1 - r_\times^k}. \quad (\text{ET25})$$

Then

$$(1/5 - \epsilon_k) \mathcal{H}_k \preceq \mathcal{O}_k \preceq (1/5 + \epsilon_k) \mathcal{H}_k. \quad (\text{ET26})$$

Indeed congruence by $|U_h(\rho)|^{-k} \text{diag}(a^{-k/2}, b_0^{-k/2})$ leaves the unprojected Gram with diagonal one and off-diagonal modulus at most $r_\times^k$. The difference $\mathcal{O}_k - \mathcal{H}_k/5$ has diagonal entries at most $4 \max(r_+^k, r_-^k)/5$ in modulus, and off-diagonal entry at most $4r_\times^k/5$, since its $j = 0$ summand cancels exactly. Its operator norm is bounded by their sum. Dividing by the lower eigenvalue $1 - r_\times^k$ proves ET26, after reversing the congruence. Every unit and coordinate factor is therefore present in the asserted original inequality.

25 Reconstructing the first-cutoff arithmetic parity coefficient

This proof recovers S2's parity coefficient from the source envelopes, without using an unavailable separate parity calculation. Set $\lambda = k\rho = c + \mu$, $\mu = k(\delta + i\gamma)$, $R = |\mu|$, and keep the terminal class

$$v = \left[\frac{\chi(S)}{(S - \lambda)\chi_\lambda(\lambda)} \right], \quad \chi_\lambda = \chi/(S - \lambda)^e.$$

At $N = q - 1$ the quotient representative is unique. Define the complete polynomial

$$B_k(y) = \frac{\chi(c + iy)}{(iy - \mu)(iy + \mu)}, \quad n = q - 2, \quad M_j = \int_{\mathbb{R}} y^j |B_k(y)|^2 m_k(y) dy. \quad (\text{ET27})$$

Its leading coefficient has modulus one; its remaining n roots, with their full remaining multiplicities, have modulus at most $R = k\sqrt{\delta^2 + \gamma^2}$. The polynomial is even. The literal representative is $(iy + \mu)B_k(y)/\chi_\lambda(\lambda)$. Evenness of the source kills the even/odd cross term exactly. Consequently the even-energy fraction is

$$\theta_{q-1} = \frac{R^2 M_0}{M_2 + R^2 M_0}. \quad (\text{ET28})$$

This derivation is valid for every complete multiplicity; it never uses $\chi'(\lambda)$.

For completeness the source estimates used here are exactly GTM1 and RTM2 [17, GTM1, RTM2]:

$$a_k(1 + y^2)^{-M} \sigma(y) \leq m_k(y) \leq A_k \sigma(y), \quad M = 21 + 4m, \quad \log(A_k/a_k) = O_h(k + \log k), \quad (\text{ET29})$$

$$c_\Gamma e^{-\alpha|y|} (1 + |y|)^{-1/2} \leq \sigma(y) \leq C_\Gamma e^{-\alpha|y|} (1 + |y|)^{-1/2}, \quad \alpha = \pi/2,$$

$$c_\Gamma = \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2}\sqrt{5}e^{2/3}.$$

In particular the full source mass remains in a_k, A_k . Their definitions are

$$a_k = \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-42-8m}}{C_\Gamma 2^{21+4m}}, \quad A_k = \frac{C_h^{\text{tilt}}}{c_\Gamma} k^{8m-7/2} \mathfrak{M}^{k-1},$$

$$\vartheta_h = \int_{-1}^1 w_h(y) dy, \quad \mathfrak{M} = \int_{\mathbb{R}} e^{\alpha y} w_h(y) dy.$$

Here are explicit finite bounds establishing concentration, including the unbounded tails. Set $x_0 = 2n/\alpha$, and impose only the eventual guard $R < x_0$. Restriction to $[x_0, x_0 + 1]$ gives

$$M_0 \geq L_k := \frac{a_k c_\Gamma e^{-\alpha(x_0+1)} (x_0 - R)^{2n}}{(1 + (x_0 + 1)^2)^M \sqrt{x_0 + 2}}. \quad (\text{ET30})$$

For $j = 0, 2$, the upper bound on either half-line is

$$|y|^j |B_k(y)|^2 m_k(y) \leq A_k C_\Gamma |y|^j (|y| + R)^{2n} e^{-\alpha|y|}. \quad (\text{ET31})$$

Put $s_* = 2/\alpha = 4/\pi$ and fix $0 < \varepsilon < s_*$. For $t = |y| + R$, the region $||y|/q - s_*| > \varepsilon$ lies in

$$t < q(s_* - \varepsilon) + R \quad \text{or} \quad t > q(s_* + \varepsilon) + R.$$

For $\nu_j = 2n + j + 1$, let T_j denote the auxiliary integral with density $\alpha^{\nu_j} t^{\nu_j-1} e^{-\alpha t} / (\nu_j - 1)!$ on $t > 0$; this auxiliary notation does not rescale ET2. Integrating ET31 yields the exact bound

$$\frac{\int_{\{|y|/q - s_*| > \varepsilon\}} |y|^j |B_k(y)|^2 m_k(y) dy}{q^j M_0} \leq \frac{2A_k C_\Gamma e^{\alpha R} (2n + j)!}{q^j \alpha^{2n+j+1} L_k} \Pr\{T_j \notin [q(s_* - \varepsilon) + R, q(s_* + \varepsilon) + R]\}. \quad (\text{ET32})$$

For $v > 1$ and $0 < v < 1$, respectively, elementary exponential Markov estimates give

$$\Pr\{T_j \geq \nu_j v / \alpha\} \leq e^{-\nu_j(v-1-\log v)}, \quad \Pr\{T_j \leq \nu_j v / \alpha\} \leq e^{-\nu_j(v-1-\log v)}. \quad (\text{ET33})$$

To prove them integrate $e^{s\alpha t}$ against the displayed density, obtaining $(1 - s)^{-\nu_j}$, and choose $s = 1 - 1/v$. This is positive for the upper tail and negative for the lower tail. All integrals follow by repeated integration by parts.

The logarithm of the prefactor in ET32 is $O_h(k + \log q)$. Indeed $\log(2n + j)! = (2n + j) \log(2n + j) - (2n + j) + O(\log n)$, obtained by bounding the sum of $\log r$ between its two adjacent integrals; $R = O_h(k)$, $R/n \rightarrow 0$, and $-2n \log(1 - R/x_0) = O_h(k)$. The remaining factors in ET30 are polynomial in q , and ET29 gives the stated source ratio. Each tail in ET33 has exponent $-c_\varepsilon q + o(q)$, since its v tends to $1 \pm \alpha\varepsilon/2$. Thus the right side of ET32 tends to zero for $j = 0$ and $j = 2$. On the complementary region $(s_* - \varepsilon)^2 \leq y^2/q^2 \leq (s_* + \varepsilon)^2$. Letting $\varepsilon \downarrow 0$ proves

$$\frac{M_2}{q^2 M_0} \rightarrow \frac{16}{\pi^2}, \quad \theta_{q-1} = \frac{\pi^2(\delta^2 + \gamma^2)}{16} \frac{k^2}{q^2} (1 + o_h(1)). \quad (\text{ET34})$$

This proof used the unchanged complete root polynomial, all roots in the degree count, both half-lines, and the actual source envelopes.

26 Exact arithmetic pair metrics and their period comparison

Reflection $\Sigma P(S) = P(k - S)$ is an isometry of the original quotient minimum because ET2 is even and $\chi(k - S) = \chi(S)$. For P_N , its minimum representative of v , put $E_N = \|P_N\|^2$ and $\theta_N = \|(P_N + \Sigma P_N)/2\|^2/E_N$. The reflected pair is independent, and its exact Gram is

$$G_N^{\text{pair}} = E_N \begin{pmatrix} 1 & h_N \\ h_N & 1 \end{pmatrix}, \quad h_N = 2\theta_N - 1, \quad 0 < \theta_N < 1. \quad (\text{ET35})$$

The decomposition into even and odd parts proves the off-diagonal entry and positivity. On this pair $A - cI = \text{diag}(\mu, -\mu)$. The spectral coordinate projector onto the first line has norm

$$\|E_+\|_{G_N^{\text{pair}}} = (1 - h_N^2)^{-1/2}. \quad (\text{ET36})$$

For a fixed first coordinate z_1 , minimization over z_2 in the denominator gives $(1 - h_N^2)|z_1|^2$, proving the formula. For real τ , $T = \exp(\tau(A - cI))$ has squared singular values with product one and sum

$$t_N(\tau) = \frac{2 \cosh(2k\delta\tau) - 2h_N^2 \cos(2k\gamma\tau)}{1 - h_N^2}. \quad (\text{ET37})$$

This follows by multiplying $G^{-1}T^*GT$; therefore its two eigenvalues are $(t_N \pm \sqrt{t_N^2 - 4})/2$. Using ET34, for each fixed $\tau \neq 0$,

$$\|T\|_{G_{q-1}^{\text{pair}}} = \frac{2q}{\pi k \sqrt{\delta^2 + \gamma^2}} e^{|\tau|k\delta} (1 + o_{h,\tau}(1)). \quad (\text{ET38})$$

The original uncentred action has the additional factor $e^{\tau k/2}$, retained exactly.

For comparison with the proposed normal-action metric in S9, any positive metric making this two-eigenvalue action normal is diagonal in the displayed eigenbasis: $H = \text{diag}(a_*, b_*)$, $a_*, b_* > 0$. Distinct eigenvectors of a normal operator are orthogonal, as follows by applying the adjoint eigenvalue equation to their inner product; conversely a diagonal metric makes this diagonal action normal. The squared generalized trace divided by the generalized determinant is

$$\frac{(a_* + b_*)^2}{a_* b_* (1 - h_N^2)}.$$

Its minimum is at $a_* = b_*$, since $(a_* - b_*)^2 \geq 0$. The positive generalized eigenvalue ratio increases with that expression. Thus, exactly,

$$\begin{aligned} \inf_{H: A \text{ normal in } H} \text{cond}((G_N^{\text{pair}})^{-1}H) &= \frac{1 + |h_N|}{1 - |h_N|}, \\ \inf_{H: A \text{ normal in } H} \text{cond}((G_{q-1}^{\text{pair}})^{-1}H) &= \frac{16q^2}{\pi^2(\delta^2 + \gamma^2)k^2} (1 + o_h(1)). \end{aligned} \quad (\text{ET38a})$$

This evaluates the metric cost of that specified construction; the calculation of the original arithmetic metric still uses ET35.

In the simple period lane the unprojected period metric on the pair is $\mathcal{H}_k$ from ET25. Its relative generalized eigenvalue condition number κ obeys the exact equation

$$\kappa + \kappa^{-1} + 2 = \frac{(a^k + b_0^k - 2h_N \Re z_0^k)^2}{(1 - h_N^2)((ab_0)^k - |z_0|^{2k})}. \quad (\text{ET39})$$

Indeed the right side is the squared generalized trace divided by the generalized determinant. Both scalar factors E_N and $|U_h(\rho)|^{2k}$ cancel only in this condition number. At every fixed admitted period, independence of b_+, b_- gives $|z_0|/\sqrt{ab_0} < 1$. Combining ET34 with ET39 gives

$$\kappa_{q-1,k} = \frac{4q^2}{\pi^2(\delta^2 + \gamma^2)k^2} \left[(a/b_0)^k + (b_0/a)^k + 2 \right] (1 + o_{h,u}(1)). \quad (\text{ET40})$$

For example, the cross term in the trace divided by $a^k + b_0^k$ is at most $(|z_0|/\sqrt{ab_0})^k$; the determinant correction is its square, and the positive solution of ET39 divided by its right side tends to one. These estimates prove ET40 also when the period norms are unbalanced. Hence S14 is recovered from explicit source envelopes and the original period provider, rather than imported as an unsupported asymptotic.

27 The exact next receiving calculation

Let $W = P(u)^{\otimes k} \eta_k E_k$ be the original cyclic image in Fourier tensor coordinates and $L = \mathcal{P}_k|_W$. The period-metric quotient norm of the original output Lv is the minimum of $\|w\|^2$ over $w \in W$

with $Lw = Lv$. As $\mathcal{P}_k$ is orthogonal, it is at least $\|Lv\|^2$ and at most $\|v\|^2$; this proves precisely the bounds in E18. ET19–26 calculate L^*L on the reflected pair, including its nonzero cross term. They do not replace the full cyclic fibre by that two-dimensional domain.

In any fixed original basis of E_k , with its full period Gram H and the original observation matrix Λ , the attained quotient matrix in a full basis of the actual image is

$$Q_{\text{per}} = (\Lambda H^{-1} \Lambda^*)^{-1}. \quad (\text{ET41})$$

To prove this, set $S = H^{-1} \Lambda^* (\Lambda H^{-1} \Lambda^*)^{-1}$. Then $\Lambda S = I$ and $S^* H I = 0$ for every kernel column I , so Sv is the unique minimum lift and $S^* H S = Q_{\text{per}}$. The arithmetic receiver has the identical exact formula with the original full arithmetic G_N . This specifies the two exact maps whose actual inverse Grams must enter the seven allocations; ET40 is the proven pair restriction of their metric comparison. No individual arithmetic projected sign follows merely from ET18 or ET26, and none is assigned here.

The incoming E19–21 additionally refer to a finite-field exponential family and a lower-boundary exact sequence. Deligne’s weight theorem is a theorem about its specified finite-field complexes and Frobenius [18, Thm. 3.3.1]. ET3–41 do not change coefficient fields or assign Frobenius to the analytic dilation. The exact identity $\pi C \partial^{\otimes k} = \pi(C \partial^{\otimes k} - \iota b)$ uses $\pi \iota = 0$; it neither constructs the comparison C nor evaluates it. The present accepted receiving results are the explicit complex maps and metrics above. The supplied family’s stationary-phase and boundary-weight providers have not been replaced by that identity.

Central path transfer through the attained quotient

Abstract

The full two-parameter arithmetic source interpolation has a uniformly small central derivative on every specified complete annihilator polynomial space. Its explicit derivative bound is the product of the original logarithmic density contrast and the central localization fraction. Integrating this derivative proves exponential relative metric bounds and an explicit bound for the actual minimum sections. For the complete relation ideal this gives a finite central-leg error and a finite rectangle error in the original four-endpoint signed determinant. The remaining source-band interaction is expressed by its exact two-variable reproducing-kernel integral. Every statement retains the full primary orders, original physical coordinate, original polynomial remainder map, complete real integration domain and both endpoint multiplicities.

28 Original source, analytic providers and fixed notation

Fix the original complete hypothetical quartet

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad m_0 \in \mathbb{Z}_{\geq 1}.$$

The divisor, its entire Taylor unit, and its arithmetic density are

$$\begin{aligned} h(z) &= \prod_{\lambda \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (z - \lambda)^{m_0}, & v_h(z) &= 2\xi(z)/h(z), \\ w_h(y) &= |v_h(1/2 + iy)|^2/(2\pi), & m_k &= w_h^{*k}, \\ k &\equiv 1 \pmod{4}, \quad k \geq 5, & e &= 1 + k(m_0 - 1), \quad q = e(k + 1)^2, \quad c = k/2, \\ \chi(S) &= \prod_{a,b=0}^k [S - c - (2a - k)\delta - i(2b - k)\gamma]^e, & E &= \mathbb{C}[S]/(\chi). \end{aligned} \tag{CPT1}$$

The symbol ξ and the zero hypothesis have their original Riemann-xi meaning. This note computes consequences on that original counterfactual source; it supplies no numerical off-critical zero. Every limit below fixes the complete packet.

The Gamma measure, full tilted mass and central endpoint are

$$\begin{aligned} \sigma(y) &= \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}, & d\sigma &= \sigma(y) dy, & c_\sigma &= \int_{\mathbb{R}} d\sigma = \sqrt{2\pi}, & \alpha &= \pi/2, \\ M_h(\theta) &= \int_{\mathbb{R}} e^{\theta y} w_h(y) dy, & \ell_h(\theta) &= \log M_h(\theta), & M &= M_h(\alpha), & M_0 &= M_h(0), \\ \mu &= \ell'_h(\alpha) > 0, & T &= k\mu, & R &= k\sqrt{\delta^2 + \gamma^2}, & C &= [-T, T], & O &= \mathbb{R} \setminus C. \end{aligned} \tag{CPT2}$$

The notation M here always denotes the full tilted mass; set $\beta_0 = 21 + 4m_0$ for the polynomial exponent, to avoid confusing these two different quantities. In particular $B_0 = 2\beta_0 = 42 + 8m_0$ and $p = 8m_0 - 9/2 > 3$.

The following original analytic estimates are the GTM1 and RTM2 providers, including their constants [21]:

$$\begin{aligned} a_k(1 + y^2)^{-\beta_0} \sigma(y) &\leq m_k(y) \leq A_k \sigma(y), \\ a_k &= \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-B_0}}{C_\Gamma 2^{B_0/2}}, & A_k &= \frac{C_h^{\text{tilt}}}{c_\Gamma} k^{p+1} M^{k-1}, & \vartheta_h &= \int_{-1}^1 w_h(y) dy, \\ c_\Gamma e^{-\alpha|y|} (1 + |y|)^{-1/2} &\leq \sigma(y) \leq C_\Gamma e^{-\alpha|y|} (1 + |y|)^{-1/2}, \\ c_\Gamma &= \sqrt{2} e^{-7/6}, & C_\Gamma &= \sqrt{2\sqrt{5}} e^{2/3}. \end{aligned} \tag{CPT3}$$

Here $c_h, C_h^{\text{tilt}} > 0$ are the original threefold convolution and boundary-tilt constants. Their original source bounds are

$$w_h^{*3}(y) \geq c_h e^{-\alpha|y|} (1 + |y|)^{-B_0}, \quad e^{\alpha y} w_h(y) \leq C_h^{\text{tilt}} (1 + |y|)^{-p}.$$

These statements, with their source derivations, are preserved in the provider file; this continuation does not select new density constants. No positive lower estimate for the one-factor zeta numerator is used.

Define the exact logarithmic contrast and the entire parameter square:

$$\begin{aligned} H(y) &= \log \frac{m_k(y)}{M^k \sigma(y)}, & H_C &= 1_C H, & H_O &= 1_O H, \\ d\nu_{s,t}(y) &= \sigma(y) e^{sH_C(y) + tH_O(y)} dy, & (s, t) &\in [0, 1]^2. \end{aligned} \tag{CPT4}$$

Thus $\nu_{0,0} = \sigma$, $\nu_{1,1} = M^{-k} m_k(y) dy$, and $\nu_{0,1}$ retains the exact arithmetic exterior while its centre is the specified Gamma density. The factor M^k has not disappeared: its exact cancellation in the particular signed determinant is proved in Section 34.

All polynomial inner products are conjugate linear in their first argument:

$$\langle P, Q \rangle_{s,t} = \int_{\mathbb{R}} \overline{P(c + iy)} Q(c + iy) d\nu_{s,t}(y).$$

The coefficient map

$$U_{c,D} : \mathbb{C}[S]_{\leq D} \longrightarrow \mathbb{C}[y]_{\leq D}, \quad U_{c,D} P(y) = P(c + iy) \tag{CPT5}$$

is an exact complex-linear isomorphism with inverse $F(y) \mapsto F((S - c)/i)$. In increasing monomial frames its diagonal is $1, i, \dots, i^D$, so $\det U_{c,D} = i^{D(D+1)/2}$. This coordinate map, with modulus-one determinant, will be displayed whenever the real-coordinate proof is returned to the original coefficient quotient.

29 Uniform bounds on the entire interpolation square

Put

$$b_* = \max\{1, A_k/M^k\}, \quad a_* = \min\{1, a_k/M^k\}, \quad a_D = \frac{a_*}{[1 + 4(D+1)^2]^{\beta_0}}. \tag{CPT6}$$

For every $(s, t) \in [0, 1]^2$,

$$a_*(1 + y^2)^{-\beta_0} d\sigma \leq d\nu_{s,t} \leq b_* d\sigma, \quad a_D \|F\|_{\sigma}^2 \leq \|F\|_{s,t}^2 \leq b_* \|F\|_{\sigma}^2 \quad (\deg F \leq D). \tag{CPT7}$$

To prove the pointwise bounds, let $r(y) = m_k(y)/(M^k \sigma(y)) > 0$. At each y , precisely one exponent in (CPT4) acts, and its value is some $u \in [0, 1]$. For any positive r ,

$$\min\{1, r\} \leq r^u \leq \max\{1, r\}.$$

Applying (CPT3) gives the first two bounds in (CPT7). In particular this proves a pointwise bound before any polynomial norm is considered.

For the polynomial lower estimate, the monic Gamma polynomials $b_j(y)$ obey

$$y b_j = b_{j+1} + j(j - \tfrac{1}{2}) b_{j-1}, \quad \gamma_j = \|b_j\|_{\sigma}^2 = \sqrt{2\pi} \frac{(2j)!}{4^j}. \tag{CPT8}$$

These are GTM7 [21]; the recurrence is also the Meixner–Pollaczek recurrence in the convention $b_j(y) = j! P_j^{(1/4)}(y/2; \pi/2)$ [26, (18.22.8)]. Indeed substituting $\lambda = 1/4, \varphi = \pi/2, x = y/2$ into that

cited recurrence and multiplying by $j!$ gives exactly (CPT8). For the orthonormal polynomials $e_j = b_j/\sqrt{\gamma_j}$,

$$ye_j = \sqrt{(j+1)(j+\frac{1}{2})} e_{j+1} + \sqrt{j(j-\frac{1}{2})} e_{j-1}.$$

The upward and downward shifts on $\text{span}(e_0, \dots, e_D)$ each have operator norm at most $D+1$, including the output coefficient at degree $D+1$. The triangle inequality therefore proves

$$\|yF\|_\sigma \leq 2(D+1) \|F\|_\sigma.$$

For a nonzero F divide the positive measure $|F|^2 d\sigma$ by its actual mass $\|F\|_\sigma^2$, solely to apply the convex scalar inequality $\mathbb{E}(1+X)^{-\beta_0} \geq (1+\mathbb{E}X)^{-\beta_0}$ to $X = y^2$. This inequality follows from the supporting tangent line to the function $(1+x)^{-\beta_0}$, whose second derivative is positive on $x \geq 0$. Multiplying the mass back gives

$$\int |F|^2 (1+y^2)^{-\beta_0} d\sigma \geq [1+4(D+1)^2]^{-\beta_0} \|F\|_\sigma^2.$$

Combining this with the pointwise lower bound proves (CPT7). No source measure or source polynomial is replaced in that argument.

The pointwise envelopes also imply

$$|H(y)| \leq |\log(a_k/M^k)| + |\log(A_k/M^k)| + \beta_0 \log(1+y^2).$$

Thus every finite polynomial moment and every finite parameter derivative of it is dominated on the closed square by a polynomial times a power of $\log(1+y^2)$ times the exponentially decreasing Gamma density. Differentiation under the integral is valid to all finite orders, with one-sided derivatives at the square's boundary. Every Gram is positive definite, since its density is positive everywhere and a nonzero polynomial has only finitely many real zeros. Its inverse and every fixed-fibre minimum section therefore depend smoothly on the square.

30 Finite central localization with every root retained

Let $g \mid \chi$ be an actual monic divisor, with its selected full primary orders, and write

$$J = \deg g, \quad d_0 = q - J, \quad Q_g(y) = i^{-d_0} (\chi/g)(c+iy), \quad L = D - d_0, \quad V_{g,D} = Q_g \mathbb{C}[y]_{\leq L}. \quad (\text{CPT9})$$

Assume the explicit finite guards

$$0 \leq J \leq q, \quad d_0 \leq D \leq 2q+1, \quad R < q, \quad T \leq q/2.$$

Each root of Q_g , counted with its remaining multiplicity, has modulus at most R . Define the finite positive bound and its cap

$$\begin{aligned} \mathfrak{e}_{k,J,D} &= \frac{b_*}{a_D} \frac{c_\sigma}{c_\Gamma} \frac{(L+1)^2 \sqrt{1+2q}}{q} e^{\pi q} 100^L \left(\frac{T+R}{q-R} \right)^{2(q-J)}, \\ \varepsilon_{k,J,D} &= \min\{1, \mathfrak{e}_{k,J,D}\}. \end{aligned} \quad (\text{CPT10})$$

Then the entire parameter square satisfies

$$\int_C |F(y)|^2 d\nu_{s,t}(y) \leq \varepsilon_{k,J,D} \|F\|_{s,t}^2, \quad F \in V_{g,D}. \quad (\text{CPT11})$$

Here is the full polynomial estimate. On $[q, 2q]$ the orthonormal Legendre basis of polynomials of degree at most L is

$$\varphi_j(y) = q^{-1/2} \sqrt{2j+1} P_j(2y/q-3), \quad 0 \leq j \leq L.$$

The Legendre norm $2/(2j+1)$ and recurrence used here are respectively [24, Table 18.3.1] and [25, Table 18.9.1]. The latter gives

$$(j+1)P_{j+1}(x) = (2j+1)xP_j(x) - jP_{j-1}(x).$$

For $|x| \leq 4$, $P_0 = 1$, $P_1 = x$ and induction prove $|P_j(x)| \leq 10^j$: the inductive upper bound is

$$\frac{2j+1}{j+1} 4 \cdot 10^j + \frac{j}{j+1} 10^{j-1} \leq 8 \cdot 10^j + 10^{j-1} < 10^{j+1}.$$

If $|y| \leq T \leq q/2$, the argument $2y/q - 3$ belongs to $[-4, -2]$. Expand the full polynomial $P = \sum_{j=0}^L d_j \varphi_j$, without omitting any coefficient. Orthogonality and Cauchy–Schwarz give

$$|P(y)|^2 \leq \left(\sum_{j=0}^L |d_j|^2 \right) \left(\frac{1}{q} \sum_{j=0}^L (2j+1) 100^j \right) \leq \frac{(L+1)^2 100^L}{q} \int_q^{2q} |P(x)|^2 dx. \quad (\text{CPT12})$$

For $F = Q_g P$, all its roots give

$$|Q_g(y)| \leq (T+R)^{d_0} \quad (y \in C), \quad |Q_g(x)| \geq (q-R)^{d_0} \quad (q \leq x \leq 2q).$$

The upper bound in (CPT7), (CPT12) and the unchanged total Gamma mass yield

$$\int_C |F|^2 d\nu_{s,t} \leq b_* c_\sigma (T+R)^{2d_0} \frac{(L+1)^2 100^L}{q} \int_q^{2q} |P(x)|^2 dx.$$

The polynomial lower bound in (CPT7) is applied on the *whole* real line before its Gamma integral is restricted to $[q, 2q]$. The exact Gamma envelope in (CPT3) gives

$$\|F\|_{s,t}^2 \geq a_D \frac{c_\Gamma e^{-\pi q}}{\sqrt{1+2q}} (q-R)^{2d_0} \int_q^{2q} |P(x)|^2 dx.$$

For $P \neq 0$, division gives $\mathfrak{e}_{k,J,D}$. The trivial central-fraction bound is one. The zero polynomial satisfies (CPT11) directly. This proves (CPT11) with both full real integration regions present in the norm.

For fixed complete packet and $D \leq 2q+1$, $\log(b_*/a_D) = O_h(k + \log q)$. Since $T+R$ is k times a fixed positive constant and $R/q \rightarrow 0$, direct logarithms in (CPT10) give, for its uncapped positive bound,

$$\log \mathfrak{e}_{k,J,2q+1} = -2(q-J) \log(q/k) + O_h(q+J+k+\log q). \quad (\text{CPT13})$$

For every fixed $0 < \vartheta < 1$ and $A > 0$, this implies

$$q^A \mathfrak{e}_{k,J,2q+1} \rightarrow 0 \quad \text{uniformly for } 0 \leq J \leq (1-\vartheta)q.$$

The finite formula (CPT10), rather than this asymptotic statement, is used in all bounds below. At $J = q$ the root factor is absent; the capped bound remains valid but supplies no such decay.

31 An explicit logarithmic central contrast

The original central-source calculation ACW11–13 [22] proves, on its explicit local-density guard $\epsilon_k^{\text{loc}} \leq 1/2$, the following formula. Let $\theta(u)$ invert ℓ'_h on $[-\alpha, \alpha]$, and put

$$V(\theta) = \ell''_h(\theta), \quad 0 < v_* \leq V(\theta) \leq V_*, \quad c_\sigma^{\text{env}}(y) = \sigma(y) e^{\alpha|y|} \sqrt{1+|y|}.$$

Then, for $|u| \leq \mu$,

$$\begin{aligned} H(ku) &= kF_h(u) + \frac{1}{2} \log(|u| + k^{-1}) - \frac{1}{2} \log(2\pi V(\theta(u))) - \log c_\sigma^{\text{env}}(ku) + \log(1 + \epsilon_{k,\theta(u)}), \\ F_h(u) &= \ell_h(\theta(u)) - u\theta(u) - \ell_h(\alpha) + \alpha|u|, \quad |\epsilon_{k,\theta}| \leq \epsilon_k^{\text{loc}}. \end{aligned} \quad (\text{CPT14})$$

For completeness, all constants and the local-density proof yielding this provider are spelled out in the appendix. In particular this formula retains the actual error, not a selected saddle coefficient.

The function F_h is even, and on $0 < u < \mu$,

$$F'_h(u) = \alpha - \theta(u) > 0, \quad F_h(\mu) = 0, \quad F_h(0) = \log M_0 - \log M.$$

Consequently $-(\log M - \log M_0) \leq F_h(u) \leq 0$. The exact argument $|u| + 1/k$ lies between $1/k$ and $\mu + 1$. Set

$$\begin{aligned} L_k &= k(\log M - \log M_0) + \frac{1}{2} \max\{\log k, |\log(\mu + 1)|\} \\ &\quad + \frac{1}{2} \max\{|\log(2\pi v_*)|, |\log(2\pi V_*)|\} \\ &\quad + \max\{|\log c_\Gamma|, |\log C_\Gamma|\} + 2\epsilon_k^{\text{loc}}. \end{aligned} \tag{CPT15}$$

Since $|\log(1 + x)| \leq 2|x|$ for $|x| \leq 1/2$, (CPT14) proves the finite estimate

$$|H_C(y)| \leq L_k, \quad L_k = O_h(k + \log k). \tag{CPT16}$$

All terms in (CPT15) are nonnegative. In particular $\log M - \log M_0 \geq 0$ by the original evenness of w_h .

There is also a uniform upper bound on the positive part:

$$H(y) \leq K_h \quad (y \in C), \quad K_h = \left[\frac{1}{2} \log(\mu + 1) - \frac{1}{2} \log(2\pi v_*) - \log c_\Gamma + \log 2 \right]_+. \tag{CPT17}$$

Indeed the kF_h term is nonpositive, and each other term of (CPT14) has the corresponding upper bound in (CPT17).

32 Central derivative, transported metrics and attained sections

Fix g, D as in Section 30, and abbreviate

$$\Lambda_{k,J,D} = L_k \varepsilon_{k,J,D}. \tag{CPT18}$$

On $V_{g,D}$ the actual derivative form is

$$\dot{h}_{s,t}^C(F, G) = \partial_s \langle F, G \rangle_{s,t} = \int_C H(y) \overline{F(y)} G(y) d\nu_{s,t}(y).$$

The bound (CPT16), Cauchy–Schwarz on C , and (CPT11) prove

$$|\dot{h}_{s,t}^C(F, G)| \leq \Lambda_{k,J,D} \|F\|_{s,t} \|G\|_{s,t}. \tag{CPT19}$$

In any fixed coefficient frame, write $A_{s,t}$ for the positive Gram and $A_{s,t}^C = \partial_s A_{s,t}$. Thus (CPT19) is the proved operator inequality

$$\left\| A_{s,t}^{-1/2} A_{s,t}^C A_{s,t}^{-1/2} \right\|_{\text{op}} \leq \Lambda_{k,J,D}.$$

For a fixed nonzero F , its squared norm satisfies $|\partial_s \log \|F\|_{s,t}^2| \leq \Lambda_{k,J,D}$. Integrating from s_0 to s_1 gives

$$e^{-\Lambda_{k,J,D}|s_1-s_0|} A_{s_0,t} \preceq A_{s_1,t} \preceq e^{\Lambda_{k,J,D}|s_1-s_0|} A_{s_0,t}. \tag{CPT20}$$

This bound holds for every $t \in [0, 1]$. It requires no smallness assumption on $(e^{L_k} - 1)\varepsilon_{k,J,D}$.

Here is the exact transfer through every fixed restriction and attained quotient of this polynomial space. A fixed injective matrix I gives the restriction $I^* A_{s,t} I$; congruence preserves (CPT20). A fixed onto matrix $J : V_{g,D} \rightarrow Y$, with a fixed value frame in Y , has minimum metric and section

$$G_{s,t}^Y = (J A_{s,t}^{-1} J^*)^{-1}, \quad R_{s,t}^Y = A_{s,t}^{-1} J^* G_{s,t}^Y. \tag{CPT21}$$

Direct multiplication gives $JR_{s,t}^Y = I_Y$. For $z \in \ker J$,

$$z^* A_{s,t} R_{s,t}^Y v = z^* J^* G_{s,t}^Y v = 0.$$

Every representative of v is $R_{s,t}^Y v + z$; orthogonality proves that its squared norm is $v^* G_{s,t}^Y v + z^* A_{s,t} z$. This proves both formulas in (CPT21) and attainment of the unique minimum. Taking the infimum over the *same* fibre in each side of (CPT20) gives the same bounds for G^Y . Repeating these two operations proves the bounds for any stated fixed sequence of restrictions and attained quotients inside this ambient polynomial space. For a rank- r resulting metric G ,

$$\left| \log \frac{\det G_{s_1,t}}{\det G_{s_0,t}} \right| \leq r \Lambda_{k,J,D} |s_1 - s_0|. \quad (\text{CPT22})$$

The rank is retained, including rank zero with determinant one.

There is a direct estimate for the actual minimizing polynomials. Let $P_s = R_{s,t}^Y v$, for $v \in Y$ fixed and t fixed. Because $JP_s = v$, differentiation gives $P'_s \in \ker J$. For every *fixed* $z \in \ker J$, differentiate its exact orthogonality to obtain

$$\langle z, P'_s \rangle_{s,t} = -\dot{h}_{s,t}^C(z, P_s).$$

This identity holds for every vector of the fixed kernel at each parameter, so it may then be evaluated at $z = P'_s$. Using (CPT19) and dividing when $P'_s \neq 0$ proves

$$\|P'_s\|_{s,t} \leq \Lambda_{k,J,D} \|P_s\|_{s,t}. \quad (\text{CPT23})$$

The minimum property and (CPT20) give $\|P_s\|_{s,t} \leq e^{s\Lambda_{k,J,D}/2} \|P_0\|_{0,t}$. The same metric comparison gives $\|P'_s\|_{0,t} \leq e^{s\Lambda_{k,J,D}/2} \|P'_s\|_{s,t}$. Integrating the resulting bound $\|P'_s\|_{0,t} \leq \Lambda_{k,J,D} e^{s\Lambda_{k,J,D}} \|P_0\|_{0,t}$ in the fixed coefficient space proves

$$\boxed{\|P_1 - P_0\|_{0,t} \leq (e^{\Lambda_{k,J,D}} - 1) \|P_0\|_{0,t}.} \quad (\text{CPT24})$$

At $\Lambda = 0$ this formula follows by the zero derivative. For arbitrary endpoints the same proof, with the oriented segment reparametrized, gives $e^{\Lambda|s_1-s_0|} - 1$ and the starting norm. Thus the original coefficients and phases of the entire minimum section, rather than only its determinant, are controlled.

33 The exact map to the original primary submodule

Let $A : E \rightarrow E$ denote multiplication by the original S . The polynomial-domain division identity gives

$$E[g] := \ker g(A) = (\chi/g)\mathbb{C}[S]/(\chi), \quad \dim_{\mathbb{C}} E[g] = J. \quad (\text{CPT25})$$

Indeed $g[P]_{\chi} = 0$ says that $\chi \mid gP$. Writing $\chi = g(\chi/g)$ and cancelling the nonzero polynomial g in $\mathbb{C}[S]$ is equivalent to $(\chi/g) \mid P$. Multiplication by χ/g identifies $\mathbb{C}[S]/(g)$ bijectively with this submodule, proving the dimension assertion and retaining every primary multiplicity.

For $N \geq q - 1$, put

$$W_{g,N} = (\chi/g)\mathbb{C}[S]_{\leq N-q+J}, \quad J_{g,N} : W_{g,N} \longrightarrow E[g], \quad J_{g,N}P = [P]_{\chi}.$$

For $J = 0$ the value space is zero and the formula has that literal meaning. The onto property follows from the degree- $< J$ representatives in $\mathbb{C}[S]/(g)$. Its kernel is exactly $\chi\mathbb{C}[S]_{\leq N-q}$, with negative degree interpreted as the zero space. Moreover every degree- $\leq N$ representative in the original remainder fibre of a class in $E[g]$ already lies in $W_{g,N}$, by the divisibility just proved. Therefore minimizing in $W_{g,N}$ is precisely the original minimum for that class.

In real coordinates $U_{c,N}W_{g,N} = V_{g,N}$, because the constant factor i^{q-J} changes no subspace. This gives the commuting diagram in explicit map form:

$$\begin{array}{ccc} W_{g,N} & \xrightarrow{U_{c,N}} & V_{g,N}, \\ J_{g,N} = (P \mapsto [P]_\chi) \big|_{W_{g,N}}, & J_{g,N} \circ U_{c,N}^{-1} : V_{g,N} \rightarrow & E[g]. \end{array} \quad (\text{CPT26})$$

If $G_N[\nu_{s,t}]$ is the original full quotient Gram on E and I_g is a fixed frame inclusion of $E[g]$ into E , the minimum metric in (CPT21) is exactly

$$G_{N;s,t}^g = I_g^* G_N[\nu_{s,t}] I_g. \quad (\text{CPT27})$$

Equations (CPT20)–(CPT24) consequently apply to the original restriction, its minimum sections and its fixed subquotients, with $D = N$, or with $D = 2q + 1$ as one common ambient degree. In particular, at all four original cutoff degrees the same explicit large-degree constant is valid.

An opposite terminal eigenpair uses the actual divisor $g = (S - \lambda)(S - (k - \lambda))$ and $J = 2$, including when the full primary length e exceeds one. The remaining $e - 1$ powers at these roots stay in χ/g . All terminal eigenlines together use $g = \text{rad } \chi$ and $J = (k + 1)^2 = q/e$. For $m_0 > 1$ this whole terminal space has the decay (CPT13). For $m_0 = 1$, this last divisor is χ , hence $J = q$; it is the full quotient.

The exact determinant relation records the complementary metric that is still present when $J < q$. Complete I_g by any fixed frame $I_\perp$ to a frame of E , and in that frame write

$$G_N = \begin{pmatrix} A_g & B_g \\ B_g^* & C_g \end{pmatrix}, \quad A_g = I_g^* G_N I_g.$$

The identity induced on the quotient $E/E[g]$ has attained metric

$$Q_g^{\text{comp}} = C_g - B_g^* A_g^{-1} B_g, \quad \det G_N = \det A_g \det Q_g^{\text{comp}}. \quad (\text{CPT28})$$

To verify it, a value z in this quotient is represented by (x, z) . Completing the square in its norm gives the unique minimum at $x = -A_g^{-1} B_g z$ and the displayed Q_g^{comp} . The block triangular elimination has determinant one, proving the determinant formula. Thus the map $E[g] \hookrightarrow E \rightarrow E/E[g]$ and its precise complementary metric remain in the calculation. A bound on A_g alone has not evaluated this second factor.

34 Original Schur identity and its signed central receiver

For any source $\nu_{s,t}$ define the original degree- N moment matrix in the S -monomial frame by

$$(H_N)_{ij} = \int_{\mathbb{R}} \overline{(c + iy)^i} (c + iy)^j d\nu_{s,t}(y), \quad 0 \leq i, j \leq N.$$

Let $\omega_n(s, t)$ be its squared monic source minimum of degree n , and let $\eta_r(s, t)$ be the squared monic relation minimum

$$\eta_r(s, t) = \min_{\substack{u \text{ monic} \\ \deg u = r}} \int_{\mathbb{R}} |\chi(c + iy) u(c + iy)|^2 d\nu_{s,t}(y).$$

Every preceding source polynomial, respectively preceding relation $\chi \mathbb{C}[S]_{<r}$, is included in these minima. Gram–Schmidt in monic triangular frames gives

$$\det H_N = \prod_{n=0}^N \omega_n, \quad \det L_r = \prod_{j=0}^{r-1} \eta_j, \quad L_r = \text{Gram}(\chi, \chi S, \dots, \chi S^{r-1}), \quad \det L_0 = 1.$$

These are identities of the actual source moments.

The concatenated coefficient frame

$$1, S, \dots, S^{q-1}, \chi, \chi S, \dots, \chi S^{N-q}$$

is monic triangular, with determinant one. Its lower Gram block is L_{N-q+1} . Completing the square in its relation coefficients yields precisely the remainder minimum metric

$$G_N = (J_N H_N^{-1} J_N^*)^{-1}, \quad J_N P = [P]_\chi.$$

Taking the block determinant proves

$$\det G_N = \frac{\prod_{n=0}^N \omega_n}{\prod_{j=0}^{N-q} \eta_j} \quad (N \geq q-1). \quad (\text{CPT29})$$

For $N = q-1$ the denominator is one. Under (CPT5) a monic degree- n polynomial becomes one of leading coefficient i^n ; multiplying it by i^{-n} returns monicity and leaves its norm unchanged. The relation χu has the corresponding factor i^{q+r} . Hence no determinant modulus or monic norm acquires a phase factor in (CPT29).

Retain the four signs, the full relation band and its weights:

$$\begin{aligned} \mathcal{B}(s, t) &= \log \det G_{q-1} + \log \det G_q - \log \det G_{2q-1} - \log \det G_{2q}, \\ c_0 &= c_q = 1, \quad c_r = 2 \quad (1 \leq r \leq q-1), \\ \mathcal{R}(s, t) &= \sum_{r=0}^q c_r \log \eta_r(s, t), \quad \mathcal{S}(s, t) = \sum_{r=0}^q c_r \log \omega_{q+r}(s, t). \end{aligned} \quad (\text{CPT30})$$

Substitute (CPT29) at each of the four cutoffs. The source prefix sums give coefficient -1 at $q, 2q$ and -2 at $q+1, \dots, 2q-1$. The relation prefix sums give coefficient $+1$ at $0, q$ and $+2$ at $1, \dots, q-1$. Thus

$$\boxed{\mathcal{B}(s, t) = \mathcal{R}(s, t) - \mathcal{S}(s, t), \quad \sum_{r=0}^q c_r = 2q.} \quad (\text{CPT31})$$

This independently proves the signs and endpoint weights of ACW20/SXR10 [22, 23].

If a source measure is multiplied by $A > 0$, all these minima and every quotient metric are multiplied by A . The quotient determinant at each cutoff is multiplied by A^q , whose four signed logarithms cancel. Equivalently each ratio η_r/ω_{q+r} is unchanged. Therefore the original arithmetic correction is exactly

$$\delta_k^\sigma := \mathcal{B}[m_k(y)dy] - \mathcal{B}[\sigma(y)dy] = \mathcal{B}(1, 1) - \mathcal{B}(0, 0). \quad (\text{CPT32})$$

This is the proved scalar cancellation of the original M^k .

Use $J = 0, D = 2q+1, g = 1$ in (CPT9), and abbreviate $\Lambda_0 = \Lambda_{k,0,2q+1}$. Every minimizing polynomial χu_r , $0 \leq r \leq q$, and each of its allowed coefficient variations belongs to this complete relation space. Differentiating η_r makes the coefficient-derivative terms vanish by its defining orthogonality to $\chi C[S]_{<r}$. The remaining derivative is exactly

$$\partial_s \eta_r = \int_C H(y) |\chi(c + iy) u_{r;s,t}(c + iy)|^2 d\nu_{s,t}(y).$$

Equations (CPT11) and (CPT16) consequently prove

$$|\partial_s \log \eta_r(s, t)| \leq \Lambda_0, \quad |\partial_s \mathcal{R}(s, t)| \leq 2q \Lambda_0. \quad (\text{CPT33})$$

The common degree $2q+1$ includes the output face needed when one original linear factor multiplies a degree- $2q$ relation. For the monic relation band alone, $D = 2q$ also suffices and gives its correspondingly sharper constant.

Splitting (CPT32) at the actual endpoint $(0, 1)$, and using (CPT31) on its central leg, gives the exact finite receiver

$$\boxed{\delta_k^\sigma = \mathcal{B}(0, 1) - \mathcal{B}(0, 0) - [\mathcal{S}(1, 1) - \mathcal{S}(0, 1)] + E_{\text{rel}}}, \quad (CPT34)$$

$$E_{\text{rel}} = \mathcal{R}(1, 1) - \mathcal{R}(0, 1), \quad |E_{\text{rel}}| \leq 2q\Lambda_0.$$

For any function F on the square, define its oriented rectangle difference $\Delta_\square F = F(1, 1) - F(0, 1) - F(1, 0) + F(0, 0)$. Both central relation legs obey (CPT33), so

$$\boxed{\Delta_\square \mathcal{B} + \Delta_\square \mathcal{S} = \Delta_\square \mathcal{R}, \quad |\Delta_\square \mathcal{R}| \leq 4q\Lambda_0.} \quad (CPT35)$$

No claim of a small mixed derivative follows from this integrated rectangle estimate; its derivation controls the two actual legs.

The sign estimate (CPT17) propagates through every monic source minimum because $d\nu_{1,t} \leq e^{K_h} d\nu_{0,t}$. Thus $\mathcal{S}(1, t) - \mathcal{S}(0, t) \leq 2qK_h$. Combining this with (CPT31), (CPT33) proves

$$\mathcal{B}(1, t) - \mathcal{B}(0, t) \geq -2qK_h - 2q\Lambda_0 \quad (0 \leq t \leq 1). \quad (CPT36)$$

At fixed packet (CPT13) implies $q^A \Lambda_0 \rightarrow 0$ for every fixed $A > 0$. In particular the lower logarithmic-scale bound

$$\liminf_{k \rightarrow \infty} \frac{\mathcal{B}(1, t) - \mathcal{B}(0, t)}{q \log k} \geq 0$$

holds uniformly for $t \in [0, 1]$. It leaves the source-band value and the exterior leg in (CPT34) explicitly present.

35 Exact mixed source-band kernel and complete cross terms

Write $\Delta_N(s, t) = \det H_N(s, t)$. The monic source-product identity above gives

$$\mathcal{S} = \log \Delta_{2q-1} + \log \Delta_{2q} - \log \Delta_{q-1} - \log \Delta_q. \quad (CPT37)$$

Let $f_N(y) = (1, c + iy, \dots, (c + iy)^N)^T$. With the conjugate-linear-first Gram convention,

$$H_N = \int \overline{f_N(y)} f_N(y)^T d\nu_{s,t}(y).$$

Define its reproducing kernel by the original coefficient formula

$$K_N^{s,t}(y, z) = f_N(y)^T H_N(s, t)^{-1} \overline{f_N(z)}. \quad (CPT38)$$

If $F(y) = f_N(y)^T a$, matrix multiplication gives

$$\int K_N^{s,t}(y, z) F(z) d\nu_{s,t}(z) = F(y).$$

Also $K_N^{s,t}(z, y) = \overline{K_N^{s,t}(y, z)}$. This verifies the kernel and its conjugations directly in the original complex coefficient frame.

At each real point $H_C H_O = 0$. Hence $\partial_s \partial_t H_N = 0$, while

$$H_{N,C} = \int_C H(y) \overline{f_N(y)} f_N(y)^T d\nu_{s,t}(y), \quad H_{N,O} = \int_O H(z) \overline{f_N(z)} f_N(z)^T d\nu_{s,t}(z).$$

Differentiating $H_N^{-1} H_N = I$ gives $\partial_t H_N^{-1} = -H_N^{-1} H_{N,O} H_N^{-1}$. The determinant derivative follows by differentiating the finite determinant polynomial and using its adjugate: $\partial_s \log \det H_N = \text{Tr}(H_N^{-1} H_{N,C})$. Consequently

$$\begin{aligned} \partial_s \partial_t \log \Delta_N &= -\text{Tr}(H_N^{-1} H_{N,C} H_N^{-1} H_{N,O}) \\ &= -\int_C \int_O H(y) H(z) |K_N^{s,t}(y, z)|^2 d\nu_{s,t}(y) d\nu_{s,t}(z). \end{aligned} \quad (CPT39)$$

For the second equality, expand the two integrals and use the trace of the rank-one products. The resulting scalar product is $K_N(y, z)K_N(z, y) = |K_N(y, z)|^2$. Every integral is absolutely convergent by the moment bounds proved after (CPT8); the finite inverse Gram introduces only finite coefficients on the compact parameter square.

Combining all four terms of (CPT37) yields

$$\begin{aligned} \partial_s \partial_t \mathcal{S}(s, t) = & - \int_C \int_O H(y) H(z) [|K_{2q-1}^{s,t}(y, z)|^2 + |K_{2q}^{s,t}(y, z)|^2 \\ & - |K_{q-1}^{s,t}(y, z)|^2 - |K_q^{s,t}(y, z)|^2] d\nu_{s,t}(y) d\nu_{s,t}(z). \end{aligned} \quad (\text{CPT40})$$

This is an exact expression for the source-band interaction, with its original two consecutive windows. Integrating over the parameter square gives $\Delta_{\square} \mathcal{S}$. Equations (CPT35), (CPT40) therefore return the full signed rectangle to this explicit mixed kernel integral with the proved remainder $4q\Lambda_0$. Disjoint scalar supports made $\partial_s \partial_t H_N = 0$; they do not make the two noncommuting compressed matrices $H_{N,C}, H_{N,O}$ have zero product.

The relation determinant has the same exact formula. If B_N is the full coefficient matrix of $\chi, \chi S, \dots, \chi S^{N-q}$, its Gram and derivatives are $B_N^* H_N B_N, B_N^* H_{N,C} B_N, B_N^* H_{N,O} B_N$. The relation version of (CPT39) therefore contains its complete cross matrix. Subtracting it from the source version in (CPT29) gives the exact quotient mixed derivative. At $N = q - 1$ the relation matrix is empty, its determinant is one, and its derivative is zero. This proves the relation-to-source and source-to-quotient maps without omitting the zero-cutoff endpoint.

36 Propagation into the earlier central statements

Equations (CPT10)–(CPT11) reproduce the root-sensitive finite annihilator localization C3–C7 from the supplied complete proofs [20], now uniformly on the entire square. For the replacement $\nu_{1,t} \rightarrow \nu_{0,t}$, the earlier endpoint estimate multiplied that localization by $e^{L_k} - 1$ and required a final smallness guard. The proved derivative argument replaces its use by

$$\Lambda = L_k \varepsilon_{k,J,D}, \quad e^{-\Lambda} G_{0,t} \preceq G_{1,t} \preceq e^{\Lambda} G_{0,t}, \quad \|P_1 - P_0\|_{0,t} \leq (e^{\Lambda} - 1) \|P_0\|_{0,t}.$$

These are valid finite inequalities for every admitted k satisfying the displayed density, degree and root guards. The old endpoint inequalities remain preserved statements on their own domains.

The previous central-leg relation radius in C17 is replaced, for this exact interpolation, by $2qL_k \varepsilon_{k,0,2q+1}$. The uniform one-sided conclusion C21–C22 now has the radius in (CPT36) for every exterior parameter. The additional rectangle identity is (CPT35), with the literal mixed kernel (CPT40). These substitutions propagate new constants and new path information; they do not evaluate an uncomputed source-band integral.

For each localized original submodule and its actual fixed subquotients, (CPT27) proves the quantitative correspondence. For the complementary quotient, (CPT28) records its precise Schur metric. For the full signed determinant, (CPT34) and (CPT40) retain the actual source-band and exterior quantities. Thus no seven absolute allocation, proper-source standard-window value, individual projected sign, or Riemann-hypothesis conclusion is supplied by a substituted rank or a two-class comparison in this note.

Appendix

37 The retained local-density constants and proof

This appendix states and verifies the particular central-density provider used in (CPT14)–(CPT16), with the exact original constants of ACW4–11 [22]. It uses the two original source envelopes

recorded below (CPT3), rather than any new analytic hypothesis. Put $\eta = 1/4$ and $C_h^{\text{tilt}} = C_{\text{tilt}}$. For $|\theta| \leq \alpha$, define the auxiliary density

$$p_\theta(y) = e^{\theta y} w_h(y) / M_h(\theta), \quad \mu(\theta) = \ell'_h(\theta).$$

Its unit mass is used only in this Fourier computation. The exact convolution identity is

$$m_k(x) = M_h(\theta)^k e^{-\theta x} p_\theta^{*k}(x). \quad (\text{CPT41})$$

It follows by pulling the convolution back to its k original variables, whose sum is x with Jacobian one. Thus the original arithmetic mass is explicitly restored.

The envelope and $M_h(\theta) \geq M_0$ give

$$V_* = \frac{2C_{\text{tilt}}}{M_0} B(3, p-3), \quad H_* = 2^{1+\eta} \left[\frac{2C_{\text{tilt}}}{M_0} B(3+\eta, p-3-\eta) + \mu^{2+\eta} \right].$$

Integrating $y^2(1+|y|)^{-p}$, and using $|x-y|^{2+\eta} \leq 2^{1+\eta}(|x|^{2+\eta} + |y|^{2+\eta})$, proves $V(\theta) \leq V_*$ and $\int |y - \mu(\theta)|^{2+\eta} p_\theta(y) dy \leq H_*$. The beta integrals converge, since $p \geq 7/2 > 3 + \eta$.

On $[-1, 1]$, (CPT41) at $k = 3$ and the original lower convolution envelope give

$$p_\theta^{*3}(x) \geq c_h e^{-2\alpha} 2^{-B_0} / M^3.$$

Define

$$a_{\text{mix}} = \min\{1/4, c_h e^{-2\alpha} 2^{-B_0} / M^3\}, \quad \beta_{\text{mix}} = 2a_{\text{mix}}, \quad v_* = \beta_{\text{mix}}/9.$$

The triple convolution is the sum of β_{mix} times the uniform density on $[-1, 1]$, and a nonnegative residual of mass $1 - \beta_{\text{mix}}$. The variance of a mixture is at least the mixture-weighted within-component variances, as follows by expanding each centred square about the full mean. It is therefore at least $\beta_{\text{mix}}/3$. The variance of the three independent summands is $3V(\theta)$, proving $V(\theta) \geq v_* > 0$. Continuity and differentiation of M_h through order two follow from the same integrable envelope. Consequently $\mu(\theta)$ is strictly increasing and maps $[-\alpha, \alpha]$ onto $[-\mu, \mu]$, establishing the inverse used in (CPT14).

For the centred characteristic function $\phi_\theta(\zeta) = \int e^{i\zeta(y-\mu(\theta))} p_\theta(y) dy$, Taylor's formula on $|u| \leq 1$, and $2 + |u| + u^2/2 \leq (7/2)|u|^{2+\eta}$ on $|u| > 1$, imply

$$|\phi_\theta(\zeta) - 1 + V(\theta)\zeta^2/2| \leq 4H_*|\zeta|^{2+\eta}.$$

Put

$$\zeta_0 = \min\{1, (2V_*)^{-1/2}, (v_*/(16H_*))^{1/\eta}\}.$$

For $|\zeta| \leq \zeta_0$, the quadratic term is nonnegative and the Taylor error is at most $v_*\zeta^2/4$. Hence $|\phi_\theta(\zeta)| \leq e^{-v_*\zeta^2/4}$. The triple-density mixture gives

$$|\phi_\theta(\zeta)|^3 \leq 1 - \beta_{\text{mix}} + \beta_{\text{mix}}|\sin \zeta/\zeta|.$$

For $|\zeta| \geq \zeta_0$ one has $|\sin \zeta/\zeta| \leq 1 - \zeta_0^2/7$. To check the constant, on $[\zeta_0, 2]$ the positive function $\sin u/u$ decreases, and its Taylor upper bound at $0 < \zeta_0 \leq 1$ is $1 - \zeta_0^2/6 + \zeta_0^4/120 \leq 1 - \zeta_0^2/7$. Beyond 2, its absolute value is at most $1/2$, which is smaller. Thus, with

$$r_* = (1 - \beta_{\text{mix}}\zeta_0^2/7)^{1/3} < 1, \quad C_2 = \frac{4\pi(C_{\text{tilt}}/M_0)^2}{2p-1},$$

we have $|\phi_\theta| \leq r_*$ off the small interval. The Fourier Plancherel identity [27, (1.14.8)] in the convention $\widehat{f}(\zeta) = \int e^{i\zeta y} f(y) dy$ gives $\int |\phi_\theta|^2 \leq C_2$. Relative to the DLME convention $\mathcal{F}f = (2\pi)^{-1/2} \int e^{i\zeta y} f(y) dy$, our transform is $\sqrt{2\pi}\mathcal{F}f$. Thus its squared L^2 norm is $2\pi \int |f|^2$; integrating the squared density envelope yields exactly C_2 .

Let

$$C_0 = 4H_* + V_*^2/8, \quad a_k^{\text{loc}} = v_*(k-1)/4,$$

and define the finite error

$$A_k^{\text{loc}} = \frac{1}{2\pi} \left[C_0 k \Gamma((3+\eta)/2) (a_k^{\text{loc}})^{-(3+\eta)/2} + C_2 r_*^{k-2} + \frac{2e^{-kv_*\zeta_0^2/2}}{kv_*\zeta_0} \right], \quad \epsilon_k^{\text{loc}} = \sqrt{2\pi k V_*} A_k^{\text{loc}}. \quad (\text{CPT42})$$

On the small interval,

$$|\phi_\theta(\zeta) - e^{-V(\theta)\zeta^2/2}| \leq C_0 |\zeta|^{2+\eta}.$$

This follows from the previous Taylor bound and the exponential remainder $V_*^2\zeta^4/8$, using $|\zeta|^4 \leq |\zeta|^{2+\eta}$ when $|\zeta| \leq 1$. Factor the difference of the k -th powers as a sum of k terms. Both factors have modulus at most $e^{-v_*\zeta^2/4}$, so its integral is bounded by

$$C_0 k \int_{\mathbb{R}} |\zeta|^{2+\eta} e^{-a_k^{\text{loc}}\zeta^2} d\zeta = C_0 k \Gamma((3+\eta)/2) (a_k^{\text{loc}})^{-(3+\eta)/2}.$$

The characteristic-function integral on the complementary region is at most $C_2 r_*^{k-2}$, by its L^2 bound. The corresponding Gaussian tail is at most $2e^{-kv_*\zeta_0^2/2}/(kv_*\zeta_0)$, by replacing the factor $|\zeta|/\zeta_0$ with its lower bound one before integrating the Gaussian derivative. Fourier inversion therefore yields

$$\left| p_\theta^{*k}(k\mu(\theta)) - (2\pi k V(\theta))^{-1/2} \right| \leq A_k^{\text{loc}}. \quad (\text{CPT43})$$

All bounds are uniform for the closed interval $|\theta| \leq \alpha$, including its two endpoints. The use of inversion is justified by $\phi_\theta^k \in L^1$, already implied by the L^2 bound and $|\phi_\theta| \leq 1$ for $k \geq 2$. Multiplying (CPT43) by $\sqrt{2\pi k V(\theta)}$ proves the relative error in (CPT14), with $\epsilon_k^{\text{loc}} = O_h(k^{-1/8})$. In particular its stated finite guard is eventually satisfied.

Finally apply (CPT41) with $\theta = \theta(u)$, insert the relative-error expression, and divide by $M^k \sigma(ku)$. Writing the Gamma density as

$$\sigma(ku) = c_\sigma^{\text{env}}(ku) e^{-\alpha k|u|} (1+k|u|)^{-1/2}$$

gives (CPT14) by exact logarithms. This completes the density-to-contrast calculation used by the new path estimates.

Joint endpoint estimates and localized two-window transport

Source and precise scope. This note verifies and supplies finite constants for the deductions labelled J1–J10 and S15–S16 in the supplied `COMPLETE_PROOFS.md`, dated 18 September 2026. The source bytes have SHA-256 `e4e54261c91ae899a78956383482650f88933024f5f3cec6b41332ceeb5a4172`. The endpoint block is at lines 799–966; the two-window block is at lines 746–770. This note retains every singular coefficient, the original Lebesgue product measure, the full polynomial relation, and both boundary windows. The incoming endpoint decomposition, the actual tilt envelope, and the old window coefficients are explicitly identified as input results. The named earlier files `TWO_WINDOW_SAME_CLASS_DEFECT.md` and the theta-endpoint note were not located in the bounded intake. Their numerical window coefficient and full theta existence theorem are therefore not claimed as independently recovered here. The finite deductions below are proved from the actual formulas recorded in the supplied source; no missing allocation or projected arithmetic sign is assumed.

38 The entire remainder has an explicit uniform norm allowance

Write the original quartet as

$$\mathcal{Z} = \{1/2 + \epsilon\delta + i\epsilon'\gamma : \epsilon, \epsilon' \in \{-1, 1\}\}, \quad 0 < \delta < 1/2, \quad \gamma > 2,$$

with common multiplicity m , $d = m - 1$, and retain the given inverse primitive

$$\psi(x) = s_h(x) + R_h(x), \quad s_h(x) = \sum_{\rho \in \mathcal{Z}} \sum_{j=0}^d a_{\rho,j} x^{-\rho} \frac{\log(1/x)^j}{j!}. \quad (\text{JL1})$$

Here R_h is the actual entire even remainder, not a freely chosen regularization. If $R_h(x) = \sum_{n \geq 0} r_{2n} x^{2n}$, put

$$B_h = \sum_{n \geq 0} |r_{2n}| < \infty, \quad C_h^{\text{reg}} = B_h^2 + 2B_h \sum_{\rho,j} \frac{|a_{\rho,j}|}{(1 - \Re \rho)^{j+1}}. \quad (\text{JL2})$$

The sum defining B_h converges absolutely because the power series is entire. One can alternatively use the possibly smaller $\sup_{0 \leq x \leq 1} |R_h(x)|$. These constants refer to that same remainder and leave all singular coefficients unchanged.

Proposition 38.1. *For every $L \geq 0$,*

$$\left| \int_{e^{-L}}^1 |\psi(x)|^2 dx - \int_{e^{-L}}^1 |s_h(x)|^2 dx \right| \leq C_h^{\text{reg}}. \quad (\text{JL3})$$

Proof. The integrand difference is $2\Re(s_h \overline{R_h}) + |R_h|^2$. Since $\Re \rho < 1$,

$$\int_0^1 x^{-\Re \rho} \frac{\log(1/x)^j}{j!} dx = \frac{1}{(1 - \Re \rho)^{j+1}}.$$

This identity follows from $x = e^{-t}$ and j integrations by parts in $\int_0^\infty e^{-(1-\Re \rho)t} t^j dt$. Take absolute values of each of the finitely many singular terms, use $|R_h| \leq B_h$, and enlarge the integration interval to $(0, 1)$. This proves (JL3). In particular the cross terms with the entire remainder do not grow with the endpoint cutoff. $\square$

39 All factorials in the oscillating endpoint leading term

Let $\rho_+ = 1/2 + \delta + i\gamma$ and $\mathcal{R} = \{\rho_+, \overline{\rho_+}\}$. The local endpoint coefficient convention in the supplied source is

$$a_{\rho,j} = [z^{d-j}] \frac{p_\delta(\rho+z)}{2z_\rho(z)}, \quad p_\delta(s) = (s-1/2)^2 + \gamma^2 - \delta^2, \quad \zeta(\rho+z) = z^m z_\rho(z). \quad (\text{JL4})$$

Thus

$$a_{\rho_+,d} = \frac{2i\delta\gamma}{2z_{\rho_+}(0)} = \frac{m! i\delta\gamma}{\zeta^{(m)}(\rho_+)} =: A_h \neq 0, \quad a_{\overline{\rho_+},d} = \overline{A_h}. \quad (\text{JL5})$$

This is the top Laurent coefficient in the recorded convention; no phase has been made real by a coordinate change.

Put $I_h(L) = \int_{e^{-L}}^1 |\psi|^2 dx$. The change of variables $x = e^{-t}$ gives the exact finite identity

$$I_h^{\text{sing}}(L) = \sum_{\rho, \sigma \in \mathcal{Z}} \sum_{j,l=0}^d \frac{a_{\rho,j} \overline{a_{\sigma,l}}}{j! l!} E_{j+l}(\rho + \bar{\sigma} - 1, L), \quad E_p(z, L) = \int_0^L e^{zt} t^p dt. \quad (\text{JL6})$$

There is no division by z in this definition, so the terms with $z = 0$, including the reflected-root pairs, remain present. For $z \neq 0$, repeated integration by parts gives

$$E_p(z, L) = e^{zL} \sum_{r=0}^p \frac{(-1)^r p! L^{p-r}}{(p-r)! z^{r+1}} - \frac{(-1)^p p!}{z^{p+1}}. \quad (\text{JL7})$$

For every ordered right/right pair and $p = j + l$ put

$$\begin{aligned} A_1 &= \sum_{\rho, \sigma \in \mathcal{R}} \sum_{j,l=0}^d \frac{|a_{\rho,j} a_{\sigma,l}|}{j! l!} \sum_{\substack{0 \leq r \leq j+l \\ (j,l,r) \neq (d,d,0)}} \frac{(j+l)!}{(j+l-r)! |\rho + \bar{\sigma} - 1|^{r+1}}, \\ A_2 &= C_h^{\text{reg}} + \sum_{\rho, \sigma \in \mathcal{R}} \sum_{j,l=0}^d \frac{|a_{\rho,j} a_{\sigma,l}|}{j! l!} \frac{(j+l)!}{|\rho + \bar{\sigma} - 1|^{j+l+1}}, \\ A_3 &= \sum_{(\rho, \sigma) \in \mathcal{Z}^2 \setminus \mathcal{R}^2} \sum_{j,l=0}^d \frac{|a_{\rho,j} a_{\sigma,l}|}{j! l! (j+l+1)}. \end{aligned} \quad (\text{JL8})$$

These are exactly J2 with (JL2) as a fully displayed valid regular allowance. Define

$$F_h(L) = \frac{1}{(d!)^2} \left(\frac{|A_h|^2}{\delta} + \Re \frac{A_h^2 e^{2i\gamma L}}{\delta + i\gamma} \right), \quad f_* = \frac{|A_h|^2}{(d!)^2} \left(\frac{1}{\delta} - \frac{1}{\sqrt{\delta^2 + \gamma^2}} \right) > 0. \quad (\text{JL9})$$

Proposition 39.1. *For every $L \geq 1$,*

$$F_h(L) \geq f_*, \quad \left| \frac{I_h(L)}{e^{2\delta L} L^{2d}} - F_h(L) \right| \leq \frac{A_1}{L} + (A_2 + A_3 L) e^{-2\delta L}. \quad (\text{JL10})$$

Proof. For the two equal right/right roots, the $j = l = d, r = 0$ terms in (JL7) sum to $|A_h|^2 / (\delta(d!)^2)$ after dividing by the displayed scale. The other two ordered right/right pairs sum to $\Re(A_h^2 e^{2i\gamma L} / (\delta + i\gamma)) / (d!)^2$. These are precisely the terms in (JL9). Every remaining upper endpoint term in (JL7), after that division, contains $L^{j+l-r-2d}$ with exponent at most -1 . The sum of its absolute coefficients is A_1 . The lower endpoint terms and (JL3) are bounded by $A_2 e^{-2\delta L}$, because $L^{-2d} \leq 1$. For every pair outside $\mathcal{R}^2$, $\Re z \leq 0$, and therefore

$$|E_{j+l}(z, L)| \leq \frac{L^{j+l+1}}{j+l+1}.$$

Its scaled contribution is at most the corresponding coefficient in $A_3 L e^{-2\delta L}$. The lower bound for F_h follows by taking the modulus of its oscillating summand. This proves all assertions, retaining all ordered cross terms. $\square$

Joint tensor/cutoff error. Put

$$\eta_h(L) = \frac{A_1/L + (A_2 + A_3L)e^{-2\delta L}}{f_*}.$$

If $\eta_h(L) \leq 1/2$, the actual ratio $I_h(L)/(e^{2\delta L}L^{2d}F_h(L))$ lies in $[1 - \eta_h(L), 1 + \eta_h(L)]$. For real $|x| \leq 1/2$, integration of $1/(1+t)$ from 0 to x proves $|\log(1+x)| \leq 2|x|$. Consequently

$$\left| \log \frac{I_h(L)^k}{e^{2\delta kL}L^{2dk}F_h(L)^k} \right| \leq 2k\eta_h(L), \quad (\text{JL11})$$

with relative error at most $e^{2k\eta_h(L)} - 1$. At $L = k^2$, the exponent is $O_h(k^{-1})$, so

$$I_h(k^2)^k = e^{2\delta k^3} k^{4dk} F_h(k^2)^k (1 + O_h(k^{-1})). \quad (\text{JL12})$$

When $m = 1$, the empty sum A_1 is zero. At $L = (\log k)/\delta \geq 1$, the exponent in (JL11) is at most

$$\frac{2}{f_*k} \left(A_2 + \frac{A_3}{\delta} \log k \right). \quad (\text{JL13})$$

Fubini's theorem gives exactly $I_h(L)^k$ for the fixed original product representative on $[e^{-L}, 1]^k$. These estimates concern that specified representative. They do not evaluate a minimum over added source corrections.

40 The literal diagonal map and the terminal factorial

For $b_\rho x^{-\rho}$, $\Re \rho = 1/2 + \delta$, the unchanged product measure gives

$$N_{\text{prod}}(k, L) = |b_\rho|^{2k} \left(\frac{e^{2\delta L} - 1}{2\delta} \right)^k. \quad (\text{JL14})$$

The literal diagonal map substitutes all $x_i = x$, hence its value is $b_\rho^k x^{-k\rho}$. Its norms in the two expressly specified measures are

$$\begin{aligned} N_{\text{diag,dx}}(k, L) &= |b_\rho|^{2k} \frac{e^{(k-1+2k\delta)L} - 1}{k - 1 + 2k\delta}, \\ N_{\text{diag,k}}(k, L) &= |b_\rho|^{2k} \frac{e^{2k\delta L} - 1}{2k\delta} \quad \text{in } x^{k-1} dx. \end{aligned} \quad (\text{JL15})$$

Integrating the respective powers of x proves these formulas. Dividing (JL14) by the second formula gives the exact discrepancy

$$\frac{N_{\text{prod}}}{N_{\text{diag,k}}} = k(2\delta)^{1-k} \frac{(1 - e^{-2\delta L})^k}{1 - e^{-2k\delta L}}. \quad (\text{JL16})$$

The common terminal factorial can also be checked without dropping the original unit. In one local algebra $\mathbb{C}[z]/(z^m)$, the actual terminal vector is z^d ; its endpoint coefficient is $b_\rho = 1/(2z_\rho(0))$ in convention (JL4). In the extremal ordered tuple, put $n = kd$ and $N_\Sigma = z_1 + \dots + z_k$. The multinomial theorem and $z_i^{d+1} = 0$ give the exact polynomial identity

$$N_\Sigma^{kd} = \frac{(kd)!}{(d!)^k} \prod_{i=1}^k z_i^d. \quad (\text{JL17})$$

Multiplying by the full $U_h^{\otimes k}$ kills only its positive-degree terms in this terminal vector, so its endpoint is

$$\frac{(kd)!}{(d!)^k} U_h(\rho)^k b_\rho^k \prod_{i=1}^k x_i^{-\rho}. \quad (\text{JL18})$$

Both product and diagonal squared norms therefore acquire $|(kd)!/(d!)^k|^2 |U_h(\rho)|^{2k}$. This verifies the sentence following J8 for that exact terminal vector. No factorial is absorbed into an unnamed choice of basis.

41 Uniform propagation of moments to the closed tilt interval

Retain the actual nonnegative one-packet density w_h and its stated envelope

$$e^{\alpha|y|}w_h(y) \leq C_h^{\text{tilt}}(1 + |y|)^{-p}, \quad \alpha = \pi/2, \quad p = 8m - 9/2 > 3. \quad (\text{JL19})$$

Let $\mu_n = \int_{\mathbb{R}} y^n w_h(y) dy$, without changing the source mass. For $j = 0, 1, 2$,

$$\sum_{r \geq 0} \frac{\alpha^r}{r!} \int_{\mathbb{R}} |y|^{r+j} w_h(y) dy = \int_{\mathbb{R}} |y|^j e^{\alpha|y|} w_h(y) dy < \infty. \quad (\text{JL20})$$

Tonelli's theorem applies to this nonnegative majorant. It proves uniform absolute convergence on $0 \leq \theta \leq \alpha$, interchange with the signed integrands, and continuity of the derivatives at the endpoint:

$$M_h^{(j)}(\theta) = \int_{\mathbb{R}} y^j e^{\theta y} w_h(y) dy = \sum_{r \geq 0} \mu_{r+j} \frac{\theta^r}{r!}. \quad (\text{JL21})$$

For an integer cutoff $L \geq 0$, let $n = L + 1$ and $T_L = n/(2e\alpha)$. For $z \geq 0$ put

$$\mathcal{Q}_n(z) = e^{-z} \sum_{r=n}^{\infty} \frac{z^r}{r!}.$$

For $0 < z \leq n/(2e) < n$, multiplication of each term by $(n/z)^{r-n} \geq 1$ yields

$$\mathcal{Q}_n(z) \leq (z/n)^n e^{n-z} \leq (ez/n)^n \leq 2^{-n}. \quad (\text{JL22})$$

For $z = 0$ the sum is zero; globally it is at most one. The absolute tail of (JL21) is bounded by

$$\int_{\mathbb{R}} |y|^j w_h(y) e^{\alpha|y|} \mathcal{Q}_n(\alpha|y|) dy.$$

Use 2^{-n} on $|y| \leq T_L$, and one elsewhere. Since

$$\int_0^{\infty} y^j (1+y)^{-p} dy = B(j+1, p-j-1), \quad \int_{T_L}^{\infty} y^j (1+y)^{-p} dy \leq \frac{(1+T_L)^{j-p+1}}{p-j-1},$$

the resulting explicit bound is

$$\sup_{0 \leq \theta \leq \alpha} \left| M_h^{(j)}(\theta) - \sum_{r=0}^L \mu_{r+j} \frac{\theta^r}{r!} \right| \leq 2C_h^{\text{tilt}} \left[2^{-(L+1)} B(j+1, p-j-1) + \frac{(1+T_L)^{j-p+1}}{p-j-1} \right]. \quad (\text{JL23})$$

The beta identity follows by $u = y/(1+y)$ in its defining integral. Thus no fourth tilted moment is used. These formulas verify J9–J10 but do not by themselves verify the separately cited arithmetic moment recurrence or evaluate a growing projection-kernel integral.

42 The localized ambient space really contains the window projections

Here restore the simple target order in S15: $a = k + 4$, $a \equiv 1 \pmod{4}$, $q = (a+1)^2$, $c = a/2$, and

$$\chi(S) = \prod_{r,s=0}^a [S - c - (2r - a)\delta - i(2s - a)\gamma]. \quad (\text{JW1})$$

The actual eigenvalue λ is one of these roots. Set $D = 2q + 1$,

$$\mathcal{W} = \frac{\chi(S)}{S - \lambda} \mathcal{P}_{D-q+1}, \quad \mathcal{R}_N = \chi \mathcal{P}_{N-q} \quad (\mathcal{P}_j = \{0\} \text{ for } j < 0). \quad (\text{JW2})$$

All norms are integrals on the original line $S = c + iy$. Let ω be the actual arithmetic density and $\tilde{\omega}$ its specified central replacement at order a , center c , and interval $[-a\mu_h, a\mu_h]$. The central-localization calculation C3–C13 gives on this *entire* fixed space the relative form bound

$$|(\tilde{\omega} - \omega)(f, f)| \leq \tau \|f\|_{\tilde{\omega}}^2 \quad (f \in \mathcal{W}), \quad 0 \leq \tau < 1. \quad (\text{JW3})$$

For clarity, the valid finite parameter is

$$\tau = B_a \min\{1, \mathfrak{e}_{a,1,2q+1}(\omega)\}, \quad B_a = e^{H_*} - 1, \quad (\text{JW4})$$

with H_* and $\mathfrak{e}$ the actual constants in C3 and C13. The guards of those expressions, including $R < q$ and $a\mu_h \leq q/2$, are retained. Self-adjointness and polarization of (JW3) yield

$$|(\tilde{\omega} - \omega)(f, g)| \leq \tau \|f\|_{\omega} \|g\|_{\omega}. \quad (\text{JW5})$$

Exact fixed relation band. Let $S_N v$ be the original attained polynomial representative of v at cutoff N . It is orthogonal to $\mathcal{R}_N$. Every representative of the chosen eigenclass belongs to $\mathcal{W}$; all its differences belong to $\mathcal{R}_N$. For $n = q - 1$ or $n = q$, put

$$F_n = S_{n+q}v - S_nv, \quad \mathcal{B}_n = \mathcal{R}_{n+q} \cap \mathcal{R}_n^{\perp_{\omega}}. \quad (\text{JW6})$$

Then $F_n \in \mathcal{B}_n$, and the ω -orthogonal projector onto this band is $B_n = P_{\mathcal{R}_{n+q}} - P_{\mathcal{R}_n}$. The analogous statement holds in the replaced metric. The *full* section-difference map from the cyclic module to $\mathcal{B}_n$ has rank q , not merely on the single eigenline. Indeed $\dim \mathcal{R}_{n+q} - \dim \mathcal{R}_n = q$. If the section difference of any class is zero, its common representative P , of degree at most n , is orthogonal to every high relation. In particular it is orthogonal to χP , whose degree is at most $n + q$.

To check the sign used here, pair each root in (JW1) with the root having opposite real displacement and the same imaginary displacement. On $S = c + iy$, each pair contributes $-[(y - (2s - a)\gamma)^2 + (2r - a)^2\delta^2]$. There are $q/2$ pairs, and $q/2$ is even because $a \equiv 1 \pmod{4}$. Every $(2r - a)\delta$ is nonzero because a is odd. Thus $\chi(c + iy) > 0$ for every real y . For $P \neq 0$, $\int \chi(c + iy)|P(c + iy)|^2 d\omega(y) > 0$, contradicting orthogonality. The map is injective and hence fills the band.

Lemma 42.1 (Explicit fixed-subspace projection bound). *For any fixed $\mathcal{R} \subseteq \mathcal{W}$, let $P, \tilde{P}$ be its orthogonal projectors in the two metrics. Put*

$$d_{\tau} = \frac{\tau}{1 - \tau}, \quad g_{\tau} = \sqrt{\frac{1 + \tau}{1 - \tau}}.$$

Then, as operators on $(\mathcal{W}, \omega)$,

$$\|\tilde{P} - P\| \leq d_{\tau}, \quad \|\tilde{P}\| \leq g_{\tau}. \quad (\text{JW7})$$

For the window band,

$$\|\tilde{B}_n - B_n\| \leq 2d_{\tau}, \quad \|\tilde{B}_n\|, \|I - \tilde{B}_n\| \leq g_{\tau}. \quad (\text{JW8})$$

Proof. For $w \in \mathcal{W}$, put $r = w - Pw$ and $h = \tilde{P}w - Pw \in \mathcal{R}$. The original orthogonality gives $\langle r, h \rangle_{\omega} = 0$; the new orthogonality gives $\langle r - h, h \rangle_{\tilde{\omega}} = 0$. Consequently

$$(1 - \tau)\|h\|_{\tilde{\omega}}^2 \leq \|h\|_{\tilde{\omega}}^2 = (\tilde{\omega} - \omega)(r, h) \leq \tau \|r\|_{\omega} \|h\|_{\omega} \leq \tau \|w\|_{\omega} \|h\|_{\omega}.$$

This proves the difference bound. For the second bound use $\|\tilde{P}w\|_{\tilde{\omega}} \leq \|w\|_{\tilde{\omega}}$ and (JW3). Subtracting the two fixed nested relation projectors proves the first part of (JW8). Their difference and its complement are orthogonal projectors in the new metric, proving its second part. $\square$

Sections, multiplication, and both projected errors. The same argument in a fixed affine observation fibre proves

$$\|\tilde{S}_N v - S_N v\|_\omega \leq d_\tau \|S_N v\|_\omega. \quad (\text{JW9})$$

Put $E_n = \|S_n v\|_\omega^2$. Nested minima and orthogonality give

$$\|S_{n+q} v\|_\omega \leq \sqrt{E_n}, \quad \|F_n\|_\omega^2 = E_n - E_{n+q} \leq E_n, \quad \|\tilde{F}_n - F_n\|_\omega \leq 2d_\tau \sqrt{E_n}. \quad (\text{JW10})$$

Multiplication by $S - \lambda$ takes all these section differences into $\chi \mathcal{P}_{D-q} \subseteq \mathcal{W}$, with degree at most D . Let M_a be a valid operator allowance for this multiplication on the input face in the original metric. The actual original Gamma comparison C2 gives the following fully explicit choice:

$$M_a = \sqrt{\frac{b_a}{a_{a,2q+1}}} (4q + 2 + |\lambda - c|). \quad (\text{JW11})$$

In fact the incoming Gamma multiplication bound on degree $2q$ is $\|yP\|_\sigma \leq 2(2q+1)\|P\|_\sigma$; the triangle inequality for $(S - \lambda)P = (iy - (\lambda - c))P$, the upper comparison for the output, and the lower comparison for the input give (JW11). Thus this bound uses only the displayed C2 provider. It does not silently replace $b_a/a_{a,2q+1}$ by a fixed constant. If the norms are written with the unscaled density m_a instead of $M_h(\alpha)^{-a}m_a$, both source/Gamma comparison constants acquire the same factor $M_h(\alpha)^a$. Their ratio in (JW11) is unchanged; all section energies in the formulas still carry their original mass.

Set $w_n = (S - \lambda)F_n$ and $\tilde{w}_n = (S - \lambda)\tilde{F}_n$. From (JW10)–(JW11),

$$\|\tilde{w}_n - w_n\|_\omega \leq 2d_\tau M_a \sqrt{E_n}. \quad (\text{JW12})$$

Equations (JW8) and (JW12), with the triangle inequality, now give

$$\begin{aligned} \|\tilde{B}_n \tilde{w}_n - B_n w_n\|_\omega &\leq 2d_\tau (g_\tau + 1) M_a \sqrt{E_n}, \\ \|(I - \tilde{B}_n) \tilde{w}_n - (I - B_n) w_n\|_\omega &\leq 2d_\tau (g_\tau + 1) M_a \sqrt{E_n}. \end{aligned} \quad (\text{JW13})$$

Every vector and every projection in these inequalities is in $\mathcal{W}$. No restricted pair-metric estimate is applied to an arbitrary projector on the full cyclic quotient.

For a squared-norm version put $\varepsilon_\tau = 2d_\tau (g_\tau + 1)$. For either projected vector V_n and its corresponding $\tilde{V}_n$,

$$\left| \|\tilde{V}_n\|_\omega^2 - \|V_n\|_\omega^2 \right| \leq [\tau(1 + \varepsilon_\tau)^2 + 2\varepsilon_\tau + \varepsilon_\tau^2] M_a^2 E_n. \quad (\text{JW14})$$

To see this, use $\|V_n\|_\omega \leq M_a \sqrt{E_n}$, (JW13), and (JW3), then expand the difference of the two squared norms. This includes the cross terms. Also, by comparing each of the two attained minima,

$$\left| \|\tilde{F}_n\|_\omega^2 - \|F_n\|_\omega^2 \right| \leq \tau(E_n + E_{n+q}) \leq 2\tau E_n. \quad (\text{JW15})$$

Consequence for the inherited leading coefficients. The supplied original two-window result records $\|F_n\|_\omega^2/E_n \rightarrow 1$ and the total coefficient $16/\pi^2$ after division by $q^2\|F_n\|_\omega^2$; it records that this coefficient is tangential at $n = q - 1$ and normal at $n = q$. Equations (JW14)–(JW15) prove stability of each of those exact existing coefficients under the specified central replacement. Indeed the displayed C2 constants give

$$\log(b_a/a_{a,2q+1}) = O_h(a + \log q), \quad \log \tau \leq -2(q - 1) \log(q/a) + O_h(q + a + \log q). \quad (\text{JW16})$$

Since $q = (a + 1)^2$ and $|\lambda - c| \leq a\sqrt{\delta^2 + \gamma^2}$, it follows that $\tau M_a^2/q^2 \rightarrow 0$. Hence the right side of (JW14), divided by $q^2 E_n$, tends to zero for each window. The denominator perturbation (JW15) also tends to zero relatively. This proves the transfer without needing an unlocated sharp $O_h(q)$ multiplication theorem.

The bare rate $O_h(\tau q \sqrt{E_n})$ stated in S16 should therefore be replaced in the received source by the finite (JW13), unless its separately named sharper multiplication provider is supplied. Its asserted leading-coefficient stability is already justified by the weaker explicit rate. The numerical coefficient of the old two-window theorem remains an inherited value: this proof does not claim to have independently calculated that absent source.

Integration boundary. The results verified here neither calculate the seven absolute native allocations nor the individual three-column projected arithmetic signs. They provide a quantitative fixed-representative endpoint limit, the exact diagonal norm discrepancy, uniform fixed-source moment input, and complete transport of two specified boundary-window projections. All original allocation and same-class response calculations remain in their original metrics.

Marked exponential image: exact connection, periods and metric receivers

This note receives the complete nine-page *Deligne marked image* calculation [30], including its finite-field endomorphism cohomology, its tensor constituents, and its characteristic-zero marked frame. Every formula used below is proved with the specified full polynomial and the original coefficient frame. The additional receiving calculations are the exact second fundamental form of the retained polynomial line, an explicit inverse to the regularized marked frame, and the transport of every original restriction and attained quotient through that frame. These are finite identities. Their statements contain no unevaluated asymptotic hypothesis.

43 Objects, coefficient models, and imported theorems

Retain the stipulated simple quartet with

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad k \equiv 1 \pmod{4},$$

and set

$$\begin{aligned} c &= k/2, & q &= (k+1)^2, & \beta_{ab} &= c + (2a-k)\delta + i(2b-k)\gamma, \\ \chi(S) &= \prod_{a,b=0}^k (S - \beta_{ab}), & \Phi'(S) &= \chi(S), & \Phi(0) &= 0, & \lambda &= k\rho, \\ p_\lambda(S) &= \frac{\chi(S)}{S - \lambda}, & b_\lambda &= \frac{1}{\chi'(\lambda)}, & v_\lambda &= b_\lambda p_\lambda. \end{aligned} \tag{MII1}$$

All the roots are distinct: equality of two roots first equates their real parts and then their imaginary parts. The numbers δ, γ are nonzero. In particular $b_\lambda \neq 0$. The arithmetic unit $v_h = 2\xi/h$, its factor $v_h(\rho)^k$ in the original inclusion, and the original convolution measure remain unchanged. If that original inclusion specifies a scalar multiple av_λ , every occurrence of b_λ in a frame determinant is replaced by ab_λ ; the present formulas use the cardinal polynomial in (MII1) and leave the inclusion itself in its original coordinates.

The marked complex is

$$\left[\mathbb{C}[u, t, S] \xrightarrow{u\partial_S + \chi(S) - t} \mathbb{C}[u, t, S] dS \right]. \tag{MII2}$$

The mark t , the period u , and the arithmetic cutoff N are three separate specified parameters. The maps connecting their uses will be given explicitly.

For the finite-field calculation choose a finitely generated coefficient ring containing the finitely many coefficients of Φ , invert u and $1, 2, \dots, q+1$, and choose a good closed fibre of characteristic $p > q+1$. Use $\mathbb{F}_b$ for its finite ground field, $Q = b^n$ for an extension cardinality, a nontrivial character $\psi : \mathbb{F}_p \rightarrow \overline{\mathbb{Q}}_\ell^*$, and $\ell \neq p$. No reduction of the analytic function ξ is used. The finite-field polynomial and the complex polynomial arise from that coefficient model; an identification of their cohomology fibres is not inserted.

We use the following results of Pierre Deligne with their exact scope [28, §3.7, (3.7.2.3), Lemma (3.7.3); (3.4.1)(iii)]. For a polynomial of degree d prime to p whose highest homogeneous part defines a smooth projective hypersurface, exponential compact cohomology in n variables is concentrated in degree n , has dimension $(d-1)^n$, and is pure of weight n . In the corresponding open coefficient family these cohomology sheaves are lisse. A lisse pointwise-pure sheaf on a normal finite-field scheme is geometrically semisimple. For the one-variable phases here, the highest projective zero locus is empty in $\mathbb{P}^0$, and the leading coefficient is invertible. Thus all mark values are included. These are the cited external theorems; the endomorphism computations below are proved here. Compact base change, Künneth and smooth Poincaré duality are used with the conventions of [28, §1.4].

44 The entire endomorphism cohomology on the mark line

Let

$$\mathcal{V} = R^1 \pi_! \mathcal{L}_\psi((\Phi(x) - tx)/u), \quad \pi : \mathbb{A}_x^1 \times \mathbb{A}_t^1 \longrightarrow \mathbb{A}_t^1. \quad (\text{MII3})$$

It is lisse of rank q , pure of weight one, and the full compact direct image is $\mathcal{V}[-1]$.

Lemma 44.1 (The exact dual and Fourier kernels). *If $\mathcal{V}_-$ is defined by the negative of the complete phase, then*

$$\mathcal{V}_- \simeq \mathcal{V}^\vee(-1). \quad (\text{MII4})$$

For $\text{pr}_a : \mathbb{A}_a^1 \times \mathbb{A}_z^1 \rightarrow \mathbb{A}_a^1$ and the zero inclusion i_0 , one has

$$R \text{pr}_{a!} \mathcal{L}_\psi(az) \simeq i_{0*} \overline{\mathbb{Q}}_\ell(-1)[-2]. \quad (\text{MII5})$$

Proof. The degree $q + 1$ of the phase is prime to p . A polar expression $g^p - g$ has highest pole order divisible by p , so the rank-one Artin–Schreier character at infinity is nontrivial on wild inertia. Its wild-inertia invariants vanish. Wild invariants are exact for ℓ -adic coefficients when $\ell \neq p$: one averages over the finite wild quotients, whose orders are units. The tame-inertia spectral sequence consequently has zero input. For the compactification $j : \mathbb{A}^1 \rightarrow \mathbb{P}^1$ this proves $j_! \mathcal{L} \simeq Rj_* \mathcal{L}$. Poincaré duality pairs the two opposite exponential degree-one groups perfectly into $\overline{\mathbb{Q}}_\ell(-1)$, proving (MII4), including its twist.

For (MII5), compact base change gives at $a = 0$ the compact complex of $\mathbb{A}^1$, namely $\overline{\mathbb{Q}}_\ell(-1)[-2]$. If $a \neq 0$, the cover $y^p - y = az$ is an affine line with coordinate y . Its compact cohomology is one Tate line in degree two, on which all deck translations act trivially. The nontrivial character summand defining $\mathcal{L}_\psi(az)$ therefore has zero compact cohomology. The relative complex is supported on the closed zero point. Recollement identifies it with the direct image of its zero-fibre complex, proving the stated relative identity. $\square$

Theorem 44.2 (Complete mark-line endomorphism complex). *Over the algebraic closure of the specified finite ground field,*

$$R\Gamma_c(\mathbb{A}_t^1, \text{End } \mathcal{V}) = \overline{\mathbb{Q}}_\ell(-1)[-2]. \quad (\text{MII6})$$

The sheaf $\mathcal{V}$ is geometrically irreducible and $H^1(\mathbb{A}^1, \text{End } \mathcal{V}) = 0$.

Proof. Compute the compact complex of $\mathcal{L}_\psi((\Phi(x) - \Phi(y) - t(x - y))/u)$ on the full ordered space $\mathbb{A}_{x,y,t}^3$. Integration first in x, y , the concentration in degree one and (MII4) give

$$R\Gamma_c(\mathbb{A}_t^1, \text{End } \mathcal{V})(-1)[-2].$$

Integration first in t applies (MII5) to $-(x - y)/u$. The result is supported exactly on $x = y$, with factor $(-1)[-2]$. The remaining phase is identically zero on that diagonal, so the same complex is $R\Gamma_c(\mathbb{A}_x^1, \overline{\mathbb{Q}}_\ell)(-1)[-2]$. Cancelling the identical twist and shift proves (MII6). The trace pairing identifies $(\text{End } \mathcal{V})^\vee$ with $\text{End } \mathcal{V}$. Duality on the smooth line now gives $H^0(\text{End } \mathcal{V}) = \overline{\mathbb{Q}}_\ell$ and $H^1(\text{End } \mathcal{V}) = 0$. Deligne’s geometric semisimplicity applies to the pure lisse sheaf $\mathcal{V}$. In a semisimple representation $\bigoplus_i W_i^{m_i}$ over the algebraically closed coefficient field, the dimension of the endomorphism algebra is $\sum_i m_i^2$. Its value one forces exactly one simple summand of multiplicity one. This proves irreducibility. $\square$

The corresponding exact trace identity is

$$\sum_{t \in \mathbb{F}_Q} \left| \sum_{x \in \mathbb{F}_Q} \psi \text{Tr}_{\mathbb{F}_Q/\mathbb{F}_p} \frac{\Phi(x) - tx}{u} \right|^2 = Q^2. \quad (\text{MII7})$$

Indeed the t -sum of the expanded square is zero for $x \neq y$ and is Q for each of the Q diagonal pairs. The complex identity (MII6) also determines the individual cohomology groups.

A morphism from $\mathcal{V}$ to any lisse target has a monodromy-invariant kernel. Thus its rank is either zero or q . The same statement holds after restricting to a nonempty open subcurve: its fundamental group surjects onto that of the original smooth curve, because every connected finite étale cover remains connected on restriction to the dense open. This is a precise restriction on a proposed lisse extension of a fibre map. The exact characteristic-zero coefficient and metric receivers of actual fibre maps are constructed below; no comparison of a complex fibre with an unspecified finite-field fibre is required by them.

45 The complete rank-four tensor-square calculation

Retain the one-packet centred family and primitive constant

$$\begin{aligned} \phi_{\beta,\eta}(x) &= x^5/5 + \beta x^3/3 + \eta x, & A(\beta, \eta) &= 1/160 + \beta/24 + \eta/2, \\ \beta &= 2(\gamma^2 - \delta^2), & \eta &= (\delta^2 + \gamma^2)^2 \end{aligned} \quad \text{at the original complex quartet.} \quad (\text{MII8})$$

In characteristic $p > 5$ with $u \neq 0$ let

$$\mathcal{U} = R^1\pi_!\mathcal{L}_\psi(\phi_{\beta,\eta}(x)/u), \quad \mathcal{A} = \mathcal{L}_\psi(A(\beta, \eta)/u), \quad \mathcal{V}^{\text{orig}} = \mathcal{A} \otimes \mathcal{U}.$$

These rank-four sheaves on $\mathbb{A}_{\beta,\eta}^2$ are lisse and pure of weight one. The constant phase is retained in $\mathcal{V}^{\text{orig}}$.

Theorem 45.1 (The three tensor constituents). *One has*

$$R\Gamma_c(\mathbb{A}^2, \text{End } \mathcal{U}) = \overline{\mathbb{Q}}_\ell(-2)[-4] \quad (\text{MII9})$$

and

$$H_c^i(\mathbb{A}^2, \text{End}(\mathcal{U}^{\otimes 2})) = \begin{cases} \overline{\mathbb{Q}}_\ell & i = 2, \\ \overline{\mathbb{Q}}_\ell(-1)^{\oplus 3} & i = 3, \\ \overline{\mathbb{Q}}_\ell(-2)^{\oplus 3} & i = 4, \\ 0 & \text{otherwise.} \end{cases} \quad (\text{MII10})$$

The following direct summands are geometrically simple, pairwise nonisomorphic, and have ranks 10, 1, 5, respectively:

$$(\mathcal{V}^{\text{orig}})^{\otimes 2} = \text{Sym}^2 \mathcal{V}^{\text{orig}} \oplus \mathcal{A}^2(-1) \oplus \left(\bigwedge^2 \mathcal{V}^{\text{orig}} \right)_0. \quad (\text{MII11})$$

The last summand is the kernel of the alternating contraction to $\mathcal{A}^2(-1)$.

Proof. For (MII9), use ordered x, y . Integration in η of the full opposite-phase product imposes $x = y$ by (MII5). The cubic and quintic differences then vanish. The remaining free coordinates are x, β . The twist (-1) and shift $[-2]$ from this Fourier integration equal those from degree-one clean duality in x, y . Cancelling them leaves exactly $R\Gamma_c(\mathbb{A}^2, \overline{\mathbb{Q}}_\ell)$.

For the tensor square use ordered x_1, x_2, y_1, y_2 . Integrating those four fibre variables first gives

$$R\Gamma_c(\mathbb{A}^2, \text{End}(\mathcal{U}^{\otimes 2}))(-2)[-4].$$

Instead integrate η, β first. Their Fourier constraints are $x_1 + x_2 = y_1 + y_2$ and $x_1^3 + x_2^3 = y_1^3 + y_2^3$, with the identical twist and shift $(-2)[-4]$. Write $x_1 = x, x_2 = s - x, y_1 = y, y_2 = s - y$. The exact polynomial identities are

$$\begin{aligned} x^3 + (s - x)^3 - y^3 - (s - y)^3 &= 3s(x - y)(x + y - s), \\ x^5 + (s - x)^5 - y^5 - (s - y)^5 &= 5s(x - y)(x + y - s)(x^2 - sx + y^2 - sy + s^2). \end{aligned} \quad (\text{MII12})$$

Their signs and all the opposite-pair solutions are retained. The second polynomial vanishes on the exact reduced zero scheme of the first. The linear map $(s, x, y) \mapsto (s, x - y, s - x - y)$ has determinant -2 , so this scheme is $Z = \{z_1 z_2 z_3 = 0\} \subset \mathbb{A}^3$. Consequently

$$R\Gamma_c(\mathbb{A}^2, \text{End}(\mathcal{U}^{\otimes 2})) = R\Gamma_c(Z, \overline{\mathbb{Q}}_\ell).$$

The complement is $(\mathbb{G}_m)^3$. Its compact cohomology in degrees 3, 4, 5, 6 is respectively $\overline{\mathbb{Q}}_\ell, \overline{\mathbb{Q}}_\ell(-1)^3, \overline{\mathbb{Q}}_\ell(-2)^3, \overline{\mathbb{Q}}_\ell(-3)$, by three Künneth products of $H_c^1(\mathbb{G}_m) = \overline{\mathbb{Q}}_\ell$ and $H_c^2(\mathbb{G}_m) = \overline{\mathbb{Q}}_\ell(-1)$. The open-closed exact sequence in $\mathbb{A}^3$ shifts the first three groups down by one to the groups of Z . The top map $H_c^6((\mathbb{G}_m)^3) \rightarrow H_c^6(\mathbb{A}^3)$ is an isomorphism of fundamental classes. This proves (MII10), including its Frobenius twists and its zero groups.

Duality on the smooth two-dimensional base gives dimensions one and three for $H^0(\text{End}\mathcal{U})$ and $H^0(\text{End}(\mathcal{U}^{\otimes 2}))$. To identify three actual summands, the odd involution $\iota(x) = -x$ identifies the negative phase with the positive phase. Clean degree-one duality gives

$$B(a, b) = \int a \smile \iota^* b, \quad B : \mathcal{U} \otimes \mathcal{U} \longrightarrow \overline{\mathbb{Q}}_\ell(-1).$$

It is nondegenerate. Pullback by ι preserves the top trace class and $\iota^2 = 1$; exchanging the two degree-one factors changes the sign. Therefore $B(b, a) = -B(a, b)$. In a symplectic basis e_1, f_1, e_2, f_2 , the invariant bivector $e_1 \wedge f_1 + e_2 \wedge f_2$ contracts to 2, so it splits a Tate line from $\bigwedge^2 \mathcal{U}$. Its contraction kernel has rank five and $\text{Sym}^2 \mathcal{U}$ has rank ten. Twisting both factors by the original $\mathcal{A}$ gives (MII11). Deligne's semisimplicity applies to the pure lisse tensor square. Its three nonzero direct summands contribute at least three to its endomorphism dimension, already proved to be exactly three. Hence each summand is simple and no two are isomorphic. Their distinct ranks give the same last conclusion directly. $\square$

For every extension field the full trace sum $S(\beta, \eta)$ satisfies

$$\sum_{\beta, \eta \in \mathbb{F}_Q} |S(\beta, \eta)|^4 = Q^2(3Q^2 - 3Q + 1). \quad (\text{MII13})$$

The constant character $\mathcal{A}$ cancels in this fourth absolute power. The two parameter sums give Q^2 times the number of points of Z . At $s = 0$ there are Q^2 pairs (x, y) ; at each $s \neq 0$ there are $2Q - 1$. The exact number is $Q^2 + (Q - 1)(2Q - 1) = 3Q^2 - 3Q + 1$, proving the formula.

For completeness the two finite fibre maps mentioned in the received continuation have elementary exact ranks. For $D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$ with $\zeta^5 = 1$ primitive, the actual cyclic average on the symmetric square is

$$P_5 = \frac{1}{5} \sum_{j=0}^4 \text{Sym}^2(D^j), \quad \text{im } P_5 = \langle x_1 x_4, x_2 x_3 \rangle. \quad (\text{MII14})$$

The characters on $x_r x_s$ are ζ^{r+s} , and precisely the two listed pairs have exponent zero modulo five. Thus the rank is two. For the original four distinct one-packet roots α_i , let e_i be their cardinal idempotents and let $Ae_i = \alpha_i e_i$. The nine distinct sums $\alpha_i + \alpha_j$ form the 3 by 3 grid with increments $2\delta, 2i\gamma$. Their sum algebra maps to the symmetric square by the exact evaluation map

$$\begin{aligned} \iota_2 : \mathbb{C}[T] / \prod_{\nu} (T - \nu) &\longrightarrow \text{Sym}^2 \mathbb{C}^4, \\ f &\longmapsto f(A \otimes 1 + 1 \otimes A)(1 \otimes 1), \\ \iota_2(e_\nu) &= \sum_{2\alpha_i = \nu} e_i \otimes e_i + 2 \sum_{i < j, \alpha_i + \alpha_j = \nu} e_i \odot e_j, \end{aligned} \quad (\text{MII15})$$

where $e_i \odot e_j = (e_i \otimes e_j + e_j \otimes e_i)/2$. The nine displayed images are nonzero and have disjoint supports in the symmetric tensor basis. This proves injectivity and rank nine, including the

multiplicity two of an off-diagonal symmetric pair. Neither this image nor (MII14) can be the image of a proper nonzero lisse projector on the simple rank-ten summand of (MII11). These exact finite maps remain available as fibre maps; the proved assertion concerns their extension as projectors of this particular finite-field family.

46 The marked coefficient module and the complete derivative frame

Return to the complex polynomial (MII1), put $R = \mathbb{C}[u, u^{-1}, t]$, and write

$$\mathcal{H} = R[S]/(u\partial_S + \chi - t)(R[S]).$$

This is a quotient of R -modules. Polynomial division in the differential expression gives a unique representative of degree less than q : subtract the leading coefficient times $(u\partial_S + \chi - t)S^{d-q}$ from a polynomial of degree $d \geq q$, and repeat. Termination follows from the strictly decreasing degree. Uniqueness follows because every nonzero polynomial f has $\deg((u\partial_S + \chi - t)f) = q + \deg f \geq q$. Thus $1, S, \dots, S^{q-1}$ is a free R -basis.

The cochain commutator and its induced connection are

$$[\partial_t - S/u, u\partial_S + \chi - t] = 0, \quad \nabla_t = \partial_t - A(t)/u. \quad (\text{MII16})$$

Here $A(t)$ is the companion matrix of $\chi - t$ on these representatives: multiplying a polynomial of degree at most $q - 1$ by S requires only the relation $(u\partial_S + \chi - t)1 = \chi - t$. This proves the displayed matrix connection. It does not define a ring multiplication on the differential quotient. At $t = 0$ the representative map identifies its underlying vector space with $E = \mathbb{C}[S]/\chi$ exactly.

Let R_j denote the representative of $[S^j p_\lambda]$. The polynomial relation $(S - \lambda)p_\lambda = \chi$ and reduction of $(\chi - t)S^{j-1}$ give

$$R_j - \lambda R_{j-1} = tS^{j-1} - u(j-1)S^{j-2}, \quad 1 \leq j \leq q. \quad (\text{MII17})$$

The last term is zero at $j = 1$. Its right side already has degree less than q , so this is equality of the literal representatives. Applying the cochain connection to the unreduced fixed polynomial before reduction gives, for every $j \geq 0$,

$$\nabla_t^j[v_\lambda] = b_\lambda(-u)^{-j}R_j. \quad (\text{MII18})$$

The commutator in (MII16) proves that differentiating the reduced representative gives exactly the same class.

Theorem 46.1 (Determinants and the entire supported cokernel). *In the increasing original S -coefficient basis, let*

$$\mathcal{O}_\lambda(t) = [v_\lambda, \nabla_t v_\lambda, \dots, \nabla_t^{q-1} v_\lambda], \quad \epsilon_q = (-1)^{(q-1)(q+2)/2}.$$

Then

$$\det \mathcal{O}_\lambda(t) = \epsilon_q b_\lambda^q u^{-q(q-1)/2} t^{q-1}, \quad \text{coker } \mathcal{O}_\lambda \simeq R/(t^{q-1}). \quad (\text{MII19})$$

The fibre matrix has rank $q-1$ at zero. The regularized original frame $\mathcal{J}_\lambda = [v_\lambda, \nabla_t^2 v_\lambda, \dots, \nabla_t^q v_\lambda]_{t=0}$ is invertible with

$$\det \mathcal{J}_\lambda = \epsilon_q (q-1)! b_\lambda^q u^{-q(q-1)/2}. \quad (\text{MII20})$$

Proof. First remove the factors $b_\lambda(-u)^{-j}$ from the columns of $\mathcal{O}_\lambda$. Perform the unipotent column operation $R_j \mapsto R_j - \lambda R_{j-1}$ simultaneously, or from right to left. Equation (MII17) gives the columns p_λ and $tS^{j-1} - u(j-1)S^{j-2}$, $1 \leq j \leq q-1$. The first is monic of degree $q-1$; the other columns have degrees at most $q-2$. Moving the first column to the last position gives the

sign $(-1)^{q-1}$. The remaining upper-bidiagonal block has diagonal entries t . Restoring the column factors gives sign $(-1)^{q-1+q(q-1)/2} = \epsilon_q$ and proves the determinant.

The first column eliminates the top coefficient generator. On the remaining generators $e_0, \dots, e_{q-2}$ the presentation is exactly

$$te_0 = 0, \quad te_j = uje_{j-1} \quad (1 \leq j \leq q-2).$$

The explicit presentation isomorphism is

$$e_j \mapsto \frac{j!}{(q-2)!} u^{-(q-2-j)} t^{q-2-j} \pmod{t^{q-1}}, \quad \text{inverse: } 1 \mapsto e_{q-2}. \quad (\text{MII21})$$

Every relation maps to zero. Conversely the displayed relations, using the invertible constants uj , express every generator in terms of e_{q-2} by precisely this formula; its sole remaining relation is $t^{q-1}e_{q-2} = 0$. This proves the isomorphism and shows that the special fibre of the cokernel has dimension one.

For the regularized frame remove the column scalars with orders $0, 2, \dots, q$. At zero $R_1 = \lambda p_\lambda$; replace R_2 by $R_2 - \lambda^2 p_\lambda$, and, from right to left, each R_j for $j \geq 3$ by $R_j - \lambda R_{j-1}$. The columns become $p_\lambda, -u, -2uS, \dots, -(q-1)uS^{q-2}$. Their determinant is $u^{q-1}(q-1)!$. The restored exponent of $-u$ is $-[q(q+1)/2 - 1]$, giving exactly (MII20). $\square$

The proof of this finite algebraic theorem also works for a terminal primary class when $\chi = (S - \lambda)^e \chi_\lambda$ and $\chi_\lambda(\lambda) \neq 0$: use $p_\lambda = \chi/(S - \lambda)$ and $b_\lambda = 1/\chi_\lambda(\lambda)$. This statement follows from the same polynomial recurrence, which used no simple-root division. It does not change the simple-quartet metric in this note.

47 The explicit inverse and the retained-line leakage

Put $v_j = (\nabla_t^j v_\lambda)|_{t=0}$. Formula (MII17) gives the following inverse reconstruction in the original coefficient space:

$$\begin{aligned} 1 &= -\frac{u}{b_\lambda} v_2 + \frac{\lambda^2}{b_\lambda u} v_0, \\ S^r &= \frac{(-u)^{r+1} v_{r+2} - \lambda(-u)^r v_{r+1}}{b_\lambda(r+1)} \quad (1 \leq r \leq q-2), \\ S^{q-1} &= b_\lambda^{-1} v_0 - \sum_{r=0}^{q-2} p_r S^r, \quad p_\lambda = S^{q-1} + \sum_{r=0}^{q-2} p_r S^r. \end{aligned} \quad (\text{MII22})$$

To verify the first identity, set $j = 2, t = 0$ in the recurrence to obtain $R_2 = \lambda^2 p_\lambda - u$. For the middle identities use $R_{r+2} - \lambda R_{r+1} = -u(r+1)S^r$ and then $R_j = b_\lambda^{-1}(-u)^j v_j$. The last identity is the monic definition of p_λ . These identities express each original basis vector in the q specified jet columns and are an explicit inverse matrix for $\mathcal{J}_\lambda$; no basis existence argument replaces the map.

The exact bundle map measuring the retained polynomial line is also explicit. Let $L = Rv_\lambda \subset \mathcal{H}$, and let $\pi_L : \mathcal{H} \rightarrow \mathcal{H}/L$ be its quotient map. Since v_λ is monic up to the unit b_λ , L is a direct summand as an R -module. The second fundamental form of this specified line is the R -linear map

$$\begin{aligned} \mathbf{b}_L : L &\longrightarrow (\mathcal{H}/L) dt, \\ f v_\lambda &\longmapsto \pi_L(\nabla_t(f v_\lambda)) dt = -\frac{b_\lambda t}{u} f \pi_L(1) dt. \end{aligned} \quad (\text{MII23})$$

The Leibniz term $f'v_\lambda$ maps to zero. The formula itself is (MII17) at $j = 1$. The class $\pi_L(1)$ is nonzero because $q \geq 2$ and a degree-zero polynomial cannot be an R -multiple of the nonconstant p_λ . In particular, the second fundamental form vanishes to exactly first order at the original mark, and

$$\pi_L(v_1) = 0, \quad \pi_L(v_2) = -\frac{b_\lambda}{u} \pi_L(1) \neq 0. \quad (\text{MII24})$$

This computes the first actual marked leakage from the original arithmetic eigenline $A(0)v_\lambda = \lambda v_\lambda$. Any connection-stable subbundle containing the specified polynomial section contains every v_j and hence, by (MII22), the entire fibre at zero.

There is also an exact formal bridge to a horizontal extension of its initial vector. Write $A(t) = A_0 + tB$, $B = e_0 e_{q-1}^\top$ in the original increasing coefficient frame. The formal matrix $H(t) = \sum_{n \geq 0} H_n t^n$ is uniquely defined by

$$H_0 = I, \quad H_{-1} = 0, \quad (n+1)H_{n+1} = u^{-1}(A_0 H_n + B H_{n-1}). \quad (\text{MII25})$$

It has invertible constant coefficient and satisfies $\nabla_t H = 0$. Thus $a(t) \mapsto H(t)a(t)$ is a connection-preserving isomorphism from the trivial formal connection to the completed marked connection. Its line through v_λ is the horizontal extension of that fibre vector. The polynomial line's actual leakage remains (MII23); these two sections are related by the explicit matrix (MII25), without identifying them.

48 Period determinant, Gamma constants, and scalar equation

The full root polynomial satisfies $\chi(c+w) = \chi(c-w)$. Its degree q is even. Write

$$\Phi(c+w) = C_\Phi + \phi(w), \quad C_\Phi = \Phi(c), \quad \phi(0) = 0, \quad \phi \text{ odd}.$$

Choose the original branch of $u^{1/(q+1)}$ and retain the ordered decay rays with angles $(\arg u + \pi + 2\pi j)/(q+1)$, $0 \leq j \leq q$. The cycles are the original differences $\Gamma_j = -\ell_0 + \ell_j$, $1 \leq j \leq q$. Moving their common finite starting point to $w = 0$ changes no integral: the same finite connecting path cancels between the two rays, and the joining arcs at infinity decay in the same sectors. Define the period matrix by

$$\Pi_{ja}(u, t) = \int_{\Gamma_j} w^{a-1} \exp((C_\Phi - ct + \phi(w) - tw)/u) dw, \quad 1 \leq j, a \leq q.$$

Let

$$\zeta = e^{2\pi i/(q+1)}, \quad F_{ja} = \zeta^{ja} - 1, \quad g_a = (q+1)^{a/(q+1)-1} e^{\pi i a/(q+1)} \Gamma\left(\frac{a}{q+1}\right). \quad (\text{MII26})$$

Theorem 48.1 (Complete original period determinant). *With the just specified branches, orientations and column order,*

$$\det \Pi(u, t) = e^{q(C_\Phi - ct)/u} (\det F) \left(\prod_{a=1}^q g_a \right) u^{q/2}. \quad (\text{MII27})$$

Proof. Separate the constant $C_\Phi - ct$ in the exponent. Vary a lower odd coefficient of the centred phase, with inserted odd power w^p , $p \leq q-1$. For the column w^j , $0 \leq j \leq q-1$, perform ordinary division $w^{p+j} = (\chi(c+w) - t)Q(w) + r(w)$. If the quotient is nonzero its degree is at most $p+j-q \leq j-1$. In the differential quotient, the inserted column is represented by $r - uQ'$. This last term has degree at most $j-2$ and needs no further reduction. The ordinary remainder map is multiplication by w^p in the quotient by the even polynomial $\chi(c+w) - t$; it reverses parity and so has zero trace. The correction $-uQ'$ is strictly lower triangular and also has zero trace. Therefore the coefficient derivative of the period matrix has the form $u^{-1}\Pi M_p$ with $\text{tr } M_p = 0$. Multilinearity, or the adjugate formula, gives zero derivative of its determinant without presuming its invertibility. This proves independence of every lower odd coefficient, including the marked coefficient $-t$ in the centred phase.

All these differentiations are dominated along the fixed rays on compact coefficient sets by $\exp[-R^{q+1}/((q+1)|u|) + O(R^{q-1})]$ times a polynomial. At the monomial phase, use $w = u^{1/(q+1)} e^{\pi i/(q+1)} \zeta^j r$ with $r > 0$ on the j th ray, followed by $s = r^{q+1}/(q+1)$. The literal integral is $\zeta^{ja} u^{a/(q+1)} g_a$. Taking the specified differences gives $F \text{diag}(u^{a/(q+1)} g_a)$. Finally $\sum_{a=1}^q a/(q+1) = q/2$, and restoring the separated constant proves (MII27). $\square$

Let $\mathbf{y}_\lambda(t)$ be the vector of integrals of the original amplitude $v_\lambda(S)$. Translation of coefficient vectors from S to w is the exact matrix

$$T_{rj} = \binom{j}{r} c^{j-r} \quad (0 \leq r \leq j \leq q-1), \quad T_{rj} = 0 \quad (r > j), \quad \det T = 1.$$

Differentiating integrals gives the cochain connection in (MII16). Hence the derivative matrix is literally $\Pi T \mathcal{O}_\lambda$, and

$$\det[\mathbf{y}_\lambda, \mathbf{y}'_\lambda, \dots, \mathbf{y}_\lambda^{(q-1)}] = \epsilon_q b_\lambda^q u^{-q(q-1)/2} t^{q-1} \det \Pi(u, t). \quad (\text{MII28})$$

The regularized determinant at the original mark is

$$\left| \det[\mathbf{y}_\lambda, \mathbf{y}''_\lambda, \dots, \mathbf{y}_\lambda^{(q)}]_{t=0} \right|^2 = (2\pi)^q ((q-1)!)^2 |b_\lambda|^{2q} |u|^{-q(q-2)} e^{2q\Re(C_\Phi/u)}. \quad (\text{MII29})$$

To check every constant, summing characters gives $F^*F = (q+1)(I + \mathbf{1}\mathbf{1}^*)$, whence $|\det F|^2 = (q+1)^{q+1}$. Gauss's multiplication formula [29, (5.5.6)–(5.5.7)] gives $\prod_{a=1}^q \Gamma(a/(q+1)) = (2\pi)^{q/2}/\sqrt{q+1}$. Consequently $|\det F \prod g_a|^2 = (2\pi)^q$. Combining this with (MII20) and (MII27) proves (MII29).

For any one original cycle let Y be its period with amplitude v_λ , let F_0 be its period with amplitude one, and put $D = -u\partial_t$. Integration by parts in S gives $(D - \lambda)Y = b_\lambda t F_0$ and $Y = b_\lambda p_\lambda(D)F_0$. For $t \neq 0$, $tDt^{-1} = D + u/t$, so

$$[p_\lambda(D + u/t)(D - \lambda) - t]Y = 0. \quad (\text{MII30})$$

The operator factors have this order. The leading two coefficients are $D^q + [\chi_{q-1} + (q-1)u/t]D^{q-1}$, where $\chi_{q-1} = -qc$. Abel's determinant equation therefore gives $W'/W = (q-1)/t - qc/u$, as in (MII28). Every component period is holomorphic at zero. Its apparent singularity is represented by the explicit cyclic cokernel (MII21) and the regularized frame, without discarding its multiplicity $q-1$.

49 Exact receiving maps for the original arithmetic metrics

Fix $u \neq 0$ in the original branch. Let G_N be the unchanged positive Hermitian metric on the original q -dimensional space $E = \mathbb{C}[S]/\chi$, using its actual masses, cutoffs, root factors, and full arithmetic unit. Set $J = \mathcal{J}_\lambda$ with its explicit inverse (MII22). The diagram of vector spaces and metrics is the literal pullback

$$J : (\mathbb{C}^q, J^*G_N J) \longrightarrow (E, G_N), \quad \det(J^*G_N J) = ((q-1)!)^2 |b_\lambda|^{2q} |u|^{-q(q-1)} \det G_N. \quad (\text{MII31})$$

The map J sends standard vectors to the original specified jet columns; it introduces no source normalization. The determinant identity follows by multiplying determinants and using (MII20). For the actual positive extremal root the normalizing constant itself is completely specified:

$$|b_\lambda|^{-1} = \prod_{\substack{0 \leq i, j \leq k \\ (i, j) \neq (0, 0)}} 2\sqrt{\delta^2 i^2 + \gamma^2 j^2}. \quad (\text{MII32})$$

This is the product of all differences $\lambda - \beta_{ab} = 2(k-a)\delta + 2i(k-b)\gamma$.

The map also transports every specified arithmetic subquotient, not only its total determinant. Let $I_X : \mathbb{C}^r \hookrightarrow E$ be any actual injective column map and $Q_X : E \twoheadrightarrow \mathbb{C}^s$ any actual surjection. Define their jet-coordinate maps $\tilde{I}_X = J^{-1}I_X$ and $\tilde{Q}_X = Q_X J$. Then

$$\begin{aligned} \tilde{I}_X^*(J^*G_N J)\tilde{I}_X &= I_X^*G_N I_X, \\ \tilde{Q}_X(J^*G_N J)^{-1}\tilde{Q}_X^* &= Q_X G_N^{-1} Q_X^*, \\ \left[\tilde{Q}_X(J^*G_N J)^{-1}\tilde{Q}_X^* \right]^{-1} &= (Q_X G_N^{-1} Q_X^*)^{-1}. \end{aligned} \quad (\text{MII33})$$

The first equality cancels J, J^{-1} directly. For the second use $(J^*G_NJ)^{-1} = J^{-1}G_N^{-1}J^{-*}$; the third follows because surjectivity and positivity make both matrices invertible. Equivalently the change $x = Jz$ bijects the exact affine fibres $\{x : Q_Xx = y\}$ and $\{z : \tilde{Q}_Xz = y\}$ and preserves their quadratic values. Their attained minima agree. Restricting first and taking a further quotient applies these same equalities to $I_X^*G_NI_X$ and the unchanged next quotient map. Thus every finite composition of the programme's actual restrictions and attained quotients has this exact receiver. No rank-proportional allocation has been substituted for any of these matrices.

Retain the original four-endpoint signs by defining

$$\mathcal{R}_{\text{ar}}f = f(q-1) + f(q) - f(2q-1) - f(2q).$$

The multiplier in (MII31) is independent of N ; $1 + 1 - 1 - 1 = 0$. Therefore

$$\mathcal{R}_{\text{ar}} \log \det(J^*G_NJ) = \mathcal{R}_{\text{ar}} \log \det G_N = \mathcal{B}_k^{\text{ar}}. \quad (\text{MII34})$$

For a restriction or an attained quotient transported by (MII33), even its full matrix is identical in its original domain or codomain, so its signed return agrees term by term. This supplies the explicit morphisms relating the geometric marked frame to the arithmetic receivers. It does not evaluate the still-present matrices $I_X^*G_NI_X$ or $Q_XG_N^{-1}Q_X^*$. In particular the seven original allocations and the three-column current responses retain those exact matrices. Their calculation continues from these equalities with the original inverse and complete action retained.

50 Acceptance scope and provenance

The admitted conclusions are the full formulas MII6–MII13 for the specified good finite-field families, MII16–MII30 for the original complex marked family, and the exact original-metric receivers MII31–MII34. MII22–MII25 and MII33 explicitly extend the received calculation. They retain the original full root polynomial, mark, period branch, and all coefficient factors. The degree-two ranks in MII14–MII15 were proved here independently.

The received text also quotes a large- k five-orbit observation rank $q - (8k - 16)$. Its precise admission domain and its earlier proof are not an input to any theorem in this receiving note; the formula is not newly certified here. The general obstruction to an intermediate-rank lisse morphism is proved above without it. No assertion here identifies an arithmetic Frobenius, makes a cross-characteristic fibre isomorphism, or evaluates an absolute native angular allocation. In particular none asserts that the broader tau-base or $\mathbb{F}_1$ programme has failed.

The entire received PDF and transcript were read. The local full D032 source was consulted specifically at (3.4.1)(iii) and (3.7.2.3) with its following lissity argument. The primary Numdam record and the DLMP multiplication formula were checked on 19 September 2026. The source hashes, exact scope, and independent algebraic checks are recorded in the accompanying machine-readable acceptance file. Finite rational fixtures are used only to detect algebraic errors; they do not represent actual zeros of ξ .

Primitive-choice receiving identities

Primitive-choice quadratic control in the original target order. Use the unchanged PRT/CPQ coefficient maps at target degree $a = k + 4$, target size $q = (a + 1)^2$, lower-grid size $(a - 7)^2$, cutoff $N - v$, and orders $s = 1, a$. Retain the actual rank $r_\partial = \dim(T_0/T_1) \in \{0, 1, 2, 3\}$. With the literal constants of PEQ and PEI7–11 set

$$B_{a,s}^{\text{end}} = \min\{\log A_{a,s}^{\text{end}}, E_{a,s}(2q - 1)\} + \min\{\log A_{a,s}^{\text{end}}, E_{a,s}(2q)\}.$$

Then

$$\begin{aligned} 0 \leq U_A \leq 3B_{a,s}^{\text{end}}, \quad 0 \leq U_B \leq (3 - r_\partial)B_{a,s}^{\text{end}}, \quad 0 \leq U_C \leq r_\partial B_{a,s}^{\text{end}}, \\ \mathcal{L}_0 - \mathcal{L}_1 = Z_1 - Z_0 = \mathcal{P}_0^{\text{rest}} - \mathcal{P}_1^{\text{rest}} = U_C - U_A + U_B, \\ |\mathcal{L}_0 - \mathcal{L}_1| \leq 3B_{a,s}^{\text{end}} \leq 6\mathfrak{E}_{a,s}^{\text{best}} = O_{\text{actual}}(q). \end{aligned}$$

PEI16 gives all directed component intervals. The primitive-choice difference divided by $q \log a$ tends to zero. The exact allocation identities remain $4S_K = \mathcal{L}_0 + Z_0 = \mathcal{L}_1 + Z_1$ and $S_R = \mathcal{T} - S_K$. The prefix contribution remains $-V_A^{\text{prefix}}$. An absolute rank- t quotient obeys the bound with its explicit factor t ; when $t = O(a)$ the bound permits $O(aq)$. This update therefore supplies no absolute allocation coefficient. The independent degree- k proper source and each response factor in the projected-sign calculation retain their original forms and unresolved values.

Backward propagation into the complete action

51 Backward propagation of the September 19 calculations

This section updates the live status of the earlier SXR10–12, GTM20, DRU11–12 and primitive-choice receivers. Their earlier proofs and versions remain in the cumulative source. All claims here refer to the original stipulated quartet, its complete root polynomial, the physical coordinate and the attained minima specified in the receiving proofs. In particular, the dimension is not substituted for the tensor order.

51.1 The arithmetic scalar in the complete action

For the original degree $a \equiv 1 \pmod{4}$, fixed multiplicity $m_0 \geq 1$, and dimension $q_a = [1 + a(m_0 - 1)](a + 1)^2$, the new scalar theorem gives

$$\delta_{a,1}^{\text{ar}} = -(4m_0 - 2)\mathcal{C}_\Gamma q_a + \frac{\log 2}{\pi} I_h a^2 + E_{h,a}, \quad (\text{MRI1})$$

where $I_h = \int_0^{\pi/2} \mu_h(\theta)^2 d\theta$ uses the original one-factor tilted mean and $\mathcal{C}_\Gamma = 2\mathcal{L}(2, \pi) - 8\log 2 + 4$. The full definitions, convergent evaluation, original finite guards, and proofs of these constants occur in M1–M8, I1–I4, OR1–41 and SAI1–12 in this edition. The quantitative error is

$$|E_{h,a}| \leq C_h q_a \left\{ \frac{\log \log(q_a + 16)}{\log(q_a + 2)} + q_a^{-1/200} \log(q_a + 2)^2 \right\} = o_h(q_a). \quad (\text{MRI2})$$

Here C_h is the fixed-source constant from that proof, not a numerically evaluated uniform constant over all quartets.

The arithmetic receiver is the exact existing equality

$$J_a = \mathcal{B}_a^{(s)} + \delta_{a,s}^{\text{ar}} + \mathcal{W}_a^{\text{mono}} - \mathcal{P}_{a,-} - \mathcal{P}_{a,+}. \quad (\text{MRI3})$$

The source-order map, including both original masses, is

$$\delta_{a,a}^{\text{ar}} - \delta_{a,1}^{\text{ar}} = \mathcal{B}_a^{(1)} - \mathcal{B}_a^{(a)}. \quad (\text{MRI4})$$

Indeed both sides follow by subtracting the two expressions MRI3 for the same action and unchanged remaining terms. Substitution of MRI1 therefore evaluates its arithmetic contribution at order one, and transports that evaluation to the other order through the displayed actual baseline difference. In particular

$$\begin{aligned} J_a &= \mathcal{B}_a^{(1)} - (4m_0 - 2)\mathcal{C}_\Gamma q_a + \frac{\log 2}{\pi} I_h a^2 \\ &\quad + \mathcal{W}_a^{\text{mono}} - \mathcal{P}_{a,-} - \mathcal{P}_{a,+} + E_{h,a}. \end{aligned} \quad (\text{MRI5})$$

This proves the exact backward receiver, retaining both penalties and the full monic window.

In the simple-quartet domain of DRU12, $q_a = (a + 1)^2$. Define the already controlled analytic remainder by its actual value

$$\mathfrak{R}_{h,a} := J_a - \delta_{a,1}^{\text{ar}} - C_B q_a^2 - 8q_a \log a. \quad (\text{MRI6})$$

DRU12 proves $\mathfrak{R}_{h,a} = O_h(q_a)$. Equations MRI1 and MRI6 give the more informative exact decomposition

$$J_a = C_B q_a^2 + 8q_a \log a - 2\mathcal{C}_\Gamma q_a + \frac{\log 2}{\pi} I_h a^2 + \mathfrak{R}_{h,a} + E_{h,a}. \quad (\text{MRI7})$$

Thus the identified arithmetic order- q_a term has been propagated, while the pre-existing analytic $O_h(q_a)$ remainder retains its own value. In particular the proof does not assign a limiting coefficient

to $\Re_{h,a}/q_a$. The leading $C_B q_a^2$ and $8q_a \log a$ terms remain those of the proved complete action. Dividing MRI1 by its literal dimension proves

$$\lim_{a \rightarrow \infty} \frac{\delta_{a,1}^{\text{ar}}}{q_a} = \begin{cases} -2\mathcal{C}_\Gamma + (\log 2/\pi)I_h, & m_0 = 1, \\ -(4m_0 - 2)\mathcal{C}_\Gamma, & m_0 \geq 2. \end{cases} \quad (\text{MRI8})$$

The limits use $a^2/(a+1)^2 \rightarrow 1$ and $a^2/([1+a(m_0-1)](a+1)^2) \rightarrow 0$, respectively.

51.2 The original allocation maps after the scalar update

Let G_M be the unchanged source form, E the original lift matrix, $\mathbf{U}_L$ its complete relation matrix, and Z_k the literal kernel inclusion. The same attained native form is

$$H_{Y,N} = E^* G_M E - E^* G_M \mathbf{U}_L (\mathbf{U}_L^* G_M \mathbf{U}_L)^{-1} \mathbf{U}_L^* G_M E, \quad H_{K,N} = Z_k^* H_{Y,N} Z_k. \quad (\text{MRI9})$$

These are the exact receiving maps for the unresolved individual allocation values. To verify the quotient map, write every lift as $Ey + \mathbf{U}_L z$ and complete the square in its original G_M norm. The unique minimizing coefficient is $z = -(\mathbf{U}_L^* G_M \mathbf{U}_L)^{-1} \mathbf{U}_L^* G_M E y$; substitution gives MRI9. Restricting to $y = Z_k x$ gives its second formula without discarding any cross block. This calculation also proves why the relation determinant alone cannot determine the kernel restriction: the latter depends on the explicitly retained mixed Gram $E^* G_M \mathbf{U}_L$ and inclusion Z_k .

At every original endpoint, restriction followed by the attained residual quotient gives Schur determinant factorization. Applying the unchanged four-endpoint return to that factorization preserves

$$S_K + S_R = \mathcal{T}, \quad 4S_K = \mathcal{L}_e + V_b + V_{\text{rem}} + V_s + V_A. \quad (\text{MRI10})$$

The new scalar MRI1 does not alter these maps. The current task is to evaluate the seven displayed quantities on MRI9 and its actual complementary quotient maps, through every term larger than q_a . The PEI bounds below provide sharper correlated comparisons between the primitive choices on these same spaces. The rank of an absolute restriction or quotient remains in its determinant allowance.

51.3 The exact reflected-pair receiving map

Write $L : E \rightarrow W$ for the literal full cyclic observation and G_N for the original positive form. Choose coordinates on the actual image $W_0 = \text{im } L$, and denote the resulting surjective map by $L_0 : E \rightarrow W_0$. Its attained form and section are

$$Q_{W_0,N} = (L_0 G_N^{-1} L_0^*)^{-1}, \quad s_{W_0,N} = G_N^{-1} L_0^* Q_{W_0,N}. \quad (\text{MRI11})$$

Indeed $L_0 s_{W_0,N} = I$ and, for $z \in \ker L_0$, $z^* G_N s_{W_0,N} = 0$. Every other lift is this section plus such a z , so Pythagoras proves the stated minimum and its form. If C embeds the two reflected observed classes into W_0 , their attained quotient metric is exactly $C^* Q_{W_0,N} C$. ET19–26 evaluate their Fourier observation Gram and prove eventual injectivity at the displayed fixed original periods. MRI11 is the exact morphism that carries those classes into the original arithmetic quotient; its full inverse and all remaining image directions are retained. The projected arithmetic responses $U = K_{13}/F_{13}$ and $V = K_{23}/F_{23}$ still require evaluation in their original three-column receiver, including the common error multiplied by $\text{Im } P_X$.

Edition corrections and mathematical scope. The Airy exterior phase in the incoming text is corrected to $(V_n - 2g + \ell)/2$ by SAI1. The two-window perturbation is the explicit JW13 finite bound with its source-ratio norm; the earlier bare allowance is superseded by this proved formula. All original source texts are preserved, and each revised formula is linked to its proof. The newly computed scalar resolves the earlier signed baseline coefficient entry. The original proper-source standard window, individual attained allocations and retained-class signs remain active calculations with their displayed receiving maps.

Residue prices, constrained composition and nonperiodic schedules

52 Residue prices, constrained composition, and nonperiodic optimal schedules

The Clankers. 17 September 2026.

52.1 1. Source, definitions, and evidence boundary

This continuation starts from the integrated Erdős Problem 817 workbench at commit dbd0a93f2cf935103e88e5c2b2b71fe85ae7537b. The corrected 29-note edition, its $L > 3Q$ separation condition, its current cone-dual domain, and all earlier source identities remain upstream. The present addition changes none of them. Its cumulative delivery begins a new text-first series from that baseline rather than nesting the historical archives.

The original $k = 4$ mathematical construction remains credited to the anonymous/deleted Reddit contributor. The finite upper-bound Lean extension remains separately credited to **sneed-and-feed**. The present results have ordinary proofs and the specified exact computational certificates. They have not undergone new Lean elaboration or independent human mathematical review. No literature-priority claim is made.

For a finite set A of distinct positive integers, retain

$$H_q(A) = \left\{ \sum_{a \in A} c_a a : 0 \leq c_a < q \right\}, \quad S(A) = \sum_{a \in A} a.$$

Each numerical value is counted once. The original coefficient words and their representation multiplicities remain available separately. The coefficient arity q and forbidden progression length k are different parameters.

A **canonical modular- k block** is (b, A) with $b > S(A)$ and with $H_2(A) \bmod b$ containing no ordered k -term progression with nonzero modular step. Repetition later in such a modular tuple is allowed. This convention checks steps of small additive order as well as full-order steps. A sequence of these blocks gives

$$P_0 = 1, \quad P_{j+1} = b_j P_j, \quad A_w = \bigcup_{j < m} P_j A_j, \quad N(w) = \sum_{j < m} |A_j|.$$

The actual generators are distinct, because every earlier generator is below the next position. Lowest-digit reduction proves that every binary prefix image is k -AP-free. Thus all of the applications below retain actual positive integer constructions.

The first new result gives sharp ternary residue prices at ranks one through four, with no upper bound on radix or generator height. The second determines the entire composition-constrained spectrum of the original radix-ten dictionary, including an interval where the optimal infinite rate is attained but no periodic schedule attains it. A general finite-horizon linear-programming theorem gives a uniform error for all prescribed composition profiles.

52.2 2. Sharp residue prices with unbounded arithmetic parameters

Theorem 1. Let $k \in \{5, 6\}$ and let (b, A) be a canonical modular- k block. If $n = |A| \leq 4$, then

$$\boxed{|H_3(A) \bmod b| \geq L_n, \quad (L_1, L_2, L_3, L_4) = (3, 5, 13, 23).} \quad (2.1)$$

All four constants are attained. The proof uses Kneser's classical addition theorem and a complete finite list of canonical obstructions. Both ingredients are stated below.

52.2.1 2.1 The required binary counts

For distinct positive $a_1 < \dots < a_n$, adjoining a_j produces at least j new values above S_{j-1} : the values $a_j + S_{j-1}$ and $a_j + S_{j-1} - a_i$ for $i < j$. Hence

$$|H_2(A)| \geq 1 + \frac{n(n+1)}{2}. \quad (2.2)$$

At rank three, equality seven is possible only when the largest weight is the sum of the other two. Indeed, a binary collision is a signed relation on disjoint supports; positivity and distinctness leave only that possibility.

At rank four, equality eleven in (2.2) forces

$$A = a\{1, 2, 3, 4\}. \quad (2.3)$$

Here is the complete argument. Equality forces the first three weights to have exactly seven values, so write them as $a < b < c = a + b$. Their image is

$$D_3 = \{0, a, b, c, a + c, b + c, 2c\}.$$

For a new $d > c$, an intersection of D_3 with $d + D_3$ uses a lower representative in $\{0, a, b\}$. There are at most three intersection values. Equality eleven requires all three. Their ordered upper representatives are precisely $a + c, b + c, 2c$. Consequently $d = a + c$, $d + a = b + c$, and $d + b = 2c$, which give $b = 2a$, $c = 3a$, $d = 4a$.

The binary image in (2.3) contains $a[0, 10]$. It therefore contains both five- and six-term nonconstant progressions. For the blocks of Theorem 1, the rank-four binary count is consequently at least twelve. The binary counts needed below are therefore four, seven, and twelve at ranks two, three, and four.

52.2.2 2.2 Kneser's theorem and its complete case reduction

We use the following classical statement (DeVos, 2013): for finite nonempty subsets U, V of an abelian group, with $H = \text{Stab}(U + V)$,

$$|U + V| \geq |U + H| + |V + H| - |H|. \quad (2.4)$$

Apply it in $\mathbb{Z}/b\mathbb{Z}$ to $D = H_2(A)$. Put

$$v = |D + D| = |H_3(A) \bmod b|, \quad h = |H|, \quad z = |D + H|.$$

Then z is a multiple of h , $|D| \leq z \leq v$, and

$$v \geq 2z - h. \quad (2.5)$$

If $h = 1$, the binary lower counts immediately give the rank-three bound thirteen and the rank-four bound twenty-three.

If $D + H$ has only one coset, it equals H because $0 \in D$. Let $g = b/h$. Every actual generator is a multiple of g , and the maps

$$\iota : \mathbb{Z}/h\mathbb{Z} \xrightarrow{\sim} H, \quad [x] \mapsto [gx], \quad A' = \{a/g : a \in A\} \quad (2.6)$$

preserve the original modular progression equations. The inverse divides the displayed representatives by g . The positive scale g , original weights, and original modulus are retained. In particular $S(A') < h$, and A' is again a canonical modular- k block of the same rank. This is a specified subgroup isomorphism, not an assumption that every nontrivial stabilizer contains the original source.

For rank two, D has four elements. If $v \leq 4$, then $D + D = D$, making D a subgroup of order four. Its full cyclic orbit gives a forbidden ordered progression.

For rank three, assume $v \leq 12$. The complete possibilities with $h \geq 2$ in (2.5) are

$$(h, z) = (4, 8), \quad (h, h) \quad (7 \leq h \leq 12). \quad (2.7)$$

In the first case, seven source points in two cosets of size four force a full coset of order four. It gives an ordered six-term progression, and hence a five-term prefix. The remaining cases reduce by (2.6) to rank-three blocks with modulus at most twelve.

For rank four, assume $v \leq 22$. The complete possibilities are

$$(h, z) \in \{(2, 12), (3, 12), (4, 12), (6, 12), (7, 14)\}, \quad (h, h) \quad (12 \leq h \leq 22). \quad (2.8)$$

In the first four cases, $|D| \geq 12 = z$ forces a full nontrivial coset of order at most six. In the $(7, 14)$ case, one seven-element coset contains at least six of the source points. Removing at most one point from a cyclic seven-element orbit leaves a six-term consecutive progression in that orbit. Each case is forbidden for both $k = 5$ and $k = 6$. The one-coset cases again reduce by (2.6).

The lists (2.7) and (2.8) follow simply by enumerating the integer inequalities $z \geq 7$ or $z \geq 12$, $h \mid z$, and $2z - h \leq 12$ or 22 . The proof data list every pair, including all one-coset cases.

52.2.3 2.3 The finite obstruction lemma

The remaining finite statement is:

- every positive distinct triple with sum below its modulus $b \leq 12$ has a modular six-term progression in its binary image;
- every positive distinct quadruple with sum below $b \leq 22$ has such a progression.

There are exactly 41 triples and 448 quadruples in these complete domains. The certificate additionally includes the one canonical pair at modulus at most four, for 490 source blocks in total.

For every source block the text certificate records its actual weights, modulus, start, nonzero step, six residue values, and six original binary subset masks. The following terminating enumeration specifies both the domain and witness test:

1. For each stated modulus, enumerate all increasing positive tuples with sum below that modulus.
2. Form all binary subset sums, keeping their masks.
3. For every two distinct values x, y , set $d = y - x \pmod{b}$ and test all six values $x + id \pmod{b}$.
4. Retain the first successful tuple and its masks.

All 490 sources have a recorded witness. The separate auditor generates the increasing tuples recursively and checks every original mask. It also checks modular freeness in an additional complete domain by rotating residue bit sets, rather than using the producer's pair enumeration. This is finite proof data for the stated lemma; it is not a search extrapolation over larger moduli.

Equations (2.2)–(2.8), with this finite lemma, prove Theorem 1. At rank one the generator's residue has order at least three: order two would itself supply a forbidden alternating modular tuple. Thus its three ternary residues are distinct.

The sharp blocks are

$$(3, \{1\}), \quad (5, \{1, 2\}), \quad (13, \{1, 3, 4\}), \quad (23, \{1, 3, 4, 7\}). \quad (2.9)$$

Their binary modular freeness and their residue sections are checked directly. Their fifth-arity images are full integer intervals, whose repeated-image growth radii are their radices. This proves sharpness as both a residue count and a homogeneous image-growth bound.

52.3 3. An all-rank generator-profile inequality

For a level (b, A) , choose one actual ternary value $d_r \in H_3(A)$ for each occurring residue r . The map

$$\{d_r\} \times Y \longrightarrow H_3(A) + bY, \quad (d_r, y) \longmapsto d_r + by \quad (3.1)$$

is injective: reduction modulo b recovers the residue and hence the chosen actual value; subtraction and division then recover y . No requirement that all chosen ternary values be below b is used.

Iterating (3.1) through any changing canonical modular- k schedule proves

$$|H_3(A_w)| \geq \prod_{j < m} |H_3(A_j) \bmod b_j|. \quad (3.2)$$

For arbitrary rank, the usual one-shift argument gives

$$|H_2(A)| \geq \gamma_k^{|A|}, \quad \gamma_k = \frac{k-1}{k-2}. \quad (3.3)$$

For clarity, before adjoining a generator a , decompose the existing binary image into maximal a -chains. Each has length at most $k-2$: adjoining a extends its end, so a longer chain would yield a forbidden k -AP. The union with its translate has size at least γ_k times the preceding size. Iteration proves (3.3). The canonical binary values remain distinct modulo the radix, so (3.3) also lower-bounds the number of ternary residues. This elementary general-bound mechanism belongs to the existing workbench and surrounding literature.

Let m_j count levels having exactly j generators, and let $N_{\geq 5}$ count all generators belonging to larger levels. Combining Theorem 1 and (3.2)–(3.3) gives the new explicit rank profile:

$$\boxed{|H_5(A_w)| \geq |H_3(A_w)| \geq 3^{m_1} 5^{m_2} 13^{m_3} 23^{m_4} \gamma_k^{N_{\geq 5}}, \quad k = 5, 6.} \quad (3.4)$$

It applies at every finite horizon and at unbounded changing heights and radices. It makes no assertion that the original generator list has a unique block presentation.

For the family in which every level is individually modular- k -free and has rank at most r , the exact optimum fifth-image rate per generator is

$$\boxed{3, \quad \sqrt{5}, \quad \sqrt{5}, \quad 23^{1/4} \quad (r = 1, 2, 3, 4).} \quad (3.5)$$

The lower bound is (3.4); repeated sharp blocks (2.9) attain it. With exactly r generators on every level, the optimum is instead $L_r^{1/r}$.

This domain is an explicitly specified subclass of general carry-safe schedules. For example the earlier singleton schedule with radices 2, 4 has a first level whose binary image is the full group of order two and is individually modular-unsafe. Its two-level aggregation is the actual rank-two block $\{1, 2\}$ in radix eight. The aggregation keeps both generators and product radix eight. Thus the broader singleton optimum $\sqrt{8}$ and (3.5)'s singleton optimum three concern different domains connected by a rank-changing aggregation map.

Set $c = \log(23)/4$, $N = N(w)$, and $\theta = N_{\geq 5}/N$. All four low-rank prices per generator are at least c . Therefore

$$\boxed{\frac{\log |H_5(A_w)|}{N} \geq (1 - \theta)c + \theta \log \gamma_k.} \quad (3.6)$$

Along any sequence of such words whose logarithmic image cost is at most $\log u + o(1)$, with $u < 23^{1/4}$,

$$\liminf \theta \geq \frac{c - \log u}{c - \log \gamma_k}. \quad (3.7)$$

For the existing comparison rates $u = 97^{1/6}$ at $k = 5$ and $u = 1651^{1/10}$ at $k = 6$, the right sides lie respectively in

$$\left(\frac{431}{10000}, \frac{432}{10000} \right), \quad \left(\frac{766}{10000}, \frac{767}{10000} \right). \quad (3.8)$$

The certificate checks these intervals by integer powers after clearing rational logarithmic coefficients. Consequently at least about 4.31% or 7.66% of the generators must lie in blocks of rank at least five in any individually modular-safe presentation approaching those respective costs. This is a necessary profile constraint, not a new numerical bound on the unrestricted λ_k .

52.4 4. Composition-constrained capacity for any finite dictionary

Let $\mathcal{D} = \{(b_i, A_i) : 1 \leq i \leq d\}$ be any finite canonical dictionary with nonempty positive distinct generator sets. Put $n_i = |A_i|$. Modular admissibility is needed for arithmetic applications, but the count statements in this section hold for every such dictionary.

For a word w retain

$$F_q(w) = |H_q(A_w)|, \quad \mathbf{n}(w) = (\text{number of each letter in } w).$$

The numerical cut map $(x, y) \mapsto x + P_u y$ is onto. Its first value lies in $[0, (q-1)(P_u-1)]$, so each fibre has at most $C = q-1$ elements. Hence

$$\boxed{C^{-1}F_q(u)F_q(v) \leq F_q(uv) \leq F_q(u)F_q(v).} \quad (4.1)$$

This is the integrated finite-period theorem's original counting mechanism, retained with its exact cut constant.

For a profile $p = (p_i)$ in the probability simplex, define $\mathcal{C}_q(p)$ as the infimum of

$$\liminf_{m \rightarrow \infty} \frac{\log F_q(w_0 \cdots w_{m-1})}{m}$$

over infinite words whose empirical letter frequencies converge to p . The mean generator reward is

$$\nu(p) = \sum_i n_i p_i > 0.$$

Thus the corresponding per-generator rate is $\exp(\mathcal{C}_q(p)/\nu(p))$. Frequency in this section counts original levels; it is not silently identified with generator frequency when the n_i differ.

For each $m \geq 1$ solve the finite linear program

$$E_m(p) = \min \left\{ \frac{1}{m} \sum_{w \in \mathcal{D}^m} \lambda_w \log F_q(w) : \lambda_w \geq 0, \sum_w \lambda_w = 1, \sum_w \lambda_w \mathbf{n}(w) = mp \right\}. \quad (4.2)$$

The program is feasible because the pure words supply the simplex vertices. An optimum uses at most d words: the d count constraints include the probability constraint, since each word has length m .

Theorem 2. For every profile,

$$\boxed{E_m(p) - \frac{\log(q-1)}{m} \leq \mathcal{C}_q(p) \leq E_m(p).} \quad (4.3)$$

The convergence is uniform on the complete simplex. In particular $\mathcal{C}_q$ is continuous and convex. Every profile has an infinite schedule attaining its value with a full logarithmic cost limit. Exact periodic attainment is a separate property.

Proof. Partition a long word into length- m chunks and a final shorter piece. Applying the lower inequality in (4.1) gives the sum of chunk log counts minus at most one $\log C$ per chunk. Along a subsequence realizing the lower limit, extract a convergent empirical distribution of the finitely many chunks. Its mean letter profile is p . Its mean chunk log count is at least $mE_m(p)$, proving the lower bound. The final piece has bounded length and reward, so it does not change the limit.

For the upper bound, take an optimal mixture in (4.2). Concatenate its finitely many words in frequencies tending to that mixture. The upper inequality in (4.1) bounds the log count by the sum of their actual log counts. The resulting profile is p , proving the upper bound.

The finite LP value is a continuous convex piecewise-affine function of p . The uniform bound in (4.3) therefore gives continuity and convexity of the limit.

For actual attainment, choose increasing horizons and rational approximations to the corresponding optimal word mixtures, with profile and mean-cost errors tending to zero. Each rational approximation gives a finite period V_j . Concatenate many copies of V_j before moving to V_{j+1} . Choose the repetitions at stage j so the accumulated length is at least $j|V_{j+1}|$. All partial-period frequency errors then tend to zero. The upper multiplicative bound gives a limiting upper cost at most $C_q(p)$; the already-proved lower bound gives the reverse lower limit. The full cost limit exists and equals $C_q(p)$. Every period and transition word is an actual original dictionary word. $\square$

Dividing (4.3) by $\nu(p)$ gives a uniform per-generator logarithmic error at most $\log(q - 1)/(m \min_i n_i)$.

The finite dual is also explicit. A vector of logarithmic prices y_i satisfying

$$\sum_i y_i n_i(w) \leq \log F_q(w) \quad \text{for every length-}m \text{ word} \quad (4.4)$$

gives the all-infinite lower certificate

$$C_q(p) \geq y \cdot p - \frac{\log C}{m}. \quad (4.5)$$

A primal and dual pair with equal values certifies the finite optimum. At rational profiles, all basis coefficients and all coefficients expressing y_i as linear combinations of integer logarithms are rational. Every inequality is checked by integer powers. These prices are auxiliary count budgets; they change no original radix, generator, or representation mass.

52.5 5. Exact minima at every composition in the radix-ten source

Retain the original dictionary

$$A = (10, \{1, 3\}), \quad B = (10, \{2, 4\}). \quad (5.1)$$

Its binary residues are $\{0, 1, 3, 4\}$ and $\{0, 2, 4, 6\}$. Neither contains an order-two pair differing by five; a step of order five or ten would require at least five distinct values. Thus both are modular-five-free. Every schedule has actual five-admissible binary prefixes and two generators per level.

On the seven integrated fifth-image coordinates

$$\{0\}, \{0, 1\}, \{0, 2\}, \{0, 1, 2\}, \{0, 3\}, \{0, 1, 3\}, \{0, 1, 2, 3\},$$

the unchanged arithmetic actions are

$$M_A f = (3f_0 + 7f_1, 2f_0 + 8f_1, f_0 + 9f_1, f_0 + 9f_1, \\ 10f_1, 10f_1, 10f_1),$$

$$M_B f = (2f_1 + 3f_3, 4f_1 + 6f_3, f_1 + 4f_3, 3f_1 + 7f_3, 4f_1 + 6f_3, 3f_1 + 7f_3, 2f_1 + 8f_3). \quad (5.2)$$

The initial tail vector is $(1, 2, 2, 3, 2, 3, 4)^\top$. These matrices and the following rank-one identity are inherited from the integrated local-flow source and are rechecked independently here by interpolation on actual finite numerical tails:

$$M_A M_B = uv, \quad u = (17, 18, 19, 19, 20, 20, 20)^\top, \quad v = (0, 2, 0, 3, 0, 0, 0). \quad (5.3)$$

For $a, b \geq 1$, define

$$T(a, b) = v M_B^{b-1} M_A^{a-1} u = \frac{8 \cdot 10^{a+b} + 4 \cdot 10^b - 3}{9}. \quad (5.4)$$

Both vector sequences satisfy the recurrence with characteristic polynomial $(X - 1)(X - 10)$. The four original values at $a, b \in \{1, 2\}$ prove (5.4) at every pair of lengths.

For a mixed cyclic word with successive runs $A^{a_i} B^{b_i}$, its nonzero spectral value is

$$\rho(M_w) = \prod_i T(a_{i+1}, b_i), \quad a_{s+1} = a_1. \quad (5.5)$$

This follows by multiplying the original rank-one factors. Cyclic rotation preserves the nonzero eigenvalues, not the original finite image count. The latter retains the bound

$$\rho(M_w) \leq F_5(w) \leq 4\rho(M_w). \quad (5.6)$$

For completeness, the weighted norm $\|x\| = \max_R |x_R|/|R|$ gives $\|M_w\| = F_5(w)$: translated copies supply the upper bound and the singleton coordinate supplies equality. Repetition and (4.1) prove (5.6).

Let $F(s, t)$ be the minimum actual fifth-image count among words having exactly s letters A and t letters B .

Theorem 3. For $s, t \geq 0$,

$$F(0, 0) = 1, \quad F(s, 0) = \frac{16 \cdot 10^s - 7}{9}, \quad F(0, t) = \frac{4 \cdot 10^t - 1}{3}. \quad (5.7)$$

For $s \geq t > 0$, write $s = at + r$, $0 \leq r < t$. Then

$$\boxed{F(s, t) = T(a, 1)^{t-r} T(a+1, 1)^r.} \quad (5.8)$$

For $0 < s < t$,

$$\boxed{F(s, t) = 93^{s-1} T(1, t-s+1).} \quad (5.9)$$

52.5.1 5.1 The lower optimization

Put $G(a, b) = 10^{a+b}/T(a, b)$. Its exact bounds are

$$\frac{15}{14} < G(a, b) < \frac{9}{8}. \quad (5.10)$$

Aligning large a with large b does not decrease the product of the gains. To check this directly, put $x = 10^{-a} \leq x' = 10^{-a'}$ and $y = 4 - 3 \cdot 10^{-b} \geq y' = 4 - 3 \cdot 10^{-b'}$. The relevant numerator difference between the two gain products is

$$(8 + xy')(8 + x'y) - (8 + xy)(8 + x'y') = 8(x' - x)(y - y') \geq 0. \quad (5.11)$$

If the number of crossed run pairs is below $\min(s, t)$, some $a \geq 2$ and some $b \geq 2$ exist. Re-pair them using (5.11), and split that pair into two positive pairs. The gain product increases strictly, since

$$(15/14)^2 > 9/8.$$

Every such re-pairing is realized by a cyclic run word with the same original letter counts. Hence an optimal spectral product has exactly $\min(s, t)$ run pairs.

When $s \geq t$, all b -lengths are one. Put $K_a = T(a, 1) = (80 \cdot 10^a + 37)/9$. The identity

$$K_a K_{a+2} - K_{a+1}^2 = 2960 \cdot 10^a > 0 \quad (5.12)$$

proves strict logarithmic convexity. Transferring one unit from a run at least two longer to a shorter run decreases the cost. Thus the lengths are a and $a + 1$ with the multiplicities in (5.8).

When $s < t$, all a -lengths are one. Put $J_b = T(1, b) = (28 \cdot 10^b - 1)/3$. Here

$$J_b J_{b+2} - J_{b+1}^2 = -252 \cdot 10^b < 0. \quad (5.13)$$

Strict logarithmic concavity moves all excess B -length to one run and gives (5.9) as the minimum spectral product. Inequality (5.6) transfers both lower bounds to the original counts.

52.5.2 5.2 Actual finite attainers

The required endpoint equalities are supplied by original numerical images:

$$H_5(BA^a) = 2[0, K_a - 1], \quad H_5(B^b A) = 2[0, J_b - 1]. \quad (5.14)$$

For the first identity, the upper A -levels have interval image $[0, 16(10^a - 1)/9]$. The low B -image is $2[0, 12]$. A single division of the whole numerical set by two gives

$$[0, 12] + 5[0, 16(10^a - 1)/9] = [0, K_a - 1].$$

For the second, the lower B -levels, after that same whole-image division, give $[0, 12(10^b - 1)/9]$. Adding the upper A -level fills consecutive intervals and gives $[0, J_b - 1]$. The map $x \mapsto x/2$ has inverse $x \mapsto 2x$ on these specified images. No individual source coordinate is dropped.

The divided digit maxima in (5.14) are below the actual macro radices 10^{a+1} and 10^{b+1} . Thus concatenating the macros gives exact multiplication of their counts. The attaining words are

$$(BA^a)^{t-r} (BA^{a+1})^r \quad (s \geq t), \quad (5.15)$$

$$B^{t-s+1} A (BA)^{s-1} \quad (s < t). \quad (5.16)$$

Their original generator sets and radices are the original concatenations. Equations (5.14)–(5.16) establish equality in the spectral lower bounds and complete the proof of Theorem 3.

52.6 6. The entire frequency spectrum and an unattained periodic regime

Let p be the limiting fraction of B -levels. The mean generator reward is two. Let $\mathcal{C}(p)$ denote the optimal logarithmic fifth-image cost per level.

Theorem 4. At $p = 0, 1$, $\mathcal{C}(p) = \log 10$. For $0 < p \leq 1/2$, write

$$a = \left\lfloor \frac{1-p}{p} \right\rfloor, \quad u = \frac{1-p}{p} - a.$$

Then

$$\boxed{\mathcal{C}(p) = p((1-\nu) \log K_a + \nu \log K_{a+1})}. \quad (6.1)$$

For $1/2 \leq p \leq 1$,

$$\boxed{\mathcal{C}(p) = (1-p) \log 93 + (2p-1) \log 10.} \quad (6.2)$$

The per-generator image rate is $\exp(\mathcal{C}(p)/2)$.

Proof. Theorem 3 bounds every finite prefix below by the formula at its actual composition. Passing to the prescribed limiting frequency yields (6.1)–(6.2); at the junctions the expressions agree. The attainable schedules below supply the reverse bounds.

For $0 < p \leq 1/2$, concatenate BA^a and BA^{a+1} with macro frequencies $1-\nu$ and ν . The number of B -levels per macro is one and the mean macro length is $a+1+\nu = 1/p$. The exact product in (5.14) proves (6.1). Rational macro frequencies can be periodic. For irrational frequencies, the difference sequence of $\lfloor j\nu \rfloor$ supplies a specific nonperiodic choice with bounded discrepancy.

For $1/2 < p < 1$, at stage j use

$$(BA)^{\lfloor (1-p)j \rfloor} B^{\lfloor (2p-1)j \rfloor}. \quad (6.3)$$

Each stage has length $j + O(1)$, while accumulated length is $j^2/2 + O(j)$. The empirical frequencies converge to p at every prefix, including prefixes inside a stage. The stage count is at most

$$93^{\lfloor (1-p)j \rfloor} \frac{4 \cdot 10^{\lfloor (2p-1)j \rfloor} - 1}{3}.$$

Submultiplicativity makes its total logarithmic overhead $O(j)$, which vanishes per accumulated length. This gives (6.2). Pure schedules attain the endpoints. $\square$

The function is continuous and convex. On $[1/(a+2), 1/(a+1)]$ it is affine with slope

$$s_a = (a+2) \log K_a - (a+1) \log K_{a+1}. \quad (6.4)$$

Identity (5.12) gives $s_a > s_{a+1}$, and $s_a \rightarrow -\log(9/8)$. The final affine segment has slope $\log(100/93)$. The unconstrained minimum is at $p = 1/3$, recovering the earlier rate $893^{1/6}$ rather than improving it.

For a fixed rational $p \in (1/2, 1)$, no periodic schedule attains (6.2). Indeed any period with s letters A and $t > s$ letters B has spectral radius at least $F(s, t)$. With $b = t - s + 1 \geq 2$,

$$\boxed{\frac{F(s, t)}{93^s 10^{t-s}} = \frac{280}{279} - \frac{1}{279 \cdot 10^{b-1}} > 1.} \quad (6.5)$$

Repeating the period retains this strict logarithmic excess per period. Yet (6.3) attains the constrained optimum. Irrational profiles of course also exclude a periodic frequency, but (6.5) gives the stronger obstruction at every rational profile in this open interval.

The excess has a local explanation. For a crossed run pair (a, b) , relative to the linear branch (6.2), its logarithmic defect is

$$\epsilon(a, b) = a \log(100/93) - \log G(a, b) \geq 0. \quad (6.6)$$

Equality holds only at $(1, 1)$. For $a = 1, b \geq 2$ it lies between $\log(311/310)$ and $\log(280/279)$. For $a \geq 2$, it is bounded below by a positive constant times $a - 1$, since $\log G(a, b) < \log(9/8) < 2 \log(100/93)$. It is also bounded above by a constant times $a - 1$ in that range. Consequently optimal schedules on the open linear branch have a vanishing density of merged A -runs and of nontrivial B -runs. Their excess B mass remains positive, so the lengths of the latter runs must grow. Formula (6.3) realizes that precise regime.

An auxiliary price ξ per B -level makes the optimal value $\min_p (\mathcal{C}(p) + \xi p)$. Equations (6.4) give infinitely many transition prices $-s_a$, accumulating at $\log(9/8)$, and periodic phases BA^a with unbounded lengths. The parameter is a specified composition price; it does not change the arithmetic radices or add generators. This observation explains why a parameter-independent

bound for exact optimal periods would discard real information, even though the finite-horizon approximation in Theorem 2 is uniform.

52.7 7. Original receiving maps and a supported empty position

The numerical cut inequality (4.1) comes from the original onto map

$$\pi_{u,v} : H_q(A_u) \times H_q(A_v) \rightarrow H_q(A_{uv}), \quad (x, y) \mapsto x + P_u y. \quad (7.1)$$

If q_u, q_v are the original coefficient-word to local-value quotients, the relation sequence is

$$0 \rightarrow \ker(q_u \otimes q_v) \rightarrow \ker(\pi_{u,v}(q_u \otimes q_v)) \xrightarrow{q_u \otimes q_v} \ker \pi_{u,v} \rightarrow 0. \quad (7.2)$$

Surjectivity at the right follows by choosing an actual coefficient-word representative of each local value. The original local boundary space remains separate from the extra cut kernel.

For positive original pair masses $w(x, y) = \mu_u(x)\mu_v(y)$, put

$$W_z = \sum_{x+P_u y=z} w(x, y).$$

The minimum-norm section in the original quotient metric is

$$s(e_z) = \sum_{x+P_u y=z} \frac{w(x, y)}{W_z} e_{(x, y)}, \quad G_{\text{receiving}} = \text{diag}_z(1/W_z). \quad (7.3)$$

Neither the cut cardinality constant nor a dual composition price replaces this metric.

A simple original example distinguishes removing generators from erasing an active radix position. In the word ABA , retain only the two A -levels. Keeping the actual positions gives generators

$$\{1, 3, 100, 300\}, \quad |H_5| = 289.$$

Erasing the middle radix gives instead

$$\{1, 3, 10, 30\}, \quad |H_5| = 177.$$

On the distinct local values $[0, 16]^2$, the first evaluation $(x, y) \mapsto x + 100y$ is injective. The second $(x, y) \mapsto x + 10y$ is onto $[0, 176]$. The additional receiving kernel of the free linearized map has dimension $289 - 177 = 112$. All original pair values remain before taking that quotient.

The full block word after deleting the middle generators still has an active empty position with radix ten. Its source has one empty choice and a present numerical zero. Absorbing its radix into the preceding level keeps the first numerical set exactly. Simply omitting its cost produces the second set and the explicitly calculated new kernel.

For comparison, if u is obtained from a word w by erasing d original letters of total generator reward E , repeated use of (4.1) gives the fully costed inequality

$$-2d \log(q - 1) \leq \log F_q(w) - \log F_q(u) \leq E \log q + d \log(q - 1). \quad (7.4)$$

It follows by inserting one removed block between its two retained neighbours and bounding both original cuts. In a dictionary of individually modular-safe levels, the contracted word is again admissible; in a more general carry-controlled language that preservation requires its own witness. The two operations are never identified without their maps and costs.

The source framework for (7.2) is the original SplitZero internal quotient and extra receiving-kernel square, equations D6–D8 in the inspected support-diagram source. On a nonbottom label, a boundary maps to that label's coefficient zero. A count or residue observation can introduce an additional kernel, recorded above, rather than changing the old relation module or global absence. This contribution transfers those interfaces to explicit finite arithmetic sources; it supplies no new analytic theta estimate or Riemann-hypothesis conclusion.

52.8 8. Verification, reproducibility, and remaining general question

The producer retains all 490 finite canonical obstruction blocks and their original six-term masks. A separate scan covers all 36,310 canonical blocks of ranks at most four and radix at most forty. The auditor checks 31,083 admissible source/length cases and their actual finite Kneser stabilizers, the sharp residue sections and all 44 weighted residue fibres, and 84 original mixed-rank image joins. These bounded checks corroborate the general reduction; the all-height theorem rests on Section 2 and the complete exceptional certificate.

For composition, all 524,286 binary words of lengths one through eighteen are accounted for, with their literal multiplicities. The 189 fixed-composition classes agree with Theorem 3. The producer uses prefix rows; the auditor recovers the two matrices by interpolation on original numerical sets and propagates tail columns. It checks 126 original-generator images directly, all 961 attainments with $0 \leq s, t \leq 30$, and the actual phase-separated prefixes in the stated finite domain.

There are 84 finite composition LP certificates, including unequal generator rewards and boundary profiles. Each stores its complete count-signature domain, rational primal weights, logarithmic dual prices, and exact cut-factor correction. The separate auditor checks 570 dual inequalities and 360 further original-generator images. Eight deliberately corrupted records are rejected. It imports neither the producer nor its helper library.

The command pair is

```
python research/residue_composition/certificates/verify_residue_composition.py
python research/residue_composition/certificates/audit_residue_composition.py
```

All check functions remain active with `python -O`. Normal and optimized receipts are compared byte-for-byte in the delivery record. The one compiled PDF is a reading copy of this full text, not a substitute for the source or certificate. The package contains no earlier ZIP, compressed proof payload, font file, or compiled Lean binary.

The unbounded problem remains the best arithmetic cost over all admissible block ranks. The rank-profile theorem imposes a uniform constraint on original individually modular-safe presentations approaching the known costs. It does not prove that enough generators must remain in the low-rank part to obtain a stronger unrestricted lower bound. The profile spectrum solves all prescribed mixtures in its original two-block family and supplies uniform constrained approximation for every finite dictionary; it does not assert a rank cutoff for the entire problem.

The continuation therefore contributes an all-height arithmetic lower certificate and a complete composition optimization, while keeping the remaining all-rank inequality visible. Existing results, benchmark rates, and mathematical credits retain their earlier scopes.

52.9 References

DeVos, M. (2013). *A short proof of Kneser's addition theorem for abelian groups* (arXiv:1303.3539v1). The primary abstract states the exact stabilizer theorem used in (2.4). The classical theorem is imported with attribution; no claim is made to reprove it or newly formalize it here. <https://arxiv.org/abs/1303.3539>

The Clankers. (2026, September 16). *Exact local-pattern quotients and switching image capacity*. Integrated workbench source, `research/local_flow/notes/local-flow-and-switching.md`, at commit `dbd0a93f2cf935103e88e5c2b2b71fe85ae7537b`. Role: original seven matrices and crossed-run identity; these finite inputs are independently replayed in this contribution.

The Clankers. (2026, September 16). *Uniform finite-period approximation and exact finite-word switching minima*. Integrated workbench source, `research/finite_period/notes/finite-period-control.md`, at the same commit. Role: cut-fibre estimate and preceding unconstrained approximation theorem. The composition LP and fixed-count spectrum above extend those results.

KokunoYumeto. (2026). *Reconstruction with changes of support index* [TeX source]. Zeta Function Research Reader, `workbenches/splitzero-tandem/tex/support_diagrams.tex`,

revision 58626a62cd5648e8ae7450fb54d4a4aab2330981, blob d3493f891291ee6e94dbf2c77649f7d85d240df2, equations D6–D8. Role: original internal quotient and additional receiving-kernel interface. No new analytic statement from that workbench is assumed.

Exact supported-quotient and metric interface for EP817

This note integrates the supplied *Residue prices, constrained composition, and nonperiodic optimal schedules* of 17 September 2026 [31]. Its full proof, original LaTeX, and exact certificates are preserved without alteration in the accompanying source subtree. The maps below implement the internal quotient and receiving-kernel construction of the original Split-Zero support diagram, equations (D6)–(D8) [32]. All word multiplicities, actual integer values, moduli, support labels, and Euclidean word metrics are retained. In the composition results the coefficient arity is an integer $q \geq 2$; the applications at $q = 3, 5$ therefore have the original cut constant $q - 1$.

53 The actual source and its two cochain complexes

Let A be a fixed nonempty finite set of distinct positive integers and let $b > \sum_{a \in A} a$ be the original radix. Set

$$\Omega = \{0, 1, 2\}^A, \quad v(c) = \sum_{a \in A} c_a a, \quad \bar{v}(c) = v(c) \bmod b. \quad (\text{E1})$$

For every $S \subseteq \Omega$ use the actual images

$$X_S = v(S) \subset \mathbb{Z}, \quad R_S = \bar{v}(S) \subset \mathbb{Z}/b\mathbb{Z}, \quad V_S = \mathbb{Z}[S], \quad E_S = \mathbb{Z}[X_S], \quad \bar{E}_S = \mathbb{Z}[R_S]. \quad (\text{E2})$$

Here square brackets denote free abelian groups on the indicated distinct bases, including a basis vector e_0 whenever the image contains zero. The maps on these displayed bases are

$$q_S(e_c) = e_{v(c)}, \quad \pi_S(e_x) = e_{x \bmod b}, \quad p_S = \pi_S q_S. \quad (\text{E3})$$

All three have their stated codomains by (E2), and both q_S, p_S are surjective. Their kernels are

$$B_S = \ker q_S, \quad B'_S = \ker p_S, \quad B_S \subseteq B'_S. \quad (\text{E4})$$

For an empty image its free group is zero and the same formulas apply.

Define two complexes concentrated in degrees $-1, 0$:

$$C_S = [B_S \xrightarrow{\iota_S} V_S], \quad D_S = [B'_S \xrightarrow{\iota'_S} V_S]. \quad (\text{E5})$$

The map $f_S : C_S \rightarrow D_S$ is inclusion in degree -1 and identity in degree zero. It is a cochain map because $\iota'_S(B_S \hookrightarrow B'_S) = \iota_S$.

Theorem 53.1 (The exact receiving kernel in the original complexes). *The maps in (E3) induce the isomorphisms and exact sequence*

$$\begin{aligned} H^0(C_S) &\xrightarrow{\sim} E_S, \quad [c] \mapsto q_S c, & H^0(D_S) &\xrightarrow{\sim} \bar{E}_S, \quad [c] \mapsto p_S c, \\ H^{-1}(C_S) &= H^{-1}(D_S) = 0, & H^0(f_S) &= \pi_S, \\ 0 &\longrightarrow B_S \longrightarrow B'_S \xrightarrow{q_S} \ker \pi_S \longrightarrow 0, & B'_S/B_S &\xrightarrow{\sim} \ker H^0(f_S), \quad [c] \mapsto q_S c. \end{aligned} \quad (\text{E6})$$

Proof. Injectivity of the two differentials gives the degree- -1 statements. Their degree-zero cohomologies are V_S/B_S and V_S/B'_S . The maps q_S, p_S are onto with exactly those kernels, proving the two displayed isomorphisms. Their commutative square is (E3), so their induced degree-zero map is precisely π_S .

If $c \in B'_S$, then $\pi_S q_S c = p_S c = 0$, so the third map of the sequence has the asserted codomain. Its kernel is exactly B_S . To prove its surjectivity without an unspecified lift, order the original coefficient words lexicographically using the increasing order on A . For each $x \in X_S$, let c_x be the first word in S with value x . An element $h = \sum_x h_x e_x \in \ker \pi_S$ is the image under q_S of $\sum_x h_x e_{c_x}$; that lift belongs to B'_S because its p_S -image is $\pi_S h = 0$. This proves the exact sequence and the final isomorphism. $\square$

For $S \subseteq T$, use basis inclusions in each of $V, E, \bar{E}$. The squares with q and π commute on every basis vector. Consequently $B_S \rightarrow B_T$ and $B'_S \rightarrow B'_T$ are the restrictions of $V_S \rightarrow V_T$, and the complexes and (E6) form actual diagrams. Use the join-semilattice

$$L = \{\perp\} \sqcup \mathcal{P}(\Omega), \quad S \vee T = S \cup T, \quad (\text{E7})$$

with a distinct bottom $\perp$ and zero fibre there. The label $\emptyset \in \mathcal{P}(\Omega)$ remains distinct from $\perp$, despite its zero amplitude group. Over each displayed integral diagram the split carrier consists of (S, x) , with addition by transport to $S \cup T$, supported scalar multiplication $r^\bullet(S, x) = (S, rx)$, and $\tau(S, x) = (\perp, 0)$. These are the original operations of [32, (D2)]. Commutativity of the above squares proves that the reconstructed q, π, f are split-linear.

In particular the all-label zero-amplitude preimage of the reconstructed π is

$$\prod_{S \in L} \ker \pi_S. \quad (\text{E8})$$

The preimage of its global zero contains only the bottom fibre, since its label map is identity. The exact comparison is the inclusion of that bottom fibre into (E8); (E6) specifies each nonbottom fibre of (E8). A supported boundary (S, c) goes to $(S, 0)$ at the same label. Also the residue-amplitude map $\bar{E}_S \rightarrow \mathbb{Z}/b\mathbb{Z}$, $\sum_r z_r e_r \mapsto \sum_r z_r r$ is integral linear; it sends the present vector e_0 to zero. These statements require no map from $\mathbb{C}$ or $\mathbb{Q}$ to the finite group.

54 Original masses and an evaluated determinant loss

For the metric statements extend the preceding groups and maps to $\mathbb{C}$. Place the orthonormal metric on the original word basis of $\mathbb{C}[S]$. Define the positive masses only on their actual images:

$$\mu_S(x) = |\{c \in S : v(c) = x\}|, \quad \bar{\mu}_S(r) = \sum_{\substack{x \in X_S \\ x \bmod b=r}} \mu_S(x). \quad (\text{E9})$$

No zero-mass value is inserted into an inverse diagonal matrix.

Theorem 54.1 (Exact integral-section loss). *The quotient metric on $E_S \otimes \mathbb{C}$, the residue quotient metric, and the unique minimum-norm right inverse of π_S are respectively*

$$G_{E,S} = \text{diag}_{x \in X_S} \frac{1}{\mu_S(x)}, \quad G_{\bar{E},S} = \text{diag}_{r \in R_S} \frac{1}{\bar{\mu}_S(r)}, \quad s_S(e_r) = \sum_{x \bmod b=r} \frac{\mu_S(x)}{\bar{\mu}_S(r)} e_x. \quad (\text{E10})$$

Let $d_S(r)$ be the least actual integer value in the residue fibre and let $\sigma_S(e_r) = e_{d_S(r)}$. Put $K_S = \sigma_S - s_S$. Then

$$\begin{aligned} \pi_S K_S &= 0, & \sigma_S^* G_{E,S} \sigma_S &= G_{\bar{E},S} + K_S^* G_{E,S} K_S, \\ K_S^* G_{E,S} K_S &= \text{diag}_{r \in R_S} \left(\frac{1}{\mu_S(d_S(r))} - \frac{1}{\bar{\mu}_S(r)} \right), \\ \log \frac{\det(\sigma_S^* G_{E,S} \sigma_S)}{\det G_{\bar{E},S}} &= \sum_{r \in R_S} \log \frac{\bar{\mu}_S(r)}{\mu_S(d_S(r))}. \end{aligned} \quad (\text{E11})$$

Every determinant is in the displayed original residue basis. Empty determinants are one.

Proof. For a fixed x , minimizing $\sum_{v(c)=x} |z_c|^2$ subject to $\sum_{v(c)=x} z_c = u_x$ gives the unique solution $z_c = u_x / \mu_S(x)$ and cost $|u_x|^2 / \mu_S(x)$, by completing the sum of squared deviations. Different value fibres are disjoint. This proves $G_{E,S}$.

Within residue r , write $u_x = \mu_S(x)t / \bar{\mu}_S(r) + h_x$ where $\sum_{x \bmod b=r} h_x = 0$. Direct expansion gives

$$\sum_{x \bmod b=r} \frac{|u_x|^2}{\mu_S(x)} = \frac{|t|^2}{\bar{\mu}_S(r)} + \sum_{x \bmod b=r} \frac{|h_x|^2}{\mu_S(x)}. \quad (\text{E12})$$

The cross term is zero because its coefficient is $\sum_x h_x = 0$. The displayed cost is uniquely minimized by $h = 0$. This proves the last two parts of (E10), as well as orthogonality of $\text{im } s_S$ to $\ker \pi_S$. Both s_S and σ_S are right inverses, so K_S has image in that kernel. Substituting $h = K_S t$ in (E12) proves the first identity in (E11). Columns for different residues have disjoint value support. The squared norm of $\sigma_S e_r$ is $1/\mu_S(d_S(r))$, while that of $s_S e_r$ is $1/\bar{\mu}_S(r)$. This proves the diagonal formula, and taking the quotient of the diagonal determinants proves the final formula. $\square$

Thus the squared generalized singular values of the actual integral section relative to the attained quotient metric are $\bar{\mu}_S(r)/\mu_S(d_S(r))$. Formula (E11) evaluates its full loss, including every original representation multiplicity.

55 Source enlargement has an exact metric boundary

Write $i_{ST} : E_S \otimes \mathbb{C} \rightarrow E_T \otimes \mathbb{C}$ and $\bar{i}_{ST} : \bar{E}_S \otimes \mathbb{C} \rightarrow \bar{E}_T \otimes \mathbb{C}$ for the original basis inclusions when $S \subseteq T$. Extend $\mu_S(x)$ by zero for $x \in X_T \setminus X_S$. The minimum-norm section defect is the map with its stated source and target

$$K_{ST} = i_{ST} s_S - s_T \bar{i}_{ST} : \bar{E}_S \otimes \mathbb{C} \longrightarrow \ker \pi_T. \quad (\text{E13})$$

Proposition 55.1 (Full source-change norm and its composition law). *For every $r \in R_S$,*

$$\begin{aligned} K_{ST} e_r &= \sum_{\substack{x \in X_T \\ x \bmod b=r}} \left(\frac{\mu_S(x)}{\bar{\mu}_S(r)} - \frac{\mu_T(x)}{\bar{\mu}_T(r)} \right) e_x, \\ \|K_{ST} e_r\|_{G_{E,T}}^2 &= \frac{1}{\bar{\mu}_S(r)^2} \sum_{x \bmod b=r} \frac{\mu_S(x)^2}{\mu_T(x)} - \frac{1}{\bar{\mu}_T(r)}, \\ 0 \leq \|K_{ST} e_r\|_{G_{E,T}}^2 &\leq \frac{1}{\bar{\mu}_S(r)} - \frac{1}{\bar{\mu}_T(r)}. \end{aligned} \quad (\text{E14})$$

The columns for distinct residues are orthogonal in $G_{E,T}$. For $S \subseteq T \subseteq U$ the exact typed composition identity is

$$K_{SU} = i_{TU} K_{ST} + K_{TU} \bar{i}_{ST}. \quad (\text{E15})$$

Proof. The two right-inverse identities and the commutative square give $\pi_T K_{ST} = \bar{i}_{ST} - \bar{i}_{ST} = 0$. Insertion of (E10) gives the first formula. In the $G_{E,T}$ norm, expansion of its square has cross term $-2/\bar{\mu}_T(r)$ and last term $1/\bar{\mu}_T(r)$, yielding the second formula. Nonnegativity follows either from this norm or Cauchy–Schwarz. Since $S \subseteq T$ gives $0 \leq \mu_S(x) \leq \mu_T(x)$, one has $\sum_x \mu_S(x)^2/\mu_T(x) \leq \bar{\mu}_S(r)$, giving the upper bound. The disjoint residue supports prove column orthogonality. Finally,

$$i_{TU}(i_{ST} s_S - s_T \bar{i}_{ST}) + (i_{TU} s_T - s_U \bar{i}_{TU}) \bar{i}_{ST} = i_{SU} s_S - s_U \bar{i}_{SU},$$

which is (E15), with all domains as in (E13). $\square$

The integral sections have the parallel defect

$$i_{ST} \sigma_S e_r - \sigma_T \bar{i}_{ST} e_r = e_{d_S(r)} - e_{d_T(r)} \in \ker \pi_T. \quad (\text{E16})$$

Its squared $G_{E,T}$ -norm is zero if the representatives coincide, and otherwise is $1/\mu_T(d_S(r)) + 1/\mu_T(d_T(r))$, because those are two different original basis vectors. The same addition and subtraction proves its version of (E15).

56 A universal derivation of the seven arithmetic matrices

The composition source retains

$$A = (10, \{1, 3\}), \quad B = (10, \{2, 4\}), \quad D_A = H_5(\{1, 3\}) = [0, 16], \quad D_B = H_5(\{2, 4\}) = 2[0, 12]. \quad (\text{E17})$$

For A , the intervals $[3j, 3j + 4]$, $0 \leq j \leq 4$, overlap and fill $[0, 16]$. For B , the original values are $2(i + 2j)$ with $0 \leq i, j \leq 4$; the intervals $[2j, 2j + 4]$ overlap and fill $[0, 12]$. Thus (E17) is an equality of original numerical sets.

Keep the seven source states in their stated order:

$$\mathcal{R} = ((0), (0, 1), (0, 2), (0, 1, 2), (0, 3), (0, 1, 3), (0, 1, 2, 3)). \quad (\text{E18})$$

For any finite $Y \subset \mathbb{Z}$, put $f_R(Y) = |Y + R|$. For $X = A, B$, $R \in \mathcal{R}$, and $j \in \{0, \dots, 9\}$, define

$$C_{X,R,j} = \left\{ \frac{d + r - j}{10} : d \in D_X, r \in R, d + r \equiv j \pmod{10} \right\}. \quad (\text{E19})$$

If this set is nonempty, write $m_{X,R,j} = \min C_{X,R,j}$ and $R_{X,R,j} = C_{X,R,j} - m_{X,R,j}$. The change of presentation is the explicit bijection

$$Y + R_{X,R,j} \longrightarrow (D_X + 10Y + R) \cap (j + 10\mathbb{Z}), \quad z \longmapsto j + 10(m_{X,R,j} + z), \quad (\text{E20})$$

whose inverse is $t \mapsto (t - j)/10 - m_{X,R,j}$. Each shift and residue is retained in `UNIVERSAL_INTERF ACE_CHECKS.json`. Direct insertion of the finite sets in (E17)–(E19) gives the following numbers of residues of each resulting state:

$$M_A = \begin{pmatrix} 3 & 7 & 0 & 0 & 0 & 0 & 0 \\ 2 & 8 & 0 & 0 & 0 & 0 & 0 \\ 1 & 9 & 0 & 0 & 0 & 0 & 0 \\ 1 & 9 & 0 & 0 & 0 & 0 & 0 \\ 0 & 10 & 0 & 0 & 0 & 0 & 0 \\ 0 & 10 & 0 & 0 & 0 & 0 & 0 \\ 0 & 10 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad M_B = \begin{pmatrix} 0 & 2 & 0 & 3 & 0 & 0 & 0 \\ 0 & 4 & 0 & 6 & 0 & 0 & 0 \\ 0 & 1 & 0 & 4 & 0 & 0 & 0 \\ 0 & 3 & 0 & 7 & 0 & 0 & 0 \\ 0 & 4 & 0 & 6 & 0 & 0 & 0 \\ 0 & 3 & 0 & 7 & 0 & 0 & 0 \\ 0 & 2 & 0 & 8 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{E21})$$

For completeness these entries can be obtained without a numerical tail: enumerate each of the at most $25 \cdot 4$ displayed pairs (d, r) , place $(d + r - j)/10$ into its residue set in (E19), subtract its recorded minimum with inverse (E20), and count the state. The accompanying independent verifier records all 140 such original fibres and checks each entry of (E21).

The residues in (E20) are disjoint. Summing their cardinalities proves, for *every* finite numerical tail Y , the identities

$$f(D_X + 10Y) = M_X f(Y), \quad f(\{0\}) = (1, 2, 2, 3, 2, 3, 4)^\top. \quad (\text{E22})$$

The matrices thus describe the exact count observations of the retained numerical sets. They are induced through the original residue partition and bijections (E20); no change to coefficient-word amplitudes is involved. Multiplication of the displayed integer matrices gives

$$M_A M_B = (17, 18, 19, 19, 20, 20, 20)^\top (0, 2, 0, 3, 0, 0, 0). \quad (\text{E23})$$

This supplies the universal finite-source input to the supplied composition proofs, alongside their exact finite and limiting attainments.

57 The retained empty radix and its actual kernel

Retaining the two A -positions of ABA gives $\{1, 3, 100, 300\}$. The distinct local values are $[0, 16]^2$, and the two evaluations at issue are

$$e_{x,y} \longmapsto e_{x+100y}, \quad e_{x,y} \longmapsto e_{x+10y}. \quad (\text{E24})$$

The first is a bijection onto a 289-element basis: if its values coincide, $100(y - y') = x' - x \in [-16, 16]$, forcing both differences to vanish. The second has image $[0, 176]$, because the intervals $[10y, 10y + 16]$, $0 \leq y \leq 16$, overlap and have precisely these endpoints. It is therefore an onto map from a free rank-289 group to a free rank-177 group, with free kernel rank 112. An explicit kernel basis chooses for each value z one pair (x_z, y_z) evaluating to z and uses $e_{x,y} - e_{x_z, y_z}$ for every other pair in that fibre. Within each fibre these vectors are independent and span its coefficient-sum-zero group, proving the rank calculation integrally.

The first evaluation identifies its basis with the original retained image, so composing its inverse with the second evaluation gives the exact contraction map between the two value modules. Formula (E6) then gives its corresponding additional receiving kernel. The mass of each local value pair remains the product of the two original local word multiplicities. Applying (E10)–(E12) to the fibres of (E24) supplies its original minimum-norm quotient, with no unit-mass substitution.

58 Accepted return and next arithmetic scope

The preserved paper proves sharp ternary residue prices 3, 5, 13, 23 for ranks 1, 2, 3, 4 of individually modular-five- or modular-six-free canonical blocks, at unrestricted height and radix. Its classical addition-theorem input is Martin Kneser’s stabilizer theorem, in the stated form attributed to Matt DeVos’s primary proof [33]. The remaining 490 exceptional source blocks have explicit original subset-mask witnesses, replayed in this intake. The preserved composition paper also gives every fixed-composition minimum and the full limiting frequency spectrum in the actual radix-ten dictionary. All supplied credit to the original anonymous/deleted contributor and to **sneed-and-feed**’s distinct Lean extension remains in the original paper.

The present note proves the exact finite cochain realization (E6), the integral-section determinant loss (E11), the source-change boundary and composition law (E14)–(E15), and the universal original matrix return (E19)–(E23). These are usable instances of the retained Split-Zero supported quotient, with their full metrics specified. The larger-rank modular arithmetic price remains an uncomputed part of this EP817 continuation. The original theta source and its signed analytic allocations receive the algebraic construction (E6)–(E16); no matrix identifying those analytic sources with the finite residue source has been supplied or proved in this intake. Therefore no analytic allocation coefficient or statement about zeta zeros is entered in the accepted-result index from this package.

Original complete equilibrium provider

This is the preserved original EIQ/GEL provider used in the arithmetic value calculation; its exact input hash is recorded in the insertion manifest.

Exact dictionary for the equilibrium provider. The elliptic modulus k used locally in EIQ is the κ of GEL, and EIQ's local $m = (u^2 + v^2)/2$ is an endpoint midpoint. They are distinctly indexed coordinates from the original packet $k_{\text{packet}} = 4l + 1$ and multiplicity m_{packet} . GEL3 gives their exact parameter map $\alpha = a/n$, $\beta = \pi q/(2n)$, tending to $(2, \pi)$ on the original sequences. GEL's constant $C_\Gamma = \Gamma(1/4)^2/(2\pi)$ in its source bound retains that definition; the leading coefficient is the separately defined $\mathcal{C}_\Gamma$ in GEL18.

59 Exact equilibrium for the square-root logarithmic potential

Throughout this section $\alpha > 0$ and $\beta > 0$. On $(0, \infty)$ set

$$V_{\alpha,\beta}(x) = \beta\sqrt{x} - \alpha \log x, \quad \mathcal{E}_{\alpha,\beta}(\mu) = \int V_{\alpha,\beta}(x) d\mu(x) - \iint \log|x-y| d\mu(x) d\mu(y). \quad (\text{EIQ1})$$

The measures considered initially are probability measures for which $\int V_{\alpha,\beta} d\mu$ is finite. Because this potential is bounded below and tends to $+\infty$ at both ends, this implies $\int(\sqrt{x} + |\log x|) d\mu < \infty$. The positive part of $\log|x-y|$ is then integrable, by $\log^+|x-y| \leq \log(1+x) + \log(1+y)$. Thus the energy is well defined with value in $\mathbb{R} \cup \{+\infty\}$. All comparisons below also apply to the usual extended energy defined by integrating the kernel $\frac{1}{2}V_{\alpha,\beta}(x) + \frac{1}{2}V_{\alpha,\beta}(y) - \log|x-y|$: the kernel is bounded below, and finite energy for this kernel implies the same moment conditions. To see the latter assertion without cancellation, put $g(x) = \frac{1}{2}V_{\alpha,\beta}(x) - \log(1+x)$. The kernel is at least $g(x) + g(y)$; g is bounded below and controls positive constant multiples of both $\sqrt{x}$ near infinity and $|\log x|$ near zero.

59.1 Existence and uniqueness of the two endpoints

For $0 < k < 1$ define the complete elliptic integrals by their actual real integrals

$$K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1-k^2 \sin^2 \theta}}, \quad E(k) = \int_0^{\pi/2} \sqrt{1-k^2 \sin^2 \theta} d\theta. \quad (\text{EIQ2})$$

There is exactly one $k \in (0, 1)$ satisfying

$$\frac{\sqrt{1-k^2}K(k)}{E(k)} = \frac{\alpha}{\alpha+2}. \quad (\text{EIQ3})$$

For this k put

$$r = \sqrt{1-k^2}, \quad v = \frac{(\alpha+2)\pi}{\beta E(k)}, \quad u = rv, \quad a = u^2, \quad b = v^2. \quad (\text{EIQ4})$$

In particular $0 < a < b < \infty$, and the endpoint equations, with all constants retained, are

$$uK(k) = \frac{\alpha\pi}{\beta}, \quad vE(k) = \frac{(\alpha+2)\pi}{\beta}, \quad k^2 = 1 - \frac{u^2}{v^2}. \quad (\text{EIQ5})$$

Here is a proof of the assertion of uniqueness. Differentiation of the integrals in (EIQ2), and integration of the derivative of $\sin \theta \cos \theta / \sqrt{1-k^2 \sin^2 \theta}$ between its two zero endpoint values, give

$$E'(k) = \frac{E(k) - K(k)}{k}, \quad K'(k) = \frac{E(k)}{k(1-k^2)} - \frac{K(k)}{k}.$$

For completeness the second identity is equivalent, after differentiation under the integral sign, to

$$\int_0^{\pi/2} \left\{ \frac{k^2 \sin^2 \theta}{(1 - k^2 \sin^2 \theta)^{3/2}} - \frac{1 - k^2 \sin^2 \theta}{(1 - k^2) \sqrt{1 - k^2 \sin^2 \theta}} + \frac{1}{\sqrt{1 - k^2 \sin^2 \theta}} \right\} d\theta = 0;$$

the expression in braces is $-k^2(1 - k^2)^{-1} d(\sin \theta \cos \theta / \sqrt{1 - k^2 \sin^2 \theta}) / d\theta$. With $t = E(k)/K(k)$, the integral definitions show strictly $r^2 < t < 1$. Consequently, if $f(k) = rK(k)/E(k)$, then

$$\frac{k f'(k)}{f(k)} = \frac{t}{r^2} + \frac{1}{t} - 1 - \frac{1}{r^2} = \frac{(t - 1)(t - r^2)}{r^2 t} < 0. \quad (\text{EIQ6})$$

At $k = 0$ the continuous limiting value is $f(0) = 1$. As $k \uparrow 1$, the integrand of $rK(k)$ tends to zero at every $\theta < \pi/2$ and is bounded by 1, while $E(k) \rightarrow 1$ by dominated convergence. Hence $f(k) \rightarrow 0$. This proves (EIQ3), including existence. It also proves smooth dependence of k, u, v, a, b on (α, β) in $(0, \infty)^2$, by the nonzero derivative in (EIQ6).

Let $m = (a + b)/2$, $R_s = \sqrt{(a + s)(b + s)}$ for $s \geq 0$, and $R_0 = uv$. Substitution $t = a \cos^2 \theta + b \sin^2 \theta$ gives, directly from (EIQ5),

$$\int_a^b \frac{V'_{\alpha, \beta}(t)}{\sqrt{(b - t)(t - a)}} dt = 0, \quad \int_a^b \frac{t V'_{\alpha, \beta}(t)}{\sqrt{(b - t)(t - a)}} dt = 2\pi. \quad (\text{EIQ7})$$

Indeed the three integrals with numerators $t^{-1/2}, t^{-1}, t^{1/2}$ are respectively $2K(k)/v$, $\pi/(uv)$, and $2vE(k)$, and $V'(t) = \beta/(2\sqrt{t}) - \alpha/t$.

59.2 A positive density and its exact Cauchy transform

The elementary Stieltjes integral, obtained by $s = xz^2$, is

$$\int_0^\infty \frac{ds}{\sqrt{s}(x + s)} = \frac{\pi}{\sqrt{x}} \quad (x > 0). \quad (\text{EIQ8})$$

Using this identity in (EIQ7) and applying Tonelli to its positive constituents gives

$$\frac{\alpha}{R_0} = \frac{\beta}{2\pi} \int_0^\infty \frac{ds}{\sqrt{s}R_s}, \quad \int_0^\infty \frac{1 - s/R_s}{\sqrt{s}} ds = 2vE(k). \quad (\text{EIQ9})$$

Both integrals converge; their integrands are $O(s^{-1/2})$ at zero and $O(s^{-3/2})$ at infinity. The second formula also follows by averaging $\sqrt{t} = \pi^{-1} \int_0^\infty s^{-1/2} t/(t + s) ds$ against the arcsine probability measure on $[a, b]$.

Define for every $x > 0$

$$\begin{aligned} h(x) &= \frac{\alpha}{xR_0} - \frac{\beta}{2\pi} \int_0^\infty \frac{ds}{\sqrt{s}(x + s)R_s} \\ &= \frac{\beta}{2\pi x} \int_0^\infty \frac{\sqrt{s}}{(x + s)R_s} ds > 0. \end{aligned} \quad (\text{EIQ10})$$

The second equality is exactly (EIQ9), with no change of the potential. All integrals in (EIQ10) converge absolutely and locally uniformly in $x > 0$. The candidate density on the original positive half-line is

$$\rho_{\alpha, \beta}(x) = \begin{cases} \frac{\sqrt{(b - x)(x - a)}}{2\pi} h(x), & a < x < b, \\ 0, & x \notin (a, b). \end{cases} \quad (\text{EIQ11})$$

It is strictly positive on (a, b) and vanishes continuously with the square-root factor at each end-point. Positivity here follows from the last integral in (EIQ10), not from a one-interval ansatz.

The following elementary transform computes its mass and resolvent. Choose $R(z) = \sqrt{(z-a)(z-b)}$ on $\mathbb{C} \setminus [a, b]$ with $R(z)/z \rightarrow 1$ at infinity. Thus $R(x) > 0$ for $x > b$, $R(x) < 0$ for $x < a$, and $R(x+i0) = i\sqrt{(b-x)(x-a)}$ on (a, b) . Direct substitution $t = m + (b-a)\cos\theta/2$, followed by $w = \tan(\theta/2)$, proves

$$\frac{1}{\pi} \int_a^b \frac{dt}{(z-t)\sqrt{(b-t)(t-a)}} = \frac{1}{R(z)}.$$

For real $z > b$, the substitution reduces the integral to $\pi^{-1} \int_0^\infty 2dw/[(z-b) + (z-a)w^2]$; the identity elsewhere follows by analytic continuation. Polynomial division then gives

$$\frac{1}{\pi} \int_a^b \frac{\sqrt{(b-t)(t-a)}}{z-t} dt = z - m - R(z), \quad \frac{1}{\pi} \int_a^b \frac{\sqrt{(b-t)(t-a)}}{t+s} dt = m + s - R_s. \quad (\text{EIQ12})$$

For example the first identity follows from $(b-t)(t-a) = -(t-z)^2 + 2(m-z)(t-z) - (z-a)(z-b)$, the preceding arcsine transform, and the arcsine first moment m . Partial fractions in these two identities yield

$$H_s(z) := \frac{1}{\pi} \int_a^b \frac{\sqrt{(b-t)(t-a)}}{(t+s)(z-t)} dt = 1 - \frac{R_s + R(z)}{z+s}. \quad (\text{EIQ13})$$

The formula at $z = -s$ is interpreted by its removable limit.

The mass of (EIQ11), computed using (EIQ12), is

$$\begin{aligned} \int_a^b \rho_{\alpha,\beta}(t) dt &= \frac{\alpha}{2R_0}(m - R_0) - \frac{\beta}{4\pi} \int_0^\infty \frac{(m+s)/R_s - 1}{\sqrt{s}} ds \\ &= -\frac{\alpha}{2} + \frac{\beta}{4\pi} \int_0^\infty \frac{1-s/R_s}{\sqrt{s}} ds = -\frac{\alpha}{2} + \frac{\beta v E(k)}{2\pi} = 1. \end{aligned} \quad (\text{EIQ14})$$

In particular this is a probability density.

Define its Cauchy transform with the explicit sign convention $G(z) = \int_a^b \rho_{\alpha,\beta}(t)/(z-t) dt$. For z outside both $[a, b]$ and $(-\infty, 0]$, formulas (EIQ10) and (EIQ13) give

$$\begin{aligned} G(z) &= \frac{\alpha}{2R_0} H_0(z) - \frac{\beta}{4\pi} \int_0^\infty \frac{H_s(z)}{\sqrt{s} R_s} ds \\ &= -\frac{\alpha}{2z} + \frac{\beta}{4\sqrt{z}} - \frac{R(z)h(z)}{2} = \frac{V'_{\alpha,\beta}(z) - R(z)h(z)}{2}. \end{aligned} \quad (\text{EIQ15})$$

Here the positive square root is continued with its negative-real-axis cut, and $h(z)$ is the first integral formula of (EIQ10). Every integral exchange is absolutely convergent, uniformly on compact subsets of this domain: near zero the auxiliary integrand is $O(s^{-1/2})$, and near infinity the original integral defining H_s is $O(s^{-1})$. The constant terms on the second line cancel by (EIQ9), and (EIQ8) supplies the term $\beta/(4\sqrt{z})$.

59.3 The Euler equality, its strict exterior inequality, and uniqueness

Set

$$U(x) = V_{\alpha,\beta}(x) - 2 \int_a^b \log|x-t| \rho_{\alpha,\beta}(t) dt \quad (x > 0). \quad (\text{EIQ16})$$

This function is continuous on $(0, \infty)$. The integral is continuous because the density is bounded, compactly supported, and the integral of $|\log|x-t||$ over an interval of length ε tends to zero uniformly in its centre. In the interior of $[a, b]$, differentiation of the logarithmic potential is the principal-value integral $\text{PV} \int \rho(t)/(x-t) dt$. This follows by subtracting $\rho(x)$ on a symmetric interval about x ; the remaining numerator is $O(|t-x|)$ because the density is continuously differentiable locally in the interior. The real part of (EIQ15) on the upper edge of the interval is precisely this principal value. Indeed subtracting $\rho(x)$ in $\int \rho(t)/(x+i\varepsilon-t) dt$ gives the same limit,

while the constant term has real part equal to its elementary principal value. Since $R(x + i0)$ is purely imaginary and $h(x)$ is real, (EIQ15) proves $U'(x) = 0$ on (a, b) . There is therefore a real constant ℓ such that

$$U(x) = \ell \quad (a \leq x \leq b). \quad (\text{EIQ17})$$

For positive x outside this interval, (EIQ15) instead gives the exact real derivative

$$U'(x) = R(x)h(x) \begin{cases} < 0, & 0 < x < a, \\ > 0, & x > b. \end{cases} \quad (\text{EIQ18})$$

Combining (EIQ17), continuity, and these signs proves

$$U(x) > \ell \quad \text{for } x \in (0, a) \cup (b, \infty). \quad (\text{EIQ19})$$

This verifies the entire exterior inequality on the original domain, including the interval next to zero.

Here is a complete positivity argument for the energy of the difference of two measures. If ν is a finite signed real measure of total mass zero for which $|\log|x - y||$ is integrable against $|\nu| \otimes |\nu|$, the scalar identity

$$-\log d = \frac{1}{2} \int_0^\infty \frac{e^{-td^2} - e^{-t}}{t} dt \quad (d > 0) \quad (\text{EIQ20})$$

follows by differentiating in d and evaluating at $d = 1$. One half of the integral over any finite subinterval of $(0, \infty)$ has the same sign as $-\log d$ and absolute value at most $|\log d|$. Consequently dominated convergence and the zero total mass give

$$-\iint \log|x - y| d\nu(x)d\nu(y) = \frac{1}{2} \int_0^\infty \frac{1}{t} \iint e^{-t(x-y)^2} d\nu(x)d\nu(y) dt \geq 0. \quad (\text{EIQ21})$$

The nonnegative sign is proved, rather than assumed, by the Gaussian Fourier identity

$$\iint e^{-t(x-y)^2} d\nu(x)d\nu(y) = \frac{1}{\sqrt{4\pi t}} \int_{\mathbb{R}} e^{-\xi^2/(4t)} |\widehat{\nu}(\xi)|^2 d\xi, \quad \widehat{\nu}(\xi) = \int e^{i\xi x} d\nu(x). \quad (\text{EIQ22})$$

The scalar Gaussian Fourier identity follows by completing the square in the Gaussian integral (or by differentiating its Fourier transform and its value at zero); its integration against the finite measures is justified by absolute integrability. Equality in (EIQ21) forces $\widehat{\nu} = 0$: the right side of (EIQ22) is strictly positive for every $t > 0$ if this continuous Fourier transform has a nonzero value. Finally $\widehat{\nu} = 0$ implies $\nu = 0$ without an extra uniqueness assumption. Convolve ν with a centred Gaussian approximate identity. Its value at every point is zero by that same Fourier integral. For every compactly supported continuous test function, its convolution with these Gaussians converges uniformly to the test function; integration against ν then proves that ν annihilates every such function.

Let μ be any probability measure of finite energy in (EIQ1), and write $\mu_*(dx) = \rho_{\alpha,\beta}(x) dx$ and $\nu = \mu - \mu_*$. The integrability needed for (EIQ21) holds. The $\mu \otimes \mu$ term has it by finite energy and the already established logarithmic moment; the $\mu_* \otimes \mu_*$ term has it by bounded compact support; and the cross term has it because the negative logarithmic singularity integrated against the bounded density is uniformly bounded, while the positive part is controlled by $\log(1+x) + \log(1+y)$. Expanding the energy therefore yields exactly

$$\mathcal{E}_{\alpha,\beta}(\mu) - \mathcal{E}_{\alpha,\beta}(\mu_*) = \int (U(x) - \ell) d\mu(x) - \iint \log|x - y| d\nu(x)d\nu(y) \geq 0. \quad (\text{EIQ23})$$

The first term is nonnegative by (EIQ17)–(EIQ19), and the second is strictly positive unless $\mu = \mu_*$. Infinite energy competitors cannot improve this finite value. Thus (EIQ11) is the unique global minimizing probability measure, with exactly the support $[a, b]$.

59.4 An explicit endpoint integral for the logarithmic moment

Retain m, R_s, u, v as defined above, and define

$$C = \frac{(u+v)^2}{4}, \quad \eta = \frac{v-u}{v+u}, \quad \zeta_s = \frac{b-a}{2(m+s+R_s)} \quad (s \geq 0). \quad (\text{EIQ24})$$

Thus $0 < \zeta_s \leq \eta < 1$ and $\zeta_0 = \eta$. For every $s \geq 0$ put

$$L_s = (m+s-R_s) \log C - (m-R_0) - 2R_s \log(1-\eta\zeta_s). \quad (\text{EIQ25})$$

Then the logarithmic moment of the equilibrium measure is exactly

$$\mathcal{L}(\alpha, \beta) := \int_a^b \log x \rho_{\alpha, \beta}(x) dx = \frac{\beta}{4\pi} \int_0^\infty \frac{L_0 - L_s}{\sqrt{s} R_s} ds. \quad (\text{EIQ26})$$

This is a convergent one-dimensional endpoint integral of explicitly given real elementary functions; its only endpoints are those specified uniquely in (EIQ3)–(EIQ5).

Here is a direct derivation, including the constants in (EIQ25). Set $h_0 = (b-a)/2$ and substitute $t = m + h_0 \cos \theta$ in arcsine integrals. The identities

$$t = C(1 + 2\eta \cos \theta + \eta^2), \quad t + s = \frac{m+s+R_s}{2}(1 + 2\zeta_s \cos \theta + \zeta_s^2)$$

give the absolutely convergent Fourier series

$$\log t = \log C + 2 \sum_{j \geq 1} \frac{(-1)^{j+1} \eta^j}{j} \cos(j\theta), \quad \frac{1}{t+s} = \frac{1}{R_s} \left(1 + 2 \sum_{j \geq 1} (-\zeta_s)^j \cos(j\theta) \right).$$

Orthogonality on $[0, \pi]$ gives

$$\begin{aligned} \frac{1}{\pi} \int_a^b \frac{\log t}{\sqrt{(b-t)(t-a)}} dt &= \log C, \\ \frac{1}{\pi} \int_a^b \frac{t \log t}{\sqrt{(b-t)(t-a)}} dt &= m \log C + (m - R_0), \\ \frac{1}{\pi} \int_a^b \frac{\log t}{(t+s) \sqrt{(b-t)(t-a)}} dt &= \frac{\log C + 2 \log(1-\eta\zeta_s)}{R_s}. \end{aligned} \quad (\text{EIQ27})$$

The second identity uses $h_0 \eta = m - R_0$. Polynomial division

$$\frac{(b-t)(t-a)}{t+s} = -t + 2m + s - \frac{R_s^2}{t+s}$$

now proves, exactly,

$$L_s = \frac{1}{\pi} \int_a^b \frac{\log t \sqrt{(b-t)(t-a)}}{t+s} dt. \quad (\text{EIQ28})$$

Using the positive integral expression for h in (EIQ10), and the identity $1/[t(t+s)] = (1/t - 1/(t+s))/s$, yields (EIQ26). The exchanges are absolutely convergent: $\log t$ is bounded on $[a, b]$, $L_0 - L_s = O(s)$ at zero by (EIQ28), and $L_s = O(s^{-1})$ at infinity. Thus the integrand in (EIQ26) is $O(s^{1/2})$ at zero and $O(s^{-3/2})$ at infinity.

59.5 Parameter continuity and the variational derivative

The density and its logarithmic moment depend smoothly on $(\alpha, \beta) \in (0, \infty)^2$ in the following precise sense. On a compact parameter neighbourhood, (EIQ3)–(EIQ6) put a, b in a fixed compact subset of $(0, \infty)$, with $b - a$ bounded below by a positive number. The integral for h and each of its parameter and x derivatives converge uniformly for x in this fixed compact set: their absolute integrands are bounded by constants times $s^{1/2}$ near zero and $s^{-3/2}$ near infinity. Parameter derivatives of the endpoints are bounded on a smaller compact neighbourhood. Under the fixed-interval coordinate $x = a + (b - a)y$, $0 \leq y \leq 1$, the measure is exactly

$$\rho_{\alpha, \beta}(x) dx = \frac{(b - a)^2}{2\pi} \sqrt{y(1 - y)} h(a + (b - a)y) dy. \quad (\text{EIQ29})$$

All parameter derivatives of its coefficient are bounded by a constant times $\sqrt{y(1 - y)}$. The same is true after multiplication by $\log(a + (b - a)y)$ or by $\sqrt{a + (b - a)y}$. Differentiation under this integral proves smoothness, in particular continuity, of both moments on the entire positive parameter quadrant.

Define $E_{\min}(\alpha, \beta) = \mathcal{E}_{\alpha, \beta}(\mu_{\alpha, \beta})$, where the measure is the unique minimizer proved above. Testing the minimizer for a parameter pair against the neighbouring potential, and then reversing the two tests, gives for $h > 0$

$$-\mathcal{L}(\alpha + h, \beta) \leq \frac{E_{\min}(\alpha + h, \beta) - E_{\min}(\alpha, \beta)}{h} \leq -\mathcal{L}(\alpha, \beta).$$

The corresponding inequalities for $h < 0$ and the proved continuity therefore give

$$\partial_{\alpha} E_{\min}(\alpha, \beta) = -\mathcal{L}(\alpha, \beta), \quad \partial_{\beta} E_{\min}(\alpha, \beta) = \int \sqrt{x} d\mu_{\alpha, \beta}(x). \quad (\text{EIQ30})$$

No unproved differentiability of a minimizing selection is being used. For the opposite convention $F(\alpha, \beta) = \sup_{\mu} \{ \iint \log |x - y| d\mu(x) d\mu(y) - \int V_{\alpha, \beta} d\mu \} = -E_{\min}(\alpha, \beta)$, the exact derivative is therefore $\partial_{\alpha} F(\alpha, \beta) = \mathcal{L}(\alpha, \beta)$.

The second moment in (EIQ30) has an exact value. Push the minimizing measure forward by the explicit dilation $x \mapsto c^2 x$, $c > 0$. Writing $M = \int \sqrt{x} d\mu_{\alpha, \beta}(x)$, its energy minus the energy at $c = 1$ is precisely $\beta(c - 1)M - 2(\alpha + 1) \log c$. Its derivative at the minimizing value $c = 1$ is zero. Hence

$$\int \sqrt{x} d\mu_{\alpha, \beta}(x) = \frac{2(\alpha + 1)}{\beta}. \quad (\text{EIQ31})$$

The same explicit dilation, or (EIQ3)–(EIQ5), proves that the map $x \mapsto (\pi/\beta)^2 x$ pushes $\mu_{\alpha, \pi}$ to $\mu_{\alpha, \beta}$, including the density and its two endpoints. Consequently

$$\mathcal{L}(\alpha, \beta) = \mathcal{L}(\alpha, \pi) + 2 \log(\pi/\beta). \quad (\text{EIQ32})$$

For the particular original limiting potential $V(x) = \pi\sqrt{x} - 2 \log x$, the substitutions are exactly $\alpha = 2$, $\beta = \pi$. Equations (EIQ5), (EIQ10), and (EIQ26) become

$$uK(k) = 2, \quad vE(k) = 4, \quad h(x) = \frac{1}{2x} \int_0^{\infty} \frac{\sqrt{s}}{(x + s)R_s} ds, \quad \mathcal{L}(2, \pi) = \frac{1}{4} \int_0^{\infty} \frac{L_0 - L_s}{\sqrt{s}R_s} ds. \quad (\text{EIQ33})$$

In particular the preceding proof includes positivity, unit mass, the full Euler inequality, the unique global minimizer, and the parameter continuity needed at these actual parameters.

60 The exact Gamma ensemble and the leading growing return

GEL1. Source, exponents, and the literal coordinate map. The source throughout is

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad \sigma(\mathbb{R}) = \sqrt{2\pi}, \quad d\nu(t) = \frac{|\Gamma(1/4 + i\sqrt{t}/2)|^2}{2\pi\sqrt{t}} dt. \quad (\text{GEL1})$$

Here ν is the full pushforward under $y \mapsto y^2$; both half-axes are present in its density. For integers $a, n \geq 0$, retain

$$H_n^{(a)} = \det \left[\int_0^\infty t^{a+i+j} d\nu(t) \right]_{i,j < n}.$$

The definition extends to real $a > -1/2$; all its integrals converge. For $n \geq 1$, expansion of the two Vandermonde determinants followed by integration term by term gives the exact identity

$$H_n^{(a)} = \frac{1}{n!} \int_{(0,\infty)^n} \prod_{i < j} (t_i - t_j)^2 \prod_{i=1}^n t_i^a d\nu(t_i). \quad (\text{GEL2})$$

Indeed expansion gives $n!$ copies of the Gram determinant after one of the two permutation indices is fixed. Absolute integrability follows by expanding the squared polynomial and using the existing moments.

Fix a positive real coordinate scale q , independently of a, n , and make the literal map $t_i = q^2 x_i$. Introduce the reference probability measure

$$d\omega(x) = \frac{1}{2} x^{-1/2} e^{-\sqrt{x}} dx, \quad B_q(x) = 2q e^{\sqrt{x}} \sigma(q\sqrt{x}).$$

Its mass is one by $x = u^2$. This reference measure does not replace or rescale the source: the exact Radon–Nikodym identity is $d\nu(q^2 x) = B_q(x) d\omega(x)$, including its whole mass. Consequently

$$H_n^{(a)} = q^{2an+2n(n-1)} J_{n,q}(a), \quad J_{n,q}(a) = \frac{1}{n!} \int \prod_{i < j} (x_i - x_j)^2 \prod_i x_i^a B_q(x_i) d\omega(x_i). \quad (\text{GEL3})$$

The powers in this identity come separately from the n factors t_i^a and the $n(n-1)/2$ squared differences. No determinant, source mass, or Jacobian has been omitted.

GEL2. An elementary global Gamma bound with the exact exponential. Put $\lambda = 1/4$, $b \geq 0$, and $C_\Gamma = \Gamma(1/4)^2/(2\pi)$. Euler's finite limit gives

$$\frac{|\Gamma(\lambda + ib)|^2}{\Gamma(\lambda)^2} = \prod_{j=0}^{\infty} \left(1 + \frac{b^2}{(j + \lambda)^2} \right)^{-1}.$$

For clarity, the finite limit follows by applying the beta integral to $\int_0^1 t^{z-1} (1-t)^N dt = N!/[z(z+1) \cdots (z+N)]$, replacing t by u/N , and using dominated convergence for $\text{Re } z > 0$. Taking squared moduli and dividing by the same limit at $z = \lambda$ proves the displayed product. Its logarithm converges absolutely because its summand is $O((j+1)^{-2})$.

The decreasing function $v \mapsto \log(1 + b^2/v^2)$ gives, with $S = \sum_{j \geq 0} \log(1 + b^2/(j + \lambda)^2)$,

$$I \leq S \leq I + \log(1 + b^2/\lambda^2), \quad I = \int_\lambda^\infty \log(1 + b^2/v^2) dv = \pi b - \lambda \log(1 + b^2/\lambda^2) - 2b \arctan(\lambda/b).$$

The integral is evaluated by differentiating $v \log(1 + b^2/v^2) + 2b \arctan(v/b)$; at $b = 0$ it is zero by continuity. Since $0 \leq 2b \arctan(\lambda/b) \leq 2\lambda$, substitution of $b = |y|/2$ proves the global, finite inequalities

$$C_\Gamma (1 + 4y^2)^{-3/4} e^{-\pi|y|/2} \leq \sigma(y) \leq C_\Gamma e^{1/2} (1 + 4y^2)^{1/4} e^{-\pi|y|/2}. \quad (\text{GEL4})$$

Both inequalities include $y = 0$. In particular, for $C_0 = \Gamma(1/4)^2/\pi$,

$$B_q(x) \leq C_0 q e^{1/2} (1 + 2q)^{1/2} e^{-(\pi q/2 - 3/2)\sqrt{x}} \quad (x > 0). \quad (\text{GEL5})$$

Here $(1 + 4q^2 x)^{1/4} \leq (1 + 2q\sqrt{x})^{1/2} \leq (1 + 2q)^{1/2} (1 + \sqrt{x})^{1/2} \leq (1 + 2q)^{1/2} e^{\sqrt{x}/2}$. On each fixed compact interval $[A, B] \subset (0, \infty)$, the lower bound is

$$B_q(x) \geq C_0 q (1 + 4q^2 B)^{-3/4} e^{-\pi q \sqrt{x}/2}. \quad (\text{GEL6})$$

Thus the exact exponential, and the size and position of the remaining source factor, have been proved directly from the original Gamma function.

GEL3. The energy object and its original parameter domain. For $\alpha, \beta > 0$ put

$$V_{\alpha, \beta}(x) = \beta\sqrt{x} - \alpha \log x, \quad K_{\alpha, \beta}(x, y) = \log|x - y| - \frac{1}{2}V_{\alpha, \beta}(x) - \frac{1}{2}V_{\alpha, \beta}(y),$$

$$\mathcal{F}(\alpha, \beta) = \sup_{\mu \in \mathcal{P}((0, \infty))} \iint K_{\alpha, \beta}(x, y) d\mu(x) d\mu(y). \quad (\text{GEL7})$$

The kernel is bounded above, so its integral has an unambiguous value in $\mathbb{R} \cup \{-\infty\}$ for every probability in this domain. Whenever its value is finite it equals $\iint \log|x - y| d\mu^2 - \int V_{\alpha, \beta} d\mu$. Indeed $K \leq r(x) + r(y)$, with $r = \log(1 + x) - V_{\alpha, \beta}/2$ bounded above and tending to $-\infty$ at both endpoints; finite kernel integral therefore forces the $\sqrt{x}$ and $|\log x|$ moments to be finite. The positive logarithmic part is then integrable, and finite kernel integral forces the negative part to be integrable as well. This defines the extended energy without forming an indeterminate difference of infinite quantities. The equilibrium calculation supplied with this section proves existence, uniqueness, and the full exterior variational inequality for its density $\rho_{\alpha, \beta}$, with support $[u^2, v^2]$. Its exact endpoints and positive density are

$$uK(\kappa) = \frac{\alpha\pi}{\beta}, \quad vE(\kappa) = \frac{(\alpha + 2)\pi}{\beta}, \quad \kappa^2 = 1 - u^2/v^2,$$

$$\rho_{\alpha, \beta}(x) = \frac{\beta\sqrt{(v^2 - x)(x - u^2)}}{4\pi^2 x} \int_0^\infty \frac{\sqrt{s} ds}{(x + s)\sqrt{(u^2 + s)(v^2 + s)}} \quad (u^2 < x < v^2). \quad (\text{GEL8})$$

The endpoint proof shows continuous parameter dependence, finite entropy relative to ω , finite logarithmic energy, and

$$\partial_\alpha \mathcal{F}(\alpha, \beta) = \int \log x \rho_{\alpha, \beta}(x) dx. \quad (\text{GEL9})$$

In particular $\mathcal{F}$ and this derivative are continuous on the positive parameter quadrant. The following ensemble argument proves the connection to GEL3; it does not presume a random-matrix asymptotic theorem.

GEL4. Complete free-energy limit from a compactified kernel. Let $n \rightarrow \infty$, $q/n \rightarrow \theta > 0$, and let $\alpha > 0$ be fixed. Then the actual integral in GEL3 satisfies

$$\boxed{\frac{1}{n^2} \log J_{n, q}(n\alpha) \longrightarrow \mathcal{F}(\alpha, \pi\theta/2).} \quad (\text{GEL10})$$

Here is a proof of both bounds. For the upper bound GEL5 removes the factor $[C_0 q e^{1/2}(1 + 2q)^{1/2}]^n/n!$, whose logarithm divided by n^2 tends to zero. The remaining exponent is

$$2 \sum_{i < j} \log|x_i - x_j| - n \sum_i V_{\alpha, \beta_n}(x_i), \quad \beta_n = \frac{\pi q}{2n} - \frac{3}{2n}.$$

For $n > 1$ it is exactly $\sum_{i \neq j} K_{Q_n}(x_i, x_j)$, where

$$Q_n = \frac{n}{n-1} V_{\alpha, \beta_n}, \quad K_Q(x, y) = \log|x - y| - \frac{1}{2}Q(x) - \frac{1}{2}Q(y).$$

Write $\beta = \pi\theta/2$. Choose $\varepsilon > 0$ so that $\alpha - \varepsilon > 0$ and $\beta - 3\varepsilon > 0$. For all sufficiently large n , the two coefficients of Q_n differ from (α, β) by at most ε . It follows pointwise that $Q_n \geq Q_- := V_{\alpha - \varepsilon, \beta - 3\varepsilon}$. Indeed its difference has the form $b\sqrt{x} - a \log x$, where $b \geq 2\varepsilon$ and $0 \leq a \leq 2\varepsilon$; it is nonnegative for $x < 1$, and for $x \geq 1$ use $\log x \leq \sqrt{x}$.

Compactify $(0, \infty)$ by adjoining its two endpoints. The function K_{Q_-} , assigned value $-\infty$ on the diagonal and at either added endpoint, has continuous truncations $K_M = \max\{K_{Q_-}, -M\}$. To verify the endpoint assertion uniformly in the other variable, use

$$K_{Q_-}(x, y) \leq r(x) + r(y), \quad r(x) = \log(1 + x) - Q_-(x)/2.$$

The function r is bounded above and tends to $-\infty$ at both endpoints. Continuity at a diagonal point in the interior follows from $\log|x-y| \rightarrow -\infty$ and local boundedness of Q_- . For the empirical probability $L_n = n^{-1} \sum_i \delta_{x_i}$,

$$\frac{1}{n^2} \sum_{i \neq j} K_{Q_-}(x_i, x_j) \leq \iint K_M dL_n^2 + \frac{M}{n} \leq s_M + \frac{M}{n}, \quad s_M = \max_{\mu} \iint K_M d\mu^2.$$

Since ω has mass one, integration preserves this upper bound. Moreover $s_M \downarrow \mathcal{F}(\alpha - \varepsilon, \beta - 3\varepsilon)$. For completeness, choose maximizing probabilities μ_M . Compactness gives a convergent subsequence. For every fixed L , the inequality $K_M \leq K_L$ when $M \geq L$, followed by continuity of the integral of K_L , bounds the subsequential limit of s_M by $\iint K_L d\mu^2$. Decrease L 's truncation to the untruncated kernel. Monotone convergence after subtracting a common upper bound gives $\lim s_M \leq \iint K_{Q_-} d\mu^2 \leq \mathcal{F}$. The reverse inequality follows by testing every K_M on an arbitrary probability of finite energy. Probabilities charging an added endpoint have energy $-\infty$, so they do not alter the supremum. First take $n \rightarrow \infty$, then $M \rightarrow \infty$, and finally $\varepsilon \downarrow 0$. Continuity proved in GEL9 yields the upper bound in GEL10.

For the lower bound choose exactly $d\mu(x) = \rho_{\alpha, \beta}(x)dx$, supported on $[A, B]$, and use GEL6. Write $p = d\mu/d\omega$. Restricting the integral to this support and changing measure to μ^n , Jensen's inequality gives

$$\begin{aligned} \log J_{n,q}(n\alpha) &\geq -\log n! + n \log\{C_0 q(1 + 4q^2 B)^{-3/4}\} \\ &\quad + n(n-1) \iint \log|x-y| d\mu(x) d\mu(y) \\ &\quad - n^2 \int V_{\alpha, \pi q/(2n)} d\mu - n \int \log p d\mu. \end{aligned}$$

All logarithms on the right are integrable: the equilibrium density is bounded, has square-root decay at its finite positive endpoints, and ω has a positive smooth density there. The prefactor contributes $O(n \log q + n \log n)$, and the entropy contributes $O(n)$. Division by n^2 therefore yields exactly the value of GEL7 at its maximizer. This proves GEL10.

GEL5. Finite positive shifts by a proved convex squeeze. For each fixed n, q , differentiation under the integral in GEL3 on every compact subinterval of $a > -1/2$ gives

$$\frac{d}{da} \log J_{n,q}(a) = \mathbb{E}_a \sum_i \log x_i, \quad \frac{d^2}{da^2} \log J_{n,q}(a) = \text{Var}_a \left(\sum_i \log x_i \right) \geq 0. \quad (\text{GEL11})$$

The expectation is with respect to the entire positive integrand in GEL3 divided by its integral. Extra logarithmic factors are integrable at zero because $a > -1/2$, and at infinity by GEL4. Hence the functions $F_{n,q}(z) = n^{-2} \log J_{n,q}(nz)$ are differentiable and convex. Suppose now

$$q/n \rightarrow 2, \quad a/n \rightarrow 2, \quad l > 0, \quad l/n \rightarrow 0.$$

Set

$$\mathcal{L} = \int_{u^2}^{v^2} \log x \rho_{2,\pi}(x) dx, \quad uK(\kappa) = 2, \quad vE(\kappa) = 4. \quad (\text{GEL12})$$

For fixed $0 < \eta < 2$, both endpoints of the secant from a/n to $(a+l)/n$ lie in $(2 - \eta/2, 2 + \eta/2)$ for large n . Convexity bounds that secant from below by the secant from $2 - \eta$ to $2 - \eta/2$, and from above by the secant from $2 + \eta/2$ to $2 + \eta$. Apply GEL10 at these four fixed positive arguments, then let $\eta \downarrow 0$ and use GEL9. This proves

$$\frac{\log J_{n,q}(a+l) - \log J_{n,q}(a)}{nl} \rightarrow \mathcal{L}.$$

Combining this with the exact source factors in GEL3 gives

$$\log \frac{H_n^{(a+l)}}{H_n^{(a)}} = 2nl \log q + nl\mathcal{L} + o(nl). \quad (\text{GEL13})$$

This proof covers moving starting exponents a ; it never differentiates the error in a free-energy asymptotic. The remainder assertion has the precise scope $o(nl)$ along the displayed sequences.

GEL6. Both original matrices and the low moment. Return to the actual packet

$$k = 4l + 1, \quad e = 1 + k(m - 1), \quad q = e(k + 1)^2 = 2n, \quad m \geq 1 \text{ fixed}, \quad l \rightarrow \infty.$$

Retain the exact parity identity from HCT35, $Z_a = H_n^{(a)} H_{n+1}^{(a)} (H_n^{(a+1)})^2$. Use GEL13 at the four literal factors, with sizes $n, n + 1, n, n$ and starting exponents $q, q, q + 1, q + 1$, respectively. Every sequence satisfies its stated domain. Addition gives

$$\log \frac{Z_{q+l}}{Z_q} = 2(2q + 1)l \log q + (2q + 1)l\mathcal{L} + o(ql). \quad (\text{GEL14})$$

Thus the contribution from the two original full matrices has been evaluated, with all four parity blocks still identifiable.

Write $\mu_a = \int y^{2a} d\sigma(y)$. GEL4 gives, for integer $a \geq 1$,

$$\log \mu_a = 2a \log a + O(a). \quad (\text{GEL15})$$

One elementary proof of the upper bound uses $(1 + 4y^2)^{1/4} \leq 1 + 2y$ for $y \geq 0$; its integrals are $(2a)!/(\pi/2)^{2a+1}$ and $2(2a + 1)!/(\pi/2)^{2a+2}$, multiplied by the exact constant $2C_{\Gamma}e^{1/2}$. The bound $N! \leq N^N$ suffices. For the lower bound restrict the positive-axis integral to $[a, a + 1]$: there $(1 + 4y^2)^{-3/4} \geq 5^{-3/4}(a + 1)^{-3/2}$, $y^{2a} \geq a^{2a}$, and $e^{-\pi y/2} \geq e^{-\pi(a+1)/2}$. These prove both sides of GEL15 with constants independent of a .

The real-exponent function $a \mapsto \log \mu_a$ is convex by the same variance proof as GEL11. Since q is even and $1 \leq l \leq q$, its secant on $[q, q + l]$ lies between the secants on $[q/2, q]$ and $[2q, 4q]$. GEL15 bounds both latter secants in absolute value by $O(\log q)$. Therefore

$$\left| \log \frac{\mu_{q+l}}{\mu_q} \right| = O(l \log q) = o(ql). \quad (\text{GEL16})$$

The original low moment is precisely $\mu_{q+l}/\mu_q = m_{2q+2l}/m_{2q}$, so no index is reinterpreted.

GEL7. The leading value of the unchanged signed centre. Keep the source product and its exact row intervals:

$$P_k = - \sum_{a=q}^{2q-1} \sum_{j=0}^{2l-1} \log(a + j + 1/2) - \sum_{a=q+1}^{2q} \sum_{j=0}^{2l-1} \log(a + j + 1/2).$$

The two outer sums each have q terms. Since $l/q \rightarrow 0$, direct Riemann sums, with $\log$ Lipschitz on $[1, 3]$, give

$$P_k = -4ql \log q - 4ql \int_1^2 \log x \, dx + o(ql) = -4ql \log q - 4ql(2 \log 2 - 1) + o(ql). \quad (\text{GEL17})$$

Explicitly, omitting $(j + 1/2)/q$ changes each logarithm by at most $2l/q$, hence changes the total by at most $8l^2$; the ordinary left and right Riemann sums together have error $O(l)$.

The full finite product calculation WGP10 strengthens this leading identity to the following enclosure for the very same P_k :

$$P_k = -4lq \log q - 4lq(2 \log 2 - 1) - 4l^2 \log 2 + \frac{8l^3 + l}{6q} + \mathcal{E}_{q,l}, \quad (\text{GEL17a})$$

$$-\left\{ \frac{l}{6q} + \frac{8l^3 + l}{12q^3} + \frac{8l^4 + 2l^2}{3q^2} \right\} \leq \mathcal{E}_{q,l} \leq \frac{l^2}{3q^2}.$$

Its proof uses the literal midpoint coordinates in WGP3, the signed quadrature remainders WGP4–5, and the nonnegative cubic remainder WGP6–9. The subsequent exact refinement WGP12–14 gives

$$P_k = -4lq \log q - 4lq(2 \log 2 - 1) - 4l^2 \log 2 + \frac{8l^3 + l}{6q} - \frac{l^4 + l^2/4}{q^2} + \mathcal{E}_{q,l}^{(3)}, \quad (\text{GEL17b})$$

$$-\frac{l}{6q} - \frac{8l^3 + l}{12q^3} \leq \mathcal{E}_{q,l}^{(3)} \leq \frac{l^2}{3q^2} + \frac{8l^4 + 2l^2}{6q^4} + \frac{7(192l^5 + 80l^3 - 2l)}{1440q^3}.$$

These original endpoint contributions are retained when the finer q -scale calculation is continued. In particular the term $-4l^2 \log 2$ has order q at $m = 1$, and the cubic term tends to $-1/256$ there; the leading limit does not remove either term.

The original quantity is exactly $W_k = P_k - \log(\mu_{q+l}/\mu_q) + \log(Z_{q+l}/Z_q)$. Substitute GEL14, GEL16, and GEL17. The surviving $2l \log q$ is $o(ql)$, so the leading growing-degree calculation is

$$\boxed{\frac{W_k}{ql} \longrightarrow \mathcal{C}_\Gamma := 2\mathcal{L} - 4(2 \log 2 - 1), \quad \mathcal{L} = \int_{u^2}^{v^2} \log x \, \rho_{2,\pi}(x) \, dx, \quad uK(\kappa) = 2, \quad vE(\kappa) = 4.} \quad (\text{GEL18})$$

Every measure, exponent, matrix dimension, and source factor entering this limit has been connected to its original object by GEL1–3 and GEL14. The exact positive density and unique endpoint equations in GEL8 make the constant an explicitly defined one-dimensional equilibrium integral. This leading result has remainder $o(ql)$. It supplies no claim of an $o(q)$ remainder, and hence does not erase the next finite-scale calculation required by the original arithmetic receiver.

The fully proved original-polynomial recurrence estimate RWB20–27 also applies to this identical W_k . It gives the finite bounds $lq/64 \leq W_k \leq 6lq$, and its narrower finite lower remainder gives $\liminf W_k/(lq) \geq 4 \log(27/(8\pi))$. Combining those proved inequalities with the now proved existence of GEL18 yields

$$\boxed{4 \log \frac{27}{8\pi} \leq \mathcal{C}_\Gamma \leq 6, \quad \mathcal{C}_\Gamma > 0.} \quad (\text{GEL18a})$$

The strict sign is analytic: RWB22 proves $4 \log(27/(8\pi)) > 104/365 > 0$ using rational integral bounds for π and the elementary logarithm inequality. Numerical evaluation of the equilibrium integral is unnecessary for this sign conclusion.

Finally LET20, whose error is $O(k) = o(q)$ at fixed original m, δ, γ , transfers the proved limit without changing its sign:

$$\frac{\mathfrak{T}_k}{ql} \longrightarrow \mathcal{C}_\Gamma, \quad \frac{R_k^0 - R_k^\sigma}{ql} \longrightarrow -\mathcal{C}_\Gamma. \quad (\text{GEL19})$$

The arithmetic identity remains $B_{\text{ar}} = B_0 + \delta_\sigma - \mathfrak{T}_k$, including its original δ_σ allowance. Thus GEL19 evaluates the Gamma summand and its exact signed map into that identity; it makes no unsupported growth assertion about either of the other two summands.

Original complete tilted-mass and source-envelope providers

61 The full tilted mass in the original arithmetic tail

This calculation uses the original complete hypothetical actual quartet, not a substituted arithmetic density. Keep

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad h(s) = \prod_{\lambda \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s - \lambda)^m, \quad m \geq 1,$$

and retain the entire unit $v_h = 2\xi/h$ and the measures

$$\begin{aligned} w_h(y) &= |v_h(1/2 + iy)|^2/(2\pi), \quad m_k = w_h^{*k}, \quad \alpha = \pi/2, \\ f(y) &= e^{\alpha y} w_h(y), \quad \mathfrak{M} = M_\alpha = \int_{\mathbb{R}} f(y) dy, \\ d\sigma(y) &= \sigma(y) dy = |\Gamma(1/4 + iy/2)|^2 dy/(2\pi), \quad \sigma(\mathbb{R}) = \sqrt{2\pi}. \end{aligned} \tag{RTM1}$$

The original convolution envelope at three factors, the boundary tilt bound, and the two-sided Gamma bounds are exactly TW7 at $k = 3$ and BT9–17:

$$\begin{aligned} w_h^{*3}(y) &\geq c_h e^{-\alpha|y|} (1 + |y|)^{-B}, \quad B = 42 + 8m, \\ 0 \leq f(y) &\leq C_h^{\text{tilt}} (1 + |y|)^{-p}, \quad p = 8m - 9/2 \geq 7/2, \\ c_\Gamma e^{-\alpha|y|} (1 + |y|)^{-1/2} &\leq \sigma(y) \leq C_\Gamma e^{-\alpha|y|} (1 + |y|)^{-1/2}, \\ c_\Gamma &= \sqrt{2} e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}} e^{2/3}. \end{aligned} \tag{RTM2}$$

Here c_h and C_h^{tilt} are the unchanged explicit constants from those proved providers. Their source is still

$$w_h(y) = \frac{(y^2 + 1/4)^2}{\sqrt{\pi}} \frac{|\zeta(1/2 + iy)|^2}{[(y - \gamma)^2 + \delta^2][(y + \gamma)^2 + \delta^2]} \sigma(y).$$

No positive lower bound for its one-factor zeta numerator is used.

61.1 Exact moments of the original tilted convolution

Define the following actual source integrals, preserving their masses:

$$\begin{aligned} J_h &= \int |y| f(y) dy, \quad \mu_h = \mathfrak{M}^{-1} \int y f(y) dy, \\ v_h^{\text{tilt}} &= \mathfrak{M}^{-1} \int (y - \mu_h)^2 f(y) dy, \\ 0 < J_h &\leq \frac{2C_h^{\text{tilt}}}{p-2}, \quad 0 < \mu_h \leq J_h/\mathfrak{M}, \quad 0 < v_h^{\text{tilt}} \leq \frac{2C_h^{\text{tilt}}}{\mathfrak{M}(p-3)}. \end{aligned} \tag{RTM3}$$

The symbol v_h^{tilt} is the displayed scalar variance, not the entire Taylor unit v_h . Finiteness follows from BT17 at orders one and two, which are both admitted even when $m = 1$. The mean is strictly positive because w_h is even and positive almost everywhere, and

$$\int y f(y) dy = \int_0^\infty y (e^{\alpha y} - e^{-\alpha y}) w_h(y) dy > 0.$$

The variance is strictly positive because f is not supported at one point. Expanding its centered square gives the upper bound by the second uncentered moment in RTM3.

For $n = k - 3 \geq 2$, let $F_n = f^{*n}$. The original convolution map has Jacobian one, and Tonelli, followed by the triangle inequality, gives

$$\begin{aligned} \int F_n(z) dz &= \mathfrak{M}^n, & \int |z| F_n(z) dz &\leq n J_h \mathfrak{M}^{n-1}, \\ \int (z - n\mu_h) F_n(z) dz &= 0, & \int (z - n\mu_h)^2 F_n(z) dz &= n v_h^{\text{tilt}} \mathfrak{M}^n. \end{aligned} \quad (\text{RTM4})$$

For the last identity, pull the integral back to the n original variables and expand $(\sum_j (y_j - \mu_h))^2$. Every cross term vanishes since $\int (y - \mu_h) f(y) dy = 0$. The n diagonal terms are each $v_h^{\text{tilt}} \mathfrak{M}^n$. All exchanges are justified by the finite first and second absolute moments in RTM3. These are identities for the full, unaltered tilted mass; no probability measure is substituted for an arithmetic source.

61.2 A tail lower bound with the complete mass

Put

$$H_k = \sqrt{2(k-3)v_h^{\text{tilt}}}, \quad Y_k^{\text{cen}} = \max\{2, (k-3)\mu_h + H_k\}, \quad Y_k^{\text{abs}} = \max\{2, 2(k-3)J_h/\mathfrak{M}\}. \quad (\text{RTM5})$$

The centered-square identity in RTM4 proves

$$\int_{|z - n\mu_h| \leq H_k} F_n(z) dz \geq \mathfrak{M}^n/2 :$$

on the complementary set its integrand multiplied by H_k^2 is at most the integrand of the centered second moment. The same argument with the absolute first moment proves

$$\int_{|z| \leq Y_k^{\text{abs}}} F_n(z) dz \geq \mathfrak{M}^n/2.$$

Inside every convolution product, $f^{*j}(x) = e^{\alpha x} w_h^{*j}(x)$, exactly, because the original coordinates sum to x . Thus RTM2 gives $f^{*3}(x) \geq c_h(1+x)^{-B}$ whenever $x \geq 0$. If $y \geq Y_k^{\text{cen}}$ and $|z - n\mu_h| \leq H_k$, then

$$0 \leq y - z \leq y - n\mu_h + H_k.$$

Restrict the positive convolution $f^{*3} * F_n$ to this entire retained-mass set. The preceding inequalities prove

$$m_k(y) \geq \frac{c_h}{2} \mathfrak{M}^{k-3} e^{-\alpha|y|} (1 + |y| - (k-3)\mu_h + H_k)^{-B} \quad (|y| \geq Y_k^{\text{cen}}). \quad (\text{RTM6})$$

The negative half-line follows from the original evenness of m_k . The denominator is positive on the stated domain; at its unrounded threshold it equals $1 + 2H_k$.

Alternatively the absolute-moment retained set proves the simpler bound

$$m_k(y) \geq \frac{c_h}{2} \mathfrak{M}^{k-3} e^{-\alpha|y|} (1 + 2|y|)^{-B} \quad (|y| \geq Y_k^{\text{abs}}). \quad (\text{RTM7})$$

In this proof $0 \leq y - z \leq y + Y_k^{\text{abs}} \leq 2y$ on the retained set when $y \geq Y_k^{\text{abs}}$. Both cutoffs are $O_h(k)$. For example RTM3 gives the fully explicit upper cutoff

$$Y_k^{\text{abs}} \leq \max \left\{ 2, \frac{4(k-3)C_h^{\text{tilt}}}{\mathfrak{M}(p-2)} \right\}.$$

The new feature is the exact factor $\mathfrak{M}^{k-3}$ instead of a truncated mass M_R^{k-3} . No loss of tilted mass is hidden in the cutoff or a limiting argument.

61.3 The original logarithmic shape on the growing source region

Retain the exact moving degree

$$k = 4l + 1 \geq 5, \quad e = 1 + k(m - 1), \quad q = e(k + 1)^2, \quad q \geq (k + 1)^2.$$

For $y \geq Y_k^{\text{cen}}$, divide RTM6 by the upper Gamma bound in RTM2. Divide the existing BT21 upper convolution envelope by the lower Gamma bound. These operations give the exact two-sided result

$$\begin{aligned} \log \frac{m_k(y)}{\sigma(y)} &\geq (k - 3) \log \mathfrak{M} + \log \frac{c_h}{2C_\Gamma} + \frac{1}{2} \log(1 + y) - B \log(1 + y - (k - 3)\mu_h + H_k), \\ \log \frac{m_k(y)}{\sigma(y)} &\leq (k - 3) \log \mathfrak{M} + \log \frac{C_h^{\text{tilt}}}{c_\Gamma} + 2 \log \mathfrak{M} + \log k + \frac{1}{2} \log(1 + y) - p \log(1 + y/k). \end{aligned} \quad (\text{RTM8})$$

There is no replacement of the Gamma density by its asymptotic in these finite inequalities.

Fix $0 < a < b < \infty$ and put $y = qx$ with $x \in [a, b]$. Both cutoffs are eventually satisfied, uniformly in x , since $k/q \rightarrow 0$. For sufficiently large k the numbers $1/y$, k/y , and $(1 - (k - 3)\mu_h + H_k)/y$ have absolute values at most $1/2$. The identity $\log(1 + z) = \int_0^z (1 + t)^{-1} dt$ gives $|\log(1 + z)| \leq 2|z|$ there. Hence RTM8 implies the following two distinct polynomial-log bounds, with their full constants:

$$\begin{aligned} \log \frac{m_k(qx)}{\sigma(qx)} &\geq (k - 3) \log \mathfrak{M} + (1/2 - B) \log(qx) + \log \frac{c_h}{2C_\Gamma} - O_{h,a,b}(k/q), \\ \log \frac{m_k(qx)}{\sigma(qx)} &\leq (k - 3) \log \mathfrak{M} + (1/2 - p) \log(qx) + (p + 1) \log k + \log \frac{C_h^{\text{tilt}}}{c_\Gamma} + 2 \log \mathfrak{M} + O_{h,a,b}(k/q). \end{aligned} \quad (\text{RTM9})$$

In particular, uniformly on this compact range,

$$\log \frac{m_k(qx)}{\sigma(qx)} = (k - 3) \log \mathfrak{M} + O_{h,a,b}(\log q). \quad (\text{RTM10})$$

RTM9 has different lower and upper powers. It does not identify a single polynomial-log shape with an $o(1)$ remainder for the actual ratio, nor does it evaluate its signed determinant return.

The retained tail is connected to the original convolution by the literal exact expression

$$\begin{aligned} e^{\alpha y} m_k(y) &= \int_{\mathbb{R}} f^{*3}(y - z) F_{k-3}(z) dz, \\ \int F_{k-3} &= \mathfrak{M}^{k-3}, \quad \int (z - (k - 3)\mu_h)^2 F_{k-3} = (k - 3) v_h^{\text{tilt}} \mathfrak{M}^{k-3}. \end{aligned} \quad (\text{RTM11})$$

Every complementary contribution remains in this equality. The proved lower bounds use only a positive part of it, and make no claim that the complementary part vanishes.

Exact remaining scope. These are estimates on the original density at the specified source points. Contributions from $|y| = O_h(k)$ to the original growing polynomial Grams have not been removed. In the simple-quartet case their natural k^2 scale can be the same order as q . Consequently RTM6–10 alone do not justify an order- q signed four-endpoint determinant calculation or a first-variation value. The source-coordinate map remains $P(S) \mapsto P(k/2 + iy)$ with its inverse, the actual packet relation remains χ , and all proper-source kernels and Taylor-unit observations are unchanged.

62 A global tilted-mass transfer with the central source retained

Retain the original source and constants RTM1–3. In particular $\mathfrak{M} = M_\alpha$ is its full boundary-tilted mass, J_h is its actual first absolute tilted moment, $B = 42 + 8m$, $M = B/2 = 21 + 4m$, and

$p = 8m - 9/2$. Keep every full packet order and every original tensor order $k = 4l + 1 \geq 5$, with $q = [1 + k(m - 1)](k + 1)^2$. The TW global bounds, on their unchanged measures, are

$$a_k(1 + y^2)^{-M}\sigma(y) \leq m_k(y) \leq A_k\sigma(y),$$

$$a_k = \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-B}}{C_\Gamma 2^{B/2}}, \quad A_k = \frac{C_h^{\text{tilt}}}{c_\Gamma} k^{p+1} \mathfrak{M}^{k-1}, \quad \vartheta_h = \int_{-1}^1 w_h(y) dy. \quad (\text{GTM1})$$

The following calculation retains the central source region, including its finite polynomial cost. It proves an $O_h(q \log q)$ bound, including $m = 1$; it does not evaluate an order- q signed coefficient.

62.1 An exact global lower measure

Define finite positive scalars and an auxiliary integer by

$$Y = \max\{2, 2(k-3)J_h/\mathfrak{M}\}, \quad b_\dagger = \frac{c_h}{2C_\Gamma 2^{3B/2}}, \quad \lambda_k = b_\dagger \mathfrak{M}^{k-3},$$

$$r = \max\left\{1, \left\lceil \frac{\log(\lambda_k/a_k)}{\log 2} \right\rceil\right\}, \quad Q(y) = (y + iY)^r (y + i)^M, \quad F(y) = y^r Q(y), \quad d = 2r + M. \quad (\text{GTM2})$$

These auxiliary polynomials occur only in proved comparison maps on the same source. They do not replace the actual packet relation χ or the observation on its quotient. The ceiling argument is exactly

$$\log(\lambda_k/a_k) = (k-3)\{\log(\mathfrak{M}/\vartheta_h) + \alpha\} + B \log((k-2)/2) - \log 2. \quad (\text{GTM3})$$

In particular $r = O_h(k)$ and $Y = O_h(k)$, including for a simple quartet.

RTM7 and the upper Gamma bound give, when $|y| \geq Y$,

$$m_k(y) \geq \frac{c_h}{2C_\Gamma} \mathfrak{M}^{k-3} (1 + 2|y|)^{-B} \sigma(y) \geq \lambda_k (1 + y^2)^{-M} \sigma(y).$$

The first inequality has only discarded the factor $(1 + |y|)^{1/2} \geq 1$. For the second use $(1 + 2|y|)^B \leq 2^B (1 + |y|)^B \leq 2^{3B/2} (1 + y^2)^{B/2}$. The multiplier $(y^2/(Y^2 + y^2))^r$ is at most one on this region. On $|y| \leq Y$ it is at most 2^{-r} , and GTM2 ensures $\lambda_k 2^{-r} \leq a_k$. The original global lower bound GTM1 therefore proves, on the entire real source,

$$\boxed{m_k(y) \geq \lambda_k \left| \frac{y^r}{(y + iY)^r (y + i)^M} \right|^2 \sigma(y), \quad y \in \mathbb{R}.} \quad (\text{GTM4})$$

The denominators have no real zeros. In particular GTM4 has not deleted the interval $|y| = O_h(k)$: its cost is recorded by r, Y .

62.2 A finite determinant inequality on the full polynomial space

On the same monomial frame $1, y, \dots, y^N$, put $H_{k,N} = \text{Gram}_{m_k}(y^j)_{j=0}^N$ and $H_N^\sigma = \text{Gram}_\sigma(y^j)_{j=0}^N$. They are positive definite. Write $n = N + 1$, $L = N + d$, and define the positive n by n matrices

$$\mathcal{A}_{ij} = \int y^{2r+i+j} |Q(y)|^{-2} d\sigma(y),$$

$$\mathcal{B}_{ij} = \int y^{2r+i+j} d\sigma(y), \quad \mathcal{C}_{ij} = \int y^{2r+i+j} |Q(y)|^2 d\sigma(y), \quad 0 \leq i, j \leq N. \quad (\text{GTM5})$$

The Gram of the lists $y^{r+j}/Q(y)$ and $y^{r+j}\overline{Q(y)}$ is $\begin{pmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{B} & \mathcal{C} \end{pmatrix}$. Completing the square proves $\mathcal{A} \succeq \mathcal{B}\mathcal{C}^{-1}\mathcal{B}$. Together with GTM4 this gives

$$\det H_{k,N} \geq \lambda_k^n (\det \mathcal{B})^2 / \det \mathcal{C}. \quad (\text{GTM6})$$

The order of the noncommuting matrices has not been changed.

The original Gamma monic polynomials satisfy the proved exact formulas

$$\gamma_j = \|b_j\|_\sigma^2 = \sqrt{2\pi}(2j)!/4^j, \quad yb_j = b_{j+1} + j(j-1/2)b_{j-1}, \quad b_0 = 1. \quad (\text{GTM7})$$

The missing term at $j = 0$ is zero. A monic change has determinant one, so $\det H_N^\sigma = \prod_{j=0}^N \gamma_j$. The exterior product of $y^r, \dots, y^{N+r}$ has coefficient one on $b_r \wedge \dots \wedge b_{N+r}$. Expanding in the orthogonal exterior basis, whose squared coefficients are nonnegative, proves

$$\det \mathcal{B} \geq \prod_{j=r}^{N+r} \gamma_j. \quad (\text{GTM8})$$

Let $\mathcal{J}$ take all divided derivatives at $0, -iY, -i$ with multiplicities r, r, M in that order. Its kernel in degree at most L is exactly $F\mathcal{P}_N$. Its restriction $\mathcal{V}$ to degrees below d has determinant

$$\det \mathcal{V} = (-iY)^{r^2} (-i)^{rM} \{i(Y-1)\}^{rM}, \quad |\det \mathcal{V}| = Y^{r^2} (Y-1)^{rM} \geq 1. \quad (\text{GTM9})$$

This is the divided-derivative confluent Vandermonde: replace each root by distinct nearby points, divide the ordinary Vandermonde by the three within-block factors, and expand their evaluation rows in Taylor series. The first nonzero coefficient gives the divided derivatives and exactly the cross-root factors in GTM9. In particular the observation is onto, with every auxiliary jet retained.

In the Gamma orthonormal frame $b_j/\sqrt{\gamma_j}$ its covariance is

$$(\mathcal{K}_L)_{(z,a),(z',a')} = \sum_{j=0}^L \frac{b_j^{[a]}(z) \overline{b_j^{[a']}(z')}}{\gamma_j}. \quad (\text{GTM10})$$

The exact relation determinant is

$$\det \mathcal{C} = \left(\prod_{j=0}^L \gamma_j \right) \frac{\det \mathcal{K}_L}{|\det \mathcal{V}|^2}. \quad (\text{GTM11})$$

Indeed the frame $1, \dots, y^{d-1}, F, \dots, Fy^N$ has determinant one. Its lower Gram block is $\mathcal{C}$, and its Schur complement is the minimum-observation Gram $\mathcal{V}^* \mathcal{K}_L^{-1} \mathcal{V}$. The latter follows directly by solving the constrained minimum: for source Gram H and onto map J , its section is $H^{-1} J^* (JH^{-1} J^*)^{-1}$. Subtracting it leaves a vector in $\ker J$ orthogonal to the section. Taking the determinant of this block Gram proves GTM11, with its displayed factors.

62.3 All finite auxiliary-jet and factorial costs

Define

$$Z_{L,Y} = \frac{\exp(2)(L+1)(L+2)(2L+1)}{\sqrt{2\pi}} [2(L+2)]^{Y+1}. \quad (\text{GTM12})$$

Then $\det \mathcal{K}_L \leq Z_{L,Y}^d$. Here is the coefficient estimate. GTM7 gives the generating function

$$\sum_{j \geq 0} \frac{b_j(z)}{j!} t^j = (1+t^2)^{-1/4} \exp(z \arctan t), \quad |t| < 1.$$

To derive it, multiply the recurrence by $t^j/j!$ and sum; $(1+t^2)\partial_t G = (z-t/2)G$ with $G(0, z) = 1$ determines the displayed analytic solution and its coefficients. On $|t| = (L+1)/(L+2)$, $|\arctan t| \leq \frac{1}{2} \log(2(L+2))$, $|1+t^2| \geq (L+2)^{-2}$, and $|t|^{-j} \leq \exp(1)$ for $j \leq L$. Cauchy's coefficient bound therefore gives

$$|b_j(z)|/j! \leq \exp(1)(L+2)^{1/2} [2(L+2)]^{(Y+1)/2} \quad (|z| \leq Y+1).$$

Applying the derivative formula on circles of radius one about $0, -iY, -i$ gives the same bound for every divided derivative used in GTM10. No factorial is removed. Also

$$(j!)^2/\gamma_j = 4^j/(\sqrt{2\pi}\binom{2j}{j}) \leq (2j+1)/\sqrt{2\pi},$$

because the central coefficient is at least the average of all $2j+1$ binomial coefficients. Summing squared derivatives proves that every diagonal entry of $\mathcal{K}_L$ is at most $Z_{L,Y}$. Successive orthogonal projection in its Gram frame gives the determinant bound as the product of at most these diagonal entries.

Put $g_0 = \log \sqrt{2\pi}$ and retain the complete finite cost

$$\begin{aligned} \mathcal{E}_{N,r,Y} &= 2r^2 \log(N+2r) + M\{g_0 + 2L \log L\} + 2rg_0 + 4r^2 \log \max(1, r) + (2r+M) \log Z_{L,Y}, \\ L &= N+2r+M. \end{aligned} \tag{GTM13}$$

Every term is finite and nonnegative on the present domain. GTM6–12 give

$$(N+1) \log \lambda_k - \mathcal{E}_{N,r,Y} \leq \log \frac{\det H_{k,N}}{\det H_N^\sigma} \leq (N+1) \log A_k. \tag{GTM14}$$

For clarity, the factorial expression on the lower side before estimation is exactly

$$T = 2 \sum_{j=r}^{N+r} \log \gamma_j - \sum_{j=0}^{N+2r+M} \log \gamma_j - \sum_{j=0}^N \log \gamma_j.$$

Writing these sums as differences of prefix sums, without a condition such as $r \leq N$, gives

$$T = \sum_{j=1}^r \{\log \gamma_{N+j} - \log \gamma_{N+r+j}\} - \sum_{j=N+2r+1}^{N+2r+M} \log \gamma_j - 2 \sum_{j=0}^{r-1} \log \gamma_j.$$

The exact ratio $\gamma_{j+1}/\gamma_j = (j+1)(j+1/2)$ bounds the first negative sum by $2r^2 \log(N+2r)$. The elementary bound $\log \gamma_j \leq g_0 + 2j \log \max(1, j)$, obtained from $(2j)! \leq (2j)^{2j}$, bounds the other two negative sums by the corresponding terms of GTM13. Finally subtract $d \log Z_{L,Y}$ and use $|\det \mathcal{V}| \geq 1$. This proves the lower side of GTM14 with all central-region, auxiliary-jet and factorial costs retained. The upper side is the determinant of GTM1's full matrix upper bound.

62.4 A finite deficit and the simple-quartet improvement

Define on the original source

$$\Xi_{k,N} = (N+1) \log A_k - \log \frac{\det H_{k,N}}{\det H_N^\sigma} \geq 0. \tag{GTM15}$$

The nonnegative sign follows from the full upper Gram inequality. There is no loss of tilted mass in the following exact scalar:

$$\log(A_k/\lambda_k) = (p+1) \log k + \log \frac{C_h^{\text{tilt}}}{c_\Gamma b_\dagger} + 2 \log \mathfrak{M}.$$

Thus GTM14 proves the finite bound

$$0 \leq \Xi_{k,N} \leq (N+1) \left\{ (p+1) \log k + \log \frac{C_h^{\text{tilt}}}{c_\Gamma b_\dagger} + 2 \log \mathfrak{M} \right\} + \mathcal{E}_{N,r,Y}. \tag{GTM16}$$

Replacing the last two scalar logarithms in braces by the absolute value of their sum gives a nonnegative, possibly looser, bound. The displayed unaltered expression is also valid when $\mathfrak{M} < 1$.

At $N \in \{q-1, q, 2q-1, 2q\}$, GTM3 and RTM3 give $r = O_h(k)$, $Y = O_h(k)$, $d = O_h(k)$ and $L = N + O_h(k)$. GTM12 implies $\log Z_{L,Y} = O_h(k \log(N+k+2))$. Every term of GTM13 is consequently bounded, with its actual degree, by

$$\mathcal{E}_{N,r,Y} = O_h(N \log(N+k+2) + k^2 \log(N+k+2)).$$

Since $q \geq (k+1)^2$ and $q = O_m(k^3)$, GTM16 proves

$$\boxed{\Xi_{k,N} = O_h(q \log q) \quad (N = q-1, q, 2q-1, 2q), \quad m \geq 1.} \quad (\text{GTM17})$$

In particular this includes $m = 1$, $q = (k+1)^2$. The term $O_h(k^2 \log q)$ from the central bridge has been retained, not declared $o(q)$ in that case. This is precisely why GTM17 does not assert an order- q asymptotic.

62.5 Exact transfer to the original quotient and its mixed maps

Return to $S = c + iy$. The coefficient map $P(S) \mapsto P(c + iy)$ has diagonal $1, i, \dots, i^N$ and determinant of modulus one. Applying the same congruence to both source Grams leaves GTM14–17 and every relative determinant unchanged. For the remainder map $J_N : \mathcal{P}_N \rightarrow E$ with its actual kernel $\chi \mathcal{P}_{N-q}$, put

$$G_N^\mu = (J_N(H_N^\mu)^{-1} J_N^*)^{-1}, \quad \Xi_{k,N}^E = q \log A_k - \log \frac{\det G_N^{\text{ar}}}{\det G_N^\sigma}.$$

Then

$$0 \leq \Xi_{k,N}^E \leq \Xi_{k,N}. \quad (\text{GTM18})$$

Here is the complete matrix argument. Set $D = A_k^{-1}(H_N^\sigma)^{-1/2} H_N^{\text{ar}} (H_N^\sigma)^{-1/2}$, so $0 \prec D \preceq I$, and let $V = (H_N^\sigma)^{-1/2} J_N^* (G_N^\sigma)^{1/2}$. The identity $V^* V = I$ makes this an isometry. In an orthonormal completion of its columns, the normalized quotient Gram is the Schur complement

$$S = (V^* D^{-1} V)^{-1} = A_k^{-1} (G_N^\sigma)^{-1/2} G_N^{\text{ar}} (G_N^\sigma)^{-1/2}.$$

Writing D in two blocks gives $\det D = \det D_{22} \det S$, with $0 \prec D_{22} \preceq I$ and $0 \prec S \preceq I$. Thus $-\log \det S$ is nonnegative and at most $-\log \det D$, proving GTM18. An empty complementary block has determinant one, covering $N = q-1$.

On a fixed actual original mixed observation $O : E \rightarrow Y$ onto its image frame, keep the complete kernel inclusion I_K and put

$$H_{K,N}^\mu = I_K^* G_N^\mu I_K, \quad Q_{Y,N}^\mu = (O(G_N^\mu)^{-1} O^*)^{-1}, \quad a = \dim K, \quad b = \dim Y, \quad a + b = q.$$

Define the kernel and image deficits using their own dimensions,

$$\Xi_{k,N}^K = a \log A_k - \log \frac{\det H_{K,N}^{\text{ar}}}{\det H_{K,N}^\sigma}, \quad \Xi_{k,N}^Y = b \log A_k - \log \frac{\det Q_{Y,N}^{\text{ar}}}{\det Q_{Y,N}^\sigma}.$$

Then

$$\Xi_{k,N}^K \geq 0, \quad \Xi_{k,N}^Y \geq 0, \quad \Xi_{k,N}^K + \Xi_{k,N}^Y = \Xi_{k,N}^E \leq \Xi_{k,N}. \quad (\text{GTM19})$$

To prove the identity, choose any fixed right inverse S_0 of O . The fixed coordinate frame $[I_K, S_0]$ is invertible. Its block Gram has determinant $\det H_{K,N}^\mu \det Q_{Y,N}^\mu$ by the same minimum-section Schur complement. The squared determinant of the coordinate frame cancels between the two original metrics. Nonnegativity follows by restriction and by minimization from $G_N^{\text{ar}} \preceq A_k G_N^\sigma$. This proof preserves the complete original kernel and image, including empty ones, and never identifies either with the auxiliary jet kernel. For the original QDC observed boundary, take $O = \Lambda$ and $Y = \mathcal{B}$: then the observed quotient is literally $Q_N^{\mathcal{B}} = (\Lambda G_N^{-1} \Lambda^*)^{-1}$ and the kernel metric is $I_K^* G_N I_K$. When

the actual joint observation O is selected instead, its own fixed image and kernel are used. All these statements concern an individual original G_N . The proper polynomial-boundary difference metric $G_N - G_{N'}$ is a different stated expression; subtracting two source-comparison inequalities does not give a bound for its cross-source relative determinant. No such bound is asserted here.

For every $N \leq 2q$, $\Xi_{k,N} \leq \Xi_{k,2q}$. Indeed the literal coefficient inclusion into $\mathcal{P}_{2q}$, transported by the Gamma square roots, is an isometry; its compression of the normalized positive matrix $D_{2q} \preceq I$ is precisely D_N . Complete that isometry orthogonally. The remaining Schur complement is at most the identity, so $\det D_{2q} \leq \det D_N$ and the asserted deficit inequality follows.

Take the original endpoint signs $(+, +, -, -)$ at $(q - 1, q, 2q - 1, 2q)$. Each common scalar dimension times $\log A_k$ cancels because the sign sum is zero. The remaining positive and negative sides each contain two nonnegative deficits bounded by $\Xi_{k,2q}$. Consequently, for the original signed quotient correction and its separate kernel and observable corrections,

$$\boxed{\max\{|\delta_k^\sigma|, |\Delta F_k^K|, |\Delta F_k^Y|\} \leq 2\Xi_{k,2q} = O_h(q \log q), \quad m \geq 1.} \quad (\text{GTM20})$$

Current nested transfer of the same full-source deficit. AFT3 proves the exact original primary Vandermonde congruence for this source Gram. AFT6–17 then carries this same $\Xi_{k,2q}$ through the actual conductor, kernel, common and complementary flag, with their full fixed frame determinant. The sum of the four nonnegative deficits is precisely the primary deficit; every subset of their signed returns has the single bound $2\Xi_{k,2q}$. This is the transfer used in AFT27, without discarding any proper kernel or mixed Schur block.

The finite upper bound for the right side is exactly twice GTM16 at $N = 2q$, with the integer guard GTM2 and complete cost GTM13. Every relative frame and dimension has been specified before performing the signed cancellation. For $O = \Lambda$, the third term of GTM20 is exactly the original observed-boundary correction $|\Delta F_k^B|$; it is not the norm of the proper polynomial boundary in the TCD construction.

The same improvement propagates through the existing complete scalar return. Retain its original identity $S_k = \mathcal{B}_k^\sigma + \delta_k^\sigma - 4q \log(D_h k)$, its exact GQB22 Gamma-energy bound $E_B(q)$, and its complete packet comparison C_k^{cmp} . The proved GQB7 and GQB22 give $|\mathcal{B}_k^\sigma - C_B q^2| \leq E_B(q) + (2q + 2)C_k^{\text{cmp}}$. Inserting GTM20 in this unchanged signed identity proves

$$\boxed{\begin{aligned} |S_k - C_B q^2| &\leq E_B(q) + (2q + 2)C_k^{\text{cmp}} + 2\Xi_{k,2q} + 4q|\log(D_h k)|, \\ S_k &= C_B q^2 + O_h(q \log q), \quad m \geq 1. \end{aligned}} \quad (\text{GTM21})$$

For the second line use the already proved $E_B(q) = O(q \log q)$, $C_k^{\text{cmp}} = O_h(e^{-1}) = O_h(1)$, the fixed positive D_h , and GTM17. All finite packet cutoffs required by those Gamma providers remain in the first line.

Original signed source-relation receiver

63 The signed return through the actual adjacent relation and quotient

Retain the full original hypothetical packet $\rho = 1/2 + \delta + i\gamma$, $0 < \delta < 1/2$, $\gamma > 2$, and its complete order m . Set

$$\begin{aligned} g &= 2\xi, \quad h(s) = \prod_{z \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s-z)^m, \quad v_h = g/h, \\ w_h(y) &= |v_h(1/2 + iy)|^2/(2\pi), \quad m_k = w_h^{*k}, \quad k = 4l + 1 \geq 5, \quad e = 1 + k(m-1), \\ q &= e(k+1)^2, \quad c = k/2, \quad \chi(S) = \prod_{a,b=0}^k (S - c - (2a-k)\delta - i(2b-k)\gamma)^e. \end{aligned} \quad (\text{SXR1})$$

The full Taylor unit, source masses, period parameter and tensor coordinates remain fixed. All limits fix this complete packet and use the displayed admitted tensor orders. The original zero hypothesis is the one in the total counterfactual; no numerical offcritical zero is supplied or asserted by this calculation.

63.1 The unchanged source path and its exact derivatives

The original CA14a source interpolation is, with its parameter written x to distinguish it from the period u ,

$$d\mu_x(y) = (1-x)d\sigma(y) + xm_k(y)dy, \quad d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad 0 \leq x \leq 1. \quad (\text{SXR2})$$

In particular its mass is $(1-x)\sqrt{2\pi} + x(\int w_h)^k$. There is no replacement by a probability measure. On $\mathcal{P}_d = \mathbb{C}[S]_{\leq d}$ use the basis $1, S, \dots, S^d$ and

$$(H_d(x))_{ab} = \int \overline{(c+iy)^a} (c+iy)^b d\mu_x(y), \quad 0 \leq a, b \leq d.$$

These positive matrices have $d+1$ rows. Embeddings into $\mathcal{P}_{2q}$ are literal zero extensions. Write

$$M(x) = H_{2q}(x), \quad \dot{M} = H_{2q}^{\text{ar}} - H_{2q}^{\sigma}, \quad C(x) = M(x)^{-1} \dot{M}. \quad (\text{SXR3})$$

The dot denotes this source derivative only. The operator C is self-adjoint in M , since $MC = \dot{M} = \dot{M}^*$. For any two polynomials in the ambient space,

$$\langle P, CQ \rangle_x = \int \overline{P(c+iy)} Q(c+iy) (m_k(y) - \sigma(y)) dy.$$

This identity proves its exact connection to the original density change. It is not a freely chosen coefficient operator.

Let $p_d(x)$ be the original monic source orthogonal polynomial and $u_r(x)$ the monic polynomial in the original relation norm. Thus

$$\omega_d(x) = \|p_d(x)\|_x^2, \quad \nu_r(x) = \|\chi u_r(x)\|_x^2, \quad T_r(x) = \frac{\nu_r(x)}{\omega_{q+r}(x)}, \quad d = q + r. \quad (\text{SXR4})$$

The polynomial u_r is orthogonal to degrees below r for the literal measure $|\chi(c+iy)|^2 d\mu_x(y)$. Its full S -coordinate monicity is retained. Positive finite moment Grams give unique monic minimizers, and solving their linear systems shows that their coefficients are smooth rational functions of x .

Denominators are positive on the whole compact interval. Differentiating the minimum norms gives exactly

$$\dot{\omega}_d = \langle p_d, C p_d \rangle_x, \quad \dot{\nu}_r = \langle \chi u_r, C \chi u_r \rangle_x. \quad (\text{SXR5})$$

Indeed $\dot{p}_d$ has degree below d and $\chi \dot{u}_r$ lies in $\chi \mathcal{P}_{r-1}$. Both coefficient-derivative cross terms vanish by their respective defining orthogonalities. This proves SXR5 without differentiating an asymptotic remainder.

The source is even. In real y coordinates its monic orthogonal polynomial has simple real roots by the sign-change proof in TCN4. The polynomial $\chi(c + iy)$ has nonreal roots, so χu_r cannot be that monic minimizer. Therefore $T_r(x) > 1$ for every x and $0 \leq r \leq q$. The phase change is exactly $P(S) \mapsto i^{-d} P(c + iy)$ in degree d ; no source norm changes.

63.2 An orthogonal exchange in the actual quotient section

Put $b_d(x) = [p_d(x)]_\chi$ in the original quotient $E = \mathbb{C}[S]/(\chi)$ and keep

$$G_d = (J_d H_d^{-1} J_d^*)^{-1}, \quad R_d = H_d^{-1} J_d^* G_d.$$

For $d = q + r$ define the following actual source vectors:

$$\begin{aligned} f_r &= \frac{p_d}{\sqrt{\omega_d}}, \quad g_r = \frac{\chi u_r}{\sqrt{\nu_r}}, \quad \tau_r = T_r^{-1}, \quad z_r = \frac{b_d}{\sqrt{\omega_d(1 - \tau_r)}}, \\ w_r &= \frac{f_r - \sqrt{\tau_r} g_r}{\sqrt{1 - \tau_r}} = R_d z_r. \end{aligned} \quad (\text{SXR6})$$

All square roots here are the positive square roots of the original positive real norms. To prove the equalities, monicity and the degree- d orthogonality give $\langle p_d, \chi u_r \rangle_x = \omega_d$, hence $\langle g_r, f_r \rangle_x = \sqrt{\tau_r}$ with this exact positive phase. The complete relation space in degree d is $\chi \mathcal{P}_r$. Both p_d and χu_r are orthogonal to $\chi \mathcal{P}_{r-1}$, and the coefficient of the last orthogonal relation vector in p_d is ω_d/ν_r . Consequently

$$R_d b_d = p_d - \frac{\omega_d}{\nu_r} \chi u_r, \quad b_d^* G_d b_d = \omega_d(1 - T_r^{-1}). \quad (\text{SXR7})$$

This is the actual full relation projection, not a deletion of its highest row. It proves SXR6 and $\|g_r\|_x = \|w_r\|_x = 1$, $\langle g_r, w_r \rangle_x = 0$. The vector g_r belongs to the original relation space and w_r to the original minimum quotient section in the same degree.

For a unit vector v write $P_v = v v^* M$ for its source-orthogonal projector. SXR5 gives the exact identity

$$\frac{d}{dx} \log T_r = \text{Tr}((P_{g_r} - P_{f_r})C) = \langle g_r, C g_r \rangle_x - \langle w_r, C w_r \rangle_x + \text{Tr}((P_{w_r} - P_{f_r})C). \quad (\text{SXR8})$$

In the orthonormal basis (g_r, w_r) , the last projector difference is exactly

$$P_{w_r} - P_{f_r} = \begin{pmatrix} -\tau_r & -\sqrt{\tau_r(1 - \tau_r)} \\ -\sqrt{\tau_r(1 - \tau_r)} & \tau_r \end{pmatrix}.$$

Its eigenvalues are $\sqrt{\tau_r}, -\sqrt{\tau_r}$; all remaining ambient eigenvalues are zero. If $\text{osc}_x C = \lambda_{\max}(C) - \lambda_{\min}(C)$, the Rayleigh quotients at the two unit eigenvectors therefore prove

$$|\text{Tr}((P_{w_r} - P_{f_r})C)| \leq T_r^{-1/2} \text{osc}_x C. \quad (\text{SXR9})$$

Every scalar part of C cancels exactly because this projector difference has trace zero. A bound by an uncentered source norm is neither needed nor substituted for the oscillation.

63.3 All four endpoint signs before the estimate

Let $V_N(x) = \det G_N(x)$ and retain precisely

$$\begin{aligned} \mathcal{B}_k(x) &= \log V_{q-1}(x) + \log V_q(x) - \log V_{2q-1}(x) - \log V_{2q}(x), \\ \delta_k^\sigma &= \mathcal{B}_k(1) - \mathcal{B}_k(0), \quad c_0 = c_q = 1, \quad c_r = 2 \quad (1 \leq r \leq q-1). \end{aligned} \quad (\text{SXR10})$$

The monic source/relation change of basis has determinant one, so $V_{q+r-1} = \prod_{d=0}^{q+r-1} \omega_d / \prod_{j=0}^{r-1} \nu_j$. Taking consecutive ratios and then the four displayed endpoint signs gives the exact finite equality $\mathcal{B}_k(x) = \sum_{r=0}^q c_r \log T_r(x)$. In particular $\sum c_r = 2q$, with coefficient one at both endpoints.

Define the original relation-versus-quotient receiving integral

$$\mathcal{Q}_k = \int_0^1 \sum_{r=0}^q c_r (\langle g_r, Cg_r \rangle_x - \langle w_r, Cw_r \rangle_x) dx. \quad (\text{SXR11})$$

This uses the original source path, actual minimum sections, full relation rows and positive monic phases of SXR6. It is an explicitly constructed scalar of that source. SXR8 gives

$$\delta_k^\sigma = \mathcal{Q}_k + \mathcal{E}_k, \quad \mathcal{E}_k = \int_0^1 \sum_{r=0}^q c_r \text{Tr}((P_{w_r} - P_{f_r})C) dx. \quad (\text{SXR12})$$

There is no assertion that either summand has a necessary sign.

Let b_-, b_+ be the extreme eigenvalues of the unchanged relative Gram $(H_{2q}^\sigma)^{-1} H_{2q}^{\text{ar}}$. Diagonalizing its self-adjoint representative in the original Gamma metric gives the exact eigenvalues of $C(x)$: $(b_j - 1)/(1 - x + xb_j)$. This expression increases in b_j , with derivative $(1 - x + xb_j)^{-2} > 0$. Direct integration gives

$$\int_0^1 \text{osc}_x C dx = \log(b_+/b_-). \quad (\text{SXR13})$$

The formula at $b_j = 1$ follows by continuity. It preserves even a large common source mass until the ratio cancels it exactly.

63.4 A finite certificate and an exponentially small error

Keep the original TW constants, without changing their values:

$$\frac{a_k}{D_d} \|P\|_\sigma^2 \leq \|P\|_{\text{ar}}^2 \leq A_k \|P\|_\sigma^2, \quad \deg P \leq d, \quad L_{k,d} = \log(A_k D_d / a_k).$$

Their full definitions are TCN10 and the unchanged TW/BT providers in the source closure; $L_{k,2q} = O_h(k + \log q)$. On SXR2 they give lower and upper factors $l_d(x) = 1 - x + xa_k/D_d$, $u(x) = 1 - x + xA_k$. The ratio $u(x)/l_d(x)$ increases from 1 to $A_k D_d / a_k$: its derivative has numerator $A_k - a_k/D_d \geq 0$. The last inequality follows already by applying TW to a nonzero polynomial. Taking the actual monic minima in numerator and denominator therefore proves, simultaneously for every x ,

$$T_r(x) \geq e^{-L_{k,2q}} T_r^\sigma, \quad 0 \leq r \leq q, \quad \log(b_+/b_-) \leq L_{k,2q}. \quad (\text{SXR14})$$

Here T_r^σ uses the same complete χ and its monic relation norm in the original Gamma source, not a replacement arithmetic packet.

It is a finite explicit determinant quantity. With $L_r^\sigma = \text{Gram}_\sigma(\chi S^j)_{0 \leq j < r}$, $\det L_0^\sigma = 1$, define

$$\Theta_k = e^{-L_{k,2q}} \min_{0 \leq r \leq q} \frac{\det L_{r+1}^\sigma}{\det L_r^\sigma} \frac{4^{q+r}}{\sqrt{2\pi}(2q+2r)!}. \quad (\text{SXR15})$$

Every determinant here is in its original monic frame and retains the full Gamma mass. For example its moments are obtained exactly by differentiating $\int e^{ity} d\sigma(y) = \sqrt{2\pi}(\cosh t)^{-1/2}$ at zero. This is the same Gamma Fourier identity used in the retained source provider, with its

stated mass. No arithmetic zero is needed to compute these reference matrices at fixed packet parameters.

Combining SXR9, SXR12–15 and the strict inequality $T_r(x) > 1$ proves the finite bound

$$\boxed{|\delta_k^\sigma - \mathcal{Q}_k| \leq 2q \min\{1, \Theta_k^{-1/2}\} \log(b_+/b_-) \leq 2qL_{k,2q} \min\{1, \Theta_k^{-1/2}\}.} \quad (\text{SXR16})$$

All $q+1$ adjacent exchanges and both original endpoint weights were included before this estimate was taken.

For completeness the new TCN/TCX estimates evaluate the size of this finite bound. They give

$$\min_{0 \leq r \leq q} \log T_r^\sigma = 2qa_1 + O_h(k + \log q), \quad a_1 = \psi(1) > 0. \quad (\text{SXR17})$$

Here is the required uniform argument rather than an inference from a four-endpoint assertion. For $0 \leq r \leq \lfloor q/64 \rfloor$, TCX2–9’s original-root interval and monic arcsine bound apply to σ with comparison factor one. They give a lower rate c_*q , $c_* > 1/3 > 2a_1$, with error $O_h(k + \log q)$. For the remaining integers, TCN13 and TCN17–26 give $\log T_r^\sigma = 2q\psi(r/q) + O_h(k + \log q)$ uniformly on the fixed compact ratio interval $[1/128, 1]$, retaining the original packet cutoff. Since ψ decreases, this is at least $2qa_1 - O_h(k + \log q)$. At $r = q$ the same formula gives a matching upper bound for the minimum. The original cutoff holds eventually because $q \geq (k+1)^2$ and $R = k\sqrt{\delta^2 + \gamma^2}$. This proves SXR17 and $\log \Theta_k = 2qa_1 + O_h(k + \log q)$. Consequently the exact error in SXR12 satisfies

$$\boxed{|\mathcal{E}_k| \leq \exp\{-qa_1 + O_h(k + \log q)\} = o(q).} \quad (\text{SXR18})$$

This is a bound on an actual signed correction term, not a claimed evaluation of the still-present $\mathcal{Q}_k$.

63.5 Return to the original joint kernel and observable image

At fixed original nonzero period and branch, keep the complete observation $O : E \rightarrow \mathbb{C}^b$ onto its fixed image frame, with its full Taylor units, reduced mixed map and ordered periods. It does not vary with the source interpolation. Let I_K be the fixed kernel frame. At each actual degree $d = q + r$ set

$$\begin{aligned} Q_d &= (OG_d^{-1}O^*)^{-1}, \quad P_{K,d} = I_K(I_K^*G_dI_K)^{-1}I_K^*G_d, \\ P_{Y,d} &= G_d^{-1}O^*Q_dO, \quad w_{K,r} = R_dP_{K,d}z_r, \quad w_{Y,r} = R_dP_{Y,d}z_r. \end{aligned} \quad (\text{SXR19})$$

Empty kernels or images use their unique empty matrices. These are the original source-Gram minimum sections, at their written degrees; no boundary metric is substituted. Multiplying the formulas gives $OP_{K,d} = 0$, $OP_{Y,d} = O$ and orthogonality of their ranges in G_d . Hence $P_K + P_Y = I$ and $w_r = w_{K,r} + w_{Y,r}$, with the two summands orthogonal in $M(x)$. The literal receiving integrand is therefore

$$\begin{aligned} \langle g_r, Cg_r \rangle_x - \langle w_r, Cw_r \rangle_x &= \langle g_r, Cg_r \rangle_x - \langle w_{K,r}, Cw_{K,r} \rangle_x - \langle w_{Y,r}, Cw_{Y,r} \rangle_x \\ &\quad - 2\Re \langle w_{K,r}, Cw_{Y,r} \rangle_x. \end{aligned} \quad (\text{SXR20})$$

Orthogonality for M does not remove the last pairing with C . It is retained, with its original sign, inside $\mathcal{Q}_k$. The quotient coefficient is exactly z_r in SXR6, and its observed value is exactly Oz_r . There is no assertion that it is the distinguished primary or that its observable fraction is uniformly nonzero. On the primary, the earlier nonvanishing subsequence and the undivided JQC identity retain their proved domains.

The relation χ_{u_r} has its original ordered theta primitive at W_* or a containing source. Its proper-source class and full h^2 -jet keep the TCD3–5 map and image restrictions. Applying the linear source maps coefficientwise retains every coefficient face, outer source label and nonempty leg mask. A zero remains at its present support and is not the external τ . Thus the controlled

exchange has been returned to the original mixed construction with its exact morphisms; it has not erased that construction's kernel.

Equations SXR12, SXR16 and SXR20 leave a precisely specified original mixed scalar with an exponentially bounded exchange error. The sign and order- q value of this scalar have not been computed here. Neither the new error estimate nor the separate positive exterior energy proves RH or certifies an offcritical zero.

Original polynomial quotient and tensor maps

64 The original period vectors and their Gamma generating map

The observation in this calculation is the literal BRU1–3 matrix. We retain its entire amplitude unit, its primary multiplicities, its period contours, its cyclic projector and its kernel. The only source orders used are the already present orders $s \in \{1, k\}$. The letter s in a source-order subscript is an integer parameter; the coordinate on the one-factor polynomial algebra will be denoted z_1 below to keep it distinct. The generating variable is z .

64.1 The original finite algebras and the observation

Retain

$$\begin{aligned} \rho &= \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad \mathcal{R} = \{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}, \\ h(z_1) &= \prod_{\alpha \in \mathcal{R}} (z_1 - \alpha)^m, \quad n = 4m, \quad v_h = g/h, \quad g = 2\xi, \quad m \geq 1, \\ E_h &= \mathbb{C}[z_1]/(h), \quad A_h = M_{z_1} : E_h \rightarrow E_h, \quad U = M_{j_h v_h} : E_h \rightarrow E_h. \end{aligned} \tag{OPG1}$$

Here m is the original complete zero order at each of the four points. Consequently $v_h(\alpha) \neq 0$ for every $\alpha \in \mathcal{R}$. The entire-function remainder $j_h v_h$ records its derivatives of orders $0, \dots, m-1$ at every one of these points. In each primary algebra it has nonzero constant term, and its inverse is the finite geometric series for that constant plus a nilpotent. Thus U is the same invertible unit as in BRU, with none of its values prescribed.

For the original positive integer k , set

$$\begin{aligned} c &= k/2, \quad \ell_k = 1 + k(m-1), \quad q = (k+1)^2 \ell_k, \\ \chi(S) &= \prod_{a,b=0}^k (S - c - (2a-k)\delta - i(2b-k)\gamma)^{\ell_k}, \\ E &= \mathbb{C}[S]/(\chi), \quad A = M_S : E \rightarrow E, \quad Y = (A - cI)/i, \\ A_\Sigma &= \sum_{j=1}^k I_{E_h}^{\otimes(j-1)} \otimes A_h \otimes I_{E_h}^{\otimes(k-j)}, \quad \iota : E \hookrightarrow E_h^{\otimes k}, \quad [P] \mapsto [P(z_1 + \dots + z_k)]. \end{aligned} \tag{OPG2}$$

For clarity, the stated map is injective with exactly this defining polynomial. On an ordered primary tensor the eigenvalue of A_Σ is the sum of the ordered roots. Its nilpotent part is the sum of the k independent nilpotents of orders m . Its power $k(m-1)$ has nonzero coefficient $(k(m-1))!/((m-1)!)^k$ on their top monomial, and its next power is zero. The distinct sums are exactly the $(k+1)^2$ displayed roots of χ , since the signs of the real and imaginary parts can be selected independently in every factor. Distinct pairs (a, b) give distinct sums by $\delta, \gamma > 0$. Their common nilpotency index is ℓ_k . These facts prove that the minimal polynomial of A_Σ is χ , and therefore prove the kernel of polynomial evaluation and the asserted injection. In particular

$$\iota A = A_\Sigma \iota, \quad \iota Y = \frac{A_\Sigma - cI}{i} \iota, \quad \iota[1] = [1]^{\otimes k}. \tag{OPG3}$$

Keep the original $u \neq 0$, its logarithm branch, and $t = 0$ in the period construction. With $\Phi'_h = h$, $\Phi_h(0) = 0$, let ℓ_j be the ray of angle $(\arg u + \pi + 2\pi j)/(n+1)$ and let $\Gamma_j = -\ell_0 + \ell_j$, $1 \leq j \leq n$, with the original orientations. The original period map is

$$\Pi : E_h \rightarrow \mathbb{C}^n, \quad (\Pi[P])_j = \int_{\Gamma_j} e^{\Phi_h(z_1)/u} \text{rem}_h P(z_1) dz_1. \tag{OPG4}$$

The representative on the right has degree less than n . These are exactly the original convergent polynomial period integrals; their existence and full determinant are proved in PDE1–13 and

IPM1–7 in the retained source closure. This definition does not assign a period to an arbitrary representative of a quotient class. Let C be the original matrix with rows corresponding to $\Gamma_j \mapsto \Gamma_{j+1} - \Gamma_1$ for $j < n$ and $\Gamma_n \mapsto -\Gamma_1$. The reduced endpoint basis gives $C^{n+1} = I$. Hence the literal finite average below is idempotent:

$$P_k = \frac{1}{n+1} \sum_{a=0}^n (C^{-a})^{\otimes k}, \quad L = P_k \Pi^{\otimes k} U^{\otimes k} \iota, \quad (OPG5)$$

$$B = \text{im } L, \quad j : B \hookrightarrow (\mathbb{C}^n)^{\otimes k}, \quad \Lambda : E \twoheadrightarrow B, \quad j\Lambda = L, \quad K = \ker \Lambda.$$

The basis inclusion j , the inclusion $I : K \hookrightarrow E$, and a coefficient section $S_* : B \rightarrow E$ with $\Lambda S_* = I_B$ are the fixed BRU choices. The full actual image is used at every rank. Empty boundary matrices use the original empty conventions.

64.2 An exact generating identity before and after observation

Put $b_s = s/2$, $M_s = (2\pi)^{s/2}$, and retain the monic Gamma polynomials and their full norms from GMB1:

$$p_0(y) = 1, \quad p_1(y) = y, \quad p_{d+1}(y) = yp_d(y) - d(b_s + d - 1)p_{d-1}(y), \quad (OPG6)$$

$$h_d^{(s)} = M_s d! (b_s)_d, \quad r_d = p_d(Y)[1], \quad e_d = r_d / \sqrt{h_d^{(s)}}, \quad \lambda_d = \Lambda e_d \in B.$$

The square root is the positive real square root. In particular $e_d = [u_d]$ in the original notation, including the phase i^{-d} in its leading coefficient. Define the holomorphic germs of $\log(1+z^2)$ and $\arctan z$ by their values zero at $z = 0$. For $|z| < 1$, the exact algebra-valued identity is

$$\sum_{d=0}^{\infty} r_d \frac{z^d}{d!} = (1+z^2)^{-s/4} \exp((\arctan z)Y)[1]. \quad (OPG7)$$

Indeed the right side is holomorphic in that disc, is $[1]$ at zero, and satisfies $(1+z^2)F'(z) = (Y - b_s z I)F(z)$. Comparing its Taylor coefficients gives exactly the recurrence in OPG6: the contribution of $z^2 F'$ to coefficient $z^d/d!$ is $d(d-1)r_{d-1}$, and the contribution of $b_s z F$ is $b_s d r_{d-1}$. Thus its derivatives at zero are the r_d , which proves the convergent identity and not merely a formal coefficient equality.

All factors of A_Σ act on different tensor coordinates and therefore commute. Their matrix exponential power series are absolutely convergent in finite dimension. The multinomial expansion of every power consequently proves

$$\exp\left(w \frac{A_\Sigma - cI}{i}\right) [1]^{\otimes k} = \left(\exp\left(w \frac{A_h - \frac{1}{2}I}{i}\right) [1]\right)^{\otimes k}. \quad (OPG8)$$

The centre on the left is exactly $c = k/2$; each of the k centres on the right is exactly $1/2$. Applying OPG3–5 to OPG7, and then using OPG8, gives the original observed generating map

$$\boxed{j \sum_{d=0}^{\infty} \sqrt{h_d^{(s)}} \lambda_d \frac{z^d}{d!} = (1+z^2)^{-s/4} P_k \left[\Pi U \exp\left((\arctan z) \frac{A_h - \frac{1}{2}I}{i}\right) [1] \right]^{\otimes k}}. \quad (OPG9)$$

The scalar outside P_k has exponent $-s/4$. It has not acquired an extra factor of k : OPG7 concerns one original Gamma source on the sum coordinate, while OPG8 is the exact factorization of that coordinate action in the already existing tensor algebra. For $s = 1$ and $s = k$ this gives the two stated source versions of the same original observation. No additional order is selected.

64.3 All primary coefficients and the original period integrals

Here is a finite formula for every vector in OPG9. For $\alpha \in \mathcal{R}$ write

$$\begin{aligned} h_\alpha(z_1) &= h(z_1)/(z_1 - \alpha)^m, & t_\alpha(z_1) &= \sum_{j=0}^{m-1} \frac{(h_\alpha^{-1})^{(j)}(\alpha)}{j!} (z_1 - \alpha)^j, \\ \varepsilon_\alpha &= [h_\alpha t_\alpha] \in E_h, & E_{\alpha,a}(z_1) &= \text{rem}_h(h_\alpha(z_1)t_\alpha(z_1)(z_1 - \alpha)^a), \quad 0 \leq a < m, \\ \mathbf{b}_{\alpha,a} &= \Pi[E_{\alpha,a}], & (\mathbf{b}_{\alpha,a})_j &= \int_{\Gamma_j} e^{\Phi_h(z_1)/u} E_{\alpha,a}(z_1) dz_1. \end{aligned} \quad (\text{OPG10})$$

The Taylor inverse exists because $h_\alpha(\alpha) \neq 0$. Reduction modulo $(z_1 - \alpha)^m$ gives $\varepsilon_\alpha = 1$ on its primary factor and zero on every other primary factor. These observations prove $\varepsilon_\alpha \varepsilon_\beta = 0$ for $\alpha \neq \beta$, $\varepsilon_\alpha^2 = \varepsilon_\alpha$, and $\sum_\alpha \varepsilon_\alpha = 1$: their differences are divisible by all four coprime factors and hence by h . They also prove that $[E_{\alpha,a}] = \varepsilon_\alpha(z_1 - \alpha)^a$ is a basis of E_h . Every representative in OPG10 has degree less than n , so each integral is precisely of the original polynomial type.

For $w \in \mathbb{C}$ put $\mathbf{v}(w) = \Pi U \exp(w(A_h - \frac{1}{2}I)/i)[1]$. The primary nilpotent $N_\alpha = (z_1 - \alpha)\varepsilon_\alpha$ satisfies $N_\alpha^m = 0$. Taylor multiplication in that algebra therefore proves, with all amplitude derivatives retained,

$$\mathbf{v}(w) = \sum_{\alpha \in \mathcal{R}} e^{w(\alpha-1/2)/i} \sum_{a=0}^{m-1} \mathbf{b}_{\alpha,a} \sum_{\nu=0}^a \frac{v_h^{(\nu)}(\alpha)}{\nu!} \frac{(w/i)^{a-\nu}}{(a-\nu)!}. \quad (\text{OPG11})$$

For a direct verification before integration, both $U \exp(w(A_h - \frac{1}{2}I)/i)[1]$ and $j_h(v_h(z_1)e^{w(z_1-1/2)/i})$ have the same first m Taylor coefficients at every α . Their difference is therefore zero in E_h . The representative to which Π is applied is exactly the finite polynomial in the basis $E_{\alpha,a}$ whose coefficients are displayed in OPG11. In particular this proof does not replace the integrand in OPG10 by the unreduced entire function $v_h(z_1)e^{w(z_1-1/2)/i}$. The precise map from that entire function to the period vector is its finite Taylor remainder j_h , followed by the degree-less-than- n representative, followed by the original polynomial period integral.

The cyclic projector is also fully evaluated as a finite operation:

$$j \sum_{d \geq 0} \sqrt{h_d^{(s)}} \lambda_d \frac{z^d}{d!} = \frac{(1+z^2)^{-s/4}}{n+1} \sum_{a=0}^n (C^{-a} \mathbf{v}(\arctan z))^{\otimes k}. \quad (\text{OPG12})$$

This is obtained by applying each tensor power in OPG5 to a pure tensor. It retains every term of the original average, its factor $1/(n+1)$, and its inverse powers of C .

There is also a finite coefficient formula requiring no series extraction. For an ordered k -tuple $\alpha \in \mathcal{R}^k$ set $\omega_\alpha = (\sum_i \alpha_i - c)/i$. For $0 \leq a_i < m$ and $0 \leq \nu_i \leq a_i$ put $D(\mathbf{a}, \boldsymbol{\nu}) = \sum_i (a_i - \nu_i)$ and

$$T_d(\alpha, \mathbf{a}) = \sum_{\substack{0 \leq \nu_i \leq a_i \\ 1 \leq i \leq k}} \frac{i^{-D(\mathbf{a}, \boldsymbol{\nu})} p_d^{(D(\mathbf{a}, \boldsymbol{\nu}))}(\omega_\alpha)}{\prod_{i=1}^k (a_i - \nu_i)!} \prod_{i=1}^k \frac{v_h^{(\nu_i)}(\alpha_i)}{\nu_i!}. \quad (\text{OPG13})$$

As usual the polynomial derivative is zero when its order exceeds d . Multivariate Taylor expansion in the independent nilpotents $(z_i - \alpha_i)$ proves the exact formula

$$j \lambda_d = \frac{1}{\sqrt{h_d^{(s)}}} \frac{1}{n+1} \sum_{a=0}^n \sum_{\alpha \in \mathcal{R}^k} \sum_{\substack{0 \leq a_i < m \\ 1 \leq i \leq k}} T_d(\alpha, \mathbf{a}) \bigotimes_{i=1}^k C^{-a} \mathbf{b}_{\alpha_i, a_i}. \quad (\text{OPG14})$$

Indeed on that primary tensor, expand each $v_h(z_i)$ first. For the remaining multi-degree $a_i - \nu_i$, Taylor expansion of $p_d((\sum_i z_i - c)/i)$ supplies its derivative of the total order D , the factor i^{-D} , and the factorials $\prod_i (a_i - \nu_i)!$. These are exactly the coefficients in OPG13. Apply the k polynomial period maps and the literal cyclic average to get OPG14. All sums are finite. Ordered tuples have not been replaced by just their eigenvalue sum, so both the amplitude data and every independent primary nilpotent survive.

64.4 The actual observation increment without a coordinate inverse

Retain the GMB maps $C_s = [e_0, \dots, e_{q-1}] : \mathbb{C}^q \rightarrow E$, $v_j = C_s^{-1}e_{q+j}$, $K_j = I_q + \sum_{a < j} v_a v_a^*$, and $R_s = \Lambda C_s$. The low-degree leading coefficients prove C_s invertible as in GMB2. On each column, applying the maps gives

$$R_s v_j = \lambda_{q+j}, \quad R_s K_j R_s^* = \sum_{d=0}^{q+j-1} \lambda_d \lambda_d^*. \quad (\text{OPG15})$$

For the second equality the identity term in K_j gives exactly the first q columns of R_s , and each remaining rank-one term gives the next displayed column. The low vectors λ_d , $0 \leq d < q$, span B because C_s is invertible and Λ is onto. Thus, in the fixed original basis of B , the matrix

$$\mathcal{M}_D = \sum_{d=0}^{D-1} \lambda_d \lambda_d^* \quad (D \geq q) \quad (\text{OPG16})$$

is positive definite whenever B has positive dimension: for a nonzero covector its quadratic form is the sum of its squared values on spanning vectors. Consequently the exact original boundary increment of GMB8 is

$$\boxed{\xi_j = \lambda_{q+j}^* \mathcal{M}_{q+j}^{-1} \lambda_{q+j}, \quad 1 + \xi_j = \frac{\det \mathcal{M}_{q+j+1}}{\det \mathcal{M}_{q+j}}.} \quad (\text{OPG17})$$

The determinant equality follows by conjugating the rank-one update by $\mathcal{M}_{q+j}^{-1/2}$; its sole possible nonzero added eigenvalue is the displayed quadratic form. If $B = 0$, the vector is empty, $\xi_j = 0$ and both determinants equal one. Thus OPG11–14 compute the boundary increment from the original period vectors, with no C_s^{-1} in OPG17. This cancellation is an identity of the proved maps, rather than a change of the observation or the source measure.

At the four original endpoints the boundary contribution is therefore the exact finite expression

$$F_B^{(s)} = \log \frac{\det \mathcal{M}_{2q} \det \mathcal{M}_{2q+1}}{\det \mathcal{M}_q \det \mathcal{M}_{q+1}} = \log(1 + \xi_0) + 2 \sum_{j=1}^{q-1} \log(1 + \xi_j) + \log(1 + \xi_q). \quad (\text{OPG18})$$

Expanding the last sum of consecutive log determinant differences leaves coefficient -1 at $\mathcal{M}_q, \mathcal{M}_{q+1}$ and $+1$ at $\mathcal{M}_{2q}, \mathcal{M}_{2q+1}$, proving every sign. GMB13 proves that the other original factor is $(1 + t_j)/(1 + \xi_j)$, with $t_j = v_j^* K_j^{-1} v_j$ and $0 \leq \xi_j \leq t_j$. Thus OPG18 is the actual BRU boundary contribution, and the full original kernel contribution remains $\mathcal{W}_q(\log((1 + t_j)/(1 + \xi_j)))$.

64.5 A finite recurrence retaining the exact action defect

The period formula also has a finite recurrence on the fixed original coefficient spaces. Define the uniquely typed projection $\kappa : E \rightarrow K$ by $I\kappa = I_E - S_*\Lambda$. It exists because the right side has image in $\ker \Lambda = \text{im } I$, and I is an inclusion. It satisfies $\kappa I = I_K$ and $\kappa S_* = 0$. In particular $E = I(K) \oplus S_*(B)$ as coefficient spaces, without an assertion that this splitting is preserved by Y . Set

$$\begin{aligned} Y_B &= \Lambda Y S_* : B \rightarrow B, & D &= \Lambda Y I : K \rightarrow B, \\ C &= \kappa Y S_* : B \rightarrow K, & Y_K &= \kappa Y I : K \rightarrow K, & \zeta_d &= \kappa e_d \in K. \end{aligned} \quad (\text{OPG19})$$

All four maps are computable from OPG1–5 and the fixed BRU section. For $d \geq 0$ put

$$a_d = \frac{1}{\sqrt{(d+1)(b_s + d)}}, \quad b_d = \sqrt{\frac{d(b_s + d - 1)}{(d+1)(b_s + d)}} \quad (d \geq 1), \quad b_0 = 0. \quad (\text{OPG20})$$

Set $\lambda_{-1} = 0$ and $\zeta_{-1} = 0$. Then the exact coupled recurrence is

$$\boxed{\begin{aligned}\lambda_{d+1} &= a_d(\mathbf{Y}_B \lambda_d + \mathbf{D} \zeta_d) - b_d \lambda_{d-1}, \\ \zeta_{d+1} &= a_d(\mathbf{C} \lambda_d + \mathbf{Y}_K \zeta_d) - b_d \zeta_{d-1}, \\ \lambda_0 &= \Lambda[1]/\sqrt{M_s}, \quad \zeta_0 = \kappa[1]/\sqrt{M_s}.\end{aligned}} \quad (\text{OPG21})$$

To prove this, the norm ratio from OPG6 gives $\sqrt{h_d^{(s)}/h_{d+1}^{(s)}} = a_d$ and $d(b_s + d - 1)\sqrt{h_{d-1}^{(s)}/h_{d+1}^{(s)}} = b_d$. Hence $e_{d+1} = a_d Y e_d - b_d e_{d-1}$, including $d = 0$ by its initial values. Substituting $e_d = S_* \lambda_d + I \zeta_d$ and applying Λ and κ proves both lines of OPG21. It retains the kernel coordinate at every degree and supplies the period vectors through degree $2q$, exactly the range used in OPG17–18.

The failure of a putative boundary action has the precise map

$$\Lambda Y - \mathbf{Y}_B \Lambda = \mathbf{D} \kappa : E \rightarrow B, \quad j\mathbf{D} = \frac{1}{i} P_k \Pi^{\otimes k} U^{\otimes k} (A_\Sigma - cI) \iota I : K \rightarrow (\mathbb{C}^n)^{\otimes k}. \quad (\text{OPG22})$$

The first equality follows from $I_E = S_* \Lambda + I \kappa$; the second follows from OPG3–5. These equations preserve the cyclic average and the full unit. They also prove exactly that $Y(K) \subseteq K$ holds if and only if $\mathbf{D} = 0$. No such vanishing is used anywhere in OPG7–21. Thus every possible boundary action defect is retained inside a proved coupled action and inside the explicit original period formula.

For completeness the multiplication action does descend to a canonical quotient that remembers every observed iterate. Define

$$\mathcal{K}_\infty = \bigcap_{a=0}^{q-1} \ker(\Lambda Y^a), \quad \mathcal{O} : E \rightarrow B^{\oplus q}, \quad x \mapsto (\Lambda x, \Lambda Y x, \dots, \Lambda Y^{q-1} x). \quad (\text{OPG23})$$

The polynomial $\chi(c + iT)$ annihilates Y by OPG2, has degree q and leading coefficient $i^q \neq 0$. It therefore expresses Y^q as a linear combination of $I, Y, \dots, Y^{q-1}$, with the exact coefficients of this original polynomial. If $x \in \mathcal{K}_\infty$, each component of $\mathcal{O}(Yx)$ vanishes by this relation. Thus $\mathcal{K}_\infty$ is Y -invariant and is precisely $\bigcap_{a \geq 0} \ker(\Lambda Y^a)$. Moreover any Y -invariant subspace contained in K lies in this intersection. Consequently the first isomorphism map

$$\begin{aligned} E/\mathcal{K}_\infty &\xrightarrow{\overline{\mathcal{O}}} \text{im } \mathcal{O}, \quad [x] \mapsto \mathcal{O}(x), \\ \chi_j &= [T^j] \chi(c + iT), \quad \chi_q = i^q, \\ \text{Shift}(b_0, \dots, b_{q-1}) &= \left(b_1, \dots, b_{q-1}, -i^{-q} \sum_{j=0}^{q-1} \chi_j b_j \right). \end{aligned} \quad (\text{OPG24})$$

is a bijection intertwining multiplication by Y with the shift of the q components, whose last component is the stated exact linear combination. It projects to the original observation by the first component, with retained kernel $K/\mathcal{K}_\infty$. This proves an action-preserving enlargement of the observed data on the original algebra. It neither discards the original kernel in OPG21 nor presumes a vanished defect in OPG22.

The original period isomorphism also locates that defect in the cyclic projection. Its inverse exists by the retained period determinant theorem used in OPG4. On the unchanged ambient tensor space put

$$\mathcal{V} = \Pi^{\otimes k} U^{\otimes k} \iota, \quad \mathcal{T} = \sum_{j=1}^k \left(\Pi \frac{A_h - \frac{1}{2} I}{i} \Pi^{-1} \right)_j. \quad (\text{OPG25})$$

The subscript j means that the displayed factor acts in position j and identities act in all other positions. Multiplication by $j_h v_h$ commutes with A_h . Equations OPG3 and OPG25 therefore give

$\mathcal{V}Y = \mathcal{T}\mathcal{V}$ exactly. Since $L = P_k\mathcal{V}$ and $P_k^2 = P_k$, splitting the ambient identity as $P_k + (I - P_k)$ now proves

$$\begin{aligned} LY &= P_k\mathcal{T}L + P_k\mathcal{T}(I - P_k)\mathcal{V}, \\ jD &= P_k\mathcal{T}(I - P_k)\mathcal{V}I. \end{aligned} \tag{OPG26}$$

For the second equality, $P_k\mathcal{V}I = LI = 0$, so the first term of the first equality vanishes on $I(K)$. This is the same defect as OPG22 on its original domain. The two ambient terms in the first line sum to a vector in $j(B)$; neither individual term is asserted to preserve that image. OPG19–21 give the full recurrence within $K \oplus B$ without such an assertion. Thus the preserved unit and period isomorphism carry the multiplication action exactly, and OPG26 identifies precisely the retained contribution of the original cyclic projection.

Equations OPG9, OPG11–14, OPG17 and OPG21 now give the same original boundary increments by analytic generating functions, finite primary coefficients, and finite coupled recurrences. Their derivations establish the exact maps connecting these calculations. In particular the new input into the original mixed-control problem is the fully specified sequence $\lambda_0, \dots, \lambda_{2q}$ with its actual unit and periods, rather than a freely selected observation or a source-order replacement. The formulas themselves do not assign a growing- parameter estimate to the individual weighted sums; they supply their unchanged mathematical input for that calculation.

Original period-coefficient and pivot restriction provider

65 The literal original conductor limit and an invertible pivot restriction

Retain all original hypotheses and maps of PQO, RPC, OCF and CEP. In particular $0 < \delta < 1/2$, $\gamma > 2$, the original unit values are $(a, \bar{a}, \bar{a}, a)$ with $a \neq 0$, and the same logarithm branch is used throughout. Write $\beta = 2(\gamma^2 - \delta^2)$, $\eta = (\delta^2 + \gamma^2)^2$, and $\mathfrak{a} = 1/160 + \beta/24 + \eta/2$. No source order is changed.

65.1 An entire coefficient matrix in the inverse original period

Let P_0 be RPC3's centered Fourier period matrix with its factor $e^{a/u}$ removed by the stated exact identity, and retain $G_u = \text{diag}(u^{a/5} g_a)_{a=1}^4$ with RPC8's full Gamma constants. Introduce the coefficient matrix $\mathcal{R}$ by the following identity, rather than replacing the original period:

$$P = e^{a/u} P_0 = e^{a/u} G_u \mathcal{R}(u^{-1}). \quad (\text{PCL1})$$

The row index below is $1 \leq a \leq 4$, and the column index is $0 \leq r \leq 3$. The complete series is

$$\begin{aligned} \mathcal{R}_{ar}(z) &= \sum_{n=0}^{\infty} R_{ar,n} z^n, \\ R_{ar,n} &= \sum_{\substack{p,h \geq 0 \\ 2p+4h=5n+r+1-a}} \frac{\beta^p \eta^h}{3^p p! h!} (-1)^L 5^L (a/5)_L, \quad L = p + h - n. \end{aligned} \quad (\text{PCL2})$$

Every sum at a fixed n is finite. For each contributing pair, $r + 3p + h + 1 = a + 5L$ and $L \geq 0$. To prove the latter assertion, the left side is positive and $1 \leq a \leq 4$, so a negative multiple of five cannot be their difference. The exponent n is nonnegative: its numerator $2p + 4h + a - r - 1$ is an integer multiple of five and is at least -3 . Thus a negative value of n is impossible.

For a direct derivation, divide every term of the absolutely convergent original series PQO9 by $u^{a/5} g_a$. The quotient of its Gamma factors is exactly

$$\frac{5^{(a+5L)/5-1} e^{\pi i(a+5L)/5} \Gamma(a/5 + L)}{g_a} = (-1)^L 5^L (a/5)_L. \quad (\text{PCL3})$$

Its remaining power of u is $u^{L-p-h} = u^{-n}$. These identities prove PCL2 including its signs, factorials and all unit-independent period coefficients. Absolute convergence of PQO9 permits grouping these terms. For each prescribed complex $z \neq 0$, put $u = 1/z$ on any of its original branches. The resulting sum of absolute values is finite. Consequently PCL2 converges absolutely for every complex z , and defines an entire matrix. This also proves that this matrix is independent of the branch; all branch factors remain in G_u in PCL1.

The complete constant matrix in PCL2 is

$$J_{\infty} = \begin{pmatrix} 1 & 0 & -\beta/3 & 0 \\ 0 & 1 & 0 & -2\beta/3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad J_{\infty}^{-1} = \begin{pmatrix} 1 & 0 & \beta/3 & 0 \\ 0 & 1 & 0 & 2\beta/3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (\text{PCL4})$$

Indeed $n = 0$ forces $2p + 4h = r + 1 - a \leq 3$, so either $p = h = 0$ or $p = 1, h = 0, r + 1 - a = 2$. PCL3 gives respectively the diagonal entries and the two negative entries displayed. In particular, the original row-scaled period limit includes both cubic-phase corrections. It is not a diagonal coefficient map.

RPC's determinant proof holds on the complete coefficient family and gives $\det P_0 = \det G_u$. Alternatively divide PQO11 by the product of the four unchanged Gamma ray factors. Therefore

$$\det \mathcal{R}(z) = 1, \quad \mathcal{R}(z)^{-1} = \text{adj } \mathcal{R}(z) \quad (z \in \mathbb{C}). \quad (\text{PCL5})$$

The equality first holds for $z \neq 0$ by PCL1 and extends to zero by continuity. Both the matrix and its inverse are entire. For any fixed $R > 0$, the finite, completely specified constant

$$C_R = \sum_{n \geq 1} \|R_n\|_1 R^{n-1} < \infty \quad (\text{PCL6})$$

gives $\|\mathcal{R}(z) - J_\infty\|_1 \leq C_R |z|$ on $|z| \leq R$. Here $\|\cdot\|_1$ is the induced column-sum matrix norm and $R_n = (R_{ar,n})$. Convergence follows from the entire power series in finite dimension, also at any larger radius. Thus this is a bound in the actual coefficients, with no uncomputed nonzero minor used as a premise.

65.2 All eighty-one actual conductor coefficients and their limit

Let V be the original centered evaluation Vandermonde in PQO12, $U = \text{diag}(a, \bar{a}, \bar{a}, a)$, and $D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$. The complete period matrix from primary to Fourier coordinates is

$$H(u) = e^{a/u} G_u \mathcal{R}(u^{-1}) V^{-1} U. \quad (\text{PCL7})$$

For $1 \leq j \leq 4$ define the entire matrix and biform

$$\begin{aligned} L_j(z) &= U^{-1} V \mathcal{R}(z)^{-1} D^{-j} \mathcal{R}(z) V^{-1} U, \\ t(Z, W) &= (ZW, Z, W, 1)^\top, \\ A_z(Z, W) &= \prod_{j=1}^4 Q_0(L_j(z) t(Z, W)) = \sum_{r,s=0}^8 a_{rs}(z) Z^r W^s. \end{aligned} \quad (\text{PCL8})$$

These are precisely all original OCF4 coefficients at $z = u^{-1}$. To prove this, compute

$$H(u)^{-1} D^{-j} H(u) = L_j(u^{-1}) : \quad (\text{PCL9})$$

the scalar $e^{a/u}$ cancels with its inverse, and the two diagonal matrices G_u, D^{-j} commute. OCF's definition $Q_j(x) = Q_0(H^{-1} D^{-j} x)$ followed by $x = Ht$ then proves PCL8. The homogeneous version uses $t = (z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)$ and retains the bidegree $(8, 8)$. Thus no unit or period entry has been omitted by the affine notation.

For an explicit finite formula for the individual coefficients, write $L_{j,be}$ for a matrix entry, and associate the four pairs $(\rho_e, \sigma_e) = (1, 1), (1, 0), (0, 1), (0, 0)$ to its columns. Put

$$\begin{aligned} c_{j,rs}(z) &= \sum_{\substack{1 \leq e, f \leq 4 \\ \rho_e + \rho_f = r, \sigma_e + \sigma_f = s}} (L_{j,1e} L_{j,4f} - L_{j,2e} L_{j,3f}), \\ a_{rs}(z) &= \sum_{\substack{0 \leq r_j, s_j \leq 2 \\ \sum_j r_j = r, \sum_j s_j = s}} \prod_{j=1}^4 c_{j,r_j s_j}(z). \end{aligned} \quad (\text{PCL10})$$

Expanding the two products defining Q_0 proves the first line; expanding its four original orbit factors proves the second. Both lines are finite and retain all ordered-pair multiplicities. Equations PCL2, PCL5 and PCL10 are a convergent formula for every actual coefficient, with no freely assigned amplitude.

Their exact leading values are obtained by the explicit matrices

$$L_j(0) = U^{-1} V J_\infty^{-1} D^{-j} J_\infty V^{-1} U, \quad a_{rs}(u^{-1}) = a_{rs}(0) + O_{\text{actual}}(|u|^{-1}). \quad (\text{PCL11})$$

The equality for $L_j(0)$ follows from PCL4, and the bound follows from the entire coefficient series. More explicitly, if $a_{rs}(z) = \sum_{n \geq 0} a_{rs,n} z^n$, the finite constant $\sum_{r,s} \sum_{n \geq 1} |a_{rs,n}| R^{n-1}$ bounds the sum of all eighty-one errors divided by $|z|$ on $|z| \leq R$. The coefficient functions are single-valued entire functions of u^{-1} even though the original period itself retains its logarithm branch. This assertion follows from the exact conjugation PCL9, not from a change of the original branch.

At any specified k , the complete central convolution section is therefore the explicitly evaluated entire matrix

$$C_{\text{in}}(z)_{(a,b),(i,j)} = a_{i+4-a,j+4-b}(z), \quad 0 \leq a, b, i, j \leq k-8, \quad (\text{PCL12})$$

with all out-of-range coefficients zero. PCL10–12 give its full constant matrix and all successive coefficients. They do not discard off-diagonal coefficients or replace this matrix by its central coefficient a_{44} . In particular the calculation of the limit has not asserted a sign or diagonal-dominance inequality for this complete original matrix.

65.3 An exact positivity slice for the literal positive period ray

For $u > 0$ on any of its original logarithm lifts, PQO21 gives $P_0 = E_{\text{br}} R$ with R real invertible and E_{br} diagonal unitary commuting with D . Its exact centered quadratic coefficient matrix is S of PQO13; all its entries are purely imaginary. Consequently, for real $y \in \mathbb{R}^4$,

$$\overline{Q_j(E_{\text{br}} y)} = -Q_{5-j}(E_{\text{br}} y) \quad (1 \leq j \leq 4). \quad (\text{PCL13})$$

Indeed the original quadratic matrix in these coordinates is $e^{-2a/u} R^{-\top} S R^{-1}$, which is purely imaginary. Complex conjugation changes its sign and changes D^{-j} into D^j ; this is precisely the displayed formula.

Let P_{conj} be the permutation matrix exchanging primary indices 1, 2 and 3, 4. For real Z and $|W| = 1$, direct substitution gives $P_{\text{conj}} t(Z, W) = W^{-1} t(Z, W)$. The original roots and unit satisfy $\bar{V} = P_{\text{conj}} V$ and $P_{\text{conj}} \bar{U} = U P_{\text{conj}}$. Thus

$$\overline{V^{-1} U t(Z, W)} = W^{-1} V^{-1} U t(Z, W). \quad (\text{PCL14})$$

Choose either square root w of W . Then $p = w^{-1} V^{-1} U t(Z, W)$ is real by PCL14, and $\text{Ht} = w E_{\text{br}} y$ for the real vector $y = e^{a/u} R p$. Homogeneity and PCL13 prove the exact nonnegative slice identity

$$\boxed{W^{-4} A_{u^{-1}}(Z, W) = |Q_1(E_{\text{br}} y) Q_2(E_{\text{br}} y)|^2 \geq 0 \quad (Z \in \mathbb{R}, |W| = 1).} \quad (\text{PCL15})$$

The choice of sign of w has no effect because the quadratic forms have even degree. In particular, comparing the Laurent coefficients on the unit circle, and then the polynomial coefficients in real Z , gives the exact symmetry

$$a_{rs}(u^{-1}) = \overline{a_{r,8-s}(u^{-1})}. \quad (\text{PCL16})$$

PCL11 extends this identity and PCL15 to the coefficient limit at $u^{-1} = 0$ by continuity. The slice in PCL15 has one real coordinate and one unit-circle coordinate; it has not changed either coordinate into a second unit circle.

65.4 A complete invertible restriction supplied by the original pivot

At each actual u in the proved five-orbit locus, use precisely OCF5's lexicographically largest nonzero coefficient (r_*, s_*) , $a_* = a_{r_* s_*} \neq 0$, and its original pivot rectangle $\mathcal{P}_k = \{(r_* + i, s_* + j) : 0 \leq i, j \leq k-8\}$. Let $\mathcal{E}_k$ be its complement and S_k the original insertion of its coordinates. No new coefficient has been selected. The full original maps satisfy

$$C N_k^\top = 0, \quad S_k^\top N_k^\top = I_\Delta, \quad \text{im } N_k^\top = V_A. \quad (\text{PCL17})$$

The first is the transpose of $N_k C^\top = 0$ in OCF7; the second is the transpose of $N_k S_k = I$. Dimension Δ then proves the last equality, since C has full row rank. It follows that restriction to the actual strip is the exact isomorphism

$$S_k^\top : V_A \longrightarrow \mathbb{C}^{\mathcal{E}_k}, \quad (S_k^\top|_{V_A})^{-1} = N_k^\top. \quad (\text{PCL18})$$

Equivalently its square pivot block has nonzero diagonal a_* and is triangular in the elimination order; OCF's complete division supplies its inverse. This is an alternative restriction to the central one in PCL12, with the exact changed set of roots.

Let $P_{\mathcal{P},k}(y) = \prod_{(a,b) \in \mathcal{P}_k} (y - \omega_{ab})$ and $P_{\mathcal{E},k}(y) = \prod_{(a,b) \in \mathcal{E}_k} (y - \omega_{ab})$, where $\omega_{ab} = (2b - k)\gamma - i(2a - k)\delta$. Thus $P_k = P_{\mathcal{P},k} P_{\mathcal{E},k}$ exactly. Let $V_{\mathcal{E}}$ evaluate $1, y, \dots, y^{\Delta-1}$ at those actual outer roots and set $D_{\mathcal{P},\mathcal{E}} = \text{diag}(P_{\mathcal{P},k}(\omega_{ab}))_{\mathcal{E}_k}$. These two matrices are invertible by distinctness of the original roots. The full polynomial-target restriction and its exact inverse are

$$\begin{aligned} \rho_{\mathcal{E}} &= V_{\mathcal{E}}^{-1} D_{\mathcal{P},\mathcal{E}}^{-1} S_k^\top|_{V_A}, \\ \rho_{\mathcal{E}}^{-1} &= N_k^\top D_{\mathcal{P},\mathcal{E}} V_{\mathcal{E}}. \end{aligned} \quad (\text{PCL19})$$

Multiplying these matrices and using PCL17 proves both inverse identities, with all inner products of root differences retained. In particular the original residual becomes the following specified onto map, with its exact complete kernel:

$$\Xi_{\mathcal{E}} = \Xi_k^{\text{fix}} N_k^\top D_{\mathcal{P},\mathcal{E}} V_{\mathcal{E}}, \quad \ker \Xi_{\mathcal{E}} = \rho_{\mathcal{E}}(K). \quad (\text{PCL20})$$

This follows by composition with the proved isomorphism, not by deleting any common-kernel vector.

Finally the original metric and its entire source graph are carried by this isomorphism as follows. Write $J_{\mathcal{E}} = N_k^\top D_{\mathcal{P},\mathcal{E}} V_{\mathcal{E}}$. At every original endpoint the actual conductor metric in these outer coordinates is exactly

$$H_{\mathcal{E},N} = J_{\mathcal{E}}^* G_N J_{\mathcal{E}}, \quad Q_{\Xi,N}^{\text{fix}} = (\Xi_{\mathcal{E}} H_{\mathcal{E},N}^{-1} \Xi_{\mathcal{E}}^*)^{-1}. \quad (\text{PCL21})$$

The first formula is the pullback along $\rho_{\mathcal{E}}^{-1}$; the minimum-section proof of GIP6 gives the second. On the full GIP graph its target map is $\rho_{\mathcal{E}} E_N^{\text{graph}}$, with the same complete t ideal columns, and composing with $\Xi_{\mathcal{E}}$ gives exactly B_N^{graph} . Its residual kernel is therefore all $t + r$ columns in GIP4, including the low- v coordinates. The coefficient-space bound for N_k does not remove the displayed root-value and Gamma matrices from PCL21. These formulas retain the full metric transport needed for the next comparison.

65.5 The exact inverse-period reflection of the central section

Let $E_{\text{par}} = \text{diag}(1, -1, 1, -1)$ in the centered coefficient order, and let P_{opp} exchange primary indices 1, 4 and 2, 3. Equation PCL2 implies

$$\mathcal{R}(-z) = E_{\text{par}} \mathcal{R}(z) E_{\text{par}}, \quad V E_{\text{par}} = P_{\text{opp}} V, \quad U P_{\text{opp}} = P_{\text{opp}} U. \quad (\text{PCL22})$$

For the first equality a contributing coefficient satisfies $5n + r + 1 - a = 2p + 4h$, so $n \equiv a - r - 1 \pmod{2}$. The signs from the two parity matrices on entry (a, r) are $(-1)^{a-1+r} = (-1)^n$. Summing the entire series proves the identity. The second equality evaluates each monomial at the opposite original root. The third uses the full original unit values $(a, \bar{a}, \bar{a}, a)$, which agree at opposite roots.

The two diagonal matrices E_{par} and D commute. Substitution into PCL8 and use of PCL22 therefore give the exact maps

$$L_j(-z) = P_{\text{opp}} L_j(z) P_{\text{opp}}, \quad A_{-z}(Z, W) = Z^8 W^8 A_z(Z^{-1}, W^{-1}). \quad (\text{PCL23})$$

For the second identity, $Q_0(P_{\text{opp}} t) = Q_0(t)$ by its two explicit products. Also $P_{\text{opp}} t(Z, W) = ZW t(Z^{-1}, W^{-1})$ when $ZW \neq 0$. Each of the four quadratic factors therefore contributes $(ZW)^2$.

This proves the identity on that open set; both sides are polynomials after multiplication by $Z^8 W^8$, so it holds identically including both coordinate axes.

Comparing every coefficient in PCL23 proves

$$a_{rs}(-z) = a_{8-r,8-s}(z), \quad C_{\text{in}}(-z) = C_{\text{in}}(z)^T, \quad \det C_{\text{in}}(-z) = \det C_{\text{in}}(z). \quad (\text{PCL24})$$

Indeed the $(a, b), (i, j)$ entry on the left of the middle identity is $a_{4+a-i, 4+b-j}(z)$, precisely the transposed entry of PCL12. Out-of-range coefficients are zero on both sides. The determinant equality follows by the permutation expansion of a determinant and its transpose. Thus the exact central determinant is an even entire function of u^{-1} , with every coefficient supplied by PCL2 and PCL10–12. This retains its actual constant matrix and does not assign that determinant a value by deleting any of its entries.

65.6 The evaluated zero-order domain in the original period

The complete PZ, PZX and PZU calculations below evaluate the previously retained value $A_{u^{-1}}(1, 1)$ of this same conductor. In their exact notation put $r = a/\bar{a}$, $T = (\delta + i\gamma)^2/(\delta - i\gamma)^2$, $A_e = (2T - 1)/(3(T - 1))$ and $B_e = (2 - T)/(3(T - 1))$. For the four actual factors $d_j(\tau) = Q_0(L_j(\tau)\mathbf{1})$, the first nonzero inverse-period degree is

$$\ell_A = 0 \quad (r \notin \{1, A_e/B_e\}), \quad \ell_A = 2 \quad (r \in \{1, A_e/B_e\}), \quad b_j = [\tau^{\ell_A}]d_j(\tau) \neq 0. \quad (\text{RMR1})$$

The complete second-order matrices and polynomial nonvanishing certificate at both exceptional phases are PZX1–6; no phase is excluded. Each d_j is even entire by PCL22. Hence, with the actual finite-valued convergent sum $C_j = \sum_{n \geq 1} |[\tau^{\ell_A + 2n}]d_j|$, define $R_A = \max\{R_*, 1, \max_j(2C_j/|b_j|)^{1/2}\}$. For $|u| \geq R_A$ summing that Taylor tail gives

$$\frac{\prod_j |b_j|}{16|u|^{4\ell_A}} \leq |E_{A,u}(0)| \leq \frac{81 \prod_j |b_j|}{16|u|^{4\ell_A}}, \quad v = 0, \quad \alpha_A = 1/E_{A,u}(0). \quad (\text{RMR2})$$

Indeed each $u^{\ell_A} d_j(u^{-1})$ lies within $|b_j|/2$ of b_j , and multiplication of the four complete factors proves both inequalities. The nonzero entire product has only finitely many zeros in $|\tau| \leq 1/R_*$; therefore $v = 0$ also on the complement of the explicitly defined finite set PZU5 in the original domain $|u| \geq R_*$. At its exceptional periods the complete PZ12 derivative test and $v \leq 32$ remain in force.

On this proved $v = 0$ domain, substitution into the original maps gives

$$\begin{aligned} \mathbb{C}[S]_{<v} = 0, \quad t_{0,n} = 0, \quad m_n = 16n - 48, \quad t_n^{\text{EFP}} = 8n - 16, \\ j_{0,n} = 8n - 32, \quad \kappa_e = d_{K,e} = d_{T,e} = 0. \end{aligned} \quad (\text{RMR3})$$

The last three equalities follow from BTK25 and the exact RRC14 sequence. Thus the full low-cycle and connecting-cycle coefficient spaces coincide; their original quotient maps have identical fibres and identical attained metrics. All RRC16 tail corrections vanish. The ROG low-coordinate shear is the identity. The conductor, α_A , every coefficient and the complete CPQ mixed forms remain present: PZU9–11 gives the exact size of this scalar and its unchanged division recursion, including the exceptional phases. In particular these equalities determine the low-order correction; the connecting rank remains the exact CPR18 intersection rank.

Original period determinant and its constants

66 The original period quadric as a residue commutator

Retain $0 < \delta < 1/2$, $\gamma > 2$, the ordered simple quartet $\alpha_1 = 1/2 + \delta + i\gamma$, $\alpha_2 = 1/2 + \delta - i\gamma$, $\alpha_3 = 1/2 - \delta + i\gamma$, $\alpha_4 = 1/2 - \delta - i\gamma$, and $h(s) = \prod_a (s - \alpha_a)$. Put

$$w = s - \frac{1}{2}, \quad r_a = \alpha_a - \frac{1}{2}, \quad \beta = 2(\gamma^2 - \delta^2), \quad \eta = (\delta^2 + \gamma^2)^2, \quad H(w) = w^4 + \beta w^2 + \eta. \quad (\text{RPC1})$$

Thus $r_4 = -r_1$, $r_3 = -r_2$, $r_2 = \bar{r}_1$ and $r_1^2 - r_2^2 = 4i\delta\gamma$. No root or original coordinate is discarded by this translation. Its coefficient isomorphism is $J_c : \mathbb{C}[w]_{<4} \rightarrow E_h$, $J_c p = [p(s - 1/2)]$; its inverse is substitution $s = w + 1/2$ in the unique original representative. The primitive with its original constant is

$$\Phi_h(s) = A + \phi(w), \quad \phi(w) = \frac{w^5}{5} + \frac{\beta w^3}{3} + \eta w, \quad A = \Phi_h(1/2) = \frac{1}{160} + \frac{\beta}{24} + \frac{\eta}{2}. \quad (\text{RPC2})$$

The last equality is forced by $\Phi_h(0) = 0$ and follows by evaluating the displayed odd polynomial at $w = -1/2$.

Let $u \neq 0$ carry the specified original branch of its logarithm. Let Π be the original period map on the rays and orientations of PDE2, and let $F_{ja} = \zeta^{ja} - 1$, $\zeta = e^{2\pi i/5}$, $1 \leq j, a \leq 4$. Its inverse has entries $(F^{-1})_{aj} = \zeta^{-ja}/5$: multiplying the two matrices and using $\sum_{j=1}^4 \zeta^{jb} = 4$ for $5 \mid b$ and -1 otherwise proves this entry by entry. The literal centered coefficient-to-Fourier map is

$$P = F^{-1} \Pi J_c, \quad P_0 = e^{-A/u} P. \quad (\text{RPC3})$$

These are invertible by the complete original period determinant PDE3–13. The contours for P_0 may be taken to be the rays based at $w = 0$ with the original five angles. Indeed translating the old rays gives rays based at $-1/2$. Join corresponding finite endpoints and close each contour at radius R . The integrand is entire. On the bounded-length outer joining curves its exponent is $-R^5/(5|u|) + O(R^4)$, uniformly along the joining curves, so the outer integrals tend to zero. The common finite joining path occurs with opposite coefficients in $-\ell_0 + \ell_j$ and cancels exactly. Cauchy's integral theorem therefore proves this contour identity, including its signs and the factor $e^{A/u}$.

66.1 A flat alternating pairing with all constants

On $\mathbb{C}[w]/(H)$ define the bilinear, rather than sesquilinear, pairing

$$\mathcal{J}(p, q) = \sum_{H(r)=0} \frac{p(r)q(-r)}{H'(r)}. \quad (\text{RPC4})$$

The roots are simple at the original packet. The equivalent polynomial formula is the coefficient of w^3 in the remainder of $p(w)q(-w)$ modulo H . To prove it, use Lagrange interpolation: the coefficient of w^3 in the idempotent at r is $1/H'(r)$. This polynomial formula extends the pairing to every complex value of β, η , including repeated roots, without a division by a repeated-root value. In the increasing frame $1, w, w^2, w^3$, multiplication by w and this pairing are the exact matrices

$$M = \begin{pmatrix} 0 & 0 & 0 & -\eta \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -\beta \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & \beta \\ 1 & 0 & -\beta & 0 \end{pmatrix}. \quad (\text{RPC5})$$

For the last matrix, the coefficient functional is zero through degree two, one in degree three, zero in degree four and six, and $-\beta$ in degree five. Inserting the sign $(-1)^j$ from the second

argument gives every displayed entry. Thus $J^\top = -J$, $\det J = 1$, and direct multiplication gives $M^\top J + JM = 0$.

Allow the two original odd-phase coefficients to vary complexly, solely to calculate the existing period map. The ray domination $e^{-r^5/(5|u|)+O(r^3+r)}$ on compact parameter sets justifies all coefficient derivatives and integrations by parts below. Ordinary division of w^{j+3} by H , followed by differential reduction, gives

$$\partial_\eta P_0 = \frac{1}{u} P_0 M, \quad \partial_\beta P_0 = \frac{1}{3u} P_0 (M^3 - uN), \quad N = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{RPC6})$$

For clarity, the quotient derivatives for columns $j = 0, 1, 2, 3$ are $0, 0, 1, 2w$, respectively; hence this is the full correction matrix. The two outer boundary terms vanish, and the two values at the common origin cancel. No differential remainder was replaced by an arithmetic remainder in RPC6.

There is also a direct nonvanishing proof on this entire two-parameter family. The displayed matrices satisfy $\text{Tr } M = 0$, $\text{Tr } M^3 = 0$ and $\text{Tr } N = 0$. The first and third follow from their diagonals; the second follows by multiplying RPC5 three times, or from the vanishing third Newton sum of the even monic polynomial H . Multilinearity of the determinant in its columns then gives $\partial_\eta \det P_0 = \partial_\beta \det P_0 = 0$ from RPC6, without first assuming an inverse exists. At the origin the literal Gamma ray integrals give the diagonal matrix G_u defined in RPC8, whose determinant is nonzero. Hence P_0 is invertible at every point of $\mathbb{C}^2$, justifying the inverses throughout this family.

The matrix identity $M^\top J + JM = 0$ also gives $(M^3)^\top J + JM^3 = 0$. Direct multiplication of RPC5–6 gives

$$\partial_\beta J + \frac{1}{3}(N^\top J + JN) = 0. \quad (\text{RPC7})$$

In particular $\partial_\beta J$ has entries $(2, 3) = 1$, $(3, 2) = -1$ when indices begin at zero, while $N^\top J + JN$ has the opposite threefold entries. Differentiate the inverse matrices in $P_0^{-\top} J P_0^{-1}$. RPC6–7 show that both its β and its η derivatives vanish. This matrix is consequently constant on the connected coefficient space $\mathbb{C}^2$.

Define the unchanged positive-Gamma ray constants

$$g_a = 5^{a/5-1} e^{\pi i a/5} \Gamma(a/5), \quad G = \text{diag}(g_1, g_2, g_3, g_4), \quad G_u = \text{diag}(u^{a/5} g_a)_{a=1}^4. \quad (\text{RPC8})$$

At $\beta = \eta = 0$, literal substitution in each original ray integral gives $P_0 = G_u$. Every g_a is nonzero, since its Gamma factor is the integral of a strictly positive function. We have therefore proved the complete flat-pairing identity

$$\begin{aligned} \Omega_0 &= P_0^{-\top} J P_0^{-1} = G_u^{-\top} J|_{\beta=0} G_u^{-1}, \\ (\Omega_0)_{14} &= -\frac{1}{u g_1 g_4}, \quad (\Omega_0)_{23} = \frac{1}{u g_2 g_3}, \quad (\Omega_0)_{ba} = -(\Omega_0)_{ab}, \\ \Omega &= P^{-\top} J P^{-1} = e^{-2A/u} \Omega_0. \end{aligned} \quad (\text{RPC9})$$

All entries not listed are zero; the indices in RPC9 begin at one. This formulation retains the Gamma constants directly and requires no unevaluated normalization of a period or source mass. For $D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$, the two paired sums of weights are five. It follows entry by entry that

$$D^\top \Omega D = \Omega, \quad D^\top \Omega_0 D = \Omega_0. \quad (\text{RPC10})$$

66.2 The full arithmetic unit and the exact quadratic form

Retain the original $v_h = 2\xi/h$ and its complete simple-root unit values. Reflection and conjugation of this original function give

$$v_h(\alpha_1) = v_h(\alpha_4) = a, \quad v_h(\alpha_2) = v_h(\alpha_3) = \bar{a}, \quad a \neq 0. \quad (\text{RPC11})$$

Indeed both numerator and denominator are invariant under $s \mapsto 1-s$ and conjugate under $s \mapsto \bar{s}$, first away from the roots and then by the removable values. The complete zero orders give nonzero values at these simple roots. Write

$$c_3 = \frac{a^{-2} - \bar{a}^{-2}}{4i\delta\gamma}, \quad c_1 = \frac{a^{-2} + \bar{a}^{-2}}{2} - (\delta^2 - \gamma^2)c_3, \quad \mathfrak{r}(w) = c_1w + c_3w^3. \quad (\text{RPC12})$$

Both coefficients are real, and $(c_1, c_3) \neq (0, 0)$. Directly, $c_1 + c_3r_1^2 = a^{-2}$ and $c_1 + c_3r_2^2 = \bar{a}^{-2}$; the same equalities hold at the respective opposite roots. Thus $\mathfrak{r}(M) = M v_h(M + 1/2)^{-2}$ in the original finite algebra, with the entire function evaluated by its complete original arithmetic unit. RPC12 is an exact interpolation of this specific multiplier.

For $Q_0(t) = t_1t_4 - t_2t_3$, let J_{prim} be the original Lagrange coordinate isomorphism and set

$$B = F^{-1}\Pi U J_{\text{prim}}, \quad Q(x) = Q_0(B^{-1}x), \quad \mathcal{H} = P\mathfrak{r}(M)P^{-1}. \quad (\text{RPC13})$$

If $p = P^{-1}x$, the primary components of $B^{-1}x$ are exactly $p(r_1)/a, p(r_2)/\bar{a}, p(r_3)/\bar{a}, p(r_4)/a$. Since $H'(r) = 4r(r^2 + \gamma^2 - \delta^2)$, its values give $r/H'(r) = 1/(8i\delta\gamma)$ on roots one and four, and $r/H'(r) = -1/(8i\delta\gamma)$ on roots two and three. Inserting these four values into RPC4 proves the exact identity

$$Q(x) = 4i\delta\gamma p^\top \mathfrak{r}(M)^\top J p = 4i\delta\gamma x^\top \mathcal{H}^\top \Omega x. \quad (\text{RPC14})$$

There is no conjugate transpose in this quadratic identity. Because $\mathfrak{r}$ is odd, $\mathfrak{r}(M)^\top J + J\mathfrak{r}(M) = 0$; hence the matrices in RPC14 are symmetric. This also proves $\mathcal{H}^\top \Omega + \Omega \mathcal{H} = 0$. The four eigenvalues of $\mathfrak{r}(M)$ are exactly $r_1/a^2, \bar{r}_1/a^2, -\bar{r}_1/a^2, -r_1/a^2$. They are nonzero, although they need not be distinct. Both B and Q_0 are invertible as linear and quadratic matrices, respectively, so Q always has rank four, including the possible eigenvalue collisions.

By RPC10, substituting Dx into RPC14 transports $\mathcal{H}$ to $D^{-1}\mathcal{H}D$. The resulting symmetric quadratic matrices are equal if and only if the corresponding $\mathcal{H}$ matrices are equal, because Ω is invertible. Consequently

$$Q(Dx) = Q(x) \iff [\mathcal{H}, D] = 0 \iff \mathcal{H}_{ab} = 0 \text{ for every } a \neq b. \quad (\text{RPC15})$$

This is an exact test in the actual original period matrix and unit. The final equivalence uses the four distinct diagonal entries of D . It makes no assumption that their off-diagonal tests are independent.

For completeness, a rank-four quadratic semi-invariant of cyclic weight $j \neq 0$ is impossible: its coefficient in row j could pair only with the missing weight zero, so that row is zero. Thus any semi-invariant rank-four quadratic must have weight zero. Since five is prime, the projective orbit of Q under D has size one or five, and size one is exactly RPC15. Every failure of any displayed off-diagonal test proves size five.

66.3 An explicit original-parameter radius forcing orbit five

Here we prove that RPC15 fails for all sufficiently large $|u|$, with a completely specified finite radius for the given original quartet and unit. The argument evaluates the original periods; it does not choose or insert a replacement amplitude.

Set $b = \beta u^{-2/5}$, $c = \eta u^{-4/5}$, and let $T(b, c)$ be the Fourier period matrix for $z^5/5 + bz^3/3 + cz$ at $u = 1$, in columns $1, z, z^2, z^3$ and on the five rays of angles $(\pi + 2\pi j)/5$. Substitution $w = u^{1/5}z$, with exactly the original branch, gives

$$P_0 = T(b, c)S_u, \quad S_u = \text{diag}(u^{a/5})_{a=1}^4, \quad S_u M S_u^{-1} = u^{1/5}M(b, c), \quad T(0, 0) = G. \quad (\text{RPC16})$$

Every column includes its original differential factor. The matrix $M(b, c)$ is RPC5 with β, η replaced by b, c ; in particular $N_0 := M(0, 0)$ has its three subdiagonal entries equal to one.

The following positive constants are explicit finite real integrals or finite expressions in the original Gamma constants:

$$\begin{aligned}
E_j &= \frac{8}{5} \int_0^\infty r^j (r^3/3 + r) e^{-r^5/5 + r^3/3 + r} dr \quad (0 \leq j \leq 3), \\
C_T &= 2 \left(\sum_{j=0}^3 E_j^2 \right)^{1/2}, \quad g_+ = \max_a |g_a|, \quad g_-^{-1} = \max_a |g_a|^{-1}, \\
\kappa &= \max_{1 \leq a < 4} |g_{a+1}/g_a|, \quad K_0 = GN_0 G^{-1}, \\
C_K &= 4C_T g_-^{-1} + 2g_+ g_-^{-1} + 2g_+ g_-^{-2} C_T, \\
C_{K,3} &= C_K(3\kappa^2 + 3\kappa C_K + C_K^2), \quad \epsilon_0 = \min\{1, (2C_T g_-^{-1})^{-1}\}.
\end{aligned} \tag{RPC17}$$

Here g_-^{-2} means $(\max_a |g_a|^{-1})^2$. The negative fifth degree exponent proves convergence of every integral. All norms in the following estimates are the Euclidean operator norm, with the Frobenius norm used explicitly as an upper bound where stated.

Write $\epsilon = |b| + |c|$. When $|b|, |c| \leq 1$, on each ray

$$|e^{bz^3/3 + cz} - 1| \leq \epsilon(r^3/3 + r)e^{r^3/3 + r}.$$

Each contour has two rays, and each entry of F^{-1} has absolute value $1/5$ with four summands. The entry in column j of $T - G$ is therefore at most $E_j \epsilon$. Summing the squares over all four rows proves

$$\|T - G\| \leq \|T - G\|_F \leq C_T \epsilon. \tag{RPC18}$$

For $\epsilon \leq \epsilon_0$, the convergent inverse series for $I + G^{-1}(T - G)$ gives $\|T^{-1}\| \leq 2g_-^{-1}$ and $\|T^{-1} - G^{-1}\| \leq 2g_-^{-2} C_T \epsilon$. Also $\|M(b, c) - N_0\| \leq \epsilon$, $\|M(b, c)\| \leq 2$, and $\|N_0\| = 1$. Subtract the products $K := TM(b, c)T^{-1}$ and K_0 in three terms, changing the left factor, then the middle factor, then the inverse factor. The preceding inequalities give, with the complete RPC17 constants,

$$\|K - K_0\| \leq C_K \epsilon, \quad \|K\| \leq \kappa + C_K \epsilon, \quad \|K^3 - K_0^3\| \leq C_{K,3} \epsilon. \tag{RPC19}$$

For the last assertion use the exact noncommutative identity $K^3 - K_0^3 = (K - K_0)K^2 + K_0(K - K_0)K + K_0^2(K - K_0)$. Its norm is at most $C_K \epsilon$ times $(\kappa + C_K)^2 + \kappa(\kappa + C_K) + \kappa^2$, as claimed.

The original multiplication and full unit in RPC13 now give exactly

$$\mathcal{H} = c_1 u^{1/5} K + c_3 u^{3/5} K^3, \quad (K_0^3)_{41} = g_4/g_1, \quad (K_0)_{21} = g_2/g_1. \tag{RPC20}$$

The scalar $e^{A/u}$ cancels from this conjugation, while it remains in the quadratic form through RPC9. Put $B_* = \beta + \eta > 0$. For $|u| \geq 1$, $\epsilon \leq B_* |u|^{-2/5}$.

If $c_3 \neq 0$, define the completely specified radius

$$T_* = \max \left\{ 1, \frac{B_*}{\epsilon_0}, \frac{2(|c_3| C_{K,3} B_* + |c_1|(\kappa + C_K))}{|c_3 g_4/g_1|} \right\}, \quad R_* = T_*^{5/2}. \tag{RPC21}$$

For every u on its original branch with $|u| \geq R_*$, RPC19–20 imply

$$\left| u^{-3/5} \mathcal{H}_{41} - c_3 g_4/g_1 \right| \leq |u|^{-2/5} (|c_3| C_{K,3} B_* + |c_1|(\kappa + C_K)) \leq \frac{1}{2} |c_3 g_4/g_1|. \tag{RPC22}$$

In particular $\mathcal{H}_{41} \neq 0$. In the quadratic matrix RPC14 this is the nonzero $(1, 1)$ entry $4i\delta\gamma \mathcal{H}_{41} \Omega_{41}$, whose cyclic weight is two.

If $c_3 = 0$, then $c_1 \neq 0$. In this case define

$$T_* = \max \left\{ 1, \frac{B_*}{\epsilon_0}, \frac{2C_K B_*}{|g_2/g_1|} \right\}, \quad R_* = T_*^{5/2}. \tag{RPC23}$$

For every $|u| \geq R_*$, the same estimates give

$$\left| u^{-1/5} \mathcal{H}_{21} - c_1 g_2 / g_1 \right| \leq |c_1| C_K B_* |u|^{-2/5} \leq \frac{1}{2} |c_1 g_2 / g_1|. \quad (\text{RPC24})$$

Thus $\mathcal{H}_{21} \neq 0$. The $(1, 3)$ entry of the quadratic matrix is then $4i\delta\gamma\mathcal{H}_{21}\Omega_{23}$, which has cyclic weight four; symmetry retains its $(3, 1)$ partner. Both cases prove the unconditional conclusion for the actual original parameters and the explicit radius belonging to those parameters:

$$\boxed{|u| \geq R_* \implies \#\{[Q(D^j x)] : 0 \leq j < 5\} = 5.} \quad (\text{RPC25})$$

The bracket denotes the projective class of the quadratic polynomial. RPC21 or RPC23 supplies its radius for every nonzero original unit; neither case assumes a condition on an uncomputed period coefficient.

Finally, the original periods are holomorphic on the connected logarithm cover of $\mathbb{C}^*$, and on each branch domain: locally the contours may be held in common decay sectors and differentiated under the dominated integral, and contour deformation identifies these local definitions. Their determinant never vanishes. Hence all entries of $\mathcal{H}$ are holomorphic there. RPC22 or RPC24 proves that its respective decisive entry is not identically zero on the logarithm cover and, by the identity theorem on that connected cover, on any open branch domain. The zeros of that one entry are isolated, by its convergent local power series. The exact exceptional set

$$\mathcal{E} = \{u : \mathcal{H}_{ab}(u) = 0 \text{ for all } a \neq b\} \quad (\text{RPC26})$$

is a subset of those isolated zeros and satisfies $|u| < R_*$. Thus it has no accumulation inside the branch domain; accumulation at $u = 0$ or a branch boundary has not been excluded. RPC15 is the full calculation specifying this set. No assertion that it is empty at arbitrary finite complex u is made or used.

Deligne: the general mixed-weight direct-image theorem

The following complete supplied section is from P. Deligne, *La conjecture de Weil. II*, section 3.3, with proof. The full source is retained in volume 16 of the programme source collection. See [18].

(3.3) The general case

Theorem 66.1 (3.3.1). *Let $f : X \rightarrow Y$ be a morphism of schemes of finite type over $\mathbb{Z}$, and let $\mathcal{F}$ be a sheaf on X . If $\mathcal{F}$ is mixed of weights $\leq n$, then, for each i , $R^i f_! \mathcal{F}$ is mixed of weights $\leq n + i$.*

The proof uses successive devissages. Exact sequences reduce the assertion in the coefficient sheaf; an open-closed decomposition reduces it in X ; base change to an open part and its complement reduces it in Y ; Leray spectral sequences give transitivity. Universal homeomorphisms do not affect étale cohomology. Finally, morphisms of relative dimension zero preserve pointwise purity and have no higher direct images. These reductions reduce to the case where f has relative dimension one, $\mathcal{F}$ is lisse and pointwise pure, and X is the complement in a smooth projective relative curve of a divisor finite étale over Y , with tame ramification along the divisor.

On each geometric fiber, Theorem (3.2.3) gives purity for $R^i f_{!} j_* \mathcal{F}$. The local-monodromy theorem (1.8.4) gives the weights of the graded pieces of the local monodromy filtration on $i^* j_* \mathcal{F}$. The finite part is handled by the relative-dimension-zero case, and the exact sequence

$$0 \rightarrow j_! \mathcal{F} \rightarrow j_* \mathcal{F} \rightarrow i_* i^* j_* \mathcal{F} \rightarrow 0$$

finishes the proof.

(3.3.2). Lower bounds for weights are obtained from the p -adic estimates of SGA 7 XXI, 5. A sheaf is called integral if all Frobenius eigenvalues are algebraic integers. The weights of a mixed integral sheaf are necessarily non-negative, since the norm of a pure algebraic integer of weight n has absolute value $q^{dn/2}$.

Corollary 66.2 (3.3.3). *Let $f : X \rightarrow Y$ be of finite type over $\mathbb{Z}$, and let $\mathcal{F}$ be an integral sheaf on X . If $\mathcal{F}$ is mixed of weights $\leq n$, then $R^i f_! \mathcal{F}$ is mixed of weights between 0 and $n + i$. If i is larger than the maximum dimension d of the fibers, the weights lie between $2(i - d)$ and $n + i$.*

For $Y = \text{Spec } \mathbb{F}_q$ this becomes:

Corollary 66.3 (3.3.4). *Let X_0 be of finite type over $\mathbb{F}_q$, and let $\mathcal{F}_0$ be mixed of weights $\leq n$ on X_0 . Then $H_c^i(X, \mathcal{F})$ is mixed of weights $\leq n + i$. If $\mathcal{F}_0$ is integral, the weights are non-negative; if moreover $i > d = \dim X$, they are $\geq 2(i - d)$.*

Corollary 66.4 (3.3.5). *If X_0 is smooth of finite type over $\mathbb{F}_q$ and $\mathcal{F}_0$ is lisse mixed of weights $\geq n$, then $H^i(X, \mathcal{F})$ is mixed of weights $\geq n + i$.*

This is Poincaré duality applied to the dual sheaf. Hence:

Corollary 66.5 (3.3.6). *If X_0 is smooth of finite type over $\mathbb{F}_q$ and $\mathcal{F}_0$ is lisse pure of weight n , then the image of*

$$H_c^i(X, \mathcal{F}) \longrightarrow H^i(X, \mathcal{F})$$

is pure of weight $n + i$.

(3.3.7)–(3.3.9). If an algebraic number α is pure of weight n relative to q , normalize p -adic valuations by $v(q) = 1$ and attach to α the pair $(v(\alpha), v(q^n \alpha^{-1}))$. For integral eigenvalues both coordinates are non-negative and sum to n . For the constant sheaf, the preceding bounds localize these pairs in the corresponding triangles in the (r, s) -plane. In particular, for X_0 proper and smooth one obtains again the main theorem of I, with projective replaced by proper:

$$\det(1 - Ft, H^i(X, \mathbb{Q}_\ell)) \in \mathbb{Z}[t]$$

is independent of $\ell \neq p$, and all complex roots have absolute value $q^{i/2}$.

(3.3.10)–(3.3.11). The theorem remains valid with “mixed” replaced by “ ι -mixed” and real-valued weights. The only weights in $R^i f_! \mathcal{F}$ are those $\leq \beta + i$ congruent modulo $\mathbb{Z}$ to a weight already occurring in $\mathcal{F}$. The smoothness assumption is used only to invoke Poincare duality, and may therefore be replaced by the corresponding rational-homology-manifold condition

$$Ra^! \mathbb{Q}_\ell = \mathbb{Q}_\ell(n)[2n]$$

for the structural morphism $a : X_0 \rightarrow \mathrm{Spec} \mathbb{F}_q$.

Complete attained-image localization proof

67 Signed arithmetic baseline: consolidation and localization

18 September 2026. A written mathematical continuation on the original fixed-packet domain. The two incoming continuations are retained verbatim in `inputs/`. This note identifies their overlap exactly, preserves the stronger statements, and proves additional endpoint and degree-band estimates. It does not claim that the full signed arithmetic coefficient has been evaluated.

67.1 Result reached

Write the original signed baseline as

$$\delta_k = \mathcal{B}_k[m_k] - \mathcal{B}_k[\sigma], \quad \mathcal{B}_k[\nu] = \log \det G_{q-1}[\nu] + \log \det G_q[\nu] - \log \det G_{2q-1}[\nu] - \log \det G_{2q}[\nu].$$

Let $C_k = [-kb, kb]$, with b the original boundary-tilted mean. Define the outer-only density, including its exact central value, by

$$\omega_k(y) = \begin{cases} \sigma(y), & y \in C_k, \\ \mathfrak{M}^{-k} m_k(y), & y \notin C_k. \end{cases}$$

The new decomposition, with all four original signs, is

$$\boxed{\delta_k = \delta_k^{\text{out}} + \widehat{\Phi}_{q,k} + \mathfrak{e}_k, \quad \delta_k^{\text{out}} = \mathcal{B}_k[\omega_k] - \mathcal{B}_k[\sigma],} \quad (\text{R1})$$

$$\boxed{\widehat{\Phi}_{q,k} \geq 0, \quad |\mathfrak{e}_k| \leq E_k = O_h(k(\log q)^2) = o_h(q).} \quad (\text{R2})$$

Here $\widehat{\Phi}_{q,k}$ is given in Section 8 by four explicit Gamma determinants of size

$$r = O_h(k \log q) = o(q).$$

Its entries use only the fixed one-factor tilted mass $M(\theta)$, the original Gamma polynomials, and finite integrals on C_k . The actual convolution density outside C_k remains solely in δ_k^{out} . No original arithmetic kernel is substituted by rank or by an asserted isometry.

There is a further bound on the central endpoint contribution

$$\delta_k^{\text{cen,end}} := \mathcal{B}_k[m_k] - \mathcal{B}_k[\omega_k] :$$

$$\boxed{\delta_k^{\text{cen,end}} = O_h(k^2), \quad \liminf \frac{\delta_k^{\text{cen,end}}}{k^2} \geq 0.} \quad (\text{R3})$$

Consequently its coefficient at the $q \log k$ scale is zero, with an $O_h(q)$ remainder, on every fixed multiplicity stratum. On each fixed stratum $m_0 \geq 2$, this entire central endpoint contribution is $o_h(q)$. On the simple stratum $m_0 = 1$, its order- q value is reduced to $\widehat{\Phi}_{q,k}$ with an $o(q)$ error, but that limiting value is not calculated in this note.

The outer term in (R1) is still unevaluated at the requested scales. In particular this note does not assign the full baseline's $q \log k$ coefficient or its order- q coefficient.

67.2 1. Original source, exact coordinates, and consolidation

Retain the original stipulated quartet, its complete multiplicity, its full arithmetic amplitude, and its original tensor orders:

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad h(s) = \prod_{z \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s-z)^{m_0},$$

$$v_h = 2\xi/h, \quad w_h(y) = |v_h(1/2 + iy)|^2/(2\pi), \quad m_k = w_h^{*k},$$

$$k = 4l + 1 \geq 5, \quad e = 1 + k(m_0 - 1), \quad q = e(k + 1)^2, \quad c = k/2,$$

$$\chi_k(S) = \prod_{a, b_0=0}^k (S - c - (2a - k)\delta - i(2b_0 - k)\gamma)^e.$$

This retains the programme's stipulated packet; it does not supply or assert an actual numerical off-critical zero.

The reference source is

$$\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}, \quad \sigma(\mathbb{R}) = \sqrt{2\pi}, \quad \alpha = \pi/2.$$

The original change of polynomial coordinates is

$$\mathcal{U}_N : P(S) \mapsto P(c + iy), \quad Q(y) = i^{-q} \chi_k(c + iy). \quad (\text{C1})$$

Its inverse is substitution $y = (S - c)/i$; its coefficient determinant has modulus one. Moreover

$$\mathcal{U}_N(\chi_k P) = i^q Q \mathcal{U}_{N-q} P.$$

Thus it carries the complete relation ideal, every remainder class and the degree cutoffs to their displayed real-coordinate versions. The phase has modulus one. We use the corresponding source norm, not an independent norm on the transported coefficients. For monic norms, the additional phase $i^{-\deg P}$ restores real-coordinate monicity exactly. The original full-unit inclusion is transported by this coefficient isomorphism; its arithmetic coefficients are not set to one.

Let $H_N[\nu]$ be the Gram of $1, y, \dots, y^N$, J_N the remainder map modulo Q , and

$$G_N[\nu] = (J_N H_N[\nu]^{-1} J_N^*)^{-1}.$$

The original monic source/relation frame gives

$$\det G_N[\nu] = \frac{\det H_N[\nu]}{\det H_N^{\text{rel}}[\nu]}, \quad (\text{C2})$$

where the relation frame is $Q, yQ, \dots, y^{N-q}Q$. Empty determinants are one. Its frame determinant is one; both cross blocks remain in the Schur complement. Equivalently,

$$\mathcal{B}_k[\nu] = \sum_{j=0}^q c_j \log \frac{\nu_j[\nu]}{\gamma_{q+j}[\nu]}, \quad c_0 = c_q = 1, \quad c_j = 2 \ (0 < j < q), \quad \sum c_j = 2q. \quad (\text{C3})$$

Here ν_j and γ_{q+j} are the attained monic relation and source norms, respectively. We use (C2) and (C3), rather than redefining the original baseline.

67.2.1 Exact dictionary between the incoming calculations

Use

$$M(\theta) = \int e^{\theta y} w_h(y) dy, \quad L = \log M, \quad \mu = L', \quad V = L'', \quad \mathfrak{M} = M(\alpha), \quad b = \mu(\alpha),$$

and let $\theta(t)$ invert μ on $[-\alpha, \alpha]$. For $0 \leq t \leq b$, define

$$D(t) = L(\alpha) - L(\theta(t)) - (\alpha - \theta(t))t. \quad (\text{C4})$$

The first continuation's $\ell_h, a_h, v_h^{\text{tilt}}, \mu_h$ are respectively the second continuation's L, μ, V, b . Their rate functions satisfy the literal equality

$$\boxed{F_h^{\text{cen}}(t) = -D(|t|)}. \quad (\text{C5})$$

The integral coefficients are also equal, without a limiting identification:

$$\boxed{\mathcal{C}_h = \mathfrak{I}_h^{\text{cen}} = \int_0^\alpha \mu(\theta)^2 d\theta =: I_h,}$$

$$\boxed{\mathcal{B}_h = \mathfrak{J}_h^{\text{cen}} = \int_0^\alpha V(\theta) \log \frac{\mu(\theta)}{4\pi V(\theta)} d\theta =: J_h.} \quad (\text{C6})$$

The second integral converges at zero because $\mu(\theta)$ is comparable to θ there. Both incoming notes therefore calculate the same central density profile and the same two coefficients. The first retains a stronger explicit remainder and a source-to-quotient comparison; the second supplies a complete standalone proof and the sharper relation-kernel weight count $2q$ instead of the valid larger count $2q + 1$. All are retained below, once.

67.2.2 Consolidated source estimate

The local estimate in the incoming ACW10–11 is

$$p_\theta^{*k}(k\mu(\theta)) = \frac{1 + \varepsilon_{k,\theta}}{\sqrt{2\pi k V(\theta)}}, \quad p_\theta(y) = \frac{e^{\theta y} w_h(y)}{M(\theta)}, \quad |\varepsilon_{k,\theta}| \leq \epsilon_k = O_h(k^{-1/8}). \quad (\text{C7})$$

The exact inverse accounting is

$$m_k(x) = M(\theta)^k e^{-\theta x} p_\theta^{*k}(x). \quad (\text{C8})$$

This follows from the original addition map $\sum y_j = x$. It retains the mass $M(\theta)^k$. At the auxiliary Hilbert-space level the isometry is $F \mapsto M(\theta)^{1/2} e^{-\theta y/2} F$ from $L^2(w_h dy)$ to $L^2(p_\theta dy)$; it is not asserted to preserve polynomial degree.

Put $c_\sigma(y) = \sigma(y) e^{\alpha|y|} \sqrt{1 + |y|}$. On $|t| \leq b$,

$$\ell_k(kt) := \log \frac{m_k(kt)}{\mathfrak{M}^k \sigma(kt)} = -kD(|t|) + \frac{1}{2} \log(|t| + k^{-1}) - \frac{1}{2} \log(2\pi V(\theta(t))) - \log c_\sigma(kt) + \log(1 + \varepsilon_{k,\theta(t)}). \quad (\text{C9})$$

The full origin term is present. With $L_k^{\text{loc}} = -\log(1 - \epsilon_k)$ on $\epsilon_k \leq 1/2$, the consolidated finite bound is

$$\left| \int_{-kb}^{kb} \ell_k(y) dy + k^2 I_h - k J_h \right| \leq 2kb L_k^{\text{loc}} + (kb + 1) \log(kb + 1) - kb \log(kb) + 2C_\Gamma^0 + 4 \log^+(kb), \quad (\text{C10})$$

$$C_\Gamma^0 = \max\{|\log(c_\Gamma/\sqrt{2})|, |\log(C_\Gamma/\sqrt{2})|\}.$$

In particular the error in (C10) is $O_h(k^{7/8} + \log k)$, which implies the second note's $o(k)$ remainder. To recover (C10) from the standalone note, use the complex Stirling bound $|R_\Gamma(z)| \leq 1/(6|z|)$ for $\Re z > 0$ in the principal term at $z = 1/4 + iy/2$. Direct substitution gives

$$|\log c_\sigma(y) - \tfrac{1}{2} \log 2| \leq 2/y \quad (y \geq 1).$$

The original Gamma envelopes control $[0, 1]$. The exact integral $k \int_0^b \log(1 + (kt)^{-1}) dt = (kb + 1) \log(kb + 1) - kb \log(kb)$ and (C9) prove the bound. The rate-area identity is $2 \int_0^b D(t) dt = I_h$, proved by Fubini or integration by parts using $\mu' = V$.

The source-coordinate approximation in the first note is retained as well. Define $\bar{m}_k$ from (C7)–(C8) by omitting $1 + \varepsilon$ on C_k , and by setting $\bar{m}_k = m_k$ outside C_k . The identity on every original affine remainder fibre gives

$$(1 - \epsilon_k) G_N[\bar{m}_k] \preceq G_N[m_k] \preceq (1 + \epsilon_k) G_N[\bar{m}_k],$$

$$|\mathcal{B}_k[m_k] - \mathcal{B}_k[\bar{m}_k]| \leq 2q \log \frac{1 + \epsilon_k}{1 - \epsilon_k} = o(q). \quad (\text{C11})$$

(C10) is not assigned as a determinant coefficient. The additional work below controls an actual endpoint determinant response.

67.3 2. Retained source constants and local-error formula

The only arithmetic envelope inputs used are RTM1–3 and the original three-factor lower bound:

$$\begin{aligned} e^{\alpha|y|} w_h(y) &\leq C(1 + |y|)^{-p}, \quad p = 8m_0 - 9/2, \\ w_h^{*3}(y) &\geq c_h e^{-\alpha|y|} (1 + |y|)^{-B}, \quad B = 42 + 8m_0, \quad M_* = B/2, \end{aligned} \quad (\text{L1})$$

$$c_\Gamma e^{-\alpha|y|} (1 + |y|)^{-1/2} \leq \sigma(y) \leq C_\Gamma e^{-\alpha|y|} (1 + |y|)^{-1/2}, \quad c_\Gamma = \sqrt{2} e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}} e^{2/3}.$$

The global upper convolution envelope is

$$m_k(y) \leq \frac{C}{c_\Gamma} k^{p+1} \mathfrak{M}^{k-1} \sigma(y). \quad (\text{L2})$$

For clarity, the finite local-error constants from the standalone note are reproduced. Let $\eta = 1/4$, $M_0 = M(0)$, and

$$V_* = \frac{2C}{M_0} B(3, p-3), \quad H_* = 2^{1+\eta} \left[\frac{2C}{M_0} B(3 + \eta, p-3-\eta) + b^{2+\eta} \right],$$

$$a_* = \min \left\{ \frac{1}{4}, \frac{c_h e^{-2\alpha} 2^{-B}}{\mathfrak{M}^3} \right\}, \quad \beta = 2a_*, \quad v_* = \beta/9,$$

$$\xi_0 = \min \left\{ 1, (2V_*)^{-1/2}, (v_*/(16H_*))^{1/\eta} \right\}, \quad r_* = (1 - \beta \xi_0^2/7)^{1/3},$$

$$C_2 = \frac{4\pi(C/M_0)^2}{2p-1}, \quad C_0 = 4H_* + V_*^2/8.$$

For an integer $n \geq 2$, put $a_n = v_*(n-1)/4$ and

$$A_n^{\text{loc}} = \frac{1}{2\pi} \left[C_0 n \Gamma((3+\eta)/2) a_n^{-(3+\eta)/2} + C_2 r_*^{n-2} + \frac{2e^{-nv_*\xi_0^2/2}}{nv_*\xi_0} \right], \quad \epsilon_n = \sqrt{2\pi n V_*} A_n^{\text{loc}}. \quad (\text{L3})$$

These retain $v_* \leq V(\theta) \leq V_*$. They are local-limit constants, not the arithmetic upper-envelope symbol in (L2).

The Fourier proof of (C7) establishes slightly more than its displayed mean value. The same absolute error holds at every shifted evaluation point:

$$\left| p_\theta^{*n}(n\mu(\theta) + x) - \frac{e^{-x^2/(2nV(\theta))}}{\sqrt{2\pi n V(\theta)}} \right| \leq A_n^{\text{loc}}, \quad x \in \mathbb{R}. \quad (\text{L4})$$

Indeed, Fourier inversion inserts the factor $e^{-i\xi x}$, of modulus one, into the very same integrable difference estimated in ACW6–10. No additional moment or large-deviation hypothesis is needed.

67.4 3. Closing the transition gap and extracting the full central rate

Define

$$W_k(y) = \begin{cases} kD(|y|/k), & |y| \leq kb, \\ 0, & |y| > kb, \end{cases} \quad D_0 = D(0) = L(\alpha) - L(0) > 0, \quad \Lambda_k = kD_0. \quad (\text{L5})$$

Then $0 \leq W_k \leq \Lambda_k$. Define the exact remaining factor

$$\psi_k(y) = \frac{m_k(y)}{\mathfrak{M}^k \sigma(y)} e^{W_k(y)}. \quad \text{Thus} \quad \boxed{\mathfrak{M}^{-k} m_k = \psi_k e^{-W_k} \sigma}. \quad (\text{L6})$$

This factorization changes neither the source nor the packet. Its significance is the new whole-line polynomial bound proved next.

Set $n = k - 3$, $V_b = V(\alpha)$, and $a_G = e^{-1/2}/(2\sqrt{2\pi})$. Work on the explicit cofinal range

$$\epsilon_k \leq 1/2, \quad \epsilon_{k-3} \leq e^{-1/2}/2, \quad k \geq V_b/b^2. \quad (\text{L7})$$

By (L4), the tilted convolution places mass at least a_G in

$$I_n = [nb - \sqrt{nV_b}, nb].$$

On this interval the Gaussian density is at least $e^{-1/2}/\sqrt{2\pi n V_b}$, and the error is at most half this value. Integrating over the length $\sqrt{nV_b}$ proves the mass bound.

For every $y \geq kb$ and $z \in I_n$,

$$0 < y - z \leq y + \sqrt{nV_b} \leq 2y.$$

Apply the original three-factor lower bound within the exact convolution and retain the full tilted mass. This gives

$$\boxed{m_k(y) \geq c_h a_G \mathfrak{M}^{k-3} e^{-\alpha y} (1+2y)^{-B}, \quad y \geq kb.} \quad (\text{L8})$$

Evenness gives the negative half-line. In contrast to the old positive-width guard beyond the boundary-tilted mean, this bound starts at $|y| = kb$ itself. In particular there is no untreated interval between the saddle window and the retained tail.

Divide (L8) by the upper Gamma envelope and use $1 + 2y \leq \sqrt{5}\sqrt{1+y^2}$. Define

$$c_{\text{tail}} = \frac{c_h a_G}{\mathfrak{M}^3 C_\Gamma 5^{M_*}}, \quad c_{\text{core}} = \frac{1}{2\sqrt{2\pi V_*} C_\Gamma}, \quad C_{\text{core}} = \frac{3\sqrt{b+1}}{2\sqrt{2\pi v_*} C_\Gamma}.$$

From (C9), on the core, $c_{\text{core}}k^{-1/2} \leq \psi_k \leq C_{\text{core}}$. Set

$$c_- = \min\{1, c_{\text{tail}}, c_{\text{core}}\}, \quad c_+ = \max\{1, C_{\text{core}}, C/(c_{\Gamma}\mathfrak{M})\}.$$

Then the exact whole-line result is

$$\boxed{c_-k^{-1/2}(1+y^2)^{-M_*} \leq \psi_k(y) \leq c_+k^{p+1}.} \quad (\text{L9})$$

The factor ω_k/σ satisfies the same bounds: it is one on C_k , and equals ψ_k off C_k . This statement retains the actual zeta-dependent density on the whole outer line. The original exponential-in- k variation is now in the explicit factor e^{-W_k} , not hidden in a comparison constant.

67.5 4. Polynomial comparison also holds after inserting the rate

Write $d = 2q$ and $d\tau_t = e^{-tW_k}\sigma(y)dy$, $0 \leq t \leq 1$. Impose the explicit eventual guard

$$\Lambda_k \leq d \log 16. \quad (\text{L10})$$

For every polynomial P of degree at most d ,

$$\boxed{\int y^2 |P|^2 d\tau_t \leq 272(d+1)^2 \|P\|_{\tau_t}^2.} \quad (\text{L11})$$

Here is a proof retaining the whole tail. The original Gamma recurrence bounds multiplication by y on degree j by $2(j+1)$. Hence

$$\|y^{d+1}P\|_{\sigma}^2 \leq [4(d+1)]^{2d+2} \|P\|_{\sigma}^2.$$

On $|y| > 16(d+1)$, this bounds the second-moment tail by

$$16(d+1)^2 16^{-d} \|P\|_{\sigma}^2 \leq 16(d+1)^2 e^{\Lambda_k} 16^{-d} \|P\|_{\tau_t}^2.$$

The inner part is at most $256(d+1)^2 \|P\|_{\tau_t}^2$. Guard (L10) proves (L11). This argument does not infer a tail estimate merely by dropping the core suppression.

Jensen's inequality for $x \mapsto (1+x)^{-M_*}$, applied to the measure $|P|^2 d\tau_t / \|P\|_{\tau_t}^2$, now proves

$$L_k \|P\|_{\tau_t}^2 \leq \|P\|_{\psi_k \tau_t}^2 \leq U_k \|P\|_{\tau_t}^2,$$

$$L_k \|P\|_{\tau_t}^2 \leq \|P\|_{(\omega_k/\sigma)\tau_t}^2 \leq U_k \|P\|_{\tau_t}^2, \quad (\text{L12})$$

where

$$L_k = c_-k^{-1/2}[1 + 272(d+1)^2]^{-M_*}, \quad U_k = c_+k^{p+1}, \quad \kappa_k = U_k/L_k \geq 1. \quad (\text{L13})$$

Thus $\log \kappa_k = O_h(\log q)$. The same scalar inequalities hold on every fixed restriction and every attained quotient of these polynomial spaces: apply them to every representative in its specified affine fibre before taking the minimum. This is the comparison used in the next section; no metric equality is asserted.

67.6 5. A fixed Taylor-jet kernel sees exponentially little central mass

Let b_j be the original Gamma monic polynomials, and

$$\gamma_j = \sqrt{2\pi} j! (1/2)_j = \sqrt{2\pi} (2j)! / 4^j, \quad \varphi_j = b_j / \sqrt{\gamma_j}.$$

Their exact generating function, equivalently derived by summing their recurrence, is

$$\sum_{j \geq 0} \frac{b_j(z)}{j!} t^j = (1 + t^2)^{-1/4} \exp(z \arctan t). \quad (\text{L14})$$

This is the original-coordinate Meixner–Pollaczek generating function; it is also DLMF 18.23.7 after $\lambda = 1/4, x = z/2, \varphi = \pi/2$.

Cauchy's estimate on $|t| = (d+1)/(d+2)$, together with $|\arctan t| \leq \frac{1}{2} \log(2(d+2))$, gives, for $|z| \leq Z$,

$$\sum_{j=0}^d |\varphi_j(z)|^2 \leq \frac{e^2 (d+1)^2 \sqrt{d+2}}{\sqrt{2\pi}} [2(d+2)]^Z. \quad (\text{L15})$$

The factorial estimate used here is $(j!)^2 / \gamma_j \leq (2j+1) / \sqrt{2\pi}$, and $\sum_{j=0}^d (2j+1) = (d+1)^2$. This preserves the original Gamma mass.

Let $K = kb$, and let $\mathcal{J}_r P$ be the first r Taylor coefficients at zero. Its kernel is the literal subspace

$$Z_{N,r} = y^r \mathcal{P}_{N-r} \subset \mathcal{P}_N. \quad (\text{L16})$$

For $P \in Z_{N,r}$, $N \leq d$, apply Cauchy's coefficient formula on $|z| = 2K$ and sum the geometric tail on $|y| \leq K$. Equation (L15) and the full mass $\sqrt{2\pi}$ give

$$\boxed{\int_{C_k} |P|^2 d\sigma \leq a_{k,r} \|P\|_{\sigma}^2, \quad a_{k,r} = 4^{1-r} \mathcal{A}_{q,k},} \quad (\text{L17})$$

$$\mathcal{A}_{q,k} = e^2 (d+1)^2 \sqrt{d+2} [2(d+2)]^{2K}.$$

There is also an operator version. Let B_N restrict a polynomial in the Gamma orthonormal frame to $L^2(C_k, \sigma dy)$, and let $\tilde{B}_{N,r}$ first truncate its Taylor series below degree r , then restrict. The same calculation proves

$$\|B_N - \tilde{B}_{N,r}\| \leq \sqrt{a_{k,r}}, \quad \text{rank } \tilde{B}_{N,r} \leq r. \quad (\text{L18})$$

Choose the explicit integer

$$r = \left\lceil 1 + \frac{\log \mathcal{A}_{q,k} + 2\Lambda_k + 2 \log \kappa_k + 10 \log q + \log 256}{\log 4} \right\rceil. \quad (\text{L19})$$

Use its explicit eventual guard $r \leq q$. Then

$$a := a_{k,r} \leq \frac{e^{-2\Lambda_k}}{256 \kappa_k^2 q^{10}}, \quad r = O_h(k \log q) = o(q). \quad (\text{L20})$$

This Taylor map is an auxiliary estimate and computational compression. It is not the programme's period observation and is not identified with its kernel. Its exact relation to each determinant is the Schur factorization used next.

67.7 6. Transfer the nonlinear central loss through the actual metrics

For $g = 1$ or $g = \psi_k$, define the full source loss

$$D_N(g) = \log \det H_N[g\sigma] - \log \det H_N[ge^{-W_k}\sigma] \geq 0. \quad (\text{L21})$$

Decompose $\mathcal{P}_N$ by the exact sequence

$$0 \longrightarrow Z_{N,r} \longrightarrow \mathcal{P}_N \xrightarrow{\mathcal{J}_r} \mathbb{C}^r \longrightarrow 0.$$

The quotient metric is the attained one, $G_{N,r}^{\text{jet}}[\nu] = (\mathcal{J}_r H_N[\nu]^{-1} \mathcal{J}_r^*)^{-1}$. Its Schur complement retains both cross factors and the full high-monomial Gram inverse. The determinant factors into the restriction determinant and this quotient determinant, in a fixed monic frame.

On the restriction $Z_{N,r}$, insertion of e^{-W_k} loses at most a of the Gamma norm, and at most $\kappa_k a$ of the $\psi_k \sigma$ norm, by (L12) and (L17). Put $f(x) = -\log(1-x)$ for $0 \leq x < 1$. The two restriction losses lie between zero and $(N+1-r)f(\kappa_k a)$.

On the jet quotient, (L12) bounds the relative determinant at both $t=0$ and $t=1$ in the same interval $[r \log L_k, r \log U_k]$. Subtracting those endpoints, before taking absolute values, gives

$$|D_N(\psi_k) - D_N(1)| \leq r \log \kappa_k + (N+1-r)f(\kappa_k a). \quad (\text{L22})$$

This is a comparison of the nonlinear, finite-amplitude central loss. It is not a first variation at the Gamma source.

Define

$$\Phi(g) = D_{2q-1}(g) + D_{2q}(g) - D_{q-1}(g) - D_q(g).$$

Adding the four estimates proves

$$|\Phi(\psi_k) - \Phi(1)| \leq 4r \log \kappa_k + (6q+2)f(\kappa_k a). \quad (\text{L23})$$

67.7.1 Remove the central prefactor, without changing the outer density

The two densities $\psi_k \sigma$ and ω_k agree off C_k . On C_k their difference has absolute value at most $2U_k \sigma$. On $Z_{N,r}$, (L17) and (L12) therefore bound the relative change by $2\kappa_k a$. On the jet quotient the relative spectrum lies in $[\kappa_k^{-1}, \kappa_k]$, giving absolute log determinant at most $r \log \kappa_k$. Thus their full source determinant difference, at each endpoint, is bounded by

$$r \log \kappa_k + (N+1-r)f(2\kappa_k a). \quad (\text{L24})$$

The original relation part is included in Section 7. Equations (L22)–(L24) are why a polynomial bound on the remaining factor suffices: it is charged on $r = O(k \log q)$ quotient coordinates, not on all q coordinates independently.

67.8 7. Retain the original relation determinant and its sign

Set

$$R = k\sqrt{\delta^2 + \gamma^2}, \quad Y = 2^{-16}q, \quad B_\chi = \frac{qR^2}{Y^2 - R^2},$$

$$\eta_0 = \frac{\sqrt{2\pi}}{c_\Gamma} \frac{(2q+1)(4q+1)\sqrt{1+2q}}{q} 16^{-q}, \quad \eta_\Gamma = \frac{e^{B_\chi} \eta_0}{1 - \eta_0}. \quad (\text{L25})$$

Use the original cofinal guards $q \geq 100, R \leq Y/4, K \leq Y$. For every polynomial P of degree at most q , the retained full-primary estimate is

$$\int_{C_k} |QP|^2 d\sigma \leq \eta_\Gamma \|QP\|_\sigma^2. \quad (\text{L26})$$

It follows by pairing the original opposite roots outside $[-Y, Y]$ and by the full shifted-Legendre expansion on $[q, 2q]$ inside. The inner polynomial bound is

$$|P(y)|^2 \leq \frac{(2q+1)(4q+1)100^{2q}}{q} \int_q^{2q} |P(t)|^2 dt,$$

while the reference power norm is at least $c_\Gamma q^{2q} e^{-\pi q} (1+2q)^{-1/2} \int_q^{2q} |P|^2$. The inner multiplier is at most $[(9/4)2^{-32}q^2]^q$, and $(9/4)e^\pi 10^4 2^{-32} < 1/16$. These bounds give (L26) with both inner contributions and the full mass retained. This is also ACW27–29 in the frozen standalone note.

By (L12), the central fraction in the $\psi_k \sigma$ relation norm is at most $\kappa_k \eta_\Gamma$. Hence insertion of e^{-W_k} changes each attained monic relation norm by a factor in $[1 - \kappa_k \eta_\Gamma, 1]$. Its full weighted loss is

$$0 \leq \Psi^{\text{rel}}(\psi_k) := \sum_{j=0}^q c_j \log \frac{\nu_j[\psi_k \sigma]}{\nu_j[\psi_k e^{-W_k} \sigma]} \leq 2q f(\kappa_k \eta_\Gamma). \quad (\text{L27})$$

The exact source/relation identity gives the signed equation

$$\mathcal{B}_k[\psi_k e^{-W_k} \sigma] - \mathcal{B}_k[\psi_k \sigma] = \Phi(\psi_k) - \Psi^{\text{rel}}(\psi_k). \quad (\text{L28})$$

Thus the relation contribution has not been assigned zero or the wrong sign.

For the central prefactor comparison between $\psi_k \sigma$ and ω_k , the analogous relative relation-Gram error is at most $2\kappa_k \eta_\Gamma$. The sum of the four relation dimensions is $2q + 2$. Its absolute signed log-determinant contribution is bounded by

$$(2q + 2)f(2\kappa_k \eta_\Gamma). \quad (\text{L29})$$

Use the explicit eventual guard $2\kappa_k \eta_\Gamma \leq 1/2$. Both (L27) and (L29) are exponentially small: $\eta_\Gamma = \exp[-q \log 16 + O_h(\log q + 1)]$, and κ_k is polynomial in q, k .

67.9 8. Four explicit Gamma matrices, with a finite total error

The Gamma loss $D_N(1)$ can be evaluated with no source interpolation. Define the central moment matrix

$$M_{W,k}[i, j] = \int_{-K}^K (1 - e^{-W_k(y)}) y^{i+j} \sigma(y) dy, \quad 0 \leq i, j < r. \quad (\text{L30})$$

Let

$$T_{r,N}[j, n] = [y^j] \varphi_n(y), \quad E_{r,N} = T_{r,N} T_{r,N}^*, \quad 0 \leq n \leq N. \quad (\text{L31})$$

All entries in T come from (L14) or the exact Gamma recurrence. All entries in $M_{W,k}$ come from the fixed function (C4) on a finite interval. Define

$$\widehat{D}_N = -\log \det(I_r - M_{W,k}^{1/2} E_{r,N} M_{W,k}^{1/2}),$$

$$\boxed{\widehat{\Phi}_{q,k} = \widehat{D}_{2q-1} + \widehat{D}_{2q} - \widehat{D}_{q-1} - \widehat{D}_q.} \quad (\text{L32})$$

The square root can be avoided in a determinant computation using $\det(I_r - M_{W,k} E_{r,N})$, by Sylvester's determinant identity. Its positive self-adjoint representative is the displayed one; a non-Hermitian coordinate matrix is not silently treated as Hermitian.

To prove the approximation, let C_N be the Gamma compression of multiplication by $(1 - e^{-W_k})1_{C_k}$. Then $D_N(1) = -\log \det(I - C_N)$, and $I - C_N \succeq e^{-\Lambda_k}I$. By (L18), its Taylor approximation $\tilde{C}_N = T^*M_{W,k}T$ satisfies

$$\|C_N - \tilde{C}_N\| \leq 2\sqrt{a} + a.$$

Put $z_k = e^{\Lambda_k}(2\sqrt{a} + a)$. Equation (L20) gives $z_k < 1/2$, so the approximate matrix is positive and

$$|D_N(1) - \hat{D}_N| \leq (N+1)f(z_k). \quad (\text{L33})$$

For example, conjugate the difference by $(I - C_N)^{-1/2}$; its norm is at most z_k , so every eigenvalue of the relative matrix lies in $[1 - z_k, 1 + z_k]$. This proves (L33) without assuming a small inverse norm independent of the rate depth.

Both $D_N(1)$ and $\hat{D}_N$ increase with N . For the latter, $E_{r,N}$ increases in the positive order. For the former, write $I - C_N$ in successive Gamma coordinates; its new Schur complement is at most one, or use compression interlacing. Hence $\Phi(1) \geq 0$ and $\hat{\Phi}_{q,k} \geq 0$.

Combining (L23), (L24), (L27)–(L29), and (L33) yields (R1) with the fully explicit radius

$$\boxed{E_k = 8r \log \kappa_k + (6q + 2)\{f(\kappa_k a) + f(2\kappa_k a) + f(z_k)\} + (2q + 2)f(2\kappa_k \eta_\Gamma) + 2qf(\kappa_k \eta_\Gamma).} \quad (\text{L34})$$

Every quantity on this right-hand side is given above. It contains no original growing Gram inverse. The prescribed rank gives

$$E_k = O_h(k(\log q)^2) = o_h(k^2) = o_h(q). \quad (\text{L35})$$

The pointwise relative approximation in (C11) is not being charged again; (L34) is a direct estimate on the actual source. In particular it is stronger at this stage than multiplying the earlier local-limit error by an assumed constant kernel density.

67.10 9. The central degree-band response is only $O_h(k^2)$

The rank $r = O(k \log q)$ establishes the transfer, but a separate exact recurrence argument improves the size of the response itself. This is essential for the scale conclusion in (R3).

The Gamma orthonormal recurrence is

$$y\varphi_n = a_{n+1}\varphi_{n+1} + a_n\varphi_{n-1}, \quad a_n = \sqrt{n(n-1/2)}. \quad (\text{L36})$$

Consider either original degree band $n = n_0, \dots, n_0 + q - 1$, where $n_0 = q$ or $q + 1$. Relative to the two initial functions $\varphi_{n_0}, \varphi_{n_0-1}$, write

$$\varphi_n(y) = A_n(y)\varphi_{n_0}(y) + B_n(y)\varphi_{n_0-1}(y). \quad (\text{L37})$$

These coefficients are obtained by the actual transfer matrices

$$\begin{pmatrix} \varphi_{n+1} \\ \varphi_n \end{pmatrix} = \begin{pmatrix} z/a_{n+1} & -a_n/a_{n+1} \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \varphi_n \\ \varphi_{n-1} \end{pmatrix}.$$

At $z = 0$ each transfer has norm one because $a_n < a_{n+1}$. Its norm at z is at most $1 + |z|/a_{n+1}$. There are fewer than q factors, and $a_{n+1} \geq q$, so

$$|A_n(z)|, |B_n(z)| \leq e^{|z|}. \quad (\text{L38})$$

This removes the common high-degree oscillation by an exact two-function representation, not by an asymptotic kernel substitution.

Truncate A_n, B_n below Taylor degree s , applying Cauchy's formula on $|z| = 2K$. The restriction map from this Gamma-orthonormal degree band to $L^2(C_k, \sigma dy)$ then has a rank-at-most $2s$ approximation, with operator error at most

$$\nu_s = 2^{2-s} \sqrt{q} e^{2K}. \quad (\text{L39})$$

Indeed each coefficient tail is at most $2^{1-s} e^{2K}$; the q coefficient columns cost $\sqrt{q}$, and the two initial functions each have full Gamma norm one. The approximation range is spanned by $y^j \varphi_{n_0}, y^j \varphi_{n_0-1}$, $j < s$. No orthogonality of these range functions is assumed. These are auxiliary output functions for the restriction operator, not replacement representatives at the original degree cutoff.

67.10.1 Retain the complete lower-degree Schur inverse

Let V_L, V_H be the low-degree and band restriction maps multiplied by $\sqrt{1 - e^{-W_k}}$. The full suppressed source Gram is $I - V^*V$. Its attained band Gram is exactly

$$\boxed{S = I - V_H^* \mathcal{A}_L V_H, \quad \mathcal{A}_L = I + V_L(I - V_L^* V_L)^{-1} V_L^*} \quad (\text{L40})$$

Here the complete preceding source Gram inverse remains. It satisfies $\|\mathcal{A}_L\| \leq e^{\Lambda_k}$, and the minimum definition gives $S \succeq e^{-\Lambda_k} I$. Thus both possible amplifications of a band approximation must be paid.

Choose

$$\boxed{s = \left\lceil 2 + \frac{2K + 2\Lambda_k + (11/2) \log q + \log 8}{\log 2} \right\rceil} \quad (\text{L41})$$

Then $\nu_s \leq e^{-2\Lambda_k} q^{-5}/8$. Replacing only V_H by its rank- $2s$ approximation changes the matrix in (L40) by at most $e^{\Lambda_k} \nu_s (2 + \nu_s)$. Relative to S , its norm is at most

$$u_k = e^{2\Lambda_k} \nu_s (2 + \nu_s) \leq 3q^{-5}/8 < 1/2.$$

The approximate band Gram is positive, at least $e^{-\Lambda_k} (1 - u_k) I$, and differs from the identity in at most $2s$ directions. Therefore

$$0 \leq -\log \det S \leq 2s\{\Lambda_k + f(u_k)\} + qf(u_k). \quad (\text{L42})$$

The lower bound follows from $S \preceq I$. For the upper bound, bound the at most $2s$ approximate eigenvalue losses by $\Lambda_k + f(u_k)$, then use the full q -dimensional relative log-determinant bound. This argument explicitly preserves the lower-space coupling in (L40).

Each of the two differences making up $\Phi(1)$ is the loss (L42) on its original band. Hence

$$\boxed{0 \leq \Phi(1) \leq 4s(\Lambda_k + 1) + 2qf(u_k) = O_h(k^2)}. \quad (\text{L43})$$

Equations (L33)–(L35) prove (R3). More quantitatively,

$$\boxed{0 \leq \liminf \frac{\delta_k^{\text{cen, end}}}{k^2} \leq \limsup \frac{\delta_k^{\text{cen, end}}}{k^2} \leq \frac{8D_0(b + D_0)}{\log 2}}. \quad (\text{L44})$$

This is a signed asymptotic interval, not a computed limiting coefficient. It follows directly from $s/k \rightarrow (2b + 2D_0)/\log 2$, $\Lambda_k/k = D_0$, and $E_k/k^2 \rightarrow 0$.

67.10.2 The scale conclusions on the original multiplicity strata

For every fixed m_0 , $q \geq (k+1)^2$, so

$$\boxed{\frac{\delta_k^{\text{cen},\text{end}}}{q \log k} \rightarrow 0, \quad \delta_k^{\text{cen},\text{end}} = O_h(q).} \quad (\text{L45})$$

On the simple stratum, $q = (k+1)^2$, (L44) is an explicit nonnegative limiting interval at the q scale. On every fixed stratum $m_0 \geq 2$, $q \asymp_{m_0} k^3$, and

$$\boxed{\delta_k^{\text{cen},\text{end}} = o_h(q), \quad \delta_k = \delta_k^{\text{out}} + o_h(q).} \quad (\text{L46})$$

Unlike the incoming unweighted area comparison, (L45)–(L46) are proved for the actual endpoint quotient-determinant contribution with its full relation determinant retained.

67.11 10. Endpoint accounting, the earlier spatial split, and the receiver

The outer-only density is explicitly ω_k , and

$$\delta_k^{\text{out}} + \delta_k^{\text{cen},\text{end}} = \mathcal{B}_k[m_k] - \mathcal{B}_k[\sigma]. \quad (\text{L47})$$

The mass cancellation used here is exact: $G_N[A\nu] = AG_N[\nu]$, so four signs cancel $q \log A$. It is equivalently the scalar coefficient map $P \mapsto \sqrt{A}P$, with the same map on remainder values, between source measures ν and $A^{-1}\nu$.

An isolated spatial central integral along the earlier diagonal interpolation is not identified with this central endpoint difference. Their exact relation can be given. Let

$$f_C = 1_{C_k} \ell_k, \quad f_O = 1_{\mathbb{R} \setminus C_k} \ell_k, \quad F(a, b_0) = \mathcal{B}_k[\sigma e^{a f_C + b_0 f_O}].$$

The earlier central spatial term is $\int_0^1 F_a(t, t) dt$, whereas the endpoint central term is $\int_0^1 F_a(t, 1) dt$. Consequently

$$\boxed{\delta_k^{\text{cen},\text{end}} - \delta_k^{\text{cen},\text{diag}} = \int_0^1 \int_t^1 F_{ab_0}(t, b_0) db_0 dt.} \quad (\text{L48})$$

The mixed derivative is determined by the original source and relation kernels. For a single full source determinant, differentiation twice gives

$$\partial_a \partial_{b_0} \log \det H_N = \text{Tr}(H_N^{-1} H_{N,ab_0}) - \text{Tr}(H_N^{-1} H_{N,b_0} H_N^{-1} H_{N,a}).$$

The first term is zero since $f_C f_O = 0$. The second is the negative of

$$\iint f_C(y) f_O(z) |K_N(y, z)|^2 d\nu(y) d\nu(z).$$

The relation expression uses its full relation kernel; (C2) and the four signs give F_{ab_0} . This explicitly records the path-interchange contribution, rather than presuming that the earlier two spatial pieces are individually path independent.

The complete action receives the new result as

$$\boxed{J_k^{\text{action}} = \mathcal{B}_k[\sigma] + \delta_k^{\text{out}} + \widehat{\Phi}_{q,k} + \mathfrak{e}_k + W_k^{\text{mono}} - P_{k,-} - P_{k,+}.} \quad (\text{L49})$$

Here W_k^{mono} denotes the programme's original monic-window term; it is distinguished from the explicitly defined rate function $W_k(y)$. Every complementary term and both penalty terms remain. No favourable sign has been assigned to δ_k^{out} , and the known positive canonical baseline is not removed.

67.12 11. What remains to be evaluated

There are now two precisely separated value calculations.

The outer correction is exactly $\mathcal{B}_k[\omega_k] - \mathcal{B}_k[\sigma]$, with the actual convolution outside $[-kb, kb]$. The original unequal upper and lower tail powers have not been converted into a sharp arithmetic tail coefficient. Its $q \log k$ and q terms remain to be evaluated.

On the simple stratum, the remaining central order- q value is that of the explicit matrices (L30)–(L32). This note proves an $o(q)$ transfer and a signed $O(q)$ interval, not a limit formula for those matrices. On higher fixed multiplicity strata that matrix contribution is $o(q)$, by (L43), and is no longer needed for the order- q coefficient.

In particular, the density-area value $-I_h$ is not inserted as the baseline coefficient. No unproved constant-density limit for the source band has been used. No current geometric purity theorem is applied without a constructed metric comparison. The new estimates are source and determinant calculations and do not constitute programme closure.

67.13 Source record and verification scope

Imported arithmetic inputs are the original RTM envelopes, the full even source and packet, SXR's source/relation identity, and the original Gamma recurrence and mass. Their upstream analytic proofs have not received a new exhaustive audit. The incoming central local-limit proof is retained in full, with its constants reproduced and the one additional shifted-point consequence proved above.

The generating-function convention and the complex Stirling bound were checked against the NIST DLMF, Sections 18.23 and 5.11. The degree-band transfer, whole-line prefactor comparison, Taylor-jet localization, nonlinear endpoint transfer, finite determinant error, and the $O(k^2)$ band estimate are derived in this note.

The executable checks are auxiliary exact matrix and polynomial diagnostics. They do not evaluate an actual off-critical xi packet, prove every analytic inequality by computation, or supply an independent review or formal verification of this derivation. Original inputs are frozen with hashes; their earlier check counts are not counted as new analytic results.

67.13.1 Executed auxiliary diagnostics

The new checker passed 88 exact checks and rejected five deliberately false formulas. Its ordinary and `python -O` outputs are byte-identical. The original checker was replayed unchanged: all 39 checks and four false-formula controls agree with the original JSON receipt and with its optimized run. These statements report the actual executions recorded in `checks/EXECUTION_RECEIPT.json`; they do not upgrade the written analytic derivation to an independent or formal verification.

Original central source and tilted local-limit proof

68 Signed arithmetic baseline: central-source continuation

18 September 2026. Written research derivation; not independently reviewed or formally verified.

68.1 Completion status

The assigned target is the finer value of

$$\delta_{k,1}^{\text{ar}} = \mathcal{B}_k^{\text{ar}} - \mathcal{B}_k^{\sigma}.$$

Its requested $q \log k$ - and q -scale coefficients are not computed in this note. The last integration in *Untitled 1775*, lines 13324–13328, also leaves that coefficient uncomputed. This note proves a new source-side central-window expansion and writes its exact, still unevaluated, return into the assigned determinant. Its central coefficient must not be recorded as the coefficient of $\delta_{k,1}^{\text{ar}}$.

The precision target in *Untitled 1775*, lines 9389–9397, is a value for the signed correction, not just a convergent enclosure around an unevaluated centre. That target is preserved here. No numerical hypothetical xi packet is supplied. All constants below retain their dependence on the fixed original packet.

68.2 1. Original objects and the imported source inequalities

Fix the programme's stipulated full quartet

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2,$$

$$h(s) = \prod_{z \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s-z)^{m_0}, \quad m_0 \geq 1,$$

and retain

$$v_h = 2\xi/h, \quad w_h(y) = \frac{|v_h(1/2 + iy)|^2}{2\pi}, \quad m_k = w_h^{*k},$$

$$k \equiv 1 \pmod{4}, \quad k \geq 5, \quad e = 1 + k(m_0 - 1), \quad q = e(k+1)^2, \quad c = k/2,$$

$$\chi(S) = \prod_{a,b=0}^k (S - c - (2a - k)\delta - i(2b - k)\gamma)^e.$$

The reference is exactly

$$\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}, \quad \int_{\mathbb{R}} \sigma(y) dy = \sqrt{2\pi}, \quad \alpha = \pi/2.$$

The RTM1–3 inputs used below, with their original constants, are

$$e^{\alpha y} w_h(y) \leq C(1 + |y|)^{-p}, \quad p = 8m_0 - 9/2 \geq 7/2, \quad C = C_h^{\text{tilt}}, \quad (\text{ACW1})$$

$$w_h^{*3}(y) \geq c_h e^{-\alpha|y|} (1 + |y|)^{-B}, \quad B = 42 + 8m_0,$$

$$c_{\Gamma} \frac{e^{-\alpha|y|}}{\sqrt{1 + |y|}} \leq \sigma(y) \leq C_{\Gamma} \frac{e^{-\alpha|y|}}{\sqrt{1 + |y|}}, \quad c_{\Gamma} = \sqrt{2}e^{-7/6}, \quad C_{\Gamma} = \sqrt{2\sqrt{5}}e^{2/3}.$$

The original density is even and positive almost everywhere. The constants and those properties are retained inputs, not re-audited upstream claims. The proof uses the three-fold lower bound only on the fixed interval $[-1, 1]$. This continuation does not require its large- y lower power to equal the large- y upper power.

Exact source pin: 06_Foundations_and_Original_Six_Readers(1).tex, lines 112323–112602, RTM1–11. The full original source has not been replaced by a model density.

68.3 2. Tilted mass, mean, and variance on the closed tilt interval

Define the actual one-factor integrals

$$\begin{aligned} M(\theta) &= \int_{\mathbb{R}} e^{\theta y} w_h(y) dy, \quad L(\theta) = \log M(\theta), \quad |\theta| \leq \alpha, \\ \mu(\theta) &= L'(\theta), \quad V(\theta) = L''(\theta), \quad \mathfrak{M} = M(\alpha), \quad b_h = \mu(\alpha). \end{aligned} \quad (\text{ACW2})$$

Evenness and ACW1 imply

$$e^{\theta y} w_h(y) \leq C(1 + |y|)^{-p} \quad (|\theta| \leq \alpha).$$

Take $\eta = 1/4$. Since $p > 3 + \eta$, dominated differentiation through order two is valid on the closed interval, with finite uniformly bounded absolute moments of order $2 + \eta$. Thus μ is continuously differentiable, $\mu' = V$, and

$$\begin{aligned} \mu(\theta) &= \frac{\int y e^{\theta y} w_h(y) dy}{M(\theta)}, \\ V(\theta) &= \frac{\int (y - \mu(\theta))^2 e^{\theta y} w_h(y) dy}{M(\theta)} > 0. \end{aligned}$$

Strict positivity follows because the density is not concentrated at a point. Furthermore M is even, μ odd, and V even. Consequently $b_h > 0$, and μ maps $[-\alpha, \alpha]$ bijectively onto $[-b_h, b_h]$. Write its inverse as $\theta(t)$.

No source probability normalization is made in ACW2. For the Fourier proof only, introduce

$$p_\theta(y) = \frac{e^{\theta y} w_h(y)}{M(\theta)}.$$

The exact map back to the original convolution is

$$\boxed{m_k(x) = M(\theta)^k e^{-\theta x} p_\theta^{*k}(x)}. \quad (\text{ACW3})$$

Indeed, multiplication of the k densities gives the factor $M(\theta)^{-k} e^{\theta \sum y_j}$; the original addition map has $\sum y_j = x$ and Jacobian one. Every original mass is therefore present in ACW3.

68.4 3. Explicit uniform Fourier bounds, including the boundary tilts

This section proves the local density estimate rather than invoking an unspecified uniform local limit theorem.

Put $M_0 = M(0)$, and write B for Euler's beta function. Define

$$\begin{aligned} V_* &= \frac{2C}{M_0} B(3, p-3), \\ H_* &= 2^{1+\eta} \left[\frac{2C}{M_0} B(3+\eta, p-3-\eta) + b_h^{2+\eta} \right]. \end{aligned} \quad (\text{ACW4})$$

These follow by integrating the original envelope:

$$V(\theta) \leq \int y^2 p_\theta(y) dy \leq V_*, \quad \int |y - \mu(\theta)|^{2+\eta} p_\theta(y) dy \leq H_*.$$

Here $M(\theta) \geq M_0$, and $|x - y|^{2+\eta} \leq 2^{1+\eta}(|x|^{2+\eta} + |y|^{2+\eta})$.

On $[-1, 1]$, the exact tilted-convolution identity gives

$$p_\theta^{*3}(x) \geq \frac{c_h e^{-2\alpha} 2^{-B}}{\mathfrak{M}^3}.$$

Set

$$a_* = \min \left\{ \frac{1}{4}, \frac{c_h e^{-2\alpha} 2^{-B}}{\mathfrak{M}^3} \right\}, \quad \beta = 2a_*, \quad v_* = \beta/9. \quad (\text{ACW5})$$

The probability density p_θ^{*3} is the sum of β times the uniform probability density on $[-1, 1]$ and a nonnegative residual of mass $1 - \beta$. Variance of this mixture is at least $\beta/3$. Its variance is also exactly $3V(\theta)$, by expansion of the centred square of the sum. Hence

$$0 < v_* \leq V(\theta) \leq V_*.$$

Let

$$\phi_\theta(\xi) = \int e^{i\xi(y - \mu(\theta))} p_\theta(y) dy.$$

Taylor's formula and ACW4 give

$$|\phi_\theta(\xi) - 1 + \frac{1}{2}V(\theta)\xi^2| \leq 4H_*|\xi|^{2+\eta}. \quad (\text{ACW6})$$

For completeness, the scalar remainder is at most $|u|^3/6$ for $|u| \leq 1$, and at most $2 + |u| + u^2/2 \leq (7/2)|u|^{2+\eta}$ for $|u| > 1$; the stated factor four bounds both regions.

Define

$$\xi_0 = \min \left\{ 1, (2V_*)^{-1/2}, \left(\frac{v_*}{16H_*} \right)^{1/\eta} \right\}.$$

For $|\xi| \leq \xi_0$, the nonnegative quadratic principal term and ACW6 imply

$$|\phi_\theta(\xi)| \leq e^{-v_*\xi^2/4}. \quad (\text{ACW7})$$

The fixed-interval mixture also gives

$$|\phi_\theta(\xi)|^3 \leq 1 - \beta + \beta \left| \frac{\sin \xi}{\xi} \right|.$$

For $|\xi| \geq \xi_0$, $|\sin \xi/\xi| \leq 1 - \xi_0^2/7$. To verify this bound, use monotonicity of $\sin x/x$ on $[0, 2]$, its Taylor bound at $0 < \xi_0 \leq 1$, and $|\sin x/x| \leq 1/2$ beyond two. Consequently

$$|\phi_\theta(\xi)| \leq r_* := \left(1 - \frac{\beta \xi_0^2}{7} \right)^{1/3} < 1 \quad (|\xi| \geq \xi_0). \quad (\text{ACW8})$$

The envelope and Plancherel give the uniform bound

$$\int_{\mathbb{R}} |\phi_\theta(\xi)|^2 d\xi \leq C_2 := \frac{4\pi(C/M_0)^2}{2p-1}. \quad (\text{ACW9})$$

In particular Fourier inversion of the k -fold density is valid for every $k \geq 2$.

68.4.1 A fully specified error

Put

$$C_0 = 4H_* + V_*^2/8, \quad a_k = v_*(k-1)/4,$$

$$A_k^{\text{loc}} = \frac{1}{2\pi} \left[C_0 k \Gamma\left(\frac{3+\eta}{2}\right) a_k^{-(3+\eta)/2} + C_2 r_*^{k-2} + \frac{2e^{-kv_*\xi_0^2/2}}{kv_*\xi_0} \right], \quad \epsilon_k = \sqrt{2\pi k V_*} A_k^{\text{loc}}. \quad (\text{ACW10})$$

These local symbols are not the programme's arithmetic upper-envelope constant A_k .

For $|\xi| \leq \xi_0$,

$$|\phi_\theta(\xi) - e^{-V(\theta)\xi^2/2}| \leq C_0 |\xi|^{2+\eta}.$$

Telescoping the difference of the two k -th powers, and using ACW7, bounds its integral by the first term of ACW10. ACW8–9 bound the remaining characteristic-function integral by $C_2 r_*^{k-2}$. Integration of the remaining Gaussian tail gives the last term. Thus

$$\left| p_\theta^{*k}(k\mu(\theta)) - \frac{1}{\sqrt{2\pi k V(\theta)}} \right| \leq A_k^{\text{loc}}.$$

Equivalently,

$$\boxed{p_\theta^{*k}(k\mu(\theta)) = \frac{1 + \varepsilon_{k,\theta}}{\sqrt{2\pi k V(\theta)}}, \quad \sup_{|\theta| \leq \alpha} |\varepsilon_{k,\theta}| \leq \epsilon_k = O_h(k^{-1/8}).} \quad (\text{ACW11})$$

The exponent is $-\eta/2 = -1/8$. All constants are independent of θ , so the two endpoint tilts are included. For logarithms, use the eventual explicitly testable range $\epsilon_k \leq 1/2$. Such a range exists by ACW10.

68.5 4. The full central density profile

Define the source contrast, with the full mass explicitly recorded,

$$\ell_k(y) = \log \frac{m_k(y)}{\mathfrak{M}^k \sigma(y)}.$$

Define on $[-b_h, b_h]$

$$F_h^{\text{cen}}(t) = L(\theta(t)) - t\theta(t) - L(\alpha) + \alpha|t|. \quad (\text{ACW12})$$

The function is even. On $0 < t < b_h$, direct differentiation using $L'(\theta(t)) = t$ gives

$$(F_h^{\text{cen}})'(t) = \alpha - \theta(t), \quad (F_h^{\text{cen}})''(t) = -\frac{1}{V(\theta(t))}, \quad F_h^{\text{cen}}(b_h) = 0.$$

In particular $F_h^{\text{cen}}(t) < 0$ for $|t| < b_h$. Its cusp at zero comes from the explicitly retained term $\alpha|t|$; the original convolution itself remains smooth.

Write

$$c(y) = \sigma(y) e^{\alpha|y|} \sqrt{1+|y|}, \quad c_\Gamma \leq c(y) \leq C_\Gamma.$$

Substitute $\theta = \theta(t)$ in the exact identity ACW3 and in ACW11. There is then the exact expression

$$\boxed{\ell_k(kt) = kF_h^{\text{cen}}(t) + \frac{1}{2} \log(|t| + k^{-1}) - \frac{1}{2} \log(2\pi V(\theta(t))) - \log c(kt) + \log(1 + \varepsilon_{k,\theta(t)})}. \quad (\text{ACW13})$$

The last scalar is the actual relative local-density error, not a freely selected endpoint value. It satisfies $|\log(1 + \varepsilon_{k,\theta(t)})| \leq 2\epsilon_k$ on the stated logarithmic range. This proves the uniform central profile including $t = 0$ and $t = \pm b_h$, without a saddle point beyond the original boundary of exponential integrability.

68.6 5. The integrated central source contrast

The scalar now evaluated is explicitly

$$C_{h,k}^{\text{cen}} := \int_{-kb_h}^{kb_h} \ell_k(y) dy.$$

It is a Lebesgue-integrated log-density statistic. It is not, by definition or by any established metric identification, the programme's signed quotient determinant.

Define the fixed original-source functional

$$\mathfrak{J}_h^{\text{cen}} = \int_0^\alpha \mu(\theta)^2 d\theta > 0. \quad (\text{ACW14})$$

It has finite explicit bounds

$$\frac{v_*^2 \alpha^3}{3} \leq \mathfrak{J}_h^{\text{cen}} \leq \frac{V_*^2 \alpha^3}{3},$$

because $\mu(0) = 0$ and $v_* \leq \mu' \leq V_*$. In particular, this coefficient has a determined nonzero sign.

The area of the rate function is evaluated exactly:

$$\boxed{\int_{-b_h}^{b_h} F_h^{\text{cen}}(t) dt = -\mathfrak{J}_h^{\text{cen}}}. \quad (\text{ACW15})$$

Indeed,

$$F_h^{\text{cen}}(t) = - \int_t^{b_h} (\alpha - \theta(u)) du \quad (0 \leq t \leq b_h).$$

Integrating this formula gives

$$\int_{-b_h}^{b_h} F_h^{\text{cen}} = -2 \int_0^{b_h} u(\alpha - \theta(u)) du.$$

The substitution $u = \mu(\theta)$ followed by integration by parts gives ACW15. The boundary term is zero at $\theta = 0$, because $\mu(0) = 0$, and at $\theta = \alpha$, because of $\alpha - \theta$.

68.6.1 Finite remainder

Set

$$C_h^{(0)} = \frac{1}{2} \max\{|\log(2\pi v_*)|, |\log(2\pi V_*)|\} + \max\{|\log c_\Gamma|, |\log C_\Gamma|\}.$$

Integrating ACW13 and using ACW15 gives the entirely explicit bound

$$\boxed{|C_{h,k}^{\text{cen}} + k^2 \mathfrak{J}_h^{\text{cen}}| \leq k \left[1 + b_h \log(1 + b_h) + 2b_h C_h^{(0)} + 4b_h \epsilon_k \right]}. \quad (\text{ACW16})$$

The elementary bound used here is

$$\int_0^{b_h} |\log(t + k^{-1})| dt \leq 1 + b_h \log(1 + b_h), \quad k \geq 1.$$

The negative part is bounded by $\int_0^1 -\log t dt = 1$; the positive part is at most $b_h \log(1 + b_h)$. No interval around zero has been removed.

Consequently

$$\boxed{C_{h,k}^{\text{cen}} = -k^2 \mathfrak{J}_h^{\text{cen}} + O_h(k)}. \quad (\text{ACW17})$$

68.6.2 Its next coefficient

The precise Gamma asymptotic also gives

$$\sigma(y) e^{\alpha|y|} |y|^{1/2} \longrightarrow \sqrt{2}.$$

This is the substitution $z = 1/4 + iy/2$ in DLMF 5.11.9 (or the modulus of Stirling's formula 5.11.1). Therefore $c(y) \rightarrow \sqrt{2}$. Define

$$\mathfrak{J}_h^{\text{cen}} = \int_0^{b_h} \log \frac{t}{4\pi V(\theta(t))} dt = \int_0^\alpha V(\theta) \log \frac{\mu(\theta)}{4\pi V(\theta)} d\theta. \quad (\text{ACW18})$$

The logarithmic endpoint is integrable, since $\mu(\theta)$ is bounded above and below by positive constants times θ . In ACW13, the variance term is bounded, the Gamma term is uniformly bounded and converges away from zero, and the local-density error tends uniformly to zero. Also $\log(t + 1/k) \rightarrow \log t$ in $L^1(0, b_h)$. Dominated convergence consequently proves

$$\boxed{C_{h,k}^{\text{cen}} = -k^2 \mathfrak{J}_h^{\text{cen}} + k \mathfrak{J}_h^{\text{cen}} + o_h(k)}. \quad (\text{ACW19})$$

The fixed integrals in ACW14 and ACW18 contain only the original one-factor tilted mass and its first two derivatives. They do not contain an unknown growing quotient Gram or an assigned projected allocation.

On the simple-quartet family $q = (k + 1)^2$,

$$C_{h,k}^{\text{cen}}/q \longrightarrow -\mathfrak{J}_h^{\text{cen}}.$$

For every fixed $m_0 \geq 2$, the same scalar remains of order k^2 , while $q \asymp k^3$; its ratio to q tends to zero. These statements concern ACW19, not the determinant target.

68.7 6. Exact return to the original signed determinant

The source/relation identity in SXR10 gives, for any one of the original positive source metrics,

$$\mathcal{B}_k^\mu = \sum_{r=0}^q c_r \log \frac{\nu_r^\mu}{\omega_{q+r}^\mu}, \quad c_0 = c_q = 1, \quad c_r = 2 \quad (0 < r < q). \quad (\text{ACW20})$$

Here ν_r^μ is the full monic minimum in $|\chi|^2 d\mu$, and ω_{q+r}^μ is the full monic source minimum. Every preceding relation is retained.

Multiplying a source by a scalar $D > 0$ multiplies its quotient Gram by D and its quotient determinant by D^q . Thus the four signs give the exact cancellation

$$\mathcal{B}_k^{D\mu} = \mathcal{B}_k^\mu, \quad (+1) + (+1) + (-1) + (-1) = 0.$$

Equivalently, both monic norms in each ratio in ACW20 are multiplied by D . Hence

$$\mathcal{B}_k^{m_k/\mathfrak{M}^k} = \mathcal{B}_k^{m_k}. \quad (\text{ACW21})$$

This is the precise account of the scalar in ℓ_k ; it does not alter the recorded original mass or the arithmetic class.

For an exact evaluation path, introduce on the same polynomial coefficient spaces

$$d\nu_u(y) = \sigma(y)e^{u\ell_k(y)} dy, \quad 0 \leq u \leq 1. \quad (\text{ACW22})$$

Its endpoints are σ and $m_k/\mathfrak{M}^k$. This is a new path used to compute the same endpoint difference; it is not identified with the original affine source path at intermediate parameter values.

The original TW/GTM global source envelopes imply, at each fixed k ,

$$|\ell_k(y)| \leq C_{h,k} + M \log(1 + y^2), \quad M = 21 + 4m_0.$$

All polynomial moments and their parameter derivatives are consequently finite, locally uniformly in u . The moment matrices are positive definite. Their monic minimizers depend smoothly on u . Differentiating their attained norms, the coefficient-derivative cross terms vanish in the full preceding polynomial or relation space, exactly as in SXR5.

Let $p_{q+r,u}$ and $u_{r,u}$ be these actual monic minimizers along ACW22. Define

$$\mathcal{K}_{k,u}(y) = \sum_{r=0}^q c_r \left[\frac{|\chi(c + iy)u_{r,u}(c + iy)|^2}{\nu_{r,u}} - \frac{|p_{q+r,u}(c + iy)|^2}{\omega_{q+r,u}} \right]. \quad (\text{ACW23})$$

Each of its two nonnegative sums has integral $2q$ in $d\nu_u$. In particular $\int \mathcal{K}_{k,u} d\nu_u = 0$. Differentiating ACW20 and integrating from zero to one gives the exact receiver

$$\boxed{\delta_{k,1}^{\text{ar}} = \int_0^1 \int_{\mathbb{R}} \ell_k(y) \mathcal{K}_{k,u}(y) d\nu_u(y) du.} \quad (\text{ACW24})$$

Both endpoints, the full relation ideal, and the monic coordinate frames agree with the assigned target. The scalar $\mathfrak{M}^k$ cancels by ACW21, before estimates. No eigenclass, observation kernel, proper-source metric, or quotient denominator has been replaced.

68.8 7. The actual relation kernel has exponentially small central mass

The central density calculation can also be returned to one of the two actual kernels in ACW24. This step does not replace the remaining source kernel by a reference kernel.

Retain GTM1's original constants

$$M = 21 + 4m_0, \quad \vartheta_h = \int_{-1}^1 w_h(y) dy, \\ a_k^{\text{TW}} = \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-B}}{C_{\Gamma} 2^{B/2}}, \quad A_k^{\text{TW}} = \frac{C}{c_{\Gamma}} k^{p+1} \mathfrak{M}^{k-1}.$$

Thus

$$a_k^{\text{TW}} (1 + y^2)^{-M} \sigma(y) \leq m_k(y) \leq A_k^{\text{TW}} \sigma(y).$$

Define

$$D_n = [1 + 4(n + M)^2]^M, \quad L_k^- = D_{2q}^{-1} \min\{1, a_k^{\text{TW}}/\mathfrak{M}^k\}, \quad L_k^+ = \max\{1, A_k^{\text{TW}}/\mathfrak{M}^k\}. \quad (\text{ACW25})$$

On the entire geometric path ACW22,

$$d\nu_u \leq L_k^+ d\sigma, \quad \|P\|_{\nu_u}^2 \geq L_k^- \|P\|_\sigma^2 \quad (\deg P \leq 2q). \quad (\text{ACW26})$$

Here is a direct verification of the lower norm inequality. The Gamma recurrence in GTM7 bounds multiplication by y on polynomials of degree at most n , including its output of degree $n+1$, by $2(n+1)$. Hence

$$\|(y+i)^M P\|_\sigma^2 \leq D_n \|P\|_\sigma^2.$$

Cauchy–Schwarz, applied to the factors $|P|(1+y^2)^{-M/2}$ and $|P|(1+y^2)^{M/2}$, proves

$$\int |P|^2 (1+y^2)^{-M} d\sigma \geq D_n^{-1} \|P\|_\sigma^2.$$

For $0 \leq u \leq 1$, raising the positive lower density ratio to the power u bounds it below by $\min\{1, a_k^{\text{TW}}/\mathfrak{M}^k\}(1+y^2)^{-M}$. This proves ACW26. In particular $\log(L_k^+/L_k^-) = O_h(k + \log q)$.

Put

$$\widehat{\chi}(y) = i^{-q} \chi(c + iy), \quad R = k\sqrt{\delta^2 + \gamma^2}, \quad Y = 2^{-16}q, \quad B_\chi = \frac{qR^2}{Y^2 - R^2},$$

$$\eta_0 = \frac{\sqrt{2\pi}}{c_\Gamma} \frac{(2q+1)(4q+1)\sqrt{1+2q}}{q} 16^{-q}. \quad (\text{ACW27})$$

Use the eventual finite domain

$$q \geq 100, \quad R \leq Y/4, \quad kb_h \leq Y.$$

It contains all sufficiently large original tensor orders at every fixed packet and fixed complete multiplicity. In particular $\eta_0 < 1/2$ there: the bound $\eta_0 \leq 540q^2 16^{-q}$ holds, its value at $q = 100$ is below one half, and its consecutive ratio decreases thereafter.

For every polynomial P of degree at most q , the original Gamma central estimate is

$$\int_{|y| \leq Y} |\widehat{\chi}(y)P(y)|^2 d\sigma \leq \frac{e^{B_\chi} \eta_0}{1 - \eta_0} \int_{\mathbb{R}} |\widehat{\chi}(y)P(y)|^2 d\sigma. \quad (\text{ACW28})$$

For completeness, pair every opposite root of $\widehat{\chi}$, retaining repetitions. Outside $[-Y, Y]$, its squared modulus is at least $e^{-B_\chi} |y|^{2q}$. Inside that interval it is at most $[(9/4)2^{-32}q^2]^q$. Expand P in the complete shifted Legendre basis on $[q, 2q]$. Its argument on $[-Y, Y]$ has modulus below four. The elementary recurrence bounds the degree- j Legendre polynomial there by 10^j . Cauchy–Schwarz therefore gives the enlarged valid bound

$$|P(y)|^2 \leq \frac{(2q+1)(4q+1)100^{2q}}{q} \int_q^{2q} |P(t)|^2 dt.$$

The reference integral $\int |y|^{2q} |P|^2 d\sigma$ is at least

$$\frac{c_\Gamma q^{2q} e^{-\pi q}}{\sqrt{1+2q}} \int_q^{2q} |P(t)|^2 dt.$$

Together with the full reference mass $\sqrt{2\pi}$ and $(9/4)e^\pi 10^4 2^{-32} < 1/16$, this bounds both the central root-weighted and central reference-power integrals by η_0 times the complete reference-power integral. The outside lower bound then proves ACW28. This is the same coefficient-space central comparison used in the supplied EC calculation; all its intervals and factors have been retained.

Since $\deg(\widehat{\chi}P) \leq 2q$, ACW26 returns this estimate to the actual entire path:

$$\int_{|y| \leq kb_h} |\hat{\chi}P|^2 d\nu_u \leq \eta_k^{\text{rel}} \int_{\mathbb{R}} |\hat{\chi}P|^2 d\nu_u, \quad \eta_k^{\text{rel}} = \frac{L_k^+ e^{B_x} \eta_0}{L_k^- 1 - \eta_0}. \quad (\text{ACW29})$$

The exact polynomial coordinate change $P(S) \mapsto P(c+iy)$ changes no degree or absolute norm; thus ACW29 applies to every actual monic relation minimizer in ACW23.

Write $\mathcal{R}_{k,u}$ for ACW23's nonnegative relation sum and $\mathcal{P}_{k,u}$ for its nonnegative source sum. Their weights sum to $2q$, so

$$\int_{|y| \leq kb_h} \mathcal{R}_{k,u} d\nu_u \leq 2q\eta_k^{\text{rel}}.$$

ACW13 bounds the absolute contrast on this whole interval by

$$T_k = k[L(\alpha) - L(0)] + \frac{1}{2} \log k + \frac{1}{2} \log(1 + b_h) + C_h^{(0)} + 2\epsilon_k.$$

Consequently the central part of the original signed correction has the exact directed enclosure

$$\left| \delta_{k,1}^{\text{ar, cen}} + \int_0^1 \int_{|y| \leq kb_h} \ell_k(y) \mathcal{P}_{k,u}(y) d\nu_u(y) du \right| \leq 2qT_k\eta_k^{\text{rel}}. \quad (\text{ACW30})$$

The right side is

$$\exp[-q \log 16 + O_h(k + \log q)].$$

Thus the actual relation part of this central return is exponentially controlled. The remaining central term is the original degree band of source minima, not an arbitrary projection or an assigned scalar density. Its weighting has not yet been evaluated at order q .

68.9 8. Exact stopping point

ACW13–19 evaluate the source contrast on the full interval $[-kb_h, kb_h]$. They do not evaluate its weighting by $\mathcal{K}_{k,u} d\nu_u$ in ACW24. The explicit Fourier error in ACW11 cannot be multiplied by an assumed constant kernel density.

In particular, the two terms still required in the assigned calculation are the values, at the requested scales, of

$$\int_0^1 \int_{|y| \leq kb_h} \ell_k(y) \mathcal{K}_{k,u}(y) d\nu_u(y) du$$

and

$$\int_0^1 \int_{|y| > kb_h} \ell_k(y) \mathcal{K}_{k,u}(y) d\nu_u(y) du.$$

ACW30 has bounded the first expression's relation term exponentially. Its remaining source-band kernel still needs an evaluation in the central window. The second still needs more than RTM9's two different tail powers. The existing trace-mass cancellations and GTM20's bound $O_h(q \log q)$ do not evaluate either term with an $o(q)$ residual.

No value of $d_{\log}(h)$ or $d_0(h)$ is asserted here. In particular, $-\mathfrak{I}_h^{\text{cen}}$ is not asserted to equal either of them.

68.10 9. D032 check and the comparison still not established

The supplied full D032 collection was consulted at the relevant weight theorem and tensor-amplification argument. Theorem 3.3.1 on the original printed page 204 is stated for schemes of finite type over $\mathbb{Z}$, not only for a single finite field. The original paper's printed page 140 gives the Kunneth/product argument that halves an existing weight error. These passages were checked against the original IAS PDF of Pierre Deligne, *La conjecture de Weil. II*, Publications mathematiques de l'IHES 52 (1980), 137–252.

The current attempt has not constructed a comparison from the actual signed kernel expression ACW24 to a D032 cohomological object that returns its required value or metric estimate. This is an uncompleted comparison in this attempt, not a proof that no such mathematical correspondence exists. D032's weight bounds are not substituted for ACW24's unknown signed kernel correlation.

68.11 Source and verification record

Imported programme inputs: RTM1–3, RTM8–11, GTM1 and GTM20, and SXR1–12. Exact source ranges and file hashes are recorded in `sources.json`. The current status and precision target were checked in *Untitled 1775*. The source history in D032 was retained; no historical translation was silently promoted to its latest maintained revision.

Classical Gamma asymptotic: NIST DLMF 5.11.9 / 5.11.1, with the exact substitution written in Section 5. Fourier inversion and Plancherel are used after the explicit integrability bounds ACW8–9.

The accompanying verifier checks exact finite source/relation, monic-frame, endpoint-sign and mass-cancellation identities, together with the central area algebra on an explicitly artificial Gaussian source. These are auxiliary checks. They are not numerical values for the hypothetical xi packet, not an interval certificate for it, not an independent verification of all imported source estimates, and not a formal verification of this analytic derivation.

Final status: central-source profile and its integrated k^2 and k coefficients derived, with an exponential bound for the actual central relation-kernel term; the original signed arithmetic-baseline coefficients remain uncomputed. The user's requested full completion has not been achieved.

Complete original conductor transport and affine fibre proof

69 Uniform all-rank transport for the original conductor

Fix the original simple quartet, period, branch and conductor coefficients. The programme origin retains its original attribution [37, 40]. Put

$$a = 4\ell + 1 \geq 9, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad q = (a + 1)^2, \quad Q = (a - 7)^2, \\ \beta_{n,u,w} = \frac{n}{2} + (2u - n)\delta + i(2w - n)\gamma, \quad E_A(z) = \sum_{u,w=0}^8 a_{uw} e^{\beta_{8,u,w} z}, \quad (LT1)$$

$$\mu_j = E_A^{(j)}(0), \quad v = \text{ord}_0 E_A \leq 80, \quad \mu_v \neq 0, \quad A_{\text{abs}} = \sum_{u,w=0}^8 |a_{uw}|,$$

$$s \in \{1, a\}, \quad b = s/2, \quad c_0 = a/2, \quad c_1 = a/2 - 4, \quad D \in \mathbb{Z}_{\geq 0}, \quad N = D + v, \quad d = D + 1.$$

The actual coefficients, v, δ, γ are fixed as a varies. The inner product is conjugate-linear in its first argument. Retain the complete original measure and all its factors:

$$dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\ \|P\|_{s,c}^2 = \int_{\mathbb{R}} |P(c + iy)|^2 dm_s(y), \quad p_{0,s} = 1, \quad p_{1,s} = y, \quad (LT2)$$

$$p_{n+1,s} = yp_{n,s} - n(n-1 + s/2)p_{n-1,s}, \quad h_{n,s} = M_s n! (s/2)_n, \\ \phi_{n,s,c}(S) = p_{n,s}((S - c)/i)/\sqrt{h_{n,s}}, \quad [S^n]\phi_{n,s,c} = i^{-n}/\sqrt{h_{n,s}}, \quad \rho_n = \sqrt{n!/(s/2)_n}.$$

These are the original orthonormal frames. To check the classical dictionary with its mass, take $p_{n,s}(y) = n! P_n^{(s/4)}(y/2; \pi/2)$ in the Meixner–Pollaczek orthogonality formulas, DLMF (18.19.8)–(18.19.9) [36]. Their hypotheses hold because $s/4 > 0$. With $y = 2x$, the squared norm is exactly

$$\frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} 2(n!)^2 \frac{2\pi\Gamma(n + s/2)}{2^{s/2} n!} = M_s n! (s/2)_n.$$

The generating identity at these same parameters is DLMF (18.23.7) [36]:

$$\sum_{n \geq 0} \frac{p_{n,s}(y)}{n!} t^n = (1 + t^2)^{-s/4} e^{y \arctan t}, \quad |t| < 1. \quad (LT3)$$

It also follows directly by multiplying the recurrence in LT2 by $t^n/n!$: the series satisfies $(1 + t^2)\partial_t F = (y - (s/2)t)F$, $F(y, 0) = 1$. The analytic factors and arctan take their branches fixed at zero.

For $\mathcal{P}_j = \mathbb{C}[S]_{\leq j}$ and $\mathcal{P}_{-1} = 0$, the literal conductor is

$$(\mathcal{T}_A P)(S') = \sum_{u,w=0}^8 a_{uw} P(S' + \beta_{8,u,w}), \quad A_{D,s} : (\mathcal{P}_N / \mathcal{P}_{v-1}, \|\cdot\|_{s,c_0, \text{quot}}) \longrightarrow (\mathbb{C}[S']_{\leq D}, \|\cdot\|_{s,c_1}). \quad (LT4)$$

Indeed $\mathcal{T}_A S^n = \sum_{j=v}^n \binom{n}{j} \mu_j S^{n-j}$. The leading coefficient for $n \geq v$ is $\binom{n}{v} \mu_v \neq 0$, and all polynomials of degree below v vanish. Elimination by degree proves surjectivity and that the full kernel is $\mathcal{P}_{v-1}$. The quotient is represented isometrically by $\phi_{v,s,c_0}, \dots, \phi_{D+v,s,c_0}$. Set

$$w(t) = -i \arctan t = \frac{1}{2} \log \frac{1 - it}{1 + it}, \quad g(t) = t^{-v} e^{-4w(t)} E_A(w(t)), \quad g(0) = (-i)^v \mu_v / v!. \quad (LT5)$$

The zero-order definition of v removes the singularity at zero. Apply $\mathcal{T}_A$ to LT3 at $y = (S - c_0)/i$. Since $c_0 - c_1 = 4$, its image is $t^v g(t)$ times the same generating function at $y = (S' - c_1)/i$. Comparing coefficients, including $n!/\sqrt{h_{n,s}} = \rho_n/\sqrt{M_s}$, gives the exact matrix

$$A_{D,s} = [\text{diag}(\rho_i)_{i=0}^D]^{-1} T_D(g) \text{diag}(\rho_{i+v})_{i=0}^D, \quad T_D(g)_{ij} = \begin{cases} g_{j-i}, & i \leq j, \\ 0, & i > j. \end{cases} \quad (\text{LT6})$$

Both copies of M_s remain in the original norms; they cancel only in this matrix connecting two frames with the same original order. No scalar $v!/\mu_v$ is inserted.

69.1 The finite analytic constants in LRC6

Define, exactly as in the incoming calculation,

$$\begin{aligned} g_b(t) &= (1+t^2)^4 g(t), & F_0 &= 4 + 4\sqrt{\delta^2 + \gamma^2}, \\ B_0^b &= 2^{v+8} A_{\text{abs}} e^{2\pi\gamma}, & B_2^b &= B_0^b \{F_0(F_0 + 1) + 4vF_0 + 4v(v+1)\}. \end{aligned} \quad (\text{LT7})$$

The superscript on B_0^b, B_2^b only prevents collision with WCF14's different zero-free-disk constant B_0 ; their numerical values are the incoming LRC6 constants, without alteration. For $\lambda_{uw} = (\beta_{8,u,w} - 4)/2$, put $\alpha_{uw} = 4 + \lambda_{uw}$ and $\beta_{uw}^b = 4 - \lambda_{uw}$. Then

$$\begin{aligned} t^v g_b(t) &= H_b(t) := \sum_{u,w} a_{uw} (1-it)^{\alpha_{uw}} (1+it)^{\beta_{uw}^b}, \\ |\Re \lambda_{uw}| &\leq 4\delta < 2, \quad |\Im \lambda_{uw}| \leq 4\gamma, \quad |\alpha_{uw}|, |\beta_{uw}^b| \leq F_0. \end{aligned} \quad (\text{LT8})$$

For $|t| < 1$, $z = (1-it)/(1+it)$ satisfies $\Re z = (1-|t|^2)/|1+it|^2 > 0$. Each $1 \pm it$ has positive real part, and the branches fixed at zero obey $\log z = \log(1-it) - \log(1+it)$ throughout the disk. For nonnegative integers j, k with $j+k \leq 2$, the real exponents in $(1-it)^{\alpha_{uw}-j} (1+it)^{\beta_{uw}^b-k}$ are positive: the smallest possible one is $2-4\delta > 0$. Their sum is $8-j-k$, while the imaginary contribution to its modulus is $\exp[-\Im \lambda_{uw} \arg z] \leq e^{2\pi\gamma}$. Consequently

$$|(1-it)^{\alpha_{uw}-j} (1+it)^{\beta_{uw}^b-k}| \leq 2^{8-j-k} e^{2\pi\gamma}. \quad (\text{LT9})$$

For $f = (1-it)^\alpha (1+it)^\beta$ its first derivative has coefficients $-i\alpha, i\beta$, and its second derivative has coefficients $-\alpha(\alpha-1), 2\alpha\beta, -\beta(\beta-1)$. LT9 therefore proves

$$\begin{aligned} |H_b| &\leq 2^8 A_{\text{abs}} e^{2\pi\gamma}, \\ |H'_b| &\leq 2^7 (2F_0) A_{\text{abs}} e^{2\pi\gamma}, \\ |H''_b| &\leq 2^6 \{2F_0(F_0+1) + 2F_0^2\} A_{\text{abs}} e^{2\pi\gamma} \leq 2^8 F_0(F_0+1) A_{\text{abs}} e^{2\pi\gamma}. \end{aligned} \quad (\text{LT10})$$

On $1/2 \leq |t| < 1$, use $g_b = t^{-v} H_b$ and

$$g_b'' = t^{-v} H_b'' - 2vt^{-v-1} H'_b + v(v+1)t^{-v-2} H_b.$$

It follows that $|g_b| \leq B_0^b$ and $|g_b''| \leq B_2^b$ on this annulus. Both functions are holomorphic on the disk, since H_b has a zero of order v at zero. Applying the maximum-modulus principle on each circle of radius $r \in [1/2, 1)$ proves

$$\boxed{|g_b(t)| \leq B_0^b, \quad |g_b''(t)| \leq B_2^b \quad (|t| < 1).} \quad (\text{LRC6})$$

If $g_b(t) = \sum_{n \geq 0} b_n t^n$, Cauchy's integral formula on $|t| = r < 1$, followed by $r \uparrow 1$, gives

$$|b_0|, |b_1| \leq B_0^b, \quad |b_n| \leq B_2^b / (n(n-1)) \quad (n \geq 2). \quad (\text{LT11})$$

This argument has retained every interior zero of g .

69.2 Exact matrix factorization and every exterior rank

Let S_D be the upper shift, with entries 1 in positions $(i, i+1)$, and set

$$R = \text{diag}(\rho_i)_{i=0}^D, \quad W = R^{-1}S_DR, \quad V_v = \text{diag}(\rho_{i+v}/\rho_i)_{i=0}^D, \quad U_b = R^{-1}T_D(g_b)R. \quad (\text{LT12})$$

For any two series analytic at zero, $T_D(fh) = T_D(f)T_D(h)$: the (i, j) entry is exactly the finite convolution for coefficient $j - i$, and all entries below the diagonal vanish. Since $S_D^{D+1} = 0$, all power-series evaluations at S_D and at W are finite. The identity $g = g_b(1 + t^2)^{-4}$ and LT6 thus give

$$\boxed{A_{D,s} = U_b(I + W^2)^{-4}V_v.} \quad (\text{LRC7})$$

This fixes the position of V_v on the right. It is never commuted through the other two matrices.

The nonzero images of different coordinate vectors under W^n are orthogonal, so its norm is the maximum of ρ_{i+n}/ρ_i . At $s = a$, each factor $(i + j + 1)/(i + j + s/2)$ is at most one, hence $\|W^n\| \leq 1$ and $\|V_v\| \leq 1$. At $s = 1$,

$$\frac{\rho_{i+n}^2}{\rho_i^2} = \prod_{j=0}^{n-1} \frac{i + j + 1}{i + j + 1/2} \leq \frac{n!}{(1/2)_n} \leq 2n + 1. \quad (\text{LT13})$$

The first inequality follows because every factor decreases with i . For the second, induction starts at $n = 0$; multiplying $2n+1$ by $(n+1)/(n+1/2)$ gives $2n+2 \leq 2n+3$. In particular $\|V_v\| \leq \sqrt{2v+1}$. Now $U_b = \sum_{n=0}^D b_n W^n$. For $n \geq 2$,

$$\frac{\sqrt{2n+1}}{n(n-1)} \leq \sqrt{10} n^{-3/2}, \quad \sum_{n=2}^{\infty} n^{-3/2} \leq \int_1^{\infty} x^{-3/2} dx = 2.$$

LT11–13 prove the original finite constant

$$\|U_b\| \leq H := B_0^b(1 + \sqrt{3}) + 2\sqrt{10}B_2^b. \quad (\text{LT14})$$

Put

$$H_s = \begin{cases} \max\{1, H\sqrt{2v+1}\}, & s = 1, \\ \max\{1, H\}, & s = a, \end{cases} \quad \kappa_s = \begin{cases} 1 + \sqrt{8/3}, & s = 1, \\ 2, & s = a. \end{cases} \quad (\text{LT15})$$

Writing $Q_D = I + W^2$, triangularity gives $\det Q_D = 1$. For $s = 1$, the decreasing-factor argument gives the sharper $\|W^2\| \leq \rho_2/\rho_0 = \sqrt{8/3}$, including $D < 2$ when $W^2 = 0$. For $s = a$ it gives $\|W^2\| \leq 1$. Thus $\|Q_D\| \leq \kappa_s$ in both cases.

For any invertible complex $n \times n$ matrix T with singular values $\sigma_1 \geq \dots \geq \sigma_n > 0$, diagonalizing its positive squared norm, or taking its singular-value decomposition, gives

$$\|\bigwedge^r T\| = \prod_{j=1}^r \sigma_j, \quad \|\bigwedge^r T^{-1}\| = \frac{\|\bigwedge^{n-r} T\|}{|\det T|} \quad (0 \leq r \leq n). \quad (\text{LT16})$$

The second equality multiplies the largest inverse singular values; its denominator is the full determinant, not a truncated product. Apply it to $Q_D^{\oplus c}$, where $c \in \{1, 4\}$, whose dimension is cd and determinant one. Exterior functoriality and submultiplicativity give

$$\|\bigwedge^r ((Q_D^{-4})^{\oplus c})\| \leq \|\bigwedge^r ((Q_D^{-1})^{\oplus c})\|^4 = \|\bigwedge^{cd-r} (Q_D^{\oplus c})\|^4 \leq \kappa_s^{4(cd-r)}. \quad (\text{LT17})$$

This is valid at $r = cd$, where the right side equals one. For a complex matrix $A = U\Sigma V^*$, $A^T = \bar{V}\Sigma U^T$ is another singular-value decomposition; ordinary transposition therefore preserves every exterior norm. Apply LRC7 to c orthogonal copies, use LT14 for the first factor and LT13 for the last, and then LT17. For $1 \leq r \leq cd$ this proves exactly

$$\boxed{\log \|\bigwedge^r ((A_{D,s}^T)^{\oplus c})\| \leq F_{D,s,r,c} := r \log H_s + 4(cd - r) \log \kappa_s.} \quad (\text{LRC8})$$

Set $F_{D,s,0,c} = 0$, the exact logarithm of the empty exterior norm.

LT6 also proves the full complex determinant identity

$$\det A_{D,s} = ((-i)^v \mu_v / v!)^d \prod_{i=0}^D \frac{\rho_{i+v}}{\rho_i}. \quad (\text{LT18})$$

The phase is retained in this equality. At $s = 1$ the product of positive ratios is at least one. At $s = a$,

$$\log \prod_{i=0}^D \frac{\rho_i}{\rho_{i+v}} = \frac{1}{2} \sum_{i=0}^D \sum_{j=0}^{v-1} \log \left(1 + \frac{s/2 - 1}{i + j + 1} \right) \leq \frac{1}{2} (s/2 - 1) v \sum_{j=1}^d \frac{1}{j}. \quad (\text{LT19})$$

Here the elementary inequality $\log(1+x) \leq x$ applies to each nonnegative x ; the sums are empty if $v = 0$. Define $\log^+ x = \max\{0, \log x\}$, $H_d^{\text{harm}} = \sum_{j=1}^d 1/j$, and retain WCF's exact finite allowance

$$J_{D,s} = d \log^+(1/|g(0)|) + \frac{1}{2} (s/2 - 1)_+ v H_d^{\text{harm}}. \quad (\text{LT20})$$

LT18–19 give $|\det A_{D,s}|^{-1} \leq e^{J_{D,s}}$. Apply LT16 to the whole matrix $B = (A_{D,s}^T)^{\oplus c}$ and LRC8 to its complementary rank. For $1 \leq r \leq cd$,

$$\log \left\| \bigwedge^r ((A_{D,s}^{-T})^{\oplus c}) \right\| \leq I_{D,s,r,c} := cJ_{D,s} + F_{D,s,cd-r,c}. \quad (\text{LRC9})$$

Set $I_{D,s,0,c} = 0$. This does not assert that the displayed positive-rank formula extends to rank zero. At full rank it gives $I_{D,s,cd,c} = cJ_{D,s}$; at $D = 0$ the original matrix is the scalar $g(0)\rho_v$. Thus the smallest cutoff, $v = 0$, empty exterior powers and full exterior powers are all included. No assumption on absence or number of interior zeros of g has entered these estimates.

There is an explicit constant for the claimed uniform order. Suppose $D \leq C_{\text{cut}} a^2$ with fixed $C_{\text{cut}} \geq 0$, and put

$$\begin{aligned} K &= C_{\text{cut}} + 1, & G &= \log^+(1/|g(0)|), & H_* &= \max\{1, H\sqrt{2v+1}\}, & \kappa_* &= 1 + \sqrt{8/3}, \\ L_* &= \max\{\log H_*, 4 \log \kappa_*\}, & J_* &= KG + \frac{v}{4}(3 + \log K). \end{aligned} \quad (\text{LT21})$$

Then $d \leq Ka^2 \leq Kq$ and $H_d^{\text{harm}} \leq 1 + \log d \leq 1 + \log K + 2 \log a$. Since $\log a \leq a$, $a \leq q$ and $a^2 \leq q$, LT20 implies $J_{D,s} \leq J_* q$. Moreover the expression in LRC8 is bounded by cdL_* . Therefore for every retained rank, including zero,

$$0 \leq F_{D,s,r,c} \leq cKL_*q, \quad 0 \leq I_{D,s,r,c} \leq c(J_* + KL_*)q. \quad (\text{LT22})$$

All constants depend only on the actual conductor and the displayed cutoff constant. These inequalities prove uniform $O(q)$ bounds at all ranks and both original source orders, with no rank restriction beyond $0 \leq r \leq c(D+1)$.

69.3 The literal lower covariance and every attained quotient

Now impose the original endpoint domain $N \geq q - 1$, $D = N - v$. Let $\mathbf{E}_-$ and $\mathbf{E}_+$ be the full original observation matrices in the unchanged root order:

$$(\mathbf{E}_-)_{n,(u,w)} = \phi_{n,s,c_1}(\beta_{a-8,u,w}), \quad 0 \leq n \leq D, \quad (\mathbf{E}_+)_{n,(u,w)} = \phi_{n,s,c_0}(\beta_{a,u,w}), \quad 0 \leq n \leq N. \quad (\text{LT23})$$

The roots are distinct because their real and imaginary parts recover the two indices. Since $q - Q = 16a - 48 \geq 96 > v$, we have $D \geq Q - 1$. The leading coefficients in LT2 and the Vandermonde determinant prove that $\mathbf{E}_-$ has rank Q and $\mathbf{E}_+$ has rank q . Define the original conductor matrix C_a by collecting every term in the identity

$$\sum_{u,w} (C_a^T \eta)_{uw} e^{\beta_{a,u,w} z} = E_A(z) \sum_{u,w=0}^{a-8} \eta_{uw} e^{\beta_{a-8,u,w} z}. \quad (\text{LT24})$$

This uses exactly $\beta_{a-8,u,w} + \beta_{8,r,t} = \beta_{a,u+r,w+t}$. All contributions producing the same root are retained. If the left side is zero, the lower exponential sum vanishes on an open set where $E_A \neq 0$, and hence everywhere. Its derivative Vandermonde forces $\eta = 0$; thus $C_a^\top$ is injective.

For a lower functional Λ , complex linearity gives $\Lambda(\mathcal{T}_A \phi_{v+j,s,c_0}) = \sum_i (A_{D,s})_{ij} \Lambda(\phi_{i,s,c_1})$. This proves the ordinary-transpose square, including the vanishing first v rows,

$$E_+ C_a^\top = \begin{bmatrix} 0_{v \times Q} \\ A_{D,s}^\top E_- \end{bmatrix}, \quad K_{-,D,s} = E_-^* E_- > 0, \quad K_{C,N,s} = E_-^* \overline{A_{D,s}} A_{D,s}^\top E_- > 0. \quad (\text{LT25})$$

This is the original WCF22 map with the unchanged orders $s = 1, a$, the lower centre $a/2 - 4$, and the shifted cutoff $N - v$.

For independent columns $e_1, \dots, e_r$, the determinant of their Gram matrix is $\|e_1 \wedge \dots \wedge e_r\|^2$. This follows by expanding both sides as the determinant of their pairwise inner products; it is the classical exterior Gram identity, for example Lyons, §4, (4.1) [39]. Apply $\bigwedge^r B$ to this wedge and $\bigwedge^r B^{-1}$ to the image. LRC8–9 give the two directed determinant bounds. Taking $E = E_-$, $B = A_{D,s}^\top$ and $r = Q$ proves

$$\boxed{-2I_{D,s,Q,1} \leq e_{N,s}^C := \log \det K_{C,N,s} - \log \det K_{-,N-v,s} \leq 2F_{D,s,Q,1}.} \quad (\text{LRC10})$$

For completeness, the same argument controls every compatible attained quotient, rather than only a restriction. Let $K_0 \subset Y \subset H$ be subspaces of a finite-dimensional Hilbert space and $B : H \rightarrow H'$ an invertible linear map. Put $Y_0 = Y \cap K_0^\perp$ and $Y_1 = BY \cap (BK_0)^\perp$. Every affine fibre modulo K_0 has a unique minimum-norm element in Y_0 , by its orthogonal decomposition. The induced map and its inverse in these minimum representatives are exactly

$$\overline{B} = P_{Y_1} B|_{Y_0}, \quad \overline{B}^{-1} = P_{Y_0} B^{-1}|_{Y_1}. \quad (\text{LT26})$$

Indeed subtraction in the first formula lies in BK_0 and subtraction in the second lies in K_0 , proving that the compositions are identities on quotient classes and hence on their unique minimum representatives. Projections and isometric inclusions contract every exterior norm, as is immediate from their zero and unit singular values. It follows that each exterior norm of $\overline{B}$ and its inverse is bounded by the corresponding exterior norm of B and B^{-1} . For any common quotient coordinate frame of rank r , the quotient Gram determinant ratio therefore lies in

$$-2I_{D,s,r,c} \leq \log \frac{\det G_1}{\det G_0} \leq 2F_{D,s,r,c} \quad (\text{LT27})$$

when $B = (A_{D,s}^\top)^{\oplus c}$. A succession of restrictions and quotients reduces to the full preimage pair $K_0 \subset Y$ in the original ambient space: the union of the affine fibres minimized at successive steps is exactly the final affine fibre. Its minimum is attained by the same orthogonal projection. Thus the complete denominators remain, and no extra factor is paid for the dimension of a denominator or the number of steps.

Retain $N_i = (q-1, q, 2q-1, 2q)_i$, $D_i = N_i - v$ and $\mathcal{R}f = -f(N_0) - f(N_1) + f(N_2) + f(N_3)$. With $f_i^C = F_{D_i,s,Q,1}$ and $i_i^C = I_{D_i,s,Q,1}$, LRC10 gives

$$-2(f_0^C + f_1^C + i_2^C + i_3^C) \leq \mathcal{R}e^C \leq 2(i_0^C + i_1^C + f_2^C + f_3^C). \quad (\text{LT28})$$

The actual cutoffs have $D_i \leq 2(a+1)^2 \leq 3a^2$ for $a \geq 9$; LT22 proves that both finite allowances are $O_{\text{actual}}(q)$. The precise native Ξ application of LT27, including its full annihilators, coordinate section and original lower Gram denominators, is given next. All earlier independently proved upper and lower allowances may be retained by taking the minimum separately on each directed side. The symbol, its zeros and their pole directions have never been removed from the original map.

69.4 The exact analytic comparison and its four signed endpoints

Retain the actual simple quartet, original period and logarithm branch, all coefficients of the original nonzero conductor, and the original root order. Write

$$\begin{aligned} a &= 4l + 1 \geq 9, \quad q = (a + 1)^2, \quad Q = (a - 7)^2 = q', \\ m &= q - Q - v, \quad N \geq q - 1, \quad D = N - v, \quad d = D + 1, \quad t = N - q + 1, \\ s &\in \{1, a\}, \quad c_0 = a/2, \quad c_1 = a/2 - 4, \quad \beta_{n,u,w} = n/2 + (2u - n)\delta + i(2w - n)\gamma, \\ E_A(z) &= \sum_{r,h=0}^8 a_{rh} e^{\beta_{8,r,h} z}, \quad \mu_j = E_A^{(j)}(0), \quad v = \text{ord}_0 E_A \leq 80, \quad \mu_v \neq 0, \quad 0 < \delta < 1/2, \quad \gamma > 2. \end{aligned} \tag{XIM1}$$

Thus $t \geq 0$, $q - Q = 16a - 48 \geq 96$, and $m > 0$. All polynomial spaces have complex coefficients. The original Hermitian norm and frame are

$$\begin{aligned} dm_s(y) &= \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\ p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{n+1,s} = yp_{n,s} - n(n - 1 + s/2)p_{n-1,s}, \\ h_{n,s} &= M_s n! (s/2)_n, \quad \phi_{n,s,c}(S) = p_{n,s}((S - c)/i)/\sqrt{h_{n,s}}, \quad \|P\|_{s,c}^2 = \int_{\mathbb{R}} |P(c + iy)|^2 dm_s(y). \end{aligned} \tag{XIM2}$$

The frame is the original orthonormal frame, with the unchanged Meixner–Pollaczek dictionary and norm [36, eqs. (18.19.8)–(18.19.9), (18.22.8)]; its leading coefficient is $i^{-n}/\sqrt{h_{n,s}}$. Neither the mass nor this phase is changed. The duals below are complex-linear duals, with the dual Hilbert norms.

Put $\mathcal{P}_j = \mathbb{C}[S]_{\leq j}$, with $\mathcal{P}_{-1} = 0$, and use S' for the target variable. The unscaled conductor map and its matrix in the displayed orthonormal frames are

$$\begin{aligned} \mathcal{T}_A P(S') &= \sum_{r,h=0}^8 a_{rh} P(S' + \beta_{8,r,h}), \\ A &= A_{D,s} : \mathcal{P}_N / \mathcal{P}_{v-1} \longrightarrow \mathbb{C}[S']_{\leq D}, \\ \mathcal{T}_A \phi_{v+j,s,c_0} &= \sum_{i=0}^D A_{ij} \phi_{i,s,c_1}, \quad 0 \leq j \leq D, \\ \rho_n &= \sqrt{n!/(s/2)_n}, \quad w(z) = -i \arctan z, \quad g(z) = z^{-v} e^{-4w(z)} E_A(w(z)) = \sum_{j \geq 0} g_j z^j, \end{aligned} \tag{XIM3}$$

$$g_0 = (-i)^v \mu_v / v!, \quad A_{ij} = \begin{cases} g_{j-i} \rho_{j+v} / \rho_i, & i \leq j, \\ 0, & i > j. \end{cases}$$

These are the literal NCT3–5 and WCF4–6 matrices. Indeed $\mathcal{T}_A S^n = \sum_{j=v}^n \binom{n}{j} \mu_j S'^{n-j}$, with nonzero leading coefficient $\binom{n}{v} \mu_v$ when $n \geq v$. It follows by elimination in degree that the kernel is exactly $\mathcal{P}_{v-1}$ and that A is an isomorphism. The quotient norm uses its original orthogonal representatives $\phi_{v,s,c_0}, \dots, \phi_{N,s,c_0}$. For the matrix entries, multiplying the recurrence in XIM2 by $z^n/n!$ gives the differential equation $(1 + z^2)\partial_z G = (y - (s/2)z)G$, $G(y, 0) = 1$, whose solution is $G(y, z) = (1 + z^2)^{-s/4} \exp(y \arctan z)$. Applying $\mathcal{T}_A$ to this generating identity centred at c_0 multiplies the same identity centred at c_1 by $z^v g(z)$, since $c_0 - c_1 = 4$. Coefficient comparison and $n!/\sqrt{h_{n,s}} = \rho_n/\sqrt{M_s}$ give exactly XIM3.

Here are the full relation and annihilator spaces. Define

$$\chi_a(S) = \prod_{u,w=0}^a (S - \beta_{a,u,w}), \quad \chi_-(S') = \prod_{u,w=0}^{a-8} (S' - \beta_{a-8,u,w}). \tag{XIM4}$$

Use brackets $[P]_-$ for the coefficient column of P in $\phi_{0,s,c_1}, \dots, \phi_{D,s,c_1}$, and $[P]_+$ for the coefficient column of its class modulo $\mathcal{P}_{v-1}$ in $\phi_{v,s,c_0}, \dots, \phi_{N,s,c_0}$. Form the matrices

$$\begin{aligned} \mathbf{X}_N &= [[\chi_a S^j]_+]_{0 \leq j < t} \in \mathbb{C}^{d \times t}, & \mathbf{R}_N &= [[\mathcal{T}_A(\chi_a S^j)]_-]_{0 \leq j < t} = A \mathbf{X}_N \in \mathbb{C}^{d \times t}, \\ \mathbf{L}_N &= [[\chi_- S'^j]_-]_{0 \leq j < d-Q} \in \mathbb{C}^{d \times (d-Q)}. \end{aligned} \quad (\text{XIM5})$$

At $t = 0$, the first two matrices have no columns. The upper relation columns are independent modulo $\mathcal{P}_{v-1}$ because their nonzero linear combinations have degree at least $q > v - 1$. The native relation columns have distinct degrees $q + j - v$ and nonzero leading coefficients $\binom{q+j}{v} \mu_v$. The ideal columns are monic of distinct degrees $Q + j$. Thus their respective ranks are $t, t, d - Q$.

For every lower root, the identity $\beta_{a-8,u,w} + \beta_{8,r,h} = \beta_{a,u+r,w+h}$ shows that $\mathcal{T}_A(\chi_a S^j)$ vanishes at that root. Distinctness of the roots follows by comparing real and imaginary parts. Division by the monic χ_- therefore has zero remainder. Consequently $\text{im } \mathbf{R}_N \subset \text{im } \mathbf{L}_N$. For a functional with coordinate column $z = (\Lambda(\phi_{i,s,c_1}))_{i=0}^D$, evaluation on a polynomial with coefficient column p is $z^\top p$. Its norm is $\|z\|_2$. The two exact annihilator spaces are therefore

$$\begin{aligned} \mathcal{Y}_D &= \ker \mathbf{R}_N^\top = (\text{im } \overline{\mathbf{R}_N})^\perp, & \dim \mathcal{Y}_D &= d - t = q - v, \\ \mathcal{X}_D &= \ker \mathbf{L}_N^\top = (\text{im } \overline{\mathbf{L}_N})^\perp, & \dim \mathcal{X}_D &= Q, \\ \mathcal{X}_D &\subset \mathcal{Y}_D, & \dim(\mathcal{Y}_D / \mathcal{X}_D) &= m. \end{aligned} \quad (\text{XIM6})$$

Complex conjugation in the orthogonal-complement expressions is necessary because the algebraic annihilator uses ordinary transpose.

We identify these spaces with the original analytic functionals and lower evaluations, including their frames. Let $\mathbb{V}[n, \beta] = \beta^n$, $0 \leq n < q$, be the upper Vandermonde in the original root order, and set

$$\begin{aligned} \mathcal{N}_v &= \{\nu \in \mathbb{C}^q : (\mathbb{V}\nu)_n = 0 \ (0 \leq n < v)\}, \\ C_{(u,w),(i,j)} &= \begin{cases} a_{i-u,j-w}, & 0 \leq i-u, j-w \leq 8, \\ 0, & \text{otherwise,} \end{cases} & C \in \mathbb{C}^{Q \times q}, \\ \mathcal{N}_\nu(z) &= \sum_\beta \nu_\beta e^{\beta z}, & \mathcal{N}_{C^\top \eta}(z) &= E_A(z) \sum_{u,w=0}^{a-8} \eta_{uw} e^{\beta_{a-8,u,w} z}. \end{aligned} \quad (\text{XIM7})$$

The last identity collects all original shift coefficients. Its zero left side forces the lower exponential sum to vanish wherever $E_A \neq 0$, hence everywhere by analyticity; the lower Vandermonde then forces $\eta = 0$. Thus $C^\top$ is injective and its image lies in $\mathcal{N}_v$.

Let P_v insert the last $q - v$ coordinates into $\mathbb{C}^q$ and put $J_v = P_v^\top \mathbb{V} C^\top$. Retain the original first nonzero Q -row minor $\mathcal{I}$ of J_v , its complementary row set $\mathcal{J}$, and the coordinate insertion $E_{\mathcal{J}}$. Then

$$Z = \mathbb{V}^{-1} P_v E_{\mathcal{J}} \in \mathbb{C}^{q \times m}, \quad [C^\top, Z] \text{ is a frame of } \mathcal{N}_v, \quad F_b = \mathcal{N}_{Z e_b} / E_A. \quad (\text{XIM8})$$

The frame assertion follows by applying $\mathbb{V}$, taking the $\mathcal{I}$ rows first, and then taking the complementary rows. Each F_b is analytic at zero because the first v numerator derivatives vanish.

These are also exactly the two packet annihilators in TAC1g-h. Put $E = \mathbb{C}[S]/(\chi_a)$, $E_- = \mathbb{C}[S]/(\chi_-)$, $L_v = \mathbb{C}[S]_{<v} \subset E$, and $V_A = \ker(\overline{\mathcal{T}}_A : E \rightarrow E_-)$. Root addition proves that this packet map is well defined, and in root-value coordinates it is C . Since $C^\top$ is injective, the map is onto and $\dim V_A = q - Q$. The degree calculation in XIM3 gives $L_v \subset V_A$. The exact complex-linear pairing is

$$\begin{aligned} \Phi : \mathbb{C}^q &\rightarrow E^\vee, & \Phi(\nu)([P]) &= \sum_\beta \nu_\beta P(\beta), \\ \Phi(\mathcal{N}_v) &= \text{ann}(L_v), & \Phi(\text{im } C^\top) &= \text{ann}(V_A), \\ \Psi : \mathcal{N}_v / \text{im } C^\top &\rightarrow (V_A / L_v)^\vee, & \Psi([\nu])([P] + L_v) &= \sum_\beta \nu_\beta P(\beta). \end{aligned} \quad (\text{XIM8a})$$

The upper Vandermonde makes Φ invertible. Its first v moment equations are precisely vanishing on L_v . The conductor dual functionals vanish on V_A , and both that image and $\text{ann}(V_A)$ have dimension Q , proving their equality. Restriction therefore has exactly this kernel. Every functional on V_A/L_v extends linearly from V_A to E while vanishing on L_v , proving that Ψ is onto and hence an isomorphism. At $v = 0$, $L_v = 0$ and the corresponding moment conditions are empty.

For $\nu \in \mathcal{N}_v$, let $\Lambda_{\mathcal{N}_\nu/E_A}(S'^r) = (\mathcal{N}_\nu/E_A)^{(r)}(0)$. Leibniz's finite identity proves

$$\Lambda_{\mathcal{N}_\nu/E_A}(\mathcal{T}_A P) = \Lambda_{\mathcal{N}_\nu}(P), \quad \deg P \leq N. \quad (\text{XIM9})$$

It vanishes for $P = \chi_a S^j$, so the native functional belongs to $\mathcal{Y}_D$. If its first $D + 1$ derivatives vanish, XIM9 makes all numerator derivatives through N vanish. The first q such equations have invertible Vandermonde, so $\nu = 0$. The dimension $q - v$ now proves that these are all of $\mathcal{Y}_D$. Every lower evaluation annihilates the ideal generated by χ_- . Their first Q evaluation rows form an invertible triangular change of the lower Vandermonde, since $D \geq Q - 1$. They are independent, and dimension Q proves that they are all of $\mathcal{X}_D$.

For completeness, the finite analytic observation matrix can be constructed without an unevaluated division of germs. Put $n_{jb} = \sum_\beta Z_{\beta b} \beta^j$ and $f_{rb} = F_b^{(r)}(0)$. For $0 \leq r \leq D$, successively set

$$f_{rb} = \frac{n_{v+r,b} - \sum_{j=v+1}^{v+r} \binom{v+r}{j} \mu_j f_{v+r-j,b}}{\binom{v+r}{v} \mu_v}. \quad (\text{XIM10})$$

At $r = 0$ the sum is empty. All denominators are nonzero. This is exactly the derivative product identity, hence retains every original coefficient. Writing $\phi_{n,s,c_1}(S') = \sum_r \kappa_{nr} S'^r$, define

$$\begin{aligned} H &= \mathbf{E}_- \in \mathbb{C}^{d \times Q}, & H_{n,\beta'} &= \phi_{n,s,c_1}(\beta'), \\ J &\in \mathbb{C}^{d \times m}, & J_{nb} &= \sum_{r=0}^n \kappa_{nr} f_{rb}, & \mathcal{X}_D &= \text{im } H, & \mathcal{Y}_D &= \text{im}[H, J]. \end{aligned} \quad (\text{XIM11})$$

Here $[H, J]$ has full column rank $Q + m = q - v$.

Let $\mathbf{E}_+[n, \beta] = \phi_{n,s,c_0}(\beta)$, $0 \leq n \leq N$, and let $\mathbf{E}_+^{\text{red}}$ consist of rows $v, \dots, N$. The upper observation matrix has full column rank because its first q rows are an invertible triangular change of the upper Vandermonde. The first v rows vanish on $\mathcal{N}_v$. The ordinary dual matrix $B = A^\top$ therefore gives the exact equations

$$\begin{aligned} BH &= \mathbf{E}_+^{\text{red}} C^\top, & BJ &= \mathbf{E}_+^{\text{red}} Z, & \mathbf{E}_+ C^\top &= \begin{bmatrix} 0_{v \times Q} \\ BH \end{bmatrix}, & \mathbf{E}_+ Z &= \begin{bmatrix} 0_{v \times m} \\ BJ \end{bmatrix}, \\ B\mathcal{Y}_D &= \ker \mathbf{X}_N^\top, & B\mathcal{X}_D &= \ker(A^{-1} \mathbf{L}_N)^\top = \text{im}(\mathbf{E}_+^{\text{red}} C^\top). \end{aligned} \quad (\text{XIM12})$$

For the first line, apply each functional to the basis equation XIM3 and use XIM9. For the second line, use $\mathbf{R}_N = A \mathbf{X}_N$ and $B = A^\top$; invertibility proves both inclusions are equalities. This proves the full commutative square on actual finite jets. No adjoint and no scalar $\alpha_A = v!/\mu_v$ occurs.

The polynomial source of the analytic quotient is also exact. Restriction of functionals induces isometric isomorphisms

$$\frac{\text{ann}(\text{im } \mathbf{R}_N)}{\text{ann}(\text{im } \mathbf{L}_N)} \simeq \left(\frac{\text{im } \mathbf{L}_N}{\text{im } \mathbf{R}_N} \right)^\vee, \quad \frac{\text{ann}(\text{im } \mathbf{X}_N)}{\text{ann}(\text{im } A^{-1} \mathbf{L}_N)} \simeq \left(\frac{\text{im } A^{-1} \mathbf{L}_N}{\text{im } \mathbf{X}_N} \right)^\vee. \quad (\text{XIM13})$$

The primal metrics first restrict the original polynomial metric and then minimize over the full relation space. To prove the dual metric claim, extend a functional from the larger displayed polynomial subspace by composition with its orthogonal projection. This is the minimum extension, since every other extension differs on the orthogonal complement and the squared norms add. It still vanishes on the relation subspace. All such extensions differ by precisely the displayed denominator annihilator. This proves isometry as well as the algebraic quotient assertion. The primal

map is A on these two polynomial subquotients, and XIM12 is its contravariant ordinary dual. Each primal subquotient has dimension $(d - Q) - t = m$. The original packet identification is

$$\begin{aligned} V_A/L_v &\longrightarrow \text{im}(A^{-1}\mathbf{L}_N)/\text{im}\mathbf{X}_N, & [P] + L_v &\longmapsto [[P]_+], \\ Y_N &= \mathbb{C}[S']_{\leq D-Q}/\{\mathcal{T}_A(\chi_a h)/\chi_- : h \in \mathcal{P}_{t-1}\}, \\ \theta_N : V_A/L_v &\longrightarrow Y_N, & [P] + L_v &\longmapsto [\mathcal{T}_A P/\chi_-]. \end{aligned} \quad (\text{XIM13a})$$

for the upper polynomial quotient and its divided native counterpart, respectively. A representative of degree at most N exists since $N \geq q - 1$. Changing it by $\chi_a h$ gives exactly an XIM5 relation, and changing it by a polynomial of degree below v gives zero. Conversely, every polynomial in $\text{im}(A^{-1}\mathbf{L}_N)$ has a degree-at-most- N representative whose conductor vanishes on the lower roots, hence represents V_A . A zero quotient class differs from a multiple of χ_a by a polynomial of degree below v . This proves both surjectivity and injectivity. Dividing the native relation by χ_- is legitimate by XIM5; its relation map is the original $h \mapsto \mathcal{T}_A(\chi_a h)/\chi_-$. Thus XIM13a is exactly the TAC1d map, including its full relation space and scalar one.

We now give both attained coefficient minima. Put

$$\begin{aligned} K_0 &= H^* H > 0, & K_1 &= H^* B^* B H > 0, & P_0 &= H K_0^{-1} H^*, & P_1 &= B H K_1^{-1} H^* B^*, \\ W_0 &= (I - P_0) J, & W_1 &= (I - P_1) B J, \\ D_{a,D}^{(s)} &= W_0^* W_0, & \widehat{D}_{a,D}^{(s)} &= W_1^* W_1. \end{aligned} \quad (\text{XIM14})$$

The first metric is the original NCT10 native analytic metric. The second is precisely NCT8/TAC5: setting $K_+ = \mathbf{E}_+^* \mathbf{E}_+$, XIM12 gives

$$\widehat{D}_{a,D}^{(s)} = Z^* [K_+ - K_+ C^\top (\overline{C} K_+ C^\top)^{-1} \overline{C} K_+] Z. \quad (\text{XIM15})$$

For every $x \in \mathbb{C}^m$, completing the two positive squares in the Schur-complement calculation [34, Appendix A.5.5, p. 650; Appendix C.4.1, p. 673] proves

$$\begin{aligned} x^* D_{a,D}^{(s)} x &= \min_{\eta \in \mathbb{C}^Q} \|Jx + H\eta\|_2^2, & \eta_0(x) &= -K_0^{-1} H^* Jx, \\ x^* \widehat{D}_{a,D}^{(s)} x &= \min_{\eta \in \mathbb{C}^Q} \|B(Jx + H\eta)\|_2^2, & \eta_1(x) &= -K_1^{-1} H^* B^* B Jx. \end{aligned} \quad (\text{XIM16})$$

Both minima retain every lower evaluation and all their cross terms. Each minimizer is unique because K_0, K_1 are positive definite. More explicitly, the difference between the first squared norm and its displayed minimum is $(\eta - \eta_0(x))^* K_0 (\eta - \eta_0(x))$, and the difference for the second is $(\eta - \eta_1(x))^* K_1 (\eta - \eta_1(x))$. These identities follow on expanding, using the two displayed formulas for η_0, η_1 ; each cross term cancels exactly. If either minimum were zero, invertibility of B and independence of $[H, J]$ would imply $x = 0$. Thus both analytic metrics are positive definite. The two minimizing coefficient columns need not coincide.

The induced map on the quotient has the following exact domain, codomain, and inverse:

$$\begin{aligned} Q_0 &= \mathcal{D}_D \cap \mathcal{K}_D^\perp = \text{im } W_0, & Q_1 &= B \mathcal{D}_D \cap (B \mathcal{K}_D)^\perp = \text{im } W_1, \\ \overline{B} : Q_0 &\longrightarrow Q_1, & \overline{B} &= P_{Q_1} B|_{Q_0}, & \overline{B}^{-1} &= P_{Q_0} B^{-1}|_{Q_1}, & \overline{B}(W_0 x) &= W_1 x. \end{aligned} \quad (\text{XIM17})$$

The quotient identifications are isometric because each representative has a unique orthogonal minimum lift. Projection after B subtracts an element of $B \mathcal{K}_D$; applying B^{-1} turns that into an element of $\mathcal{K}_D$, which the second projection removes. This proves both inverse identities. It also proves the last equation using XIM14. In the orthonormal minimum-lift frames $W_0(D_{a,D}^{(s)})^{-1/2}$ and $W_1(\widehat{D}_{a,D}^{(s)})^{-1/2}$, the matrix is

$$[\overline{B}] = (\widehat{D}_{a,D}^{(s)})^{1/2} (D_{a,D}^{(s)})^{-1/2}, \quad [\overline{B}]^* [\overline{B}] = (D_{a,D}^{(s)})^{-1/2} \widehat{D}_{a,D}^{(s)} (D_{a,D}^{(s)})^{-1/2}. \quad (\text{XIM18})$$

Indeed $W_1^* B W_0 = W_1^* W_1 = \widehat{D}_{a,D}^{(s)}$, because the discarded part lies in $B\mathcal{K}_D$. Thus the original TAC9 correction is exactly

$$\epsilon_{a,N,s} = \log \det \widehat{D}_{a,D}^{(s)} - \log \det D_{a,D}^{(s)} = 2 \log |\det[\overline{B}]|. \quad (\text{XIM19})$$

Write the established all-rank logarithmic exterior estimates for the actual ambient matrix, including its original weights, as

$$\log \left\| \bigwedge^r ((A_{D,s}^\top)^{\oplus c}) \right\| \leq F_{D,s,r,c}, \quad \log \left\| \bigwedge^r ((A_{D,s}^\top)^{\oplus c})^{-1} \right\| \leq I_{D,s,r,c}, \quad 0 \leq r \leq cd. \quad (\text{XIM20})$$

The F and I in this display are logarithms of exterior norms, without the Gram factor two. Orthogonal projections contract every exterior power: in orthonormal bases their singular values are zero or one. Restrictions and inclusions of Hilbert subspaces also contract exterior powers. Applying XIM17 in both directions and then XIM20 at $c = 1$, $r = m$, proves the finite bound

$$\boxed{-2I_{D,s,m,1} \leq \epsilon_{a,N,s} \leq 2F_{D,s,m,1}}. \quad (\text{XIM21})$$

There is no added Q -rank allowance: the complete denominator has already been retained in the exact quotient map and in both minima. For c copies, the denominator has dimension cQ and the full analytic quotient dimension cm . Any further restriction followed by an attained quotient has its full preimages $K' \subset Y'$ in the ambient c copies; necessarily K' contains the original cQ -dimensional denominator. Nested minimization over these fibres equals minimization over K' , because the union of the original affine fibres is the final affine fibre. XIM17 applies to $K' \subset Y'$ and proves the same determinant interval $[-2I_{D,s,r,c}, 2F_{D,s,r,c}]$ with its actual final rank r , including $r = 0$ with empty determinant one. No intermediate dimension is charged a second time.

Finally retain the actual TAC endpoints and dual sign orientation:

$$N_i = (q-1, q, 2q-1, 2q)_i, \quad D_i = N_i - v, \quad \Xi_{a,s} = -\epsilon_{a,N_0,s} - \epsilon_{a,N_1,s} + \epsilon_{a,N_2,s} + \epsilon_{a,N_3,s}. \quad (\text{XIM22})$$

For these four endpoints, $d_i = (q-v, q+1-v, 2q-v, 2q+1-v)_i$ and $t_i = (0, 1, q, q+1)_i$. The annihilator dimensions remain exactly $q-v$ and Q , and the analytic quotient rank remains m at every endpoint. Set $F_i = F_{D_i,s,m,1}$ and $I_i = I_{D_i,s,m,1}$. Negating the two low-endpoint inequalities in XIM21 and adding the two high-endpoint inequalities gives

$$\boxed{-2(F_0 + F_1 + I_2 + I_3) \leq \Xi_{a,s} \leq 2(I_0 + I_1 + F_2 + F_3)}. \quad (\text{XIM23})$$

These are bounds on the same four correlated matrices appearing in TAC9. A separate radius infimum, minimum of established allowances, or improved ambient estimate can be inserted in each F_i or I_i provided it bounds XIM20; no attainment of an infimum is used. The source order remains 1 or a at the lower grid, its centre remains $a/2 - 4$, and its cutoff remains $N_i - v$. In particular no replacement by the order $a - 8$ or by an unshifted cutoff occurs. The explicit uniform bounds LT21–22 of the preceding transport proof apply because $D_i \leq 3a^2$ for $a \geq 9$. Inserting them in XIM23 proves $\Xi_{a,s} = O_{\text{actual}}(q)$ for both original source orders. The identical endpoint bound XIM21 holds at every cutoff $q-1 \leq N \leq 2q$, including each growing-prefix endpoint $N = q + R$ with $0 \leq R < q$, with the same fixed constant.

Complete absolute common-kernel and full-numerator estimates

70 The absolute common kernel: the new finite result

This complete incoming calculation sharpens the current absolute common kernel estimate. Its original comparison was against the preceding logarithmic bound; the intervening KAF chapter already established the $O(kq)$ scale. The new finite envelope reduces its uniform limsup constant from $16(4 + 10 \log 2)$ to $32 \log(25 \log 3 / (4\pi))$. AKJ below takes the pointwise minimum of both finite bounds. No limiting kernel coefficient is inferred from these upper bounds.

Retain the actual hypothetical complete quartet and its entire quotient

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad h(z) = \prod_{\alpha \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (z - \alpha)^m, \quad v_h = 2\xi/h.$$

The full primary unit is $U = M_{jh v_h}$. The original tensor order, centre, polynomial, and coefficient quotient are

$$\begin{aligned} k &= 4l + 1, \quad l \geq 1, \quad e = 1 + k(m - 1), \quad q = e(k + 1)^2, \quad c = k/2, \\ \lambda_{a,b} &= c + (2a - k)\delta + i(2b - k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \lambda_{a,b})^e, \quad E = \mathbb{C}[S]/(\chi). \end{aligned} \quad (\text{AKS1})$$

For the principal conclusion $m = 1$. The source-operator estimates below also hold with the stated repeated-root list for $m > 1$, but no simple-quartet rank is assigned to those multiplicities.

Let $\Lambda_k(u) : E \rightarrow B_k(u)$ be the original observation, $K(u) = \ker \Lambda_k(u)$, and $I : K(u) \hookrightarrow E$ its original coefficient inclusion. For a source $s \in \{1, k\}$, let $G_N^{(s)}$ be the original minimum-remainder Gram, in the frame $(1, S, \dots, S^{q-1})$. Define $H_{K,N}^{(s)} = I^* G_N^{(s)} I$ and retain exactly

$$F_K^{(s)} = \log \det H_{K,q-1}^{(s)} + \log \det H_{K,q}^{(s)} - \log \det H_{K,2q-1}^{(s)} - \log \det H_{K,2q}^{(s)}. \quad (\text{AKS2})$$

Every empty determinant is one. All conjugate transposes refer to the specified coordinate matrices, not to a bilinear residue pairing.

Here are entirely explicit positive numbers used in the bound:

$$\begin{aligned} R &= k\sqrt{\delta^2 + \gamma^2}, \quad b_s = s/2, \quad t_q = \frac{\pi}{2} - \frac{1}{q}, \quad \mathbf{a}_q = \frac{2 \log 3}{t_q}, \quad \mathbf{s}_q = \frac{\mathbf{a}_q^q - 1}{\mathbf{a}_q - 1}, \\ \mathbf{b}_{s,q} &= \left[\frac{9}{25} \sin \frac{1}{q} \right]^{-b_s} \exp \left(2R \left(\log 3 + \frac{t_q}{2} \right) \right), \\ \mathcal{Z}_{s,j} &= 1 + \mathbf{b}_{s,q} \mathbf{s}_q^2 \sum_{d=q}^{q+j-1} \frac{d!}{(b_s)_d} \left(\frac{25}{16} \right)^d, \quad \mathcal{L}_{s,j} = \log \mathcal{Z}_{s,j} \quad (0 \leq j \leq q + 1). \end{aligned} \quad (\text{AKS3})$$

The empty sum gives $\mathcal{Z}_{s,0} = 1$. In the admitted family $q \geq 36$, so $t_q \in (0, \pi/2)$ and $\mathbf{a}_q > 1$. These parameters are radii in the existing Laplace and generating functions; neither is a new source order.

Main theorem (absolute finite bound). For the original simple quartet, every original logarithm branch, and every $|u| \geq R_*$, where R_* is precisely RPC21 or RPC23 (recorded in Section 76), put $r = 8k - 16$. Then

$$0 \leq F_K^{(s)}(k, u) \leq r(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}), \quad s \in \{1, k\}. \quad (\text{AKS4})$$

There is a stronger, fully period-dependent interval in AKS28 below. In particular, with

$$\kappa_* = 32 \log \left(\frac{25 \log 3}{4\pi} \right), \quad 25.02078097649 < \kappa_* < 25.02078097650, \quad (\text{AKS5})$$

one has the uniform statement

$$\boxed{0 \leq \limsup_{\substack{k \rightarrow \infty \\ k \equiv 1 \pmod{4}}} \sup_{\substack{|u| \geq R_* \\ \text{original branches}}} \frac{F_K^{(s)}(k, u)}{kq} \leq \kappa_*}. \quad (\text{AKS6})$$

Here the source may be $s = 1$ for every k , or $s = k$. The same bound holds for the actual arithmetic kernel. Its exact finite allowances are in AKS34–36. Thus the absolute kernel is $O_h(kq)$ on this domain, not only $O_h(kq \log q)$. Neither AKS5 nor AKS6 calls κ_* the actual leading coefficient.

71 An exact coefficient realization of the actual lost directions

71.1 The period matrix and invariant restriction, with their full coefficients

For $m = 1$, order the one-factor roots as $(++)$, $(+-)$, $(-+)$, $(--)$, and let $\mathcal{E}_{\text{prim}}$ have their original Lagrange idempotents as columns. Set $\zeta = \exp(2\pi i/5)$ and $F_{jt} = \zeta^{jt} - 1$, $1 \leq j, t \leq 4$. The exact inverse is $(F^{-1})_{tj} = \zeta^{-tj}/5$. Retain

$$\begin{aligned} \Pi[P]_j &= \int_{\Gamma_j} e^{\Phi_h(z)/u} \text{rem}_h P(z) dz, & \Phi'_h &= h, & \Phi_h(0) &= 0, \\ \Gamma_j &= -\ell_0 + \ell_j, & \arg \ell_j &= (\arg u + \pi + 2\pi j)/5, \\ \mathbf{H} &= F^{-1} \Pi U \mathcal{E}_{\text{prim}}, & D &= \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4), \\ Q(x) &= Q_0(\mathbf{H}^{-1}x), & Q_0(t) &= t_1 t_4 - t_2 t_3, & Q_a(x) &= Q(D^{-a}x). \end{aligned} \quad (\text{AKS7})$$

The integral always uses the degree-less-than-four representative. Invertibility of Π , all contour phases, and its full determinant are the PDE/PQO results in the supplied proofs. The four unit values are $(a, \bar{a}, \bar{a}, a)$, with the actual $a = v_h(\rho) \neq 0$; no value of a is selected. Hence $\mathbf{H}$ is the original invertible matrix.

The following formulas make the coefficient realization directly computable. Write

$$\mathcal{I}_{k,0} = \{\nu \in \mathbb{Z}_{\geq 0}^4 : |\nu| = k, \nu_1 + 2\nu_2 + 3\nu_3 + 4\nu_4 \equiv 0 \pmod{5}\}.$$

For $\nu \in \mathcal{I}_{k,0}$ define $B_{\nu;(a,b)}$ by summing over 4×4 arrays $n_{t,\alpha} \geq 0$ with row sums ν_t , total plus-real count a , and total plus-imaginary count b :

$$B_{\nu;(a,b)} = \sum_{\substack{\sum_{\alpha} n_{t,\alpha} = \nu_t \\ \sum_{t,\alpha: \epsilon_{\alpha} = +} n_{t,\alpha} = a \\ \sum_{t,\alpha: \eta_{\alpha} = +} n_{t,\alpha} = b}} \left(\prod_{t=1}^4 \frac{\nu_t!}{\prod_{\alpha} n_{t,\alpha}!} \right) \prod_{t,\alpha} \mathbf{H}_{t,\alpha}^{n_{t,\alpha}}. \quad (\text{AKS8})$$

Let $\mathbf{V}_{(a,b),d} = \lambda_{a,b}^d$ for $0 \leq d < q$ and set $\mathbf{A} = \mathbf{B}\mathbf{V}$. All entries of $\mathbf{A}$ are now finite expressions in the actual periods and unit. There is no generic period matrix in this definition.

For completeness, put $t_{\alpha}(w) = \exp((\eta_{\alpha}\gamma - i\epsilon_{\alpha}\delta)w)$ and $x(w) = \mathbf{H}t(w)$. Expansion gives

$$x(w)^{\nu} = \sum_{a,b} B_{\nu;(a,b)} e^{\omega_{a,b} w}, \quad \omega_{a,b} = (2b - k)\gamma - i(2a - k)\delta.$$

In the original primary algebra, $e^{wY}[1] = \sum_{a,b} e^{\omega_{a,b} w} \varepsilon_{a,b}$, where $Y = (M_S - cI)/i$. The exponentials are independent by the Vandermonde of the distinct $\omega_{a,b}$. OCS9's unscaled symmetric word basis therefore identifies the original observation with

$$j\Lambda_k = \mathcal{F}_0 \mathbf{A}, \quad \ker \mathbf{A} = K(u) \quad \text{in the original coefficient frame.} \quad (\text{AKS9})$$

Every multinomial in AKS8 belongs to its actual ordered words; the word multiplicity is not inserted a second time. The exact inherited row Gram of $\mathcal{F}_0$ is the restriction to $\mathcal{I}_{k,0}$ of

$$\mathcal{G}_{\nu,\mu} = \sum_{\substack{n_{tu} \geq 0 \\ \sum_u n_{tu} = \nu_t \\ \sum_t n_{tu} = \mu_u}} \frac{k!}{\prod_{t,u} n_{tu}!} \prod_{t,u} [5(\delta_{tu} + 1)]^{n_{tu}}. \quad (\text{AKS10})$$

This follows by taking the coefficient of $\bar{x}^\nu y^\mu$ in $(\bar{x}^T F^* F y)^k$, since $F^* F = 5(I + \mathbf{1}\mathbf{1}^*)$. It is positive definite and includes every row cross term.

Here is the precise role of the orbit ideal. The map sending a homogeneous polynomial f to the covector specified by $\Theta_k(f)(e^{wY}[1]) = f(x(w))$ has kernel $Q\mathbb{C}[x]_{k-2}$ and is onto $E^\vee$. Indeed equal exponential indices correspond exactly to monomial differences divisible by $t_1 t_4 - t_2 t_3$; the displayed Vandermonde proves that there are no other relations. Its invariant image is $\text{im } \Lambda_k^\vee$. On the five-member domain, the five Q_a are distinct prime quadrics. An invariant polynomial divisible by Q is divisible by each Q_a , and therefore by their full product; its quotient is invariant by cancellation in the polynomial ring. Consequently

$$\mathbb{C}[x]^{(D)} \cap (Q) = \left(\prod_{a=0}^4 Q_a \right) \mathbb{C}[x]^{(D)}, \quad (\text{AKS11})$$

$$K(u)^\vee \simeq \mathbb{C}[x]_k / (\mathbb{C}[x]_k^{(D)} + Q\mathbb{C}[x]_{k-2}),$$

$$b := \text{rank } \mathbf{A} = d_{5,k,0} - d_{5,k-10,0} = k^2 - 6k + 17, \quad r = q - b = 8k - 16.$$

Negative degrees are zero. The count is $d_{5,d,0} = ((\binom{d+3}{3} + 4\epsilon_d)/5$, $\epsilon_d = \mathbf{1}_{d \equiv 0(5)} - \mathbf{1}_{d \equiv 1(5)}$; subtracting the cubics gives the stated expression for $k \geq 10$, and $k = 5, 9$ give 12, 44 directly. This is the inherited OQX calculation, repeated to specify which kernel AKS9 realizes. The source metric is still to be applied to this kernel, not to the invariant ring itself.

71.2 A determinant formula requiring no assumed nonzero pivot

Let $\mathcal{G}_0$ be the full positive row Gram in AKS10 and set

$$W = \mathbf{A}^* \mathcal{G}_0 \mathbf{A}, \quad \sigma_j = [z^j] \det(I_q + zW), \quad P_K = \frac{1}{\sigma_b} \sum_{j=0}^b (-1)^j \sigma_{b-j} W^j. \quad (\text{AKS12})$$

Since W has exactly b positive eigenvalues, $\sigma_b > 0$. The scalar polynomial in AKS12 is one at zero and zero at every nonzero eigenvalue: it is the normalized product of the factors $\lambda_i - z$ for the b positive eigenvalues. Hence $P_K = P_K^* = P_K^2$ and $\text{im } P_K = K(u)$. This is the projector derived from the literal observation, not a freely chosen projection. Replacing $\mathcal{G}_0$ by any other positive coordinate Gram leaves this same coefficient-orthogonal kernel projector, since its image and orthogonal complement are unchanged. No such replacement is needed here.

At every original endpoint define

$$D_{K,N}^{(s)} = [z^r] \det(I_q + z P_K G_N^{(s)} P_K). \quad (\text{AKS13})$$

These numbers are strictly positive for $r > 0$ and are one for $r = 0$. To prove their exact metric meaning, choose an orthonormal coefficient basis U_K of $\text{im } P_K$ just for the proof. In a unitary extension, $P_K G_N P_K$ has its positive block $U_K^* G_N U_K$ and a zero block. Thus $D_{K,N} = \det(U_K^* G_N U_K)$. Write the original inclusion as $I = U_K T$; then $|\det T|^2 = \det(I^* I)$ and

$$\det H_{K,N}^{(s)} = \det(I^* I) D_{K,N}^{(s)}, \quad \boxed{F_K^{(s)} = \log \frac{D_{K,q-1}^{(s)} D_{K,q}^{(s)}}{D_{K,2q-1}^{(s)} D_{K,2q}^{(s)}}}. \quad (\text{AKS14})$$

This supplies an equivalent exact coefficient determinant formula for all the $8k - 16$ lost directions, including the entire period dependence. The original frame factor is present at each endpoint and cancels only with its four actual signs. It is unnecessary to suppose any particular minor nonzero. Alternatively, selecting the first independent r columns of P_K gives an explicit coefficient frame; the determinant formula avoids even that chart choice.

For $m > 1$ replace AKS8–9 by OCS12–18’s full confluent matrix $\mathfrak{A}_{\nu;(a,b,D)} = i^D D! c_{\nu;a,b,D}$ and its primary isomorphism. AKS12–14 are then identical with the actual rank, not with the rank in AKS11. All primary nilpotents and all unit derivatives remain in that matrix.

72 A bounded exact remainder operator on the original source

72.1 The complete source norms and coefficient map

For either existing source set $M_s = (2\pi)^{s/2}$ and retain

$$dm_{0,s}(y) = \frac{M_s 2^{b_s}}{4\pi\Gamma(b_s)} |\Gamma(b_s/2 + iy/2)|^2 dy, \quad \int e^{ty} dm_{0,s} = M_s (\cos t)^{-b_s}, \quad |t| < \pi/2. \quad (\text{AKS15})$$

The supplied IGO1–3 proof derives this transform by the beta integral, Fourier inversion and a contour shift within $|\operatorname{Im} x| < \pi/2$. In particular it includes the exact masses $M_1 = \sqrt{2\pi}$ and $M_k = (2\pi)^{k/2}$ and exponential domination on every smaller strip. The original real monic polynomials satisfy

$$\begin{aligned} p_0 &= 1, \quad p_1 = y, \quad p_{d+1} = yp_d - d(b_s + d - 1)p_{d-1}, \\ \sum_{d \geq 0} \frac{p_d(y)}{d!} z^d &= (1 + z^2)^{-b_s/2} e^{y \arctan z}, \quad h_d^{(s)} = \|p_d\|^2 = M_s d! (b_s)_d. \end{aligned} \quad (\text{AKS16})$$

The generating formula follows by differentiating in z . To check the normalization directly, integrate the product of two generating functions at small real z, w . AKS15 and the cosine addition identity give $M_s(1 - zw)^{-b_s}$. Taking any finite pair of derivatives at zero, justified by the stated exponential domination, gives orthogonality and the norms in AKS16. Thus the leading phase in $u_d(S) = p_d((S - c)/i)/\sqrt{h_d^{(s)}}$ is exactly i^{-d} . The same generating function agrees with NIST DLMF 18.23.7 after the original parameter rescaling; it is derived here from the supplied recurrence, not substituted for it.

List the original roots $\omega_\nu = (\lambda_{a,b} - c)/i$ in lexicographic order, repeating each exactly e times, and put

$$P(y) = i^{-q} \chi(c + iy) = \prod_{\nu=1}^q (y - \omega_\nu), \quad \Phi : [f(y)] \mapsto [f((S - c)/i)]. \quad (\text{AKS17})$$

The inverse is substitution $S = c + iy$, and $P((S - c)/i) = i^{-q} \chi(S)$. This is an exact algebra isomorphism, including the complete confluent orders; there is no monomial comparison divisor. Let C_s have columns $[u_0], \dots, [u_{q-1}]$. Its nonzero determinant is $i^{-q(q-1)/2} \prod_{d < q} (h_d^{(s)})^{-1/2}$. Define

$$v_j = C_s^{-1} [u_{q+j}], \quad \mathbf{K}_j = I_q + \sum_{a=0}^{j-1} v_a v_a^*, \quad 0 \leq j \leq q + 1. \quad (\text{AKS18})$$

Parseval in the low-degree original orthonormal polynomial basis proves

$$\|v_j\|_2^2 = \frac{\|\operatorname{rem}_P p_{q+j}\|_{m_{0,s}}^2}{M_s (q+j)! (b_s)_{q+j}}. \quad (\text{AKS19})$$

This is the full norm of the actual remainder, not a diagonal entry of an approximated quotient Gram.

72.2 Newton interpolation with all roots and their orders

For $0 \leq a < q$ define $N_a(y) = \prod_{\nu=1}^a (y - \omega_\nu)$, with $N_0 = 1$. For every complex v , the unique entire-function remainder is

$$\text{rem}_P(e^{vy}) = \sum_{a=0}^{q-1} [\omega_1, \dots, \omega_{a+1}] e^{v \cdot} N_a(y), \quad |[\omega_1, \dots, \omega_{a+1}] e^{v \cdot}| \leq \frac{|v|^a}{a!} e^{R|v|}. \quad (\text{AKS20})$$

Here divided differences at repeated nodes mean Hermite divided differences, not evaluation with a zero denominator.

A proof covering the confluent case is useful. On the standard a -simplex with barycentric coordinates $\theta_1, \dots, \theta_{a+1}$,

$$[\omega_1, \dots, \omega_{a+1}] e^{v \cdot} = v^a \int_{\Delta_a} e^{v \sum \theta_\nu \omega_\nu} d\theta.$$

The integral of $\prod \theta_\nu^{n_\nu}$ is $\prod n_\nu! / (a + \sum n_\nu)!$, as follows by iterated integration of integer beta integrals. Expanding the exponential therefore gives $\sum_{n \geq 0} v^{a+n} h_n(\omega_1, \dots, \omega_{a+1}) / (a+n)!$. For monomials, the divided difference of y^{a+n} is exactly h_n : this follows by multiplying the geometric series $\prod (1 - \omega_\nu z)^{-1}$ and the Newton division recurrence. It is a polynomial identity in the nodes and so also holds at repeated nodes. This proves the integral formula. Iterated Newton division for polynomials, followed by the convergent exponential series, proves the first equality in AKS20, including every required derivative. Finally the simplex has volume $1/a!$ and $|\sum \theta_\nu \omega_\nu| \leq R$, proving its stated bound.

For any $0 < t < \pi/2$, put $\tau = 2/t$. Positivity of every term of the cosh series in AKS15 gives

$$\|y^a\|_{m_{0,s}} \leq \sqrt{M_s} (\cos t)^{-b_s/2} \tau^a a!.$$

Indeed its square is at most $M_s (\cos t)^{-b_s} (2a)! t^{-2a}$ and $(2a)! \leq 4^a (a!)^2$. Expanding the *whole* Newton polynomial, then using $|e_j(\omega_1, \dots, \omega_a)| \leq \binom{a}{j} R^j$, proves

$$\begin{aligned} \|N_a\|_{m_{0,s}} &\leq \sqrt{M_s} (\cos t)^{-b_s/2} \sum_{j=0}^a \binom{a}{j} R^{a-j} \tau^j j! \\ &= \sqrt{M_s} (\cos t)^{-b_s/2} \tau^a a! \sum_{n=0}^a \frac{(R/\tau)^n}{n!} \\ &\leq \sqrt{M_s} (\cos t)^{-b_s/2} \tau^a a! e^{R/\tau}. \end{aligned} \quad (\text{AKS21})$$

There are no omitted polynomial coefficients. Combining AKS20–21 gives for every complex v

$$\|\text{rem}_P(e^{vy})\|_{m_{0,s}} \leq \sqrt{M_s} (\cos t)^{-b_s/2} e^{R(|v|+t/2)} \sum_{a=0}^{q-1} (2|v|/t)^a. \quad (\text{AKS22})$$

72.3 Cauchy extraction in the actual Gamma generating function

Fix $0 < z < 1$ and write $a_z = \text{arctanh } z$. For $|w| = z$ the defining power series gives $|\arctan w| \leq a_z$ and $|1 + w^2| \geq 1 - z^2$. Apply the literal finite-dimensional remainder map to AKS16 and use Cauchy's formula for its d th coefficient. The norm triangle inequality on that circle and AKS22 give

$$\begin{aligned} \frac{\|\text{rem}_P p_d\|_{m_{0,s}}}{d!} &\leq z^{-d} (1 - z^2)^{-b_s/2} \sqrt{M_s} (\cos t)^{-b_s/2} \\ &\quad \times e^{R(a_z+t/2)} \sum_{a=0}^{q-1} (2a_z/t)^a. \end{aligned} \quad (\text{AKS23})$$

Cauchy's formula is legitimate for the finite remainder vector: its entries are holomorphic for $|w| < 1$ by the actual matrix generating identity OPG7. The source norm is taken only after this finite-vector identity; there is no unjustified integral of a nonintegrable generating function on the large circle.

Substitution of AKS23 into AKS19 retains the factorial denominator and cancels the two occurrences of the original mass, giving

$$\|v_j\|_2^2 \leq \frac{d!}{(b_s)_d} z^{-2d} [\cos t(1 - z^2)]^{-b_s} e^{2R(az+t/2)} \left[\sum_{a=0}^{q-1} (2a_z/t)^a \right]^2, \quad d = q + j. \quad (\text{AKS24})$$

Adding the *complete* rank-one positive matrices in AKS18 now proves $K_j \preceq \mathcal{Z}_{s,j} I_q$ after setting $z = 4/5$ and $t = t_q$. Indeed $a_{4/5} = \log 3$, $1 - z^2 = 9/25$, and $\cos t_q = \sin(1/q)$, so the exact scalar obtained is AKS3. The high index $j = q + 1$ includes the original polynomial of degree $2q$.

There is also a root-by-root finite refinement. Put $\mu_{s,d} = \int y^{2d} dm_{0,s}$, $R_a = \max_{\nu \leq a+1} |\omega_\nu|$, and

$$B_{s,a}^{\text{root}} = \sum_{d=0}^a |e_{a-d}(\omega_1, \dots, \omega_a)| \sqrt{\mu_{s,d}}, \quad (\text{AKS25})$$

$$J_s^{\text{root}}(z) = \frac{(1 - z^2)^{-b_s}}{M_s} \left[\sum_{a=0}^{q-1} \frac{a_z^a}{a!} e^{a_z R_a} B_{s,a}^{\text{root}} \right]^2.$$

Replace the scalar multiplying $d!z^{-2d}/(b_s)_d$ in AKS24 by $J_s^{\text{root}}(z)$ to get a valid sharper bound, by the same proof before AKS21 is relaxed. Every moment is exactly $M_s(2d)! [w^{2d}] (\cos w)^{-b_s}$. Thus this refinement keeps every actual root coefficient and a finite original moment formula. AKS21 bounds it by the explicit scalar in AKS24. The proof never replaces the source determinant by a product of diagonal moments: AKS24 is a full-vector operator bound, and the next step restricts the inverse of the full covariance.

73 The four original kernel Grams and the evaluated scale

The original orthonormal source map at degree $q + j - 1$ is $C_s[I_q, v_0, \dots, v_{j-1}]$. Its minimum lift of $C_s x$ is $[I_q, v_0, \dots, v_{j-1}]^* K_j^{-1} x$. It has that image and is orthogonal to the full relation kernel, so Pythagoras proves

$$G_{q+j-1}^{(s)} = C_s^{-*} K_j^{-1} C_s^{-1}, \quad H_j := H_{K,q+j-1}^{(s)} = T_s^* K_j^{-1} T_s, \quad T_s = C_s^{-1} I. \quad (\text{AKS26})$$

This verifies the source-minimum step in the coefficient determinant AKS13–14. Neither a cross block nor any original relation is removed.

Since $I_q \preceq K_j \preceq \mathcal{Z}_{s,j} I_q$, inversion and compression give $\mathcal{Z}_{s,j}^{-1} H_0 \preceq H_j \preceq H_0$, where $H_0 = T_s^* T_s$. Define $d_j = -\log \det(H_0^{-1/2} H_j H_0^{-1/2})$. Congruence and all r eigenvalues give

$$0 = d_0 \leq d_1 \leq \dots \leq d_{q+1}, \quad 0 \leq d_j \leq r \mathcal{L}_{s,j}, \quad F_K^{(s)} = d_q + d_{q+1} - d_1. \quad (\text{AKS27})$$

The identity uses precisely the four endpoints in AKS2. In particular it proves AKS4, as well as the sharper interval

$$d_1 \leq F_K^{(s)} \leq r(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - d_1. \quad (\text{AKS28})$$

For a directly calculable expression for its lower endpoint set $V = \|v_0\|^2$ and $\theta = v_0^* T_s (T_s^* T_s)^{-1} T_s^* v_0$. Then

$$0 \leq \theta \leq V, \quad d_1 = \log \frac{1 + V}{1 + V - \theta}. \quad (\text{AKS29})$$

Indeed the inverse of $I + v_0 v_0^*$ is $I - v_0 v_0^*/(1 + V)$; take its full compressed determinant. For $r = 0$ set $\theta = d_1 = 0$. In GMB/OPR notation $V = t_0$ and $V - \theta = \xi_0$, so this is exactly the first original section-corrected kernel increment, not an independently chosen kernel observation.

Here is an evaluated finite remainder for AKS6. Put

$$\begin{aligned} a_\infty &= \frac{4 \log 3}{\pi}, & c_0 &= \log \frac{25 \log 3}{4\pi} > 0, \\ E_{s,q}(R) &= \log 2 + \frac{4}{\pi - 2/q} + \frac{s}{2} \log \frac{25\pi q}{18} + 2R \left(\log 3 + \frac{\pi}{4} \right) \\ &\quad + \log((q+1)(4q+1)) - 2 \log(a_\infty - 1). \end{aligned} \tag{AKS30}$$

Then for both source orders, every $q \geq 36$, and the same actual kernel,

$$\boxed{\mathcal{L}_{s,q}, \mathcal{L}_{s,q+1} \leq 2c_0 q + E_{s,q}(R), \quad 0 \leq F_K^{(s)} \leq 4rc_0 q + 2rE_{s,q}(R).} \tag{AKS31}$$

To verify every factor, $b_s \geq 1/2$ gives

$$\frac{d!}{(b_s)_d} \leq \frac{4^d}{\binom{2d}{d}} \leq 2d + 1.$$

The last inequality uses that the central binomial is the largest of the $2d + 1$ positive coefficients whose sum is 4^d . Thus the sum in AKS3 is at most $(q+1)(4q+1)(5/4)^{4q}$. Also $\mathfrak{s}_q \leq \mathfrak{a}_q^q/(a_\infty - 1)$ and

$$2q \log \mathfrak{a}_q \leq 2q \log a_\infty + \frac{4}{\pi - 2/q}, \quad \sin(1/q) \geq \frac{2}{\pi q}.$$

The first follows from $-\log(1-x) \leq x/(1-x)$ at $x = 2/(\pi q)$; the second is the chord bound for sine on $[0, \pi/2]$. Use $t_q \leq \pi/2$ in the remaining exponential. The order- q terms combine to $4q \log(5/4) + 2q \log a_\infty = 2c_0 q$. The upper bound for the logarithm of the added positive scalar is positive, so $\log(1 + e^x) \leq \log 2 + \max(0, x)$ proves the first bound in AKS31. AKS28 proves the second.

On the original simple family $q = (k+1)^2$, $R = O_h(k)$ and $r = 8k - 16$. Consequently $E_{1,q}(R) = O_h(k + \log q)$ and $E_{k,q}(R) = O_h(k \log q)$. Dividing AKS31 by kq proves AKS6, with $32c_0 = \kappa_*$. In fact the fixed-source upper bound is

$$F_K^{(1)} \leq (32k - 64)c_0 q + (16k - 32)E_{1,q}(R), \tag{AKS32}$$

whose explicit second term is $O_h(k^2)$. The tensor-source version has an $O_h(k^2 \log q)$ upper remainder. These are remainders of upper bounds, not assertions about the actual signed error or leading value. The proof is uniform in all the period parameters in the stated domain: no norm or condition number of a period matrix occurs in AKS30.

The precise earlier loss. OKV15 had already retained $\|v_j\|^2 \leq q^2 2^{6q} (2D_* q)^{2d} / (d!(b_s)_d)$, $d = q + j$. Replacing its denominator by $1/2$ caused the old $q \log q$ exponent. Even without AKS20–24 one obtains the valid repair

$$\|K_{q+1}\| \leq 1 + (q+1)q^2 2^{6q} \frac{(4D_* q)^{4q}}{(4q)!}, \quad D_* \geq 3. \tag{AKS33}$$

Here $d!(b_s)_d \geq (2d)!/4^d$ and the remaining factorial ratio increases up to $d = 2q$ for $D_* \geq 3$. The elementary inequality $\log n! \geq n \log n - n + 1$ already makes the logarithm of AKS33 $O_h(q)$. The Newton calculation strengthens this repair to the packet-independent limsup constant AKS5, while retaining R in its finite remainder.

Exact enclosure of the displayed numerical ceiling. For $x > 1$ and $z = (x - 1)/(x + 1)$ the integrated geometric series gives

$$0 \leq \log x - 2 \sum_{j=0}^{39} \frac{z^{2j+1}}{2j+1} \leq \frac{2z^{81}}{81(1-z^2)}.$$

For $0 < x < 1$, the alternating sum through $j = 39$ for $\arctan x$ has error between zero and $x^{81}/81$. The tangent addition formula, with the angle in $(0, \pi/2)$, proves $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$: the tangent of $4 \arctan(1/5) - \arctan(1/239)$ is exactly one. Use the resulting rational lower and upper bounds for π and for $\log 3$, then apply the first displayed bound to the two endpoints of $25 \log 3/(4\pi)$. Multiplication by 32 gives, with outward decimal rounding, the interval

$$\kappa_* \in [25.0207809764994871, 25.0207809764994872].$$

The accompanying rational-arithmetic implementation checks these inequalities by integer cross multiplication, rather than a floating-point logarithm. This proves the less finely rounded strict interval AKS5.

74 The complete arithmetic and mixed receivers

74.1 The unchanged finite arithmetic allowances

Use exactly AMT4's constants. To prevent a loss hidden in order notation, write them here:

$$\begin{aligned} \alpha &= \pi/2, \quad p = 8m - 9/2, \quad B_h = 42 + 8m, \quad M = 21 + 4m, \\ \vartheta_h &= \int_{-1}^1 w_h(y) dy, \quad M_\alpha = \int e^{\alpha y} w_h(y) dy, \quad w_h(y) = |v_h(1/2 + iy)|^2/(2\pi), \\ c_\Gamma &= \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}, \\ a_k &= \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-B_h}}{C_\Gamma 2^{B_h/2}}, \quad A_k = \frac{C_h^{\text{tilt}}}{c_\Gamma} k^{p+1} M_\alpha^{k-1}, \\ D_N &= [1 + 4(N + M)^2]^M, \quad L_N^{\text{ar}} = \log(A_k D_N / a_k), \\ E_k^+ &= L_{q-1}^{\text{ar}} + L_q^{\text{ar}}, \quad E_k^- = L_{2q-1}^{\text{ar}} + L_{2q}^{\text{ar}}. \end{aligned} \tag{AKS34}$$

The constants c_h, C_h^{tilt} are the original TW/BT envelope constants, not new choices. Their actual inequalities and complete source construction are preserved in AMT4–6 and the current source closure. Minimize AMT6 on the same remainder fibre and then restrict to $I(K)$. AMT9–20 gives, with the original arithmetic source w_h^{*k} ,

$$-rE_k^+ \leq F_K^{\text{ar}} - F_K^{(1)} \leq rE_k^-.$$

Let $\underline{U}_s = d_1^{(s)}$ and $\overline{U}_s = r(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - d_1^{(s)}$. The new finite conclusion is

$$\boxed{\max\{0, \underline{U}_1 - rE_k^+\} \leq F_K^{\text{ar}} \leq \overline{U}_1 + rE_k^-} \tag{AKS35}$$

This improves AMT37's absolute upper bound. For the sharper correlated version retain $d_{E,N} = q \log A_k - \log(\det G_N^{\text{ar}} / \det G_N^{(1)})$ and $c_{X,N} = \min\{r_X L_N^{\text{ar}}, d_{E,N}\}$, where $r_K = r$, $r_B = q - r$. Put $C_X^- = c_{X,q-1} + c_{X,q}$, $C_X^+ = c_{X,2q-1} + c_{X,2q}$. AMT18 then gives

$$\begin{aligned} \max\{0, \underline{U}_1 - C_K^-, \underline{U}_1 + \delta_k^\sigma - C_B^+\} &\leq F_K^{\text{ar}} \\ &\leq \overline{U}_1 + \min\{C_K^+, \delta_k^\sigma + C_B^-\}. \end{aligned} \tag{AKS36}$$

One may replace $d_{E,N}$ in the caps by the original full-source $\Xi_{k,N}$ or its proved finite majorant. The scalar $\delta_k^\sigma = \mathcal{B}_k^{\text{ar}} - \mathcal{B}_k^\sigma$ retains exactly AMT21:

$$\begin{aligned} \max\{-qE_k^+, -\Xi_{k,q-1} - \Xi_{k,q}\} &\leq \delta_k^\sigma \\ &\leq \min\{qE_k^-, \Xi_{k,2q-1} + \Xi_{k,2q}\}. \end{aligned} \quad (\text{AKS37})$$

Since $E_k^\pm = O_h(k + \log q)$ by the exact AMT22 expression, AKS35 and AKS32 prove the arithmetic version of AKS6, with an explicit $O_h(k^2)$ upper remainder. Nonnegativity follows from the actual nested source minima, as in AMT15, not from comparison with a possibly negative lower endpoint.

74.2 The original Gamma multiplier, not a replacement higher source

Retain $t_j = 2j + 1/2$, $f_l(S) = \prod_{j=0}^{l-1} (S - c - t_j)$ and

$$dm_{0,k} = \beta_k |f_l(c + iy)|^2 d\sigma, \quad \beta_k = \frac{(2\pi)^{k/2}}{\sqrt{2}\Gamma(k/2)}, \quad T_l = \prod_{j=0}^{l-1} t_j^2, \quad U_{l,N} = \prod_{j=0}^{l-1} [4(N + j + 1)^2 + t_j^2]. \quad (\text{AKS38})$$

The exact source map is $\mathcal{P}_N \rightarrow f_l \mathcal{P}_N \subset \mathcal{P}_{N+l}$, $P \mapsto f_l P$, with inverse division on this specified subspace. Let $\mathbf{E}_\chi$ be the complete divided-jet map and U_f its Leibniz multiplication matrix, $(U_f)_{r,a} = f_l^{(r-a)}(\lambda)/(r-a)!$ for $r \geq a$. AMT25–28 gives the precise transported observation and kernel

$$\begin{aligned} \tilde{\Lambda} &= \Lambda \mathbf{E}_\chi^{-1} U_f^{-1}, \quad \tilde{I} = U_f \mathbf{E}_\chi I, \quad \ker \tilde{\Lambda} = U_f \mathbf{E}_\chi K, \\ C_N &= \mathcal{K}_{\chi,N+l} - \mathcal{K}_{\chi,f;N+l} \mathcal{K}_{f,N+l}^{-1} \mathcal{K}_{f,\chi;N+l}, \\ H_{K,N}^{(k)} &= \beta_k \tilde{I}^* C_N^{-1} \tilde{I}. \end{aligned} \quad (\text{AKS39})$$

Each covariance is the original finite sum of full polynomial jet outer products divided by its original monic norm. These formulas retain the extra f constraints and their full cross covariance. In particular $M_f K = K$ is not assumed. AKS26 calculates the same tensor metric directly in its own orthonormal source; the two representations agree by their identical source norm and identical constrained minimum.

Define

$$w_N = \log(U_{l,N}/T_l), \quad \Omega^{\text{low}} = w_{q-1} + w_q, \quad \Omega^{\text{high}} = w_{2q-1} + w_{2q}.$$

The already proved same-coefficient comparison AMT42–44 retains all mass factors before cancellation and gives

$$-r\Omega^{\text{low}} \leq \Delta_K^\Gamma := F_K^{(1)} - F_K^{(k)} \leq r\Omega^{\text{high}}. \quad (\text{AKS40})$$

It therefore sharpens the two independent absolute bounds by

$$\begin{aligned} \max\{U_k, \underline{U}_1 - r\Omega^{\text{high}}, 0\} &\leq F_K^{(k)} \\ &\leq \min\{\bar{U}_k, \bar{U}_1 + r\Omega^{\text{low}}\}. \end{aligned} \quad (\text{AKS41})$$

The original finite estimate $0 \leq w_N \leq l \log[1 + 16(N + l)^2]$ proves that AKS40 is $o(kq)$ on the actual rank- r domain. Along with AKS35 this preserves the completed fact that all three source kernels differ by $o(kq)$. AKS4, not AKS40, is the new absolute-scale calculation.

74.3 Boundary, QGQ and arithmetic receiving identities

At each endpoint retain the original minimum section $S_N^{(s)} = (G_N^{(s)})^{-1} \Lambda^* (\Lambda(G_N^{(s)})^{-1} \Lambda^*)^{-1}$. The full Schur identity is

$$\det(I^* G_N^{(s)} I) \det Q_N^{(s)} = |\det[I, S_*]|^2 \det G_N^{(s)}, \quad Q_N^{(s)} = (\Lambda(G_N^{(s)})^{-1} \Lambda^*)^{-1}. \quad (\text{AKS42})$$

Thus the new finite absolute boundary interval is

$$\max\{0, \mathcal{B}_k^{(s)} - \overline{U}_s\} \leq F_B^{(s)} \leq \mathcal{B}_k^{(s)} - \underline{U}_s. \quad (\text{AKS43})$$

The baselines in this formula are the *complete* original BSL/HBL objects, with their full finite relation errors. No q^2 coefficient is assigned to K ; its proved coefficient there remains zero.

Write $\mathcal{H}_k = \mathcal{B}_k^\sigma - \mathcal{B}_k^0 = -\mathfrak{T}_k$. The exact full identities still read

$$\begin{aligned} F_B^{(1)} - F_B^{(k)} &= \mathcal{H}_k - \Delta_K^\Gamma, \\ \boxed{\mathcal{B}_k^{\text{ar}} = F_K^{(k)} + F_B^{(k)} + \mathcal{H}_k + \delta_k^\sigma = F_K^{(1)} + F_B^{(1)} + \delta_k^\sigma.} \end{aligned} \quad (\text{AKS44})$$

The Gamma interval is not re-estimated in this argument. For either complete QRI1 or QRI2 interval $[\omega^-, \omega^+]$ for its already calculated centre W_k , retain its exact source errors $e_0 \in [-a_0, b_0]$ and $e_q + e_{q+1} \in [-b_k^+, b_k^-]$. Then

$$\underline{\mathcal{H}}_k = -\omega^+ - b_k^- - a_0, \quad \overline{\mathcal{H}}_k = -\omega^- + b_k^+ + b_0, \quad \mathcal{H}_k \in [\underline{\mathcal{H}}_k, \overline{\mathcal{H}}_k]. \quad (\text{AKS45})$$

These are exactly QRI3, not new unspecified error choices. The complete QRI1–9, QGT and QGQ definitions remain verbatim in each updated receiver; the new interval is intersected with all their retained tighter finite intervals. In particular the simple-quartet nonzero error radius stays

$$\limsup \left| \frac{\mathcal{H}_k + lq\mathcal{C}_\Gamma}{q} + \frac{\partial_\alpha \mathcal{L}(2, \pi)}{8} - \frac{\log 2}{4} \right| \leq \frac{A_{\text{QGT}}}{\sqrt{2}}. \quad (\text{AKS46})$$

No zero-radius or $o(q)$ arithmetic assertion is introduced.

For example the complete finite absolute defect relative to the tensor boundary, with the actual Gamma and arithmetic scalars retained, is

$$\underline{U}_k + \mathcal{H}_k + \delta_k^\sigma \leq \mathcal{B}_k^{\text{ar}} - F_B^{(k)} \leq \overline{U}_k + \mathcal{H}_k + \delta_k^\sigma. \quad (\text{AKS47})$$

Inserting AKS41 makes this sharper. Replacing $\mathcal{H}_k$ by the appropriate endpoints of AKS45 and δ_k^σ by AKS37 gives its fully inherited finite scalar enclosure with the same signs. Also $F_B^{\text{ar}} = \mathcal{B}_k^\sigma + \delta_k^\sigma - F_K^{\text{ar}}$; subtract the interval AKS35 or AKS36 to get its new absolute boundary interval. This retains the exact correlations rather than taking independent hypothetical source errors.

The existing limit is $\mathcal{H}_k/(kq) \rightarrow -\mathcal{C}_\Gamma/4$ and $\delta_k^\sigma/(kq) \rightarrow 0$. Therefore the new interval implies

$$\begin{aligned} \liminf_k \inf_{|u| \geq R_*} \frac{\mathcal{B}_k^{\text{ar}} - F_B^{(k)}}{kq} &\geq -\mathcal{C}_\Gamma/4, \\ \limsup_k \sup_{|u| \geq R_*} \frac{\mathcal{B}_k^{\text{ar}} - F_B^{(k)}}{kq} &\leq \kappa_* - \mathcal{C}_\Gamma/4. \end{aligned} \quad (\text{AKS48})$$

All original branches are included in the infima and suprema. This does not decide its sign. The already proved boundary *difference* limits of AMT49 are unchanged. Every subsequential limit of the absolute normalized kernel lies in $[0, \kappa_*]$ and is shared by its three source versions when they are evaluated on the same sequence of period parameters. Existence of a single limit is not proved by boundedness.

Finally the HC residual remains exactly $\mathcal{R}_k = \mathcal{B}_k^{\text{ar}} - 4q \log(D_h k)$, with its original D_h and original nonnegative contraction, phase and trace terms. AKS44 changes none of them. The supplied positive q^2 baseline and its complete Gamma correction are not re-presented as new results here.

75 Actions, support faces, and the actual arithmetic class

The coefficient formula AKS12 supplies all four blocks of the same sum action: $P_K Y P_K$, $P_K Y (I - P_K)$, $(I - P_K) Y P_K$ and $(I - P_K) Y (I - P_K)$. There is no claim that either off-diagonal block vanishes. In the actual minimum-section coordinates the corresponding formula remains

$$[I, S_N]^{-1} Y [I, S_N] = \begin{pmatrix} H_{K,N}^{-1} I^* G_N Y I & H_{K,N}^{-1} I^* G_N Y S_N \\ \Lambda Y I & \Lambda Y S_N \end{pmatrix}. \quad (\text{AKS49})$$

Its proof is multiplication by the inverse coordinate maps $H_{K,N}^{-1} I^* G_N$ and Λ . In the invariant-ring model, the same action is the derivation $\mathcal{D}f = (\mathbf{H} \Omega \mathbf{H}^{-1} x) \cdot \nabla f$, $\Omega = \text{diag}(\gamma - i\delta, -\gamma - i\delta, \gamma + i\delta, -\gamma + i\delta)$, with $\mathcal{D}Q = 0$. For the actual conductor $\mathcal{A}_Q = \prod_{a=1}^4 Q_a$, its full defect is $\mathcal{D}\mathcal{A}_Q = \sum_{a=1}^4 (\mathcal{D}Q_a) \prod_{b \neq a, 1 \leq b \leq 4} Q_b$, exactly OQC22. Thus this coefficient realization preserves the original action leaving the observed dual.

In particular the section-corrected vector is still OPR2's

$$C_s r_j^{\ker} = e_{q+j} - S_{q+j-1}^{(s)} \Lambda e_{q+j} = I(\zeta_{q+j} - \kappa S_{q+j-1}^{(s)} \Lambda e_{q+j}),$$

with its full denominator $1 + \xi_j$ in the kernel increment. The operator proof estimates the complete covariance and its actual restriction, so none of these coupled terms is replaced by a raw fixed-section vector. The all-iterate kernel stays $\bigcap_{a=0}^{q-1} \ker(\Lambda Y^a)$; nothing in AKS4 declares it zero.

On the existing arithmetic coefficient realization retain $\mathcal{V}_{h,k} P = P(D_1 + \dots + D_k) F_h^{\otimes k}$ and $J^{(k)} \mathcal{V}_{h,k} = \eta \text{rem}_\chi$, where $\eta = U^{\otimes k} \iota$ is the original injection. AKS17 and the Newton coefficients compute precisely this same remainder. Any two polynomial lifts differ by χQ in the original relation space, so their original arithmetic cohomology classes agree. The specified primitive of the difference is EP.48, with the full differential and cochain signs proved in AKP3–5 below; it depends on the lift difference. The source comparison to w_h^{*k} is the identity on original polynomial coefficients, followed by the actual minima as in AMT8; it is not a replacement arithmetic source.

The scalar object stays $G(R) = \{\tau\} \sqcup R^\bullet$, with supported zero $0_R^\bullet$, injective amplitude/support pair, and infinite integer target. Every coefficient map above is applied on each original present support label and outer leg. It commutes with the specified coefficient inclusions because both compositions apply the same linear map to the same coefficient; a present zero is not converted into absence. No new base object, purity condition, or identification of the sum action with a Frobenius action is used. The actual distinguished class and its higher-jet directions remain in the full fibre, even when the observation kills them.

76 Exact period domain and the interior alternative

For reference the inherited radius used above is completely specified as follows; these are AMT33–34/RPC17–23, with no new constants. Put

$$\begin{aligned} \beta &= 2(\gamma^2 - \delta^2), \quad \eta = (\delta^2 + \gamma^2)^2, \quad a = v_h(\rho) \neq 0, \\ c_3 &= (a^{-2} - \bar{a}^{-2}) / (4i\delta\gamma), \quad c_1 = (a^{-2} + \bar{a}^{-2}) / 2 - (\delta^2 - \gamma^2) c_3, \\ g_j &= 5^{j/5-1} e^{\pi i j/5} \Gamma(j/5), \quad g_+ = \max_j |g_j|, \quad g_-^{-1} = \max_j |g_j|^{-1}, \quad \kappa = \max_{j < 4} |g_{j+1}/g_j|, \\ E_j^{\text{ray}} &= \frac{8}{5} \int_0^\infty r^j (r^3/3 + r) e^{-r^5/5 + r^3/3 + r} dr, \quad C_T = 2 \left(\sum_{j=0}^3 (E_j^{\text{ray}})^2 \right)^{1/2}, \\ C_K &= 4C_T g_-^{-1} + 2g_+ g_-^{-1} + 2g_+ g_-^{-2} C_T, \\ C_{K,3} &= C_K (3\kappa^2 + 3\kappa C_K + C_K^2), \quad \epsilon_* = \min\{1, (2C_T g_-^{-1})^{-1}\}, \quad B_* = \beta + \eta. \end{aligned} \quad (\text{AKS50})$$

Then

$$R_* = \begin{cases} \left[\max \left\{ 1, \frac{B_*}{\epsilon_*}, \frac{2(|c_3|C_{K,3}B_* + |c_1|(\kappa + C_K))}{|c_3g_4/g_1|} \right\} \right]^{5/2}, & c_3 \neq 0, \\ \left[\max \left\{ 1, \frac{B_*}{\epsilon_*}, \frac{2C_KB_*}{|g_2/g_1|} \right\} \right]^{5/2}, & c_3 = 0. \end{cases} \quad (\text{AKS51})$$

The actual unit implies $(c_1, c_3) \neq (0, 0)$ and all integrals converge. RPC's full period comparison proves a nonzero $(4, 1)$ or $(2, 1)$ commutator entry, respectively, for $|u| \geq R_*$ on every branch. It therefore proves the five-member orbit needed in AKS11.

Inside this radius the same coefficient determinant AKS12–14 and the operator bound AKS24 remain valid. The orbit is decided by the original noninvariant quadratic coefficients, not by the radius alone. Since Q is nondegenerate, its projective orbit has either one or five members. A one-member orbit must have character one: its character has both fourth and fifth powers equal to one, by the quadratic determinant and $D^5 = I$. In that case $Q = a'x_1x_4 + b'x_2x_3$ with $a'b' \neq 0$, and

$$b = d_{5,k,0} - d_{5,k-2,0} = \left\lfloor \frac{(k+1)^2}{5} \right\rfloor + \mathbf{1}_{k \equiv 0 \text{ or } 3 \pmod{5}}, \quad r = q - b. \quad (\text{AKS52})$$

If a noninvariant coefficient is nonzero, AKS11 applies instead. These are the complete OQC alternatives. The bound AKS31 always uses that actual r ; it does not extend the $8k - 16$ specialization to an unevaluated exceptional point. This calculation does not assert that the interior exceptional set is empty.

77 Verification scope and the remaining absolute coefficient

The new mathematical content is AKS20–32 and its application to the actual coefficient determinant AKS12–14, followed by the finite receiving improvements AKS35–48. The complete source and orbit identifications used in that application are the supplied OCS/OQC/OQX, RPC, OPG/OPR and AMT results; their exact predecessors are preserved. AKS33 pinpoints a repair obtainable immediately from the earlier proof, whereas the Newton bound supplies the sharper explicit constant.

The bound *at* scale kq is now finite and uniform. What is not proved is a matching lower coefficient, a unique limit for $F_K^{(1)}/(kq)$, or a sign for the interval AKS48. Those questions involve the orientation of the original period kernel within the inverse remainder covariance; replacing that compression by a scalar bound loses precisely that information. This is the finite-bounds outcome allowed by the current request, not a claim to have evaluated an asymptotic coefficient. The source differences are already completed inputs and are not renamed as the remaining problem.

The accompanying exact checks distinguish finite algebraic diagnostics from the analytic proof for the full family. They do not assert that a diagnostic root is a zero of ξ , that a diagnostic matrix equals an actual period matrix, or that a new Lean theorem has been checked. No RH conclusion is drawn from this kernel bound.

Original rank-three primitive connecting maps and endpoint returns

78 The original three-equation transfer of every connecting return

Fix the simple quartet, its actual amplitude, period and logarithm branch on the original five-orbit locus. Keep $k = 4l + 1 \geq 9$, $j = k + 4$, $q = (k + 1)^2$, $Q = (j + 1)^2$, $d = Q - q = 8k + 24$, and the full nonreduced conductor

$$E_A(z) = \sum_{a,b=0}^8 a_{ab} e^{\beta_{s,a,b} z}, \quad v = \text{ord}_0 E_A, \quad \mu_v = E_A^{(v)}(0) \neq 0, \quad \alpha_A = v!/\mu_v. \quad (\text{PRT1})$$

No conductor factor is removed. Write $B = B_{\text{cof}}$, $\mathcal{U} = \mathbb{C}^{4m_j}$, $\mathbb{J} = \ker \mathbb{B}$, $\mathcal{B} = \text{im } B$ and $W = \mathcal{B} + \mathbb{J}$, as in CPQ2. Define, in this same coefficient space,

$$S_e = \text{im}[B, F_e], \quad K_e = S_e \cap W, \quad T_e = S_e + W \quad (e = 0, 1). \quad (\text{PRT2})$$

These are the CPQ spaces, with S_e abbreviated only to make the two endpoint subscripts legible. The later norm cutoff N is independent of the defining endpoint e . At every $Q - 1 \leq N \leq 2Q$ use exactly

$$D = N - v, \quad \mathcal{D}_N^{(s)} = \text{diag}(D_{j,D}^{(s)}, 4), \quad c'_j = j/2 - 4, \quad h_{n,s} = (2\pi)^{s/2} n! (s/2)_n, \quad s = 1 \text{ or } j. \quad (\text{PRT3})$$

Thus both actual source pairs $(1, 1)$ and (k, j) are covered in their target order. Their degree- k proper source retains its separate form; none of the identities below substitutes that form for A_{inv} .

78.1 The actual three rows in the original period coefficients

Retain the original polynomials $L_i = (\mathbf{H}(ZW, Z, W, 1)^\top)_i$, $b_{i,a,b} = [Z^a W^b] L_i^4$, the exponential $B_i(z)$ and the translation maps $\mathcal{T}_i, \mathcal{T}_A$ of CPR2. In particular

$$R_i(S) = \frac{\sum_{a,b=0}^4 b_{i,a,b} \chi_j(S + \beta_{4,a,b})}{\chi_k(S)}, \quad b_i = B_i(0), \quad \chi_n(S) = \prod_{a,b=0}^n (S - \beta_{n,a,b}). \quad (\text{PRT4})$$

The quotient is the complete polynomial quotient: root addition makes every numerator vanish at every distinct root of χ_k . Choose the first original index i_* with $b_{i_*} \neq 0$, and put

$$C_i = R_i - (b_i/b_{i_*}) R_{i_*} \quad (i \neq i_*), \quad V = ([S^a] C_i)_{0 \leq a < d, i \neq i_*}. \quad (\text{PRT5})$$

Existence of i_* and independence of these three columns are the complete PDS10–11/CPR17 calculation: the original Bezout identity at zero gives a nonzero b_i ; the four R_i are independent; their leading coefficients are b_i . Subtracting their exact leading multiples therefore gives three independent columns of degree at most $d - 1$. No period coefficient is replaced in this operation.

Let h_b be the original primitive germs used to define the d columns of F_0 in CIT7. They are specified modulo the entire raw analytic source. Define their relation-value matrix by

$$(M_0)_{a,b} = \Lambda_{h_b}(\mathcal{T}_A(\chi_k S^a)), \quad 0 \leq a < d, \quad L = V^\top M_0. \quad (\text{PRT6})$$

Every entry uses the ordinary, complex-linear transpose. The complete annihilator identity CPR9–10 proves that the displayed d independent relation polynomials are the full dual of the raw source quotient. The h_b are a basis of that quotient, so M_0 is invertible. Thus L has rank exactly three. The matrix is computable from the original period arrays by finite polynomial division and pairing; it contains neither a new numerical model nor a selected nonzero period minor.

Choose once a full column basis Z of $\ker L$ and a three-column complement Y so that $[Z, Y]$ is invertible. Then the literal coefficient spaces obey

$$\boxed{S_1 = \mathcal{B} + F_0 \ker L \subset S_0, \quad S_0/S_1 \text{ has the three lifts } F_0 Y.} \quad (\text{PRT7})$$

Here equality is of subspaces of the original $\mathcal{U}$, not only of abstract quotient dimensions. To prove it, use PDS11: truncation of the endpoint-one primitive quotient identifies it with exactly $\ker V^\top$ in endpoint-zero relation-value coordinates. The missing top value is uniquely $-u_*^\top x_0/b_{i_*}$ with the complete u_* from PDS10. For a representative h , set $f = E_A h$. The original target numerator in component i is

$$N_i(f) = \mathbb{V}_j^{-1}(r![z^r](B_i f))_{0 \leq r < Q}, \quad (\mathbb{V}_j)_{r,(a,b)} = \beta_{j,a,b}^r. \quad (\text{PRT8})$$

Those values use only the truncation through degree $Q - 1$. At endpoint one, the additional four equations assert exactly that the same interpolants also have the required next jet; they do not alter a coefficient in PRT8. Division by the complete E_A gives the same target analytic functions, modulo precisely the original lower exponentials. An analytic source change has numerator d and changes the four target numerators by $(L_i^4 d)_i$, hence changes their DNE coordinates by Bx with the full original scalar α_A . This is the coefficient square RRC5–9/CIT8. It proves both inclusions in PRT7, including every analytic-source and lower-gauge change. Finally F_0 induces an injection modulo $\mathcal{B}$ by CIT9; the invertible matrix $[Z, Y]$ therefore proves the asserted dimension and the stated three lifts.

The coefficient change can itself be written explicitly. For the original endpoint-one primitive columns $h_{1,b}$ put $(M_1)_{a,b} = \Lambda_{h_{1,b}}(\mathcal{T}_A(\chi_k S^a))$ for $0 \leq a \leq d$, and let $P_{<d}$ select the first d rows. Then

$$\Gamma = M_0^{-1} P_{<d} M_1, \quad \text{im } \Gamma = \ker L, \quad F_1 = F_0 \Gamma + B H \quad (\text{PRT7a})$$

for the unique matrix H with the displayed equality. Indeed PDS11 proves the first two assertions, including that Γ has rank $d - 3$. The preceding numerator interpolation calculation shows that each difference column lies in $\mathcal{B}$. The original injectivity of B gives its unique coordinate in H . Equivalently $\Gamma = ZT$ for an invertible $(d - 3)$ -square matrix T , so every original fixed frame change in PRT7 is specified by a literal coefficient equation. No conjugation occurs in M_0, M_1, L, Γ . Hermitian adjoints enter only when evaluating the actual metric below.

78.2 An exact sequence of these three directions

PRT7 gives $K_1 \subset K_0$ and $T_1 \subset T_0$. The identity on coefficient representatives gives the exact sequence

$$0 \longrightarrow K_0/K_1 \xrightarrow{[x] \mapsto [x]} S_0/S_1 \xrightarrow{[x] \mapsto [x]} T_0/T_1 \longrightarrow 0. \quad (\text{PRT9})$$

The first map is injective because $K_0 \cap S_1 = K_1$. If $x \in S_0$ maps to zero on the right, write $x = s + w$ with $s \in S_1$, $w \in W$; then $w = x - s \in K_0$, so its class is the image of K_0/K_1 . Conversely every such class maps to zero. Every element of $T_0 = S_0 + W$ has the class of its S_0 summand modulo $T_1 = S_1 + W$, proving surjectivity. Let

$$a = \dim(T_0/T_1) = \dim(\mathcal{G}_k \cap \mathcal{C}_k) \quad \text{from CPR17–18.} \quad \dim(K_0/K_1) = 3 - a, \quad 0 \leq a \leq 3. \quad (\text{PRT10})$$

The equality with CPR's actual polynomial intersection follows because $\dim(T_e/W) = c_e$ and CPR18 proves $c_0 - c_1 = a$. It can also be calculated directly in the three columns $F_0 Y$: take their images modulo T_1 , retaining the full $\mathbb{B}$, B and L of PRT2/PRT6. Their exact rank is a and their kernel has dimension $3 - a$. This proves a complete original-data rank computation without assuming a generic minor is nonzero. Rank strata, including $a = 0$, are retained.

Give the three terms of PRT9 their original attained metrics

$$A_N = \text{Gram}_{\mathcal{D}_N}(S_0/S_1), \quad B_N = \text{Gram}_{\mathcal{D}_N}(K_0/K_1), \quad C_N = \text{Gram}_{\mathcal{D}_N}(T_0/T_1). \quad (\text{PRT11})$$

Every value frame is fixed in N and in the source-order comparison. For example if X_1 is any fixed independent frame of S_1 , the first one, in the exact three lifts $X_A = F_0 Y$, is

$$A_N = X_A^* \mathcal{D}_N X_A - X_A^* \mathcal{D}_N X_1 (X_1^* \mathcal{D}_N X_1)^{-1} X_1^* \mathcal{D}_N X_A. \quad (\text{PRT12})$$

Select the first independent images of F_0Y modulo T_1 to obtain the a lifts X_C . A full basis of $\ker(F_0Y \bmod T_1)$, followed by subtracting an S_1 representative as in PRT9, gives $3 - a$ lifts X_B in K_0 . Applying PRT12 with (X_1, X_A) replaced by the frames of (K_1, X_B) and (T_1, X_C) gives B_N and C_N . All three are positive definite on their stated quotient values; the determinant of an empty quotient is one.

For completeness every fixed frame factor can be retained before taking a return. For a fixed inclusion $K \subset Y_0$, choose frames P_K, P_{Y_0} and quotient lifts X . Put

$$[P_K, X] = P_{Y_0} R_{Y_0/K}, \quad d_{Y_0/K} = |\det R_{Y_0/K}|^2, \quad g_{Y_0}(N) = \det(P_{Y_0}^* \mathcal{D}_N P_{Y_0}). \quad (\text{PRT13})$$

This is the standard positive-block Schur calculation [35, §2.1, (2.3)–(2.4)]; the determinant factorization follows from the same block elimination [38, p. 222, (6a)–(6b)]. Here the block frame is exactly $[P_K, X]$, with its retained factor $d_{Y_0/K}$; neither reference supplies the programme's connecting sequence or its endpoint comparison.

The full block Schur determinant identity is

$$\det \text{Gram}_{\mathcal{D}_N}(Y_0/K) = d_{Y_0/K} g_{Y_0}(N) / g_K(N). \quad (\text{PRT14})$$

It follows by expressing the restricted form in $[P_K, X]$, taking its determinant, and eliminating its upper-left block. This defines and evaluates every factor from the stated original frames. It does not identify a varying minimizing representative with a fixed lift. CPQ's frames for each quotient are used in PRT13 when that quotient appears below.

78.3 Evaluation of the difference of the actual CPQ returns

Use the unchanged target return $\mathcal{R}f = -f(Q-1) - f(Q) + f(2Q-1) + f(2Q)$ and write

$$U_A = \mathcal{R} \log \det A_N, \quad U_B = \mathcal{R} \log \det B_N, \quad U_C = \mathcal{R} \log \det C_N. \quad (\text{PRT15})$$

These symbols are returns of precisely the three evaluated quotient matrices PRT11–12. They are not freely assignable parameters. The exact CPQ differences are

$$\begin{aligned} V_{\text{conn},0} - V_{\text{conn},1} &= U_A, & V_{h,0} - V_{h,1} &= U_B, \\ V_{s,0} - V_{s,1} &= U_A - U_B, & V_{t,0} - V_{t,1} &= U_C, \\ V_{\text{rem},1} - V_{\text{rem},0} &= U_C, & \mathcal{L}_0 - \mathcal{L}_1 &= U_C - U_A + U_B. \end{aligned} \quad (\text{PRT16})$$

Here V_b, V_I, V_A are independent of the defining primitive endpoint e : their spaces, coefficient maps and frames are the same CPQ3/9 objects. To prove every line, apply PRT14 to $S_e/\mathcal{B}$, $K_e/\mathcal{B}$, S_e/K_e , T_e/W and $\mathcal{U}/T_e$, respectively. For example the logarithmic determinant difference of the latter two endpoint quotients is $\log g_{T_0} - \log g_{T_1}$ plus the explicitly defined fixed frame-factor difference; PRT14 for T_0/T_1 identifies its nonconstant term with $\log \det C_N$. Each frame factor is independent of N , and $\mathcal{R}1 = -1 - 1 + 1 + 1 = 0$, so it cancels only after this full return. The other four differences follow by the same stated determinant identities. Finally substitute them into CPQ10's $\mathcal{L}_e = V_I + V_{t,e} - V_{s,e} - V_A$.

This also gives the exact propagation, without deleting a complement,

$$\begin{aligned} Z_e &= V_b + V_{\text{rem},e} + V_{s,e} + V_A, & Z_1 - Z_0 &= U_C - U_A + U_B, \\ \mathcal{P}_0^{\text{rest}} - \mathcal{P}_1^{\text{rest}} &= U_C - U_A + U_B, & 4S_{K,j} &= \mathcal{L}_0 + Z_0 = \mathcal{L}_1 + Z_1. \end{aligned} \quad (\text{PRT17})$$

In particular a change of primitive endpoint changes the common loss and complementary sum by exactly opposite amounts in their allocation identity. The original total and the complete-action allocation cancellation are preserved.

78.4 Directed finite bounds with the original masses

Use the proved original reconstruction/source constant of FRE1, evaluated at degree j and order s :

$$A_{j,s}^{\text{end}} = \max\{1, Q\mathcal{B}_{2Q-v,s,j,A}^2\}(M_{j,s}^{\text{rec}})^2. \quad (\text{PRT18})$$

The two factors retain DNE13's full mass $(2\pi)^{s/2}$, the full conductor scalar and division coefficients, and LJR16's original fixed-disk reconstruction, phases and Vandermonde. FRE2-4 proves on this exact metric

$$\mathcal{D}_{Q-1}^{(s)} \preceq \mathcal{D}_N^{(s)} \preceq A_{j,s}^{\text{end}} \mathcal{D}_{Q-1}^{(s)}, \quad \log A_{j,s}^{\text{end}} \leq 2Q \log j + O_{\text{actual}}(Q). \quad (\text{PRT19})$$

Substitution in each full affine fibre defining PRT11 and subsequent minimization preserve both inequalities. Original row extension also proves that each of A_N, B_N, C_N increases with N . If G_N is one of these forms of rank r , set $f(N) = \log \det G_N - \log \det G_{Q-1}$. The generalized eigenvalues in PRT19 show $0 \leq f(N) \leq r \log A_{j,s}^{\text{end}}$ and f is increasing. Hence $\mathcal{R} \log \det G = f(2Q-1) + f(2Q) - f(Q)$ is nonnegative and at most $2r \log A_{j,s}^{\text{end}}$. This proves the actual directed intervals

$$\begin{aligned} 0 \leq U_A \leq 6 \log A_{j,s}^{\text{end}}, \quad 0 \leq U_B \leq 2(3-a) \log A_{j,s}^{\text{end}}, \quad 0 \leq U_C \leq 2a \log A_{j,s}^{\text{end}}, \\ \boxed{0 \leq V_{\text{rem},1} - V_{\text{rem},0} \leq 2a \log A_{j,s}^{\text{end}}.} \\ -2(3-a) \log A_{j,s}^{\text{end}} \leq V_{s,0} - V_{s,1} \leq 6 \log A_{j,s}^{\text{end}}, \\ |\mathcal{L}_0 - \mathcal{L}_1| \leq 6 \log A_{j,s}^{\text{end}}, \quad |Z_1 - Z_0| \leq 6 \log A_{j,s}^{\text{end}}. \end{aligned} \quad (\text{PRT20})$$

The last bound uses $\dim B + \dim C = 3$, keeping the exact sequence rather than separately charging two unrelated rank-three errors. These are finite bounds on actual CPQ expressions. For each fixed original period they are $O_{\text{actual}}(Q \log j) = o(kq)$, for both source pairs. At the actual rank stratum $a = 0$, the remaining-target forms are the same quotient in fixed value frames and their returns are exactly equal. At $a = 3$, the kernel spaces coincide and $U_B = 0$ exactly. All intermediate strata remain governed by their actual original three-column matrix. No claim is made that either endpoint's common leading return has now been evaluated; PRT16 evaluates their difference and reduces its calculation to the stated three original directions.

The proof requires the five-orbit coefficient construction, its original distinct quartet roots and $\mu_v \neq 0$. It does not use the additional pole-coordinate gate minors of RPI/RPK. Consequently a period excluded by a pole-coordinate pivot remains covered when it belongs to the original five-orbit locus: PRT6/8/12 construct all maps without that pivot. Rank changes of the three-row image are handled by PRT10. The value v is retained throughout; neither $v \leq 32$ nor a generic value of v has been substituted for an actual coefficient.

Complete boundary-tilt and Binet envelope proof

79 Actual boundary tilts and uniform convolution tails

This calculation retains the original arithmetic density and the complete hypothetical actual quartet. It does not replace that density with a reference measure. The estimates do not assume RH or a pointwise lower bound for zeta.

79.1 1. Objects and the exact retained expression

Fix $0 < \delta < 1/2$, $\gamma > 0$, and an integer $m \geq 1$, and suppose the selected full-order quartet is a quartet of zeros of the original $g = 2\xi$. Its monic polynomial is

$$h(s) = \prod_{\epsilon, \eta \in \{-1, 1\}} (s - \tfrac{1}{2} - \epsilon\delta - i\eta\gamma)^m, \quad \deg h = 4m. \quad (\text{BT1})$$

The entire function g/h , its original Taylor unit, source maps, and measure remain unchanged:

$$w_h(t) = \frac{|(g/h)(1/2 + it)|^2}{2\pi}, \quad \alpha = \frac{\pi}{2}, \quad R_0 = \sqrt{\delta^2 + \gamma^2}. \quad (\text{BT2})$$

The original completion formula gives, exactly,

$$\begin{aligned} w_h(t) &= \frac{(t^2 + 1/4)^2}{\sqrt{\pi}} \frac{|\zeta(1/2 + it)|^2}{|h(1/2 + it)|^2} \sigma(t), \\ \sigma(t) &= \frac{|\Gamma(1/4 + it/2)|^2}{2\pi}, \\ |h(1/2 + it)|^2 &= [((t - \gamma)^2 + \delta^2)((t + \gamma)^2 + \delta^2)]^{2m}. \end{aligned} \quad (\text{BT3})$$

The two factors with each fixed imaginary sign contribute a minus sign before multiplication, so their product is the positive expression in the last line. Every denominator in (BT3) is positive. Conjugation of the original g and the real coefficients of h give $w_h(-t) = w_h(t)$.

79.2 2. A centered partial-sum bound from the original floor formula

For every integer $N \geq 1$, initially for $\Re s > 1$,

$$\zeta(s) = \sum_{n=1}^N n^{-s} + \frac{N^{1-s}}{s-1} - s \int_N^\infty \{x\} x^{-s-1} dx. \quad (\text{BT4})$$

To verify (BT4), integrate the constant value $\lfloor x \rfloor$ on each interval $[n, n+1)$, or telescope integration by parts of the absolutely convergent tail. The last integral is absolutely and locally uniformly convergent for $\Re s > 0$, so the identity continues to that half-plane with $s \neq 1$.

Let $B_1(x) = \{x\} - 1/2$, so $|B_1(x)| \leq 1/2$. The identity $\int_N^\infty x^{-s-1} dx = N^{-s}/s$ gives exactly

$$\zeta(s) = \sum_{n=1}^{N-1} n^{-s} + \frac{1}{2} N^{-s} + \frac{N^{1-s}}{s-1} - s \int_N^\infty B_1(x) x^{-s-1} dx. \quad (\text{BT5})$$

No asymptotic remainder is substituted here.

On $s = 1/2 + it$, put $\varrho = \sqrt{t^2 + 1/4}$, which is both $|s|$ and $|s - 1|$. For $N \geq 2$,

$$\sum_{n=1}^{N-1} n^{-1/2} + \tfrac{1}{2} N^{-1/2} \leq 2\sqrt{N-1} - 1 + \frac{1}{2\sqrt{N}} \leq 2\sqrt{N} - 1.$$

The first inequality follows from the integral bound for the decreasing function $x^{-1/2}$. For the second, $2(\sqrt{N} - \sqrt{N-1}) = 2/(\sqrt{N} + \sqrt{N-1}) \geq 1/\sqrt{N}$. At $N = 1$, the same asserted inequality holds directly, with left side $1/2$. The absolute integral in (BT5), before its multiplier s , is at most $\frac{1}{2} \int_N^\infty x^{-3/2} dx = N^{-1/2}$. Therefore

$$|\zeta(1/2 + it)| \leq 2\sqrt{N} - 1 + \frac{\sqrt{N}}{\varrho} + \frac{\varrho}{\sqrt{N}} \quad (t \in \mathbb{R}, N \geq 1). \quad (\text{BT6})$$

The infimum over these explicitly specified integer choices is also a bound.

For $x = |t| \geq 1$, choose $N = \lfloor x \rfloor$. Then

$$x/2 \leq N \leq x, \quad \varrho \leq \frac{\sqrt{5}}{2}x, \quad \frac{\sqrt{N}}{\varrho} \leq x^{-1/2}.$$

Since $-1 + x^{-1/2} \leq 0$, (BT6) yields

$$\boxed{|\zeta(1/2 + it)| \leq Z_* \sqrt{|t|} \quad (|t| \geq 1), \quad Z_* = 2 + \sqrt{5/2}.} \quad (\text{BT7})$$

This improves the polynomial exponent in the uncentered estimate used in TG.15. It is not a claim that (BT7) is the strongest known zeta bound. On the whole real line the original $N = 1$ uncentered formula gives independently

$$|\zeta(1/2 + it)| \leq 1 + 2\sqrt{t^2 + 1/4}. \quad (\text{BT8})$$

79.3 3. Explicit two-sided Gamma bounds

The complete Binet calculation included below proves

$$\boxed{c_\Gamma e^{-\alpha|t|}(1 + |t|)^{-1/2} \leq \sigma(t) \leq C_\Gamma e^{-\alpha|t|}(1 + |t|)^{-1/2} \quad (t \in \mathbb{R}),} \quad (\text{BT9})$$

where

$$c_\Gamma = \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}. \quad (\text{BT10})$$

For completeness, its exact starting identity for $\Re z > 0$ is

$$\log \Gamma z = (z - \tfrac{1}{2}) \log z - z + \tfrac{1}{2} \log(2\pi) + R(z), \quad R(z) = \int_0^\infty e^{-zu} \left(\frac{1}{e^u - 1} - \frac{1}{u} + \frac{1}{2} \right) \frac{du}{u}.$$

The logarithmic branch is the one real on the positive real axis. The kernel is $(\coth x - 1/x)/(4x)$, with $x = u/2 > 0$. Define

$$F(x) = x \cosh x - \sinh x, \quad G(x) = (x^2 + 3) \sinh x - 3x \cosh x.$$

Both vanish at zero, and $F'(x) = x \sinh x \geq 0$, $G'(x) = xF(x) \geq 0$. Dividing their nonnegative values by the appropriate positive factors gives $0 \leq \coth x - 1/x \leq x/3$. Thus the kernel lies between zero and $1/12$. At $z = 1/4 + it/2$, integration of $e^{-u/4}/12$ proves $|R(z)| \leq 1/3$.

Twice the real part of Binet's identity, with $x = |t|$, gives exactly

$$\sigma(t) = |z|^{-1/2} \exp[-x \arctan(2x) - 1/2 + 2\Re R(z)].$$

For $x > 0$,

$$-x \arctan(2x) - \tfrac{1}{2} = -\alpha x + x \arctan(1/(2x)) - \tfrac{1}{2} \leq -\alpha x,$$

because $\arctan v \leq v$ for $v \geq 0$. At $x = 0$, the same inequality is $-1/2 \leq 0$. The lower bound uses $\arctan(2x) \leq \pi/2$. Hence the full exponent is between $-\alpha x - 7/6$ and $-\alpha x + 2/3$, including zero.

Finally $|z|^2 = 1/16 + x^2/4$, and the exact squared differences

$$\frac{(1+x)^2}{4} - |z|^2 = \frac{3+8x}{16}, \quad |z|^2 - \frac{(1+x)^2}{20} = \frac{(4x-1)^2}{80}$$

give

$$\sqrt{2}(1+x)^{-1/2} \leq |z|^{-1/2} \leq \sqrt{2\sqrt{5}}(1+x)^{-1/2}.$$

These prove (BT9)–(BT10) with no unproved Stirling remainder and no unquantified compact Gamma minimum.

79.4 4. A global explicit actual boundary-tilt bound

Set

$$p = 8m - \frac{9}{2}, \quad \beta = p - \frac{1}{2} = 8m - 5, \quad T_h = \max(1, 2R_0). \quad (\text{BT11})$$

Then $p \geq 7/2 > 1$ and $\beta > 0$. Define

$$\begin{aligned} C_{\text{comp}} &= \frac{C_\Gamma}{\sqrt{\pi}} (T_h^2 + 1/4)^2 [1 + 2\sqrt{T_h^2 + 1/4}]^2 \delta^{-8m} (1 + T_h)^\beta, \\ C_{\text{tail}} &= \frac{25C_\Gamma Z_*^2}{16\sqrt{\pi}} \left(\frac{4}{3}\right)^{4m} \left(1 + \frac{1}{T_h}\right)^\beta, \\ C_h^{\text{tilt}} &= \max(C_{\text{comp}}, C_{\text{tail}}). \end{aligned} \quad (\text{BT12})$$

These constants are explicit in the fixed quartet and elementary constants. None is an unevaluated zeta supremum, and none changes the source mass.

For $|t| \leq T_h$, (BT3) has $|h(1/2 + it)|^2 \geq \delta^{8m}$. Apply (BT8) and the upper inequality of (BT9). Multiplying by $e^{\alpha|t|}(1 + |t|)^p$ leaves the factor $(1 + |t|)^\beta$; bounding $|t|$ by T_h proves the constant C_{comp} .

For $x = |t| \geq T_h$, retain the exact identity

$$((t - \gamma)^2 + \delta^2)((t + \gamma)^2 + \delta^2) = (t^2 - R_0^2)^2 + 4\delta^2 t^2 \geq \frac{9}{16} t^4. \quad (\text{BT13})$$

The last inequality uses $t^2 \geq 4R_0^2$. Thus

$$|h(1/2 + it)|^2 \geq (3/4)^{4m} x^{8m}, \quad (t^2 + 1/4)^2 \leq (25/16)x^4.$$

Applying (BT7) and (BT9) to (BT3) gives

$$e^{\alpha x} w_h(t) \leq \frac{25C_\Gamma Z_*^2}{16\sqrt{\pi}} \left(\frac{4}{3}\right)^{4m} x^{5-8m} (1 + x)^{-1/2}.$$

Multiplication by $(1 + x)^p$ leaves $((1 + x)/x)^\beta \leq (1 + 1/T_h)^\beta$. This proves the global bound

$$\boxed{0 \leq e^{\alpha|t|} w_h(t) \leq C_h^{\text{tilt}} (1 + |t|)^{-p} \quad (t \in \mathbb{R}).} \quad (\text{BT14})$$

The exact expression (BT3), including its actual zeta numerator and complete quartet denominator, is the object bounded throughout.

79.5 5. Boundary moments and finite derivative orders

Define the original two boundary tilts and their actual masses:

$$f_\pm(t) = e^{\pm\alpha t} w_h(t), \quad M_\pm = \int_{\mathbb{R}} f_\pm(t) dt. \quad (\text{BT15})$$

The nonzero entire function g/h has only isolated zeros on the integration line, so both masses are positive. Evenness gives $M_+ = M_- =: M_\alpha$. Integration of (BT14) proves

$$\boxed{0 < M_\alpha = M_h(\alpha) = M_h(-\alpha) \leq \frac{2C_h^{\text{tilt}}}{p-1} < \infty.} \quad (\text{BT16})$$

For every real $j \geq 0$ with $j < p - 1$, the inequality $|t|^j \leq (1 + |t|)^j$ gives

$$\int_{\mathbb{R}} |t|^j e^{\pm\alpha t} w_h(t) dt \leq \frac{2C_h^{\text{tilt}}}{p-j-1}. \quad (\text{BT17})$$

More precisely, for $R \geq 0$, the part $|t| \geq R$ is at most

$$\frac{2C_h^{\text{tilt}}}{p-j-1}(1+R)^{j-p+1}. \quad (\text{BT18})$$

Thus every integer order $0 \leq j \leq 8m-6$ is absolutely integrable at both boundary tilts. For a simple quartet, this includes orders zero, one and two.

For the actual Laplace function $M_h(z) = \int e^{zt} w_h(t) dt$, differentiation in $|\Re z| < \alpha$ gives $\int t^j e^{zt} w_h(t) dt$. For each stated order, (BT14) dominates the absolute integrand uniformly on $|\Re z| \leq \alpha$ by the integrable function $C_h^{\text{tilt}} |t|^j (1+|t|)^{-p}$. Dominated convergence therefore extends those derivatives continuously to the closed strip with exactly the boundary moments above. This does not assert holomorphic continuation through the boundary or finiteness of every endpoint derivative.

79.6 6. A uniform convolution envelope with the literal tilted mass

If a nonnegative integrable function f has actual mass $M > 0$ and $f(x) \leq C(1+|x|)^{-p}$, then for every integer $k \geq 1$,

$$f^{*k}(u) \leq kCM^{k-1}(1+|u|/k)^{-p}. \quad (\text{BT19})$$

For $k=1$ this is the hypothesis. In the convolution integral with $x_1 + \dots + x_k = u$, at least one index has $|x_i| \geq |u|/k$. Bound its integrand by the sum of the k restrictions to these events. On the event for index i , the factor $f(x_i)$ is at most $C(1+|u|/k)^{-p}$. Parametrize the convolution hyperplane by the other $k-1$ coordinates; the change from any such convolution parametrization to another has absolute determinant one. Dropping the event now leaves the full product integral M^{k-1} . Summing proves (BT19), without a probability normalization.

Multiplication inside the original convolution gives exactly

$$f_{\pm}^{*k}(u) = e^{\pm \alpha u} m_{h,k}(u), \quad m_{h,k} = w_h^{*k}. \quad (\text{BT20})$$

Use the plus tilt for $u \geq 0$, the minus tilt for $u < 0$, and their same actual mass M_{α} . The result is

$$m_{h,k}(u) \leq kC_h^{\text{tilt}} M_{\alpha}^{k-1} e^{-\alpha|u|} (1+|u|/k)^{-p} \quad (k \geq 1, u \in \mathbb{R}). \quad (\text{BT21})$$

All dependence on k is displayed. The constants $C_h^{\text{tilt}}, p, M_{\alpha}$ belong to fixed h .

For every integer $n \geq 0$ with $2n > p-1$, use $(1+u/k)^{-p} \leq k^p u^{-p}$ for $u > 0$. The Gamma integral gives

$$\int_{\mathbb{R}} u^{2n} m_{h,k}(u) du \leq 2C_h^{\text{tilt}} k^{p+1} M_{\alpha}^{k-1} \frac{\Gamma(2n-p+1)}{\alpha^{2n-p+1}}. \quad (\text{BT22})$$

The condition $2n > p-1$ is precisely what makes this comparison integrable at zero. The actual raw moments exist also when the condition fails, but (BT22) is not used there.

For the retained source comparison, one direct corollary is

$$m_{h,k}(u) \leq b_k \sigma(u), \quad b_k = \frac{C_h^{\text{tilt}}}{c_{\Gamma}} k^{p+1} M_{\alpha}^{k-1} > 0. \quad (\text{BT23})$$

Indeed $1+|u|/k \geq (1+|u|)/k$; (BT21) and the lower Gamma bound leave the additional factor $(1+|u|)^{-p+1/2} \leq 1$. This is a proved identity-map norm comparison; it does not assert equality of the measures.

80 Internal derivation of the exact Binet identity

This note proves the exact identity used in the actual boundary-tilt calculation from the defining Gamma integral, elementary integration, locally uniform products, and the Beta substitution. No Stirling formula, Gamma asymptotic, or cited Binet identity is assumed.

All complex powers of positive real integration variables use their real logarithms. The notation $\text{Log } z$ on $\Re z > 0$ denotes the principal holomorphic logarithm. The holomorphic logarithm of Gamma will be constructed below and will be real on the positive real axis.

80.1 1. Defining integrals and the finite Euler formula

For $\Re z > 0$, define

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx. \quad (\text{BI1})$$

This is absolutely convergent. On a compact set with $0 < a \leq \Re z \leq b$, its integrand and each of its first finitely many derivatives in z are bounded in absolute value by

$$(x^{a-1} + x^{b-1})(1 + |\log x|^j)e^{-x},$$

with the appropriate fixed integer j . This function is integrable at zero and infinity. Differentiation under the integral is therefore valid, and Γ is holomorphic on the right half-plane. Integration by parts gives

$$\Gamma(z+1) = z\Gamma(z), \quad \Gamma(1) = 1, \quad (\text{BI2})$$

because $x^z e^{-x}$ tends to zero at both endpoints for $\Re z > 0$.

For $\Re z, \Re w > 0$, let

$$B(z, w) = \int_0^1 t^{z-1} (1-t)^{w-1} dt. \quad (\text{BI3})$$

For an integer $N \geq 1$, integration by parts gives

$$B(z, N+1) = \frac{N}{z} B(z+1, N).$$

Iterating, and using $B(z+N, 1) = 1/(z+N)$, proves the exact finite formula

$$B(z, N+1) = \frac{N!}{z(z+1) \cdots (z+N)}. \quad (\text{BI4})$$

In particular none of these denominators vanishes in the right half-plane.

Set

$$E_N(z) = N^z B(z, N+1) = \frac{N^z N!}{z(z+1) \cdots (z+N)}, \quad N \geq 1. \quad (\text{BI5})$$

The substitution $t = x/N$ gives

$$E_N(z) = \int_0^N x^{z-1} (1-x/N)^N dx. \quad (\text{BI6})$$

For $0 \leq x < N$, $\log(1-x/N) \leq -x/N$, so

$$0 \leq (1-x/N)^N \leq e^{-x}.$$

For each fixed $x > 0$, the expression on the left tends to e^{-x} , with the indicator of $x < N$ understood.

On a compact set as above, therefore

$$\begin{aligned} \sup_z |E_N(z) - \Gamma(z)| &\leq \int_0^\infty (x^{a-1} + x^{b-1}) \\ &\quad \cdot |(1-x/N)^N \mathbf{1}_{\{x < N\}} - e^{-x}| dx \longrightarrow 0 \end{aligned}$$

by dominated convergence. The same argument with an additional factor $|\log x|^j$, for $j = 1, 2$, proves locally uniform convergence of those two derivatives as well. Thus the Euler limit and the derivative limits are established directly from their integrals.

80.2 2. Nonvanishing and the holomorphic Gamma logarithm

Write $H_N = \sum_{j=1}^N 1/j$. The real numbers

$$\gamma_N = H_N - \log N$$

decrease and are bounded below by zero. Indeed

$$\gamma_{N+1} - \gamma_N = \frac{1}{N+1} - \log(1 + 1/N) < 0$$

by integrating $1/x > 1/(N+1)$ on $[N, N+1)$, while $H_N \geq \int_1^{N+1} dx/x = \log(N+1)$. Let $\gamma_E = \lim_N \gamma_N$.

For $\Re z > 0$, every $1 + z/j$ has positive real part. Its principal logarithm is holomorphic. Define

$$Q_N(z) = \text{Log } z + z(H_N - \log N) + \sum_{j=1}^N (\text{Log}(1 + z/j) - z/j). \quad (\text{BI7})$$

Exponentiating this literal sum and using (BI5) gives exactly

$$e^{Q_N(z)} = zN^{-z} \prod_{j=1}^N (1 + z/j) = \frac{1}{E_N(z)}. \quad (\text{BI8})$$

The logarithmic series in (BI7) converges locally uniformly. To prove this explicitly, for $|w| \leq 1/2$,

$$\text{Log}(1 + w) - w = - \int_0^1 \frac{tw^2}{1 + tw} dt,$$

whence $|\text{Log}(1 + w) - w| \leq |w|^2$. On a compact set with $|z| \leq R$, this bounds the j -th summand by R^2/j^2 whenever $j \geq 2R$. The finite initial part is holomorphic and the summable bound proves uniform convergence of the tail. Hence

$$Q(z) = \text{Log } z + \gamma_E z + \sum_{j=1}^{\infty} (\text{Log}(1 + z/j) - z/j) \quad (\text{BI9})$$

is holomorphic and $Q_N \rightarrow Q$ locally uniformly.

Taking limits in $E_N e^{Q_N} = 1$, using Section 1, proves

$$\Gamma(z) e^{Q(z)} = 1. \quad (\text{BI10})$$

This proves that Gamma is nowhere zero on the entire right half-plane. It also constructs the required holomorphic logarithm:

$$\mathcal{G}(z) := -Q(z), \quad e^{\mathcal{G}(z)} = \Gamma(z). \quad (\text{BI11})$$

For $x > 0$, every term in $Q(x)$ is real and the defining Gamma integral is positive. Thus $\mathcal{G}(x)$ is precisely the real logarithm of $\Gamma(x)$; there is no unspecified additive $2\pi i$ constant.

The second derivatives of the series in (BI9) converge locally uniformly, since $\sum_j |z + j|^{-2}$ does. Termwise differentiation gives

$$\boxed{\mathcal{G}''(z) = \sum_{j=0}^{\infty} \frac{1}{(z + j)^2} \quad (\Re z > 0).} \quad (\text{BI12})$$

For justification without assuming a derivative-series theorem, on any slightly larger compact subset of the same half-plane the series converges uniformly; Cauchy's integral formula on small circles then gives uniform convergence of its derivatives. Direct estimates of the derivatives of its summands give the same conclusion.

80.3 3. Beta product, duplication, and the factor $\sqrt{\pi}$

The absolutely convergent product of the two defining Gamma integrals admits the change of variables

$$x = rt, \quad y = r(1 - t), \quad r > 0, \quad 0 < t < 1.$$

Its Jacobian matrix has determinant $-r$, so the positive integration Jacobian is r . Consequently

$$\Gamma(z)\Gamma(w) = \Gamma(z + w)B(z, w) \quad (\Re z, \Re w > 0). \quad (\text{BI13})$$

All powers here are powers of positive real bases. Fubini and the change of variables are justified by the same integrals with real parts of the exponents.

In $B(z, z)$, first substitute $t = (1 + u)/2$, and then use symmetry and $v = u^2$. Every factor is retained:

$$\begin{aligned} B(z, z) &= 2^{1-2z} \int_{-1}^1 (1 - u^2)^{z-1} du \\ &= 2^{2-2z} \int_0^1 (1 - u^2)^{z-1} du \\ &= 2^{1-2z} B(1/2, z). \end{aligned} \quad (\text{BI14})$$

Finally $x = t^2$ in (BI1) gives $\Gamma(1/2) = 2 \int_0^\infty e^{-t^2} dt$. The latter is the positive integral over the whole real line. Its square is $\int_{\mathbb{R}^2} e^{-(x^2+y^2)} dx dy$. Polar coordinates, whose positive Jacobian is r , evaluate this square as

$$2\pi \int_0^\infty r e^{-r^2} dr = \pi.$$

Thus $\Gamma(1/2) = \sqrt{\pi}$. Inserting (BI13) into (BI14), and using the nonvanishing established in (BI10), proves

$$\boxed{\Gamma(z)\Gamma(z + 1/2) = 2^{1-2z} \sqrt{\pi} \Gamma(2z) \quad (\Re z > 0).} \quad (\text{BI15})$$

On the positive real axis all these factors are positive, so their real logarithms obey the corresponding additive identity without any branch ambiguity.

80.4 4. The bounded integral remainder

For $u > 0$, define

$$K(u) = \left(\frac{1}{e^u - 1} - \frac{1}{u} + \frac{1}{2} \right) \frac{1}{u}. \quad (\text{BI16})$$

We prove its exact bounds rather than assuming a Bernoulli expansion. With $x = u/2 > 0$,

$$K(u) = \frac{\coth x - 1/x}{4x}.$$

The two functions

$$F_0(x) = x \cosh x - \sinh x, \quad F_1(x) = (x^2 + 3) \sinh x - 3x \cosh x$$

vanish at zero and satisfy $F_0'(x) = x \sinh x \geq 0$ and $F_1'(x) = xF_0(x) \geq 0$. Dividing their nonnegative values by positive factors gives

$$0 \leq \coth x - \frac{1}{x} \leq \frac{x}{3}.$$

Therefore

$$\boxed{0 \leq K(u) \leq \frac{1}{12} \quad (u > 0).} \quad (\text{BI17})$$

Define the actual remainder candidate

$$R(z) = \int_0^\infty e^{-zu} K(u) du, \quad \Re z > 0. \quad (\text{BI18})$$

It is absolutely convergent and holomorphic. Differentiating j times is justified on every compact set with $\Re z \geq a > 0$ by $u^j e^{-au}/12$. In particular

$$|R(z)| \leq \frac{1}{12\Re z}, \quad 0 \leq R(x) \leq \frac{1}{12x} \quad (x > 0). \quad (\text{BI19})$$

80.5 5. Equality of second derivatives

Put $c = \frac{1}{2} \log(2\pi)$, and define

$$\mathcal{F}(z) = (z - \tfrac{1}{2}) \text{Log } z - z + c + R(z). \quad (\text{BI20})$$

Differentiation and the exact algebraic identity

$$u^2 K(u) = \frac{u}{e^u - 1} - 1 + \frac{u}{2}$$

give

$$\begin{aligned} \mathcal{F}''(z) &= \frac{1}{z} + \frac{1}{2z^2} + \int_0^\infty e^{-zu} \left(\frac{u}{e^u - 1} - 1 + \frac{u}{2} \right) du \\ &= \frac{1}{z^2} + \int_0^\infty \frac{ue^{-zu}}{e^u - 1} du. \end{aligned} \quad (\text{BI21})$$

Here $\int_0^\infty e^{-zu} du = 1/z$ and $\int_0^\infty ue^{-zu} du = 1/z^2$ follow by elementary integration on $\Re z > 0$; all cancellation factors are displayed.

The geometric series $1/(e^u - 1) = \sum_{j=1}^\infty e^{-ju}$ is valid for $u > 0$. Its absolute integral after multiplication by ue^{-zu} is bounded by the same expression with $\Re z$, which is integrable at zero and infinity. Hence termwise integration is valid, and

$$\mathcal{F}''(z) = \frac{1}{z^2} + \sum_{j=1}^\infty \int_0^\infty ue^{-(z+j)u} du = \sum_{j=0}^\infty \frac{1}{(z+j)^2} = \mathcal{G}''(z). \quad (\text{BI22})$$

The right half-plane is connected, so the holomorphic function $\mathcal{F} - \mathcal{G}$ has zero second derivative and consequently equals $az + b$ there, for constants $a, b \in \mathbb{C}$.

80.6 6. The recurrence fixes the linear constant

For positive real x , (BI2) and the real-axis branch in (BI11) give

$$\mathcal{G}(x+1) - \mathcal{G}(x) = \log x.$$

The corresponding exact difference of the explicit function (BI20) is

$$\mathcal{F}(x+1) - \mathcal{F}(x) - \log x = (x + \tfrac{1}{2}) \log(1 + 1/x) - 1 + R(x+1) - R(x). \quad (\text{BI23})$$

The left side equals a . The elementary integral bounds

$$\frac{y}{1+y} \leq \log(1+y) \leq y \quad (y > 0)$$

show that $x \log(1 + 1/x) \rightarrow 1$ and $\frac{1}{2} \log(1 + 1/x) \rightarrow 0$. The remainder bound (BI19) makes the remaining difference tend to zero. Thus $a = 0$, and $\mathcal{F} - \mathcal{G} = b$ throughout the right half-plane.

80.7 7. Duplication fixes the additive constant

For $x > 0$, the positive-real logarithm of (BI15) is

$$\mathcal{G}(x) + \mathcal{G}(x + 1/2) - \mathcal{G}(2x) = (1 - 2x) \log 2 + \frac{1}{2} \log \pi. \quad (\text{BI24})$$

Since $\mathcal{F} = \mathcal{G} + b$, the difference between the same expression using $\mathcal{F}$ and the right side of (BI24) equals exactly $b + b - b = b$.

Insert (BI20) without dropping any constant. The terms containing $\log x$ cancel, and the resulting difference is

$$\begin{aligned} b &= x \log(1 + 1/(2x)) - \frac{1}{2} \\ &\quad + \left(c - \frac{1}{2} \log(2\pi)\right) + R(x) + R(x + 1/2) - R(2x). \end{aligned} \quad (\text{BI25})$$

Our displayed $c = \frac{1}{2} \log(2\pi)$ cancels exactly. The same elementary logarithm bounds show $x \log(1 + 1/(2x)) \rightarrow 1/2$, and each remainder tends to zero by (BI19). Thus $b = 0$.

Combining Sections 5–7 proves the complete exact identity

$$\boxed{\log \Gamma(z) = (z - \frac{1}{2}) \text{Log } z - z + \frac{1}{2} \log(2\pi) + \int_0^\infty e^{-zu} \left(\frac{1}{e^u - 1} - \frac{1}{u} + \frac{1}{2} \right) \frac{du}{u}, \quad \Re z > 0.} \quad (\text{BI26})$$

The logarithm on the left is the uniquely specified holomorphic branch from (BI11). The integral is absolutely convergent, and its bound and all differentiations have been justified. This is the identity used by the actual boundary-tilt proof.

Complete original tilted-convolution envelope proof

81 Uniform original source comparison and the two growing norm windows

Retain an actual complete off-line quartet $1/2 \pm \delta \pm i\gamma$ of common order m , with $0 < \delta < 1/2$ and $\gamma > 0$. The polynomial h , the entire quotient $v_h = (2\xi)/h$, its full Taylor unit, and the original source are unchanged. Set

$$w_h(u) = \frac{|v_h(1/2 + iu)|^2}{2\pi}, \quad m_{h,k} = w_h^{*k}, \quad \mu_h = \int_{\mathbb{R}} w_h(u) du, \quad \alpha = \pi/2. \quad (\text{TW1})$$

The proved actual boundary-tilt calculation supplies

$$p = 8m - \frac{9}{2} > 1, \quad e^{\alpha|u|} w_h(u) \leq C_h(1 + |u|)^{-p}, \quad M_\alpha = \int_{\mathbb{R}} e^{\alpha u} w_h(u) du \in (0, \infty). \quad (\text{TW2})$$

Here C_h denotes exactly the explicit constant C_h^{tilt} of (BT12), with its original compact and tail expressions. Its proof retains the exact factor $-\pi^{-1/4}(u^2 + 1/4)e^{-iu \log \pi/2} \zeta(1/2 + iu)$ and the full quartet denominator. In particular (TW2) does not use a positive lower bound for ζ or an assumption about its zero locations.

The comparison measure used below is the fixed original one-factor Gamma measure

$$d\sigma(u) = \frac{|\Gamma(1/4 + iu/2)|^2}{2\pi} du, \quad \int d\sigma = \sqrt{2\pi}. \quad (\text{TW3})$$

Its proved two-sided boundary bounds are

$$c_\Gamma e^{-\alpha|u|} (1 + |u|)^{-1/2} \leq \frac{d\sigma}{du} \leq C_\Gamma e^{-\alpha|u|} (1 + |u|)^{-1/2}, \quad c_\Gamma = \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}. \quad (\text{TW4})$$

This fixed measure has mass $\sqrt{2\pi}$ for every k . It is distinct from the k -fold Gamma convolution of mass $(2\pi)^{k/2}$ in (AT4). Both comparisons, with their exact connecting constants, will be retained.

81.1 A uniform convolution bound at the actual exponential boundary

For every integer $k \geq 1$ and every $u \in \mathbb{R}$,

$$\boxed{m_{h,k}(u) \leq k C_h M_\alpha^{k-1} (1 + |u|/k)^{-p} e^{-\alpha|u|}}. \quad (\text{TW5})$$

To prove this, put $f_+(x) = e^{\alpha x} w_h(x)$. Its mass is exactly M_α , and $f_+(x) \leq C_h(1 + |x|)^{-p}$. The original product integral gives $f_+^{*k}(u) = e^{\alpha u} m_{h,k}(u)$. On the hyperplane $x_1 + \dots + x_k = u$, at least one coordinate satisfies $|x_i| \geq |u|/k$. Split this integral into the union of those k sets. On its i th set the i th factor is at most $C_h(1 + |u|/k)^{-p}$. Eliminate $x_i = u - \sum_{j \neq i} x_j$; the change from the original convolution coordinates has absolute Jacobian one. Integrate the other $k - 1$ nonnegative factors over all their original domains; the resulting bound is $C_h(1 + |u|/k)^{-p} M_\alpha^{k-1}$. Summing proves the bound with $e^{-\alpha u}$. Conjugation of the original real-entire 2ξ and of the complete quartet polynomial proves $w_h(-x) = w_h(x)$. Thus $f_-(x) = e^{-\alpha x} w_h(x)$ has the same mass, and the same argument with f_- proves the bound with $e^{\alpha u}$. Taking the smaller of these two proved expressions gives exactly (TW5). This argument also covers $k = 1$ directly. Tonelli's theorem applies because all integrands are nonnegative and the displayed masses are finite.

For integers n with $2n > p - 1$, this yields the literal raw moment bound

$$\int_{\mathbb{R}} u^{2n} m_{h,k}(u) du \leq 2 C_h k^{p+1} M_\alpha^{k-1} \frac{\Gamma(2n - p + 1)}{\alpha^{2n - p + 1}}. \quad (\text{TW6})$$

Indeed $(1 + u/k)^{-p} \leq k^p u^{-p}$ for $u > 0$, and substitution $y = \alpha u$ evaluates the remaining positive half-line integral. No arithmetic density has been assigned the value of a Gamma density.

81.2 Pointwise comparison of the original convolution

For $k \geq 3$, retain the proved original convolution lower envelope

$$m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(|u|+k-3)} (k-2+|u|)^{-B}, \quad B = 42 + 8m, \quad \vartheta_h = \int_{-1}^1 w_h(u) du > 0. \quad (\text{TW7})$$

Its complete proof is the original arithmetic envelope (AU.10), also retained as (CJ.15); it concerns this same g/h convolution. Define the following explicit constants, without altering either measure:

$$M = B/2 = 21 + 4m, \quad a_k = \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-B}}{C_\Gamma 2^{B/2}}, \quad A_k = \frac{C_h}{c_\Gamma} k^{p+1} M_\alpha^{k-1}. \quad (\text{TW8})$$

Then

$$\boxed{a_k (1+u^2)^{-M} \frac{d\sigma}{du} \leq m_{h,k}(u) \leq A_k \frac{d\sigma}{du}}. \quad (\text{TW9})$$

For the upper bound, $1+|u|/k \geq (1+|u|)/k$ in (TW5), and the lower inequality of (TW4) leaves the factor $(1+|u|)^{-(p-1/2)} \leq 1$. This gives precisely A_k . For the lower bound, $k-2 \geq 1$ gives $k-2+|u| \leq (k-2)(1+|u|)$. The upper inequality of (TW4) implies $e^{-\alpha|u|} \geq C_\Gamma^{-1} d\sigma/du$. Finally $(1+|u|)^B \leq 2^{B/2} (1+u^2)^{B/2}$. Substitution of these three inequalities in (TW7) proves its lower counterpart in (TW9), with its displayed constant and direction.

81.3 Full finite polynomial forms, with the original coordinate maps

Let b_j be the fixed Gamma monic polynomials in u . The original generating-integral calculation (TG.7)–(TG.14) proves

$$\gamma_j = \|b_j\|_\sigma^2 = \sqrt{2} j! \Gamma(j+1/2) = \sqrt{2\pi} \frac{(2j)!}{4^j}, \quad u b_j = b_{j+1} + j(j-1/2) b_{j-1}. \quad (\text{TW10})$$

The last term is absent at $j=0$. The second equality follows by induction from $\Gamma(1/2) = \sqrt{\pi}$ and the Gamma recurrence; no asymptotic formula is used there.

For $N \geq 0$ put

$$D_N = [1 + 4(N+M)^2]^M.$$

Every polynomial P of degree at most N satisfies

$$\int |P(u)|^2 (1+u^2)^M d\sigma(u) \leq D_N \int |P(u)|^2 d\sigma(u). \quad (\text{TW11})$$

Here is the full multiplication estimate. The orthonormal vectors $e_j = b_j/\sqrt{\gamma_j}$ obey

$$u e_j = \sqrt{(j+1)(j+1/2)} e_{j+1} + \sqrt{j(j-1/2)} e_{j-1}.$$

On degrees at most N , the first weighted shift has norm at most $N+1$ and the second has norm at most N ; this follows by taking squared norms of their orthogonal images term by term. The triangle inequality therefore gives $\|uP\|_\sigma \leq 2(N+1)\|P\|_\sigma$. After each multiplication the degree increases by one. Iteration gives $\|u^j P\|_\sigma \leq [2(N+M)]^j \|P\|_\sigma$ for $0 \leq j \leq M$. Expand $(1+u^2)^M$, integrate its nonnegative summands, and use the binomial theorem; the result is exactly (TW11).

Apply Cauchy–Schwarz with the two functions $|P|(1+u^2)^{-M/2}$ and $|P|(1+u^2)^{M/2}$ in $L^2(\sigma)$:

$$\left(\int |P|^2 d\sigma \right)^2 \leq \left(\int |P|^2 (1+u^2)^{-M} d\sigma \right) \left(\int |P|^2 (1+u^2)^M d\sigma \right).$$

For $P \neq 0$, divide by its positive σ norm and apply (TW11). For $P = 0$ the resulting inequality is immediate. Combining with (TW9) proves the full finite-form comparison

$$\boxed{\frac{a_k}{D_N} \int |P|^2 d\sigma \leq \int |P|^2 m_{h,k}(u) du \leq A_k \int |P|^2 d\sigma \quad (\deg P \leq N).} \quad (\text{TW12})$$

It holds for every complex coefficient vector, not just monic polynomials. In particular $A_k D_N / a_k \geq 1$ by applying (TW12) to any nonzero P .

The original sum coordinate is $S = c + iu$, $c = k/2$. Its exact algebra map is $\Psi_k : \mathbb{C}[S] \rightarrow \mathbb{C}[u]$, $\Psi_k P(u) = P(c + iu)$, with inverse $f \mapsto f((S - c)/i)$. In degree at most N its coefficient matrix has diagonal $1, i, \dots, i^N$. Thus (TW12) is a congruence statement in the original S coordinates as well. On the monic degree- n affine space, $P \mapsto i^{-n} \Psi_k P$ is a bijection to the monic u polynomials and preserves the norm because $|i^{-n}| = 1$. No degree-dependent phase is asserted to be a single algebra map. Consequently, writing $H_{k,N}$ for the actual original polynomial Gram and $H_N^{\sigma,k}$ for the fixed σ Gram evaluated at $S = c + iu$, the fully typed matrix form is

$$\boxed{(a_k/D_N) H_N^{\sigma,k} \preceq H_{k,N} \preceq A_k H_N^{\sigma,k}.} \quad (\text{TW13})$$

The identity on the original polynomial source commutes with remainder modulo the same $\chi_{h,k}$ and hence with its full unitful jet observation. Neither this comparison nor the coordinate congruence changes $\chi_{h,k}$, S , its full jets, or the original source norm.

81.4 Uniform arithmetic monic norms

Let $\omega_{h,k,n}$ be the minimum original squared norm of a monic degree- n polynomial. Minimizing (TW12) on that same affine space, using (TW10) for its fixed comparison minimum, gives

$$\boxed{\frac{a_k \gamma_n}{D_n} \leq \omega_{h,k,n} \leq A_k \gamma_n, \quad k \geq 3, n \geq 0.} \quad (\text{TW14})$$

The upper bound may equivalently be verified on the explicit trial polynomial b_n ; the lower bound applies to every monic polynomial. Thus no minimizer was transferred between different metrics without the stated inequality.

Here and below constants in O_h may depend on the fixed complete quartet and the fixed proven envelope constants, but not on k, n . Taking logarithms of the exact constants in (TW8) gives

$$\log a_k = (k - 3)(\log \vartheta_h - \alpha) - B \log(k - 2) + \log c_h - \log C_\Gamma - (B/2) \log 2,$$

$$\log A_k = (k - 1) \log M_\alpha + (p + 1) \log k + \log(C_h/c_\Gamma).$$

Also $\log D_n = M \log(1 + 4(n + M)^2) = O_h(1 + \log(n + 1))$. These displayed equalities prove, uniformly for $k \geq 3, n \geq 0$,

$$\boxed{\log \omega_{h,k,n} = \log \left(\sqrt{2\pi} \frac{(2n)!}{4^n} \right) + O_h(k + \log(n + 1)).} \quad (\text{TW15})$$

This is a high-degree norm estimate for the actual arithmetic convolution. It is not an assertion that consecutive norm ratios have an asymptotic; the error in (TW15) does not justify that differentiation in n .

The same all-polynomial comparison also controls the original raw even moments $I_{k,2n} = \int_{\mathbb{R}} u^{2n} m_{h,k}(u) du$. For $k \geq 3$ and $n \geq 1$ it gives the explicit two-sided bounds

$$\boxed{\frac{a_k \sqrt{2} c_\Gamma e^{-\alpha}}{D_n} \frac{(2n - 1)!}{\alpha^{2n}} \leq I_{k,2n} \leq 2 A_k C_\Gamma \frac{(2n)!}{\alpha^{2n+1}}.} \quad (\text{TW15a})$$

For the lower comparison moment, restrict (TW4) to $u \geq 1$ and use $(1+u)^{-1/2} \geq 1/(\sqrt{2}u)$. Both half-lines give $\int u^{2n} d\sigma \geq \sqrt{2}c_\Gamma \alpha^{-2n} \Gamma(2n, \alpha)$. Repeated integration by parts gives $\Gamma(2n, \alpha) = (2n-1)!e^{-\alpha} \sum_{j=0}^{2n-1} \alpha^j/j! \geq (2n-1)!e^{-\alpha}$. For the upper comparison moment, drop the factor $(1+|u|)^{-1/2} \leq 1$ in (TW4) and evaluate the Gamma integral. Applying (TW12) to the original degree- n polynomial u^n proves (TW15a). Its logarithms give, uniformly on these stated degrees,

$$\log I_{k,2n} = \log((2n)!\alpha^{-2n}) + O_h(k + \log(n+1)). \quad (\text{TW15b})$$

The full one-factor zeta numerator and actual tilted mass in (TW2) remain in a_k, A_k ; (TW15a) is a bound on the original moments.

81.5 The two original windows with full finite errors

Put $q = [1 + k(m-1)](k+1)^2$. For $n = q-1$ or $n = q$, division of (TW14) at the two exact degrees gives

$$\log \frac{\gamma_{n+q}}{\gamma_n} - \log \frac{A_k D_{n+q}}{a_k} \leq \log \frac{\omega_{h,k,n+q}}{\omega_{h,k,n}} \leq \log \frac{\gamma_{n+q}}{\gamma_n} + \log \frac{A_k D_n}{a_k}. \quad (\text{TW16})$$

All logarithmic error bounds here are nonnegative by (TW12). The exact factorial formula (TW10) and monotonicity of $\log x$ give

$$\log \frac{\gamma_{2q}}{\gamma_q} = 2q \log(4q/e) + \eta_q, \quad 0 \leq \eta_q \leq \log 2. \quad (\text{TW17})$$

Indeed the logarithm of $(4q)!/(2q)!$ is $\sum_{j=2q+1}^{4q} \log j$. Its excess over $\int_{2q}^{4q} \log x dx$ is nonnegative and at most $\sum_{j=2q+1}^{4q} (\log j - \log(j-1)) = \log 2$. Subtracting $q \log 4$ gives precisely (TW17). The other original window satisfies the exact identity

$$\frac{\gamma_{2q-1}}{\gamma_{q-1}} = \frac{\gamma_{2q}}{\gamma_q} \frac{q-1/2}{4q-1}, \quad \frac{1}{6} \leq \frac{q-1/2}{4q-1} < \frac{1}{4}.$$

It follows in particular that

$$4q \log(4q/e) - \log 6 \leq \log \frac{\gamma_{2q-1} \gamma_{2q}}{\gamma_{q-1} \gamma_q} \leq 4q \log(4q/e). \quad (\text{TW18})$$

For fixed h , one has $k/q \rightarrow 0$ and $\log q/q \rightarrow 0$. Thus each finite error in (TW16), divided by $2q$, tends to zero. Equations (TW16)–(TW18) prove both limits on their stated windows:

$$\boxed{\lim_{k \rightarrow \infty} \frac{1}{q} \left(\frac{\omega_{h,k,2q}}{\omega_{h,k,q}} \right)^{1/(2q)} = \lim_{k \rightarrow \infty} \frac{1}{q} \left(\frac{\omega_{h,k,2q-1}}{\omega_{h,k,q-1}} \right)^{1/(2q)} = \frac{4}{e}.} \quad (\text{TW19})$$

No limit with an unspecified ratio n/q is asserted.

81.6 The resulting actual four-window lower estimate

Retain the original relative controls ε_N , phases ϕ_N , and contraction penalties P_-, P_+ of (HC1)–(HC2). Their exact first boundary and regular products, inherited from (CJ.9)–(CJ.13), give

$$\mathcal{B}_k = \sum_{N=q-1}^{2q-1} w_N \log(\varepsilon_N^2 + \phi_N^2) - \log \frac{\omega_{h,k,2q-1} \omega_{h,k,2q}}{\omega_{h,k,q-1} \omega_{h,k,q}} + P_- + P_+,$$

where the endpoint weights are $1, 2, \dots, 2, 1$. Set

$$E_k = 2 \log(A_k/a_k) + \log D_{q-1} + \log D_q \geq 0, \quad E_k = O_h(k + \log q).$$

Adding the upper sides of (TW16) and applying (TW18) gives the actual sum-norm upper bound $4q \log(4q/e) + E_k$. Consequently, with the full original trace $\mathcal{L} = 2\delta\ell_k(k+1)\lfloor(k+1)^2/4\rfloor$,

$$\begin{aligned} \mathcal{B}_k &\geq 4q \log(\delta ek/8) - E_k + P_- + P_+ \\ &\quad + \sum_N w_N \log(1 + \phi_N^2/\mathcal{L}^2) + \sum_N w_N \log \frac{\varepsilon_N^2 + \phi_N^2}{\mathcal{L}^2 + \phi_N^2} + 4q \log \frac{2\mathcal{L}}{\delta kq}. \end{aligned} \quad (\text{TW20})$$

All terms after $-E_k$ are the displayed original nonnegative costs. The sums retain both end-point degrees and all their multiplicities. To verify the constant, $4q \log \mathcal{L} - 4q \log(4q/e)$ equals $4q \log(\delta ek/8) + 4q \log(2\mathcal{L}/(\delta kq))$. This proves the identity used in the bound with no rescaling of S . Let $\mathcal{R}_k^{\text{TW}}$ denote exactly the right side of (TW20), with every displayed contraction, phase, control and trace term. The original Gamma four-volume $\mathcal{B}_k^\Gamma$ uses the same χ and the original tensor Gamma density. Substituting (AT19), whose $\mathfrak{C}_*$ is computed on that precise pair of source Grams, gives the additional current consequence

$$\mathcal{R}_k^{\text{TW}} \leq \mathcal{B}_k \leq \mathfrak{b}_*^+, \quad \mathfrak{b}_*^+ \geq \mathcal{R}_k^{\text{TW}}, \quad \mathfrak{C}_* \geq J_* \geq \max\{0, \mathcal{R}_k^{\text{TW}} - \mathcal{B}_k^\Gamma\}. \quad (\text{TW20a})$$

Here $\mathfrak{b}_*^+$ is (AT21e) on the actual tensor Gamma path, incorporating the nonlinear radius I_* as well as the additive control. All inequalities follow by substitution of the same scalar; none of the nonnegative terms of (TW20) is discarded or assigned an asymptotic cancellation. Since $E_k/q \rightarrow 0$, the quantified refinement includes

$$\liminf_{k \rightarrow \infty} \frac{\mathcal{B}_k - 4q \log k}{q} \geq 4 \log(\delta e/8). \quad (\text{TW21})$$

The leading threshold four is inherited. The new content is the actual boundary-tilt comparison, its uniform finite-degree errors, and the sharper literal window constant. These lower estimates alone do not assign a sign to the signed arithmetic correction in (AT14)–(AT17).

Original local zeta mass and complete three-factor lower envelope

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Arithmetic endpoint bounds on the original τ -base source

81.7 1. Result and exact source boundary

The formalization session's endpoint criterion selects an actual member of the canonical arithmetic family. This continuation proves one of its previously unproved analytic inputs. For the **unchanged arithmetic measure**, a fixed finite packet h , and integers $n \geq k \geq 3$, there is a finite constant C_h , independent of n and k , such that

$$\left(\frac{\omega_{h,k,2n}}{\omega_{h,k,n}} \right)^{1/(2n)} \leq C_h n. \quad (1)$$

The norms are the original monic polynomial norms for $m_{h,k} = w_h^{*k}$; the total mass remains μ_h^k . No root-location hypothesis is used in this bound. The proof retains a compact-source mass, a lower convolution constant, and both exponential moments. These constants are defined explicitly below; they have not been numerically enclosed.

For the packet consisting exactly of a hypothetical nonreal off-line quartet, let

$$q_k = [1 + k(m-1)](k+1)^2, \quad L_{h,k} = 2\delta[1 + k(m-1)](k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor. \quad (2)$$

The existing exterior theorem gives $L_{h,k} \leq \epsilon_{h,k,N}$ at every admitted degree. Combining (1) with the other session's endpoint theorem gives

$$\min_{q_k \leq N < 2q_k} \epsilon_{h,k,N} \leq C_h q_k \sinh\left(\frac{\mathcal{B}_{h,k}}{2q_k}\right), \quad \mathcal{B}_{h,k} = \log \frac{V_{q_k-1} V_{q_k}}{V_{2q_k-1} V_{2q_k}}. \quad (3)$$

Consequently such a quartet forces

$$\mathcal{B}_{h,k} \geq 2q_k \operatorname{arsinh}\left(\frac{\delta k}{2C_h}\right), \quad \liminf_{k \rightarrow \infty} \frac{\mathcal{B}_{h,k}}{q_k \log k} \geq 2. \quad (4)$$

The upper bound for this four-volume quantity is not proved here. In particular, (1) does not establish RH. It completes the polynomial-norm input in (3) on a specific admissible window and leaves the source/relation-volume comparison with its exact maps. Section 9 expresses that remaining quantity as two positive block log-determinants of actual new relation layers.

The finite quotient, canonical source, residue action and rank-two control are inputs from the preceding analytic notes. The new proofs in Sections 3–7 use only their explicitly defined arithmetic measure. The endpoint theorem is the formalizer's written result, not attributed to this continuation. Its finite telescopes are reported kernel-checked; the analytic estimates below have written proofs, not new Lean certificates.

81.8 2. The base, source, and both kinds of quotient

Retain the original marked coefficient square

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (5)$$

Here $G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}$, $e_R = 0_R^\bullet$, $p_R(\tau) = 0_R$, and $p_R(r^\bullet) = r$. The target of $p_{\mathbb{Z}}$ is infinite. The construction remains over the original pointed base $\mathfrak{b}_\tau$; no new scalar zero is adjoined at a tensor degree or at a polynomial relation layer.

Keep $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees 0, 1, $Q = \mathcal{B}/\Theta V$, and $D = -x\partial_x$. The source spaces and the Fourier transform remain those of the earlier theta construction. In particular,

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad \mathcal{M}\Theta\phi_* = g. \quad (6)$$

Let h be a fixed finite product of zeros of g , with the full order of each selected zero retained, and write $d = \deg h$. Define

$$v_h = g/h, \quad w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad m_{h,k} = w_h^{*k}. \quad (7)$$

The quotient v_h is entire, including at the selected centres. It is not the meromorphic expression evaluated before removing its actual removable factors. For $h = 1$, the same analytic definitions are meaningful, but the finite arithmetic packet is zero. Only the nonempty case is used for the finite Gram inverses below.

The source map is

$$\mathcal{V}_{h,k} : \mathbb{C}[S] \longrightarrow \mathcal{B}^{\widehat{\otimes} k}, \quad P \longmapsto P(D_1 + \cdots + D_k)(F_h^{\otimes k}), \quad \mathcal{M}F_h = v_h. \quad (8)$$

The original cyclic module is $E_{h,k} = \mathbb{C}[S]/(\chi_{h,k})$. Its arithmetic inclusion keeps the Taylor unit:

$$\eta_{h,k}[P] = v_h^{\otimes k} P(A_k)1, \quad v_h = j_h(g/h), \quad A_k = \sum_i M_{s_i}. \quad (9)$$

The supplied source identities are

$$J^{(k)}\mathcal{V}_{h,k} = \eta_{h,k}\pi_\chi, \quad q^{(k)}\mathcal{V}_{h,k} = \sigma_h^{\otimes k}\eta_{h,k}\pi_\chi. \quad (10)$$

The actual norm of (8) is

$$\|\mathcal{V}_{h,k}P\|^2 = \int_{\mathbb{R}} |P(k/2 + iu)|^2 m_{h,k}(u) du. \quad (11)$$

For $\mathcal{P}_n = \mathbb{C}[S]_{\leq n}$, the monic norm is the quotient norm of the leading-coefficient map

$$0 \longrightarrow \mathcal{P}_{n-1} \longrightarrow \mathcal{P}_n \xrightarrow{\text{lc}_n} \mathbb{C} \longrightarrow 0, \quad \omega_{h,k,n} = \inf_{\text{lc}_n P=1} \|\mathcal{V}_{h,k}P\|^2. \quad (12)$$

The arithmetic quotient is the other specified map from the same source:

$$0 \longrightarrow \mathcal{P}_{n-q} \xrightarrow{\times\chi} \mathcal{P}_n \xrightarrow{\pi_\chi} E_{h,k} \longrightarrow 0, \quad q = \deg \chi, \quad (13)$$

for $n \geq q-1$, with the preceding relation space zero at $n = q-1$. Multiplication by the monic χ carries $\mathcal{P}_{n-q}/\mathcal{P}_{n-q-1}$ isomorphically to $\mathcal{P}_n/\mathcal{P}_{n-1}$, with leading coefficient unchanged. The map π_χ kills that very relation. Thus (12), (13), and multiplication by χ specify the relation between the degree-line norm and the arithmetic quotient metric. One norm is not substituted for the other.

At each admitted original support label λ , the lifted quotient is $(\lambda, P) \mapsto (\lambda, [P]_\chi)$. An original relation has image $(\lambda, 0)$; external absence has image τ . Its norm before this projection remains available through (11).

81.9 3. A quantitative local arithmetic mass lemma

81.9.1 3.1 Uniform polynomial growth on a fixed strip

There is a constant $C_0 \geq 1$ such that, for $|T| \geq 10$, the function

$$f_T(z) = \zeta(1/2 + iT + iz) \quad (14)$$

is holomorphic on and inside the ellipse

$$z = \frac{1}{2}(w + w^{-1}), \quad |w| = 8, \quad (15)$$

and has modulus at most $M_T = C_0(1 + |T|)^6$ there.

The ellipse has real and imaginary semiaxes $65/16$ and $63/16$. Hence its image in the s -plane lies in $-55/16 \leq \Re s \leq 71/16$ and $|\Im s - T| \leq 65/16$. The pole of ζ is outside this set when $|T| \geq 10$.

For $\Re s \geq 1/4$, use the identity

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx. \quad (16)$$

It follows by summing the steps of $[x]$ for $\Re s > 1$ and then by analytic continuation for $\Re s > 0$, $s \neq 1$. Its integral is bounded by $1/\Re s$, giving $O(1 + |T|)$ on the indicated part of the ellipse. For $\Re s \leq 1/4$, the functional equation

$$\zeta(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) \zeta(1-s) \quad (17)$$

and the vertical-strip gamma estimate give $O((1+|T|)^{3/2-\Re s})$. On the stated strip this exponent is below 5. The common power 6 in (14) is therefore a valid upper bound with one fixed constant. The gamma estimate and functional equation are the classical inputs recorded in DLMF §§5.11 and 25.4; none assumes RH.

Every disc of radius one about a point of $[-1, 1]$ lies inside the ellipse. The maximum principle and Cauchy's derivative formula therefore also give

$$\sup_{x \in [-1, 1]} |f'_T(x)| \leq M_T. \quad (18)$$

81.9.2 3.2 Propagate a nonzero value to the critical-line interval

Put

$$r_* = (3 + \sqrt{13})/2 < 4, \quad z_* = -3i/2 = \frac{1}{2}(-ir_* + (-ir_*)^{-1}), \quad c_* = \zeta(2)^{-1}. \quad (19)$$

The absolutely convergent reciprocal Dirichlet series gives $|1/\zeta(2 + iT)| \leq \zeta(2)$, so $|f_T(z_*)| \geq c_* > 0$. Let $m_T = \max_{[-1, 1]} |f_T|$.

Apply the maximum principle on $1 \leq |w| \leq 8$ to $f_T((w + w^{-1})/2)$. Equivalently subtract from its logarithmic modulus the harmonic function linear in $\log |w|$ with boundary values $\log m_T$ and $\log M_T$. Zeros are handled by the usual subharmonic maximum principle. At $w = -ir_*$ this gives

$$c_* \leq m_T^\vartheta M_T^{1-\vartheta}, \quad \vartheta = 1 - \frac{\log r_*}{\log 8} > \frac{1}{3}. \quad (20)$$

Here $0 < c_* < 1$ and $M_T \geq 1$. Consequently

$$m_T \geq c_*^3 M_T^{-2}. \quad (21)$$

This is the three-circles argument with every coordinate map and radius fixed. No polynomial approximation to ζ or assumption about its zeros is used.

Choose a point attaining m_T . By (18), on a one-sided subinterval of length $m_T/(2M_T) \leq 1/2$ inside $[-1, 1]$, the modulus is at least $m_T/2$. Thus

$$\int_{T-1}^{T+1} |\zeta(1/2 + it)|^2 dt \geq \frac{m_T^3}{8M_T} \geq \frac{c_*^9}{8C_0^7} (1 + |T|)^{-42}. \quad (22)$$

For $T \in [-10, 10]$, the integral in (22) is continuous and strictly positive: otherwise ζ would vanish on an interval and hence identically on its holomorphic domain. Its minimum on that compact interval is positive. Combining it with (22) proves the all-real statement

$$\boxed{\exists c_\zeta > 0 \quad \int_{T-1}^{T+1} |\zeta(1/2 + it)|^2 dt \geq c_\zeta (1 + |T|)^{-42} \quad (T \in \mathbb{R}).} \quad (23)$$

The exponent 42 is deliberately an explicit sufficient exponent, not an optimal local mean-square estimate. The proof is a classical analytic propagation argument specialized here to the needed arithmetic interval.

81.10 4. A lower envelope for the actual convolution, after three factors

81.10.1 4.1 Local mass of the unchanged g/h

The gamma estimate, with a positive lower constant valid also on compact sets, implies

$$|\Gamma(1/4 + it/2)|^2 \geq c_\Gamma (1 + |t|)^{-1/2} e^{-\pi|t|/2}. \quad (24)$$

Let $H_h = \prod_{\rho \in Z} (1 + |\rho - 1/2|)^{m_\rho}$, so $|h(1/2 + it)| \leq H_h (1 + |t|)^d$. The entire identity $g = hv_h$ makes this bound valid at selected centres as well: $|g| \leq H_h (1 + |t|)^d |v_h|$. The factors in (6), including $|s(s-1)|^2 = (t^2 + 1/4)^2$, then give a positive constant $c_h^{(0)}$ with

$$w_h(t) \geq c_h^{(0)} e^{-\alpha_0|t|} (1 + |t|)^{-2d} |\zeta(1/2 + it)|^2, \quad \alpha_0 = \pi/2. \quad (25)$$

Equation (25) is a weaker bound than the available factor $(1 + |t|)^{7/2-2d}$; it follows by an inequality, not by changing the density. On $[T-1, T+1]$, its exponential and polynomial weights are bounded below by $e^{-\alpha_0} e^{-\alpha_0|T|}$ and $2^{-2d} (1 + |T|)^{-2d}$, respectively. Hence (23) proves

$$\int_{T-1}^{T+1} w_h(t) dt \geq c_h^{(1)} e^{-\alpha_0|T|} (1 + |T|)^{-B_h}, \quad B_h = 42 + 2d. \quad (26)$$

81.10.2 4.2 The positive compact convolution factor

The function w_h is continuous, bounded, integrable, and positive outside a discrete set. Thus $m_{h,2} = w_h * w_h$ is continuous and strictly positive at every real point. Indeed its integrand is positive almost everywhere; continuity follows from translation continuity in L^1 and boundedness of the other factor. Define the **actual** compact quantities

$$b_h = \min_{|v| \leq 1} m_{h,2}(v) > 0, \quad \vartheta_h = \int_{-1}^1 w_h(v) dv > 0. \quad (27)$$

Both are determined by the original density; neither is assigned the value one. Restricting a nonnegative convolution integral gives

$$m_{h,3}(u) \geq b_h \int_{u-1}^{u+1} w_h(t) dt \geq c_h e^{-\alpha_0|u|} (1 + |u|)^{-B_h}, \quad c_h = b_h c_h^{(1)} > 0. \quad (28)$$

The omitted part is the nonnegative integral over $|v| > 1$ and remains part of $m_{h,3}$. We use a lower bound, not a new compactly supported density.

For $k \geq 3$, restrict the remaining $k - 3$ variables to $[-1, 1]$ and use (28). Their sum has absolute value at most $k - 3$. This gives the central analytic estimate

$$\boxed{m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha_0(|u|+k-3)} (1 + |u| + k - 3)^{-B_h} \quad (k \geq 3, u \in \mathbb{R}).} \quad (29)$$

All mass and tensor factors are explicit. For $k = 3$ the factor ϑ_h^0 is the empty product. No statement for $k = 1, 2$ is inferred by using a negative exponent. The amplification below only needs $k \geq 3$.

81.11 5. Upper moments and exact lower monic norms

Fix $0 < b < \pi/2$ and retain both one-factor Laplace values

$$M_h(\pm b) = \int e^{\pm bt} w_h(t) dt > 0. \quad (30)$$

They are finite by the original gamma decay. Fubini and positivity give

$$\int e^{b|u|} m_{h,k}(u) du \leq M_h(b)^k + M_h(-b)^k. \quad (31)$$

The elementary exponential-series inequality $x^j \leq j! b^{-j} e^{bx}$ for $x \geq 0$ yields

$$\omega_{h,k,j} \leq (2j)! b^{-2j} (M_h(b)^k + M_h(-b)^k). \quad (32)$$

The trial polynomial is $(S - k/2)^j$, whose value on $S = k/2 + iu$ is $i^j u^j$. Its phase has modulus one and remains in the source map; the generator $A = M_S$ has not been shifted to define a different spectral problem.

For every leading coefficient of modulus one, the exact least squared norm of a degree- j polynomial on $[-L, L]$ with Lebesgue measure is

$$\mathfrak{l}_j(L) = \frac{2L^{2j+1}}{2j+1} \left(\frac{2^j (j!)^2}{(2j)!} \right)^2. \quad (33)$$

To check all constants, Rodrigues' formula gives the Legendre polynomial $P_j(x)$ with leading coefficient $(2j)!/(2^j (j!)^2)$ and squared norm $2/(2j+1)$. Its scaled monic version on $[-L, L]$ is the unique minimizer: subtract it from any competing monic polynomial, and the difference has lower degree and is orthogonal to it. For the source coordinate $S = k/2 + iu$, multiply this minimizer by the retained leading phase i^j . This proves (33) at the original type. See also DLMF §§18.3 and 18.5.

Restricting (11) to $[-L, L]$ and using (29) proves

$$\boxed{\omega_{h,k,j} \geq c_h \vartheta_h^{k-3} e^{-\alpha_0(L+k-3)} (1 + L + k - 3)^{-B_h} \mathfrak{l}_j(L) \quad (k \geq 3, L > 0).} \quad (34)$$

The observation has the explicit isometric splitting into the inside and outside interval components. Addition recovers the full source function, and their squared norms add. In the split lift each zero component is the supported zero at its label. The outside component has not been declared absent.

81.12 6. The norm-window estimate is proved

Set $j = n$ and $L = n$ in (34), and use $j = 2n$ in (32). The exact finite bound is

$$\boxed{\left(\frac{\omega_{h,k,2n}}{\omega_{h,k,n}} \right)^{1/(2n)} \leq \left[\frac{(4n)! b^{-4n} (M_h(b)^k + M_h(-b)^k) e^{\alpha_0(n+k-3)} (1 + n + k - 3)^{B_h}}{c_h \vartheta_h^{k-3} \mathfrak{l}_n(n)} \right]^{1/(2n)}}. \quad (35)$$

For $n \geq k \geq 3$, this is at most $C_h n$. Here is one explicit sufficient constant, with $M_* = \max\{1, M_h(b), M_h(-b)\}$:

$$C_h = 64b^{-2} \exp(\alpha_0) 2^{B_h/2} \exp\left(\frac{B_h}{2 \exp(1)}\right) \sqrt{M_* \max(1, c_h^{-1}) \max(1, v_h^{-1})}. \quad (36)$$

Proof of the uniform estimate: $\binom{2n}{n} \leq 4^n$ in (33) gives $\mathfrak{l}_n(n) \geq 2n^{2n+1}/((2n+1)4^n)$. Also $(4n)! \leq (4n)^{4n}$, $M_h(b)^k + M_h(-b)^k \leq 2M_*^k$, $n+k-3 \leq 2n$, and $1+n+k-3 \leq 2n$. After taking the $2n$ -th root the factorial and monic factors contribute at most $64n$. The mass factors contribute at most the square root in (36); the exponential contributes at most $\exp(\alpha_0)$. Finally $n^{1/n} \leq \exp(1/\exp(1))$ gives $(2n)^{B_h/(2n)} \leq 2^{B_h/2} \exp(B_h/(2 \exp(1)))$. This proves (1).

The constant is intentionally not optimized. Its dependence on the fixed packet is allowed; its independence of the polynomial and tensor degrees is what is needed. Defining it via positive compact minima does not claim to have provided a numerical interval enclosure for those minima.

For a general source window the same two inequalities give, without asymptotic notation,

$$\left(\frac{\omega_{h,k,n+r}}{\omega_{h,k,n}}\right)^{1/(2r)} \leq \left[\frac{(2(n+r))! b^{-2(n+r)} (M_h(b)^k + M_h(-b)^k) e^{\alpha_0(L+k-3)} (1+L+k-3)^{B_h}}{c_h v_h^{k-3} \mathfrak{l}_n(L)} \right]^{1/(2r)}. \quad (37)$$

81.13 7. Compose the proved bound with the formalizer's endpoint theorem

For a nonempty cyclic packet $E_{h,k}$, let G_N be its original canonical quotient metric, $V_N = \det G_N > 0$, and $\epsilon_{h,k,N}$ the original rank-two control allowance. The new endpoint theorem supplied by PR #23 is

$$\min_{0 \leq j < r} \epsilon_{h,k,n+j} \leq \left(\frac{\omega_{h,k,n+r}}{\omega_{h,k,n}}\right)^{1/(2r)} \sinh\left(\frac{1}{2r} \log \frac{V_{n-1} V_n}{V_{n+r-1} V_{n+r}}\right), \quad n \geq q. \quad (38)$$

Its proof multiplies the local Toda bounds, applies concavity of $\log \sinh x$, and uses the two exact telescopes. The zero-contraction case is handled before taking logarithms. This result is credited to the parallel formalization/research session. The proof and the distinction between checked finite components and written Jensen assembly are retained in its source.

Choose $n \geq \max(q, k)$ and $r = n$. Equations (1) and (38) yield the actual canonical selection estimate

$$\min_{n \leq N < 2n} \epsilon_{h,k,N} \leq C_h n \sinh\left(\frac{1}{2n} \log \frac{V_{n-1} V_n}{V_{2n-1} V_{2n}}\right). \quad (39)$$

The chosen index is a member of the existing canonical family. The source representative, Taylor unit, kernel, action and trace are not chosen again after selecting this index.

For the packet consisting exactly of the quartet in (2), $q_k \geq k$ and $L_{h,k}/q_k \geq \delta k/2$. The earlier exterior bound holds at every admitted index, including the one selected by (39). Since $\sinh$ is increasing on $[0, \infty)$, (4) follows. Moreover $\operatorname{arsinh}(ak) = \log k + O_a(1)$ for every fixed $a > 0$, which proves the stated threshold 2.

A proved upper bound with $\limsup \mathcal{B}_{h,k}/(q_k \log k) < 2$ would contradict (4). Such an upper bound is **not** an output of this note. The previous sufficient target $\mathcal{B}_{h,k} = o(q_k \log k)$ remains sufficient, but (4) specifies the larger quantitative threshold against which a future upper estimate can be compared. No repeated cost-free exterior operation is used: the multiplicity correction in the new formalization is retained.

81.14 8. The remaining four volumes retain their finite jet maps

The previous source/relation determinant identity remains

$$V_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_{N-q+1}}. \quad (40)$$

In the parallel full-jet transfer coordinates (PR #24), let $a = N - q + 1$, let $\mathbf{F}_a$ be its $2q$ by $2q$ raw confluent-jet matrix, and let $\mathbf{V}$ be its raw-derivative Vandermonde matrix, with every factorial retained. Its established identity is

$$V_N = \left(\prod_{j=a}^{a+q-1} \omega_{h,k,j} \right) \frac{\det \mathbf{V}}{\det \mathbf{F}_a}. \quad (41)$$

For the window $n = r = q$, the four transfer indices are exactly $0, 1, q, q + 1$. Put $\Omega_j = \prod_{i=j}^{j+q-1} \omega_{h,k,i}$. Then

$$\exp \mathcal{B}_{h,k} = \frac{\Omega_0 \Omega_1}{\Omega_q \Omega_{q+1}} \frac{\det \mathbf{F}_q \det \mathbf{F}_{q+1}}{\det \mathbf{F}_0 \det \mathbf{F}_1}. \quad (42)$$

The common Vandermonde cancels in this explicit ratio, not inside any underlying map. The transfer matrices and their inverses still need their actual arithmetic control; the fixed width alone gives no such bound.

The empty packet has $q = 0$, $V_N = 1$ as an empty determinant, and no positive-rank inverse or quartet estimate. The analytic inequalities (23)–(37) still apply to its theta seed, but are not statements about a nonexistent finite packet.

81.15 9. The volume window is also two explicit retained-boundary costs

There is a second useful representation of the same remaining quantity, with no reference-density change. Fix $i < j$, both admissible. In the original monic polynomial basis put

$$F_{i,j} = [b_{i+1}, \dots, b_j], \quad \Omega_{i,j} = \text{diag}(\omega_{i+1}, \dots, \omega_j), \quad b_a = [p_a]_\chi. \quad (43)$$

The kernel update is the actual sum $K_j = K_i + F_{i,j} \Omega_{i,j}^{-1} F_{i,j}^*$, where $K_i = G_i^{-1}$. The matrix determinant lemma gives

$$\frac{V_i}{V_j} = \det \left(I_{j-i} + \Omega_{i,j}^{-1} F_{i,j}^* G_i F_{i,j} \right). \quad (44)$$

This determinant has positive real value. Its factors are $1 + \lambda_a$, where $\lambda_a \geq 0$ are the generalized eigenvalues of the Hermitian pair $(F_{i,j}^* G_i F_{i,j}, \Omega_{i,j})$. No non-Hermitian matrix is asserted positive in an unspecified Euclidean metric.

Let $T_{i,j}$ have columns $\mathcal{V}_{h,k} p_{i+1}, \dots, \mathcal{V}_{h,k} p_j$. The original boundary-layer isomorphism is

$$\mathfrak{b}_{i,j} = T_{i,j} - R_i F_{i,j} : \mathbb{C}^{j-i} \xrightarrow{\sim} \mathcal{D}_j \cap \mathcal{D}_i^\perp, \quad \mathfrak{b}_{i,j}^* \mathfrak{b}_{i,j} = \Omega_{i,j} + F_{i,j}^* G_i F_{i,j}. \quad (45)$$

Its jets vanish; its highest polynomial coefficients give injectivity. Any new boundary can be reduced by these highest coefficients to an old boundary, and orthogonality then gives surjectivity. These statements prove the displayed map with the same original relation spaces, not only a dimension count. The representative update is

$$R_j = R_i + \mathfrak{b}_{i,j} (\Omega_{i,j} + F_{i,j}^* G_i F_{i,j})^{-1} F_{i,j}^* G_i. \quad (46)$$

Every correction in (46) is an original theta boundary. Its split quotient image is e at the receiving fibre, and the input relation remains available in (45). Its cochain primitives are the same fixed-order divisions and tensor signs as in the previous cyclic source construction.

Apply (44) to $(i, j) = (q - 1, 2q - 1)$ and $(q, 2q)$. This proves

$$\boxed{\mathcal{B}_{h,k} = \sum_a \log(1 + \lambda_a^{(0)}) + \sum_a \log(1 + \lambda_a^{(1)})}. \quad (47)$$

In particular

$$\mathcal{B}_{h,k} \leq \text{Tr}(\Omega_{q-1,2q-1}^{-1} F^* G_{q-1} F) + \text{Tr}(\Omega_{q,2q}^{-1} F^* G_q F), \quad (48)$$

where each F has its own indicated source and target from (43). When this trace bound is too large, (47) retains the full logarithmic cost rather than replacing it by its linear upper bound. The analytic norm estimate in (1) does not bound (47); it supplies the other factor in (39).

81.16 10. Deligne reading and what has actually been transferred

The six supplied S20 parts were reassembled in this turn. Their individual chunk hashes matched, and the assembled archive has 216,580,466 bytes and SHA-256 `e2005bf31e1fcf362765f73c315c855e0f504f24f34d3a0ab188049da522b7c8`. This verifies the supplied bytes, not every transcription claim in them.

I read the TeX for Definition 1.3.5, Lemmas 3.2.10–3.2.12, the tensor argument 3.2.13, and Corollary 3.3.6 in the nested source files. I compared those exact passages with the supplied top-level French/English frozen page records at printed pages 158, 203, and 206. No PDF text extraction or OCR was used. The source review records a genuine inherited transcription fault: old `fr_seg2.tex` writes a weight bound ≤ 2 at 3.2.10, whereas the controlling page record says < 2 ; only the latter, with integrality, implies ≤ 1 . The old file also corrupts the explicit squared-eigenvalue step in 3.2.13. Those old files remain unmodified; they are not used as authority for these comparisons. No claim of a full independent audit of Weil II is made.

Deligne’s determinant/rank operation and the upper/dual-lower architecture remain attached to the programme’s specified exterior and residue maps. Here the additional estimate is a characteristic-zero analytic estimate for the actual source, (1), followed by the endpoint selection (39). This is not a statement that the new base satisfies Deligne’s finite-field hypotheses. The remaining volume estimate has not been inferred from purity, positivity of a Gram, or the mere existence of the support.

The new result is therefore a completed analytic component of the current route: its two endpoint monic norms have the required $O_h(q_k)$ scale on the actual quartet window. The other session’s four-volume theorem now leaves the explicitly specified relation-volume cost as the remaining quantitative input on that window.

81.17 11. Proof and computation status

Sections 3–7 contain the new written analytic argument. Sections 8–9 contain explicit finite comparisons and a handoff for checking them using the existing quotient and matrix library. The standard complex-analysis, gamma and Legendre facts used in the argument are proved to the extent needed or referenced below. No general novelty claim is made for these classical ingredients.

The accompanying checker tests finite exact identities: coordinate factors, Legendre extremal norms, the local algebra in the interval argument, convolution moments on declared Laplace models, source quotients, block updates, raw confluent-jet endpoint factors, the volume telescope, and retained support cases. These tests do not certify the analytic inequalities by sampling them. There is no new Lean certificate and no interval-certified value of C_h .

This continuation makes no remote changes. The integration files are add-only, and the parallel formalization branches and the reader’s existing sources are left untouched.

81.18 References and pinned inputs

- Deligne, P. *La conjecture de Weil. II*. Publications Mathématiques de l’IHÉS **52** (1980), 137–252. Supplied S20 source and current page-local records; the selected reading and discrepancies are itemized in `SOURCE_REVIEW.md`.
- NIST Digital Library of Mathematical Functions: §§25.4 (zeta functional equation), 5.11 (gamma estimates), 18.3 and 18.5 (Legendre polynomials). Public source pages consulted: <https://dlmf.nist.gov/25.4>, <https://dlmf.nist.gov/5.11>, <https://dlmf.nist.gov/18.3>, <https://dlmf.nist.gov/18.5>.
- Original programme: the delivered *Tau Gamma Convolution Descent*, *Tau Toda Volume Control*, *Tau Cyclic Sum Control*, and *Tau Exterior Trace Amplification* notes. Their definitions of the source, action, unit and canonical quotient are retained; this is not a complete replay of all their upstream analytic claims.
- Parallel endpoint criterion: PR #23, `c720f40530eed2f5969dbabe94dfd3fddc0f507f`, `workbenches/tau-toda-volume-formal/ENDPOINT_CRITERION.md`.
- Parallel confluent transfer: PR #24, `dfcbbba5cbf7fec8c9301fe242c13e741d762013e`, `workbenches/tau-confluent-transfer/RESEARCH_NOTE.md`.
- Inspected main: `b32ca2128e0deb0eec6eafb860776a5d6a28dcbb`, after its merge of the original-Gram exterior/trace contribution.

Original Gamma mass, recurrence and exact monic norms

82 The exact gamma reference and the arithmetic Gram determinants

The norm sequence in (KL.24) is attached to the actual measure

$$d\mu(t) = |g(\tfrac{1}{2} + it)|^2 \frac{dt}{2\pi}, \quad g = 2\xi. \quad (\text{TG.1})$$

We construct an explicit gamma reference for this measure, prove its polynomial norms, and retain the full arithmetic multiplier in finite Gram determinants. This provides another exact formula for the original boundary allowance; no asymptotic estimate for an arithmetic determinant is assumed.

82.1 The original measure, phase, and packet cancellation

The defining identity for the completed function gives, on the original line $s = 1/2 + it$,

$$\begin{aligned} g(\tfrac{1}{2} + it) &= \Gamma(\tfrac{1}{4} + it/2) a(t), \\ a(t) &= -\pi^{-1/4} (t^2 + \tfrac{1}{4}) e^{-it \log \pi/2} \zeta(\tfrac{1}{2} + it). \end{aligned} \quad (\text{TG.2})$$

Indeed $s(s-1) = -(t^2 + 1/4)$ and $\pi^{-s/2} = \pi^{-1/4} e^{-it \log \pi/2}$. Both the minus sign and this phase remain in the multiplier.

For $\lambda > 0$, define

$$d\sigma_\lambda(t) = |\Gamma(\lambda + it/2)|^2 \frac{dt}{2\pi}. \quad (\text{TG.3})$$

In particular

$$d\mu = |a|^2 d\sigma_{1/4}, \quad d\nu_h = |a_h|^2 d\sigma_{1/4}, \quad a_h(t) = \frac{g(1/2 + it)/h(1/2 + it)}{\Gamma(1/4 + it/2)}. \quad (\text{TG.4})$$

The last expression defines the values at every zero of the original full packet h ; away from those points it equals $a(t)/h(1/2 + it)$. The function g/h is entire because the complete order of each selected zero is removed. Gamma has no zero and has no pole on the real t axis. Thus (TG.4) retains the actual cancellation at each critical packet centre, without introducing a pole into the measure.

Multiplication gives surjective Hilbert isometries

$$\begin{aligned} U_a : L^2(\mu) &\longrightarrow L^2(\sigma_{1/4}), \quad F \longmapsto aF, \\ U_{a_h} : L^2(\nu_h) &\longrightarrow L^2(\sigma_{1/4}), \quad F \longmapsto a_h F. \end{aligned} \quad (\text{TG.5})$$

The integral identity proves the isometry. The zero set of each multiplier on the real axis is discrete, since it comes from a nonzero analytic function. Given $H \in L^2(\sigma_{1/4})$, define $F = H/a$, respectively H/a_h , off this null set and give F arbitrary finite values on it. Then $\int |F|^2 d\mu = \int |H|^2 d\sigma_{1/4}$, respectively the packet identity, proving surjectivity and the stated inverse. These are maps between the indicated Hilbert spaces. Their images of polynomial subspaces are exactly $a\mathbb{C}[t]$ and $a_h\mathbb{C}[t]$, with the displayed multipliers retained. Moreover the packet isometry from (KL.23) satisfies the exact diagram

$$U_{a_h}(h(1/2 + it)P(t)) = U_a P(t). \quad (\text{TG.6})$$

82.2 Gamma norms from a generating integral

We give the integral derivation to fix all constants independently of different conventions for Meixner–Pollaczek polynomials. The beta integral, with $r = e^x$, gives

$$\int_{\mathbb{R}} \frac{e^{\lambda x}}{(1 + e^x)^{2\lambda}} e^{iux} dx = \frac{\Gamma(\lambda + iu)\Gamma(\lambda - iu)}{\Gamma(2\lambda)}. \quad (\text{TG.7})$$

For completeness the beta integral follows by applying the change of variables $v = r_1 + r_2$, $y = r_1/(r_1 + r_2)$ to the product of the two absolutely convergent gamma integrals; the Jacobian is v . The conditions $\operatorname{Re}(\lambda \pm iu) > 0$ hold for real u . The function on the left before Fourier transformation and its second derivative are integrable: at each end they are bounded by a constant times $e^{-\lambda|x|}$. Two integrations by parts therefore make its Fourier transform integrable. Fourier inversion applies. One direct justification is to multiply the transform by $e^{-\varepsilon u^2}$, apply Fubini, and use the Gaussian approximate identity; dominated convergence removes that factor because the transform is integrable. Using its evenness and then $t = 2u$, we obtain

$$\int_{\mathbb{R}} e^{iyt} d\sigma_{\lambda}(t) = \frac{2\Gamma(2\lambda)}{(2 \cosh y)^{2\lambda}} = 2^{1-2\lambda} \Gamma(2\lambda) (\cosh y)^{-2\lambda}. \quad (\text{TG.8})$$

The factor 2 in the numerator is the Jacobian of the original change $t = 2u$.

The following integral proves the required exponential integrability.

The function $(2 \cosh(x/2))^{-2\lambda}$ extends holomorphically to $|\operatorname{Im} x| < \pi$, using the branch positive on the real axis. On each closed narrower strip it is bounded by $Ce^{-\lambda|\operatorname{Re} x|}$. In (TG.7), for $u > 0$, shift the contour to $\mathbb{R} + ic$, where $0 < c < \pi$. The vertical sides of the truncated rectangle tend to zero by that bound. For $u < 0$, shift down instead. It follows that

$$|\Gamma(\lambda + iu)|^2 \leq C_{\lambda, c} e^{-c|u|}.$$

Consequently $\int e^{v|t|} d\sigma_{\lambda}(t) < \infty$ for every $0 < v < \pi/2$, by choosing $2v < c < \pi$. This proves holomorphic continuation of both sides of (TG.8) to $|\operatorname{Im} y| < \pi/2$, where they agree by the identity theorem and use the branch inherited from the real axis.

For z near zero use the logarithms that vanish at $z = 0$ and set

$$F_{\lambda}(z, t) = (1 - iz)^{-\lambda + it/2} (1 + iz)^{-\lambda - it/2} = \sum_{n \geq 0} P_n^{(\lambda)}(t/2; \pi/2) z^n. \quad (\text{TG.9})$$

For real z near zero this is $(1 + z^2)^{-\lambda} \exp(t \arctan z)$. Apply (TG.8) at $y = -i(\arctan z + \arctan w)$, where z, w are real and sufficiently small. The identity

$$\cos(\arctan z + \arctan w) = \frac{1 - zw}{\sqrt{1 + z^2} \sqrt{1 + w^2}}$$

then proves

$$\int_{\mathbb{R}} F_{\lambda}(z, t) F_{\lambda}(w, t) d\sigma_{\lambda}(t) = 2^{1-2\lambda} \Gamma(2\lambda) (1 - zw)^{-2\lambda}. \quad (\text{TG.10})$$

All derivatives in z, w at zero can be passed through the integral: on a smaller closed bidisc their integrands are bounded by a polynomial in $|t|$ times $e^{c|t|}$, with $c < \pi/2$. Thus comparison of the Taylor coefficients in (TG.10) is justified, not formal.

Define the monic polynomial and its unchanged squared norm by

$$b_n^{(\lambda)}(t) = n! P_n^{(\lambda)}(t/2; \pi/2), \quad \gamma_n^{(\lambda)} = 2^{1-2\lambda} n! \Gamma(n + 2\lambda). \quad (\text{TG.11})$$

The coefficient of t^n in (TG.9) is $1/n!$, since $\arctan z = z + O(z^3)$. Consequently $b_n^{(\lambda)}$ is monic. Equation (TG.10) gives the full orthogonality identity

$$\int_{\mathbb{R}} b_n^{(\lambda)}(t) b_j^{(\lambda)}(t) d\sigma_{\lambda}(t) = \gamma_n^{(\lambda)} \delta_{nj}. \quad (\text{TG.12})$$

All these polynomials have real coefficients, as is also clear from the real form of the generating function. Differentiating (TG.9) gives $(1 + z^2) \partial_z F_{\lambda} = (t - 2\lambda z) F_{\lambda}$. Its coefficient of z^n proves

$$\begin{aligned} b_0^{(\lambda)} &= 1, & b_1^{(\lambda)} &= t, \\ b_{n+1}^{(\lambda)} &= t b_n^{(\lambda)} - n(n + 2\lambda - 1) b_{n-1}^{(\lambda)}, & n &\geq 1, \\ \gamma_n^{(\lambda)} / \gamma_{n-1}^{(\lambda)} &= n(n + 2\lambda - 1), & n &\geq 1. \end{aligned} \quad (\text{TG.13})$$

For the actual reference $\lambda = 1/4$, abbreviate $b_n = b_n^{(1/4)}$ and $\gamma_n = \gamma_n^{(1/4)}$. Thus

$$\boxed{\gamma_n = \sqrt{2} n! \Gamma(n + \tfrac{1}{2}), \quad \gamma_{n+1}/\gamma_n = (n+1)(n + \tfrac{1}{2}).} \quad (\text{TG.14})$$

These formulas agree, with the displayed Jacobian and monic factors, with NIST DLMF §18.19, equations (18.19.8)–(18.19.9), and §18.22, equation (18.22.8).

82.3 Finite arithmetic determinants in the retained metrics

For $N \geq 0$ let $\Gamma_N = \text{diag}(\gamma_0, \dots, \gamma_N)$, and keep both matrices

$$\begin{aligned} \mathbf{B}_N &= \left(\int_{\mathbb{R}} b_i(t) b_j(t) |a(t)|^2 d\sigma_{1/4}(t) \right)_{0 \leq i, j \leq N}, \\ \mathbf{J}_N &= \Gamma_N^{-1/2} \mathbf{B}_N \Gamma_N^{-1/2}, \quad D_N = \det \mathbf{J}_N, \quad D_{-1} = 1. \end{aligned} \quad (\text{TG.15})$$

The diagonal factors are explicit invertible coordinate maps; $\mathbf{B}_N = \Gamma_N^{1/2} \mathbf{J}_N \Gamma_N^{1/2}$ recovers every original inner product. The integrals converge with no unproved growth estimate. Indeed, integration of the step function $\lfloor x \rfloor$ gives, initially for $\text{Re } s > 1$, and then by analytic continuation for $\text{Re } s > 0$, $s \neq 1$,

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx.$$

The initial equality follows by integrating each constant step on $[n, n+1)$, or by summation by parts of the absolutely convergent Dirichlet series. The last integral converges absolutely and locally uniformly in the asserted half-plane since $0 \leq \{x\} < 1$. On $s = 1/2 + it$, it proves $|\zeta(s)| \leq 1 + 2\sqrt{t^2 + 1/4}$, because $|s/(s-1)| = 1$. Equation (TG.2) therefore bounds $|a(t)|$ by a polynomial of degree three. The exponential integrability just proved gives all its polynomial moments. The packet multiplier has the same property on the tail because h is monic, and is bounded on each compact interval by its cancelled analytic expression in (TG.4). Each matrix is positive definite: the integral of the squared modulus of a nonzero polynomial is positive, because a vanishes only at isolated real points.

Theorem 82.1 (The actual theta norms as arithmetic determinants). *For all $n \geq 0$, with the initial value included,*

$$\boxed{\mathfrak{h}_n = \gamma_n \frac{D_n}{D_{n-1}}, \quad \frac{\mathfrak{h}_{n+1}}{\mathfrak{h}_n} = (n+1)(n + \tfrac{1}{2}) \frac{D_{n+1} D_{n-1}}{D_n^2}.} \quad (\text{TG.16})$$

Proof. Let $Q_n(t)$ be the monic polynomial orthogonal to degrees below n in $L^2(\mu)$. Its existence and uniqueness follow from the positive definite Gram matrix. Expressing $Q_0, \dots, Q_N$ in $b_0, \dots, b_N$ gives an upper triangular matrix with diagonal entries one. The determinant of the Gram matrix is therefore unchanged, and orthogonality gives $\det \mathbf{B}_N = \prod_{j=0}^N \|Q_j\|_\mu^2$.

The algebra isomorphism $\Psi : \mathbb{C}[s] \rightarrow \mathbb{C}[t]$, $(\Psi p)(t) = p(1/2 + it)$, preserves the original integral inner product. On the affine space of degree- n monic polynomials the map $p \mapsto i^{-n} \Psi p$ is its monic version and preserves the norm because $|i^{-n}| = 1$. It takes all lower-degree polynomial subspaces onto themselves. Thus the original monic theta residual has squared norm $\mathfrak{h}_n = \|Q_n\|_\mu^2$. The degree-dependent phase is used on this affine space; it is not asserted to define a linear map on the entire polynomial algebra.

Finally $D_N = \det \mathbf{B}_N / \prod_{j=0}^N \gamma_j$. Divide this equality at $N = n$ by the equality at $N = n-1$, taking an empty product to be one. This proves the first formula. Dividing consecutive formulas and using (TG.14) proves the second. $\square$

Define $B_N^{(h)}, J_N^{(h)}, D_N^{(h)}$ by (TG.15) with the entire-cancelled multiplier a_h of (TG.4). The same proof, with no restriction on whether the packet centres are critical, gives

$$\kappa_n^{(h)} = \gamma_n \frac{D_n^{(h)}}{D_{n-1}^{(h)}}. \quad (\text{TG.17})$$

Combining (TG.17) and (TG.16) with (KL.24) yields the exact packet-to-reference determinant map, for every $m \geq 0$,

$$r_m^{(h)} = \frac{(d+m)! \Gamma(d+m+1/2)}{m! \Gamma(m+1/2)} \frac{D_{d+m}^{(h)}}{D_{d+m-1}^{(h)}} \frac{D_{m-1}}{D_m}. \quad (\text{TG.18})$$

Every determinant is strictly positive and the full degree $d = \sum_{\rho} m_{\rho}$ is retained.

For a packet stable under both ordinary conjugation and $\rho \mapsto 1 - \bar{\rho}$, and for every $m \geq 0$, (KL.32) becomes

$$\epsilon_m^2 = (m+1)(m+\frac{1}{2}) \frac{D_{m+1} D_{m-1}}{D_m^2} (r_m^{-1} - 1)(1 - r_{m+1}). \quad (\text{TG.19})$$

This is the exact same finite-stage allowance. Its factor $(m+1)(m+1/2)$ is explicit; the arithmetic determinant quotient remains in the formula.

82.4 Full confluent kernel and the original coordinates

Set $\hat{b}_j(s) = i^j b_j((s-1/2)/i)$. These are monic polynomials in the original variable s . With $\hat{\Gamma}_N = \text{diag}(1, i, \dots, i^N)$, their actual packet Gram matrix is

$$B_{N,s}^{(h)} = \hat{\Gamma}_N^* B_N^{(h)} \hat{\Gamma}_N. \quad (\text{TG.20})$$

Indeed $\hat{b}_j(1/2+it) = i^j b_j(t)$; taking the conjugate in the first argument gives exactly the displayed congruence, including every phase.

Let C_N have columns $[\hat{b}_j]_h \in E_h = \mathbb{C}[s]/(h)$, in the original monomial remainder coordinates. For $N \geq d-1$,

$$K_N = C_N (B_{N,s}^{(h)})^{-1} C_N^*, \quad \mathbf{G}_{N-d} = U_{\varepsilon_h}^* K_N^{-1} U_{\varepsilon_h}. \quad (\text{TG.21})$$

Here $\mathbf{G}_{-1}$ is the fixed initial-section metric, and $\varepsilon_h = (j_h(g/h))^{-1}$ is the same full local unit as before. To prove the first identity, choose the actual monic orthogonal polynomial basis through degree N . If T is its invertible coefficient matrix in the $\hat{b}_j$ basis, then

$$T^* B_{N,s}^{(h)} T = \text{diag}(\kappa_0, \dots, \kappa_N).$$

Inverting gives $(B_{N,s}^{(h)})^{-1} = T \text{diag}(\kappa_j^{-1}) T^*$. The columns of $C_N T$ are the full remainders $[p_j]_h$. Substitution proves exactly $K_N = \sum_{j=0}^N [p_j]_h [p_j]_h^* / \kappa_j$, the original interpolation kernel. The first d monic polynomials span all remainders, so C_N is onto and K_N is positive definite. The second identity is the fixed-unit minimum-lift formula already proved in (KL.27), with the exact index $N = d+m$. No single remainder column is required to be nonzero or invertible.

For a centre ρ , put $t_{\rho} = (\rho - 1/2)/i$ and write $f^{[\ell]} = f^{(\ell)}/\ell!$. The full Taylor coordinates of these columns are explicitly

$$\hat{b}_j^{[\ell]}(\rho) = i^{j-\ell} b_j^{[\ell]}(t_{\rho}), \quad 0 \leq \ell < m_{\rho}. \quad (\text{TG.22})$$

This follows by differentiating the affine argument ℓ times. CRT converts these full Taylor coordinates to the monomial remainder coordinates, and multiplication by U_{v_h} retains all derivatives of g/h . Thus (TG.21) computes the entire confluent kernel, including nilpotent orders and critical centres.

82.5 The finite map to Romik's gamma parameter

Romik's expansion uses $P_n^{(3/4)}(t/2; \pi/2)$, with the unchanged variable $t/2$; see his §3 and Appendix A.2. The exact finite change to the parameter in (TG.3) is

$$\begin{aligned} b_n^{(1/4)}(t) &= \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n!}{(n-2k)!} \binom{1/2}{k} b_{n-2k}^{(3/4)}(t), \\ b_n^{(3/4)}(t) &= \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n!}{(n-2k)!} \binom{-1/2}{k} b_{n-2k}^{(1/4)}(t). \end{aligned} \quad (\text{TG.23})$$

To prove these formulas, divide the two generating functions: $F_{1/4}(z, t) = (1 + z^2)^{1/2} F_{3/4}(z, t)$. The binomial series at zero and coefficient comparison prove the first formula. Multiplication by $(1 + z^2)^{-1/2}$ proves the inverse. Each finite coefficient matrix is triangular with diagonal one, preserves parity, and is invertible. Applied to the same measure, it transports Gram matrices by its full congruence and transports every Taylor column by the same coefficient matrix. Consequently the kernel expression (TG.21) is invariant under this change of polynomial basis.

The two reference measures themselves have the exact Hilbert-space map

$$L^2(\sigma_{1/4}) \longrightarrow L^2(\sigma_{3/4}), \quad F(t) \longmapsto \frac{\Gamma(1/4 + it/2)}{\Gamma(3/4 + it/2)} F(t). \quad (\text{TG.24})$$

The gamma quotient has neither zeros nor poles for real t . Cancellation of its squared modulus proves the isometry and its inverse; this isometry has a multiplier, whereas (TG.23) is the finite polynomial coordinate map. Their respective domains and metrics have both been given explicitly.

Romik's coefficient asymptotics concern the expansion of Ξ in a fixed gamma-orthogonal basis. Inoue's §2 similarly treats expansion coefficients and convergence. In the present construction the arithmetic norms are the determinants (TG.16). Equations (TG.20)–(TG.24) are the exact maps connecting those polynomial presentations to these determinants; no assertion that their coefficient asymptotics already estimate D_n is used.

82.6 Exact scope of the exponential-weight theorem

The original theorem of Lubinsky–Mhaskar–Saff does include tail order one. In their 1988 paper, Definition 2.1 requires $W = e^{-Q}$, with Q even and continuous, Q' existing on $(0, \infty)$, $xQ'(x)$ bounded at zero, and, eventually,

$$Q' > 0, \quad x^2|Q'''|/Q' \leq C, \quad 1 + xQ''/Q' \longrightarrow \alpha > 0. \quad (\text{TG.25})$$

Theorem 2.3 admits a finite generalized Jacobi factor, a lower-degree polynomial exponential, and a bounded nonnegative factor $\Psi(t) \rightarrow 1$. Its Theorem 2.4 replaces the polynomial exponential by V , but requires $\log V(t)/Q(t) \rightarrow 0$, as well as bounded reciprocal approximation on the expanding interval by polynomials of degree $o(n)$. The precise statements are on printed pages 67–70; their proof is on pages 80–81. These hypotheses concern the full weight, not its gamma factor alone.

We prove a concrete obstruction to inserting the actual arithmetic multiplier into those displayed hypotheses. Hardy's theorem supplies real numbers $T_j \rightarrow \infty$ with $g(1/2 + iT_j) = 0$. Removing a fixed full polynomial packet leaves an infinite subsequence of these zeros. At each remaining T_j , the multiplier a_h of (TG.4) has the local form

$$a_h(t) = (t - T_j)^{\ell_j} v_j(t), \quad \ell_j \geq 1, \quad v_j(T_j) \neq 0, \quad (\text{TG.26})$$

where v_j is analytic in a neighbourhood of T_j . This follows from the Taylor series at an isolated zero, with its entire order ℓ_j retained.

Let $B(t) > 0$ be any continuous comparison amplitude near all sufficiently large T_j , and suppose it has no zeros there. For every j and every $\eta > 0$, continuity and (TG.26) give a nonempty interval $I_{j,\eta}$ about T_j on which

$$|a_h(t)|/B(t) < \eta. \quad (\text{TG.27})$$

Thus a decomposition with this ratio as the bounded multiplier Ψ cannot satisfy $\Psi \rightarrow 1$. The obstruction holds on intervals of positive measure, so changing the value only at each zero does not repair it. A finite generalized Jacobi factor has only finitely many real zeros or singularities; choosing the subsequence beyond them leaves (TG.27) unchanged.

For the logarithmic hypothesis choose any eventually positive continuous Q . At a remaining T_j , shrink the neighbourhood so that $Q(t) \leq Q(T_j) + 1$, and choose $\eta = \exp(-j(Q(T_j) + 1))$ in (TG.27). On the resulting positive-measure interval,

$$\frac{\log(|a_h(t)|/B(t))}{Q(t)} < -j \quad (t \neq T_j). \quad (\text{TG.28})$$

Hence this ratio cannot be V with $\log V/Q \rightarrow 0$. An additional factor $\Psi \rightarrow 1$ is bounded away from zero and infinity on the tail and does not remove the contradiction. This disproves the indicated application of Theorems 2.3–2.4, including almost-everywhere modifications of the weight. It makes no claim that other asymptotic methods are unavailable. The exact positive Gram matrices (TG.15)–(TG.21) retain the arithmetic information at all those zeros and are available for such further analysis.

82.7 Primary source locators

The content-reading route and source hashes are recorded in `work/theta_norm_literature_route_20260912.json` and `work/theta_norm_literature_sources/download_receipt.json`. The principal public locators are:

- D. S. Lubinsky, H. N. Mhaskar and E. B. Saff, *A proof of Freud's conjecture for exponential weights*, Constructive Approximation 4 (1988), 65–83, Definition 2.1, Theorems 2.3–2.5 and §4: <https://math.vanderbilt.edu/saffeb/texts/81.pdf>.
- Dan Romik, *Orthogonal polynomial expansions for the Riemann xi function in the Hermite, Meixner–Pollaczek, and continuous Hahn bases*, §3.1–3.3 and Appendix A.2: <https://www.math.ucdavis.edu/~romik/data/uploads/papers/riemannxi-final.pdf>.
- Hiroto Inoue, *Expansion of Dirichlet L-function on the critical line in Meixner–Pollaczek polynomials*, arXiv:1412.1220, §2.1–2.2: <https://arxiv.org/abs/1412.1220>.
- NIST DLMF, equations (18.19.8)–(18.19.9), (18.22.8): <https://dlmf.nist.gov/18.19>, <https://dlmf.nist.gov/18.22>.
- G. H. Hardy, *Sur les zéros de la fonction $\zeta(s)$ de Riemann*, Comptes rendus 158 (1914), 1012–1014, §1–3. The exact page images and transcriptions are linked in the machine-readable source route.

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